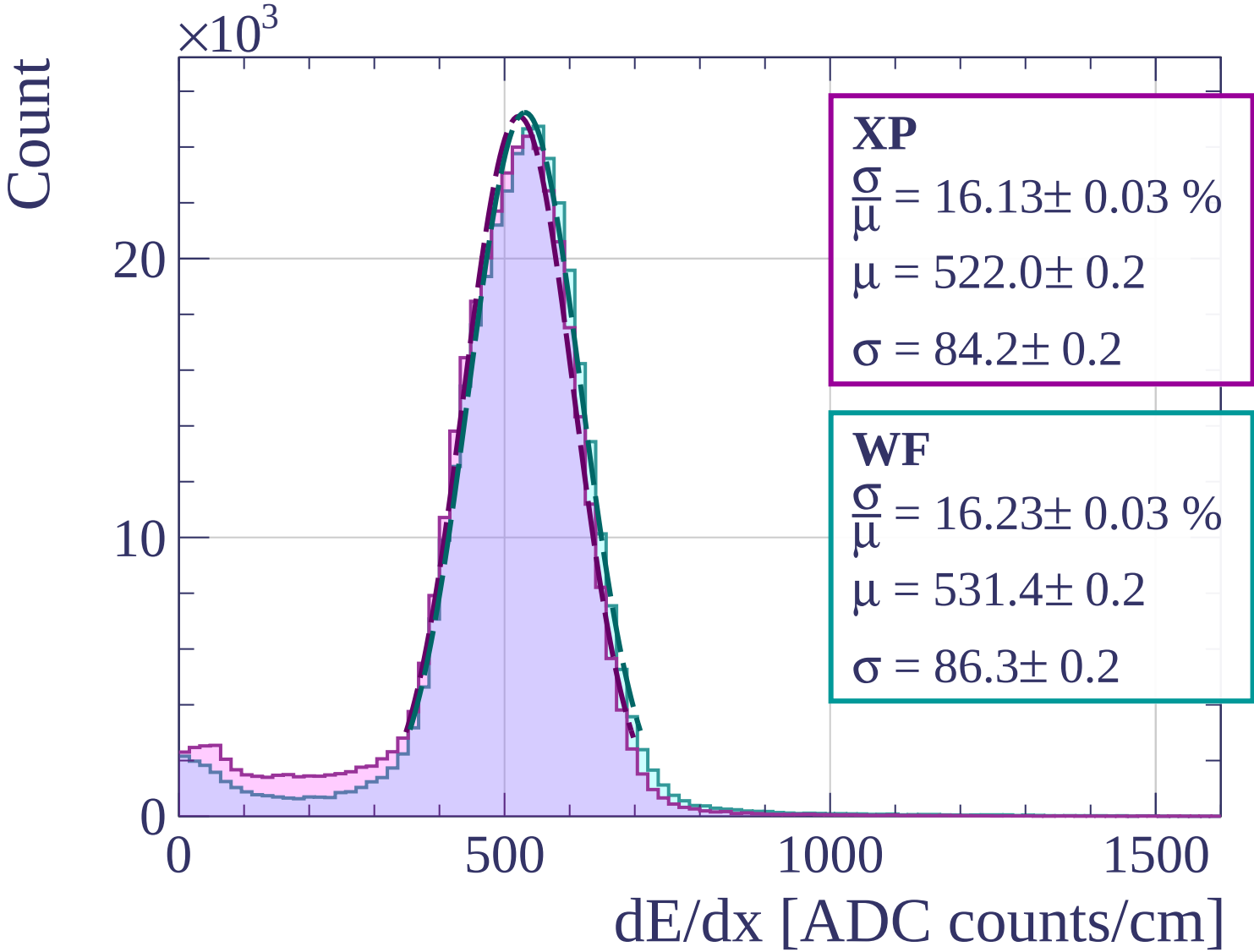
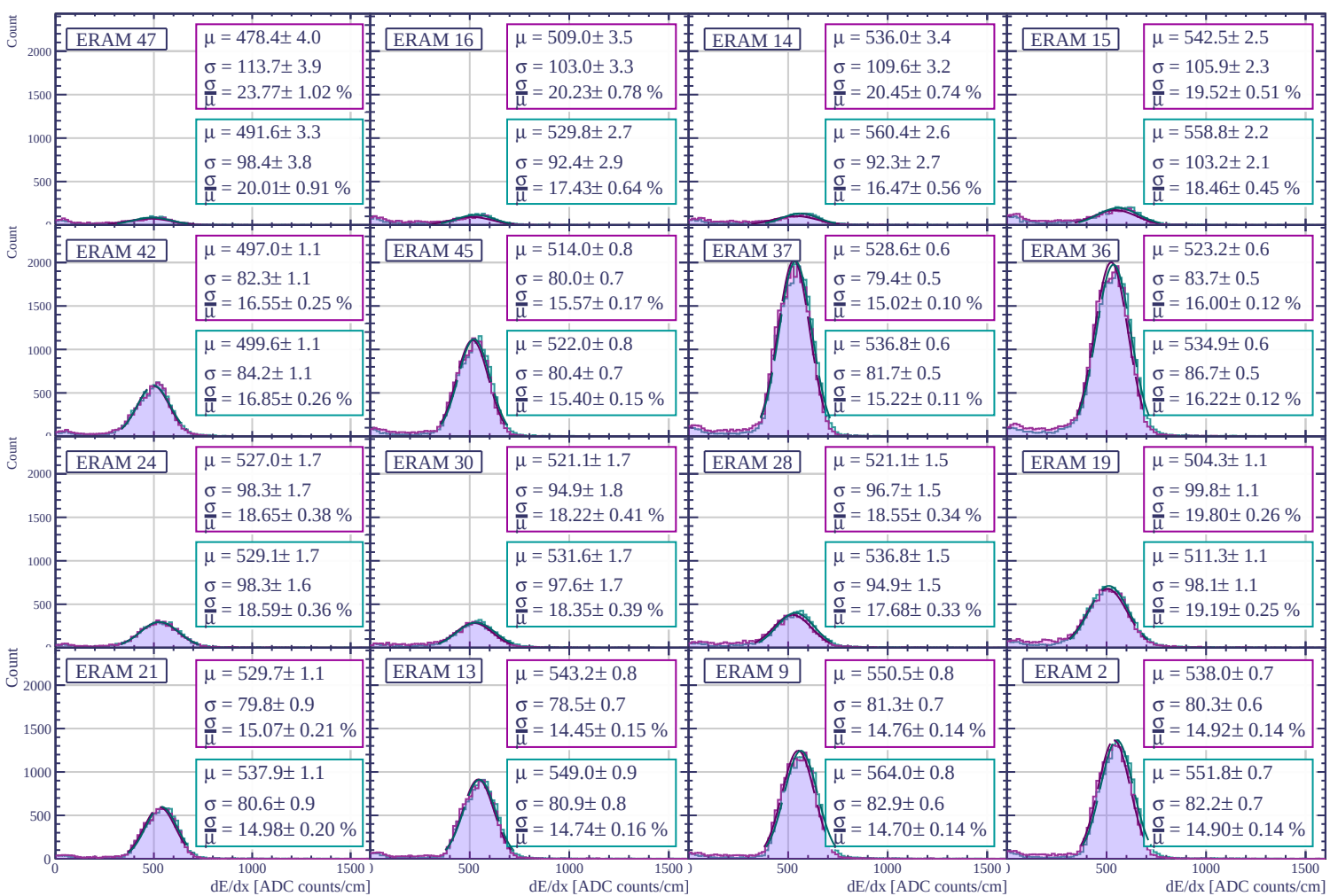
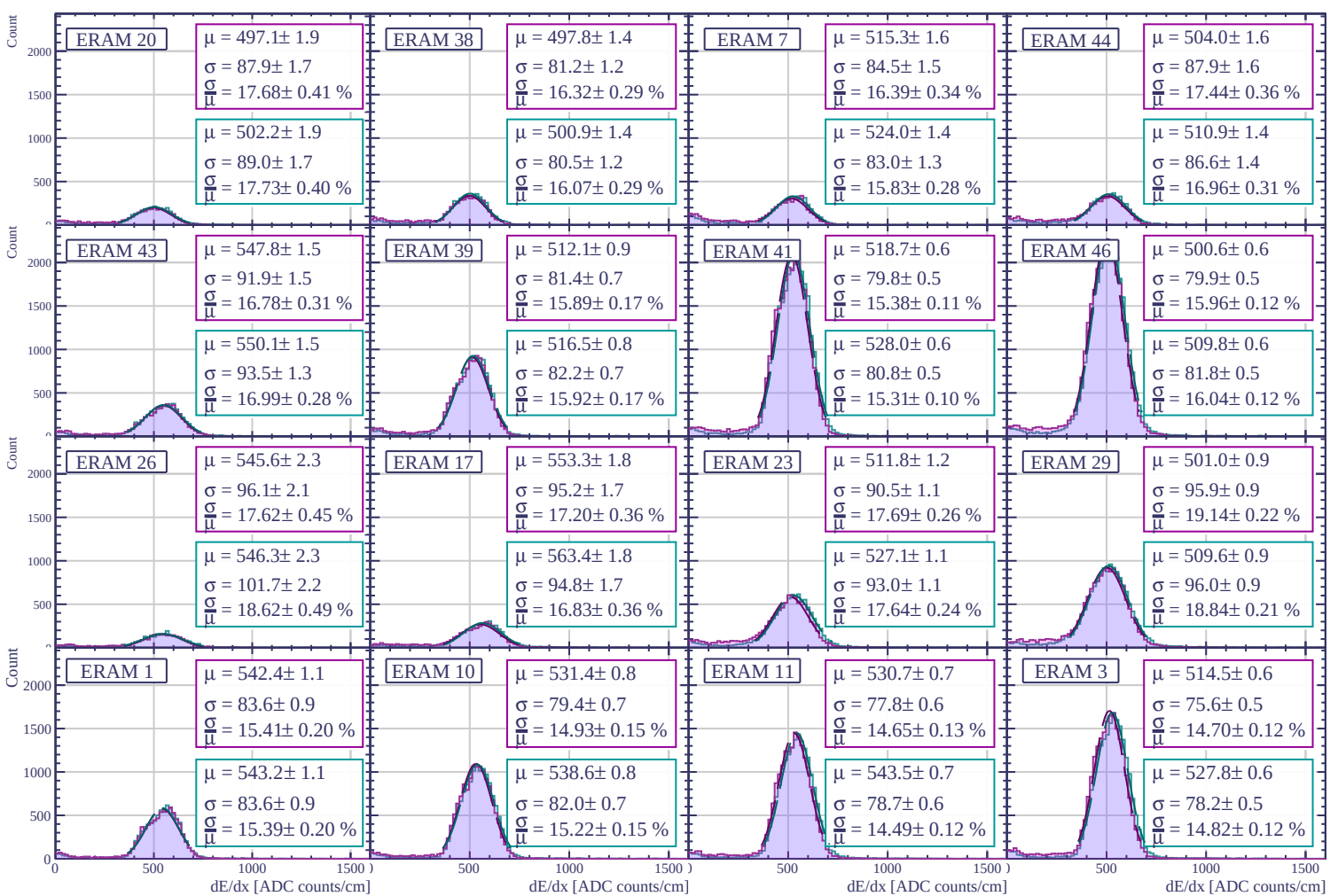
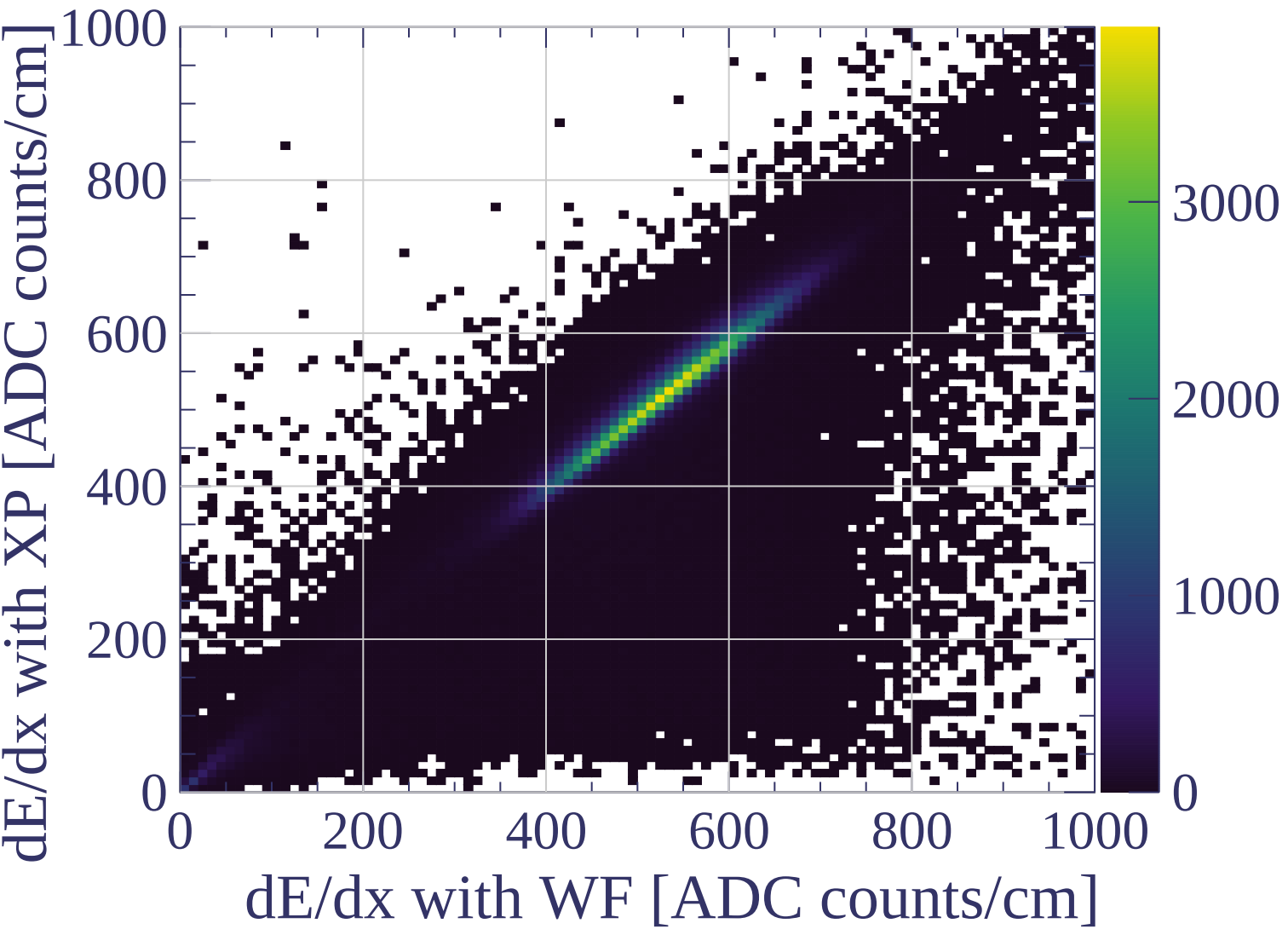
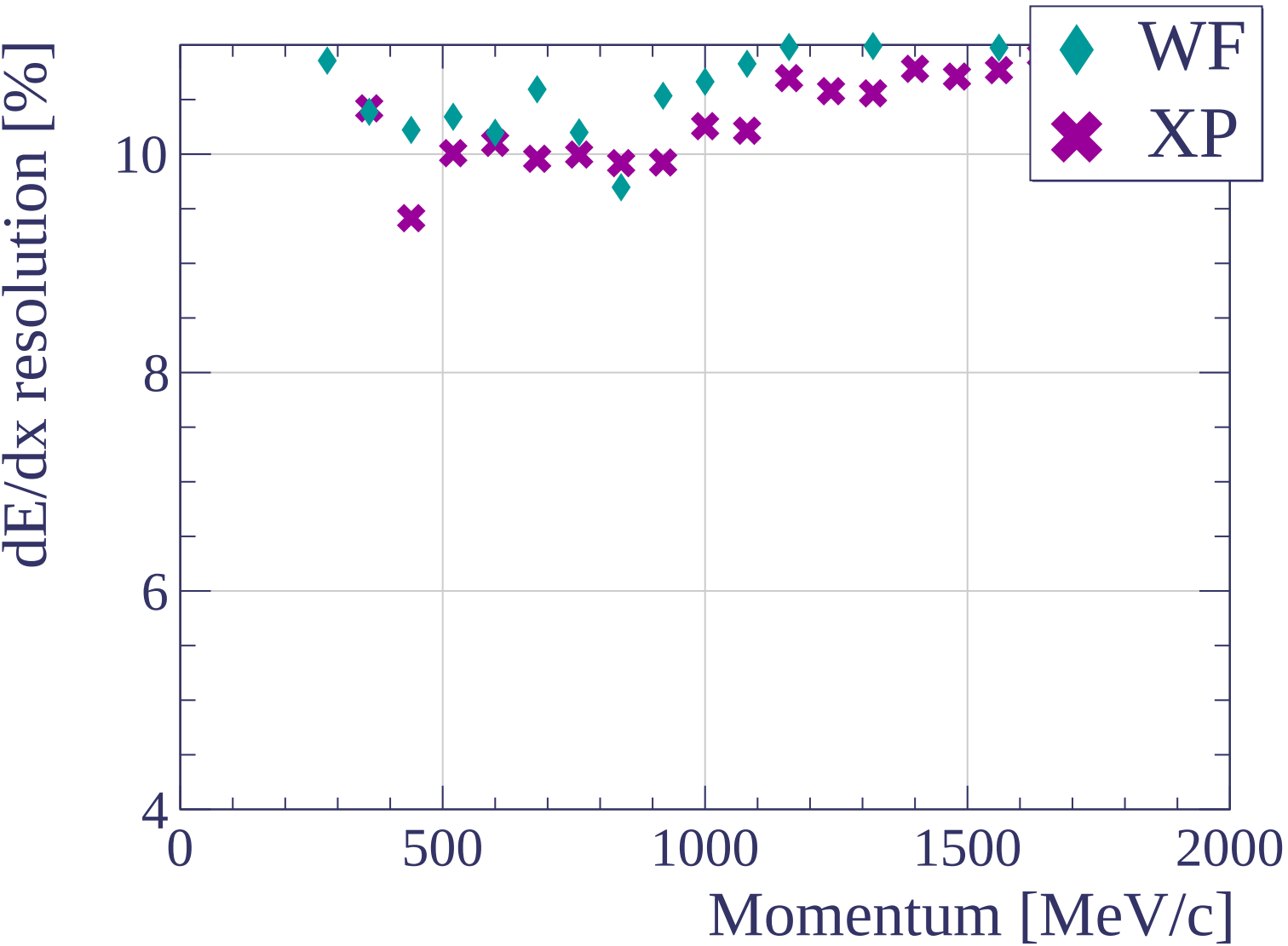


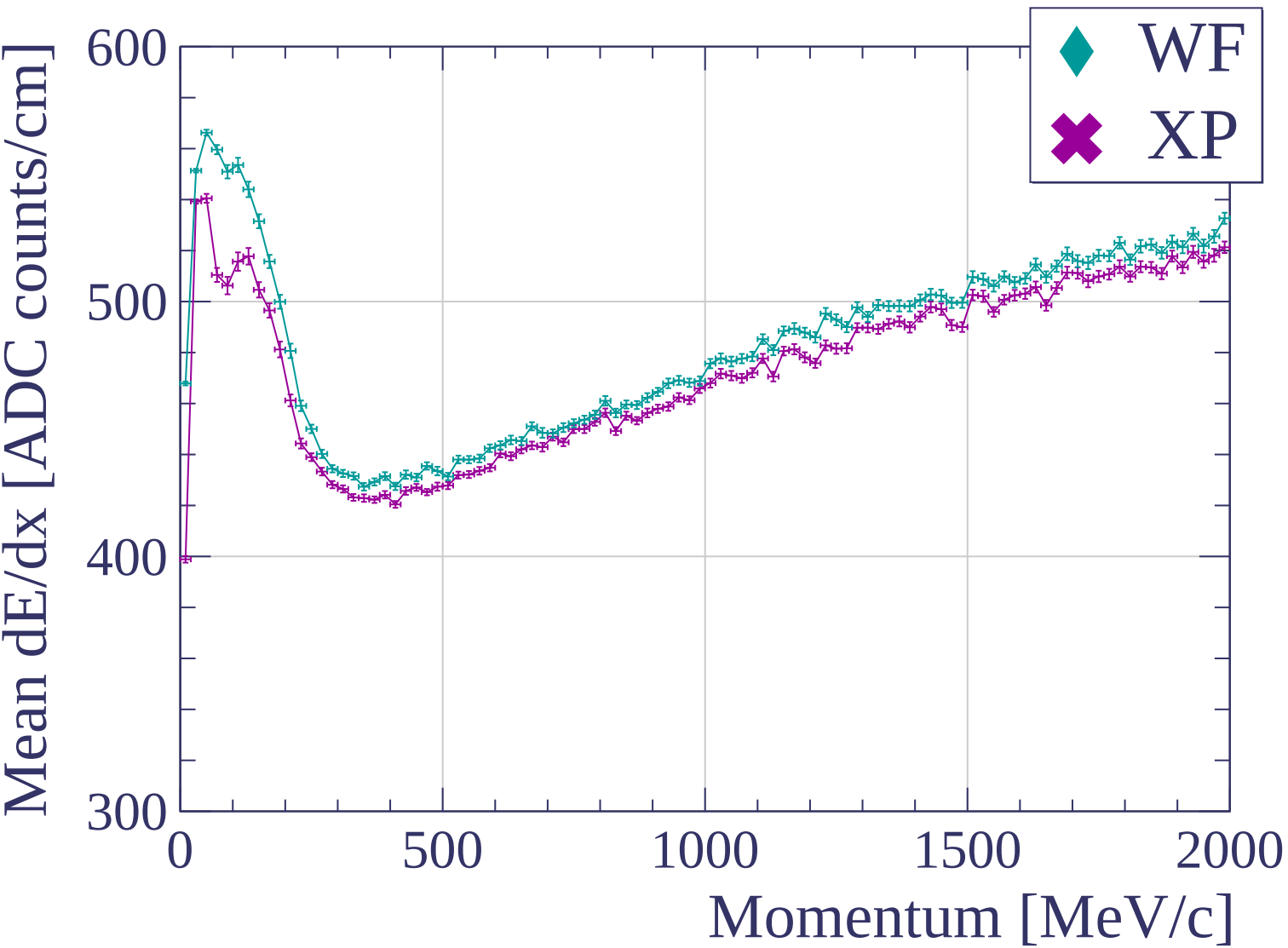


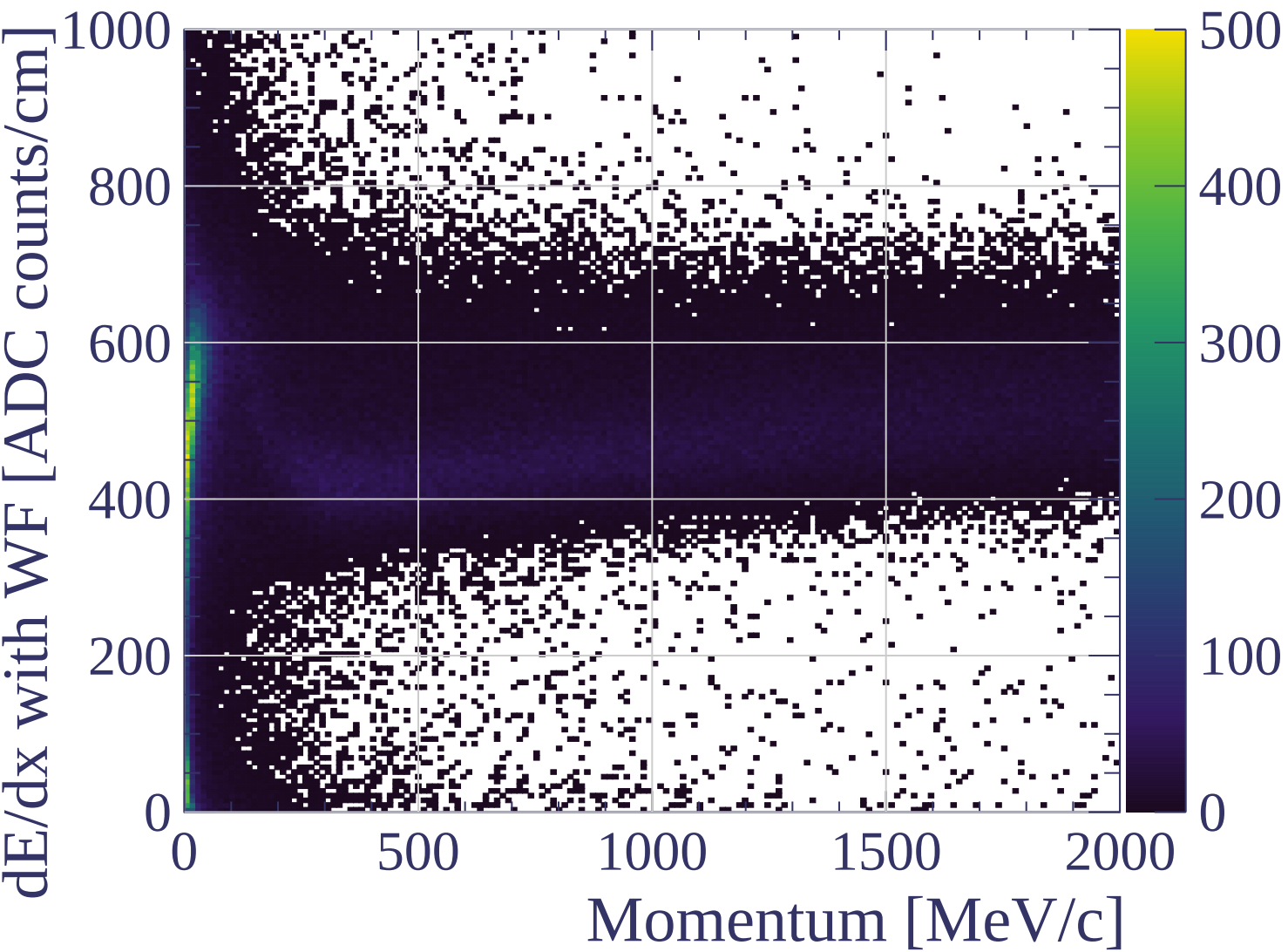


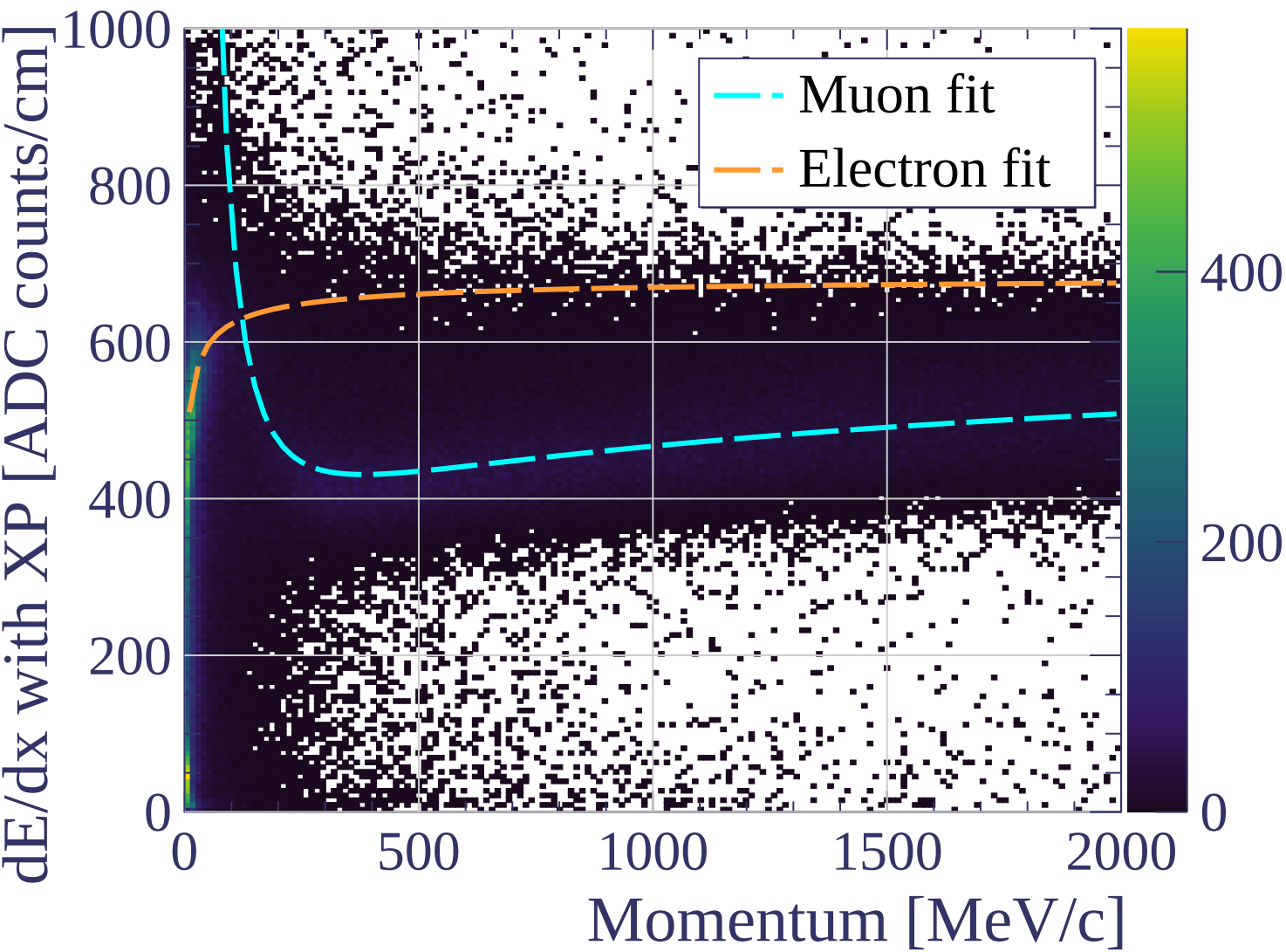




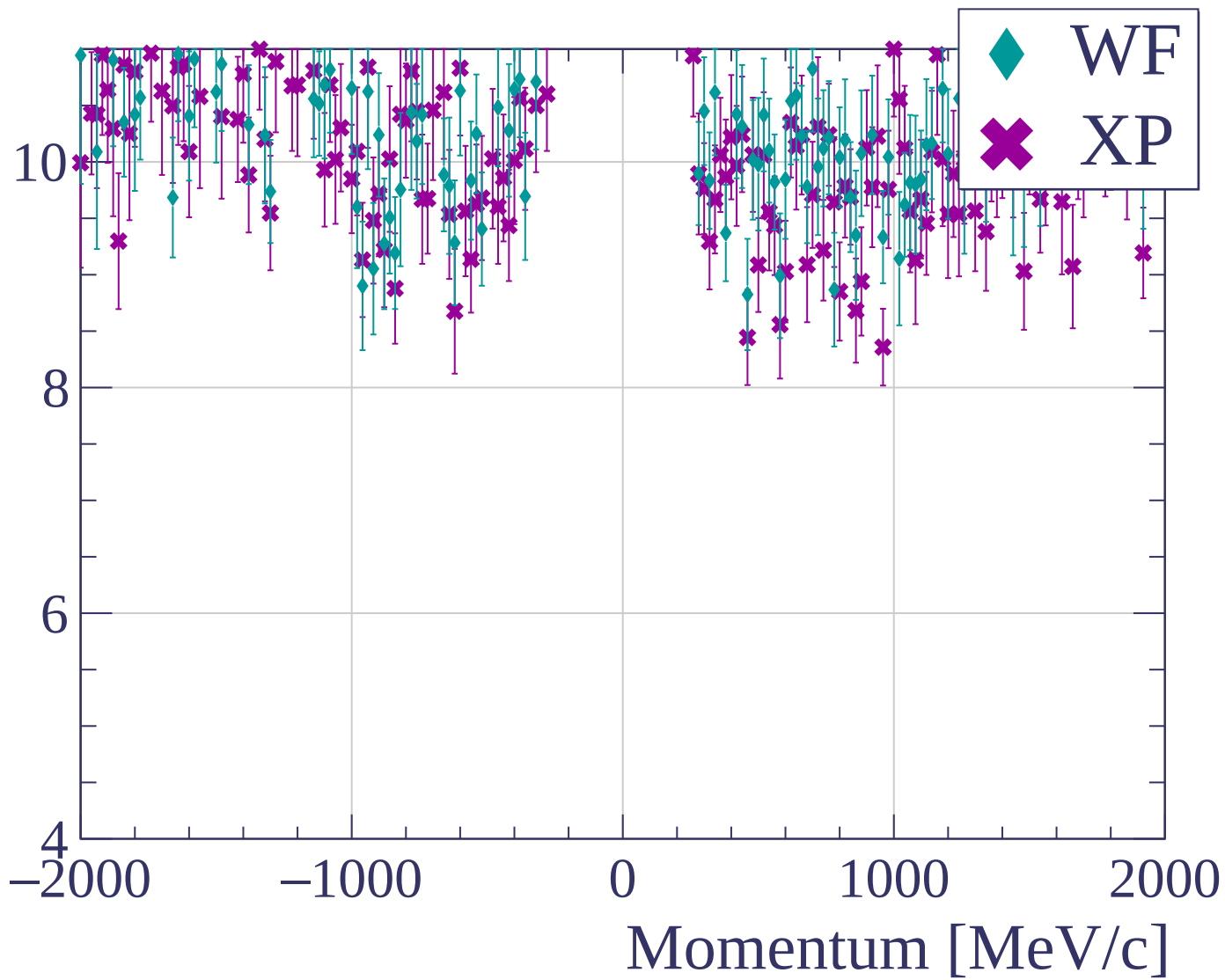


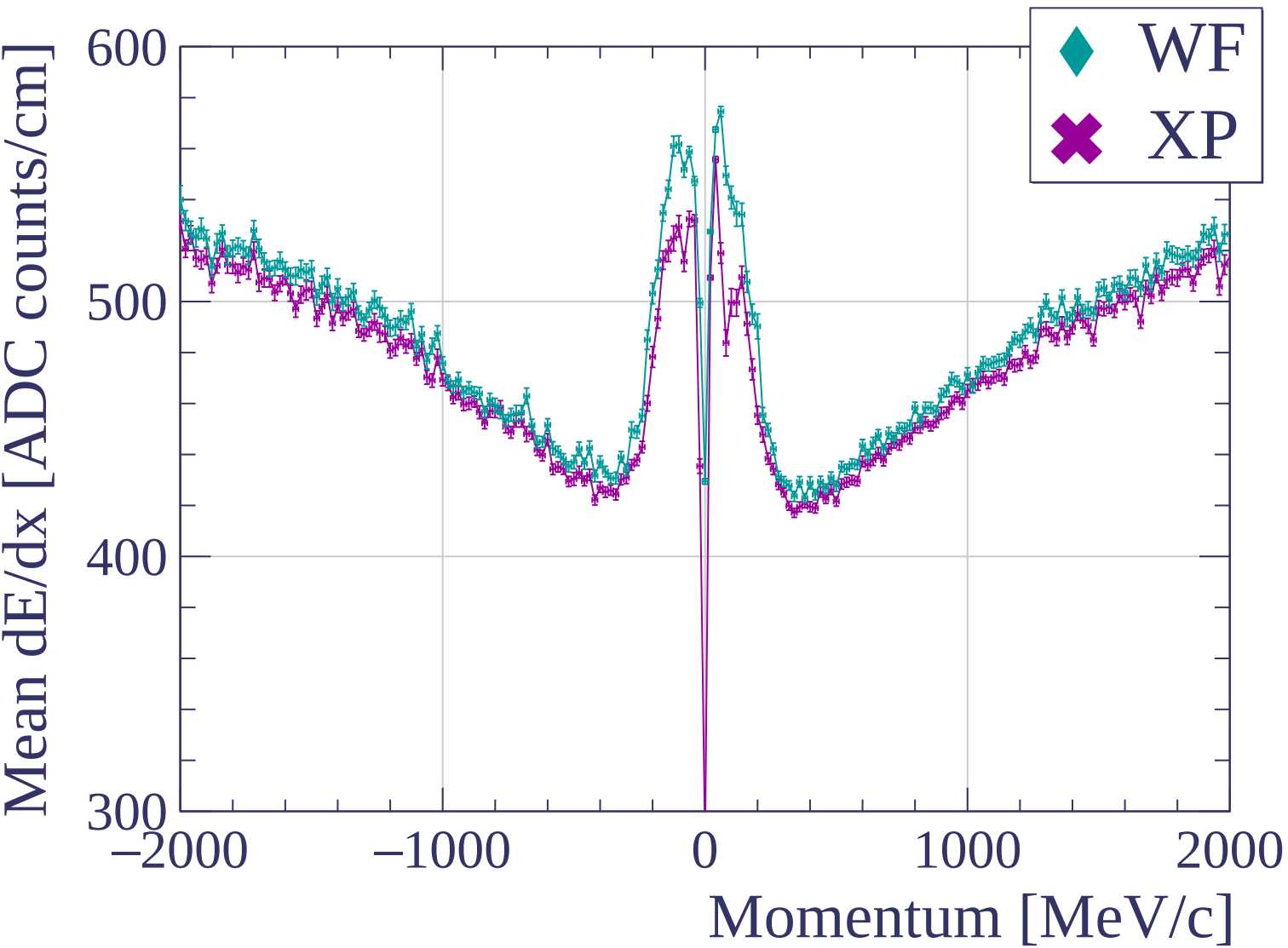


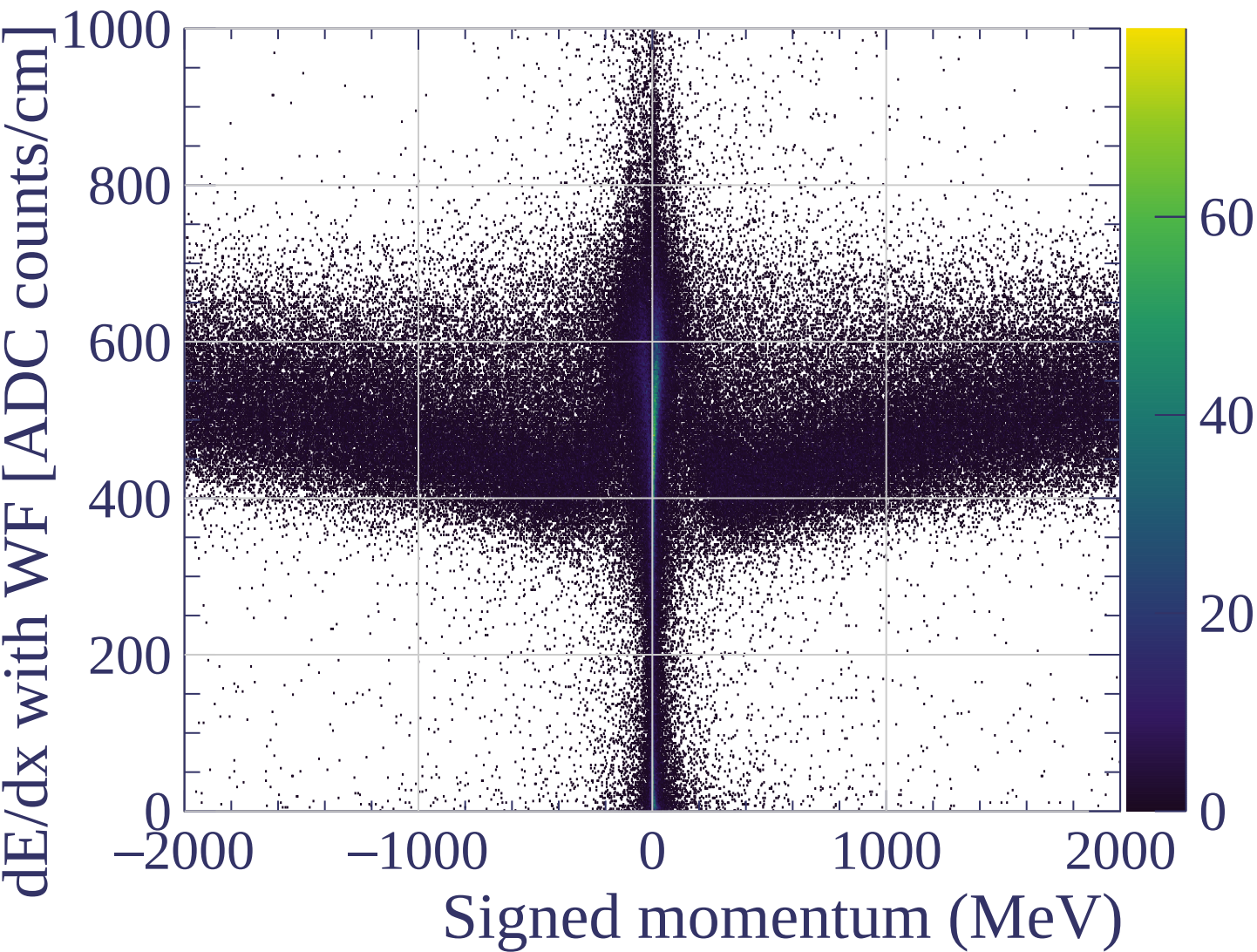


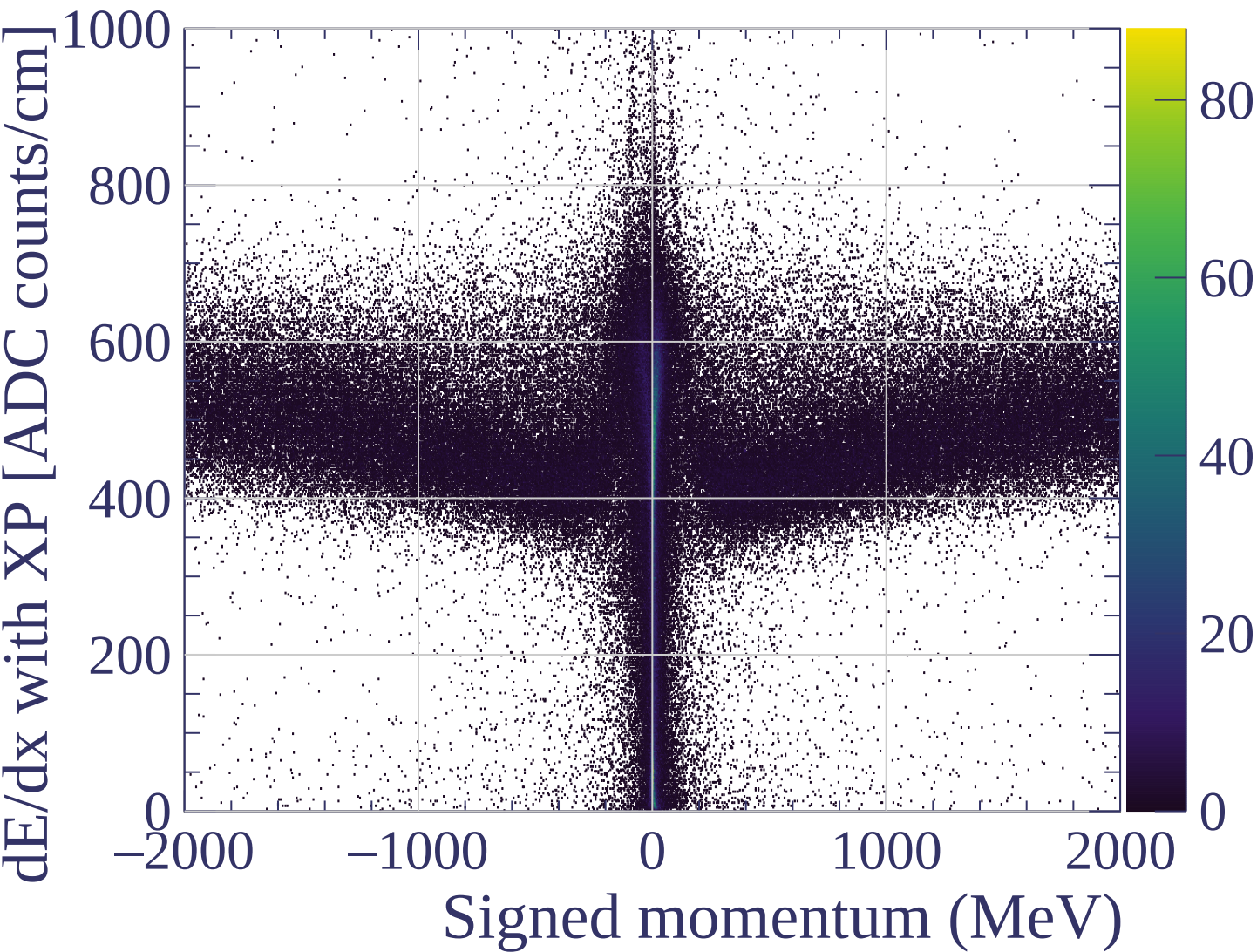


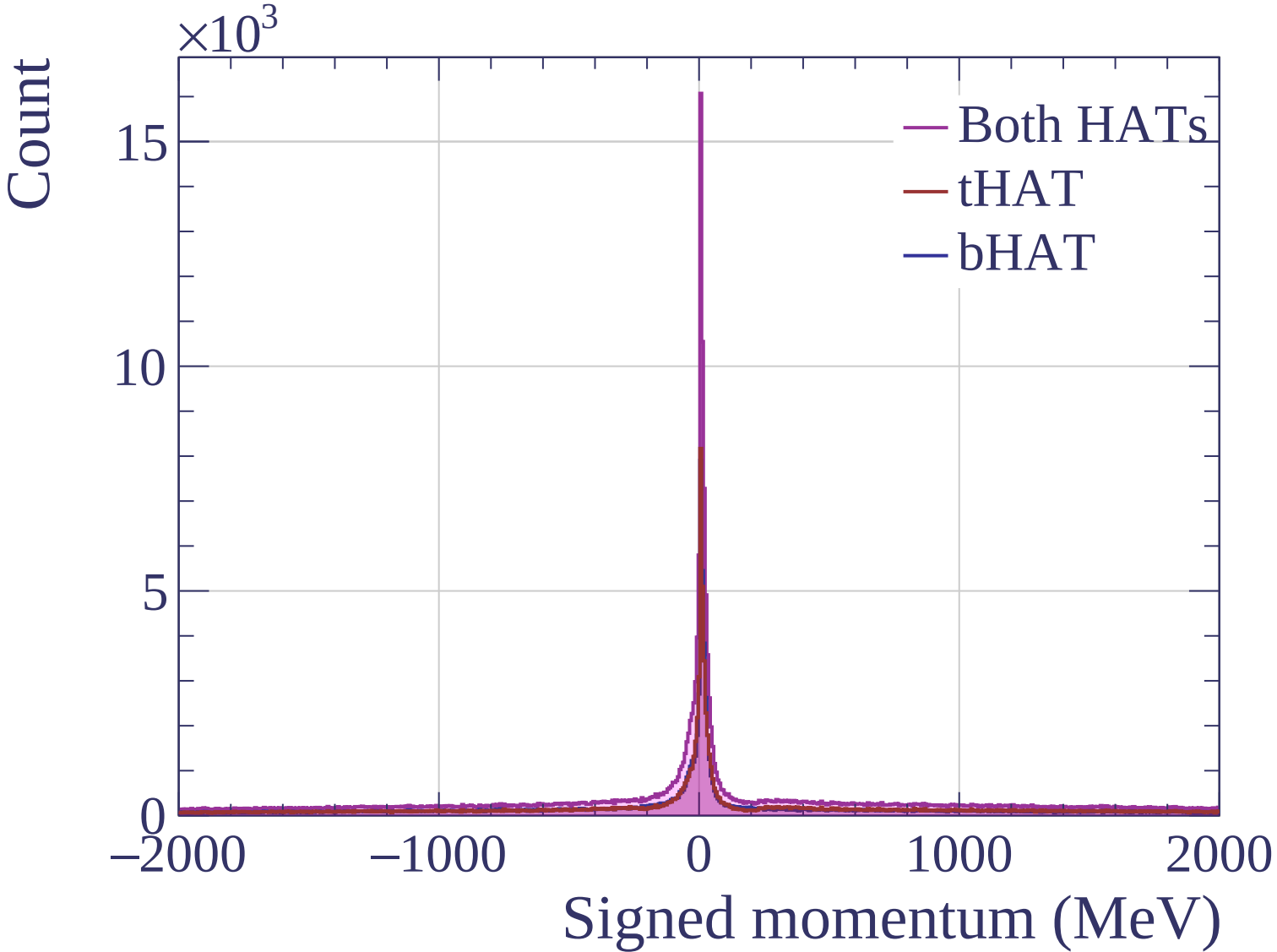
dE/dx resolution [%]

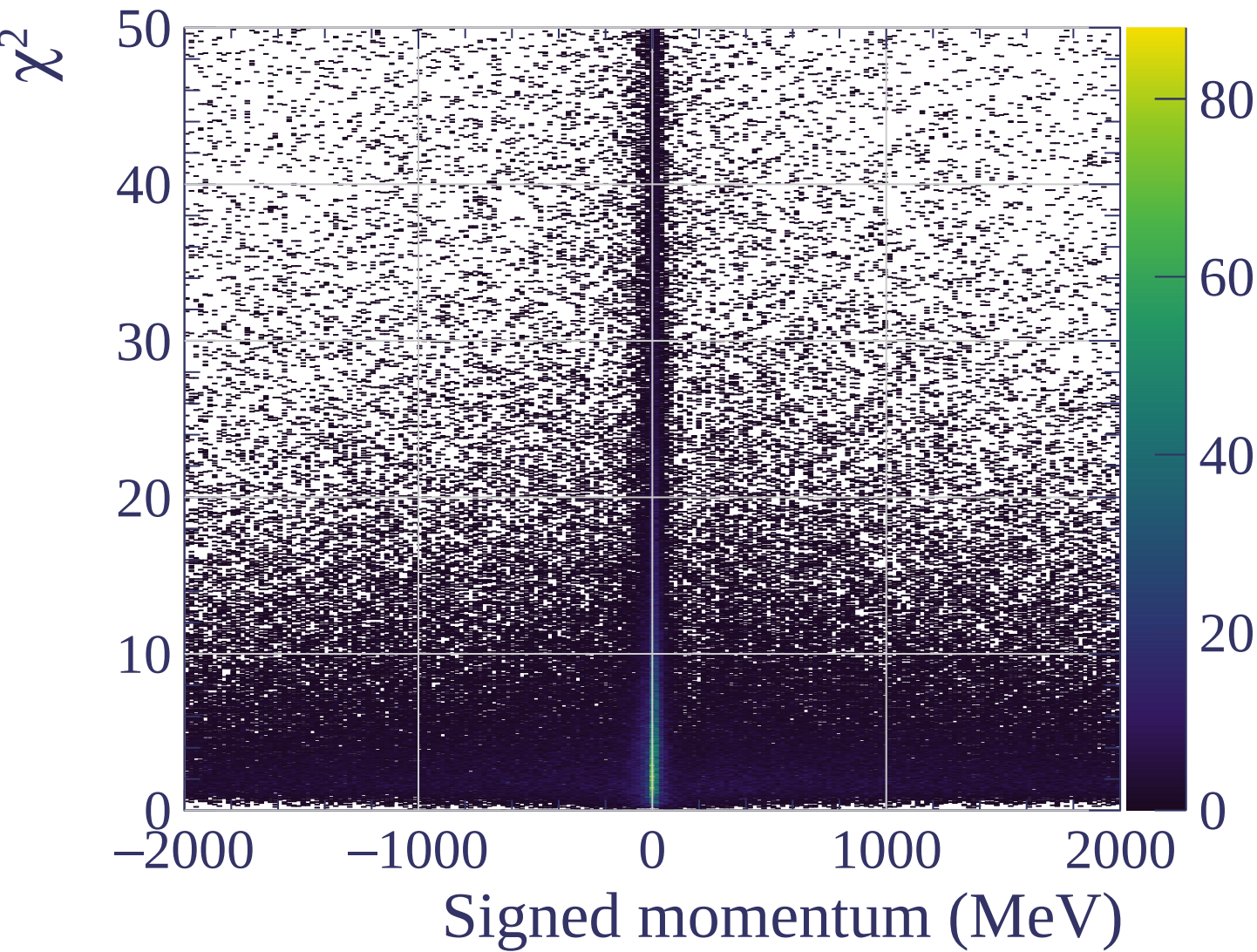


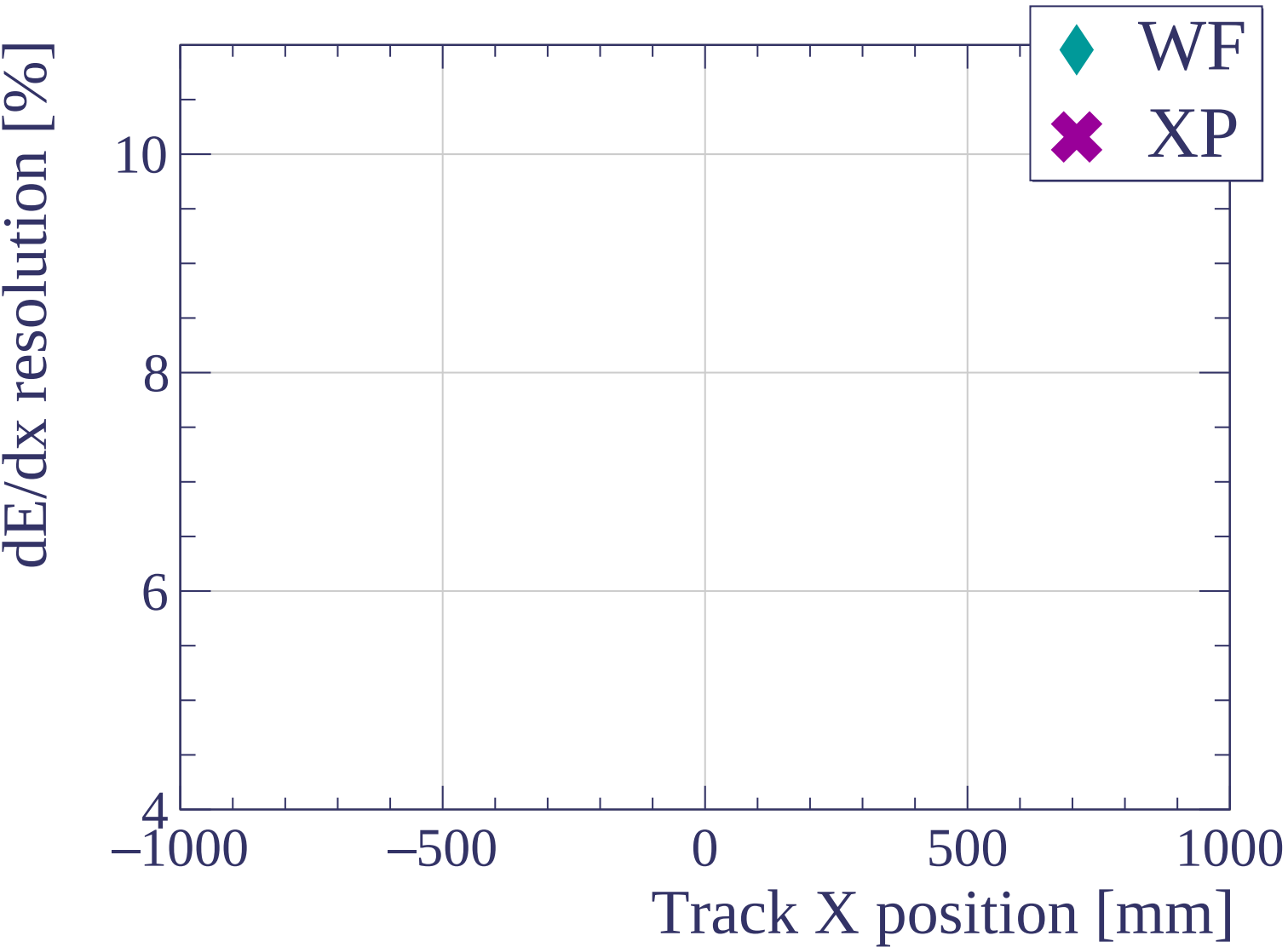


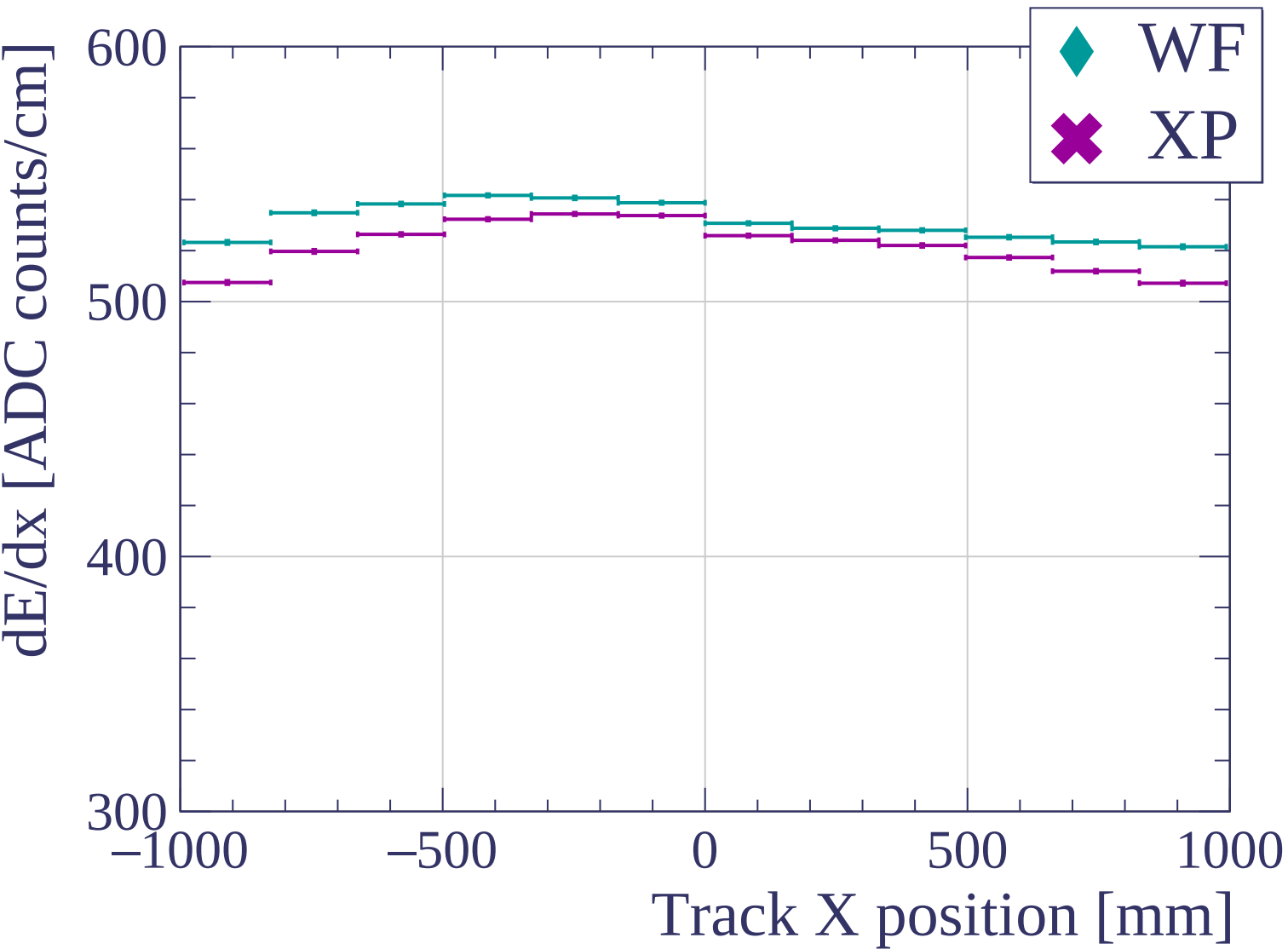


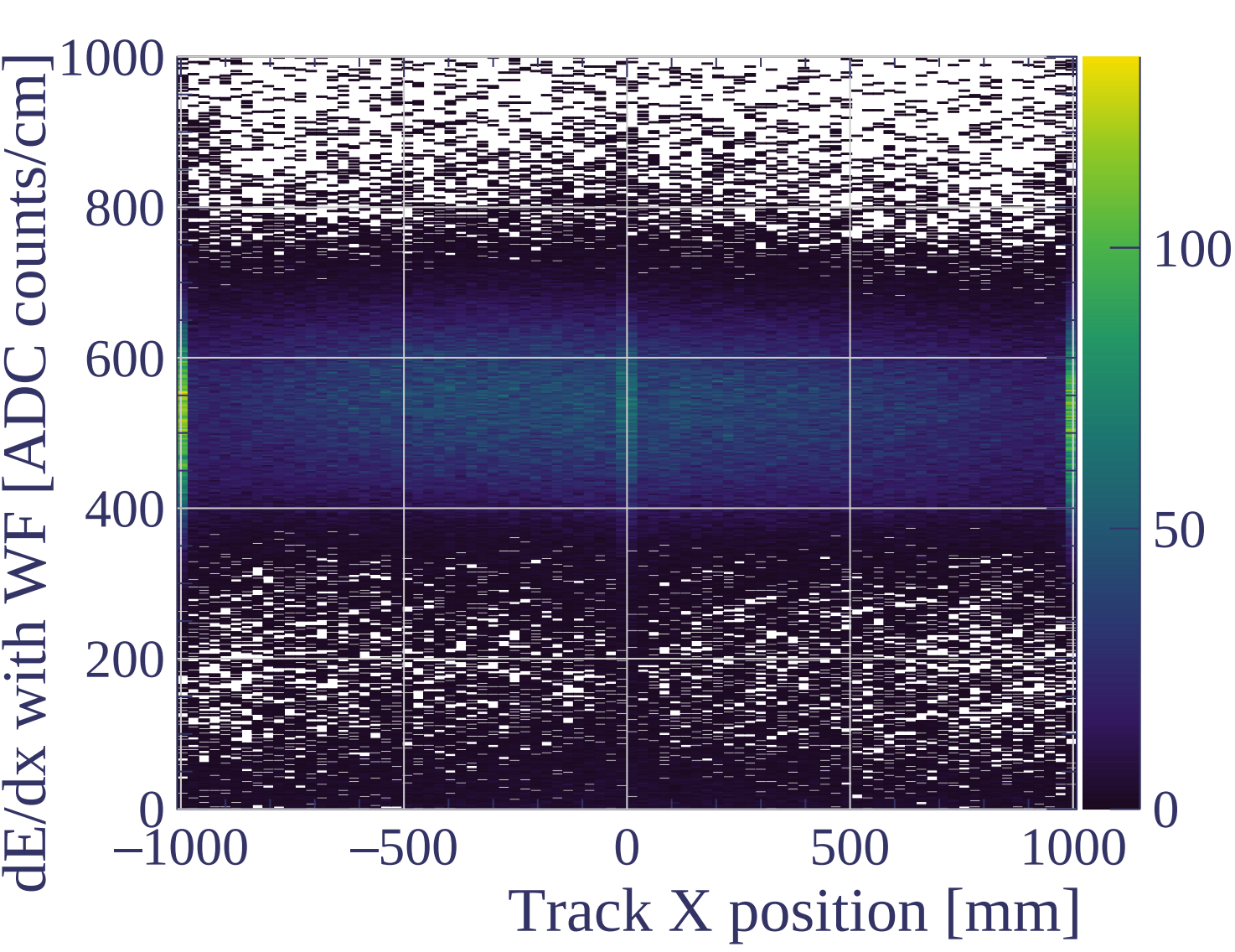


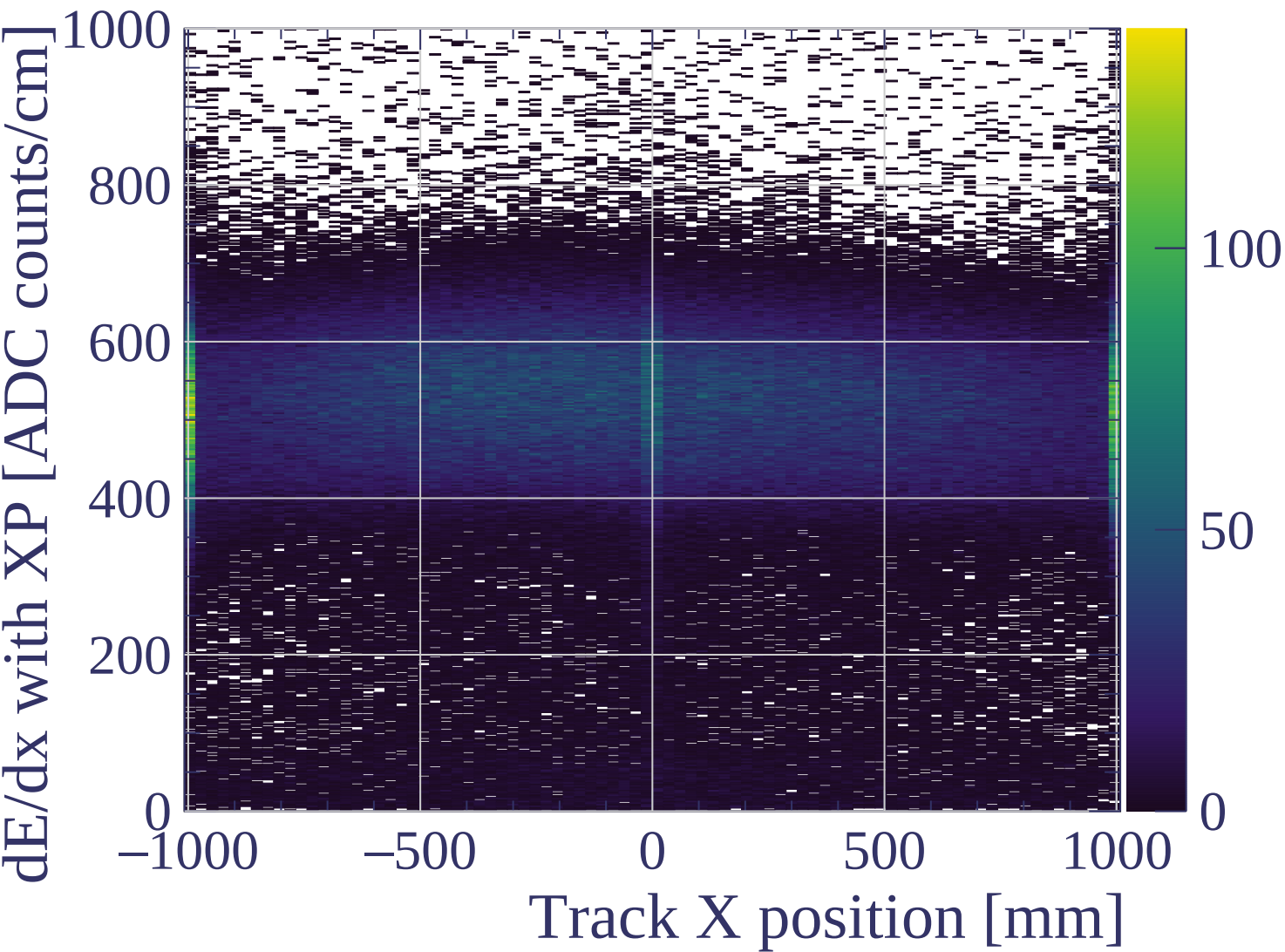


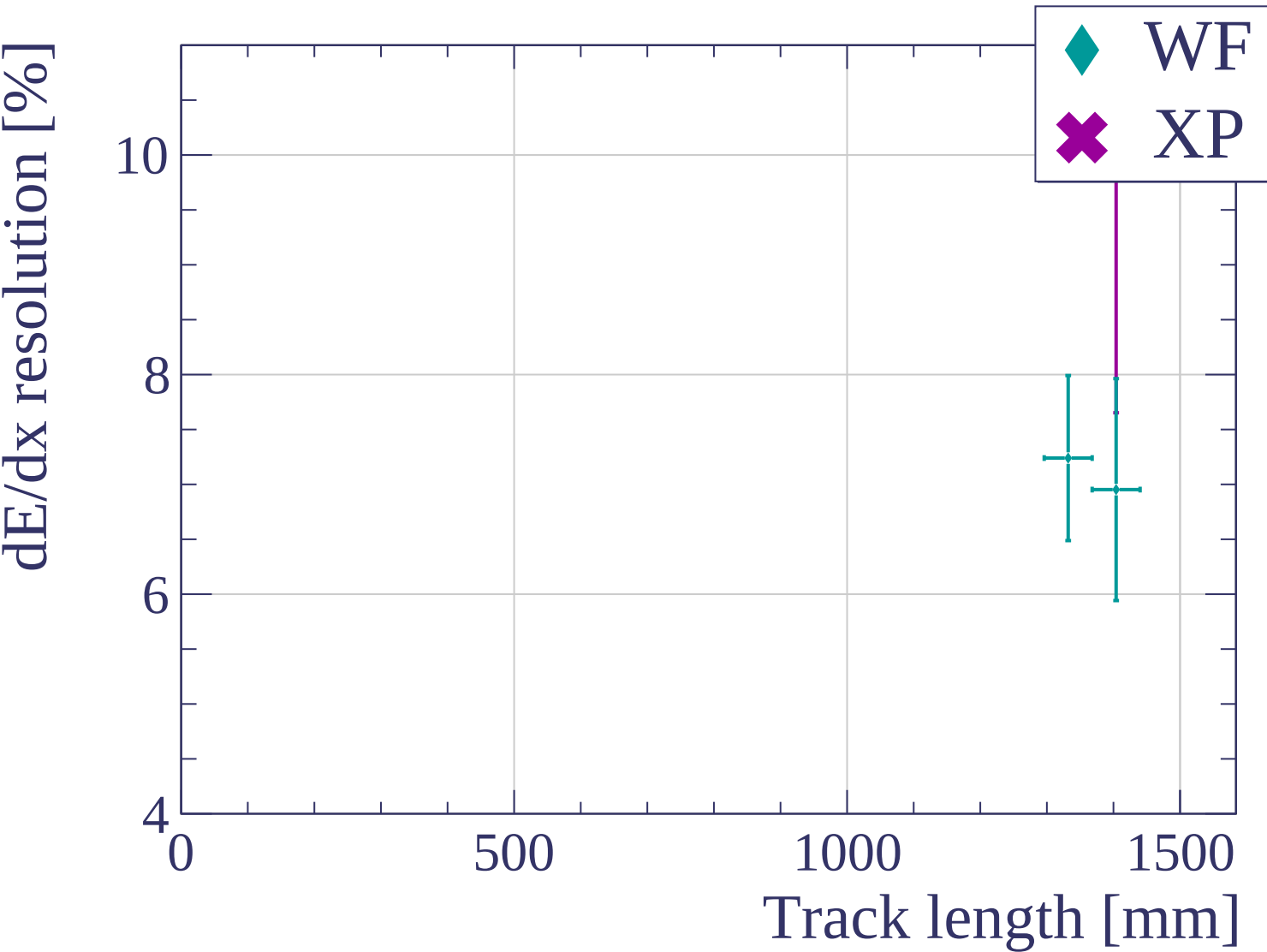


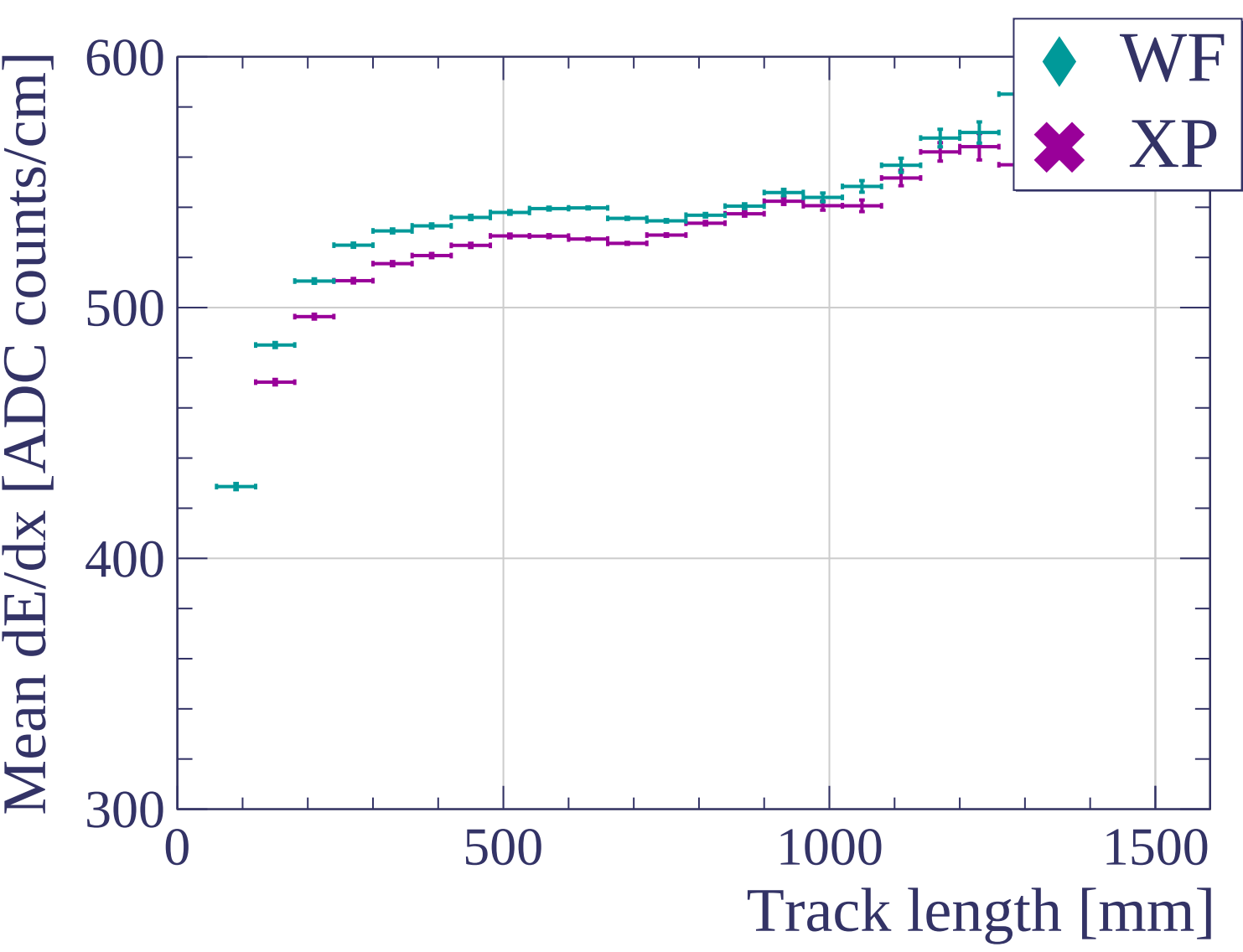


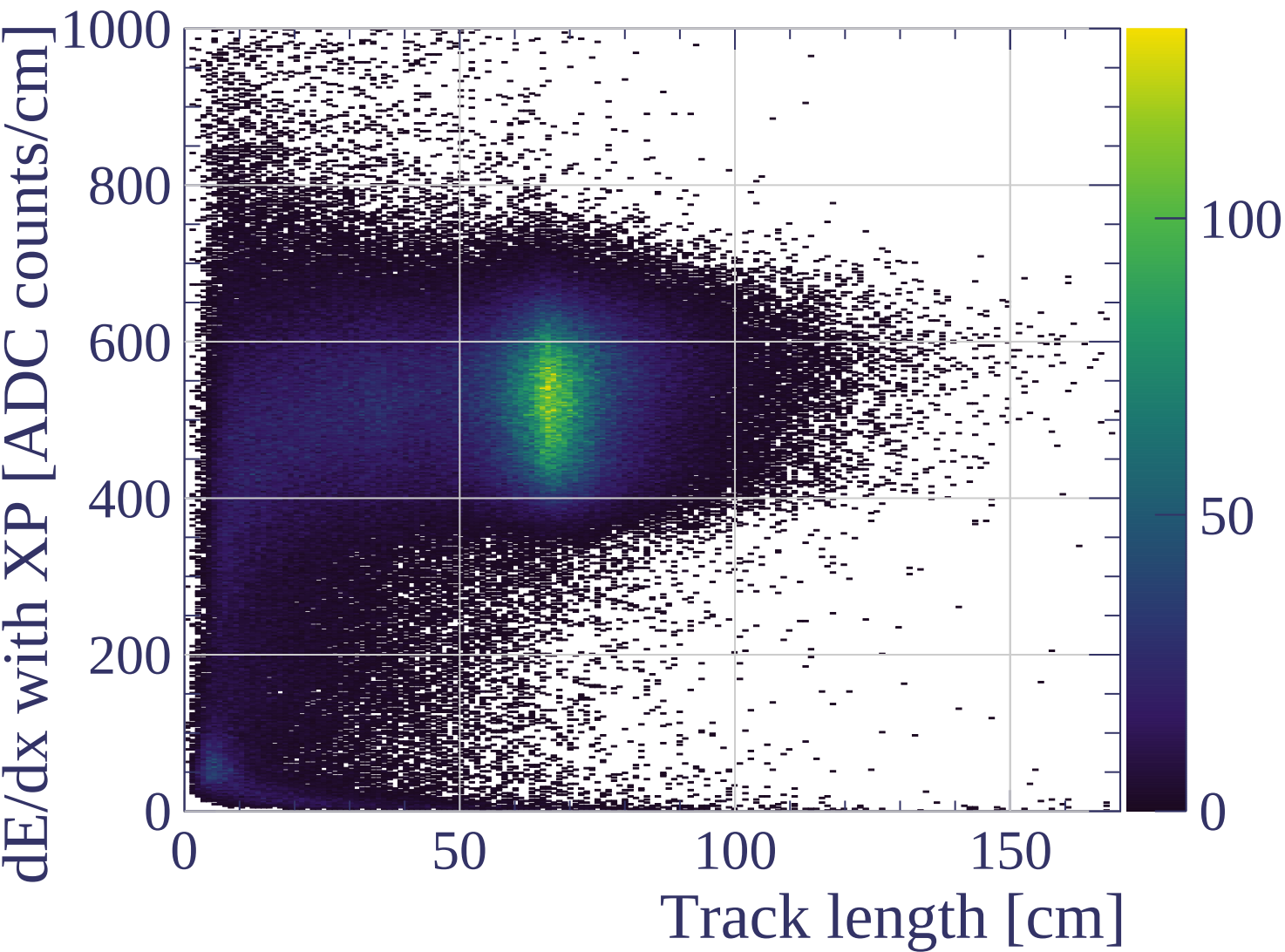


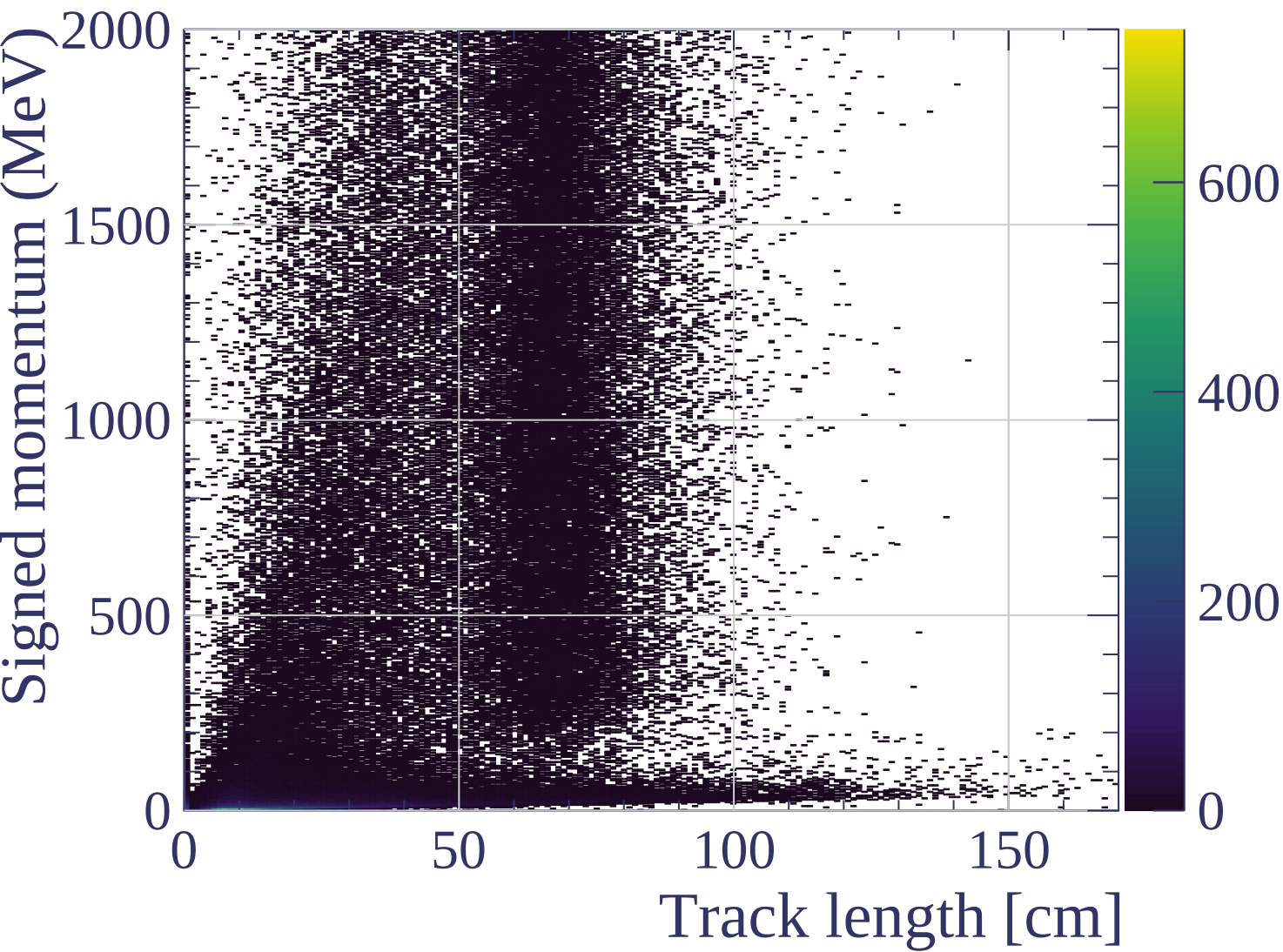


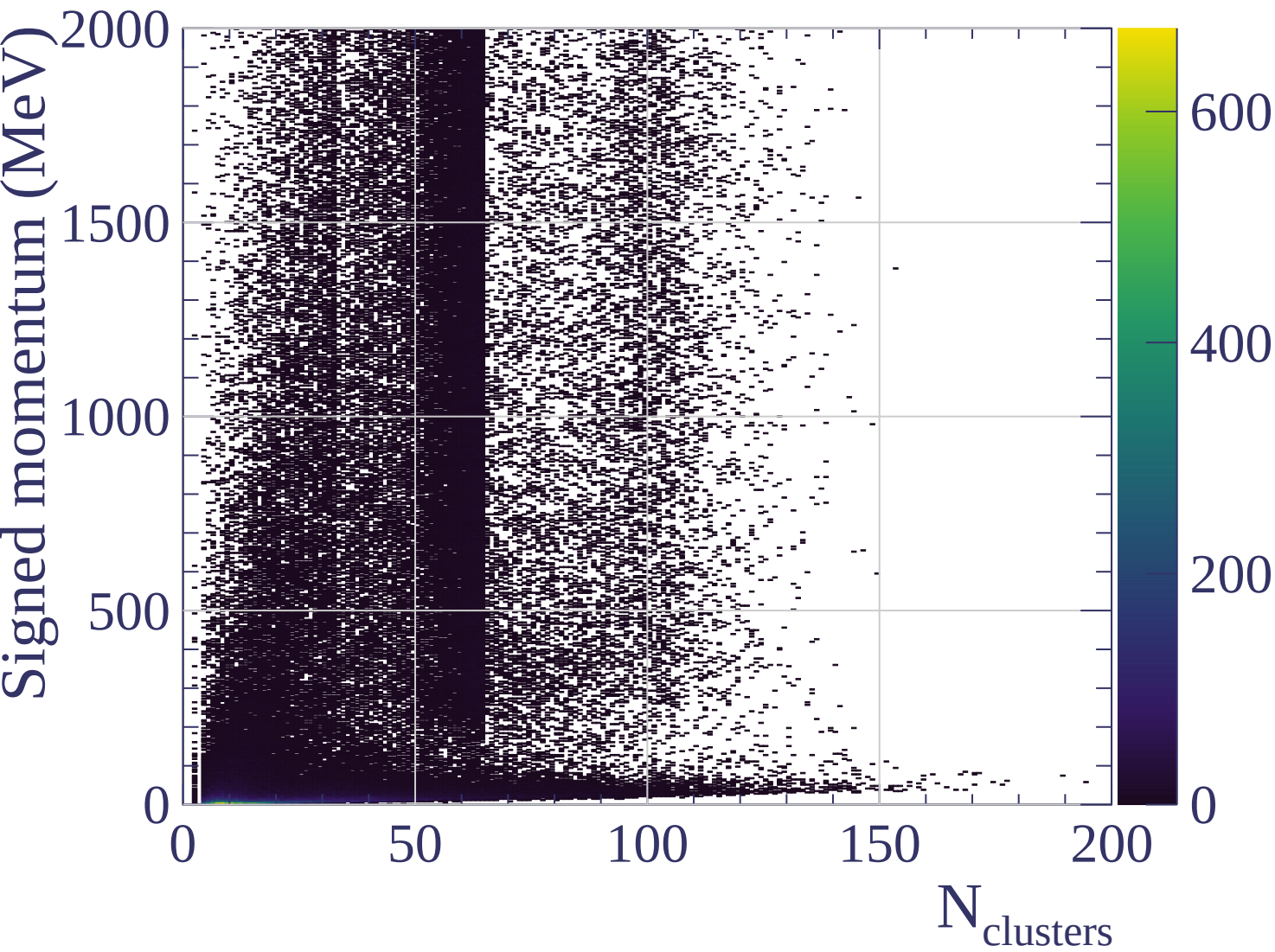


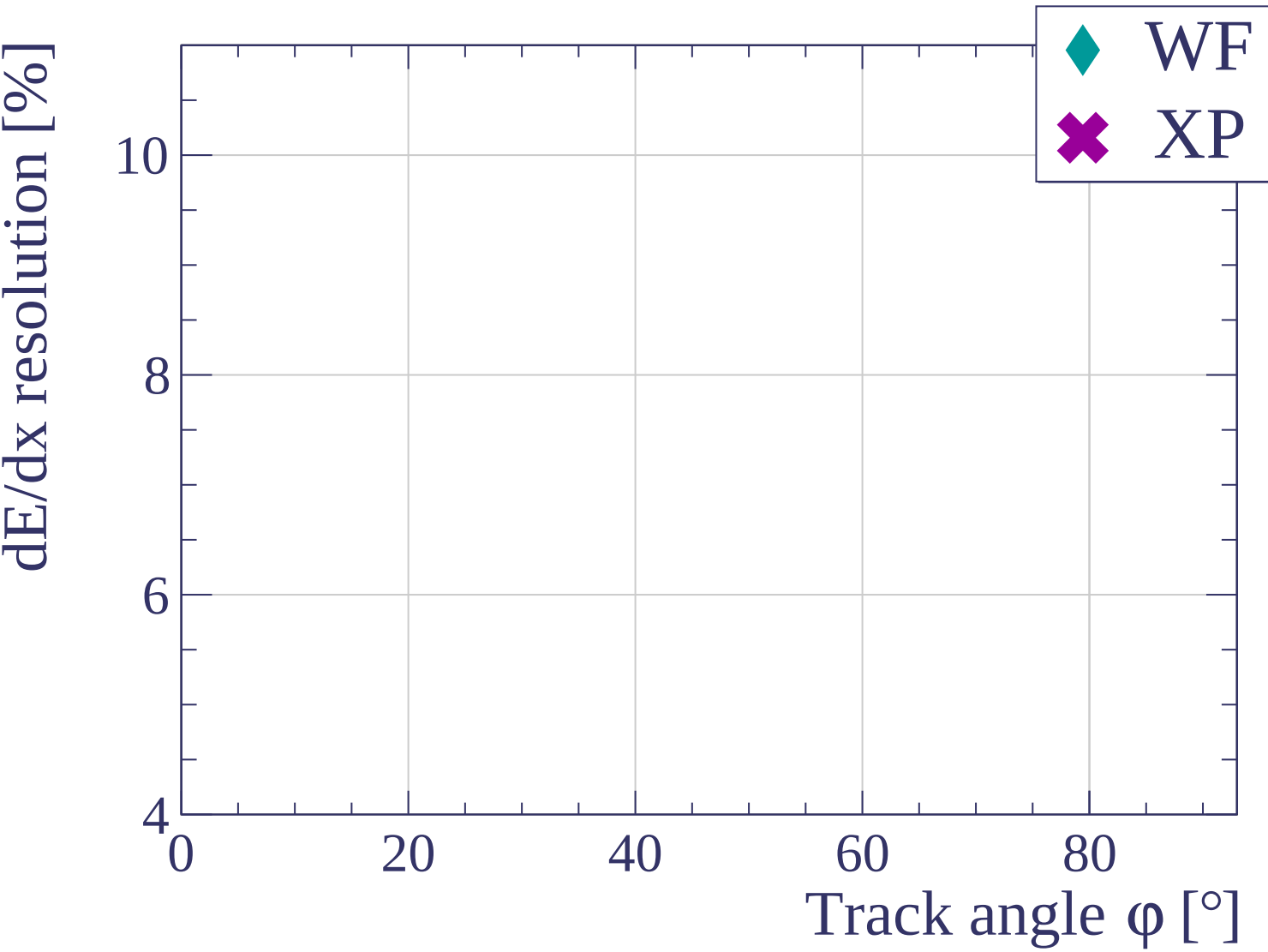


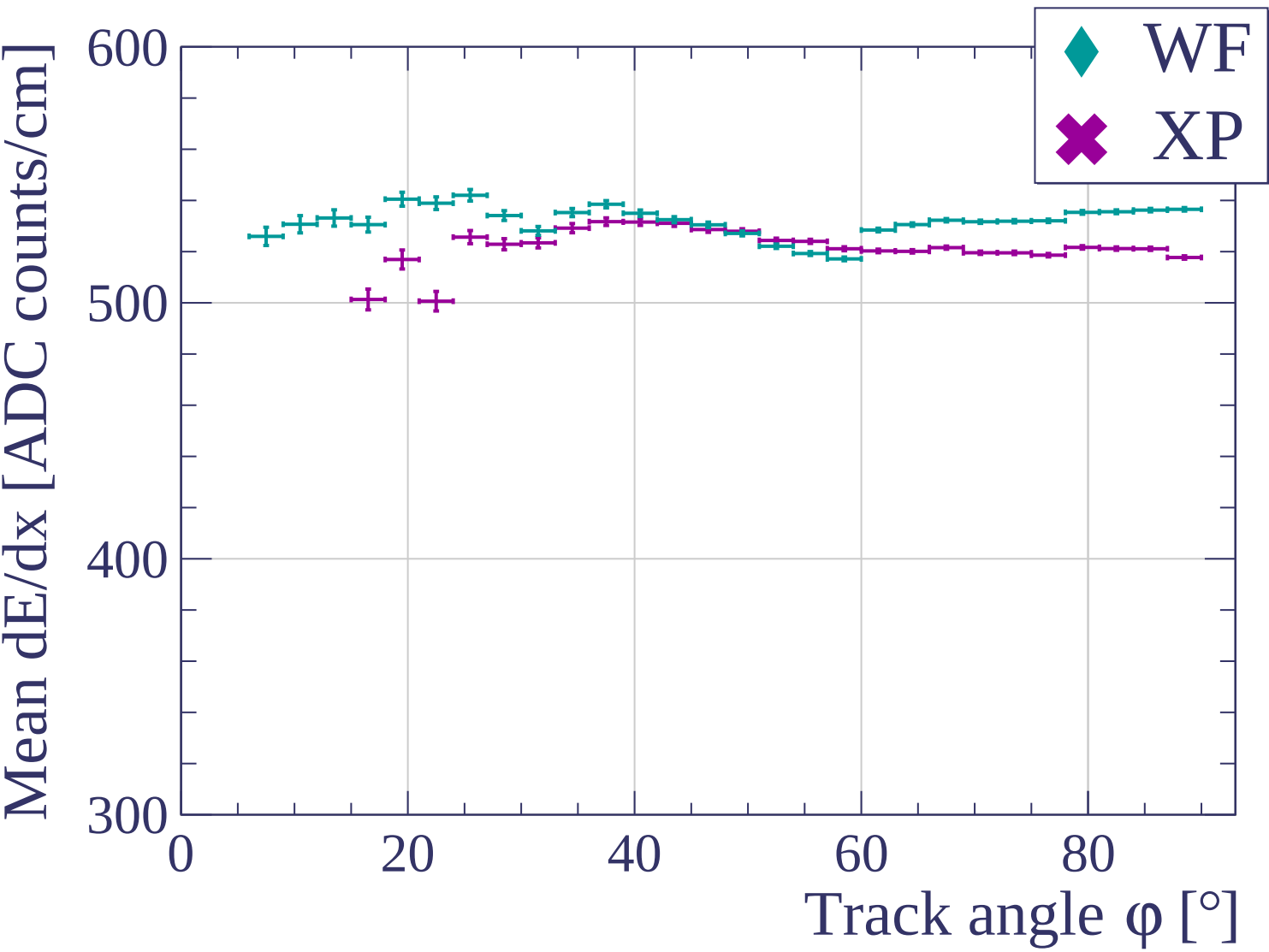


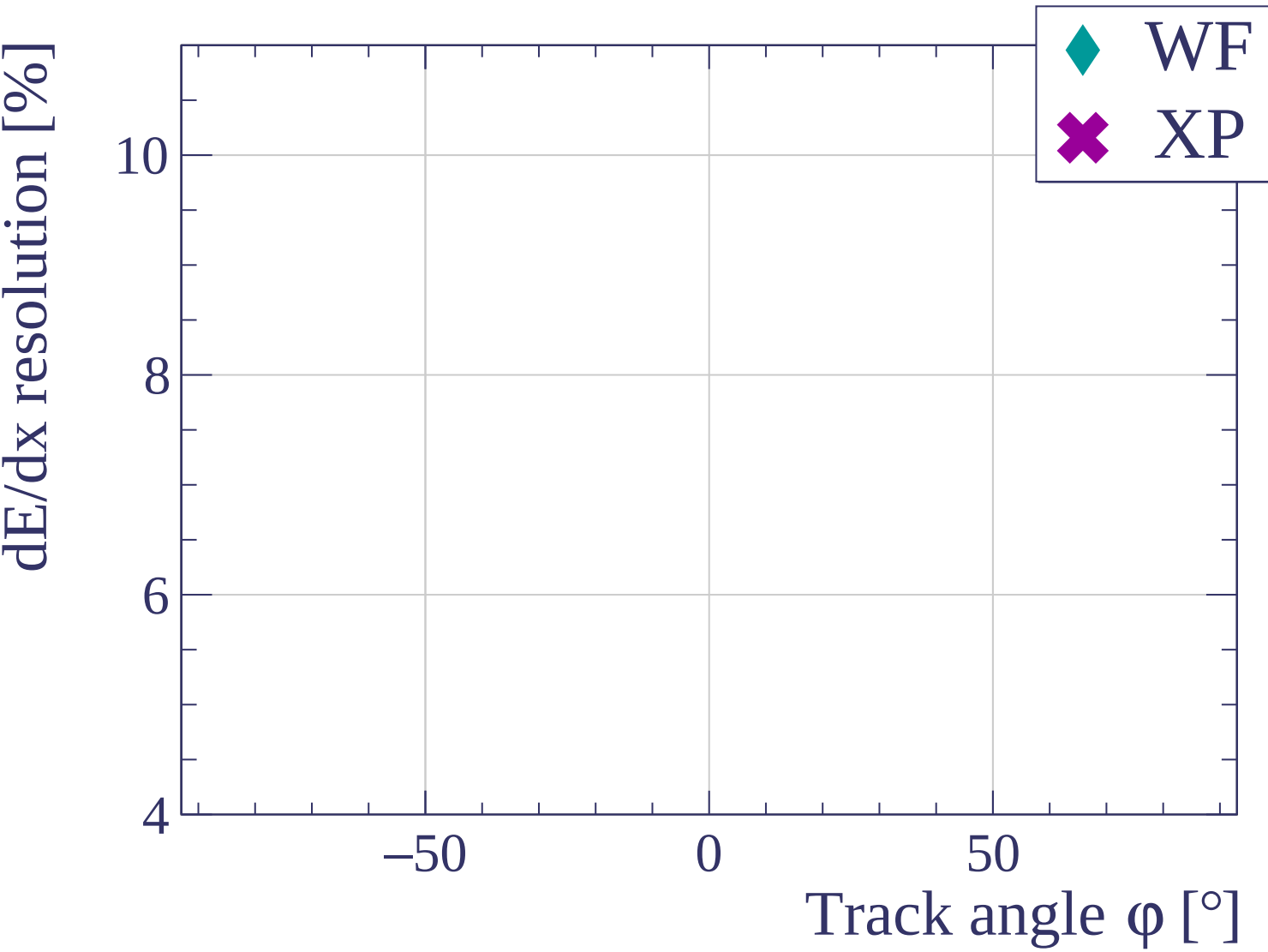


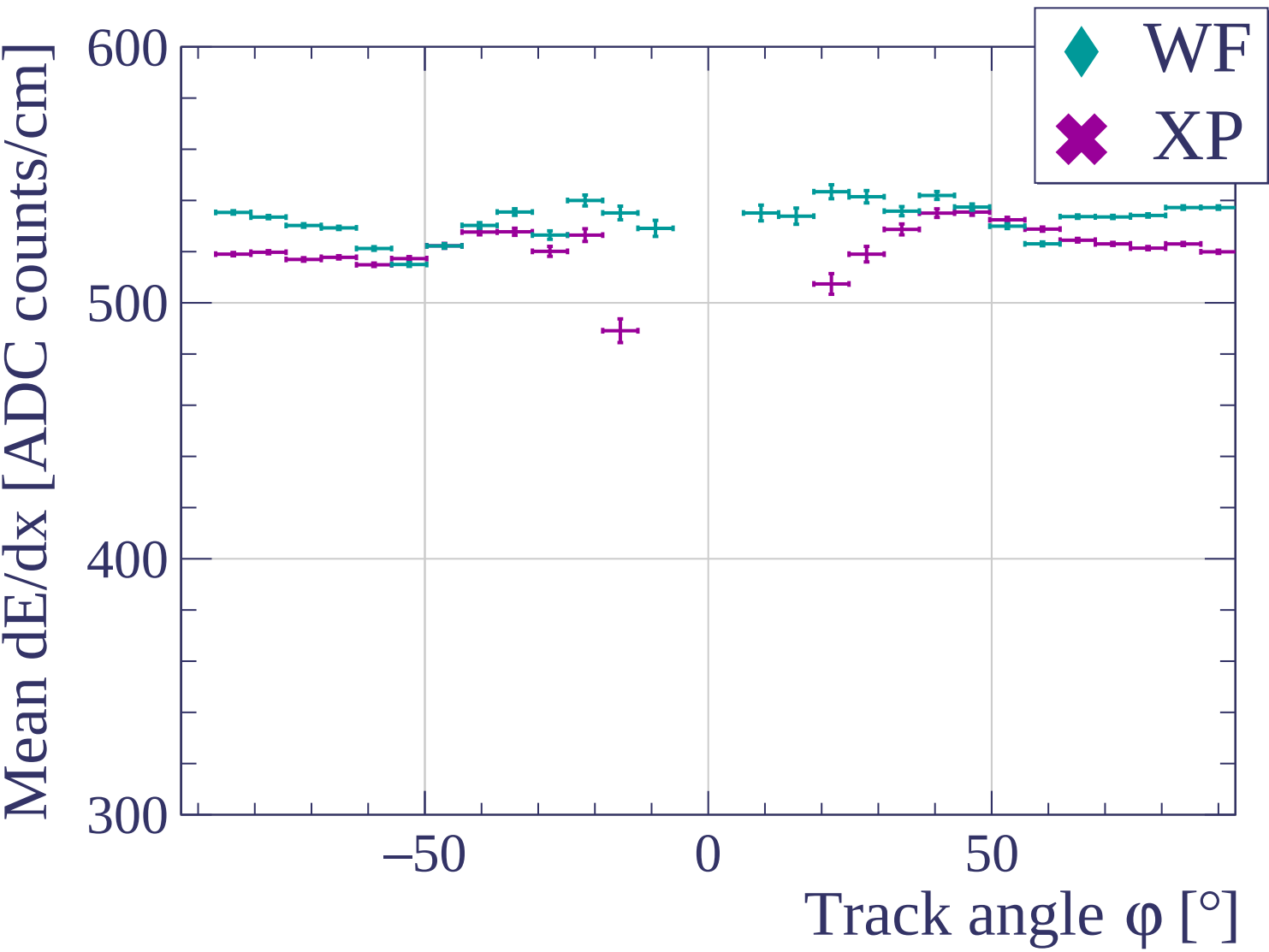


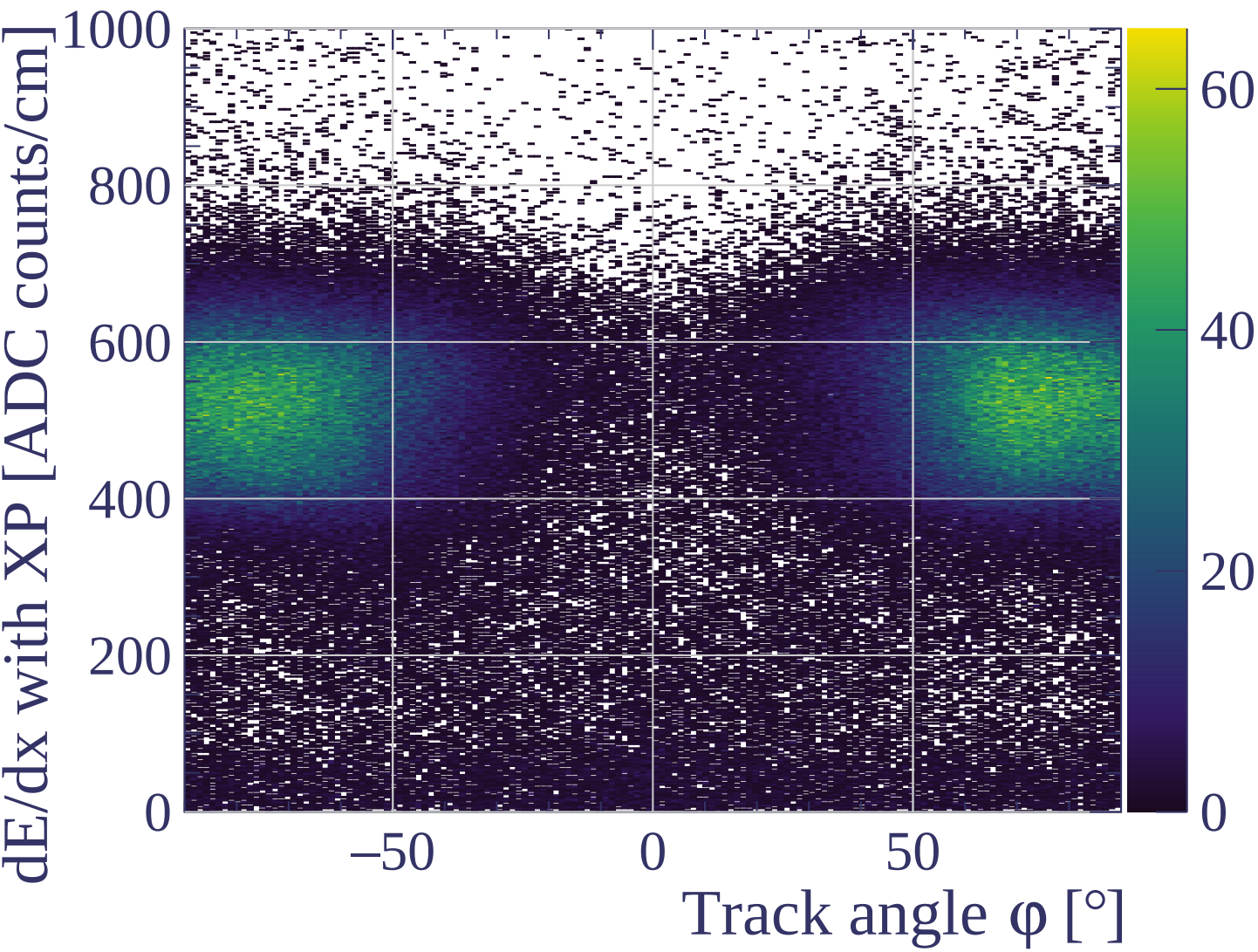




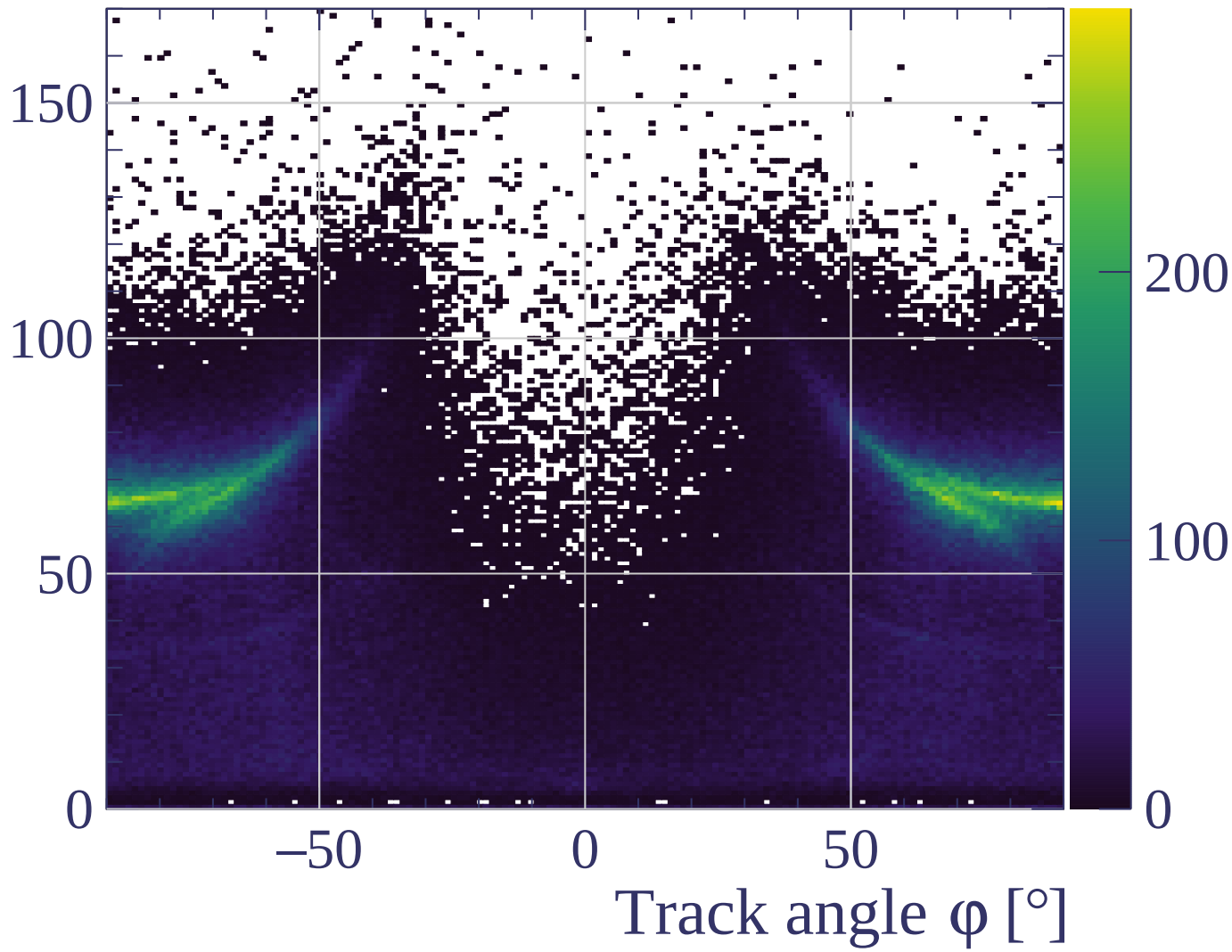


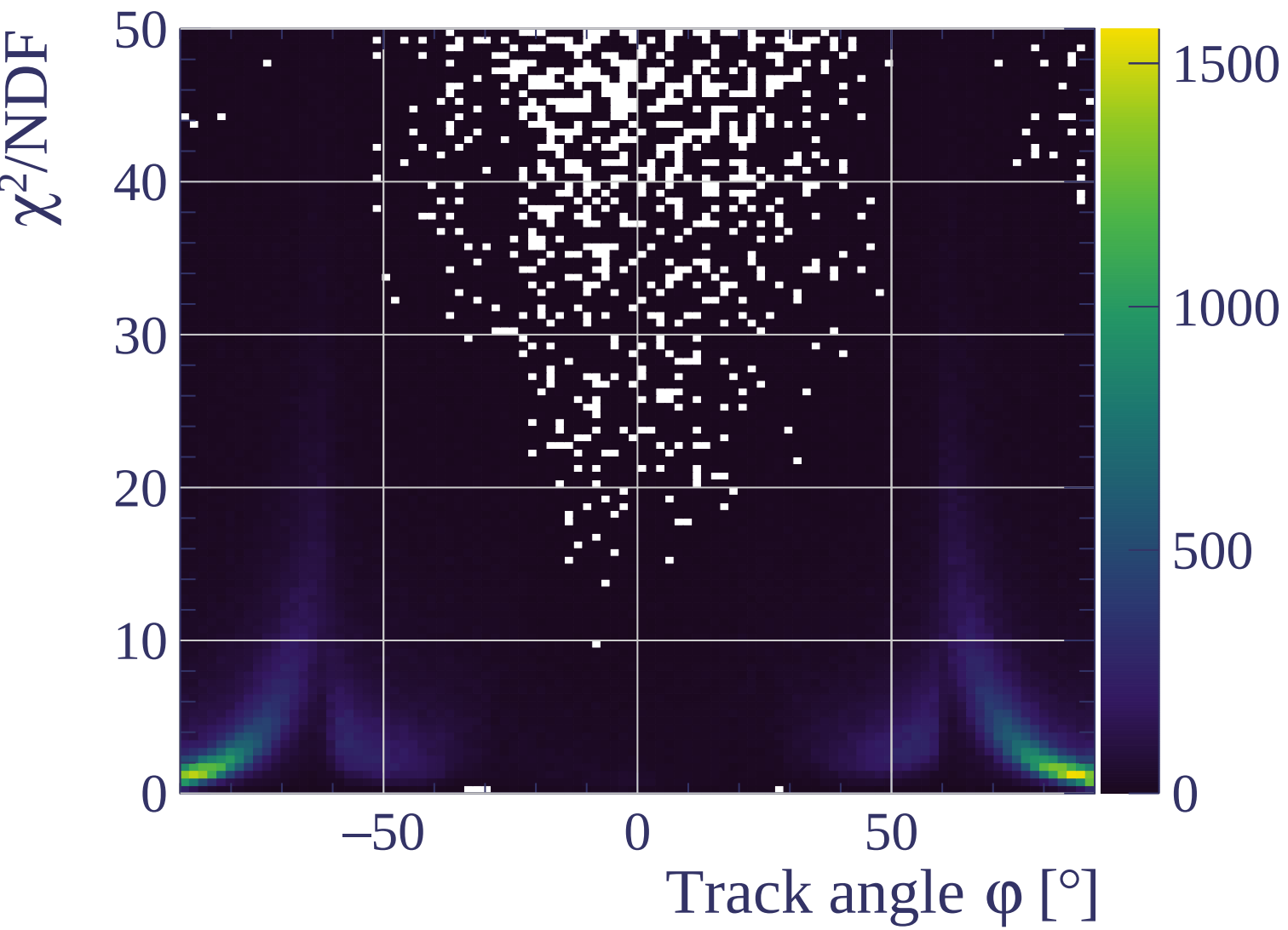


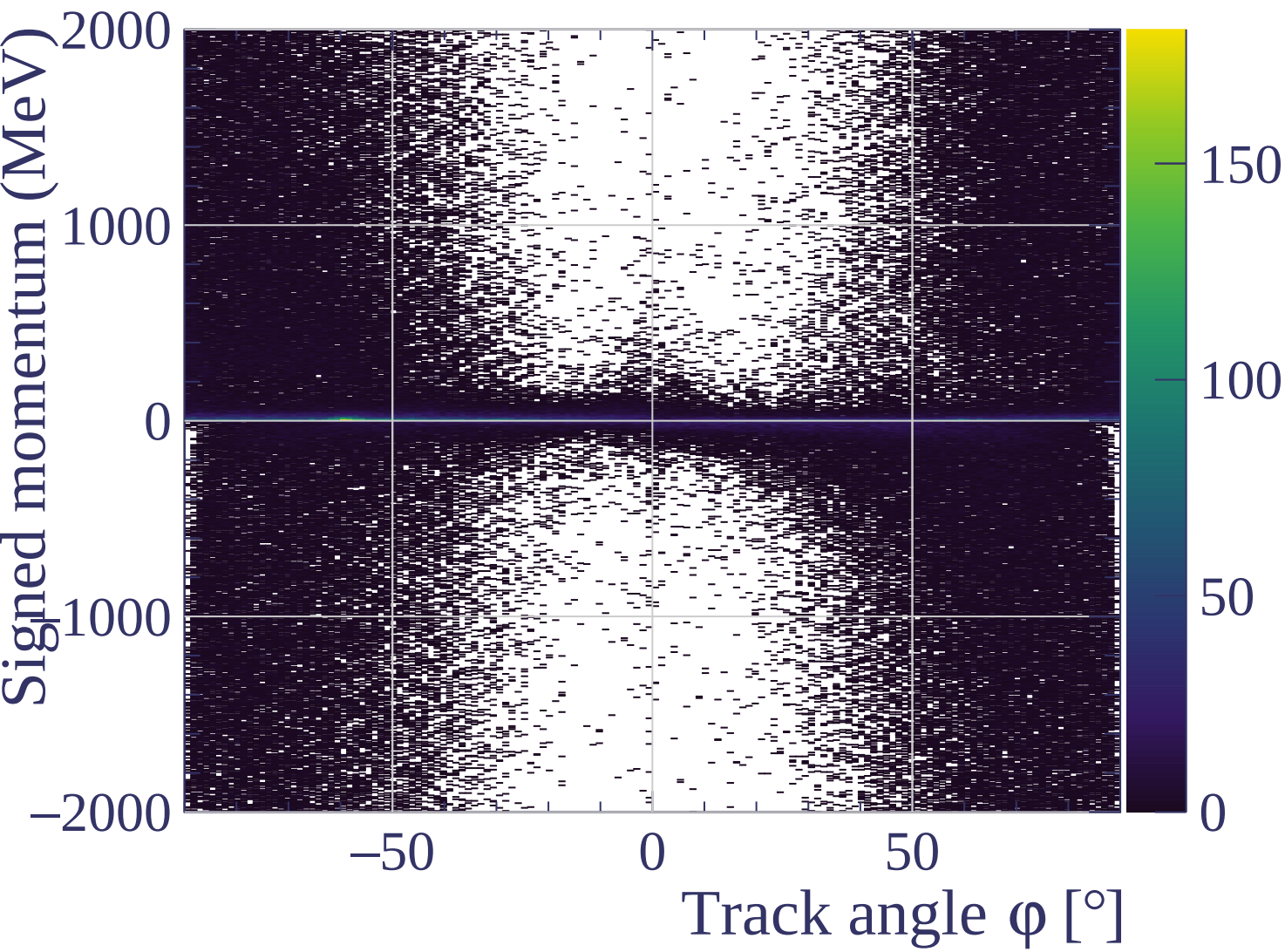


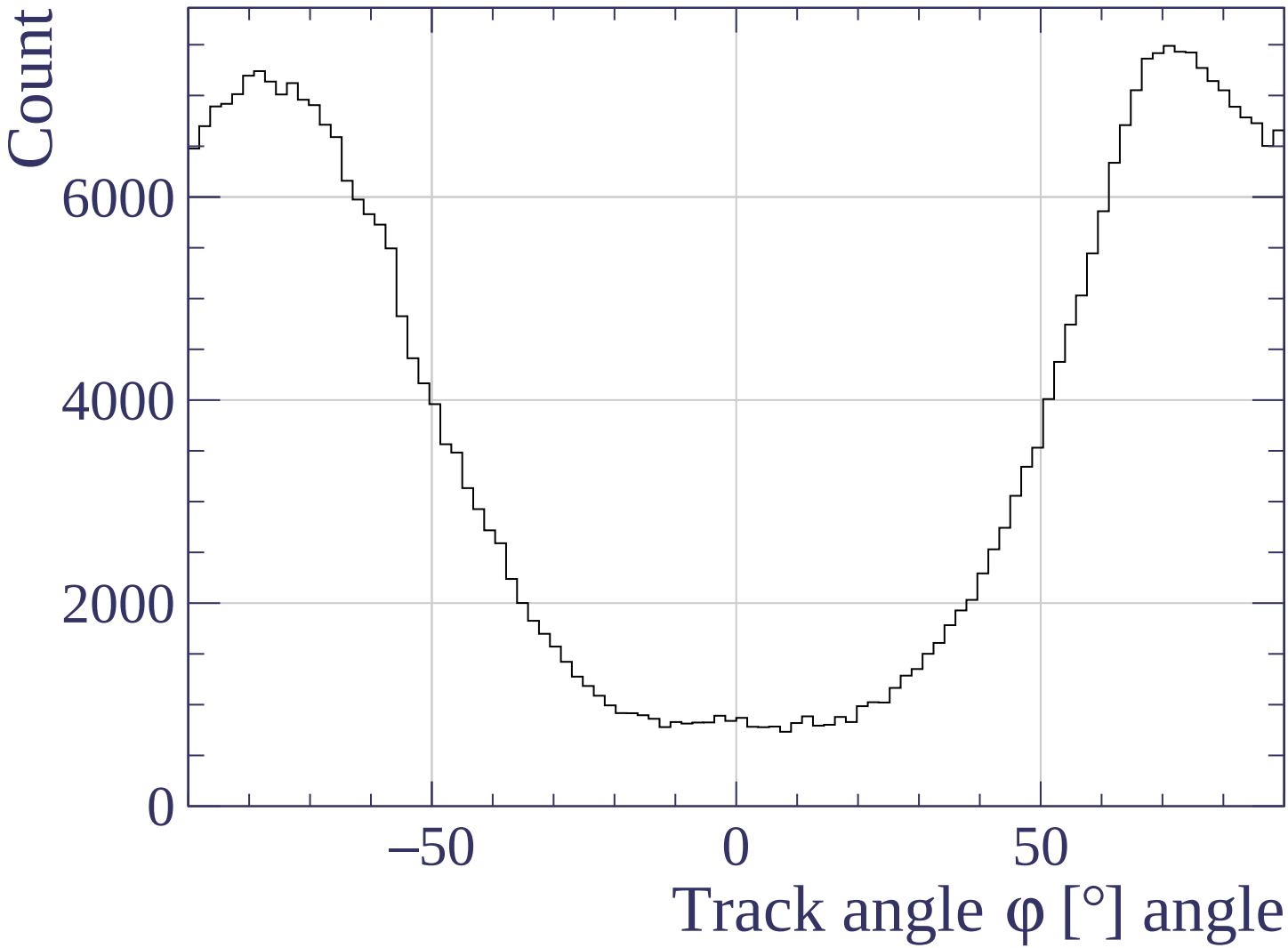


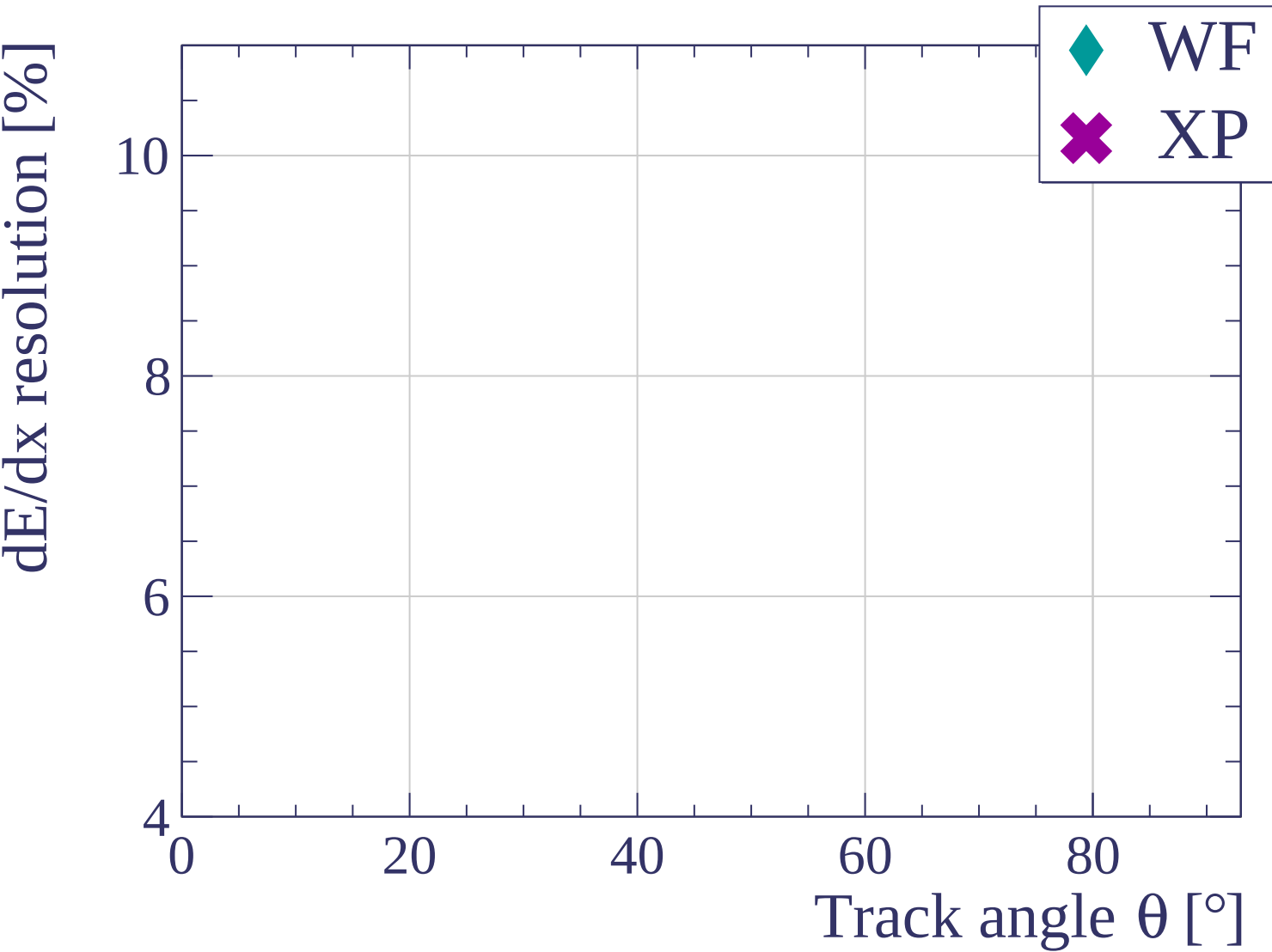
Track length [cm]

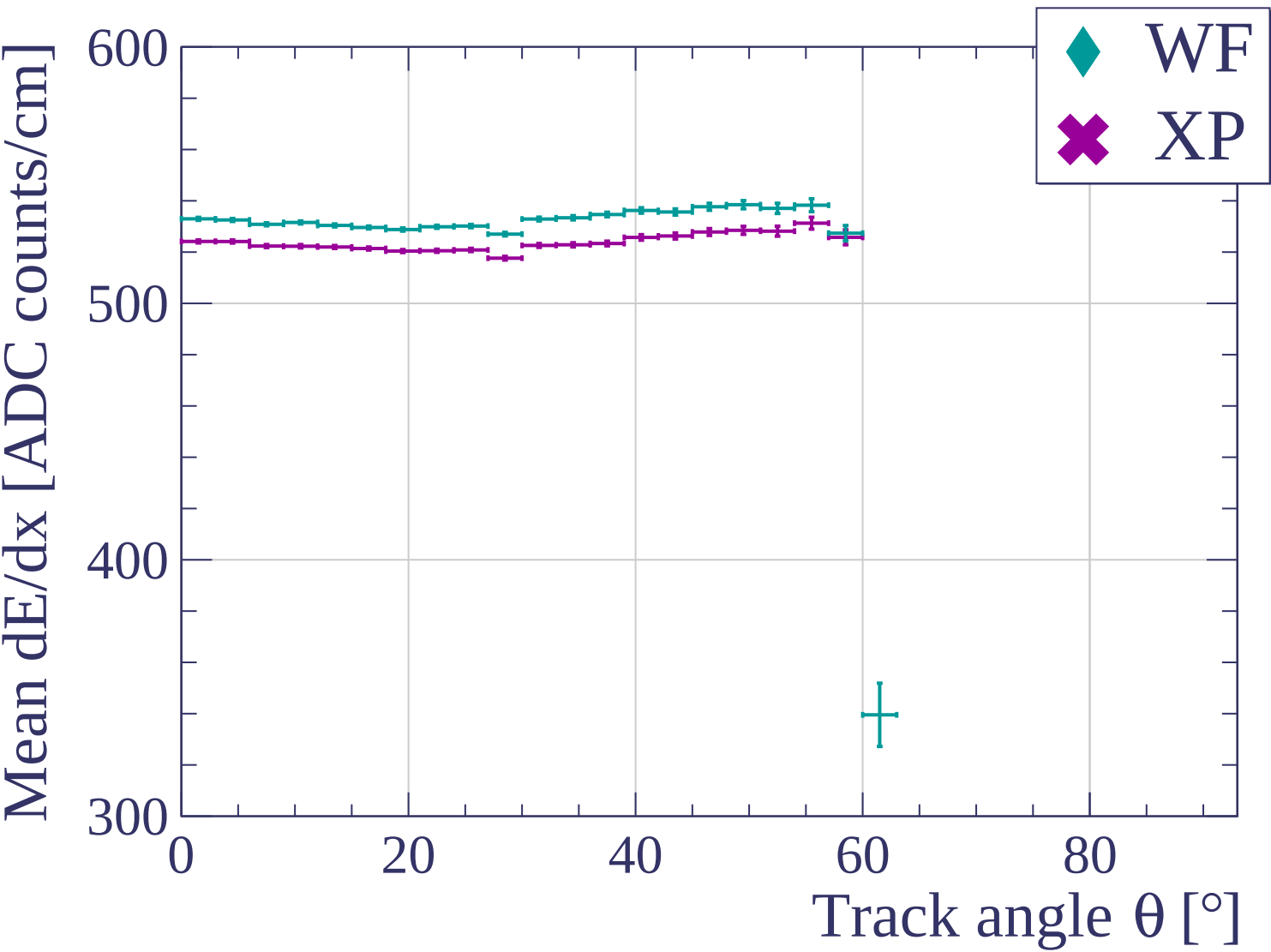


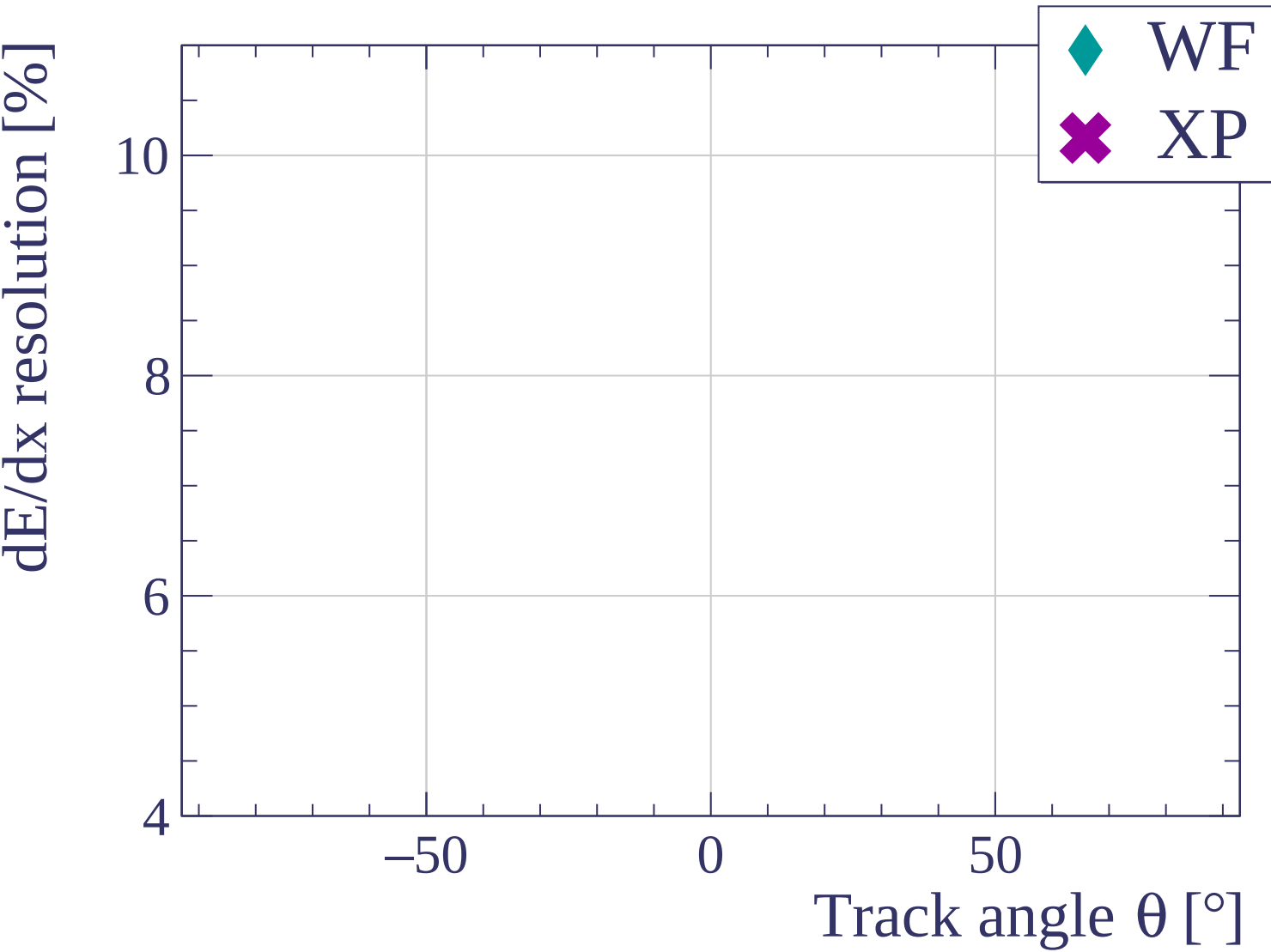


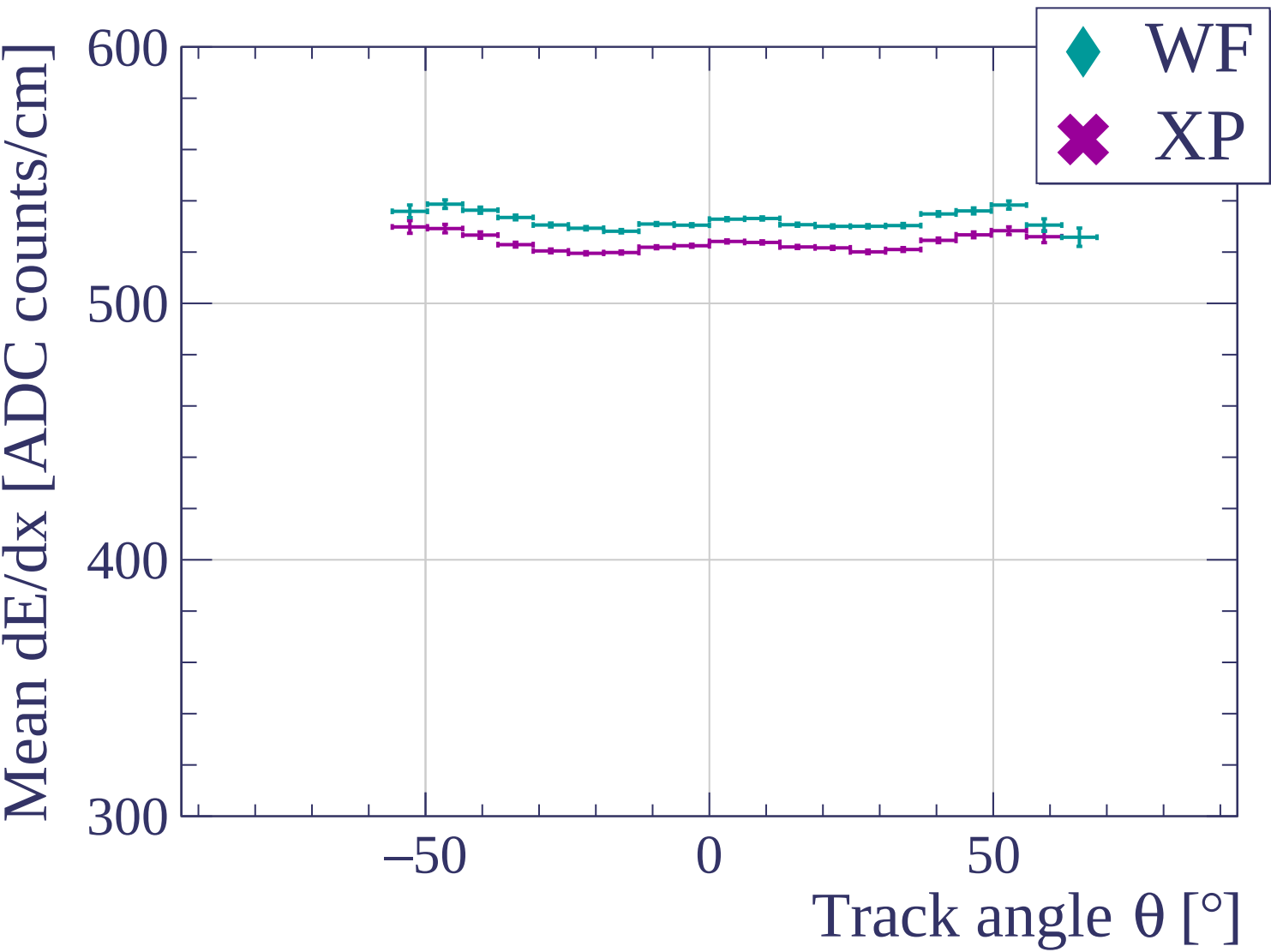


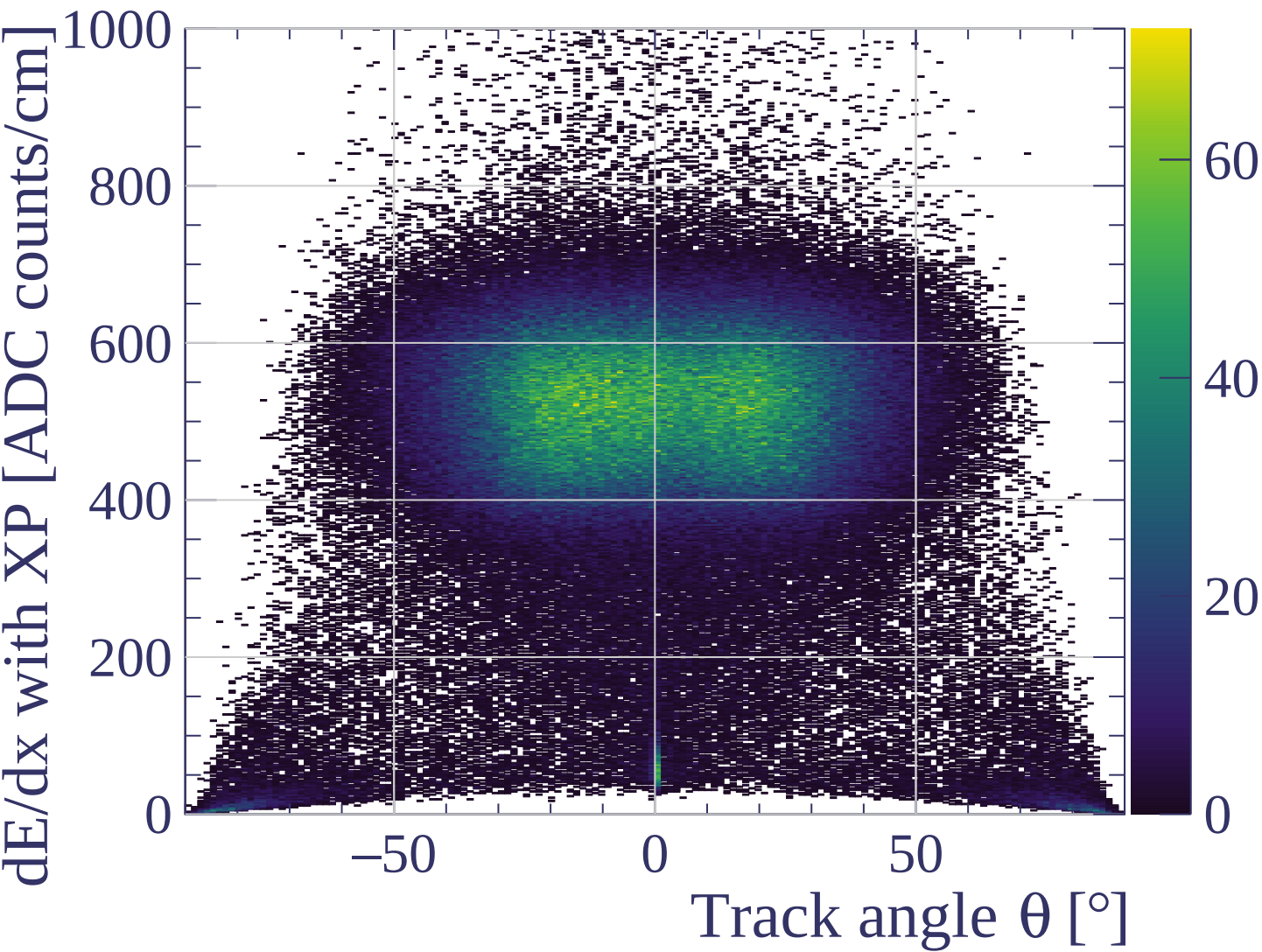




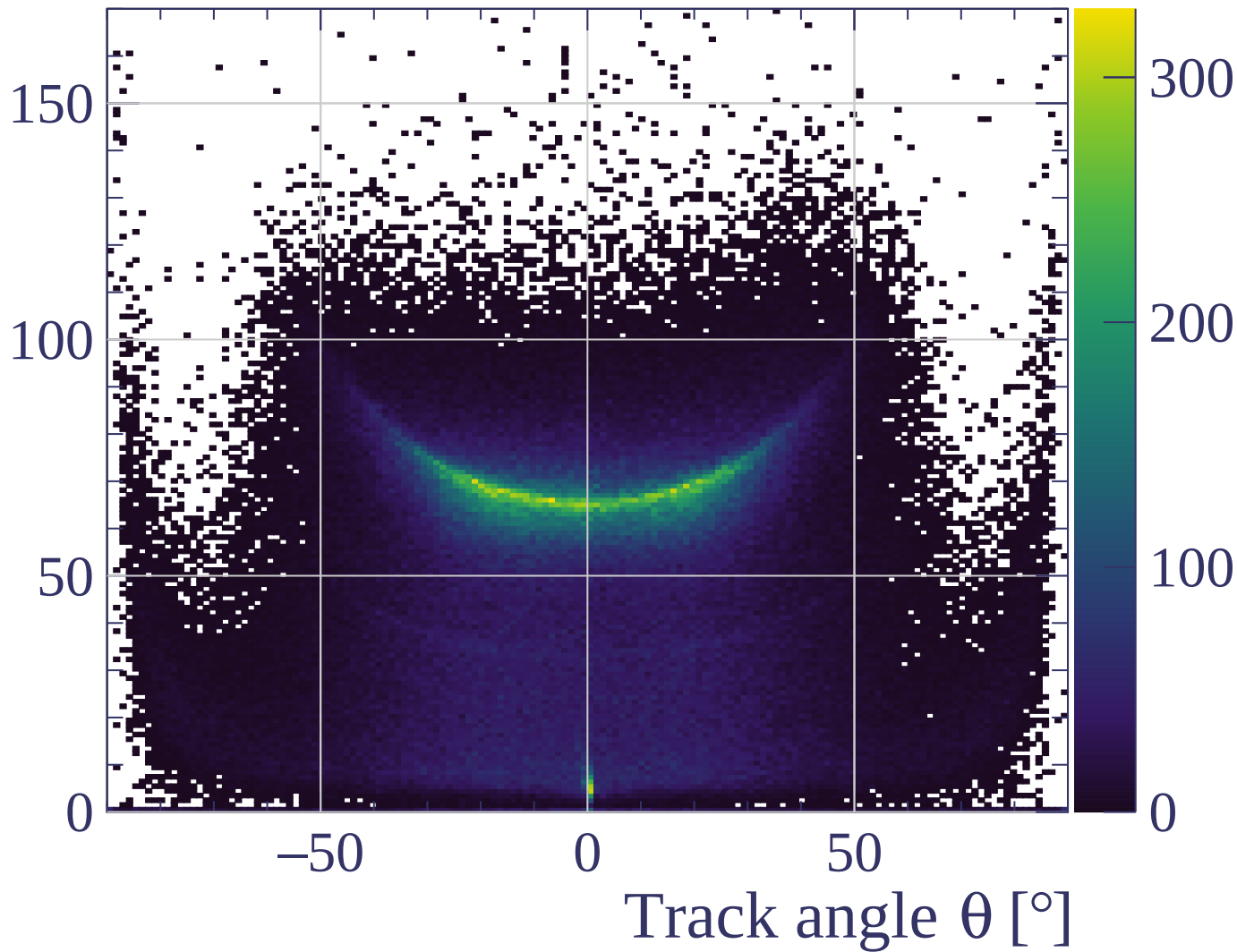


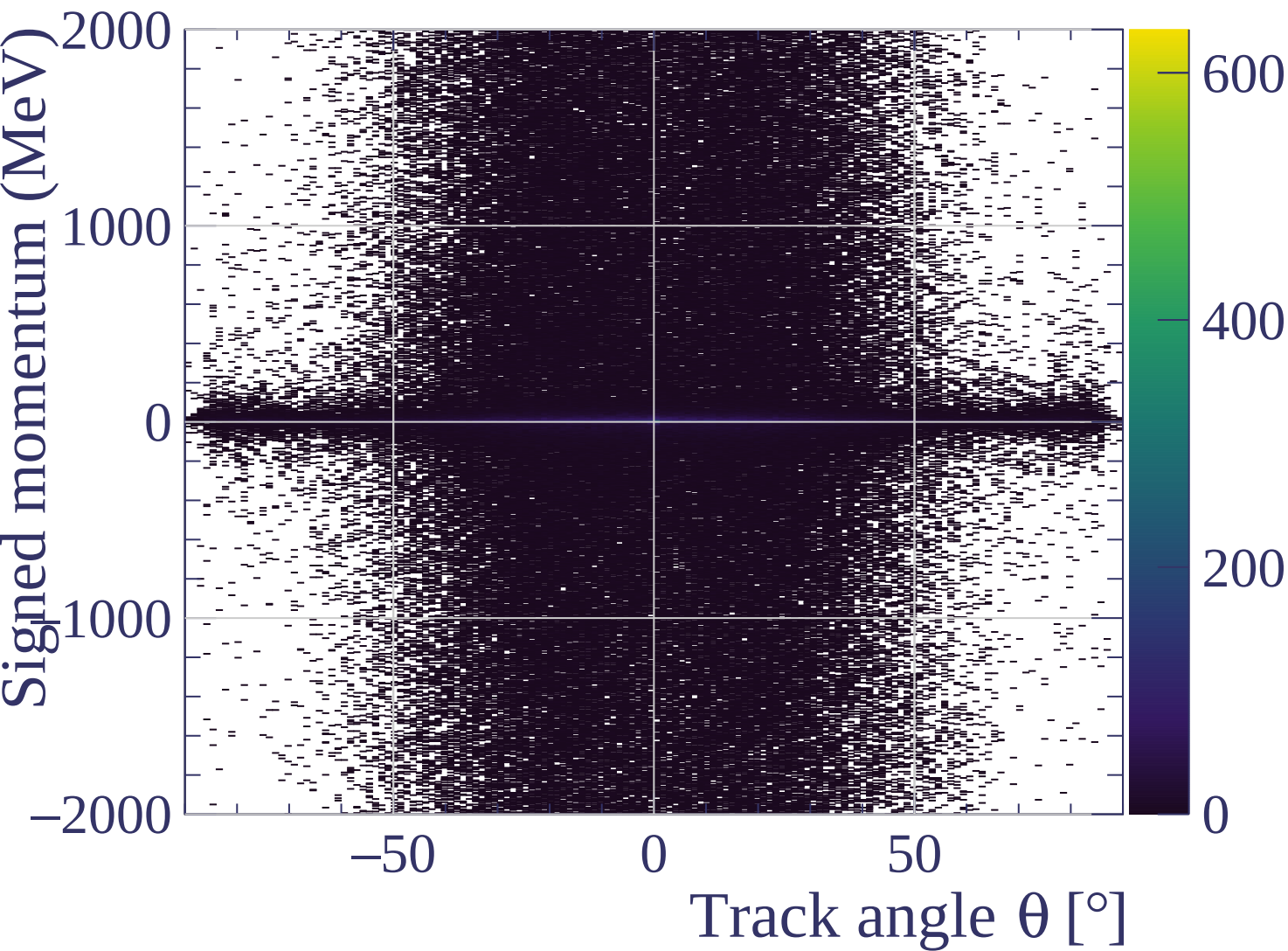


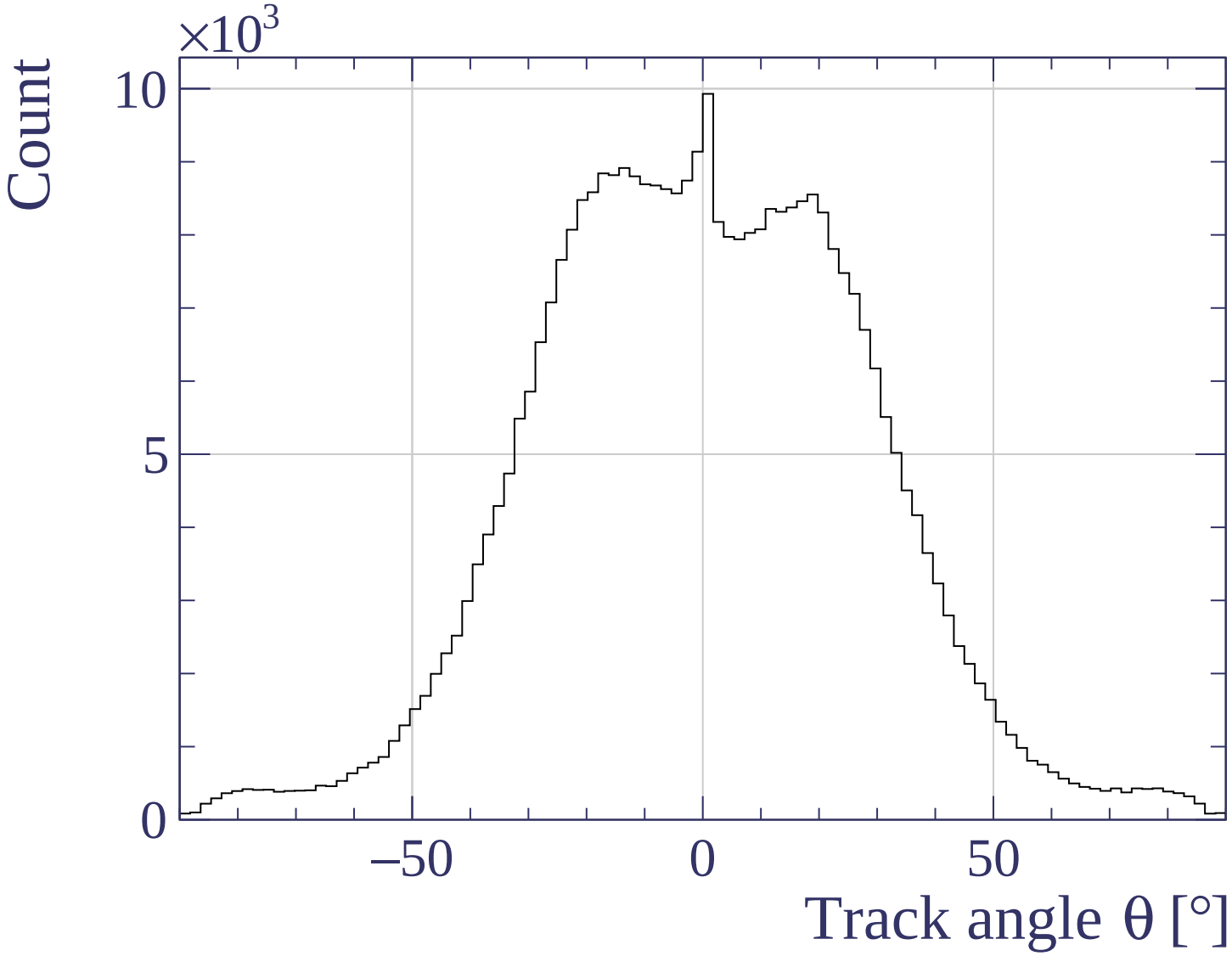




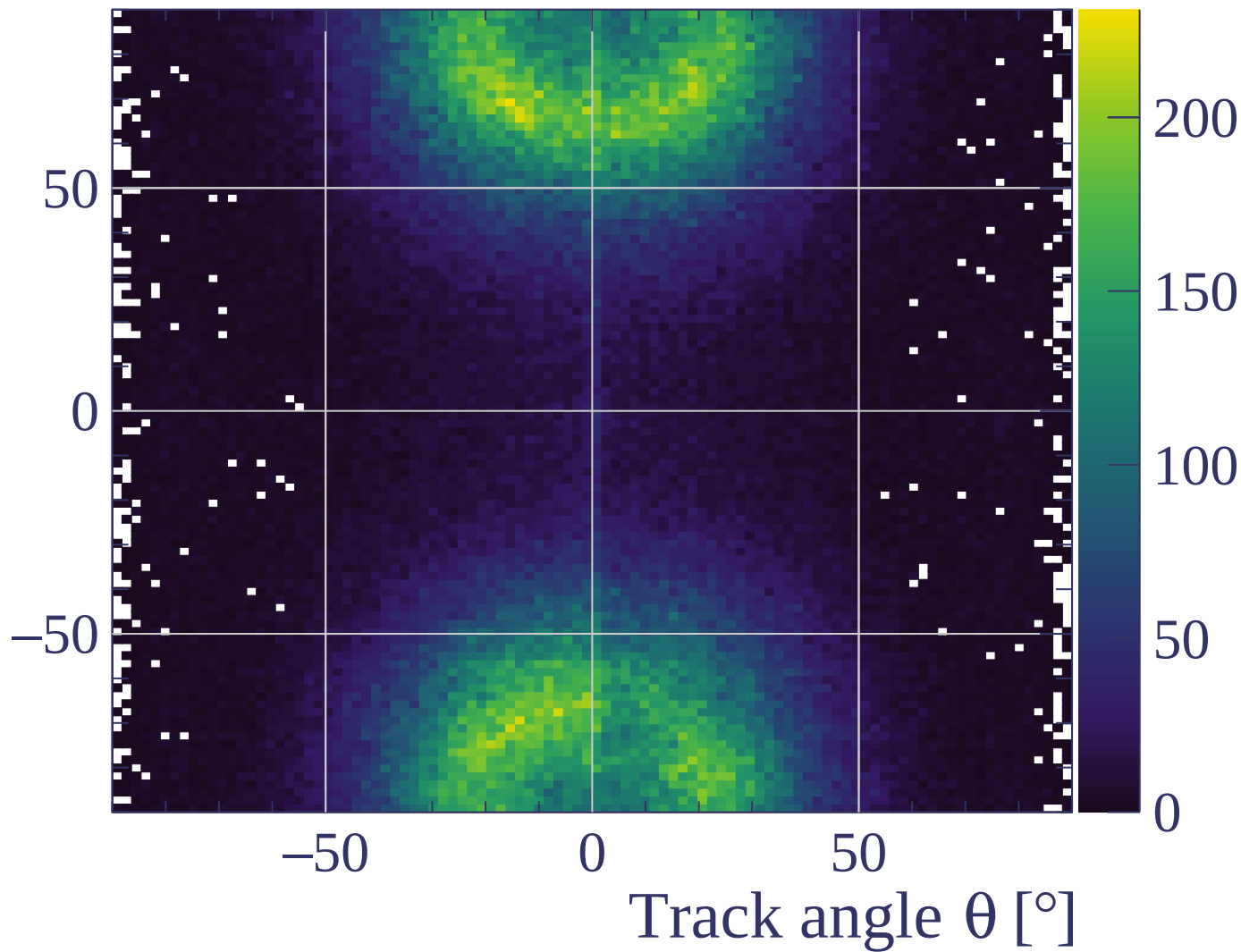
Track length [cm]



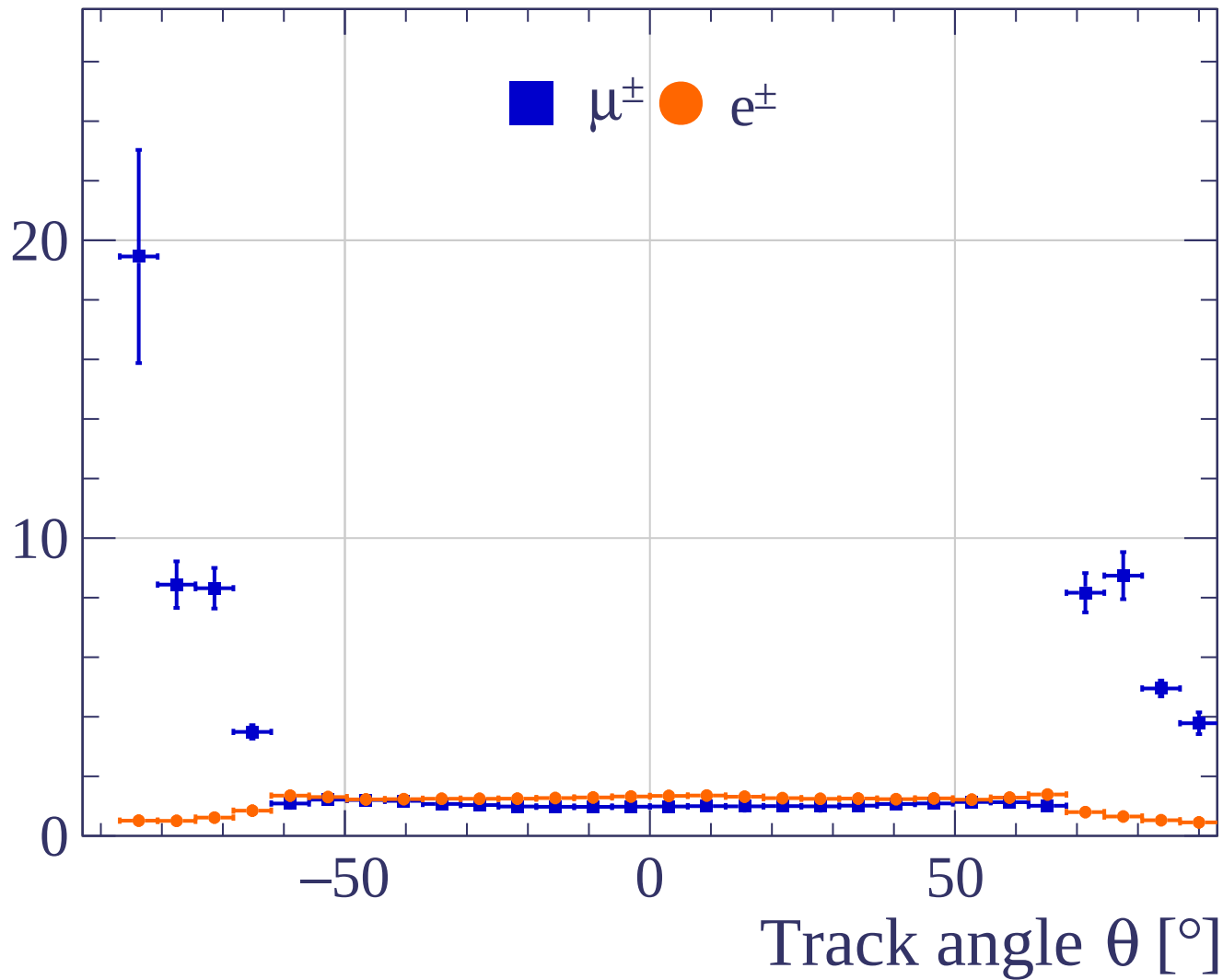




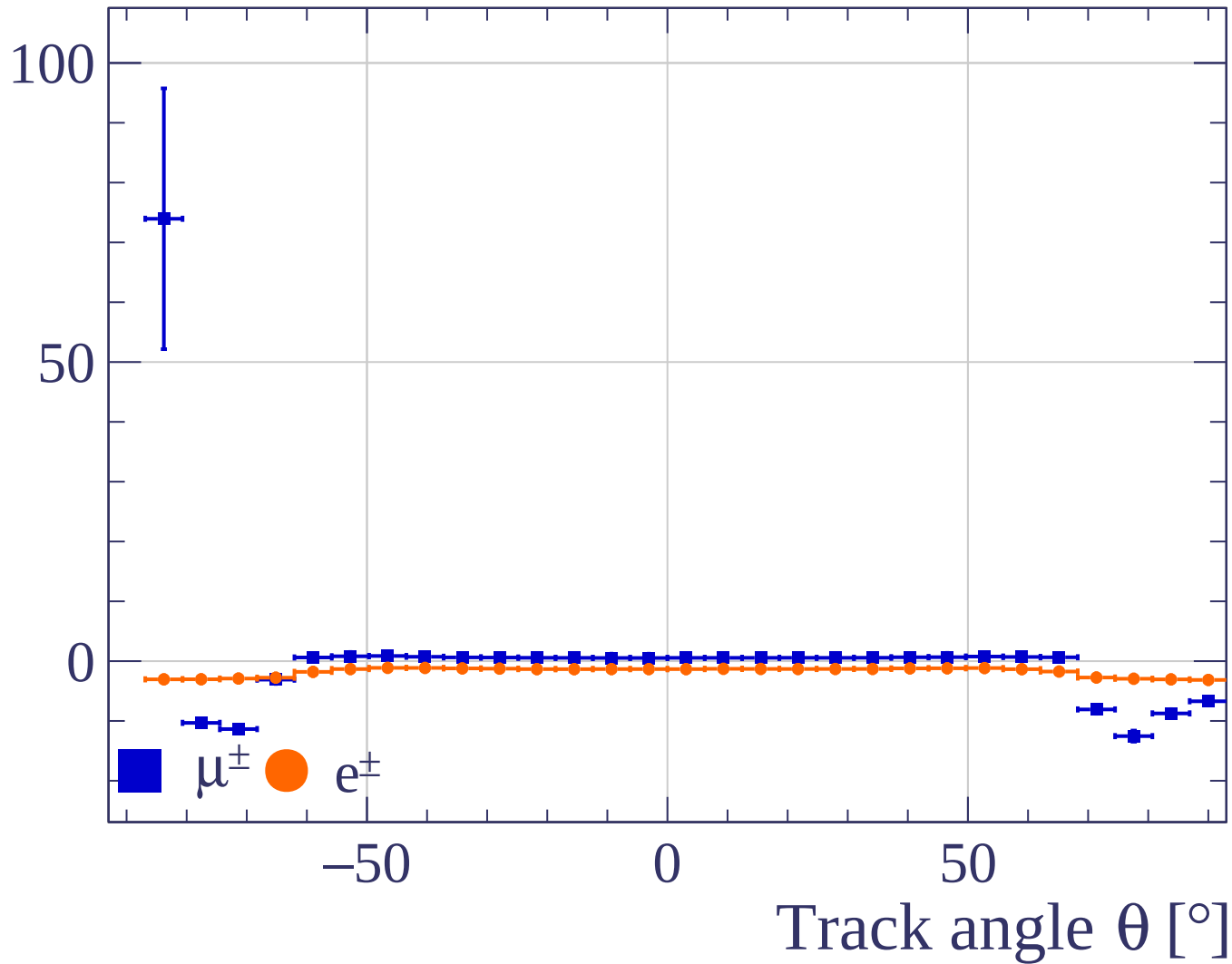
Track angle ϕ [°]



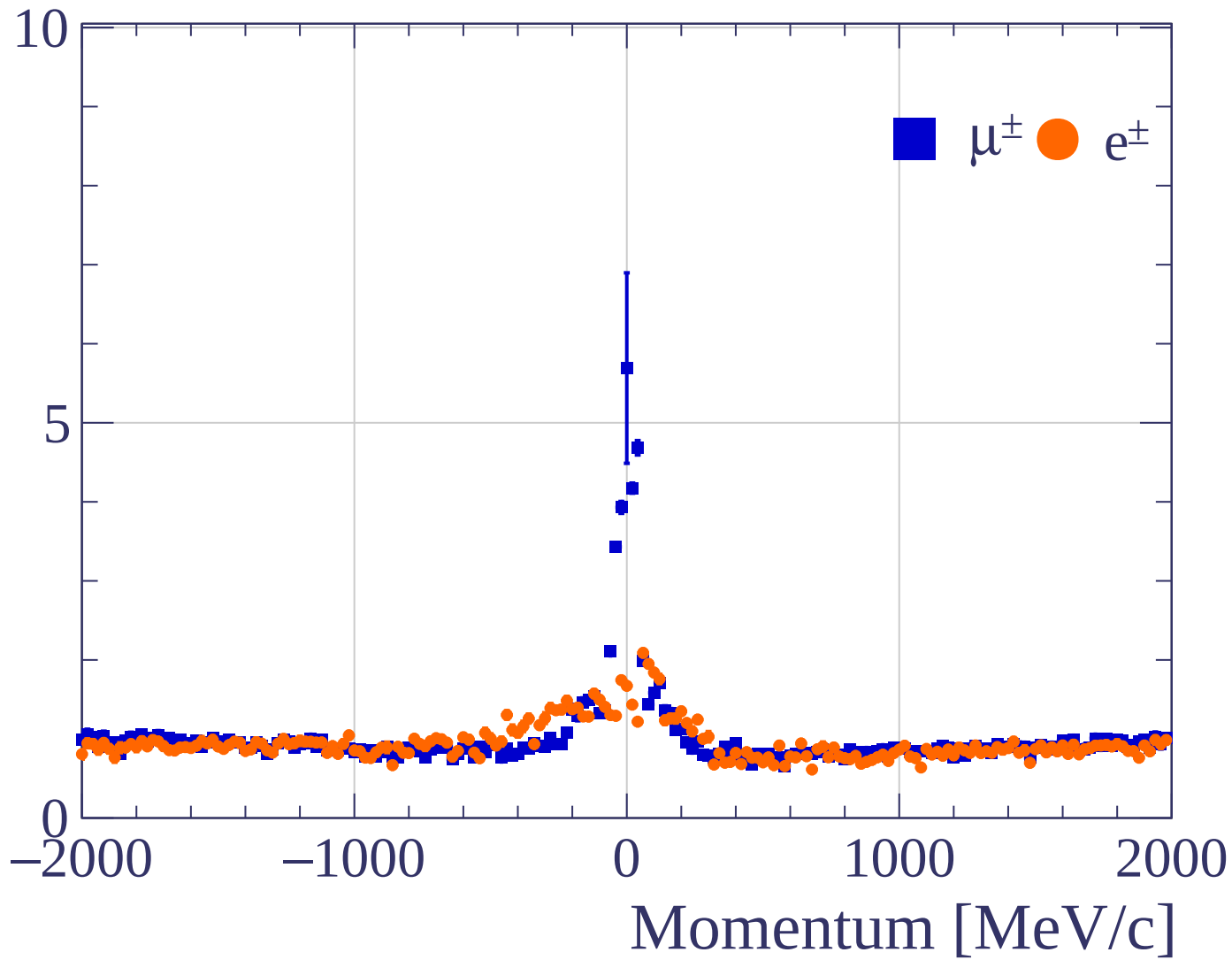
Pulls standard deviation



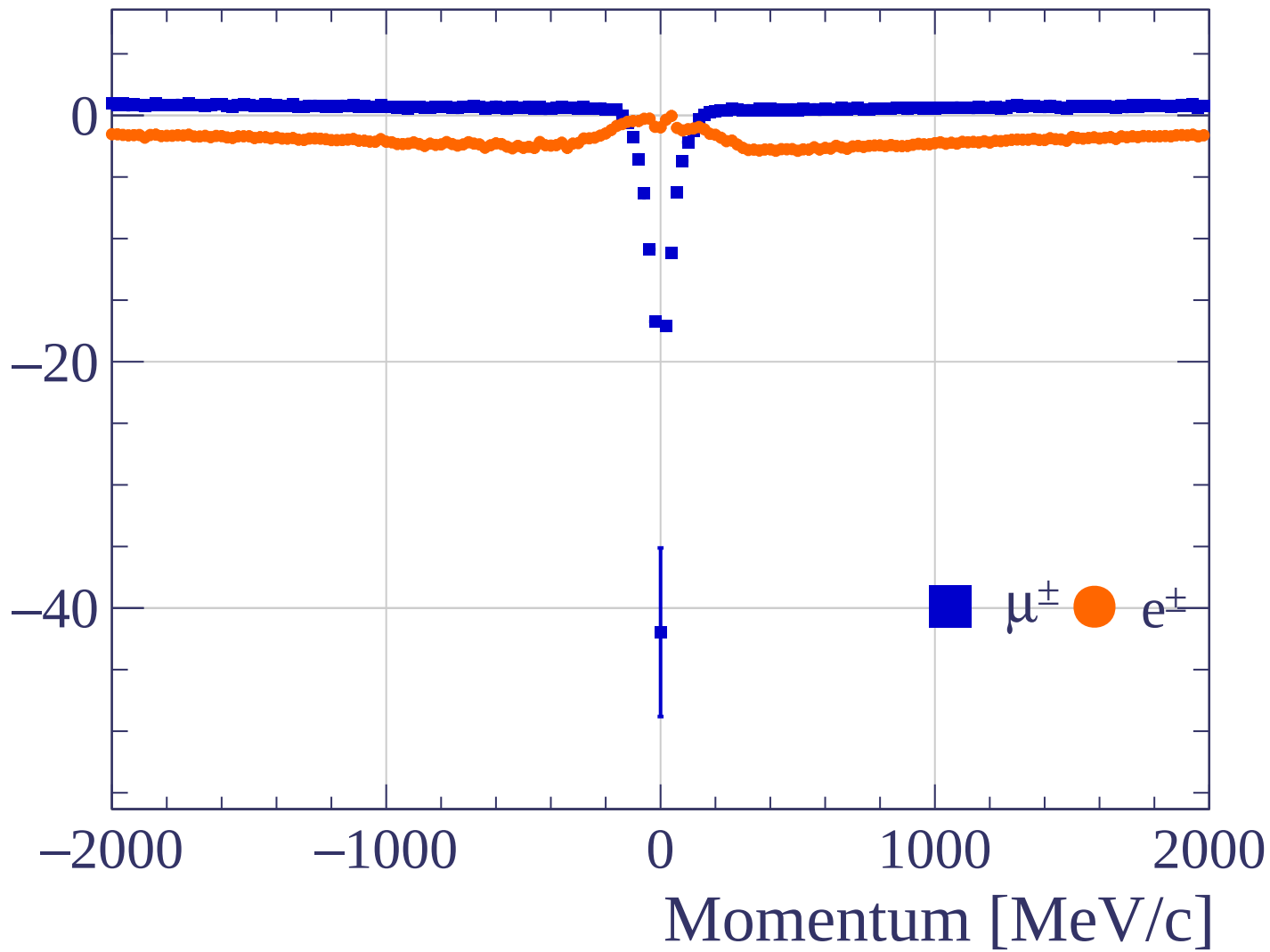
Pulls mean

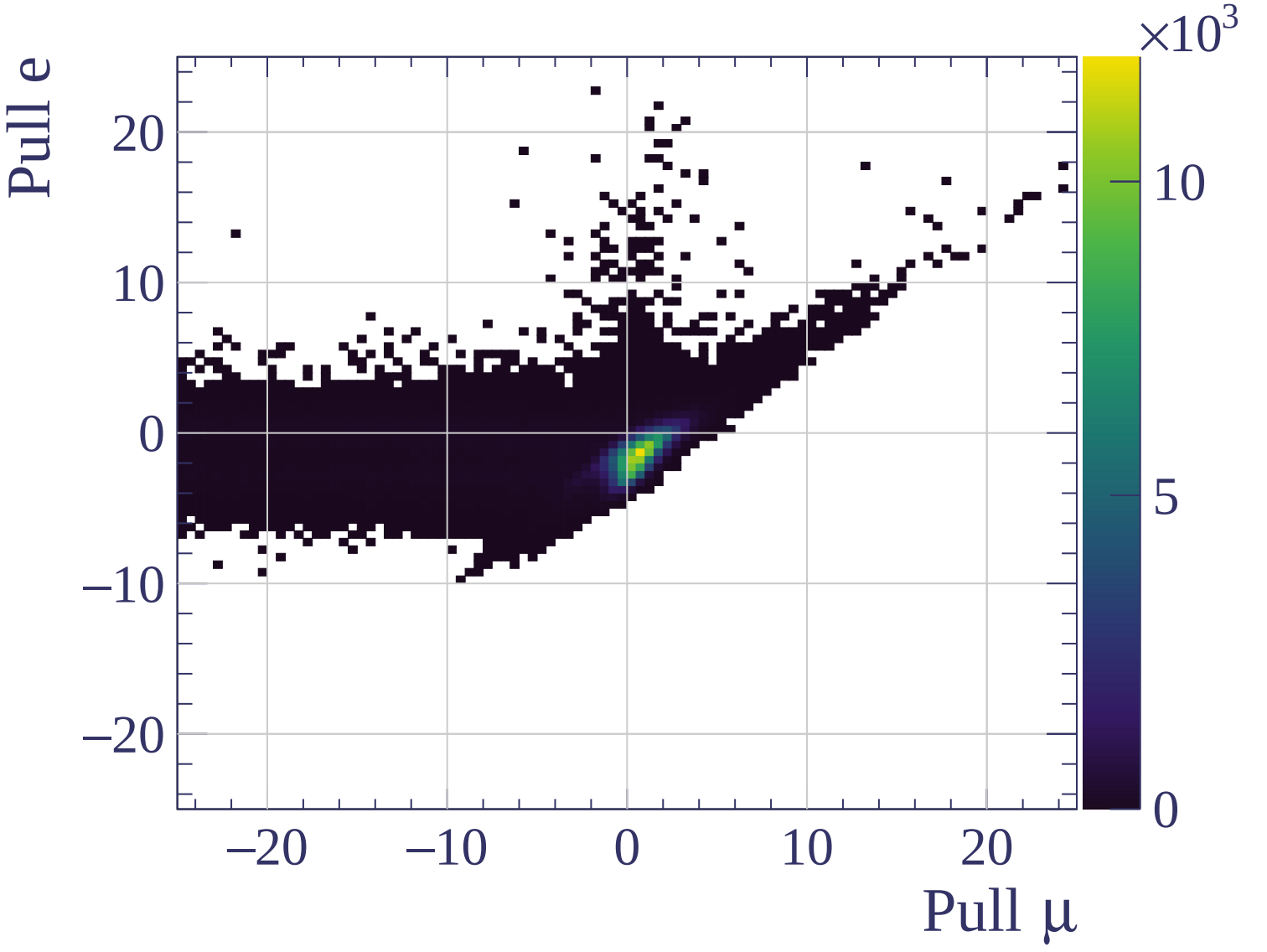


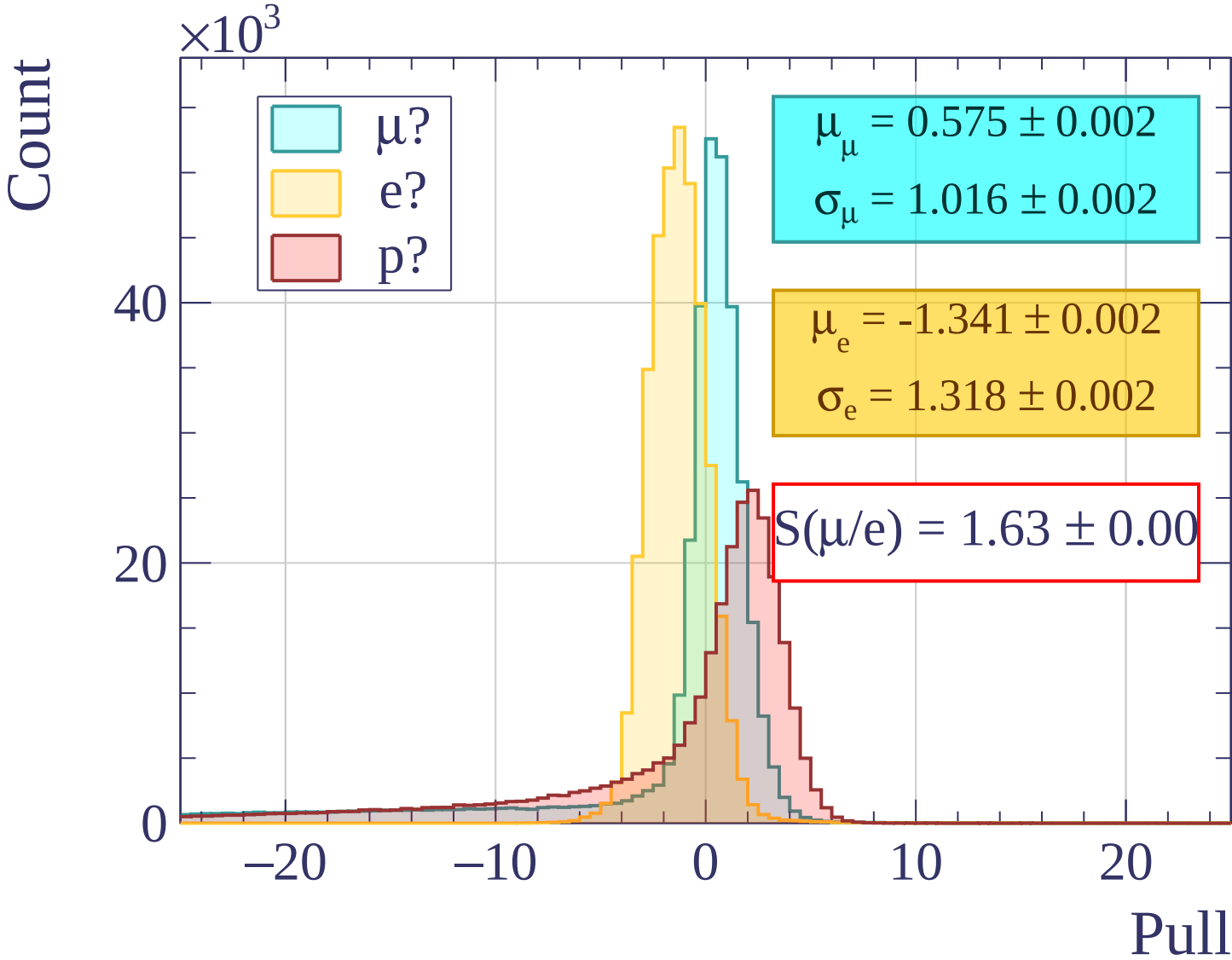
Pulls standard deviation

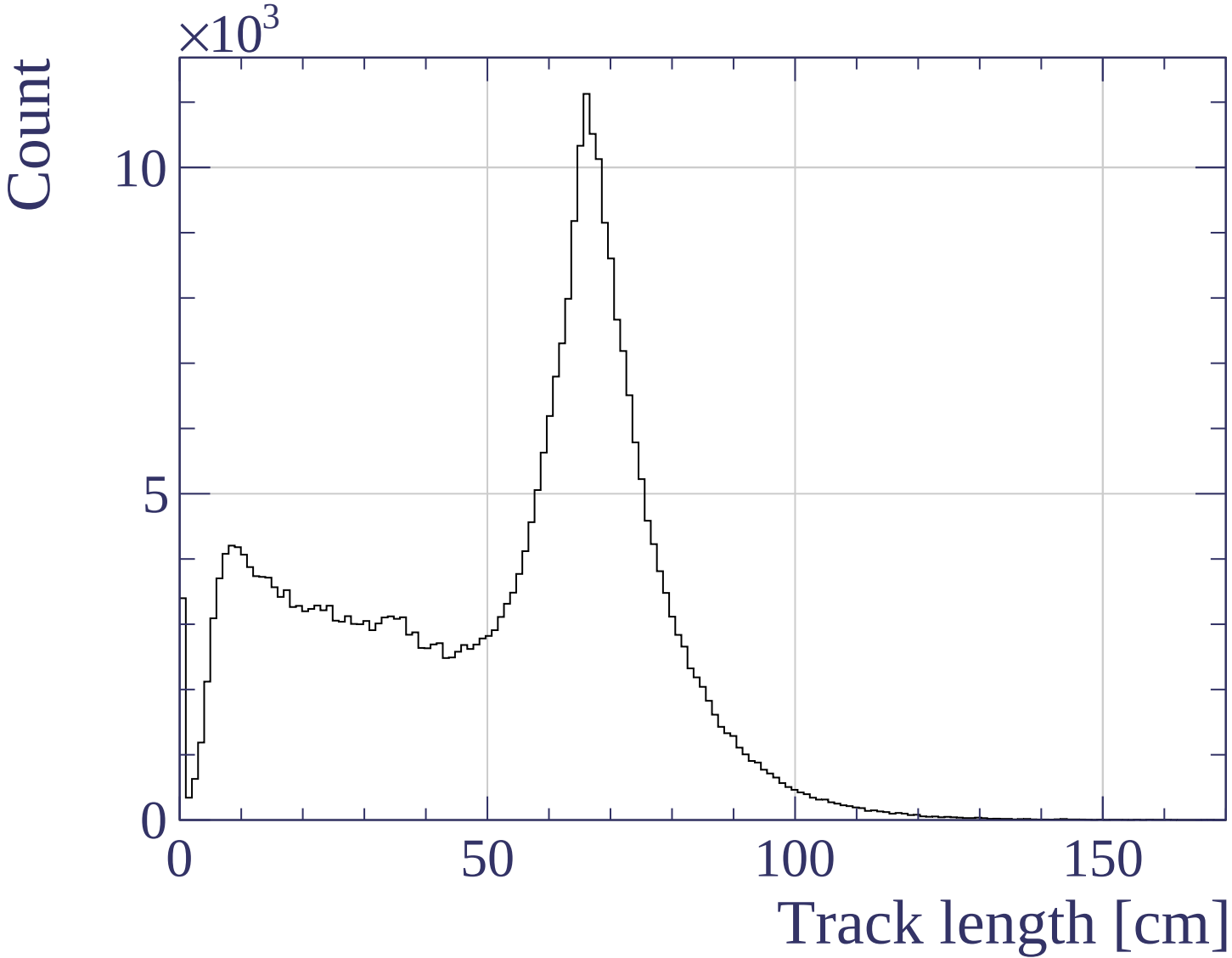


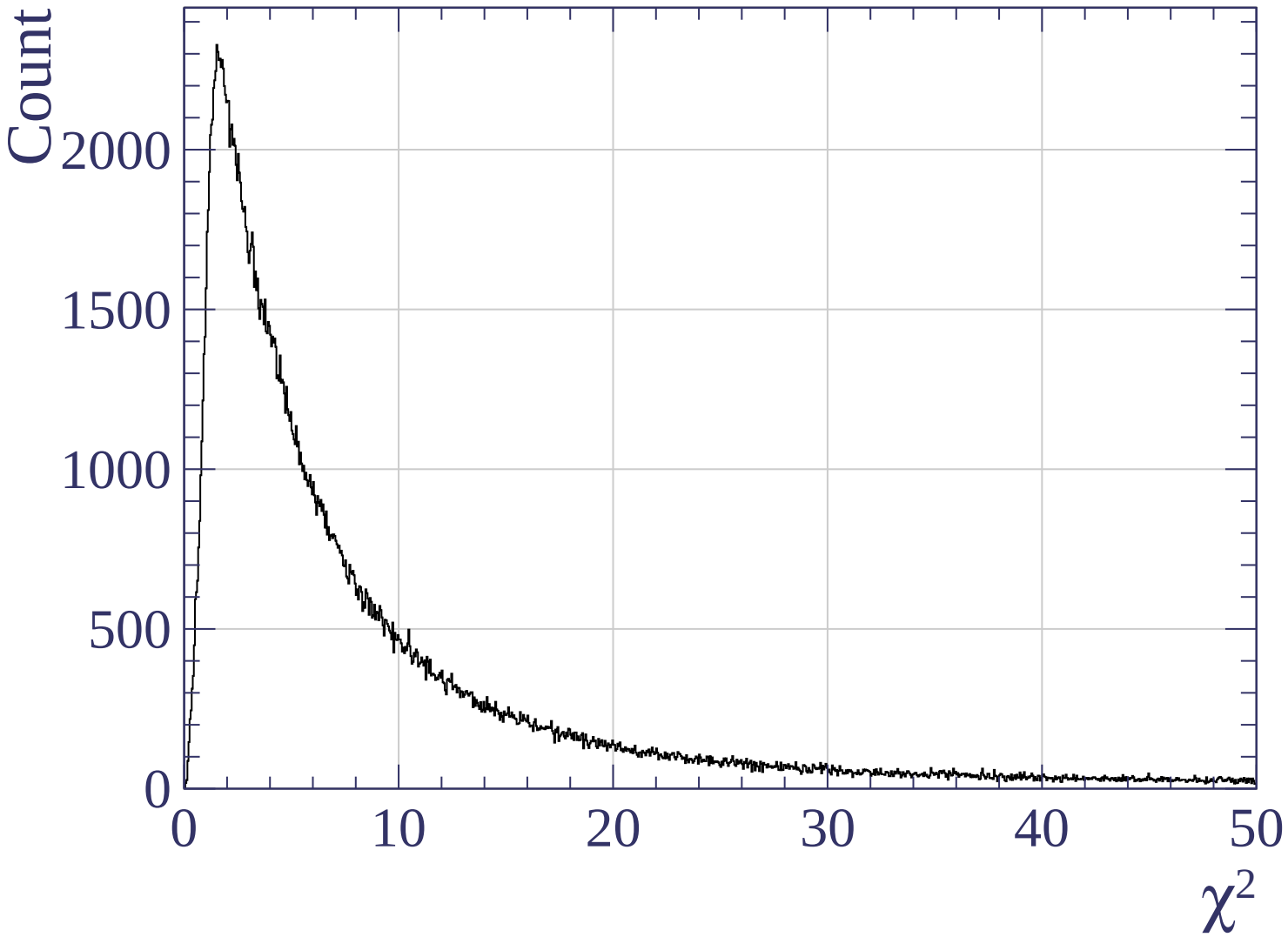
Pulls mean

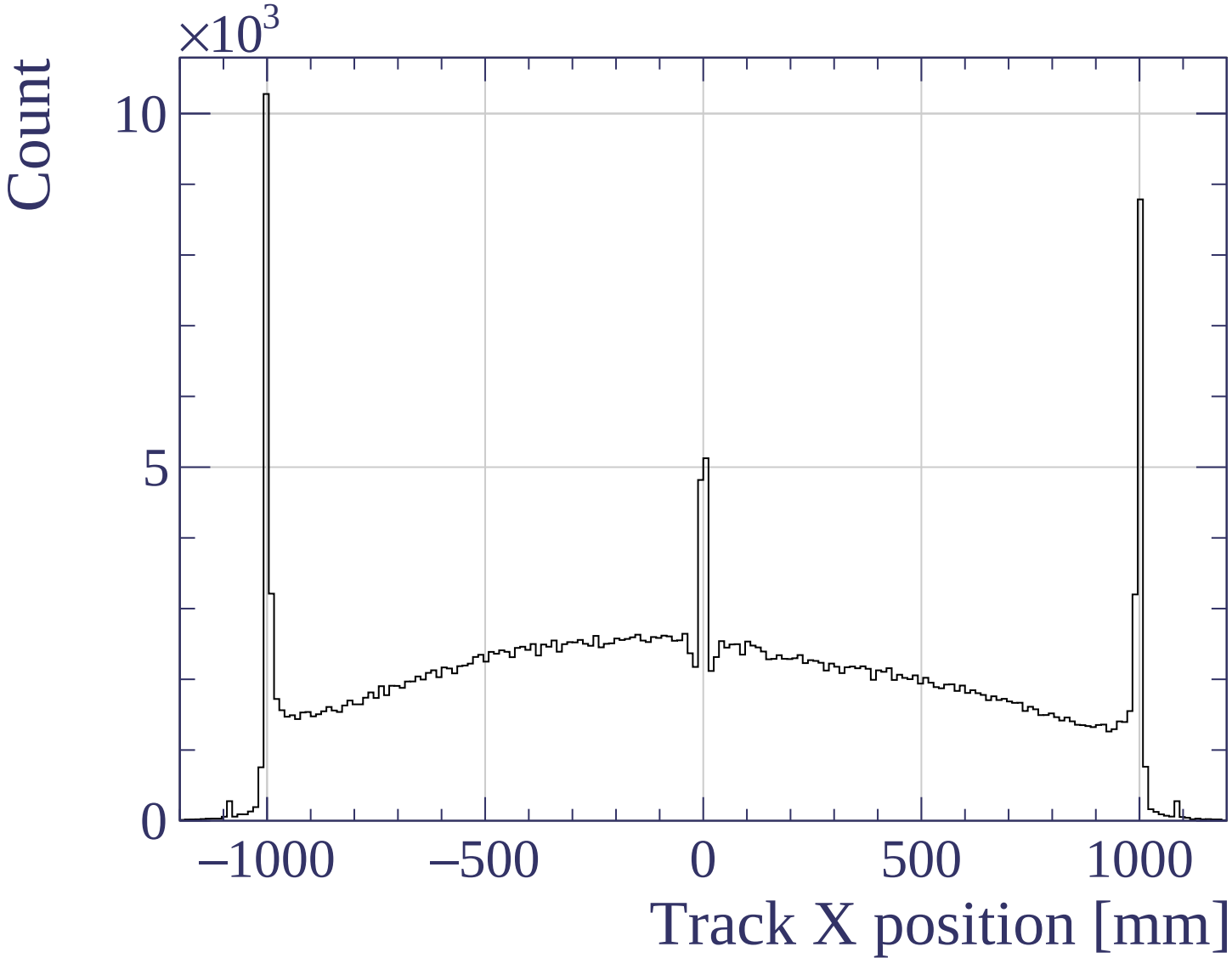


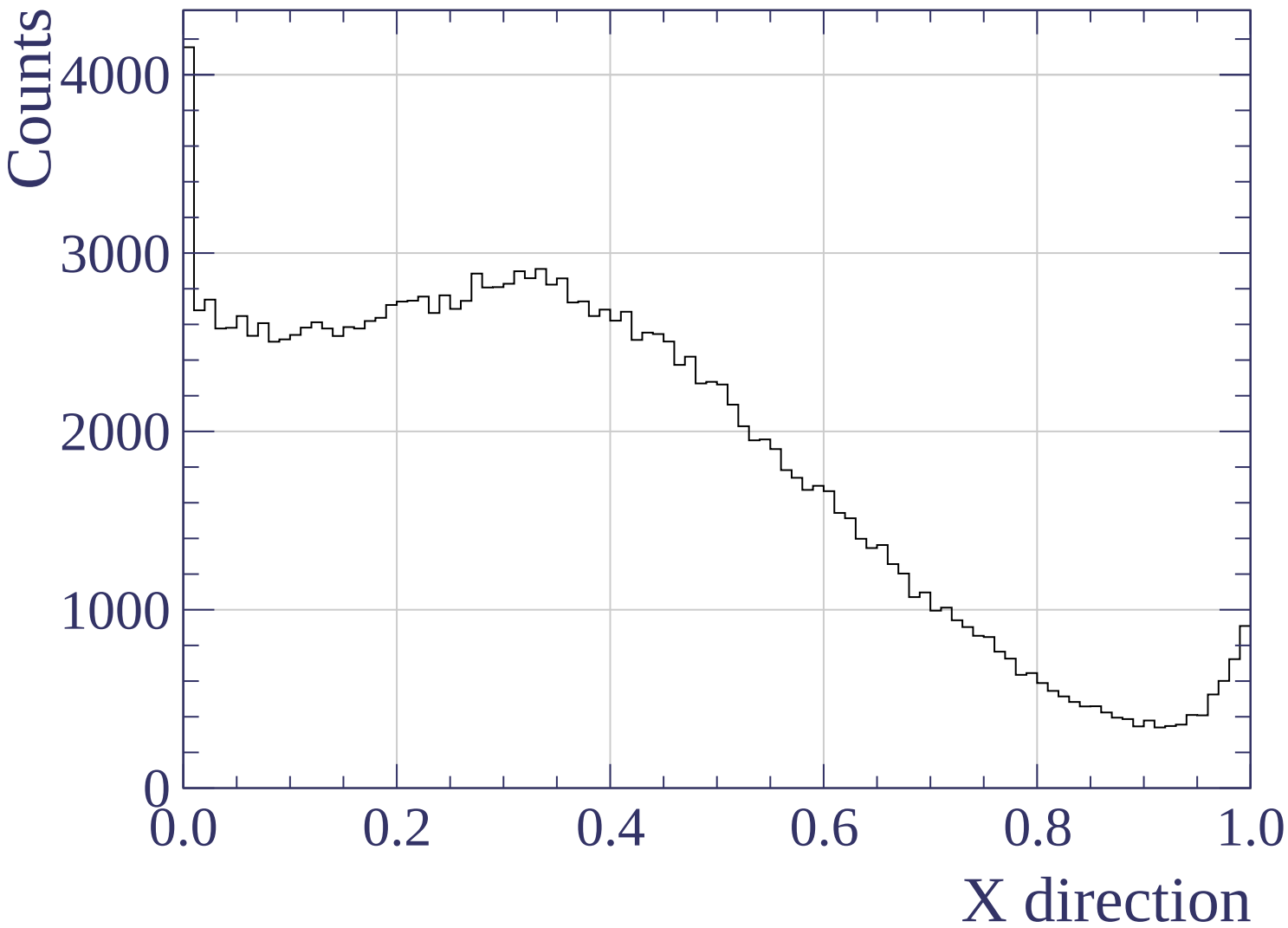


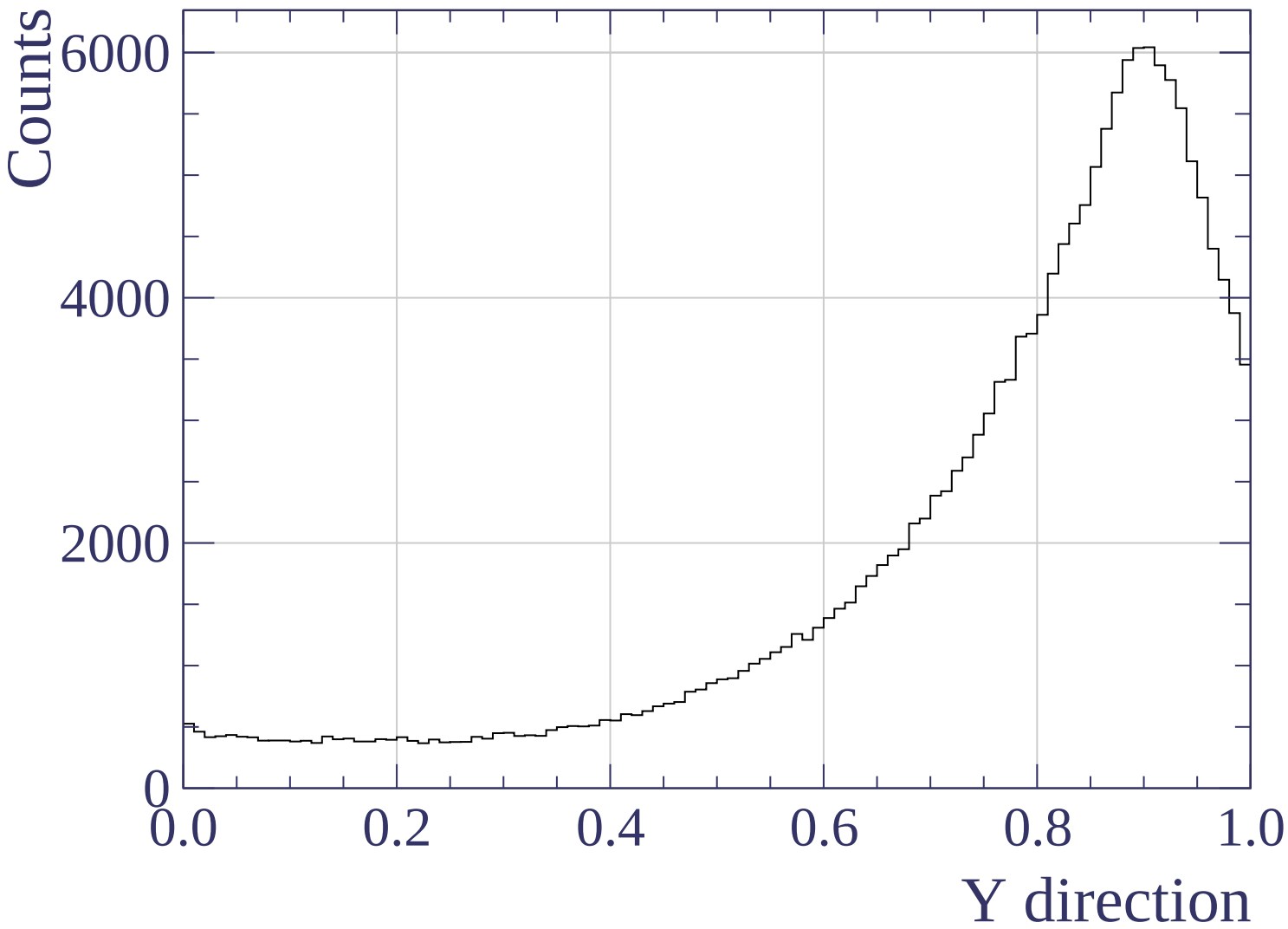




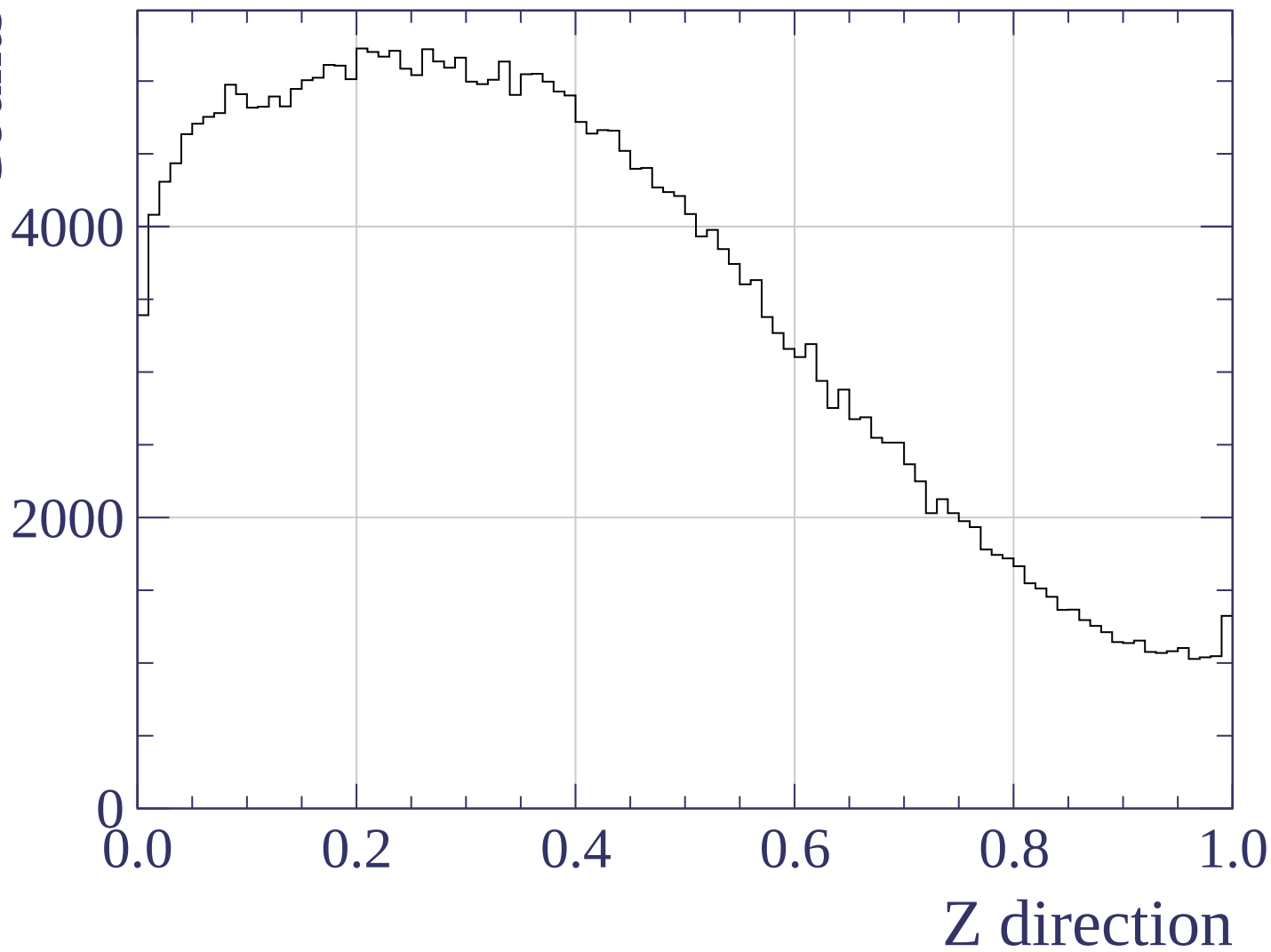


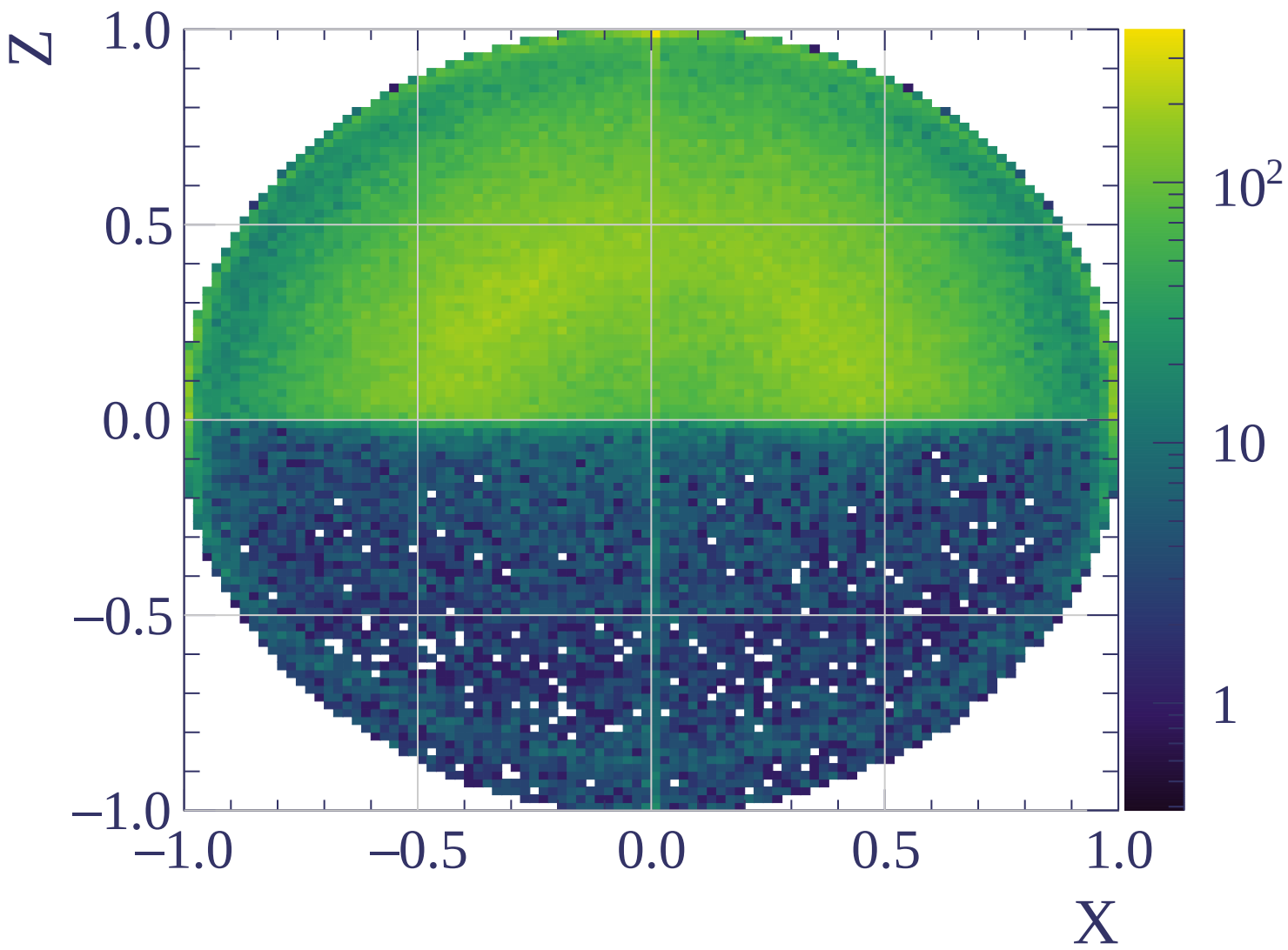


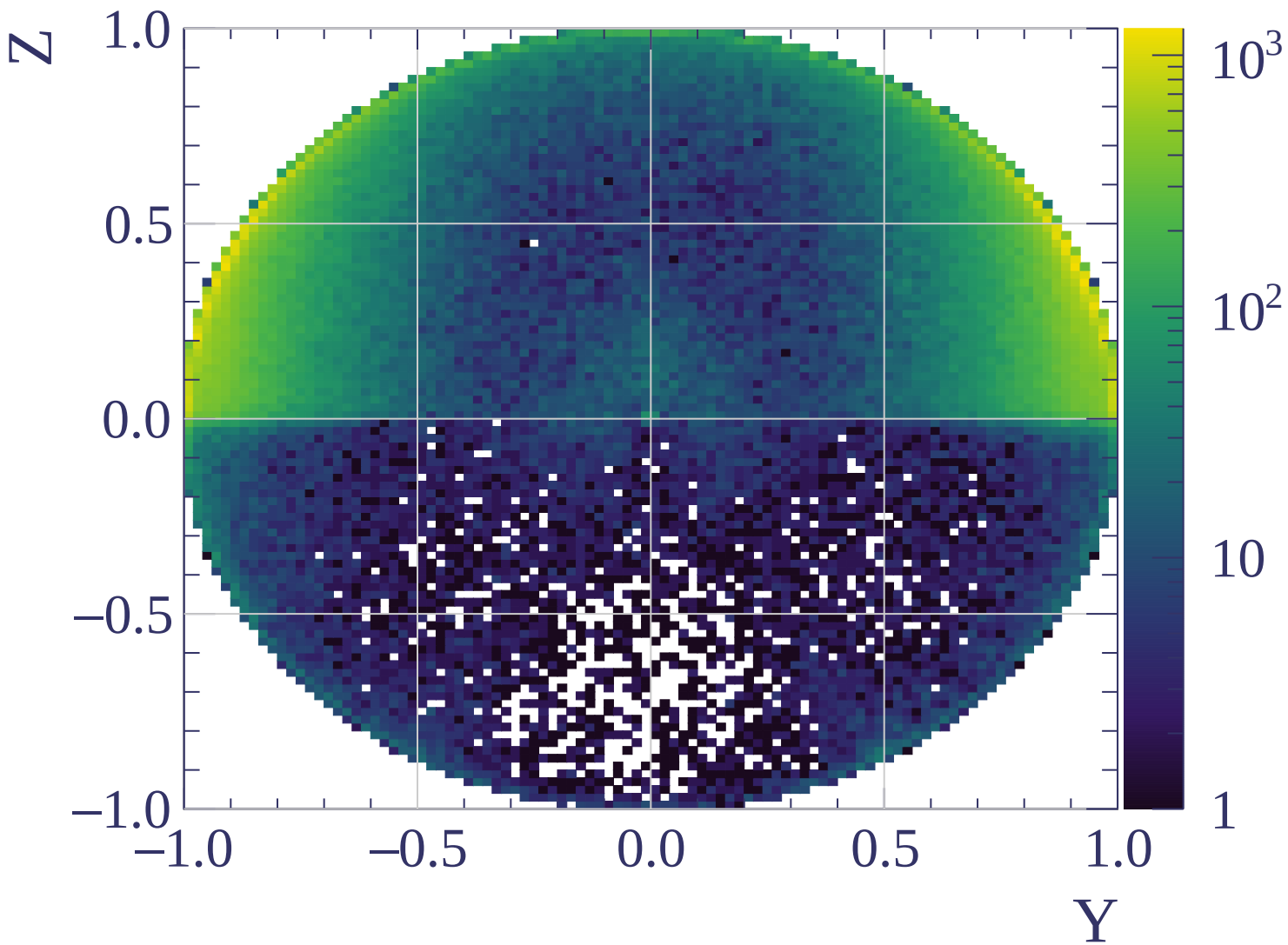


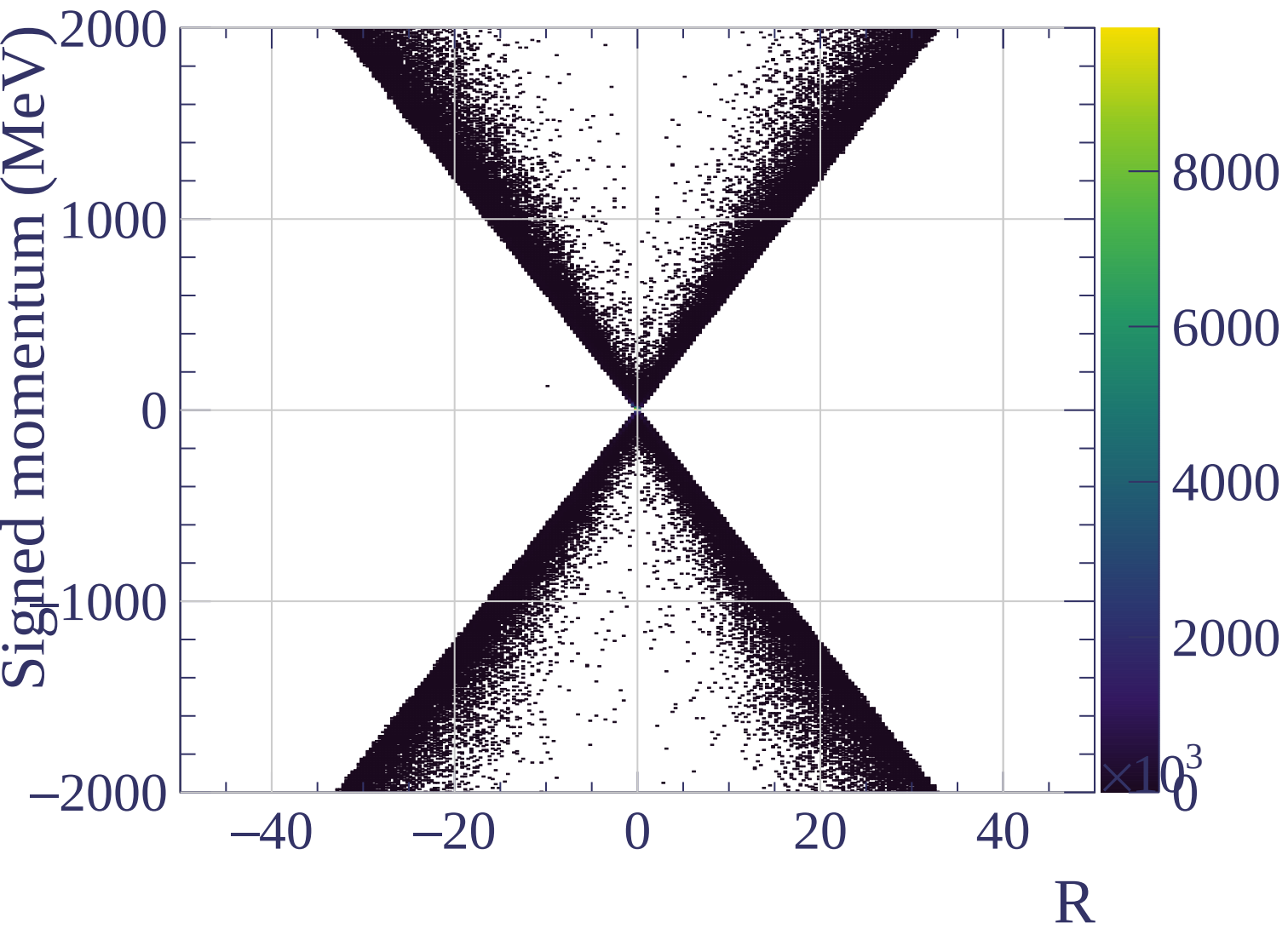


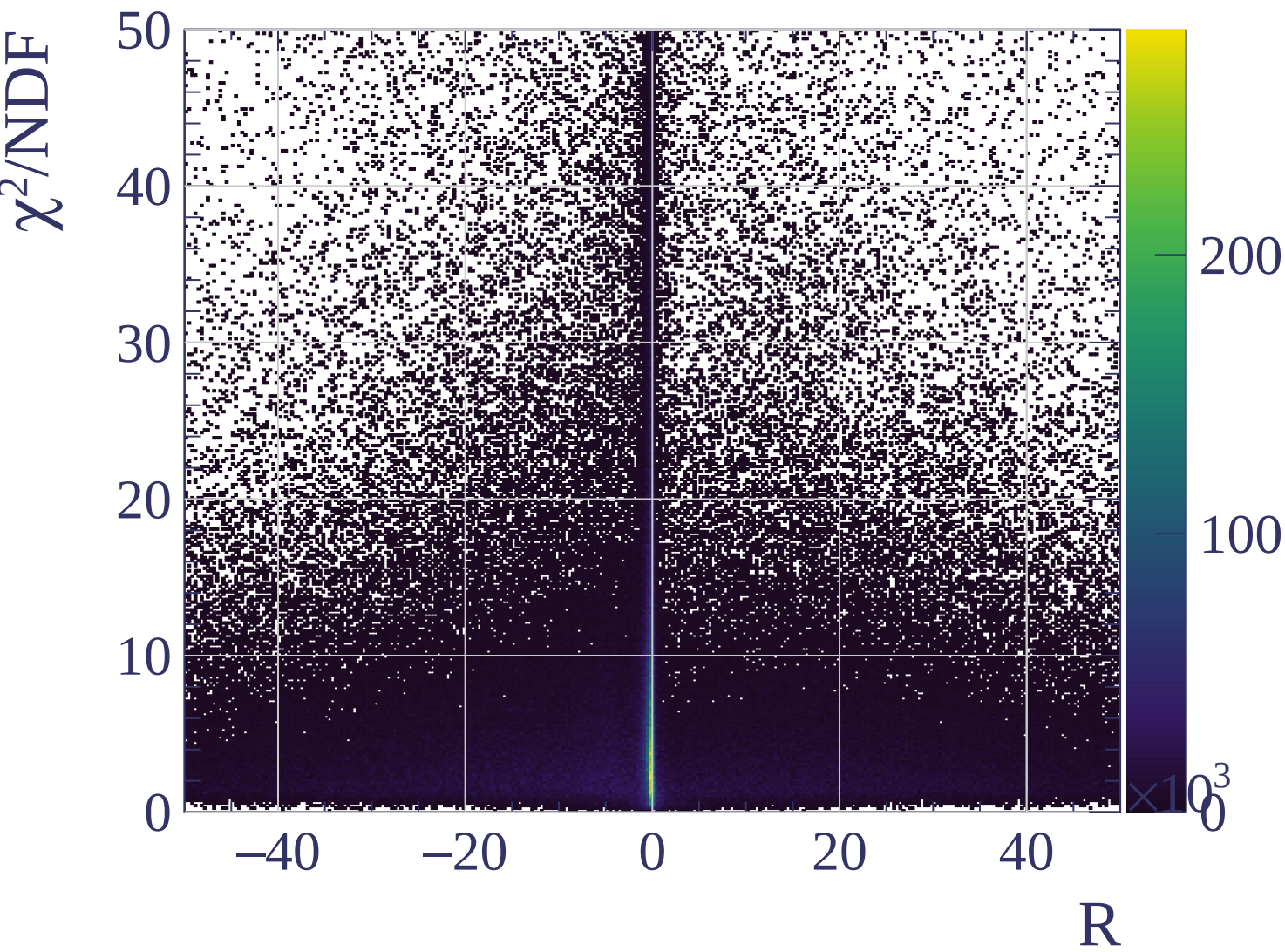
Counts

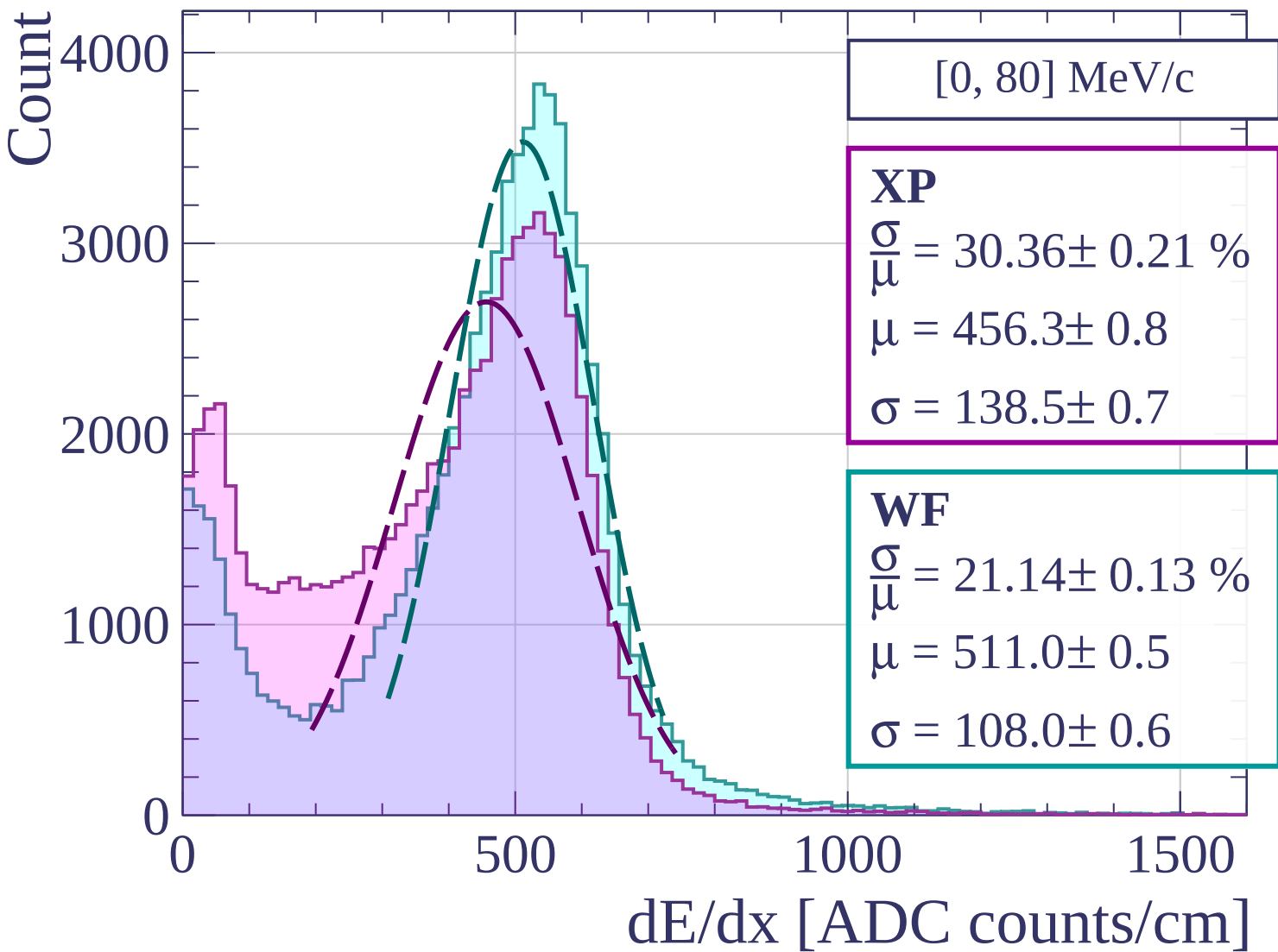












Count

[80, 160] MeV/c

XP

$$\frac{\sigma}{\mu} = 26.12 \pm 0.38 \%$$

$$\mu = 512.7 \pm 1.7$$

$$\sigma = 133.9 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 21.72 \pm 0.31 \%$$

$$\mu = 546.0 \pm 1.4$$

$$\sigma = 118.6 \pm 1.4$$

600

400

200

0

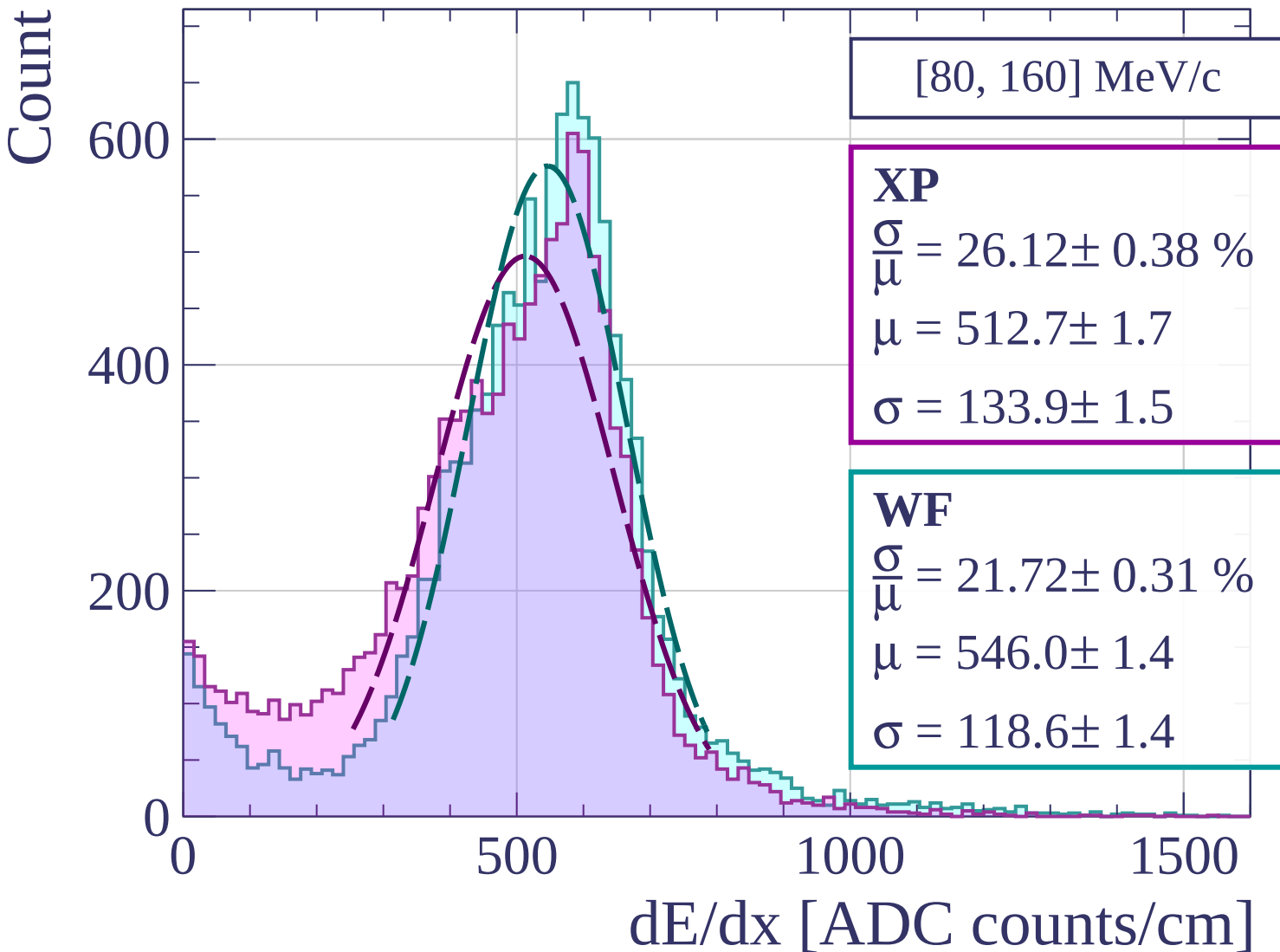
0

500

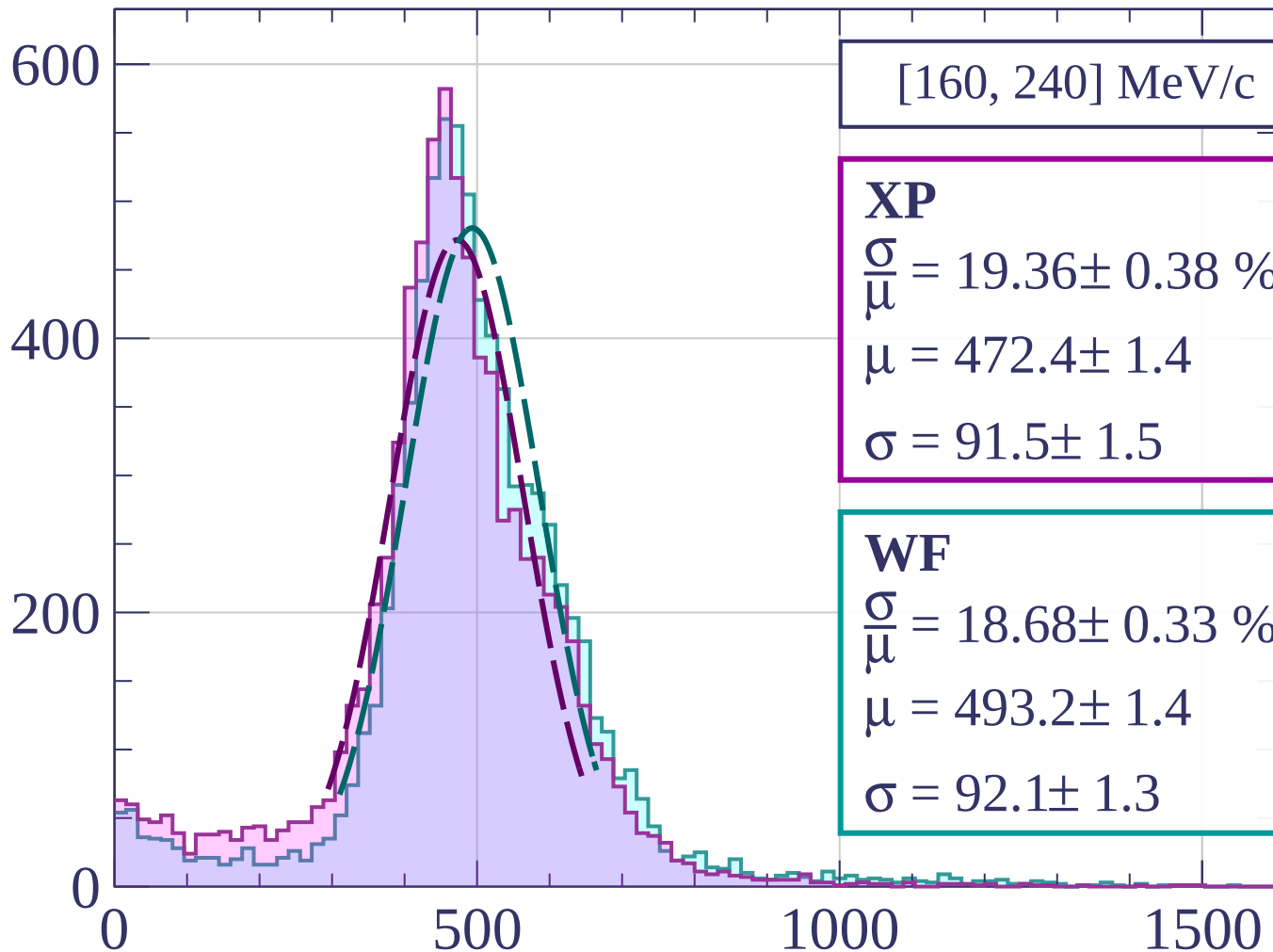
1000

1500

dE/dx [ADC counts/cm]



Count



[160, 240] MeV/c

XP

$$\frac{\sigma}{\mu} = 19.36 \pm 0.38 \%$$

$$\mu = 472.4 \pm 1.4$$

$$\sigma = 91.5 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 18.68 \pm 0.33 \%$$

$$\mu = 493.2 \pm 1.4$$

$$\sigma = 92.1 \pm 1.3$$

dE/dx [ADC counts/cm]

Count

[240, 320] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.02 \pm 0.19 \%$$

$$\mu = 432.6 \pm 0.7$$

$$\sigma = 47.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.86 \pm 0.19 \%$$

$$\mu = 437.3 \pm 0.7$$

$$\sigma = 47.5 \pm 0.7$$

800

600

400

200

0

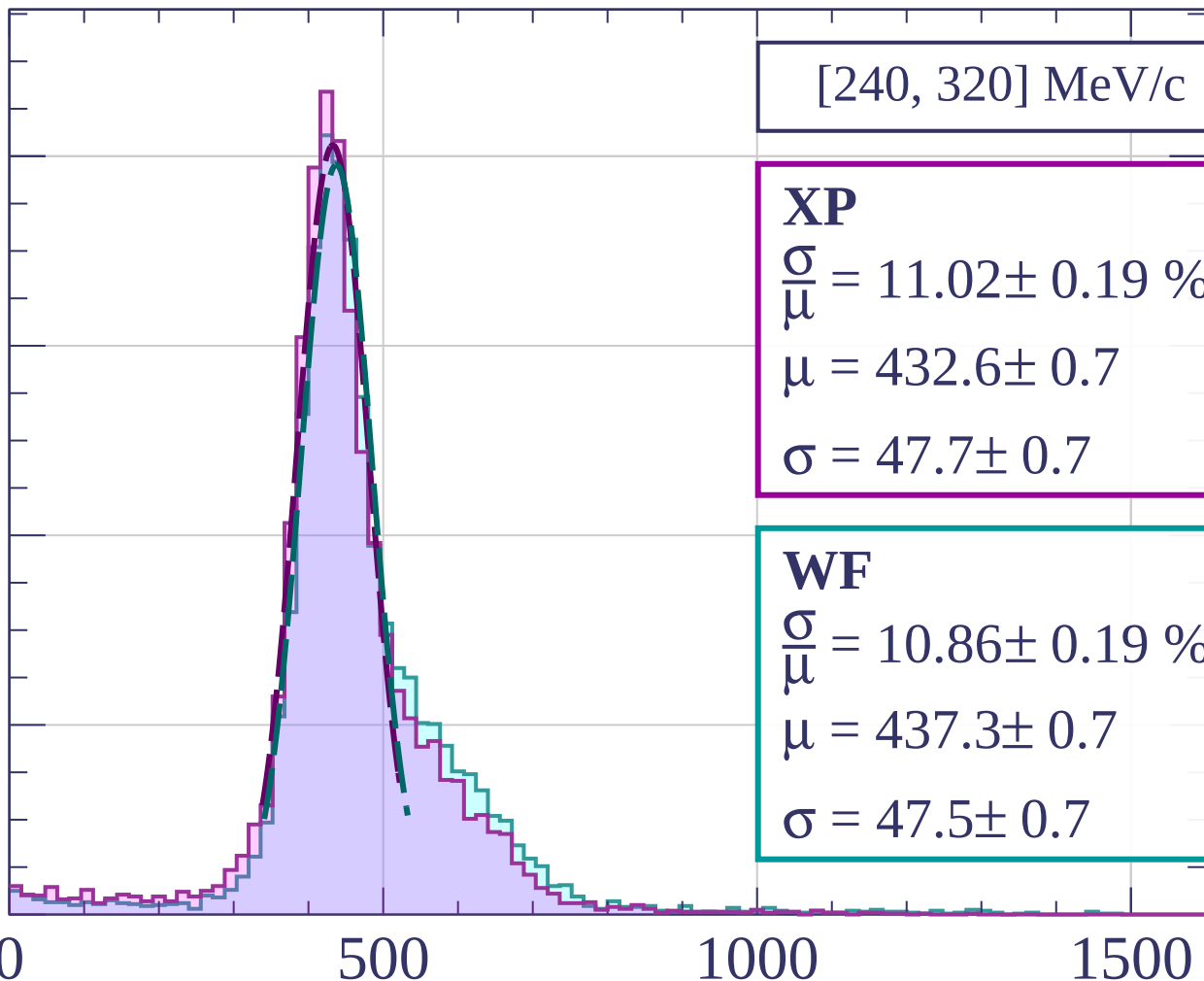
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[320, 400] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.42 \pm 0.18 \%$$

$$\mu = 423.5 \pm 0.7$$

$$\sigma = 44.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.39 \pm 0.19 \%$$

$$\mu = 429.0 \pm 0.7$$

$$\sigma = 44.6 \pm 0.7$$

800

600

400

200

0

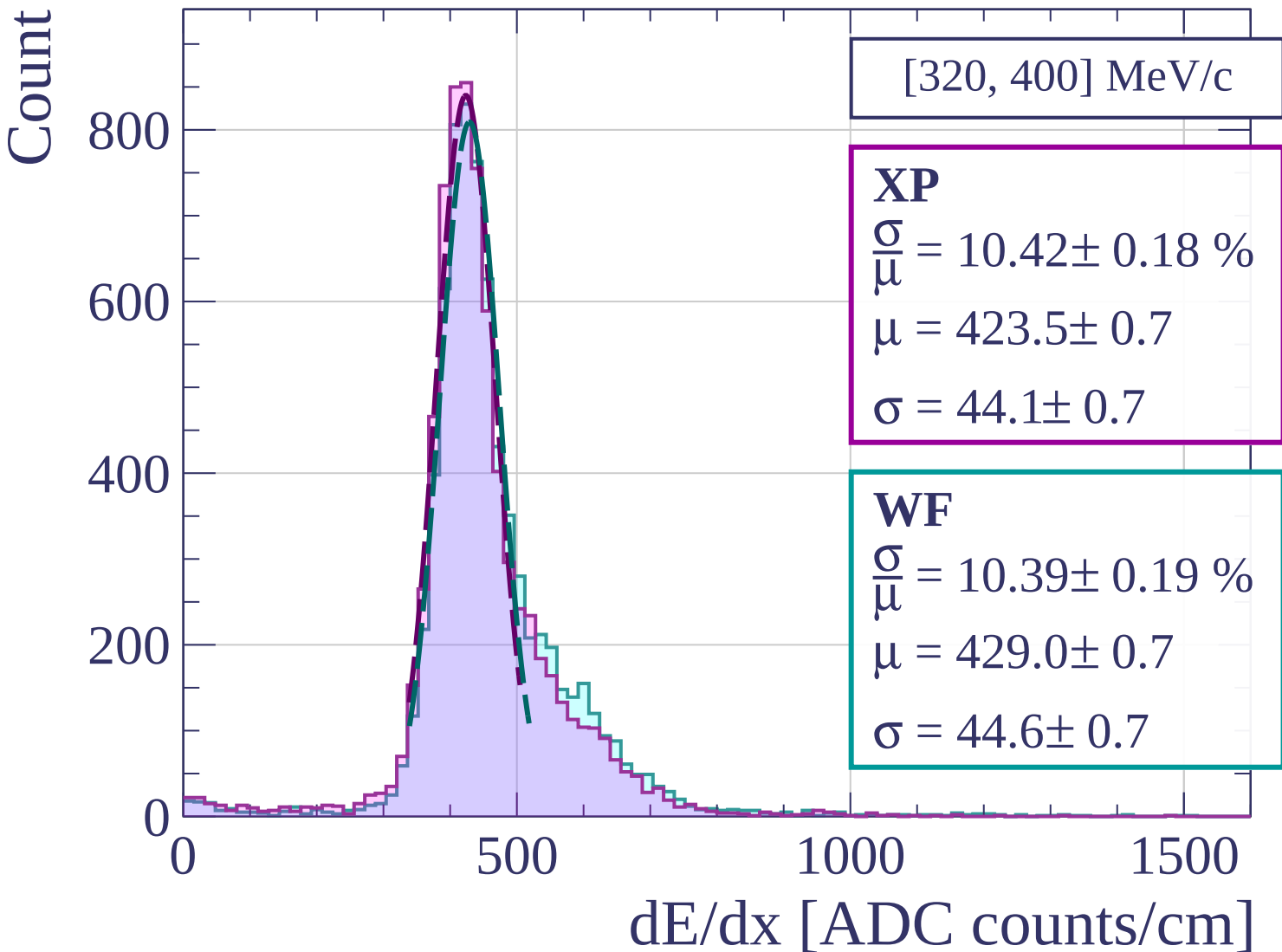
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[400, 480] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.41 \pm 0.19 \%$$

$$\mu = 423.8 \pm 0.7$$

$$\sigma = 39.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.22 \pm 0.19 \%$$

$$\mu = 430.6 \pm 0.8$$

$$\sigma = 44.0 \pm 0.7$$

800

600

400

200

0

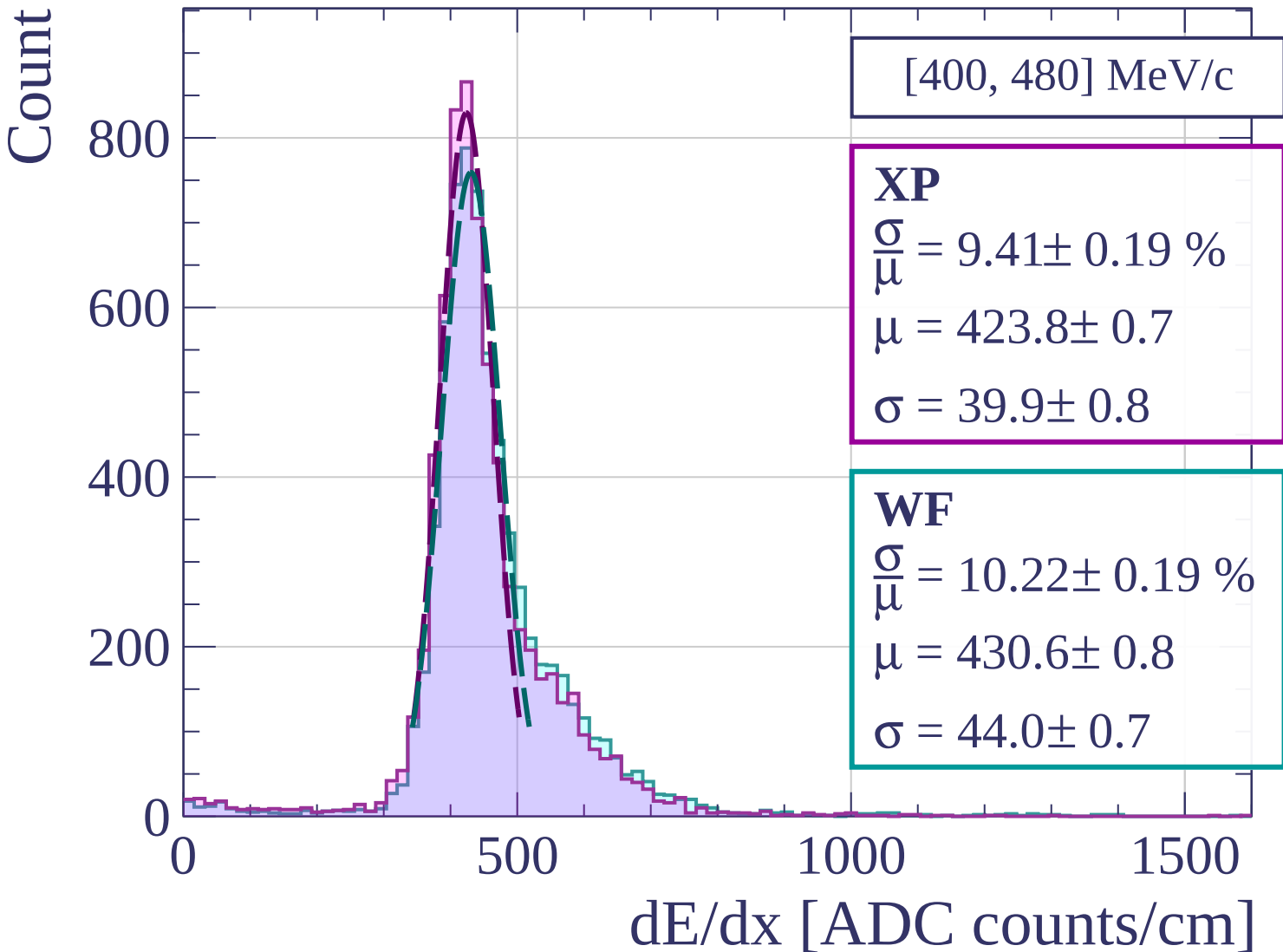
0

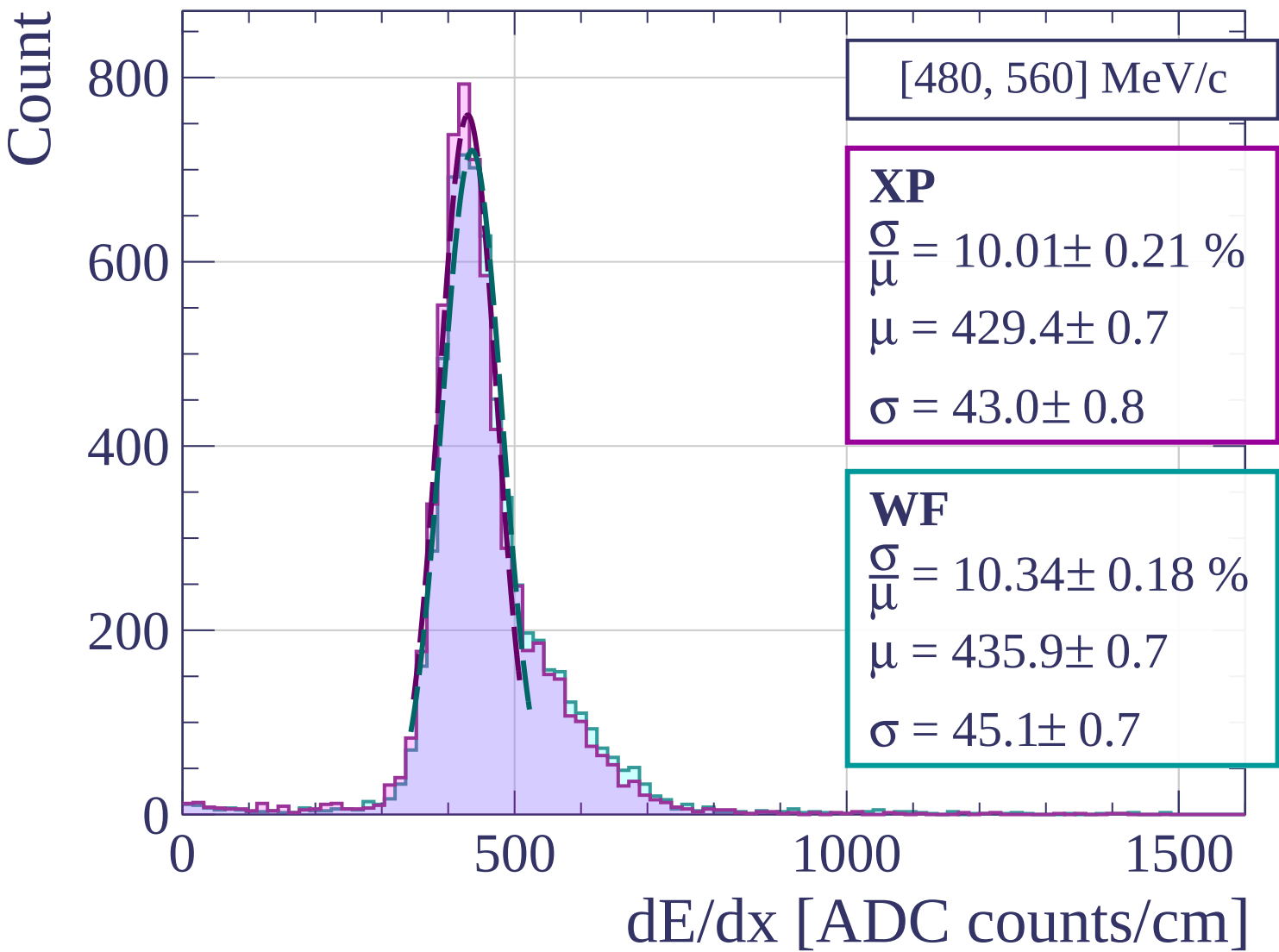
500

1000

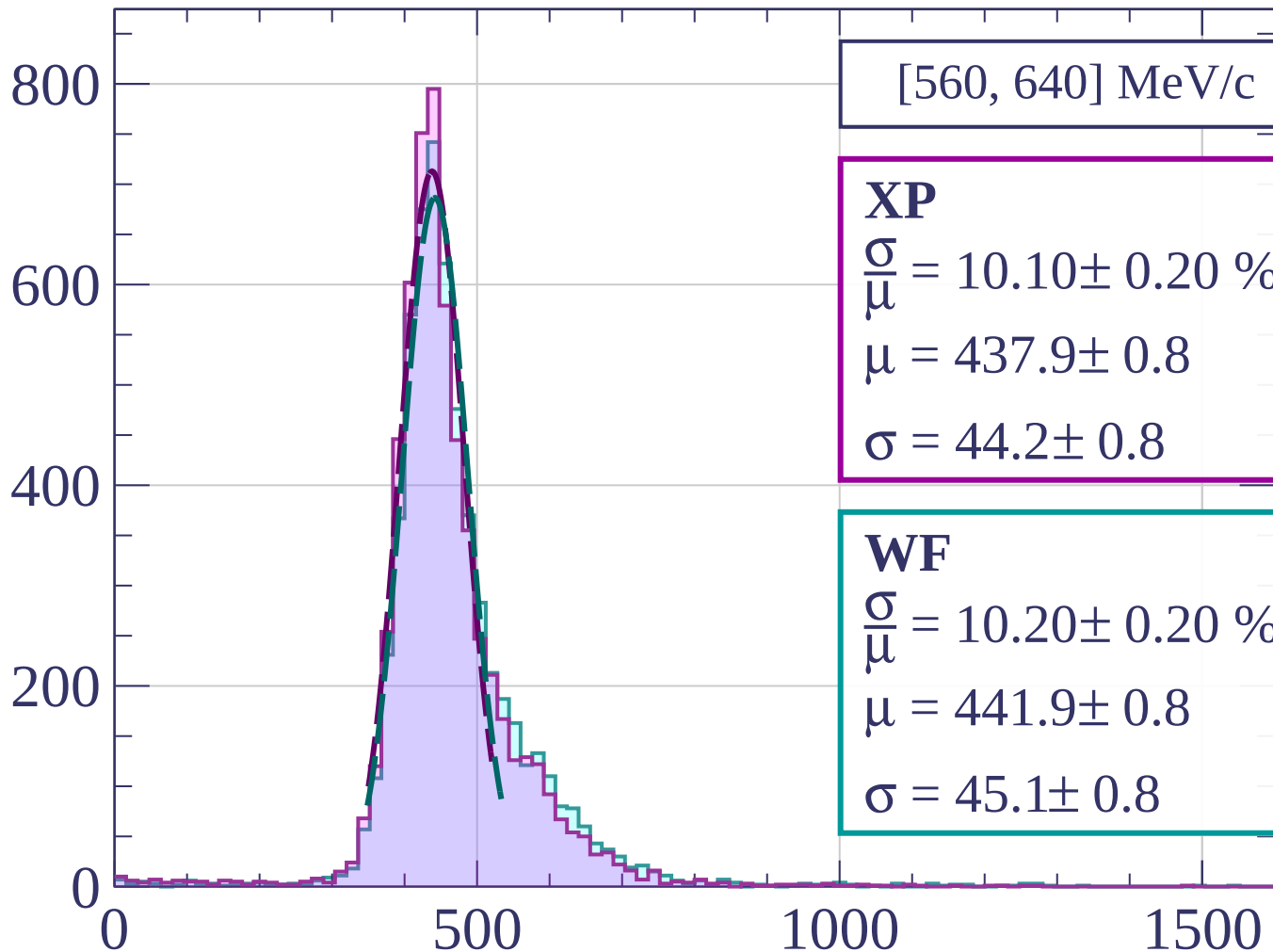
1500

dE/dx [ADC counts/cm]



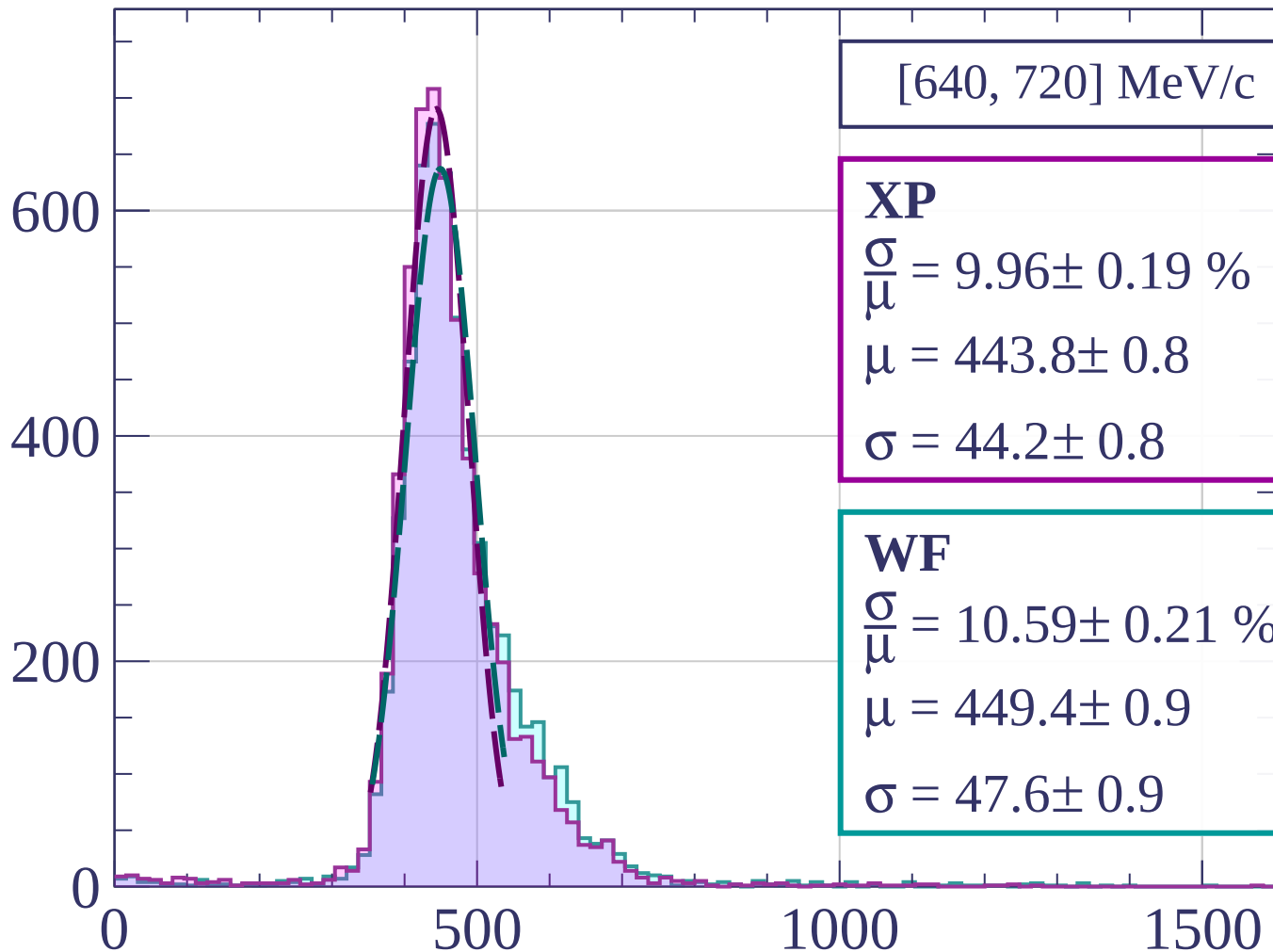


Count



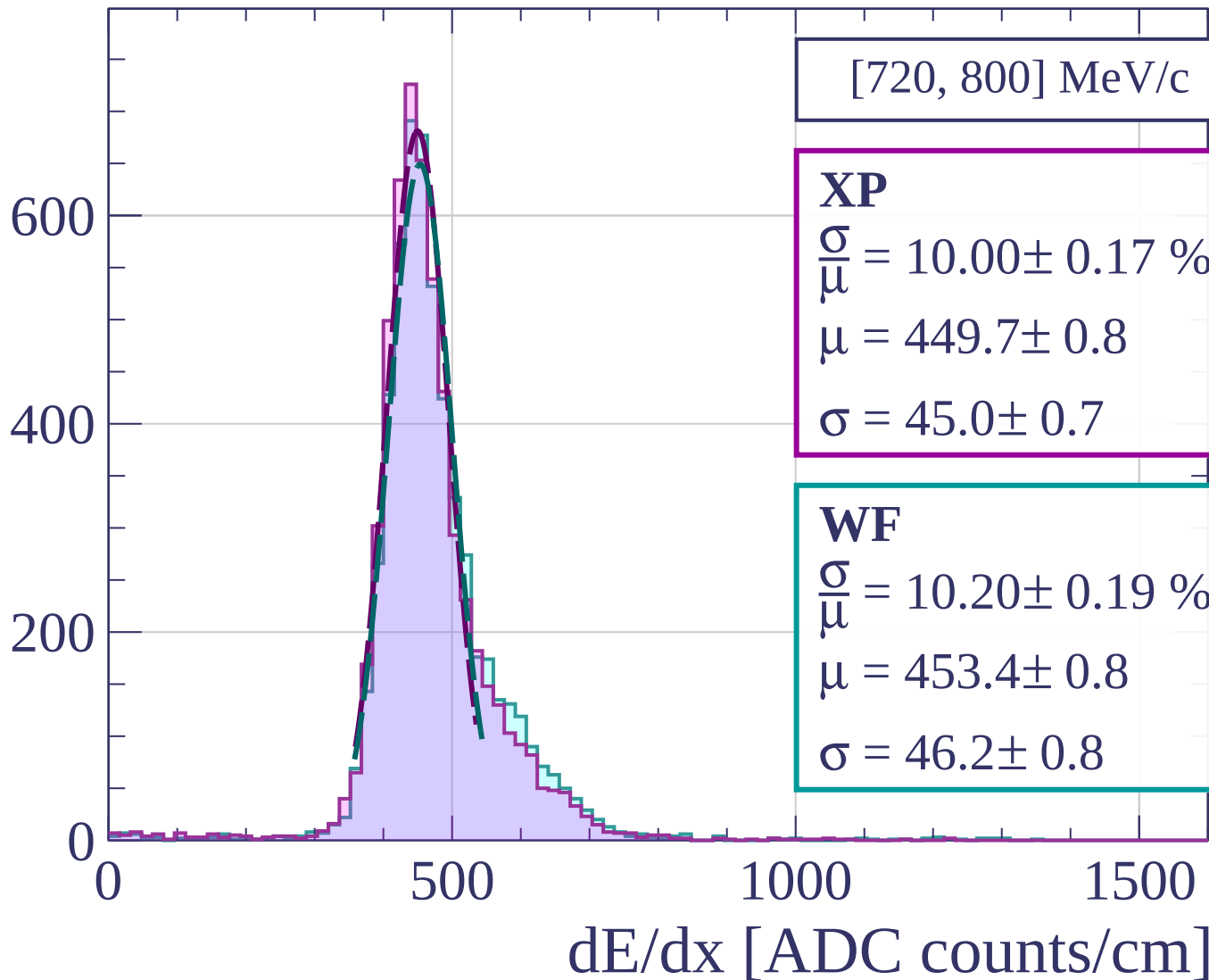
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



Count

[800, 880] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.92 \pm 0.20 \%$$

$$\mu = 454.2 \pm 0.8$$

$$\sigma = 45.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.70 \pm 0.19 \%$$

$$\mu = 458.8 \pm 0.8$$

$$\sigma = 44.5 \pm 0.8$$

600

400

200

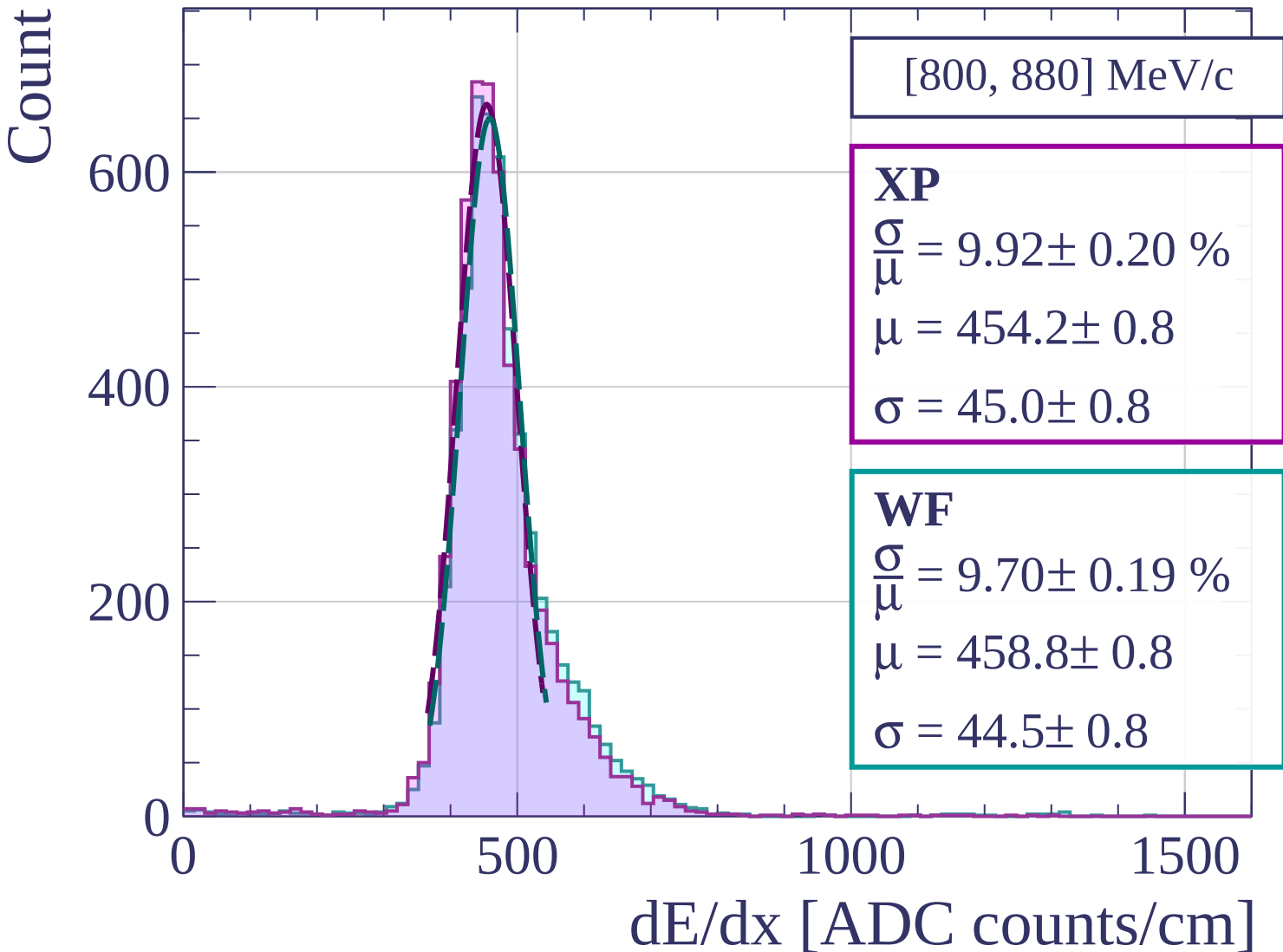
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[880, 960] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.92 \pm 0.20 \%$$

$$\mu = 458.9 \pm 0.8$$

$$\sigma = 45.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.53 \pm 0.21 \%$$

$$\mu = 465.8 \pm 0.9$$

$$\sigma = 49.1 \pm 0.9$$

600

400

200

0

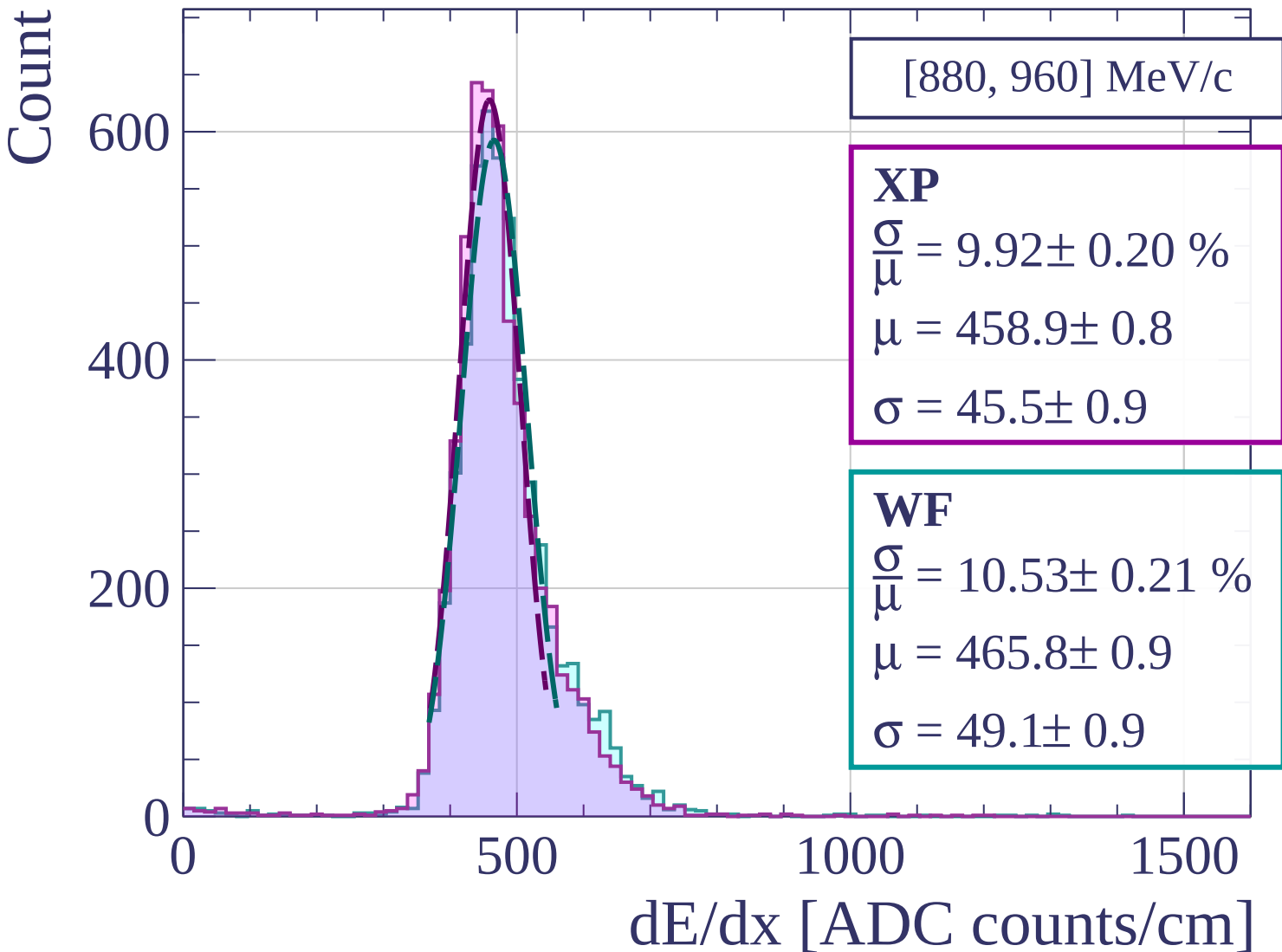
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[960, 1040] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.26 \pm 0.20 \%$$

$$\mu = 466.6 \pm 0.9$$

$$\sigma = 47.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.66 \pm 0.20 \%$$

$$\mu = 473.4 \pm 0.9$$

$$\sigma = 50.5 \pm 0.9$$

600

400

200

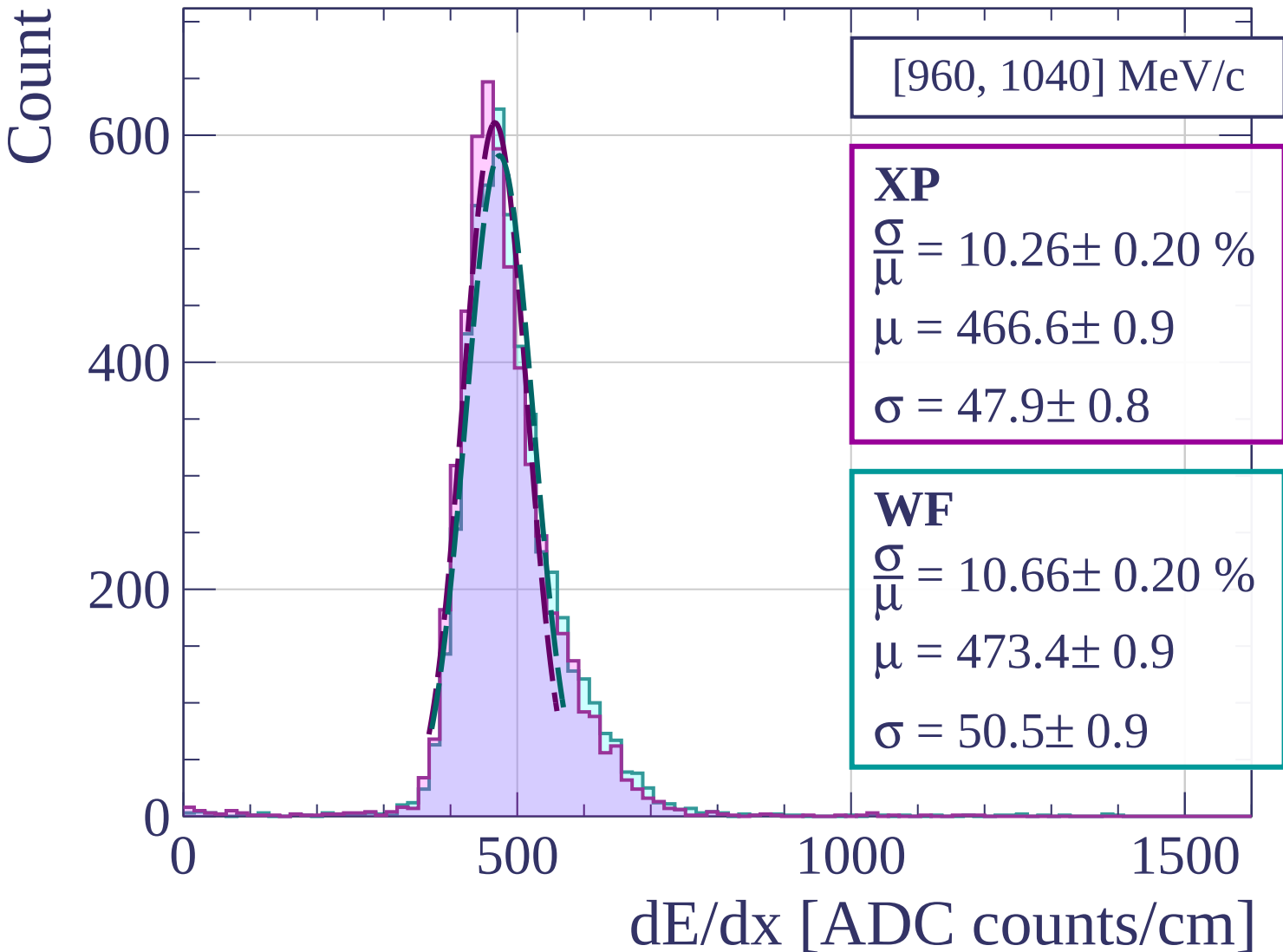
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1040, 1120] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.21 \pm 0.22 \%$$

$$\mu = 472.0 \pm 0.9$$

$$\sigma = 48.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.83 \pm 0.24 \%$$

$$\mu = 478.4 \pm 1.0$$

$$\sigma = 51.8 \pm 1.0$$

600

400

200

0

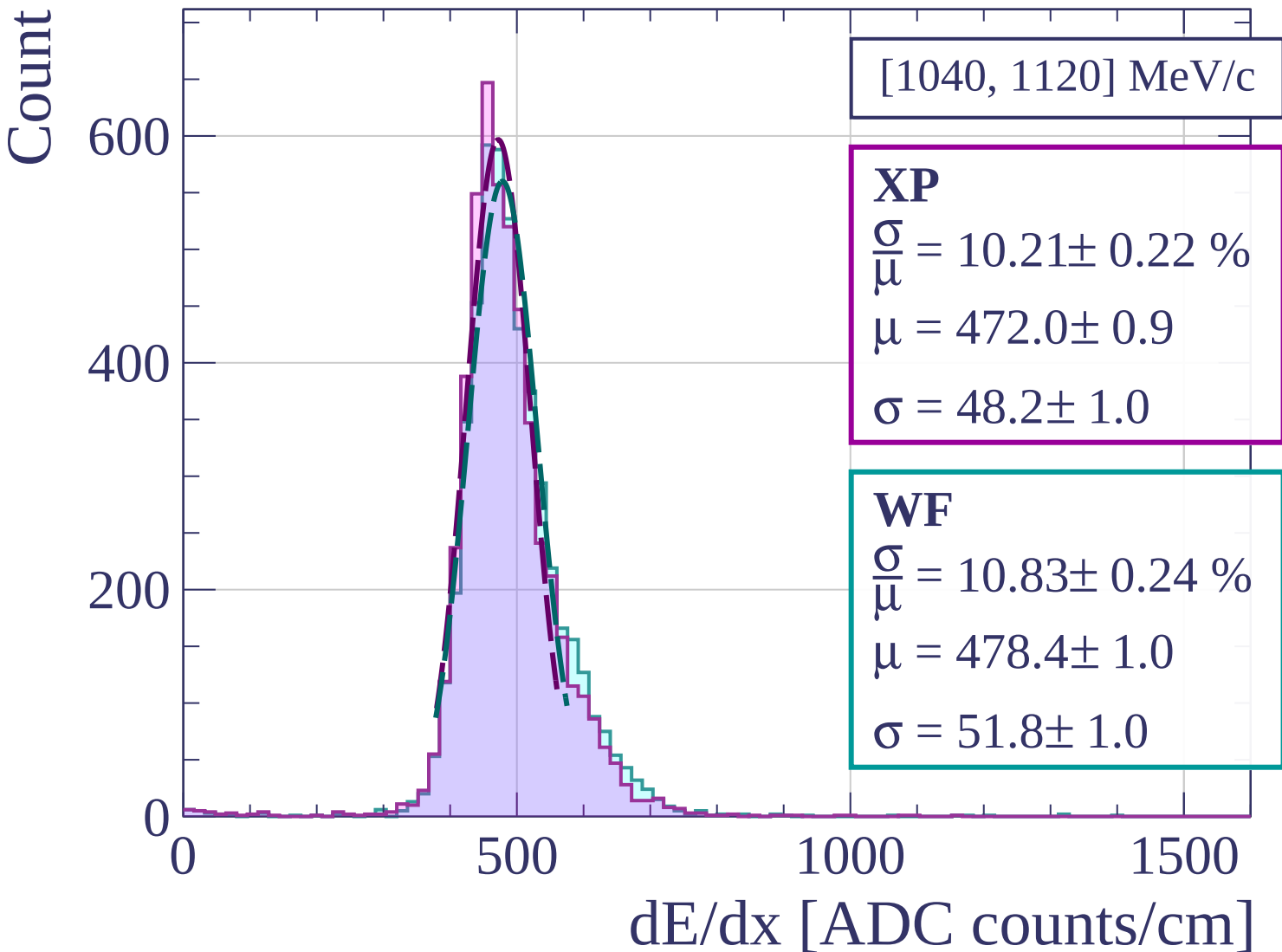
0

500

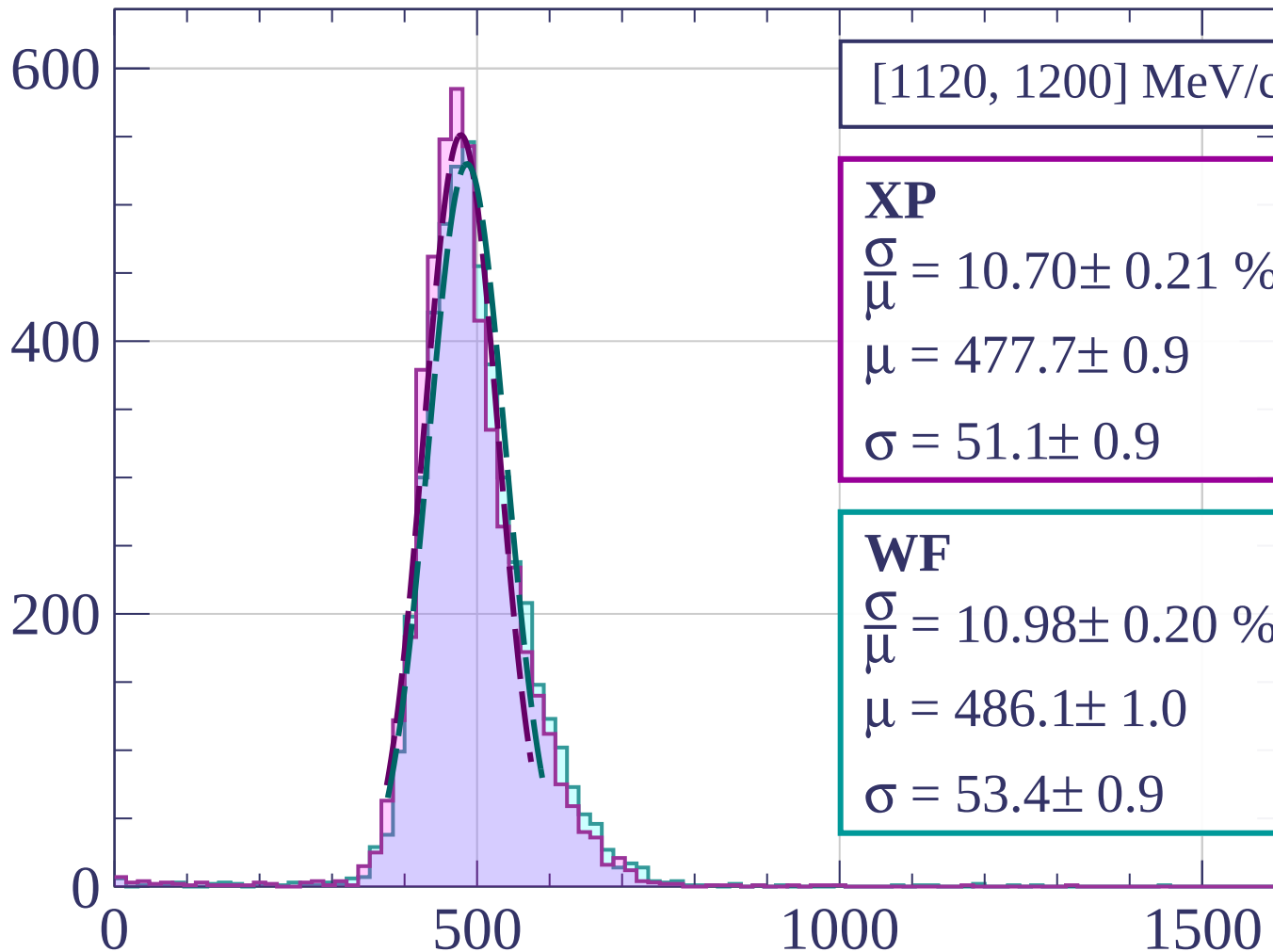
1000

1500

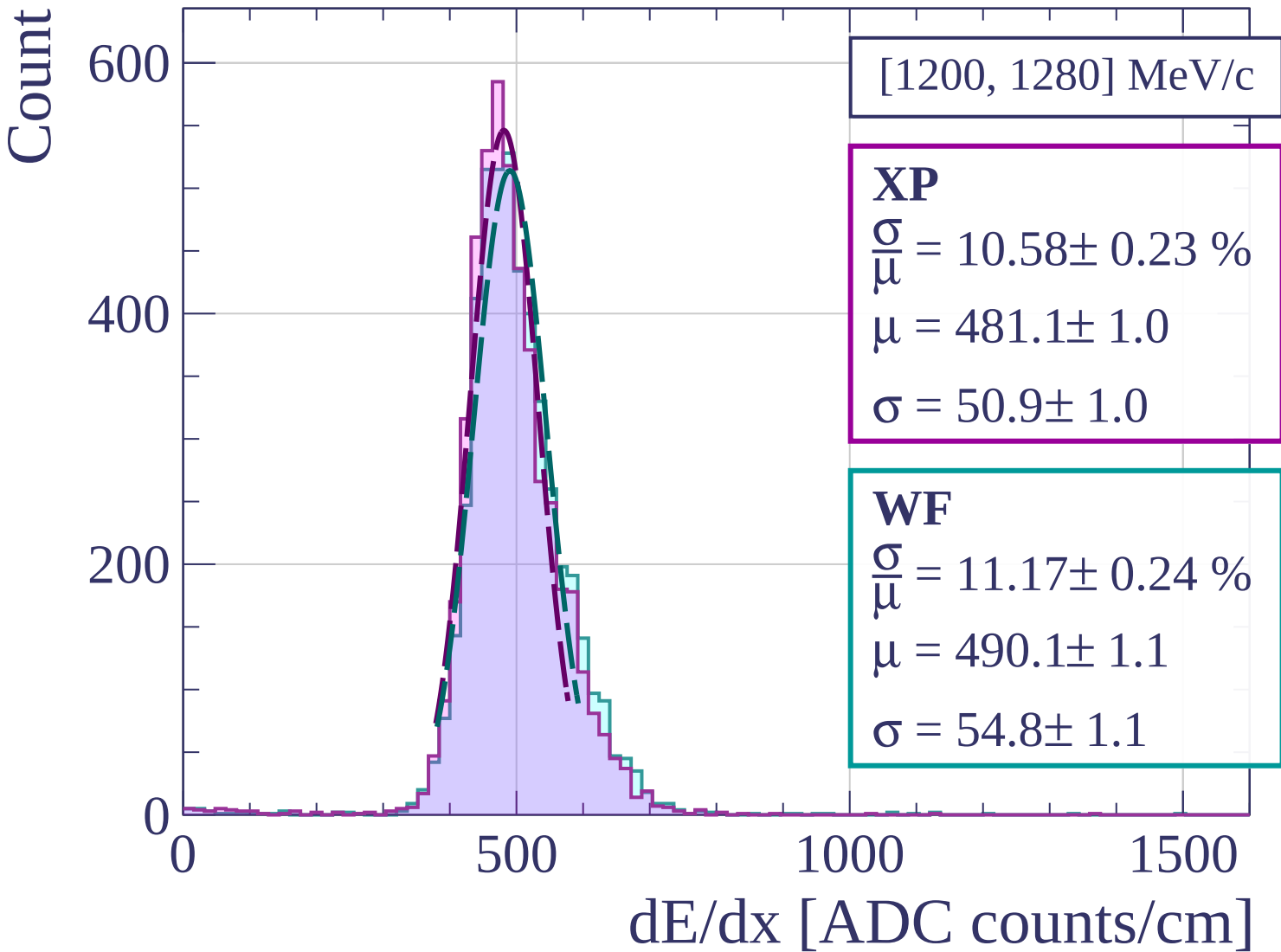
dE/dx [ADC counts/cm]

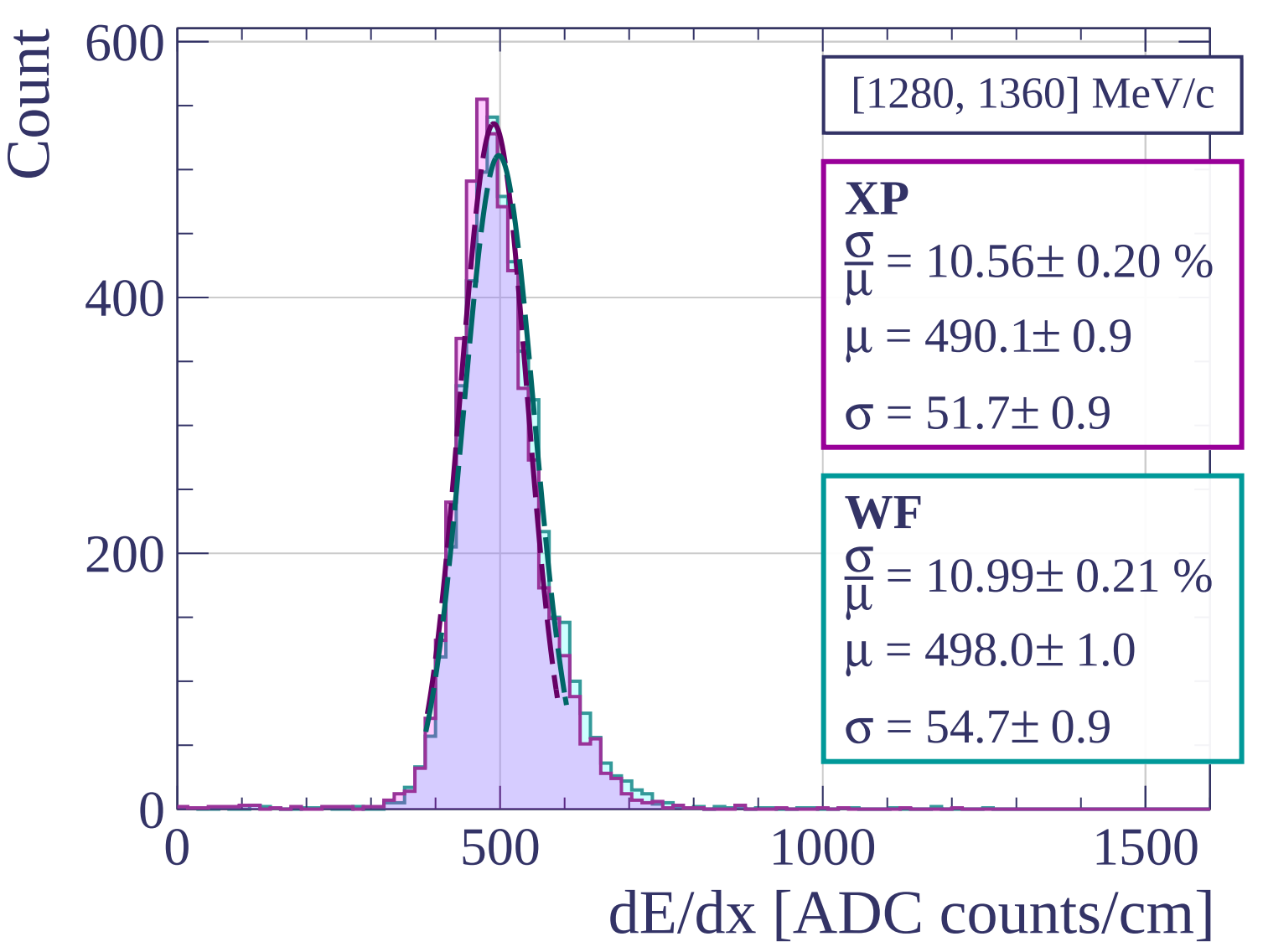


Count



dE/dx [ADC counts/cm]





Count

[1360, 1440] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.78 \pm 0.22 \%$$

$$\mu = 492.9 \pm 1.0$$

$$\sigma = 53.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.32 \pm 0.23 \%$$

$$\mu = 500.0 \pm 1.1$$

$$\sigma = 56.6 \pm 1.1$$

400

200

0

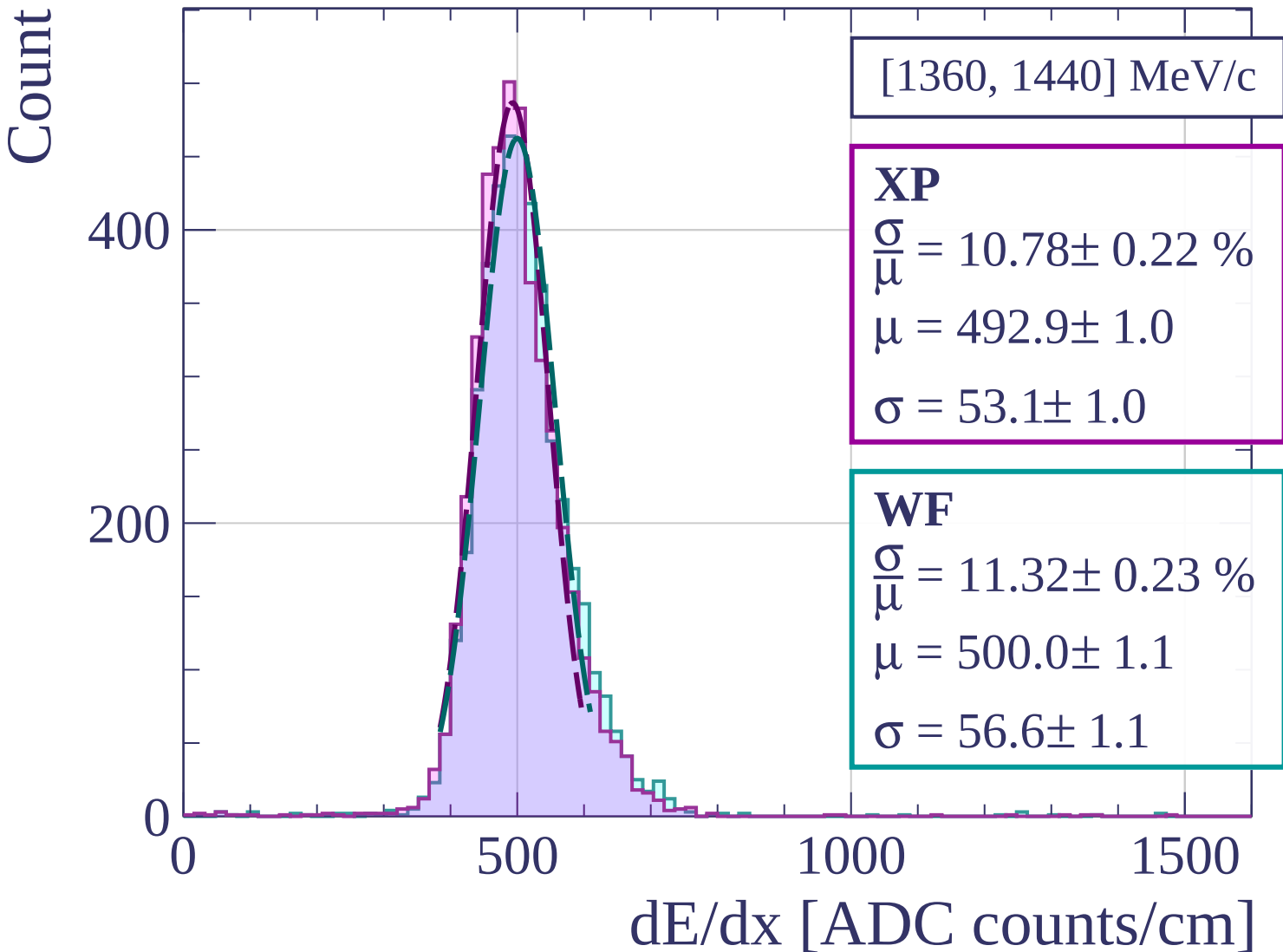
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1440, 1520] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.71 \pm 0.23 \%$$

$$\mu = 495.2 \pm 1.1$$

$$\sigma = 53.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.18 \pm 0.24 \%$$

$$\mu = 502.9 \pm 1.1$$

$$\sigma = 56.2 \pm 1.1$$

400

200

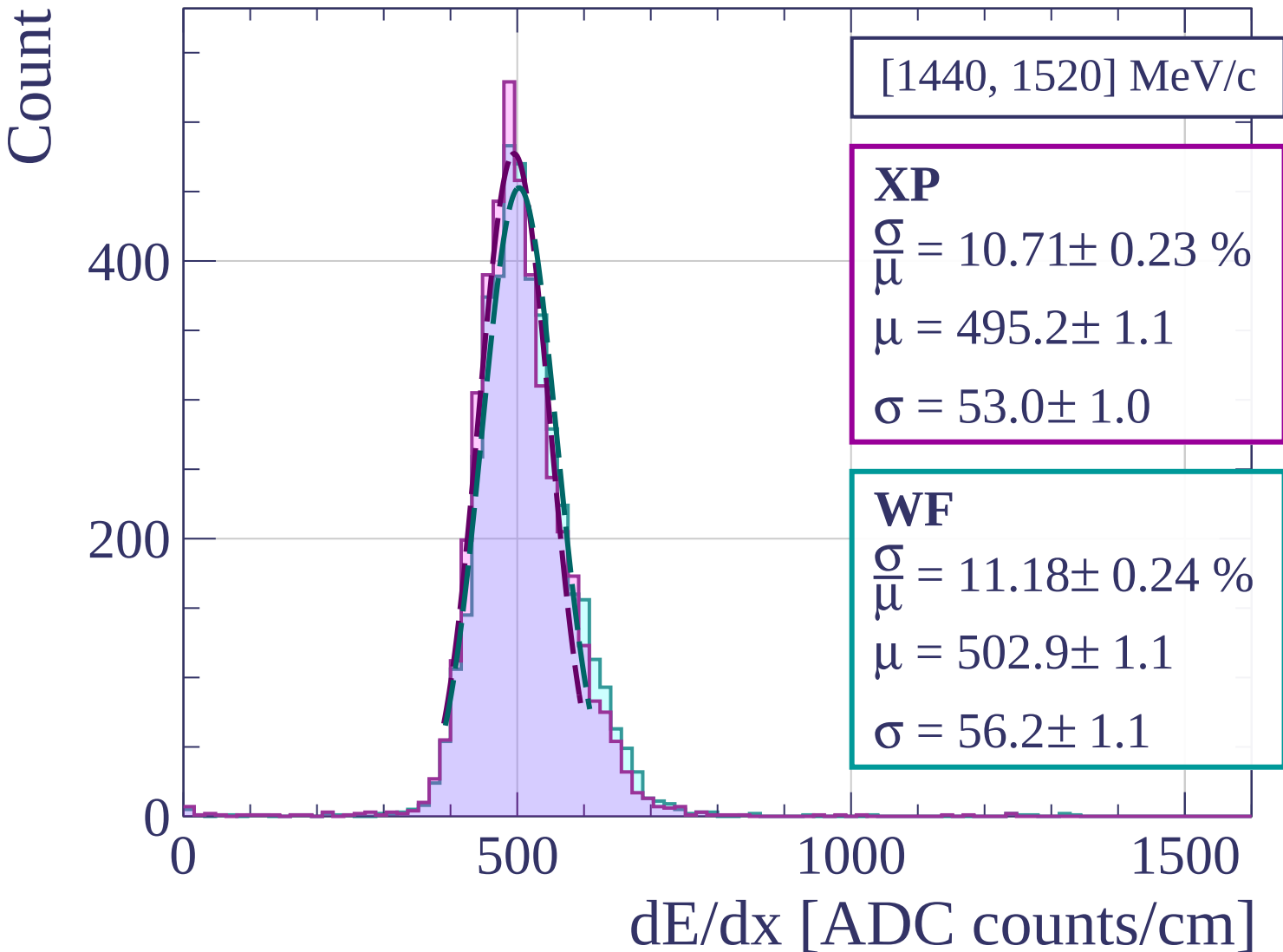
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1520, 1600] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.77 \pm 0.21 \%$$

$$\mu = 501.2 \pm 1.0$$

$$\sigma = 54.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.97 \pm 0.20 \%$$

$$\mu = 508.9 \pm 1.0$$

$$\sigma = 55.9 \pm 0.9$$

400

200

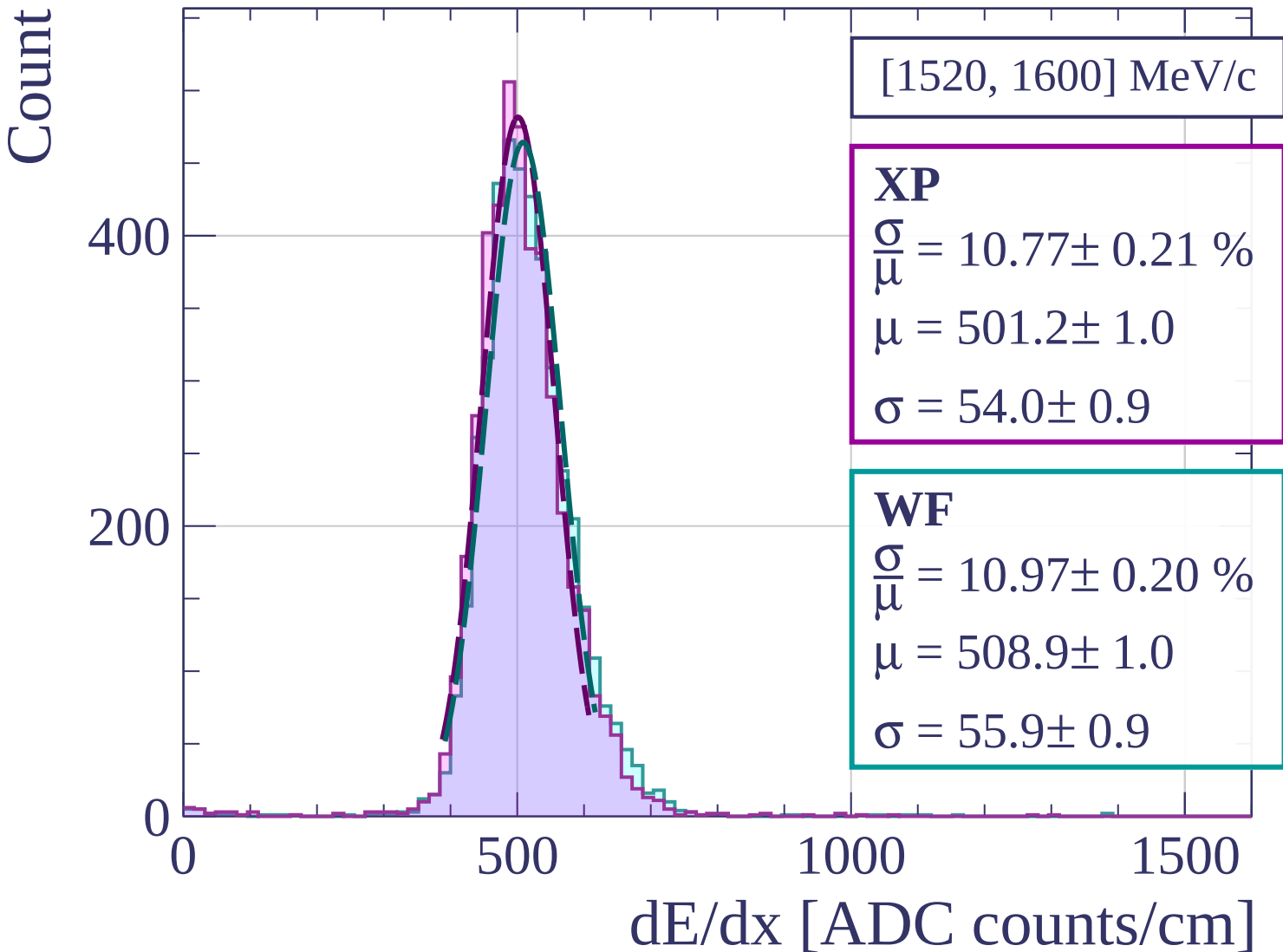
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1600, 1680] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.90 \pm 0.25 \%$$

$$\mu = 503.5 \pm 1.1$$

$$\sigma = 54.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.34 \pm 0.24 \%$$

$$\mu = 511.8 \pm 1.1$$

$$\sigma = 58.0 \pm 1.1$$

400

200

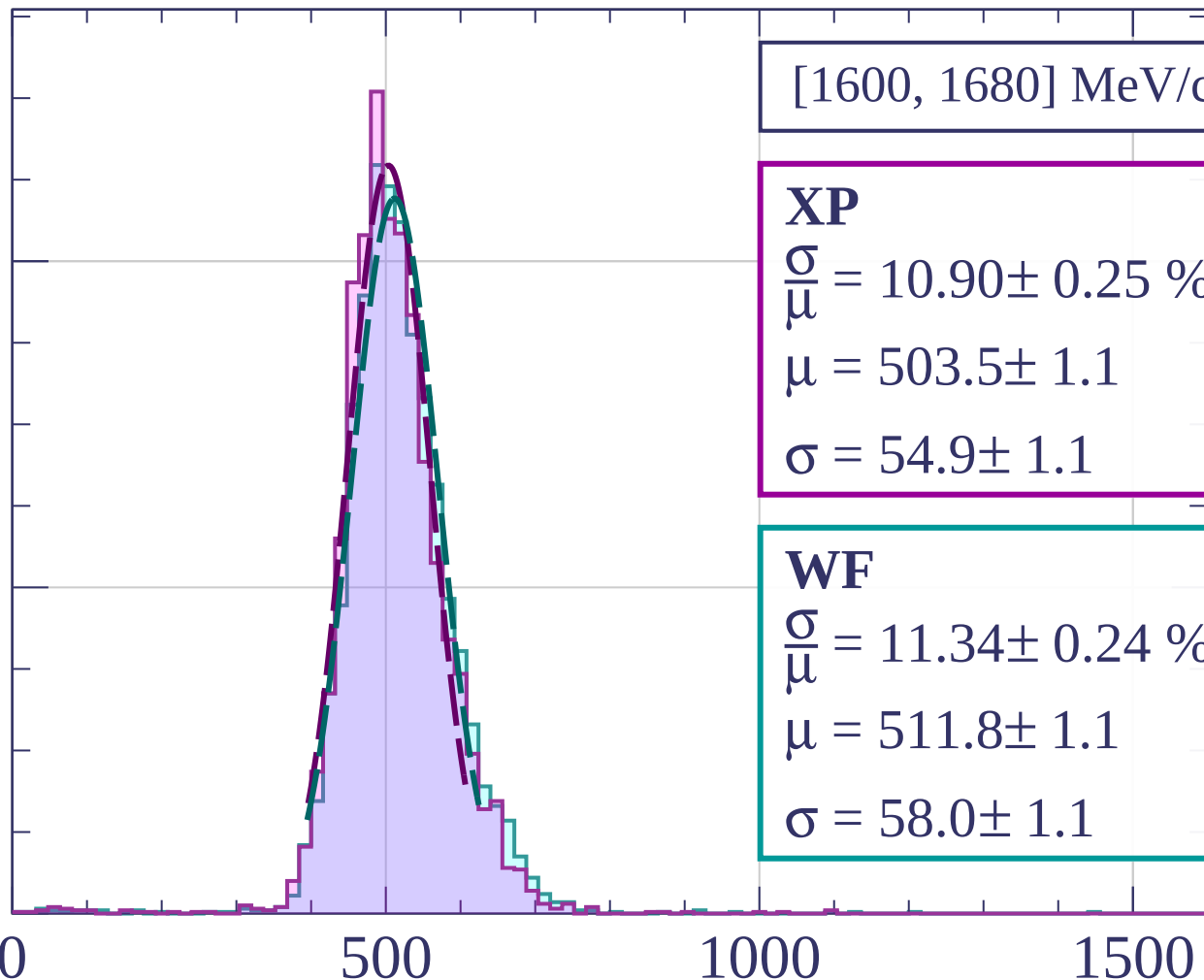
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1680, 1760] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.12 \pm 0.22 \%$$

$$\mu = 510.3 \pm 1.1$$

$$\sigma = 56.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.61 \pm 0.24 \%$$

$$\mu = 517.6 \pm 1.2$$

$$\sigma = 60.1 \pm 1.1$$

400

300

200

100

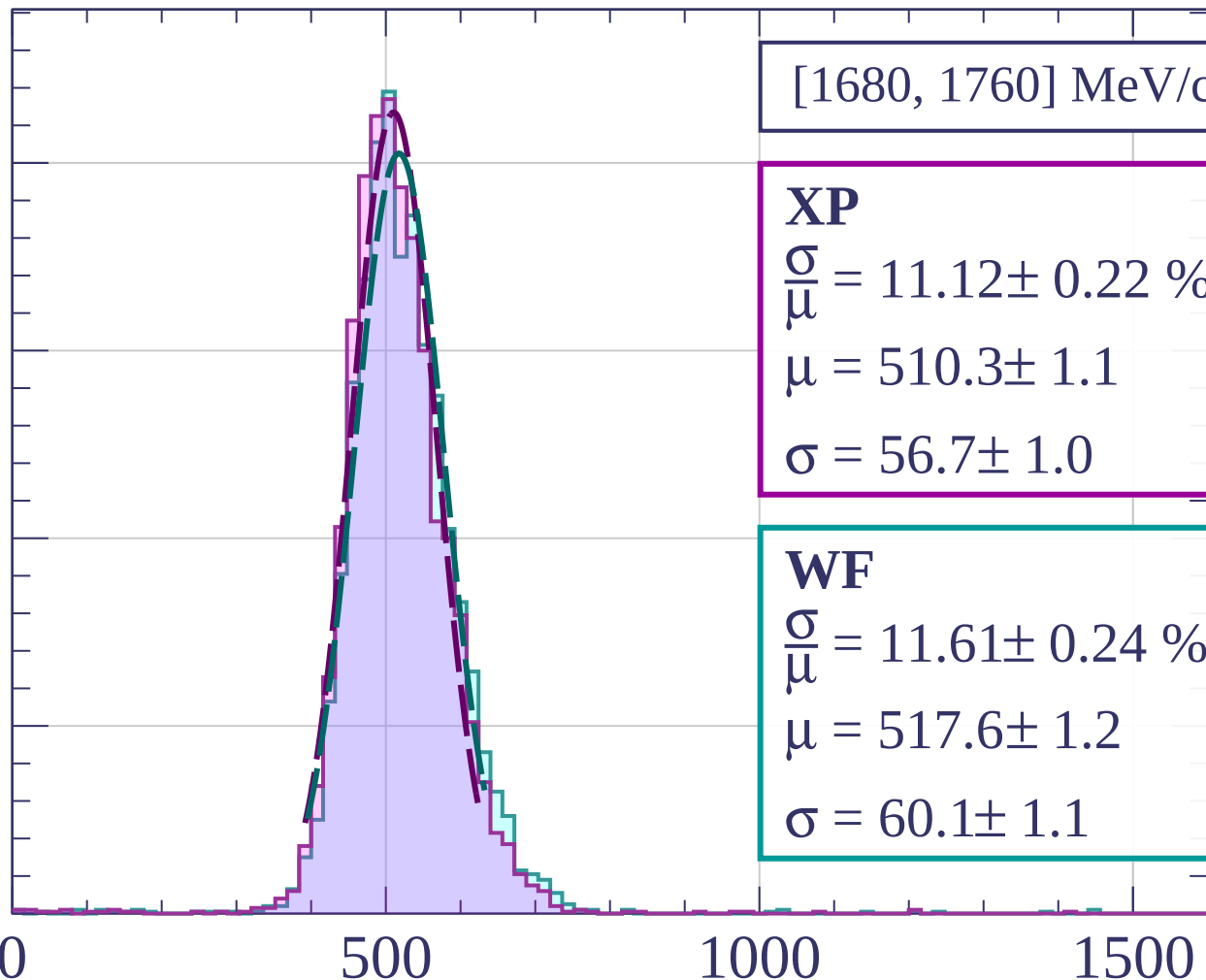
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1760, 1840] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.12 \pm 0.24 \%$$

$$\mu = 512.3 \pm 1.1$$

$$\sigma = 57.0 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.42 \pm 0.24 \%$$

$$\mu = 519.9 \pm 1.1$$

$$\sigma = 59.4 \pm 1.1$$

400

300

200

100

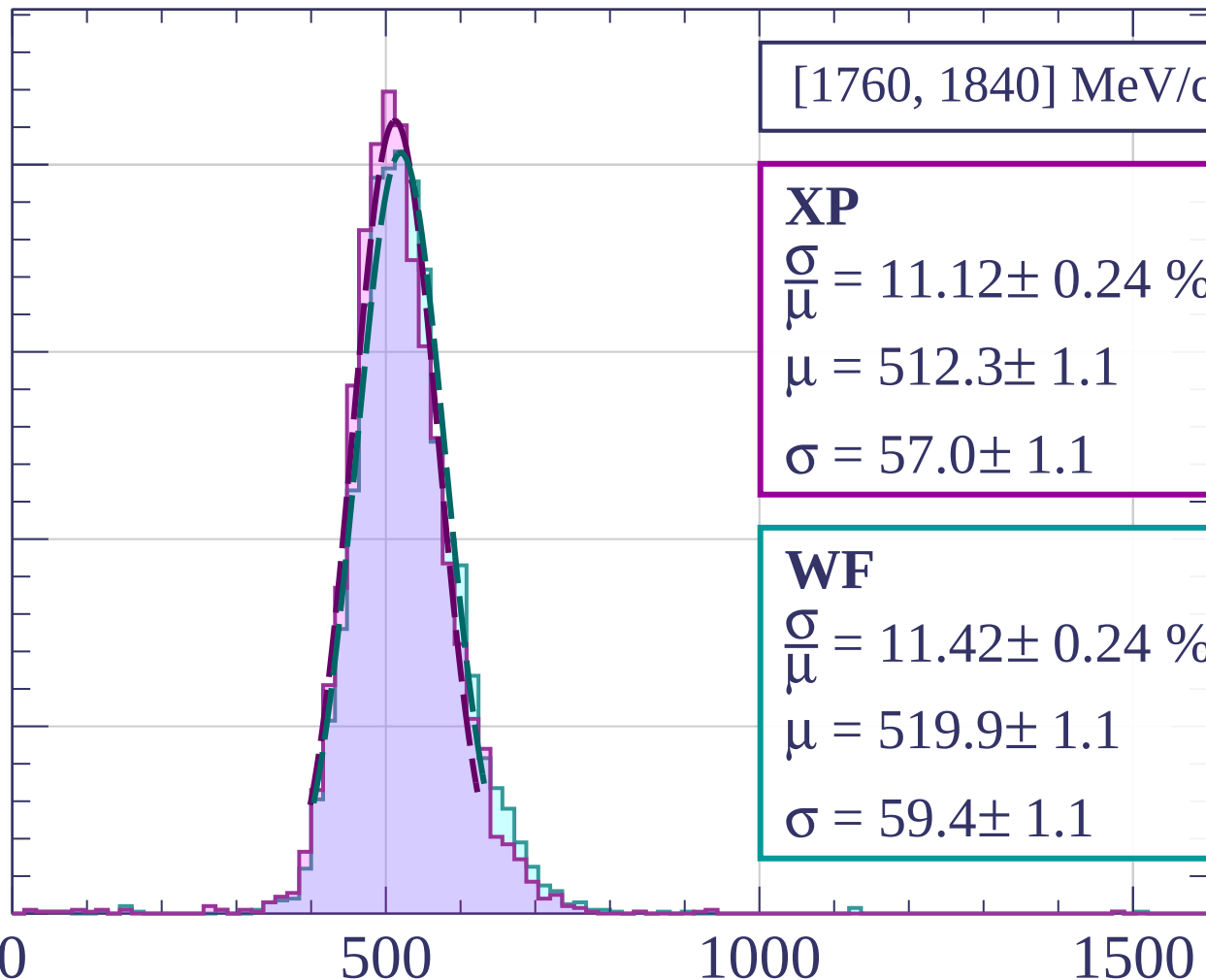
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1840, 1920] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.68 \pm 0.23 \%$$

$$\mu = 514.2 \pm 1.1$$

$$\sigma = 54.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.06 \pm 0.23 \%$$

$$\mu = 523.2 \pm 1.2$$

$$\sigma = 57.9 \pm 1.1$$

400

300

200

100

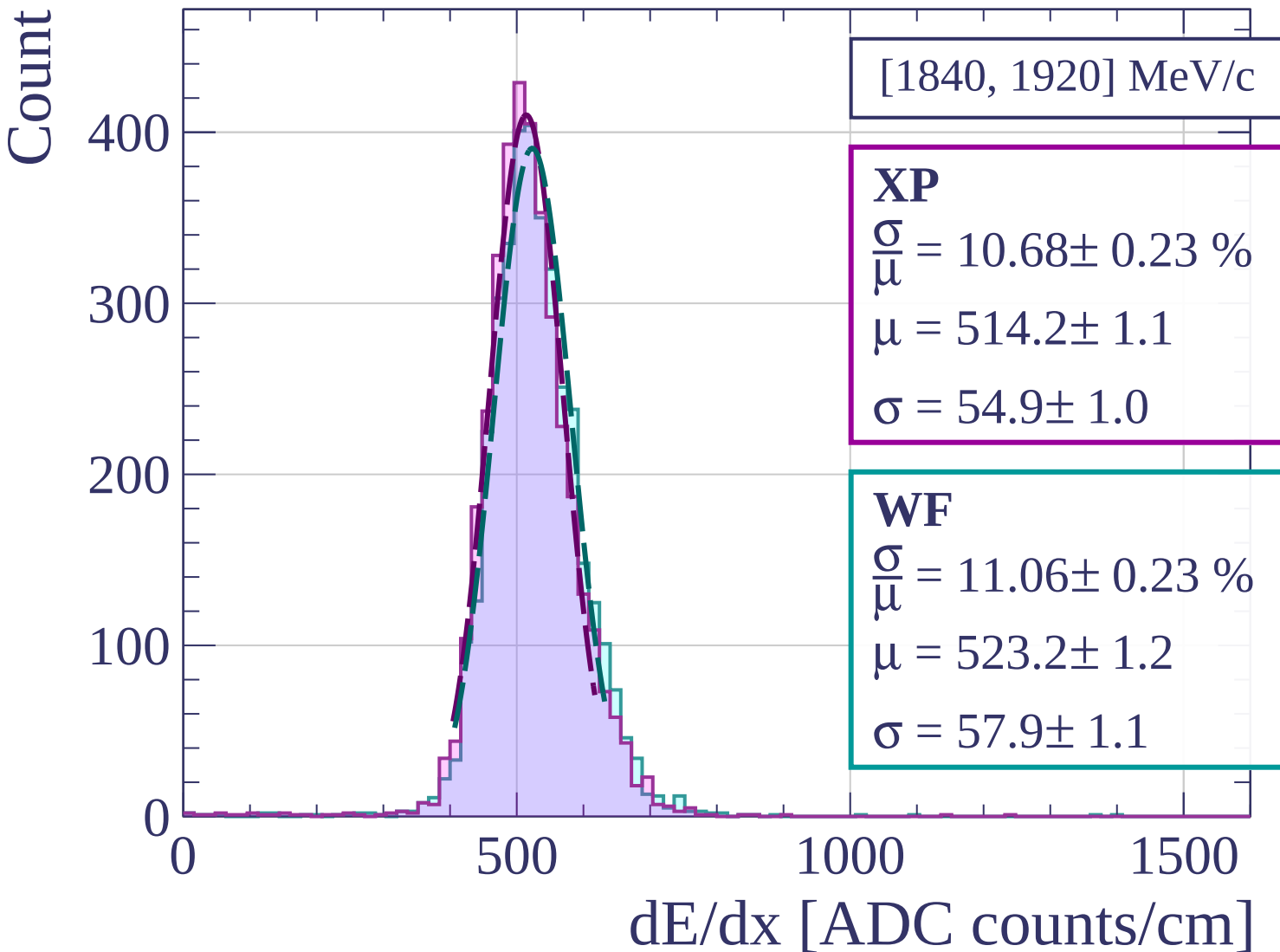
0

500

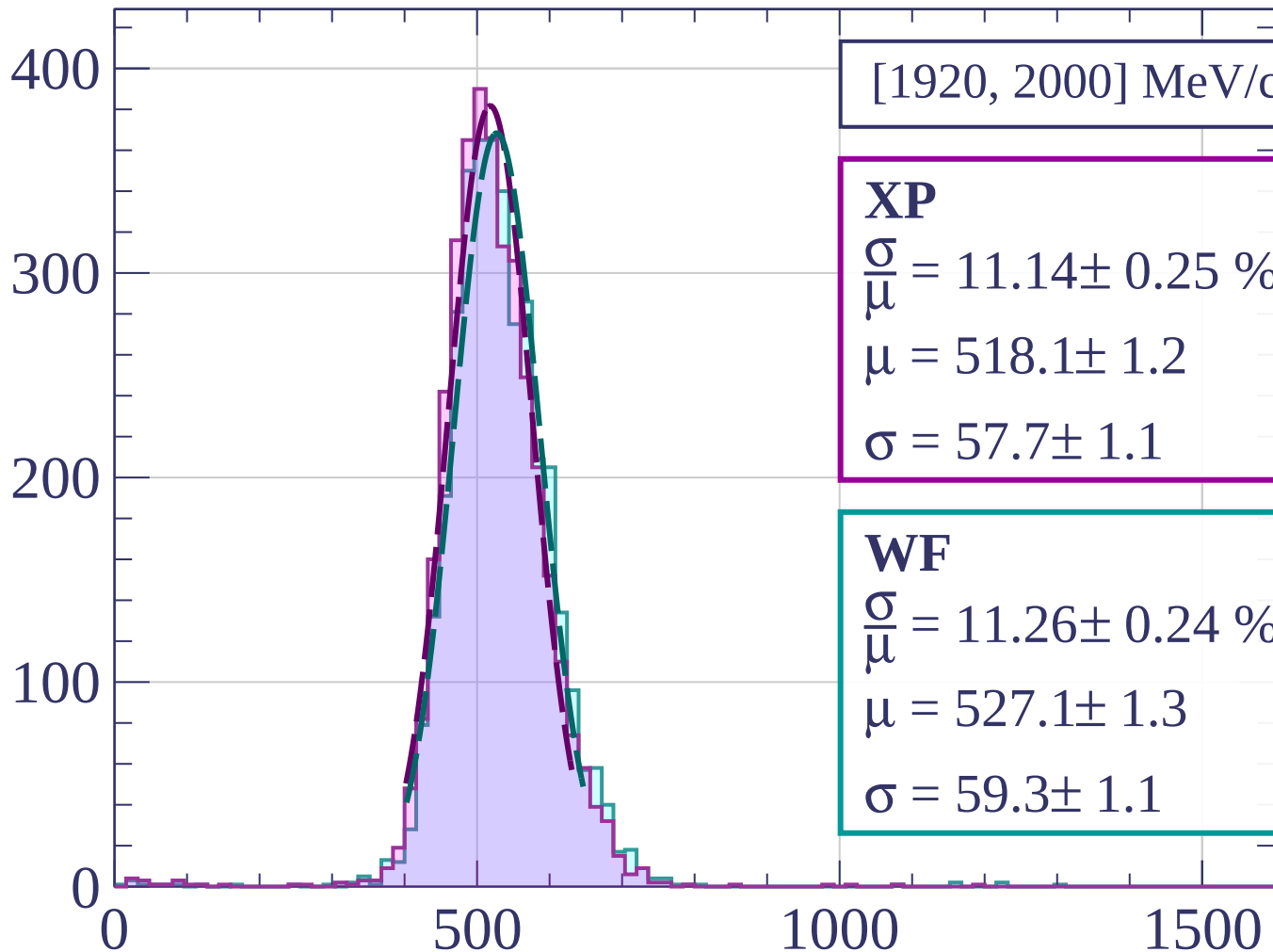
1000

1500

dE/dx [ADC counts/cm]



Count



[1920, 2000] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.14 \pm 0.25 \%$$

$$\mu = 518.1 \pm 1.2$$

$$\sigma = 57.7 \pm 1.1$$

WF

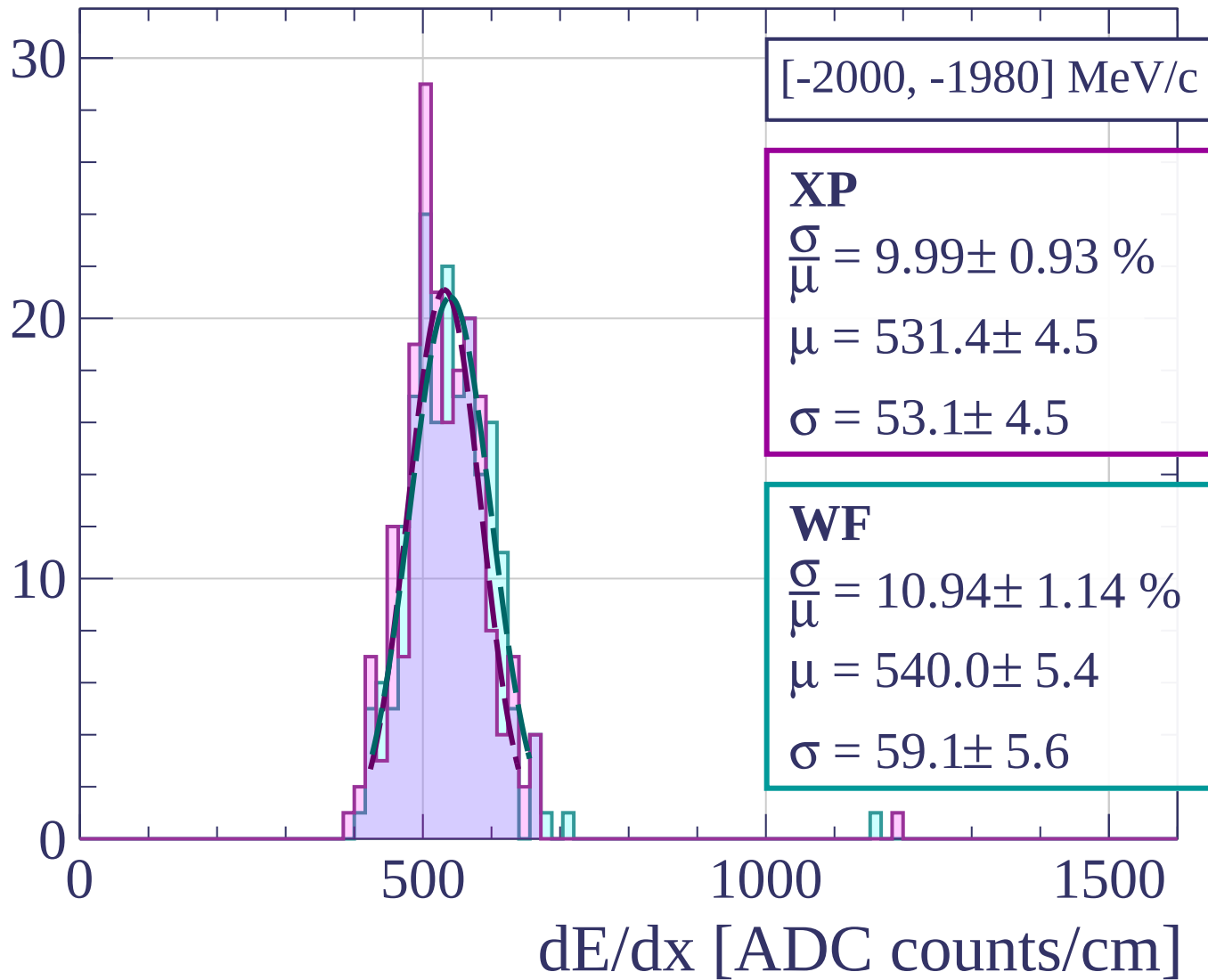
$$\frac{\sigma}{\mu} = 11.26 \pm 0.24 \%$$

$$\mu = 527.1 \pm 1.3$$

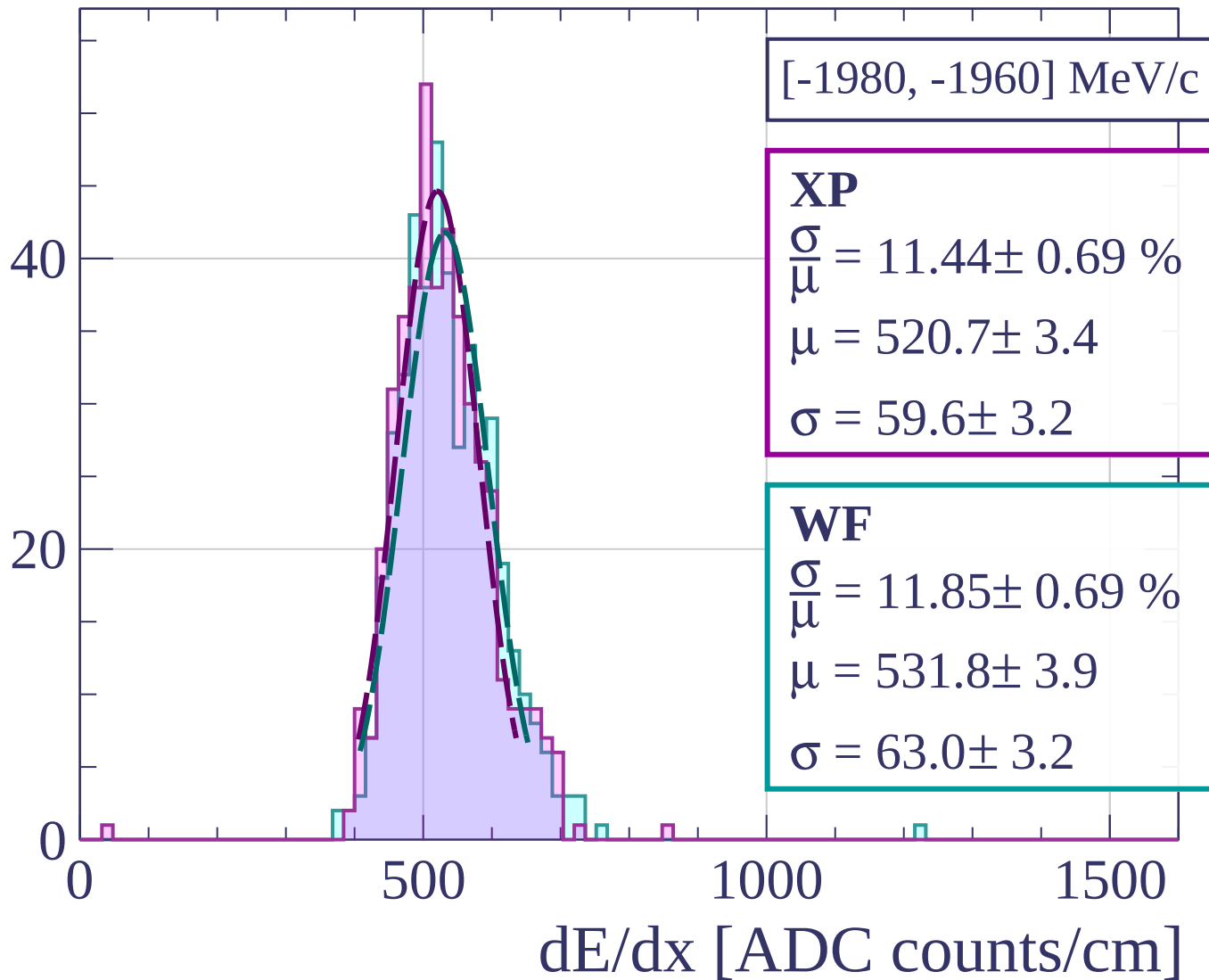
$$\sigma = 59.3 \pm 1.1$$

dE/dx [ADC counts/cm]

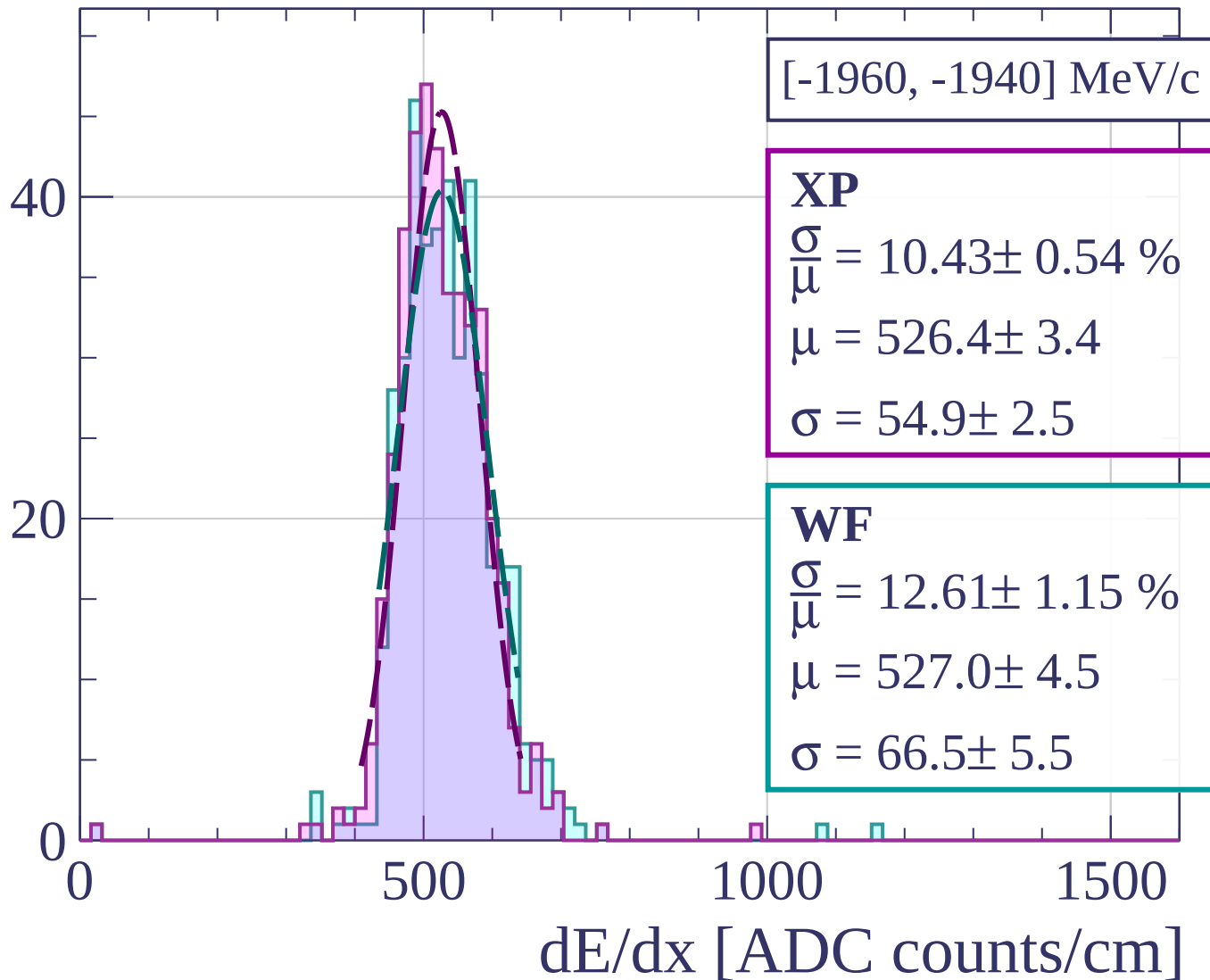
Count



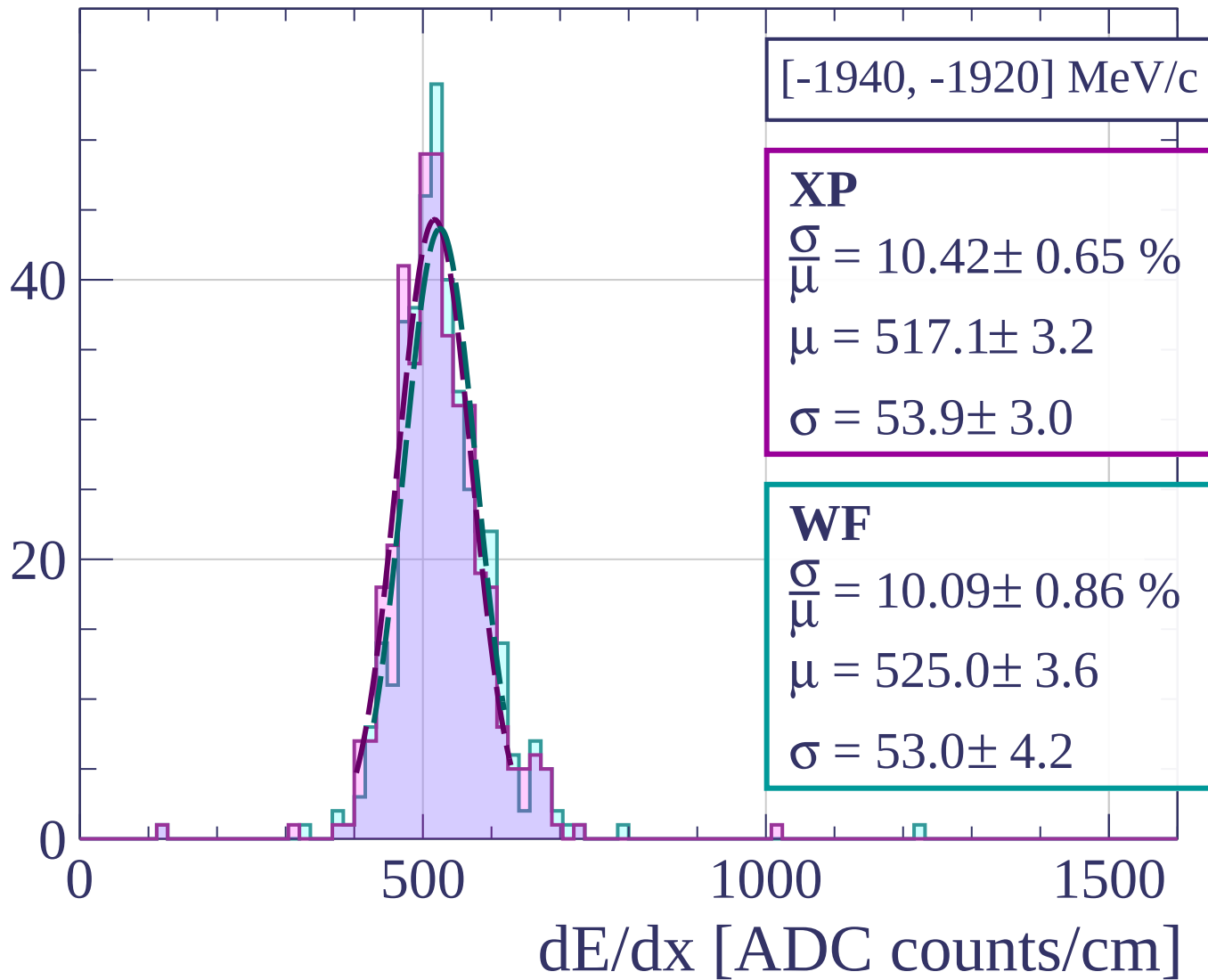
Count



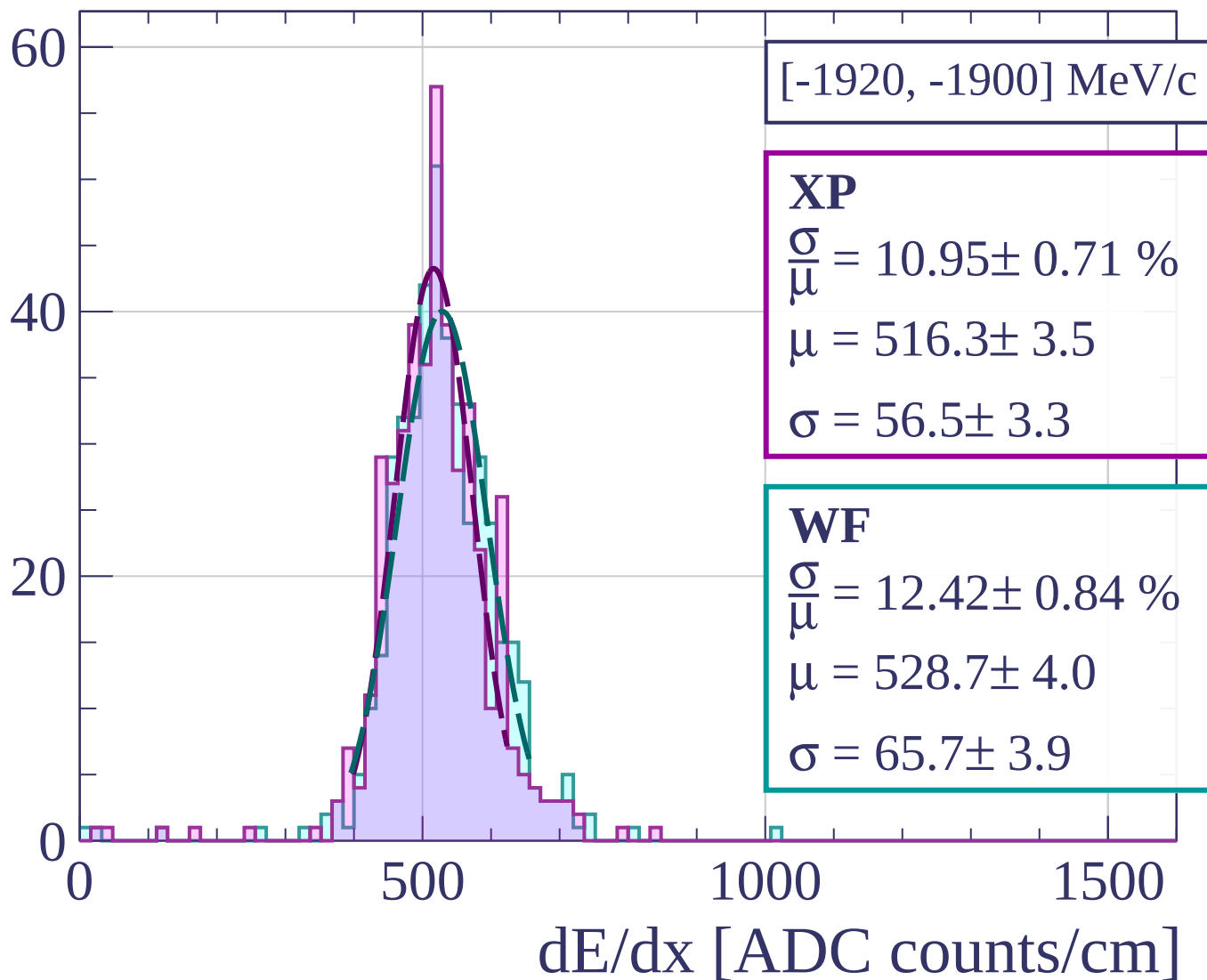
Count



Count

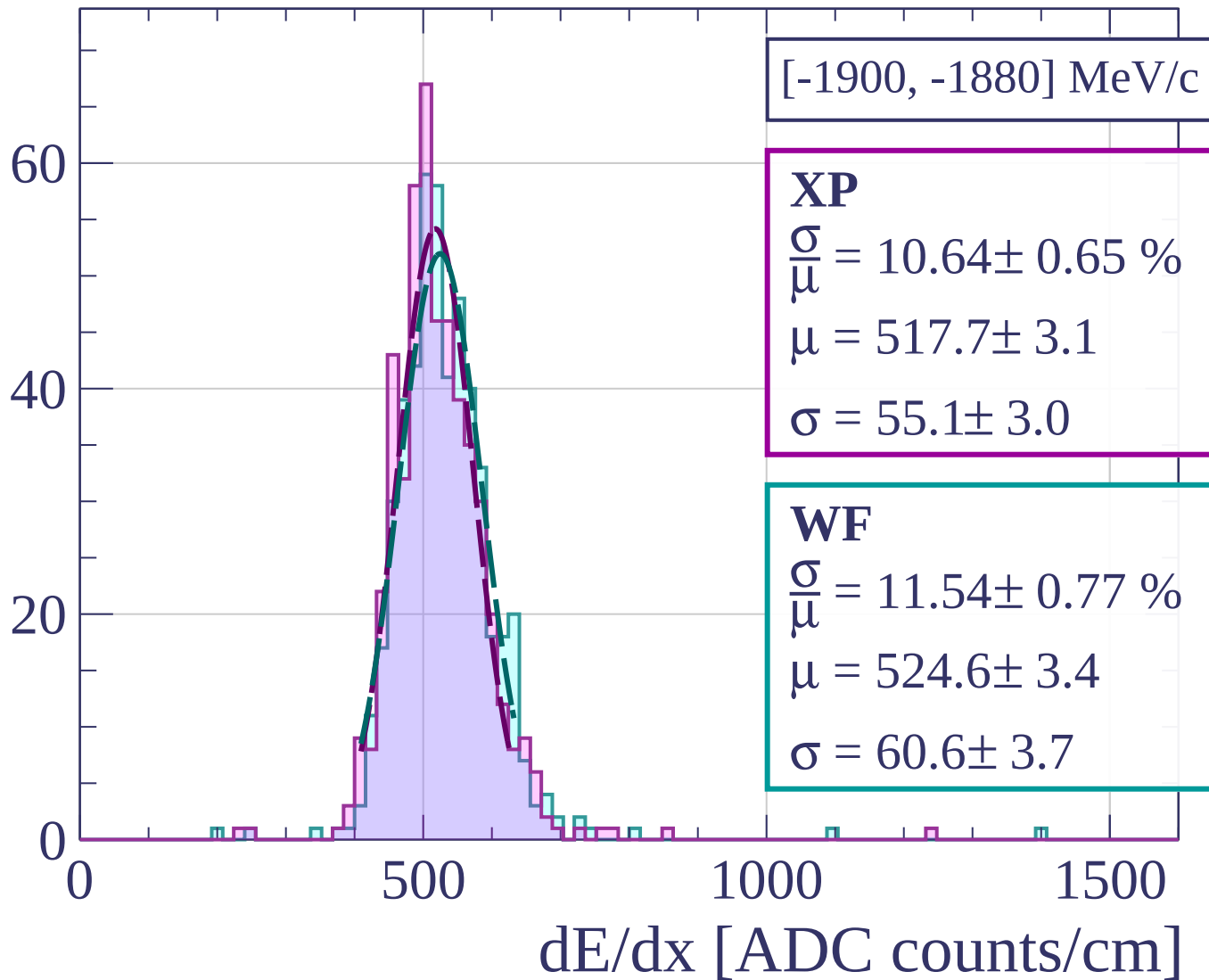


Count

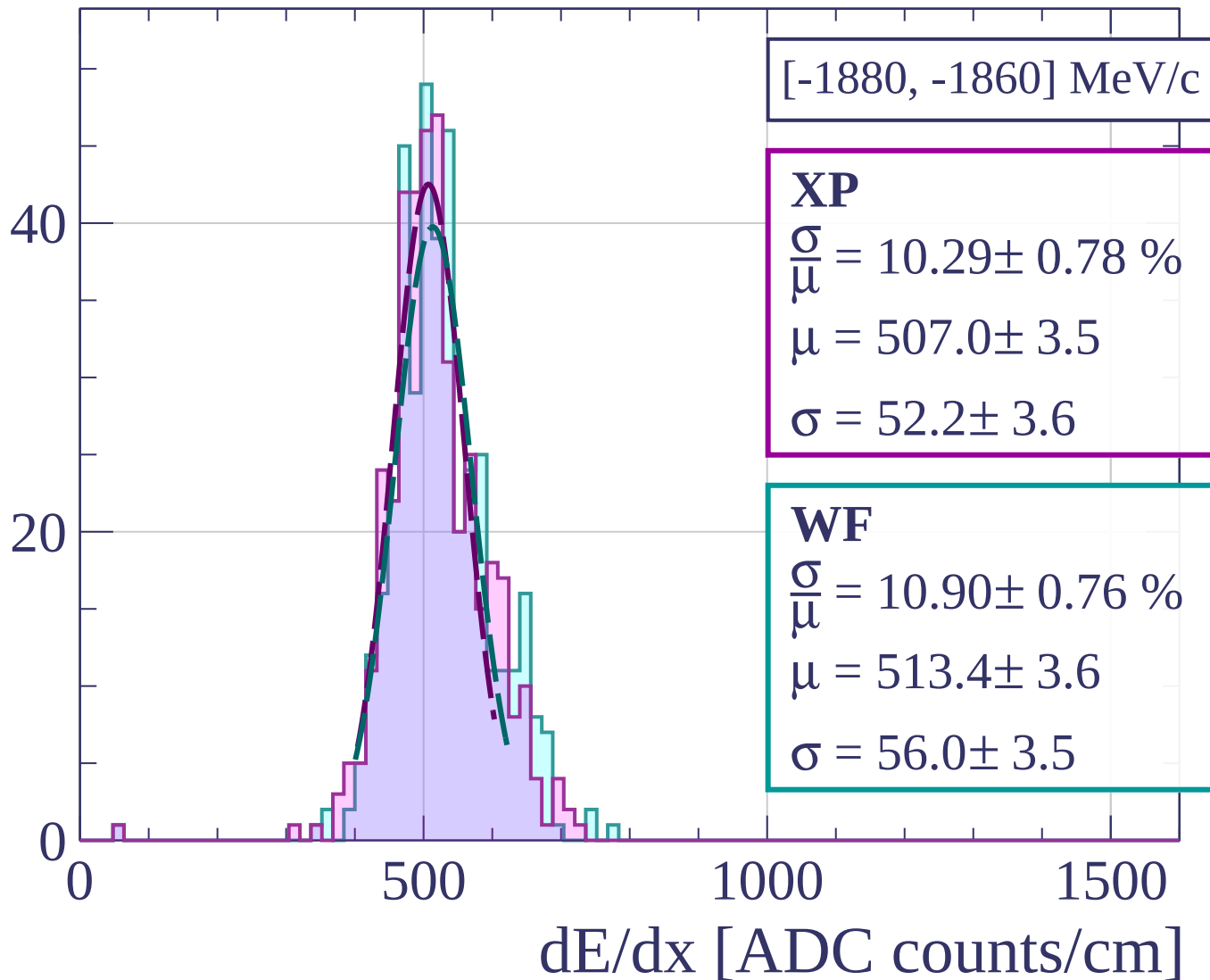


[-1920, -1900] MeV/c

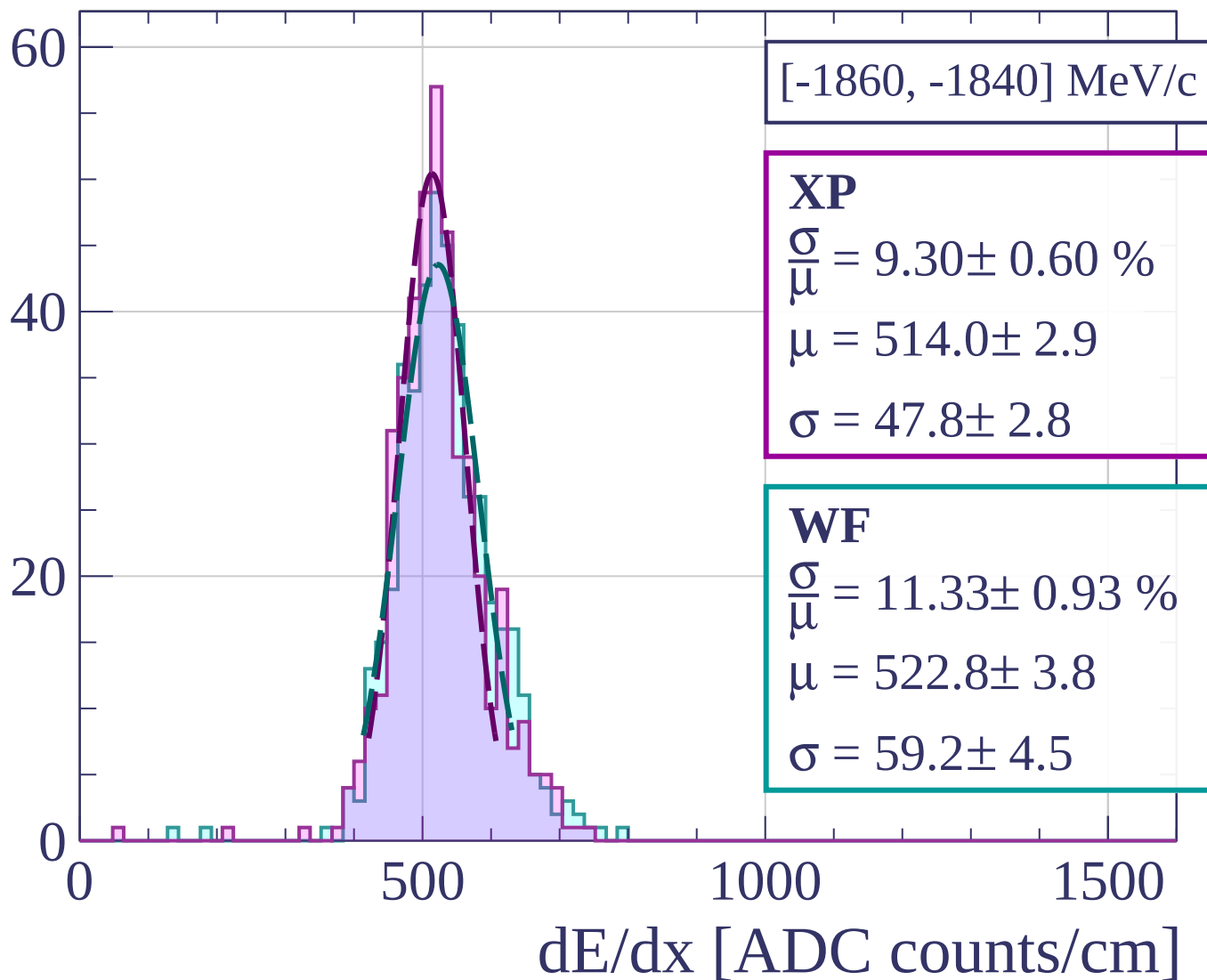
Count



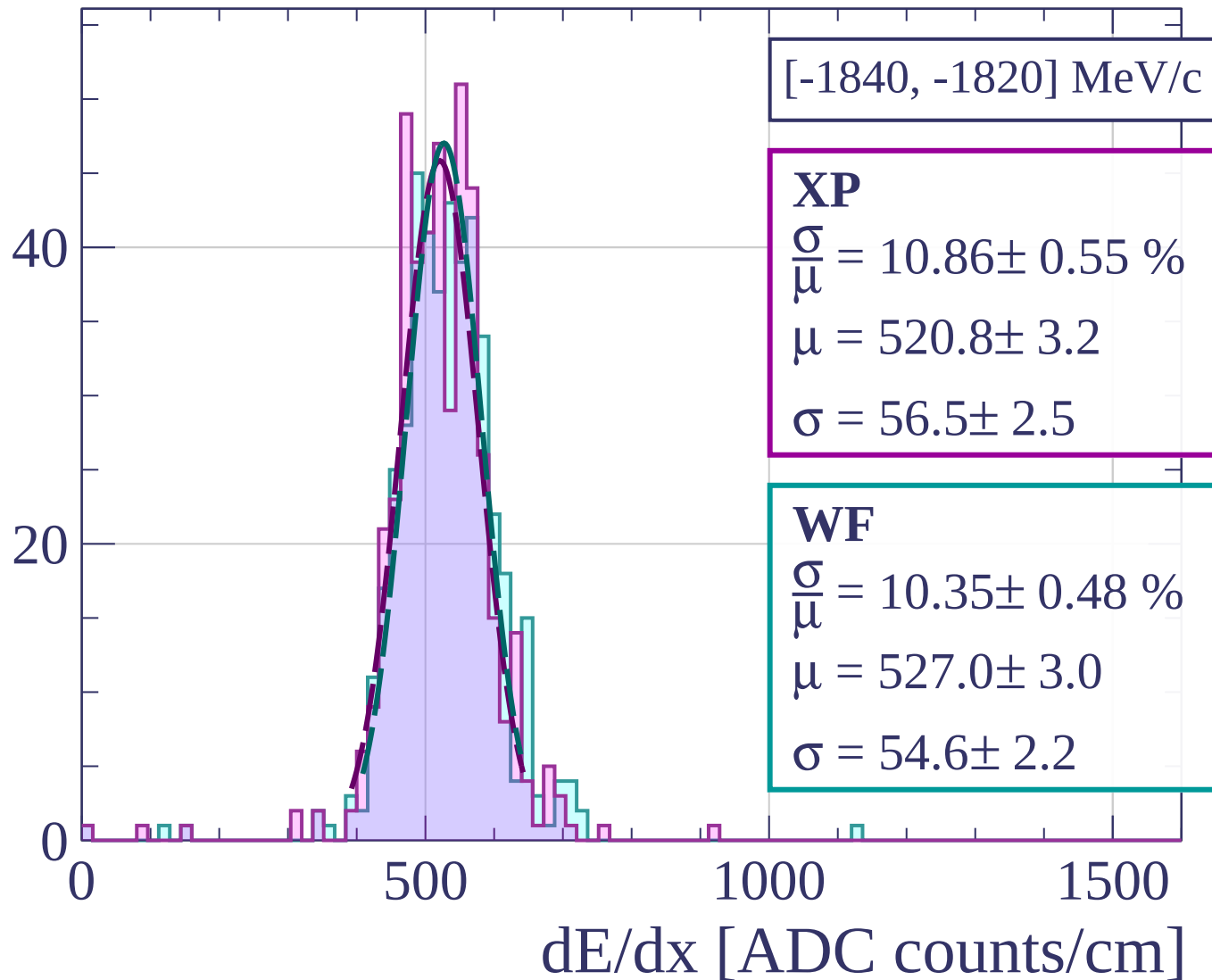
Count



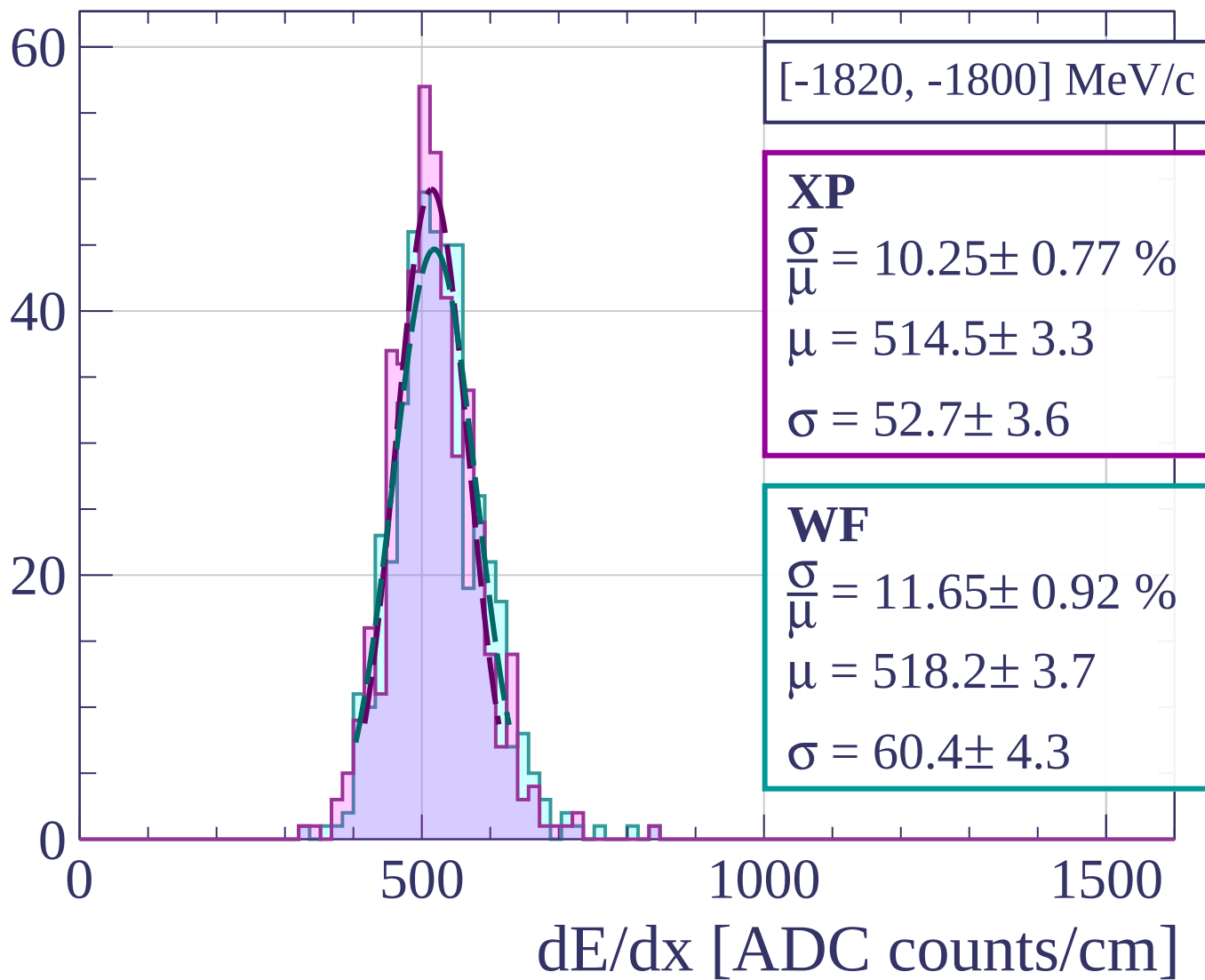
Count



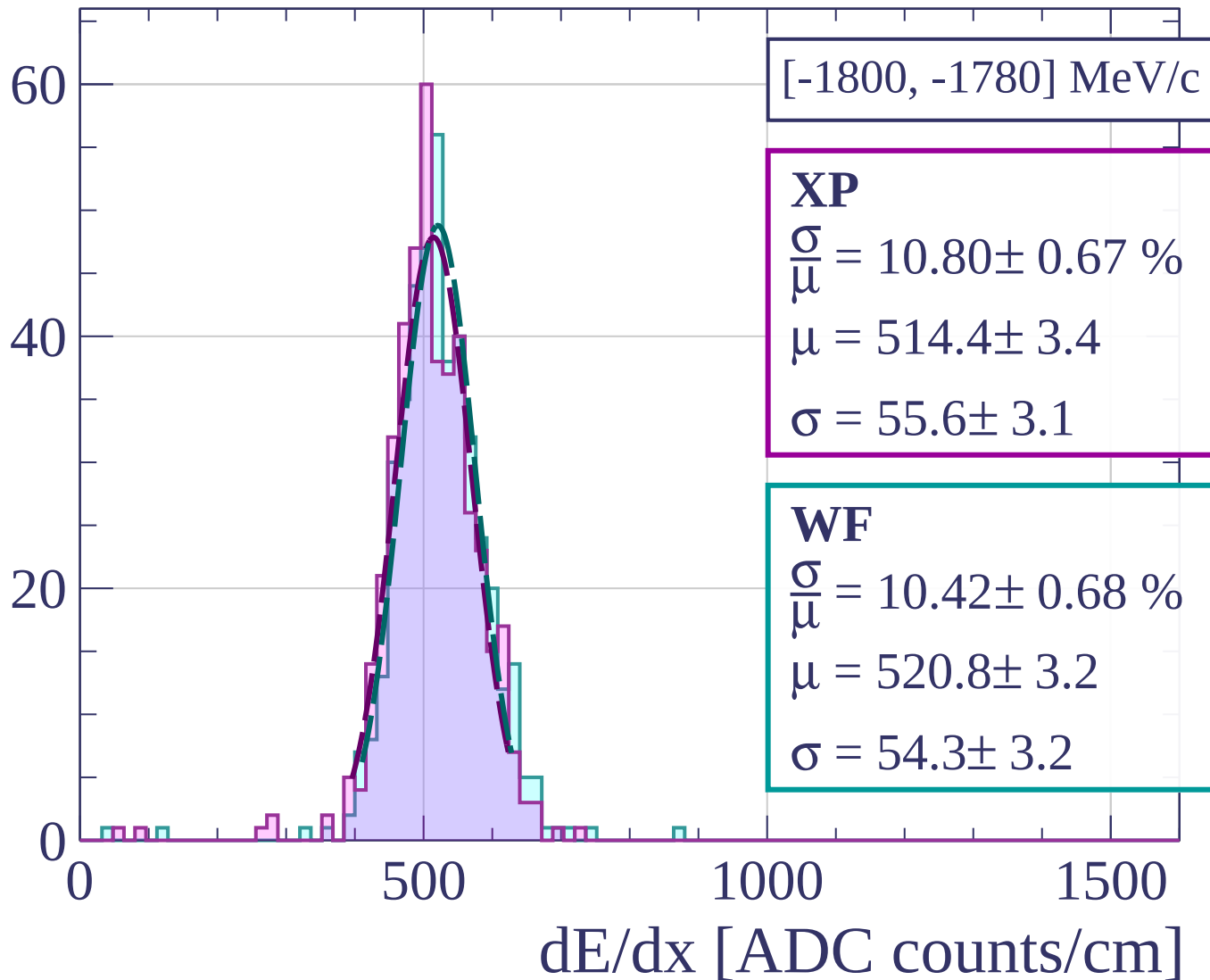
Count



Count

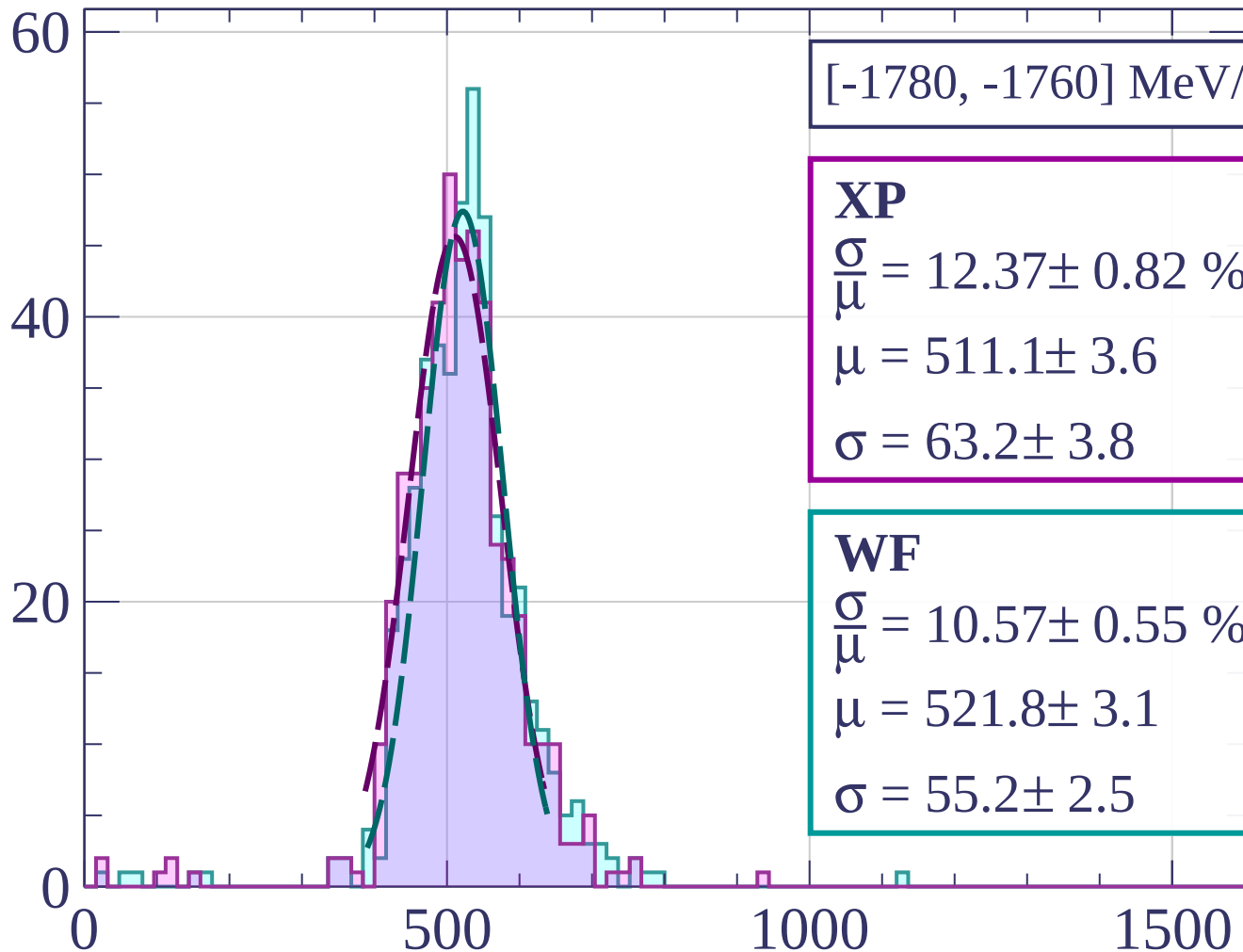


Count



[-1800, -1780] MeV/c

Count



[-1780, -1760] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.37 \pm 0.82 \%$$

$$\mu = 511.1 \pm 3.6$$

$$\sigma = 63.2 \pm 3.8$$

WF

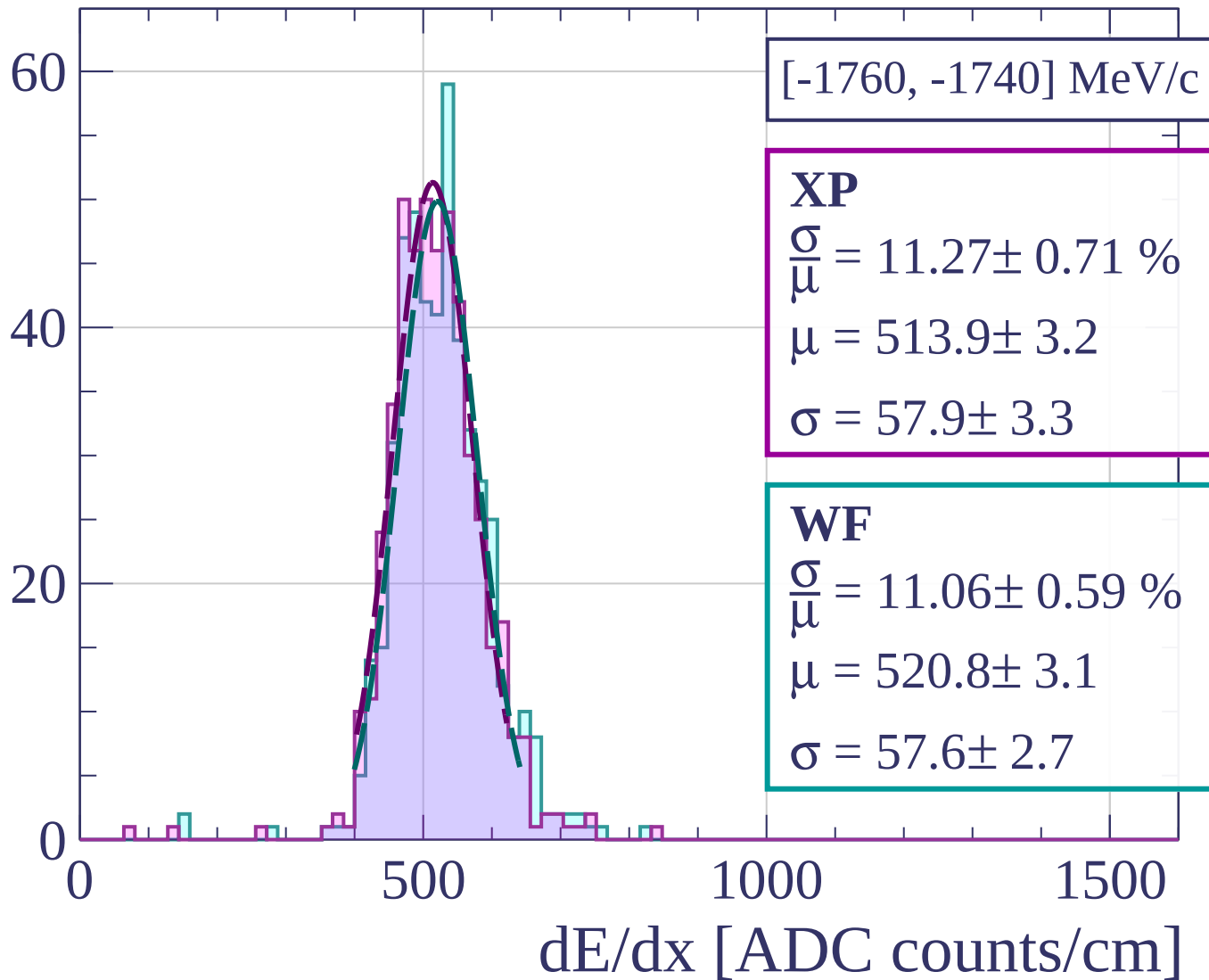
$$\frac{\sigma}{\mu} = 10.57 \pm 0.55 \%$$

$$\mu = 521.8 \pm 3.1$$

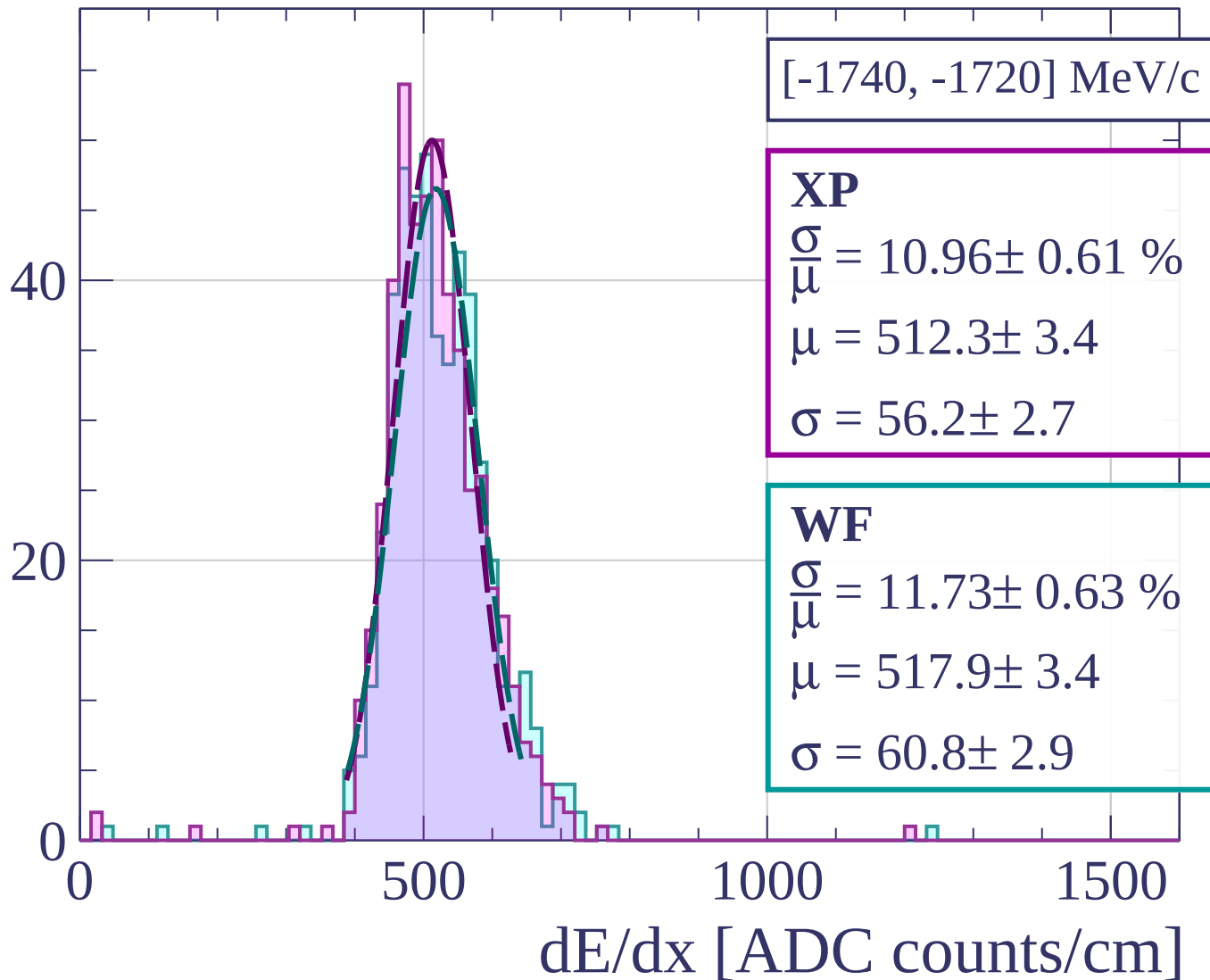
$$\sigma = 55.2 \pm 2.5$$

dE/dx [ADC counts/cm]

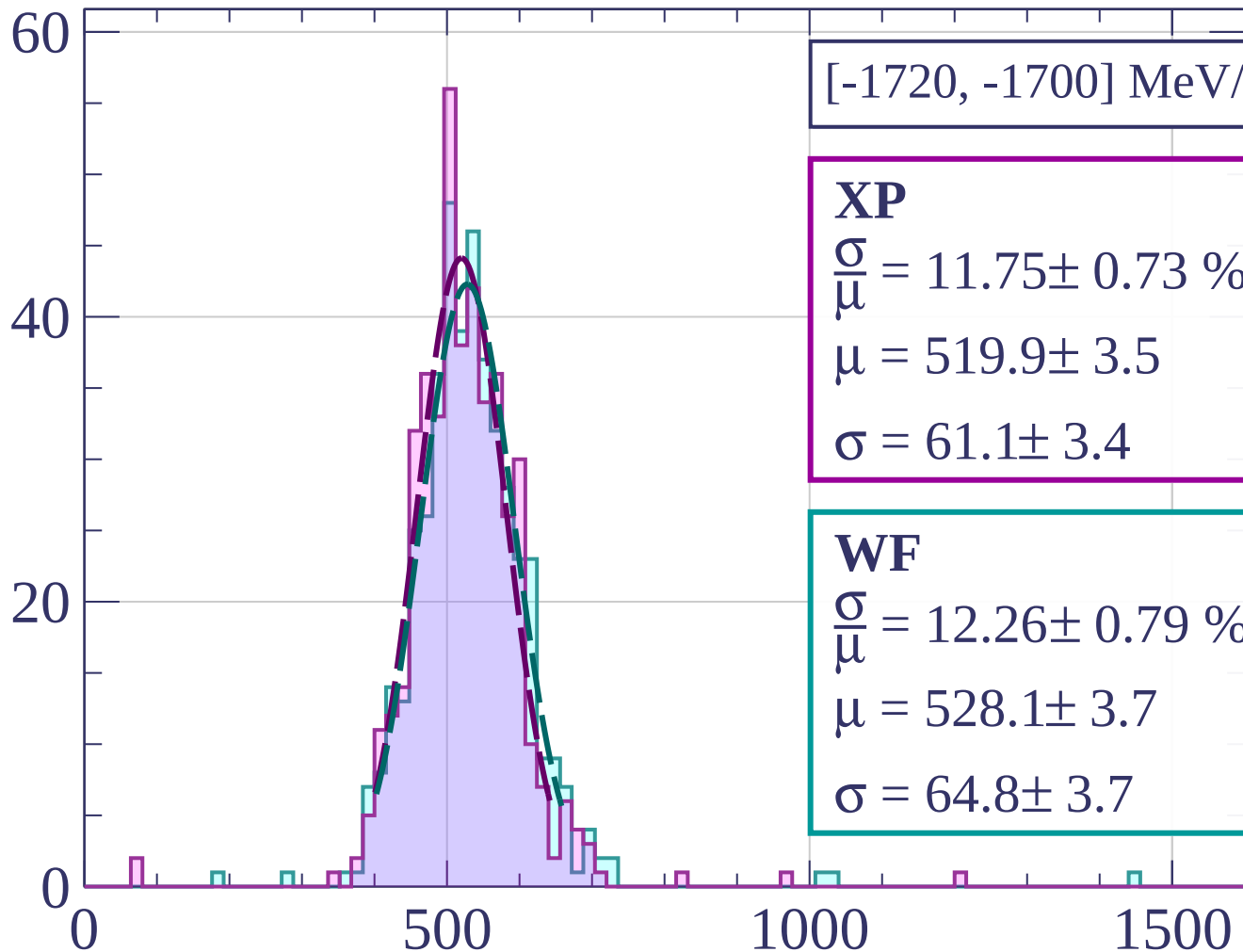
Count



Count



Count



[-1720, -1700] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 11.75 \pm 0.73 \%$$

$$\mu = 519.9 \pm 3.5$$

$$\sigma = 61.1 \pm 3.4$$

WF

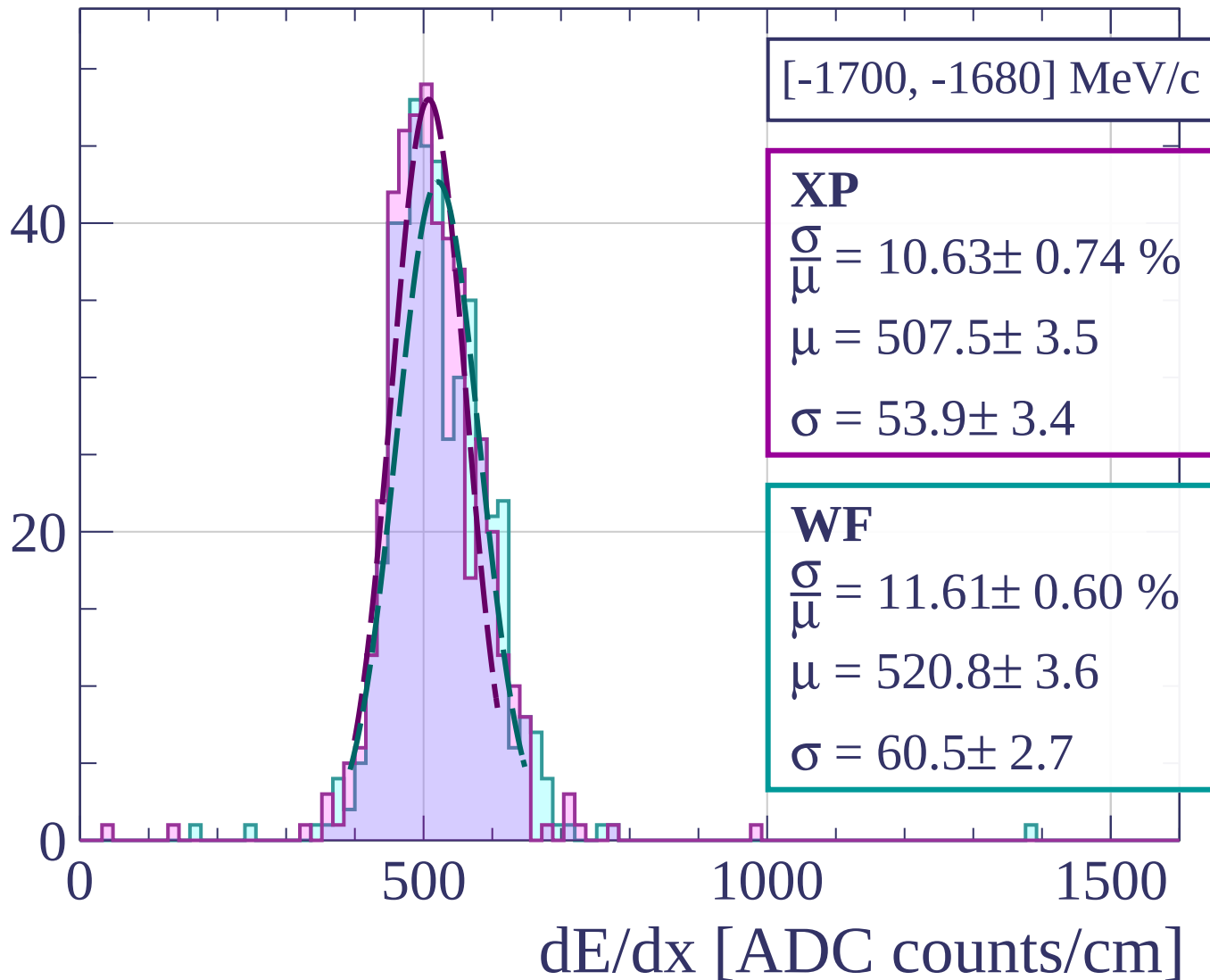
$$\frac{\sigma}{\bar{\mu}} = 12.26 \pm 0.79 \%$$

$$\mu = 528.1 \pm 3.7$$

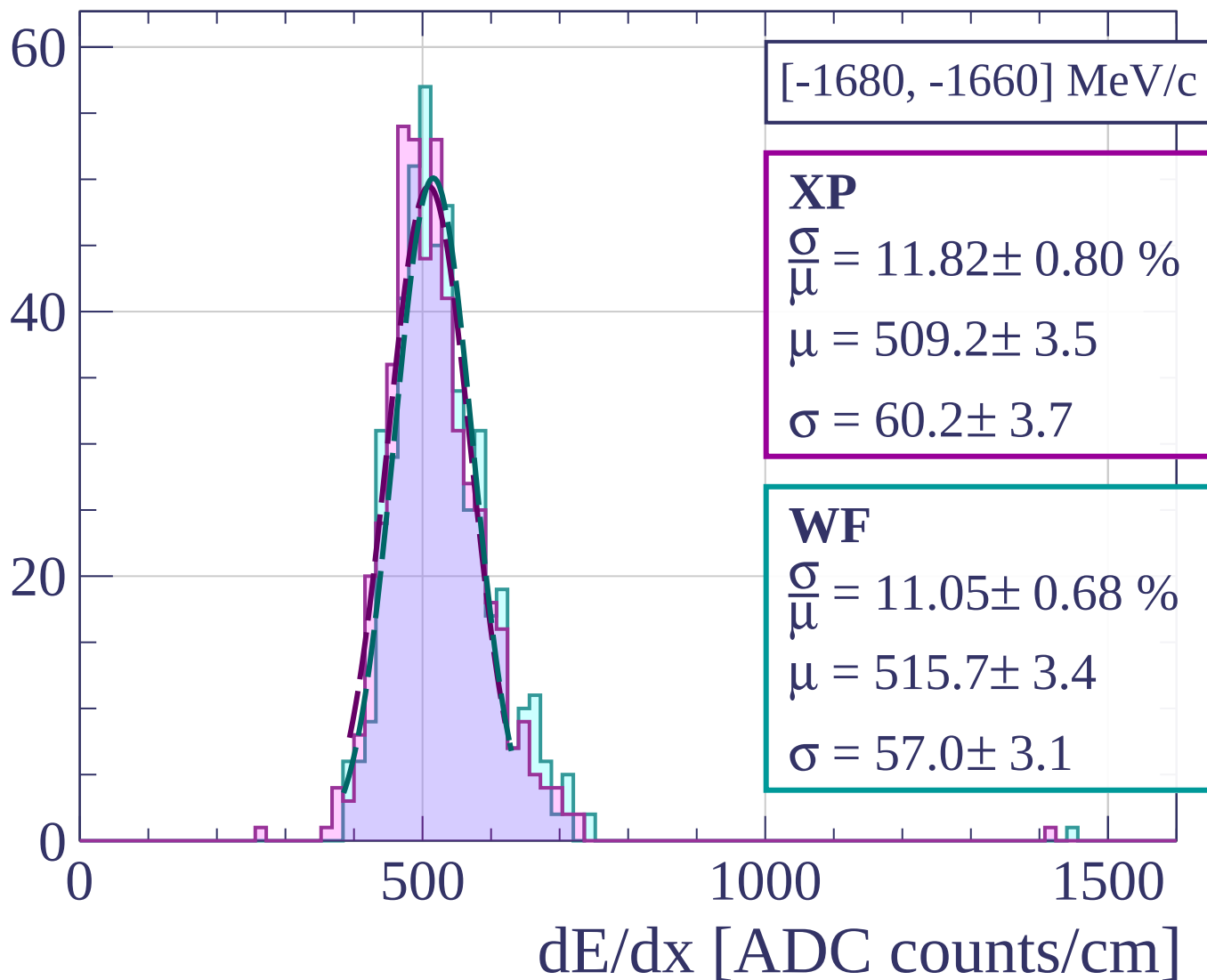
$$\sigma = 64.8 \pm 3.7$$

dE/dx [ADC counts/cm]

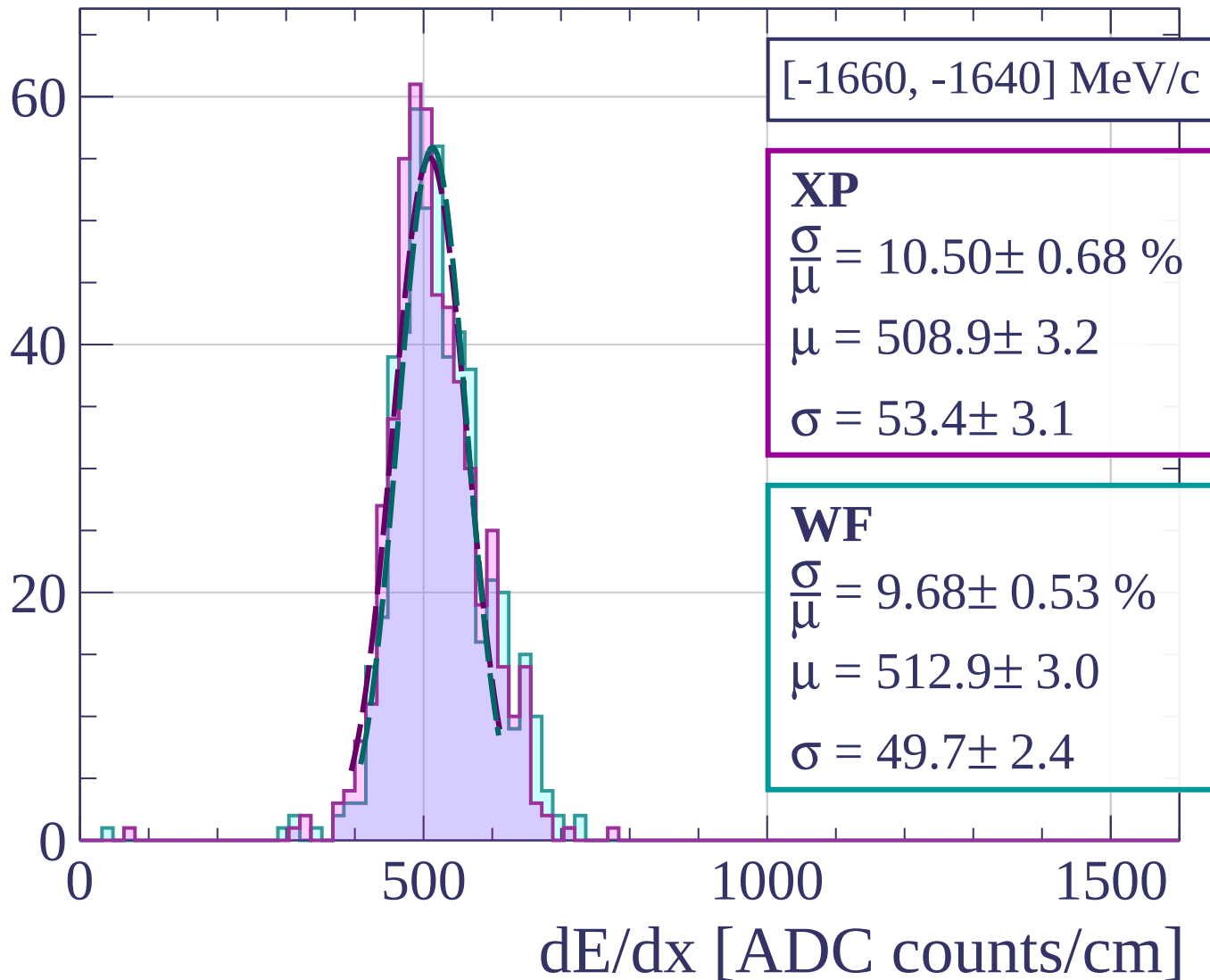
Count



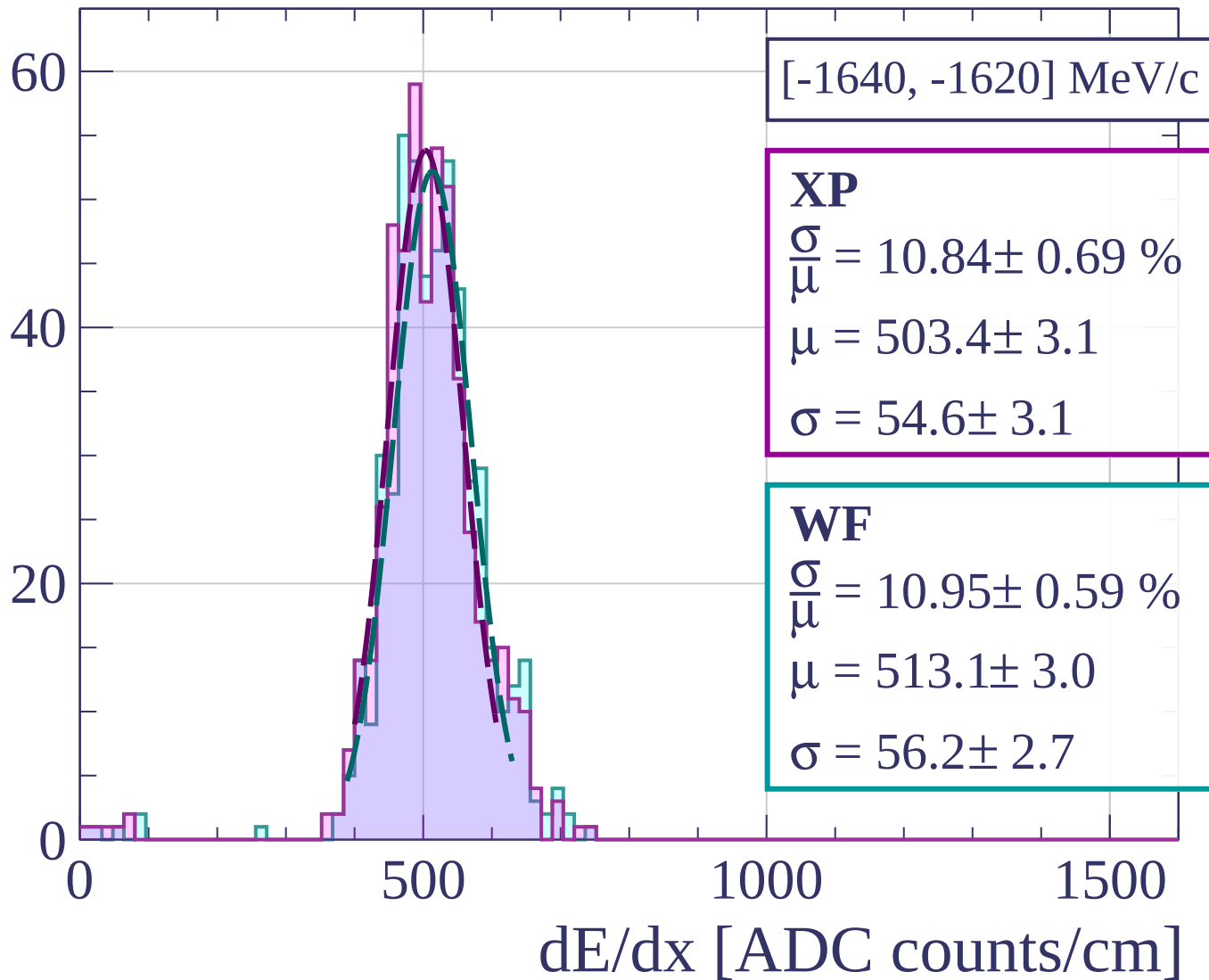
Count



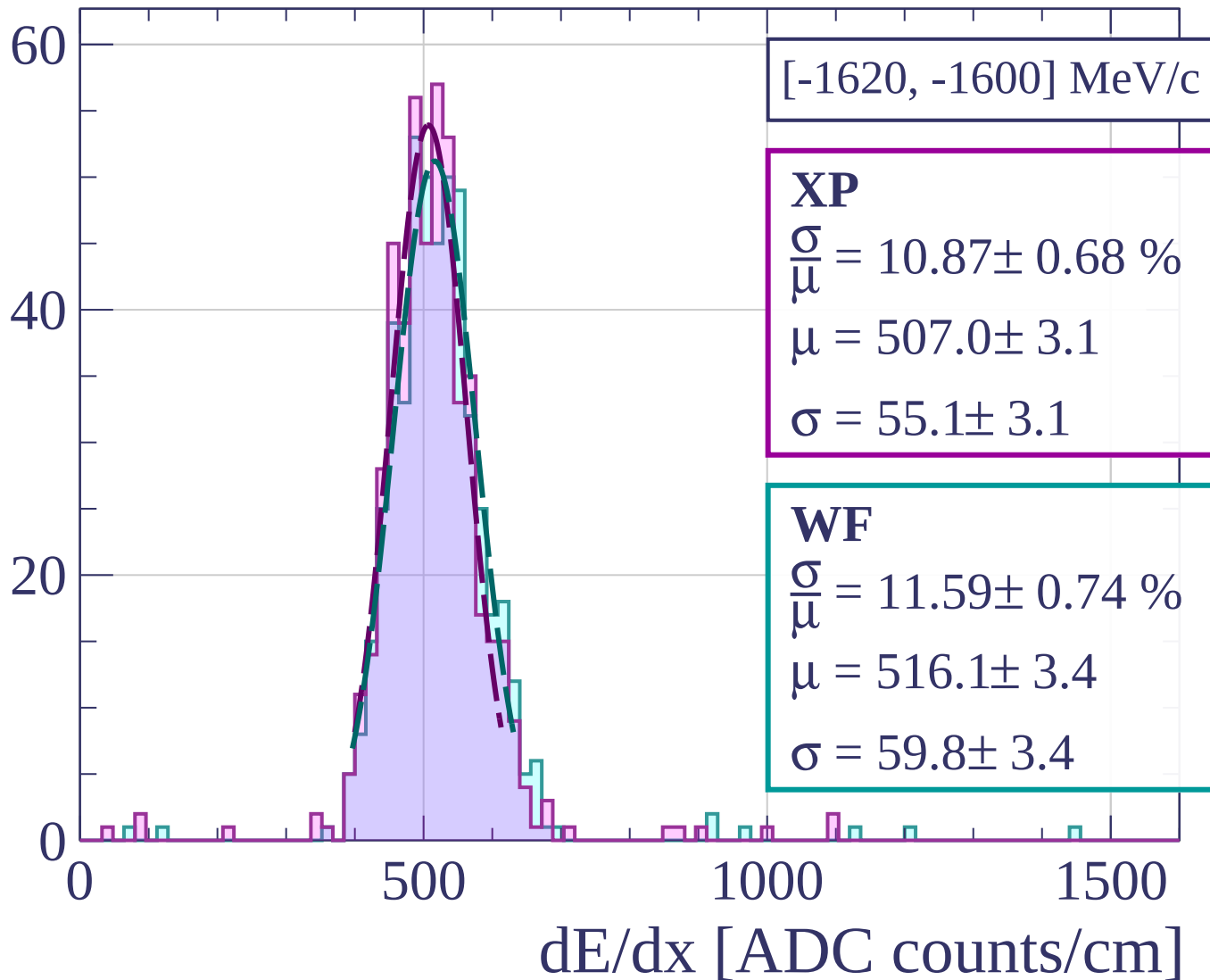
Count

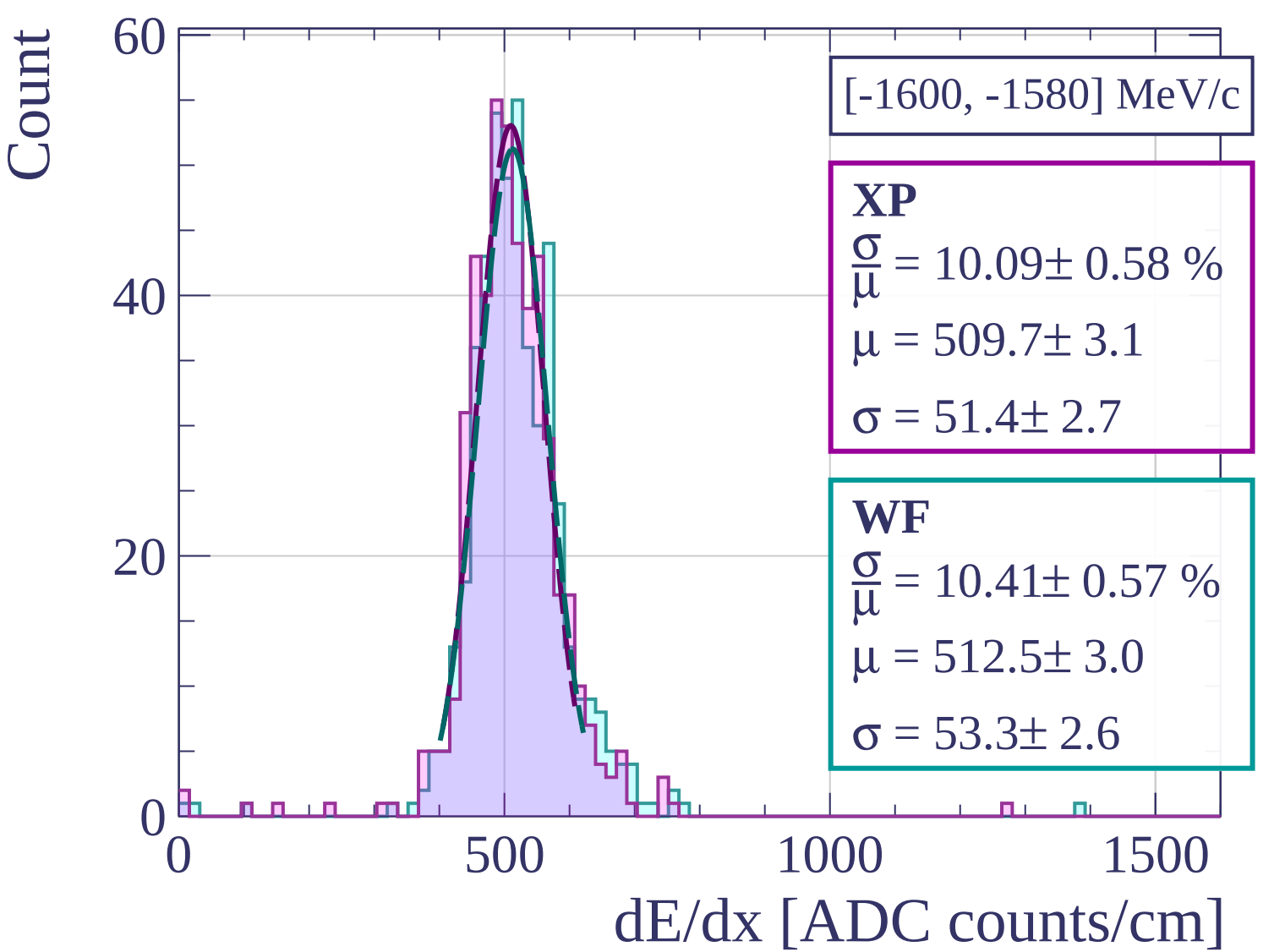


Count

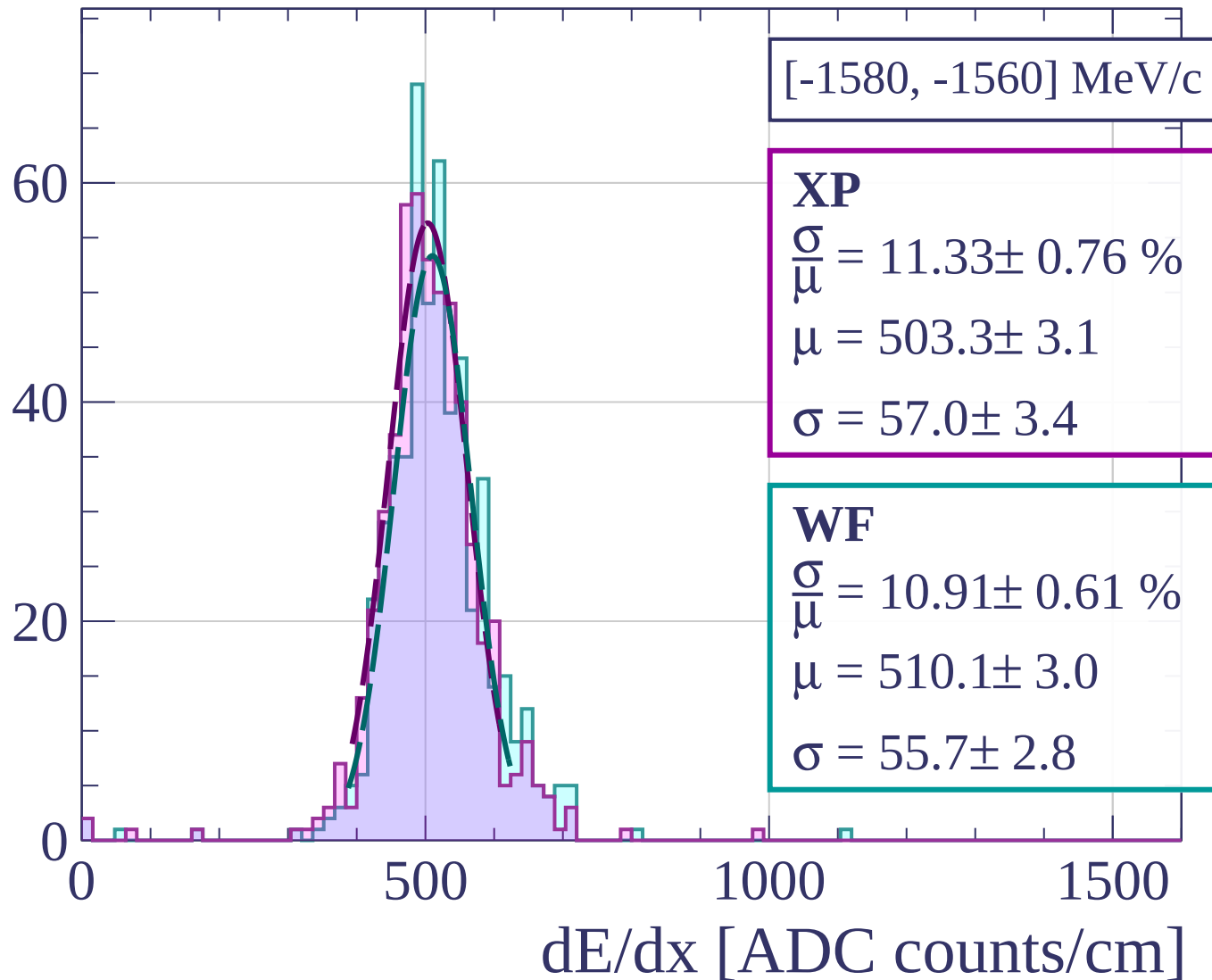


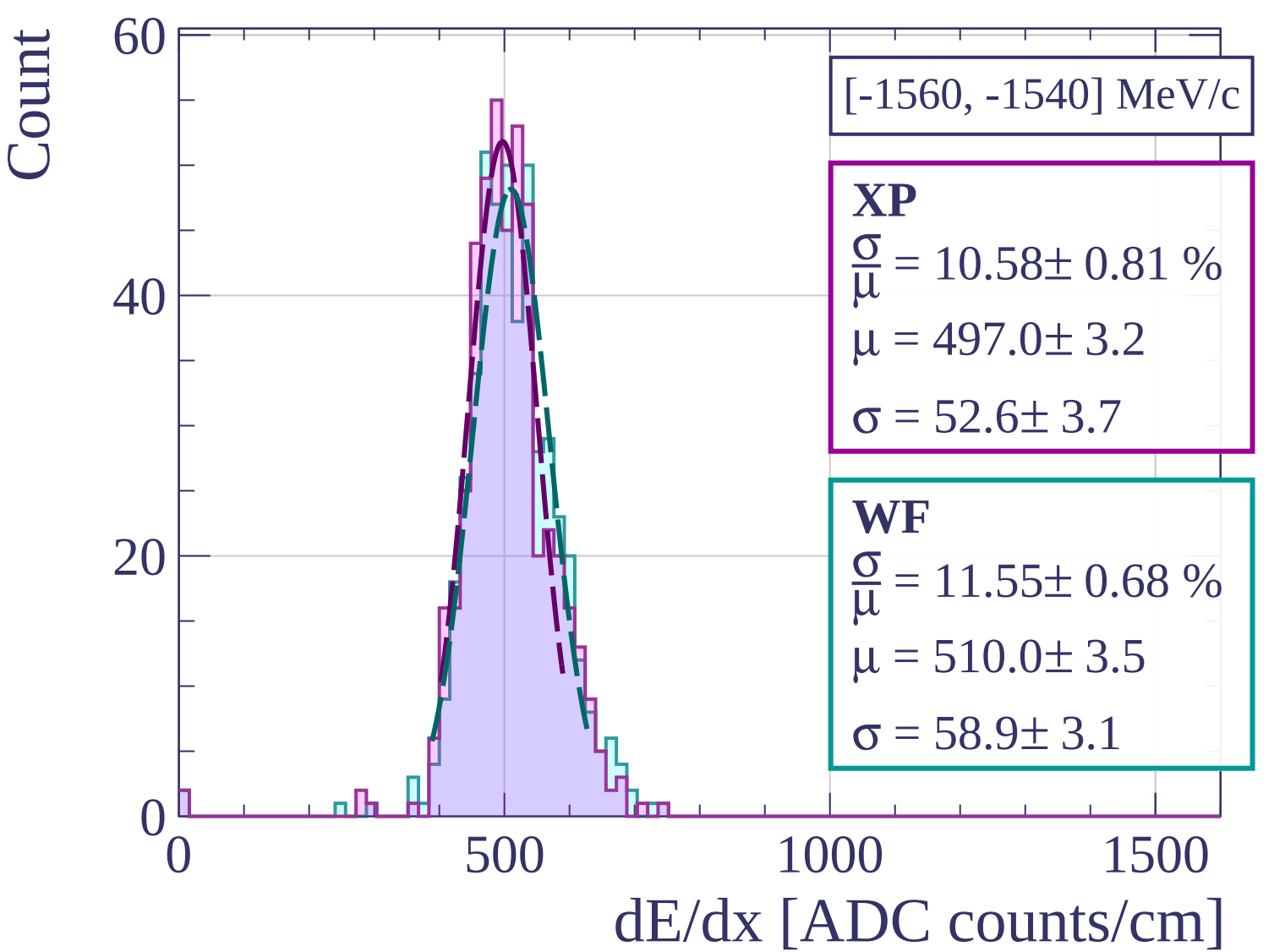
Count



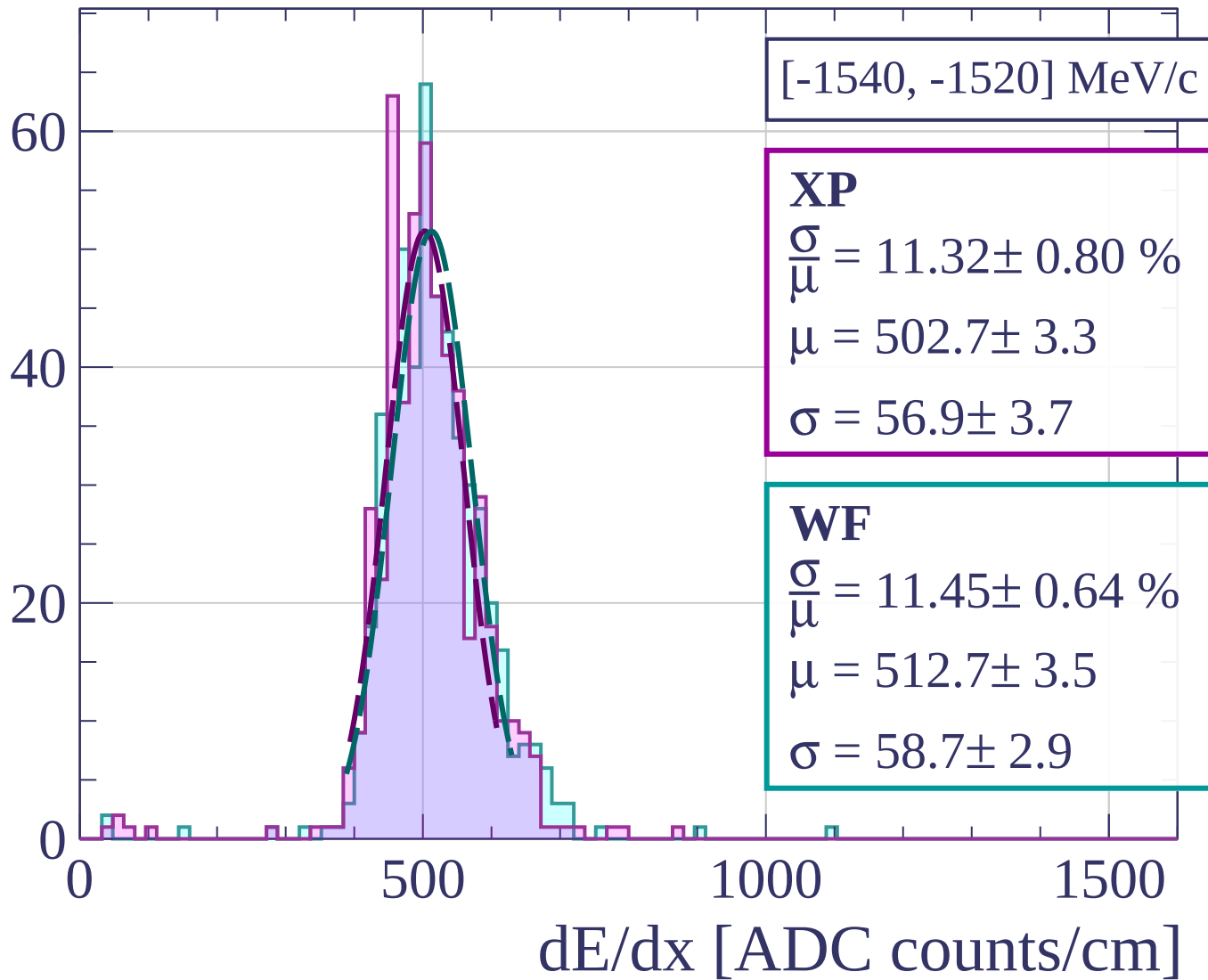


Count

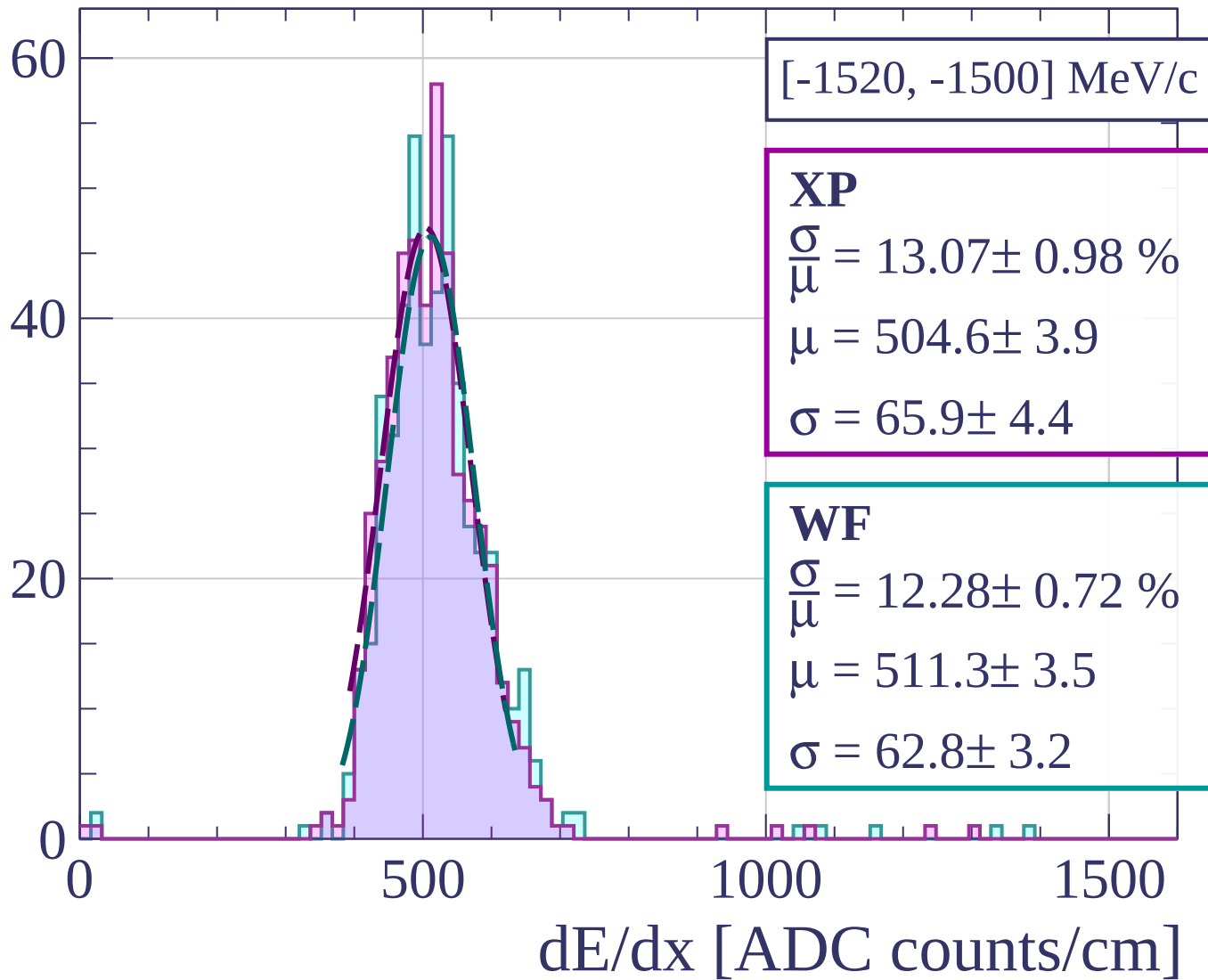


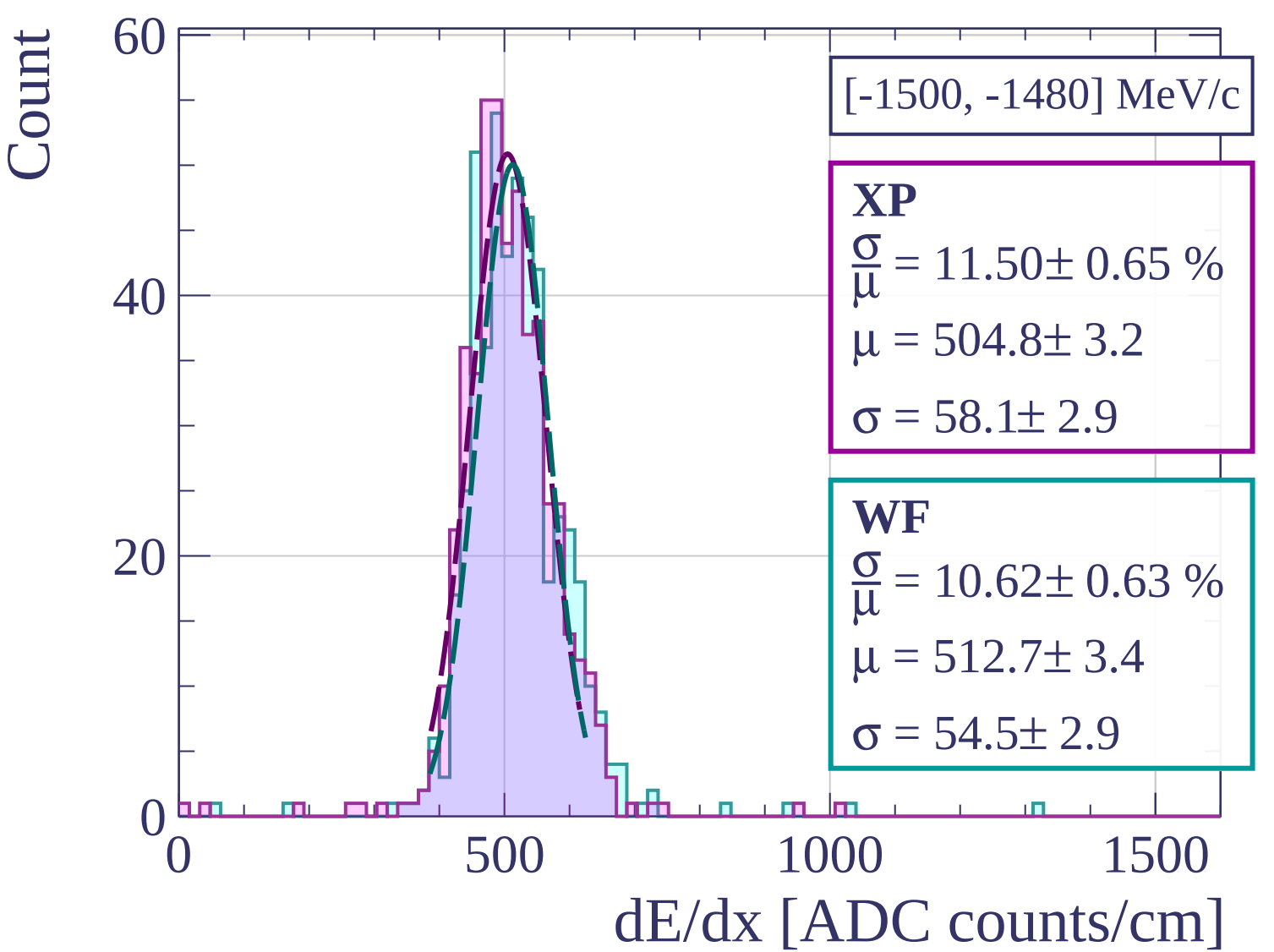


Count

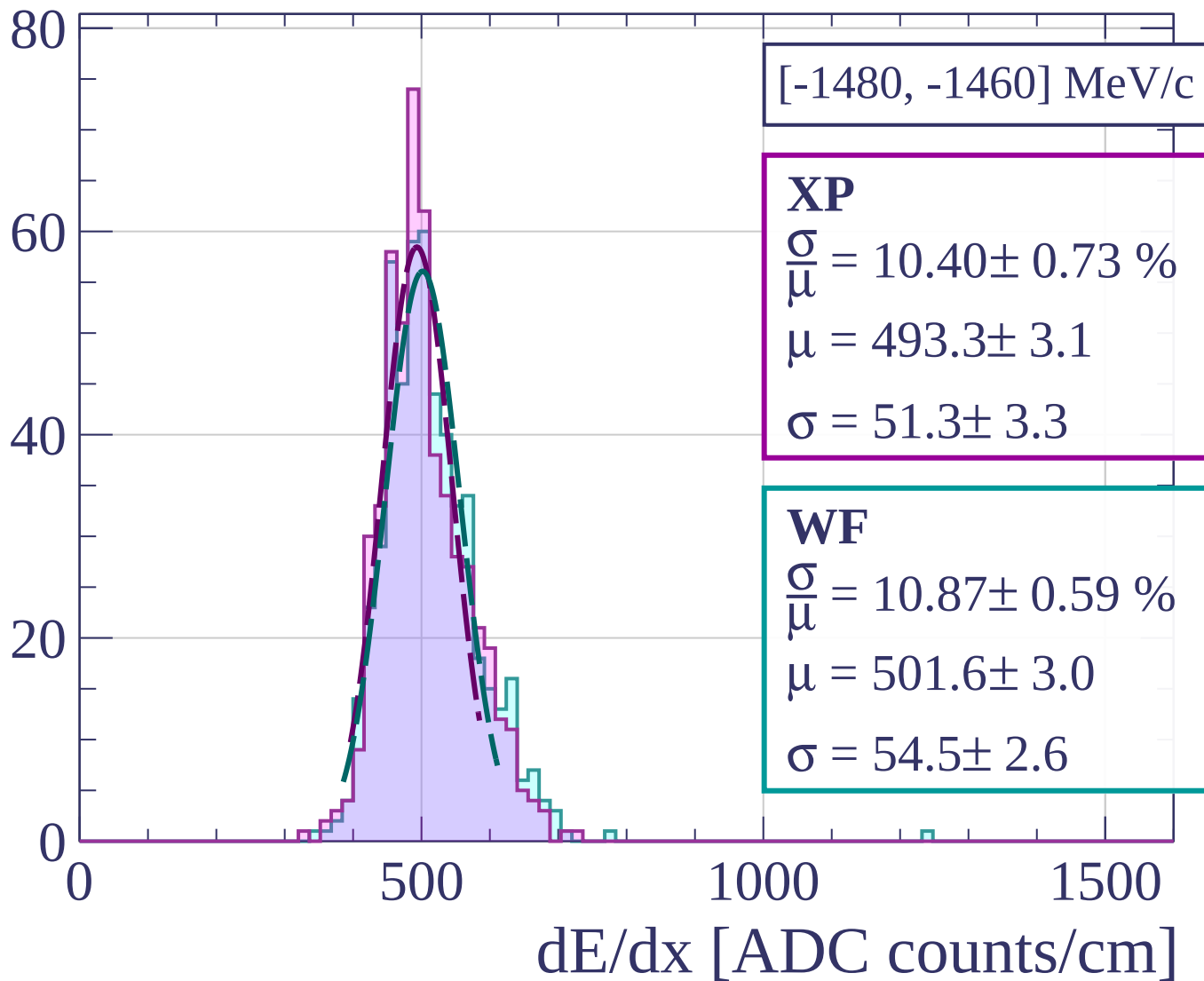


Count

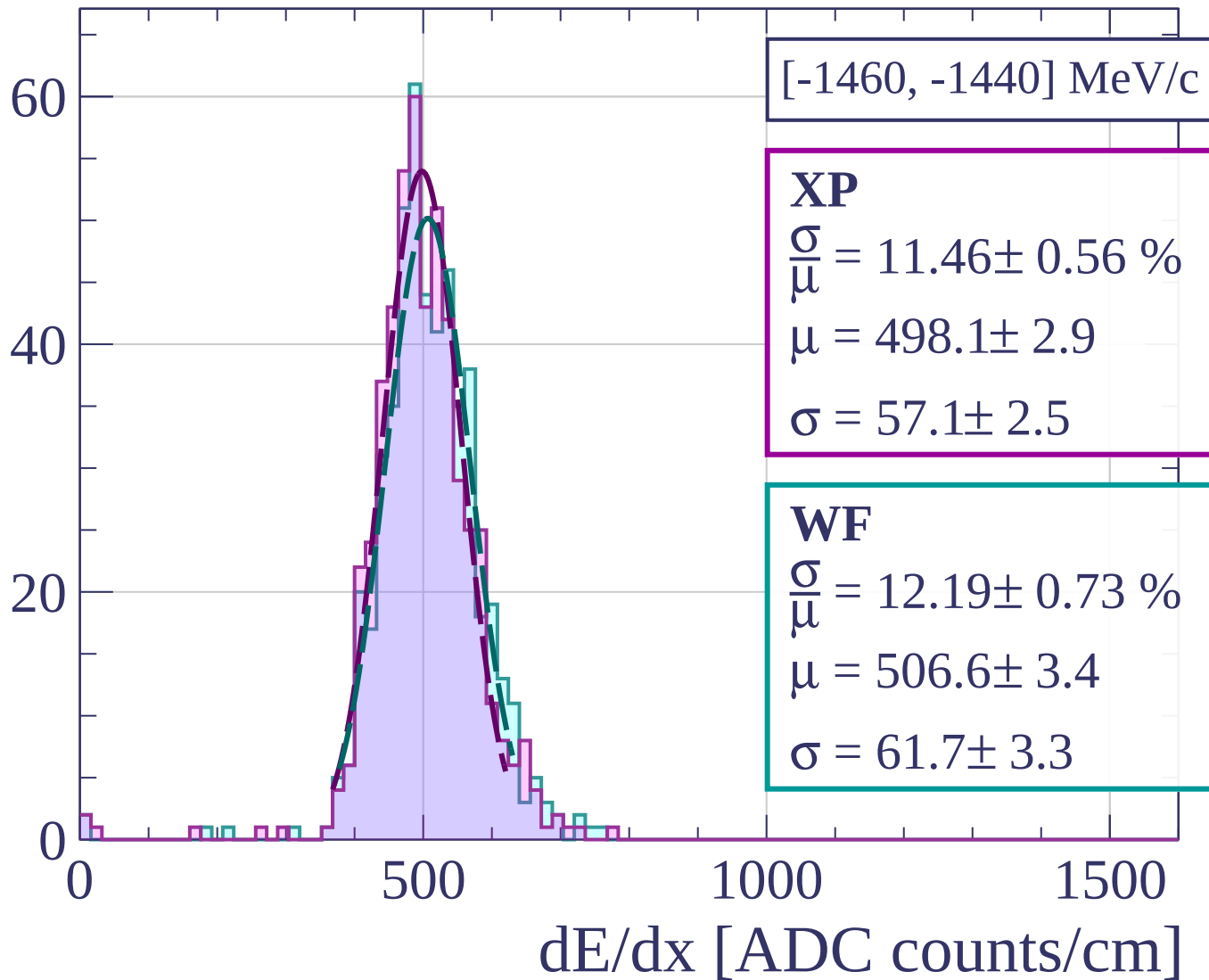




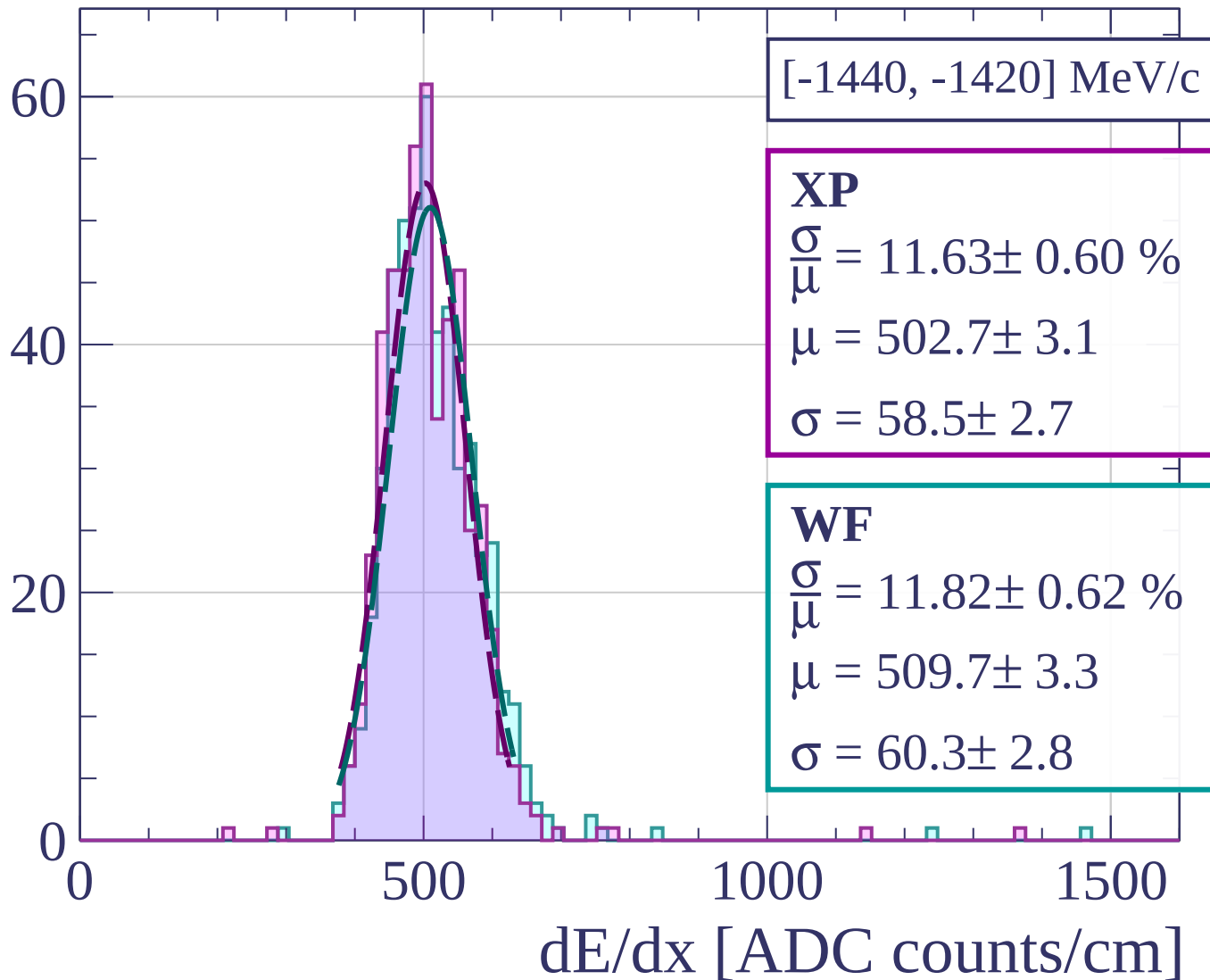
Count



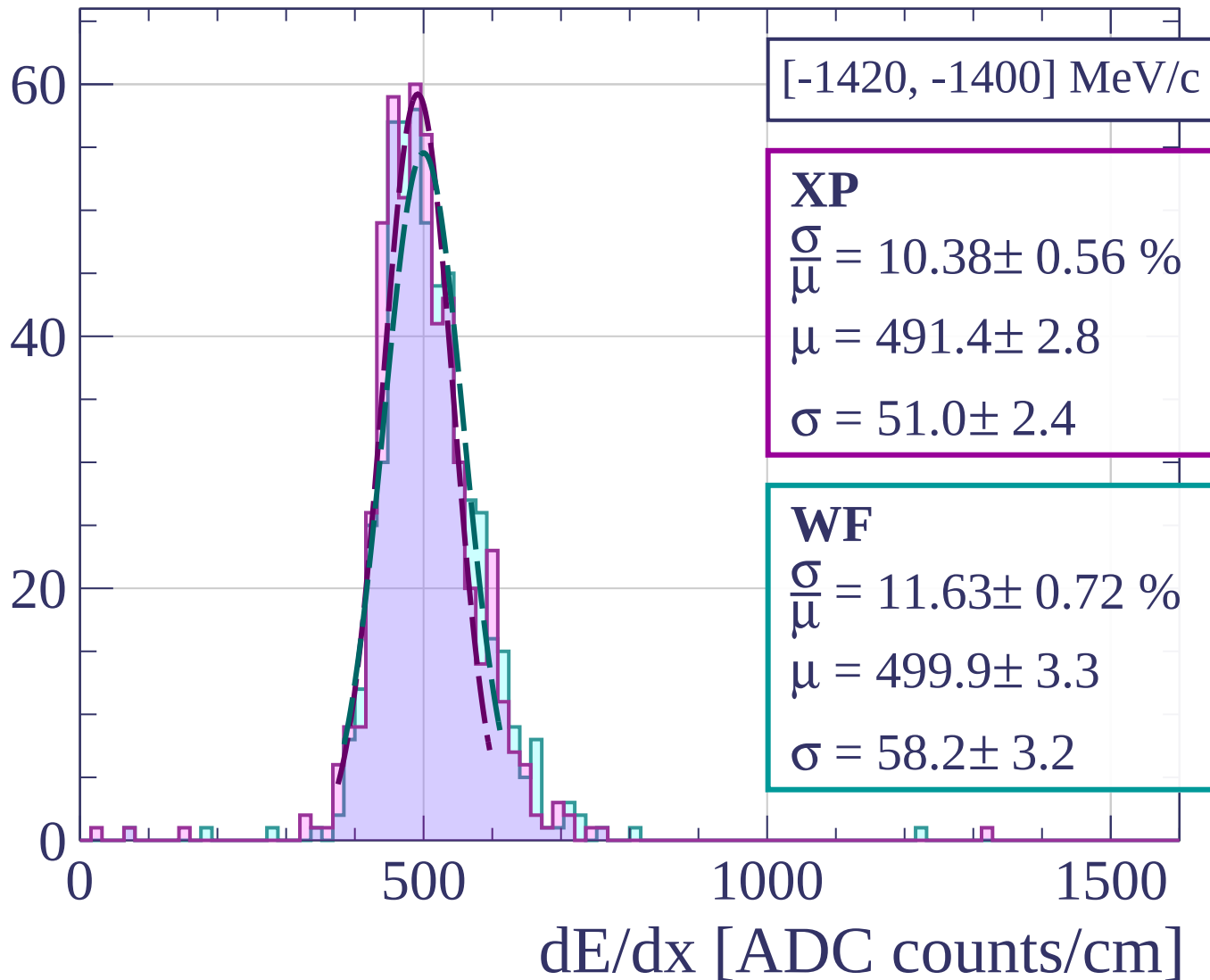
Count



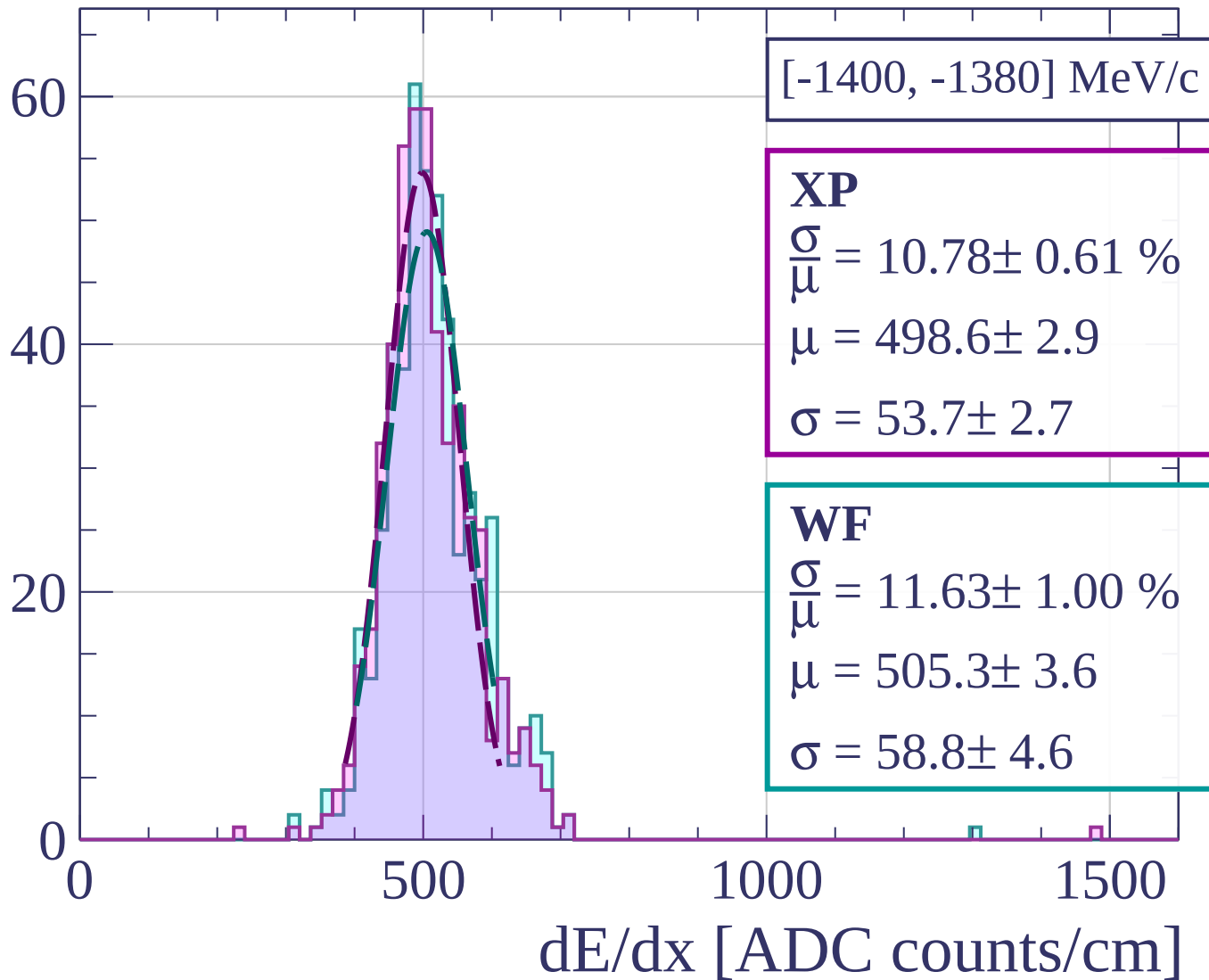
Count



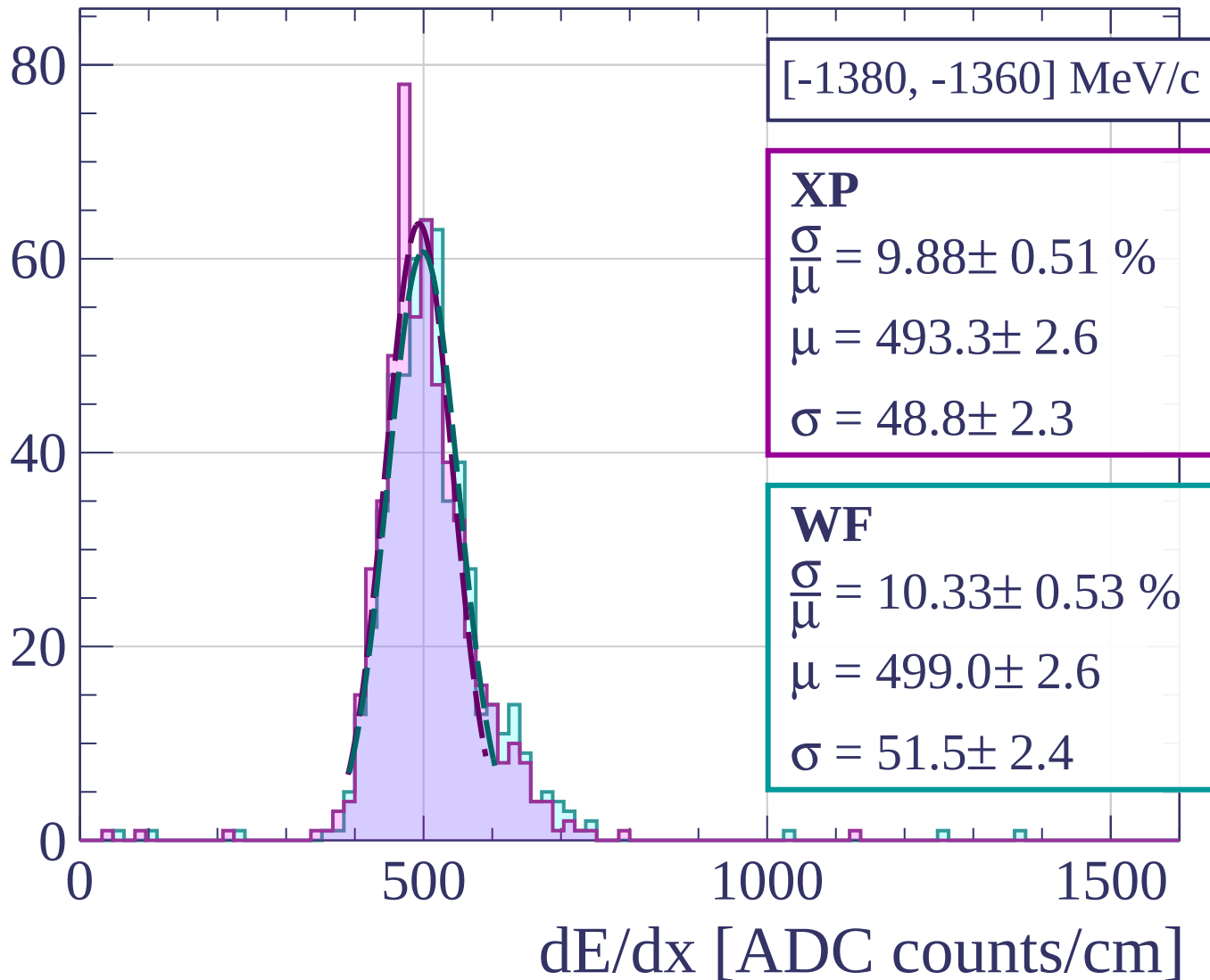
Count



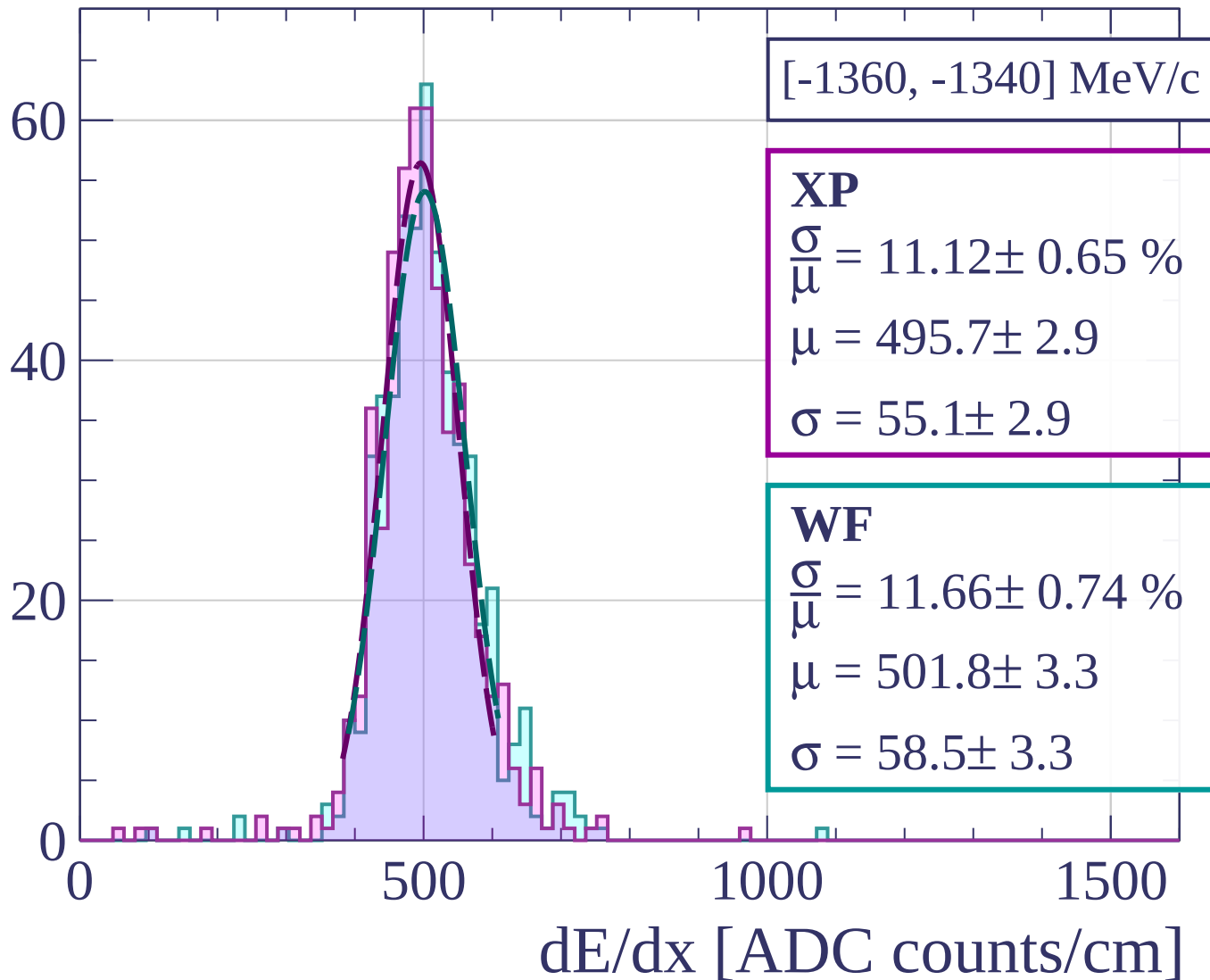
Count



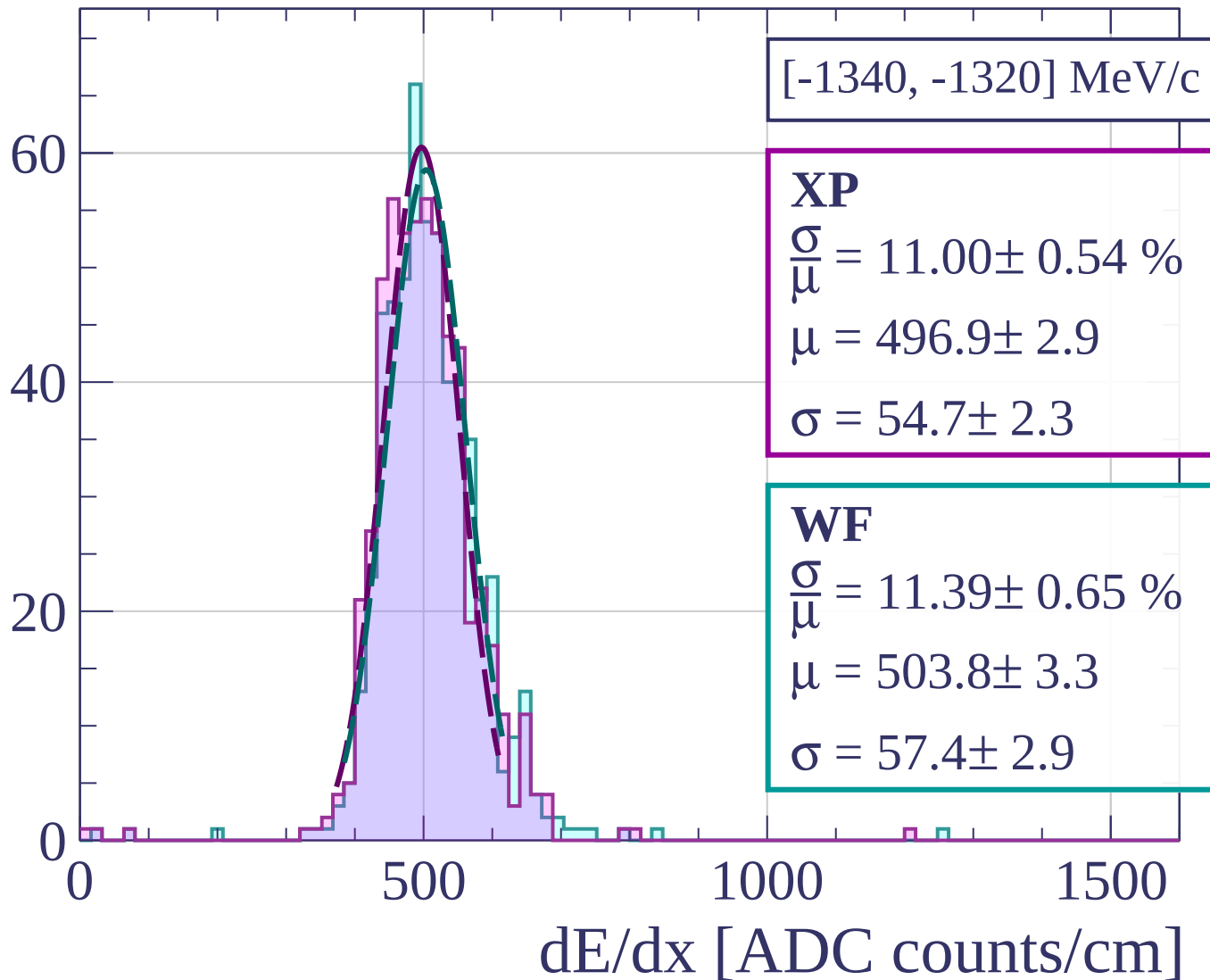
Count



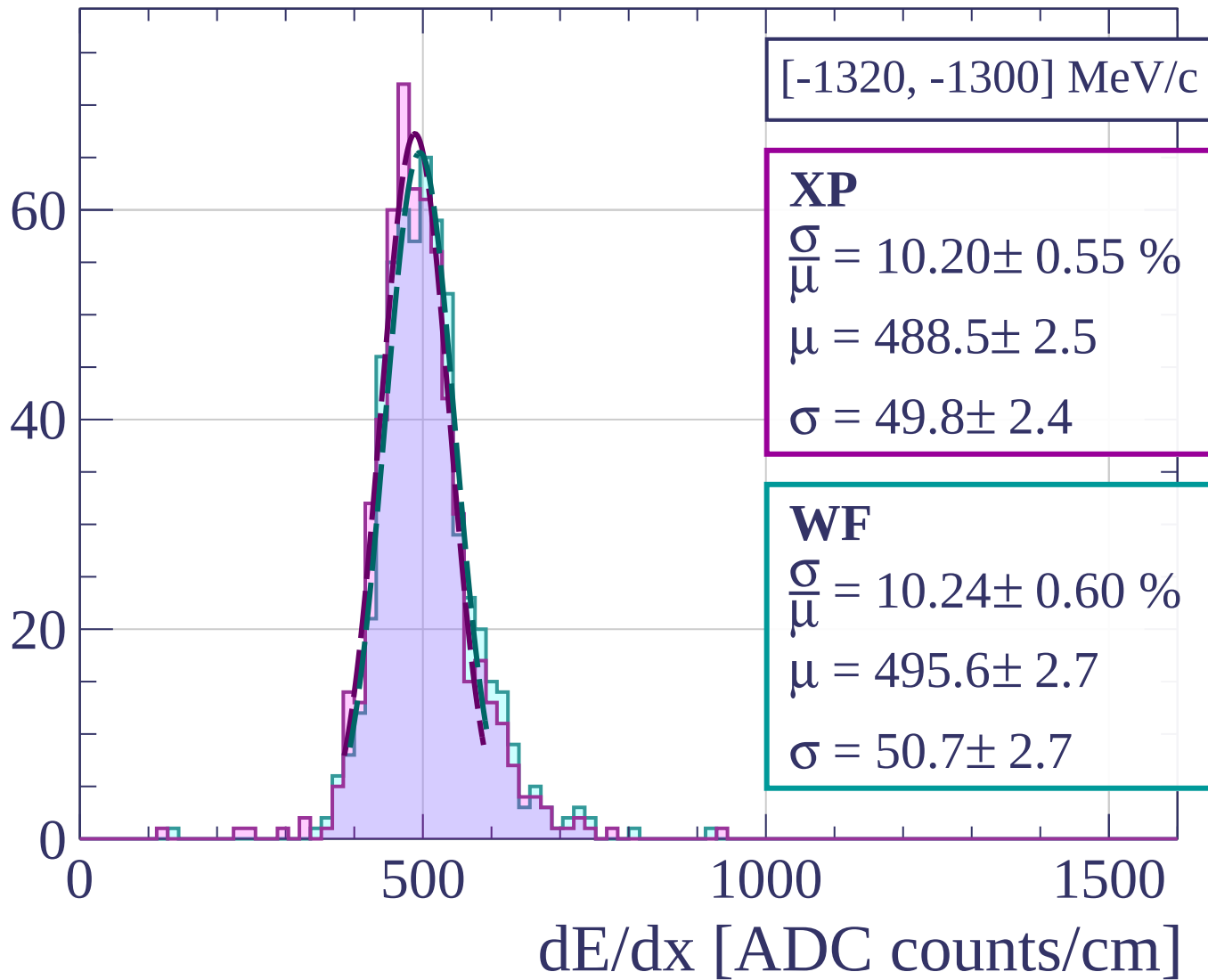
Count



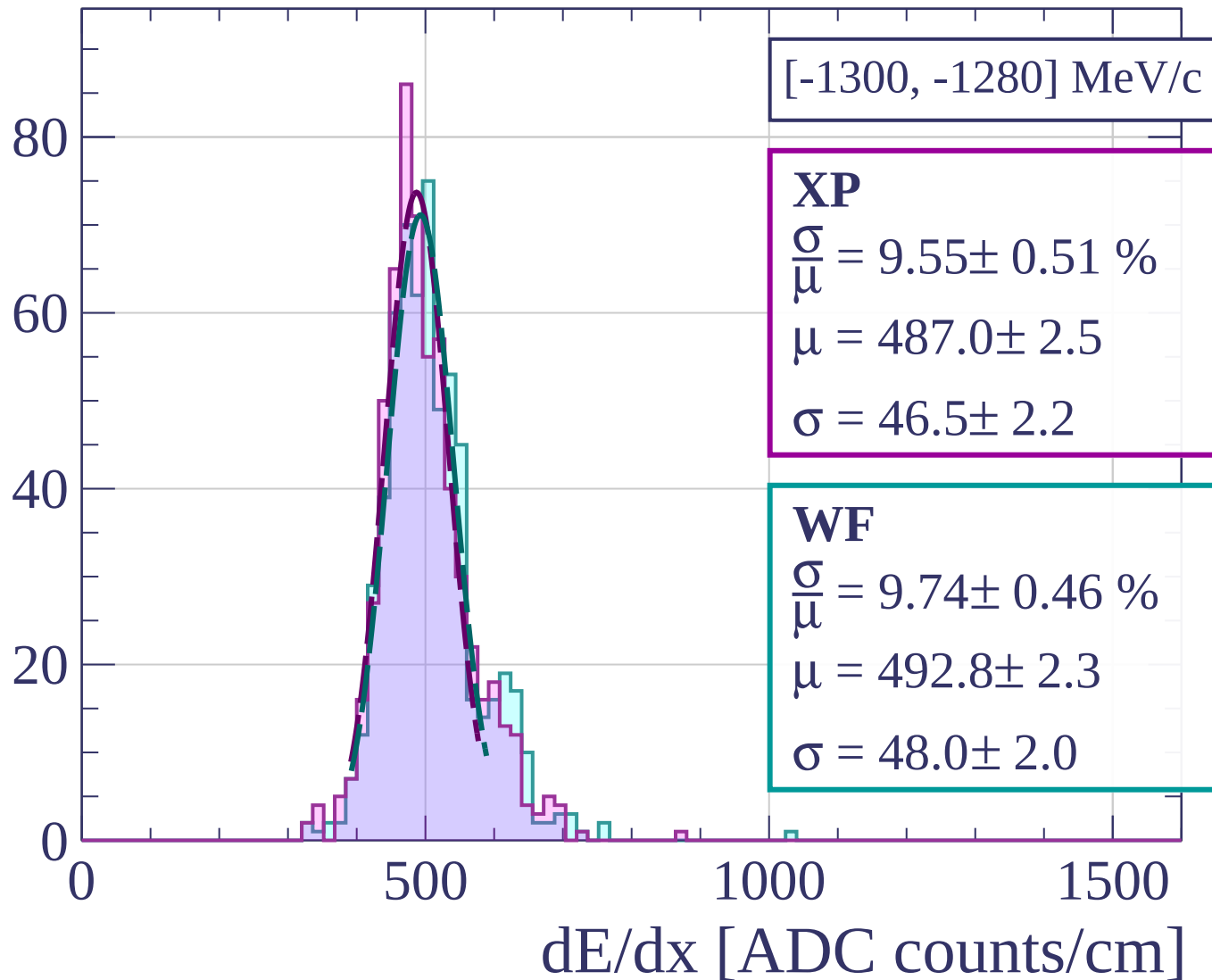
Count



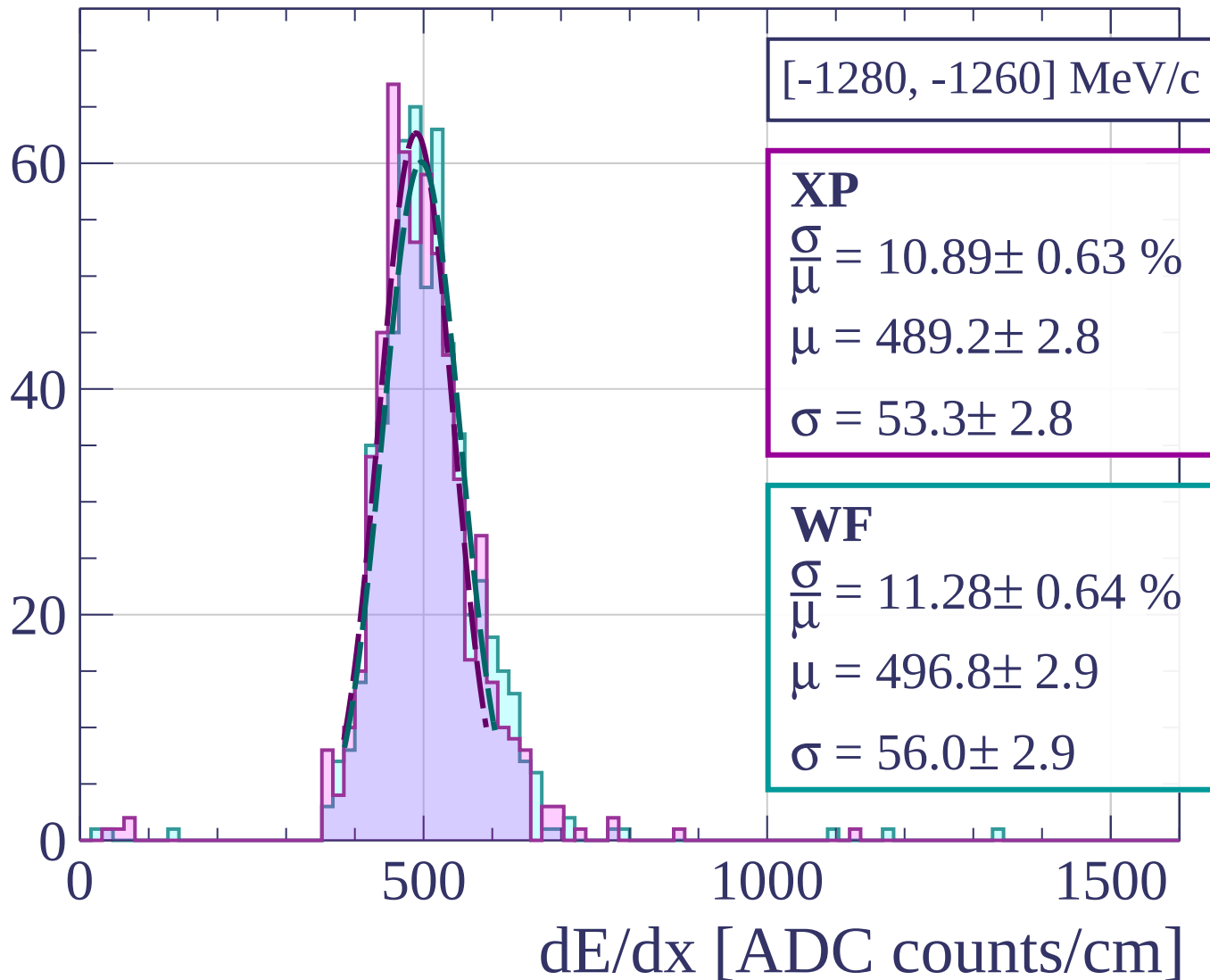
Count



Count

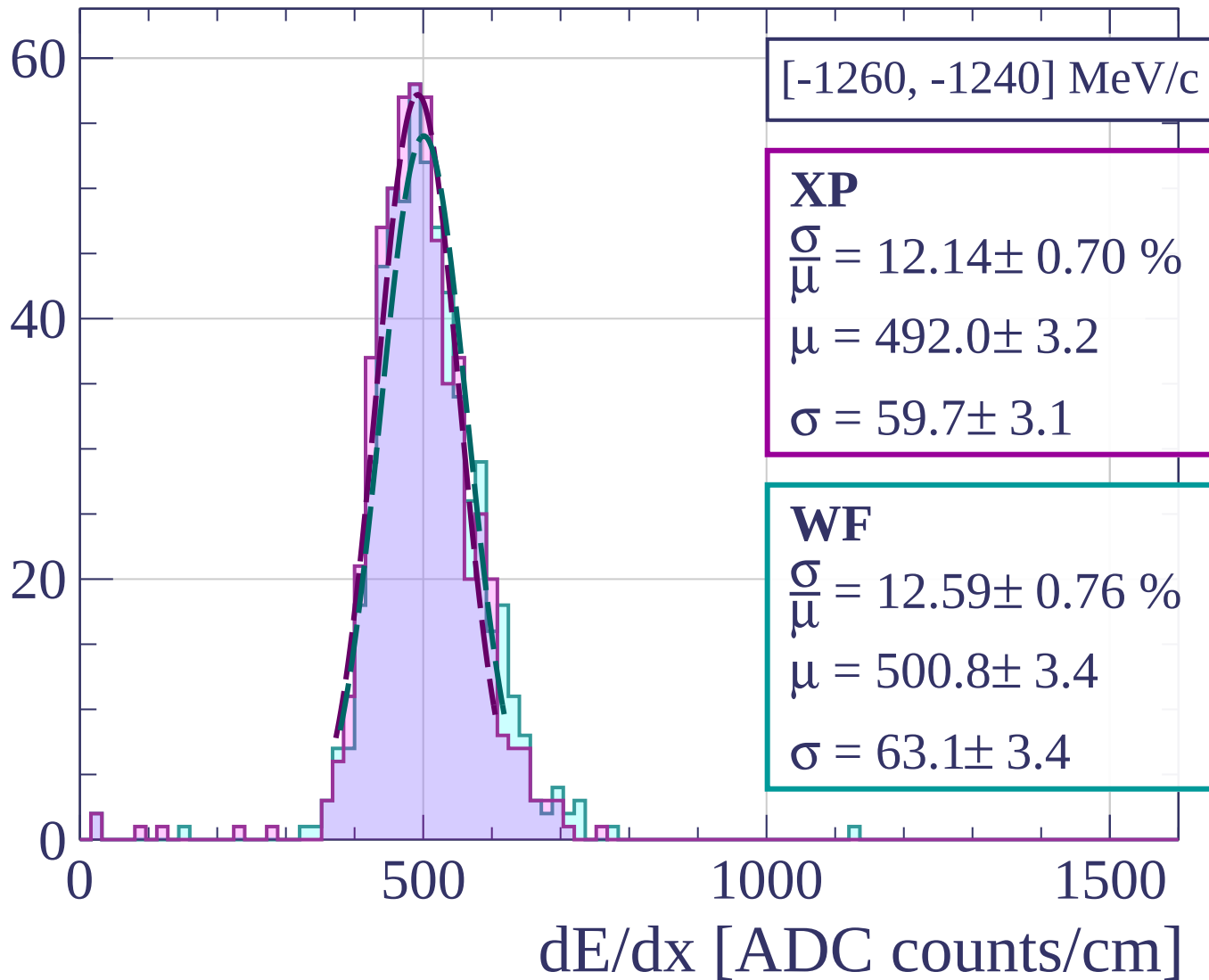


Count

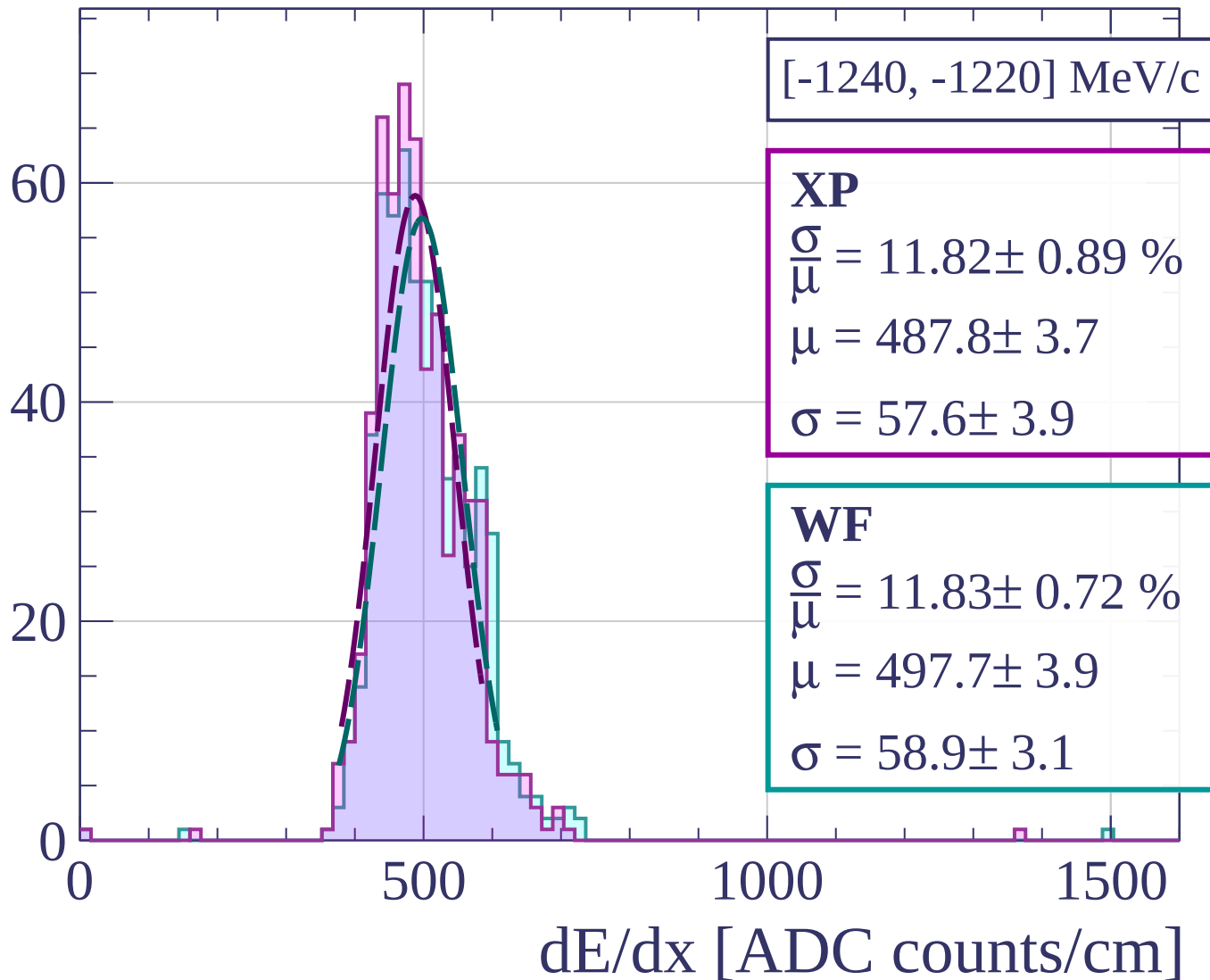


[-1280, -1260] MeV/c

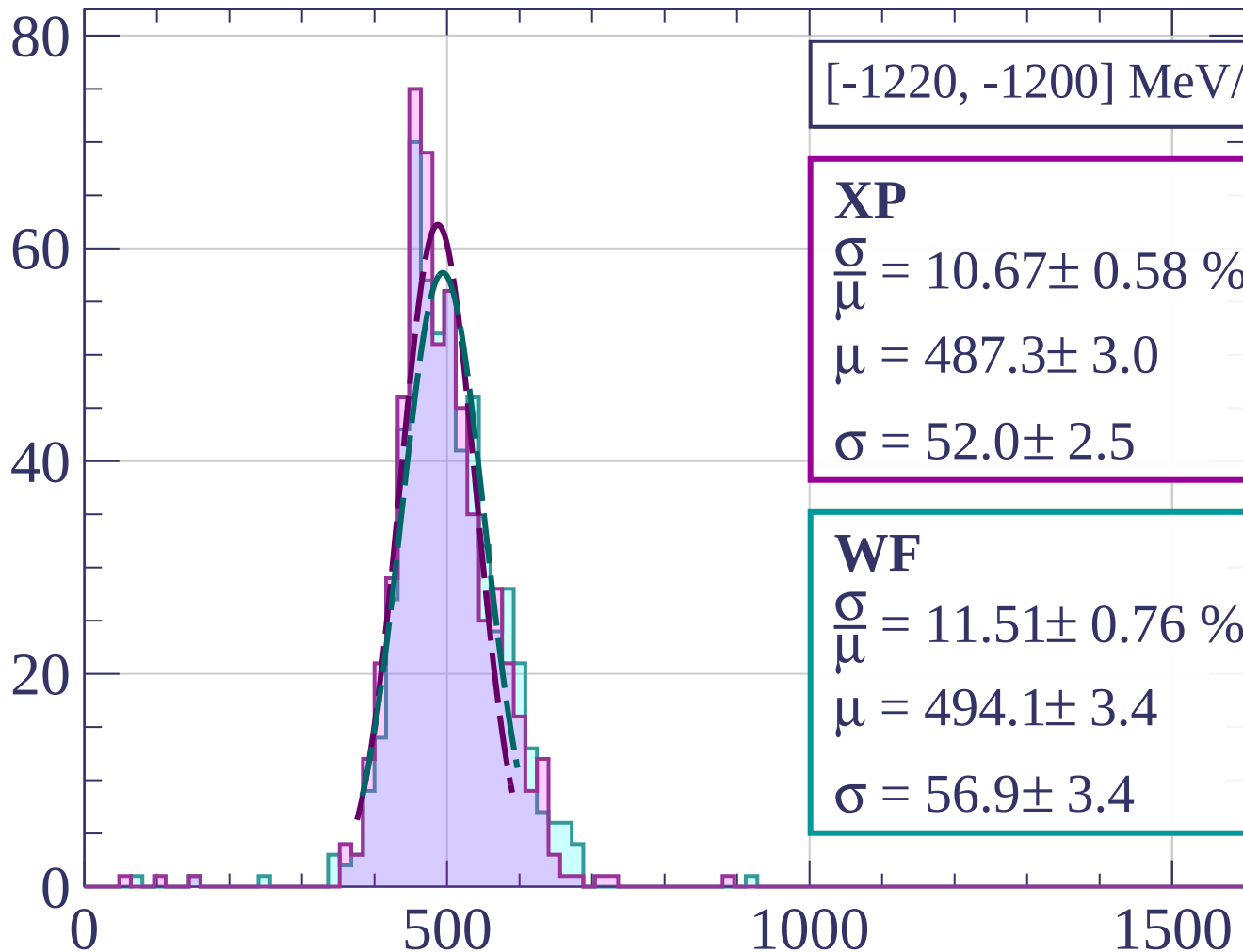
Count



Count



Count



[-1220, -1200] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 10.67 \pm 0.58 \%$$

$$\mu = 487.3 \pm 3.0$$

$$\sigma = 52.0 \pm 2.5$$

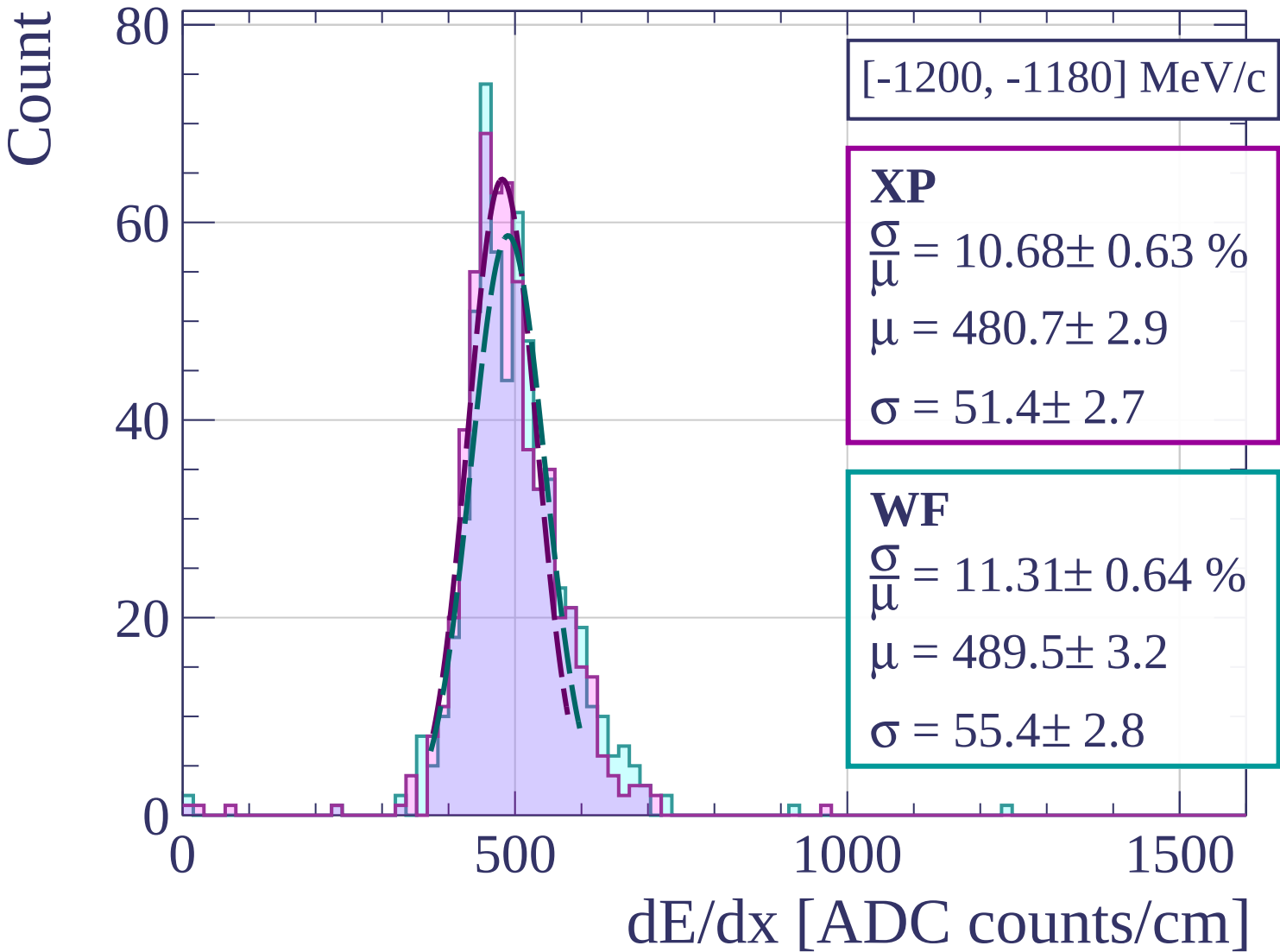
WF

$$\frac{\sigma}{\bar{\mu}} = 11.51 \pm 0.76 \%$$

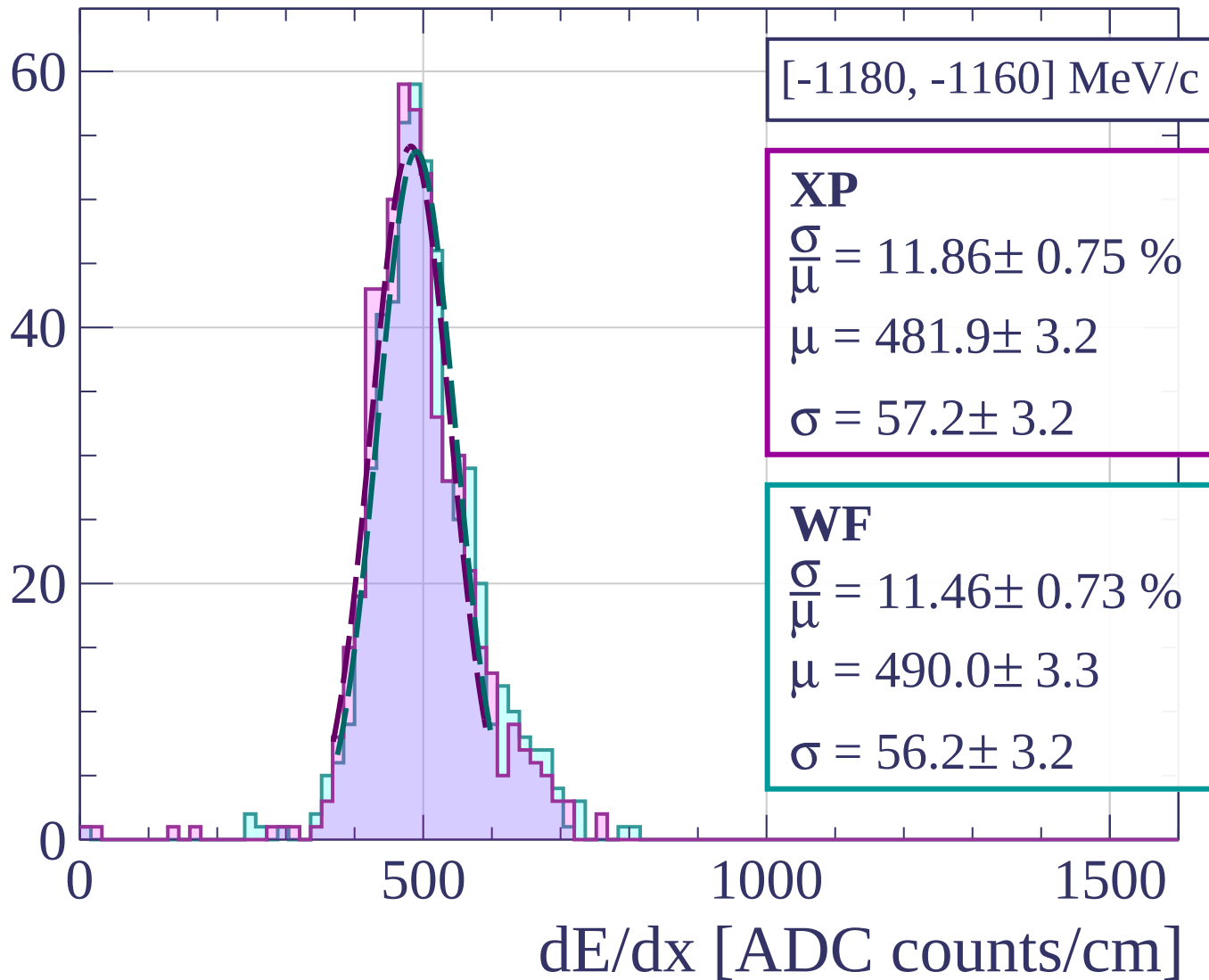
$$\mu = 494.1 \pm 3.4$$

$$\sigma = 56.9 \pm 3.4$$

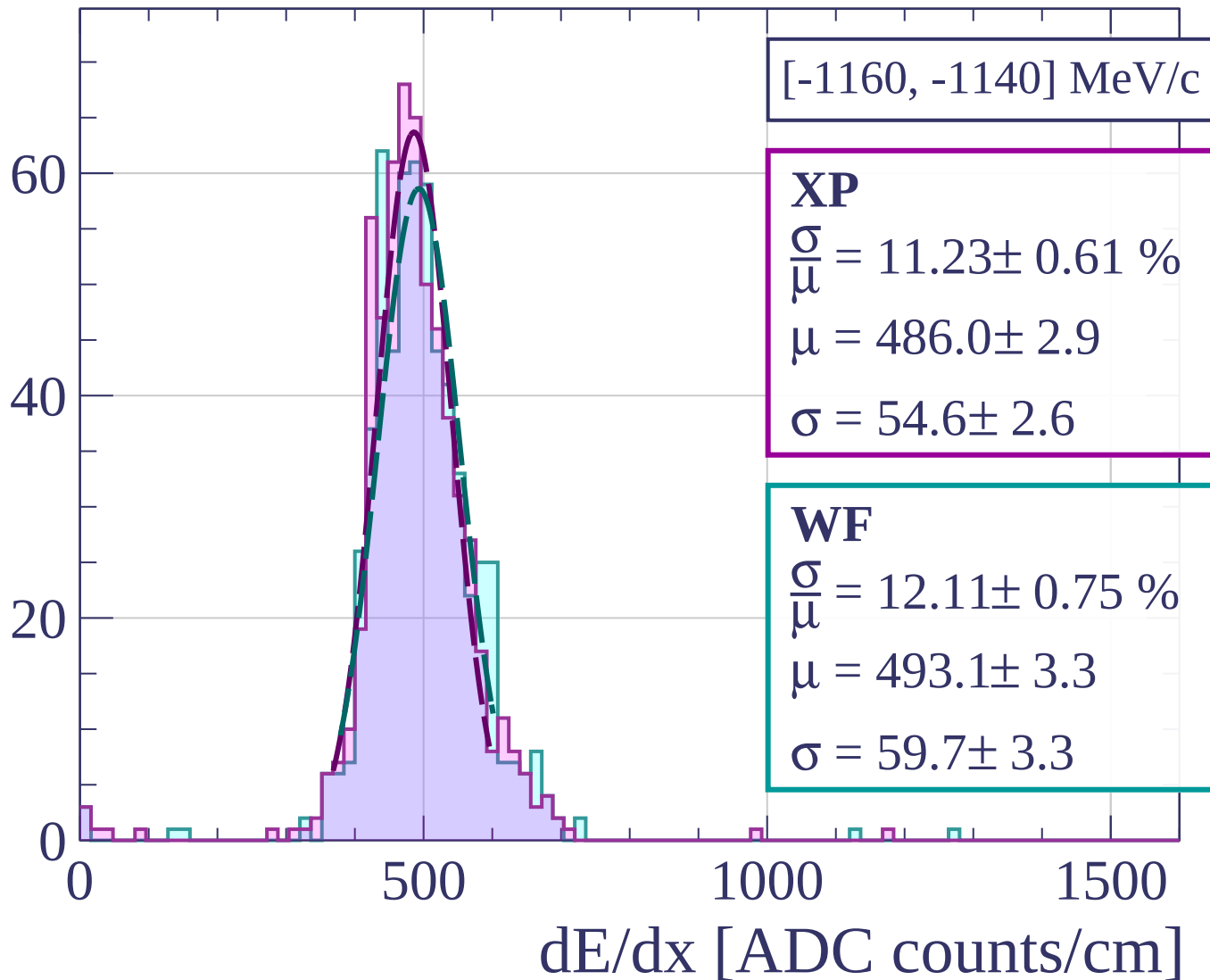
dE/dx [ADC counts/cm]

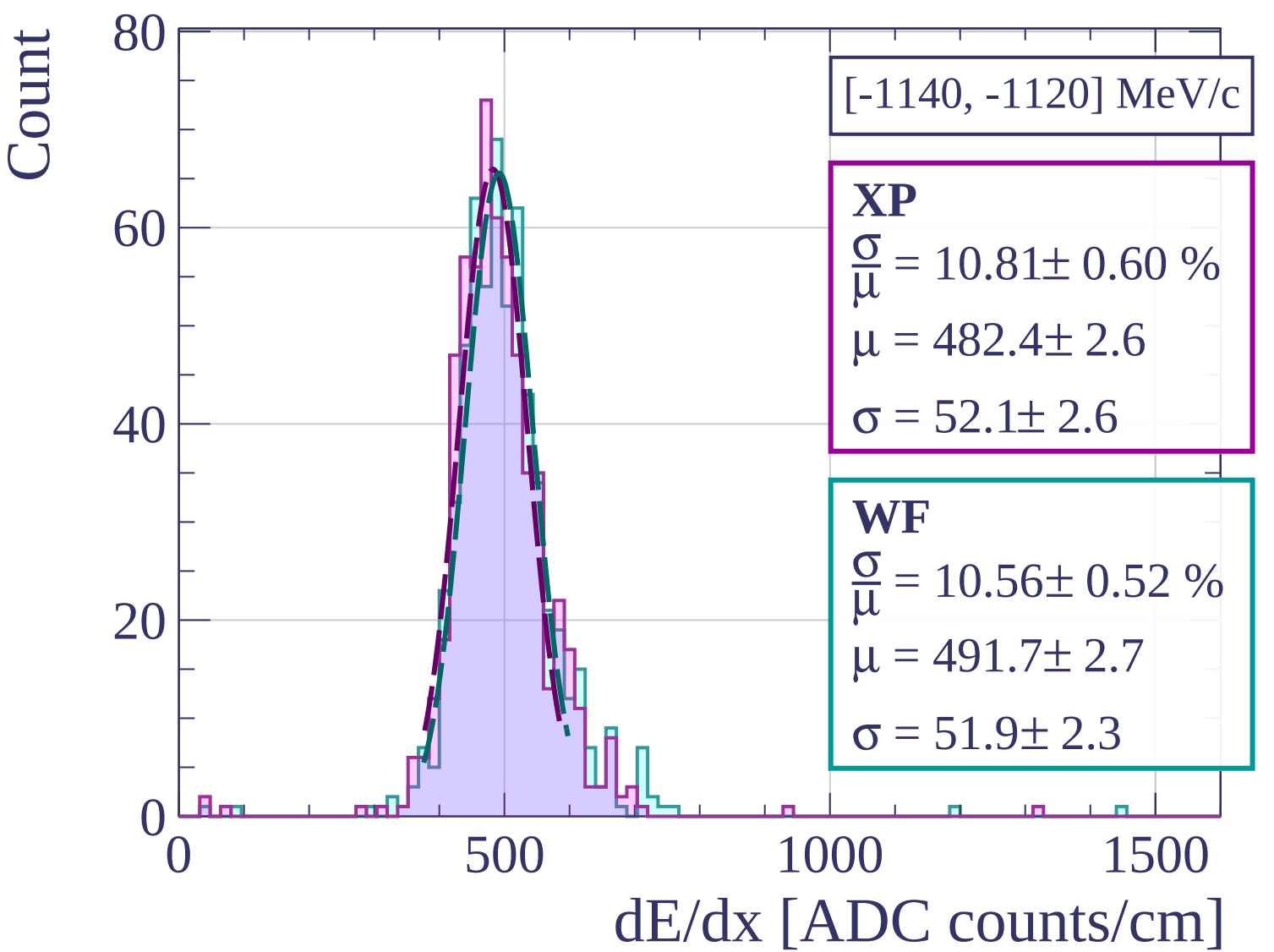


Count

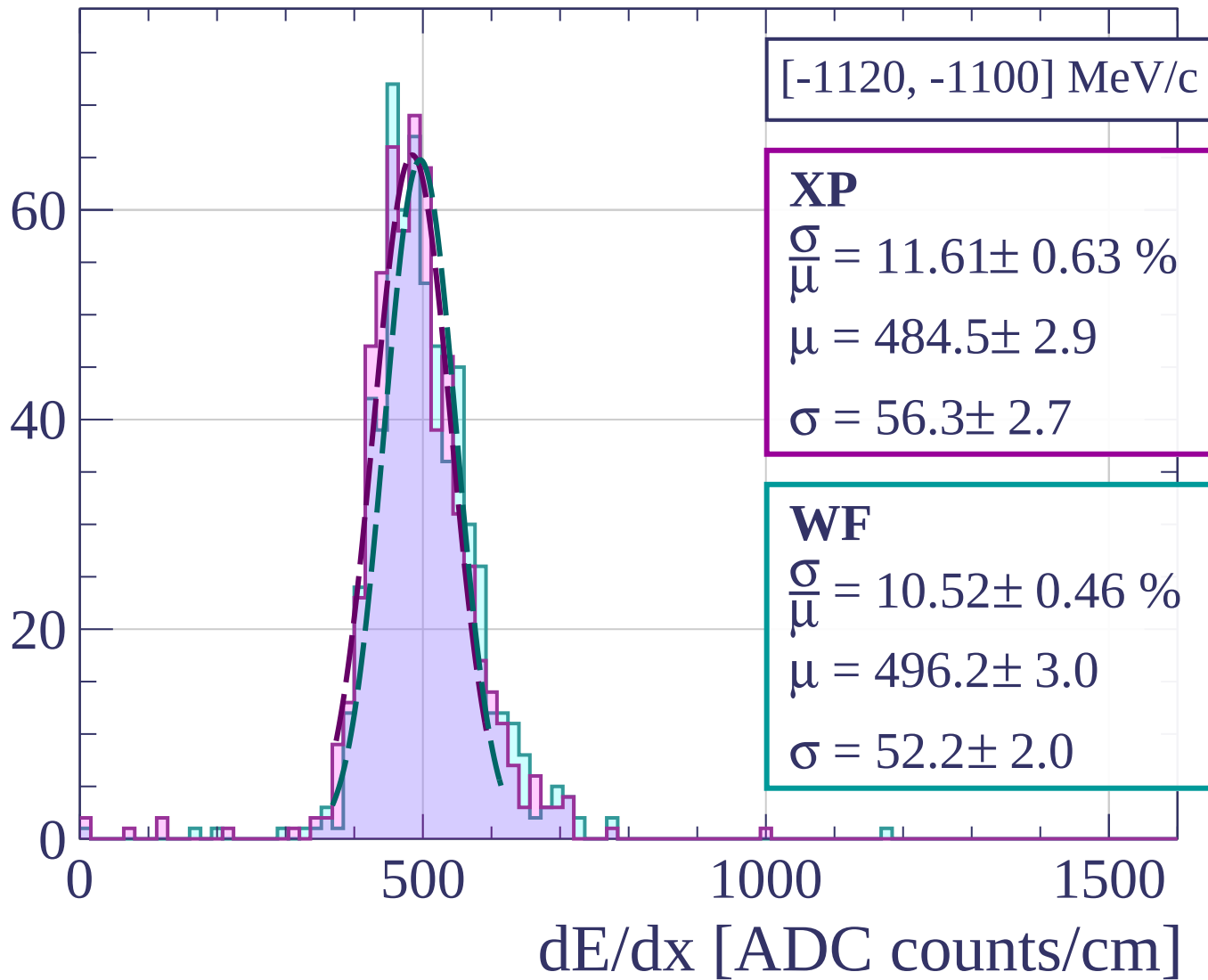


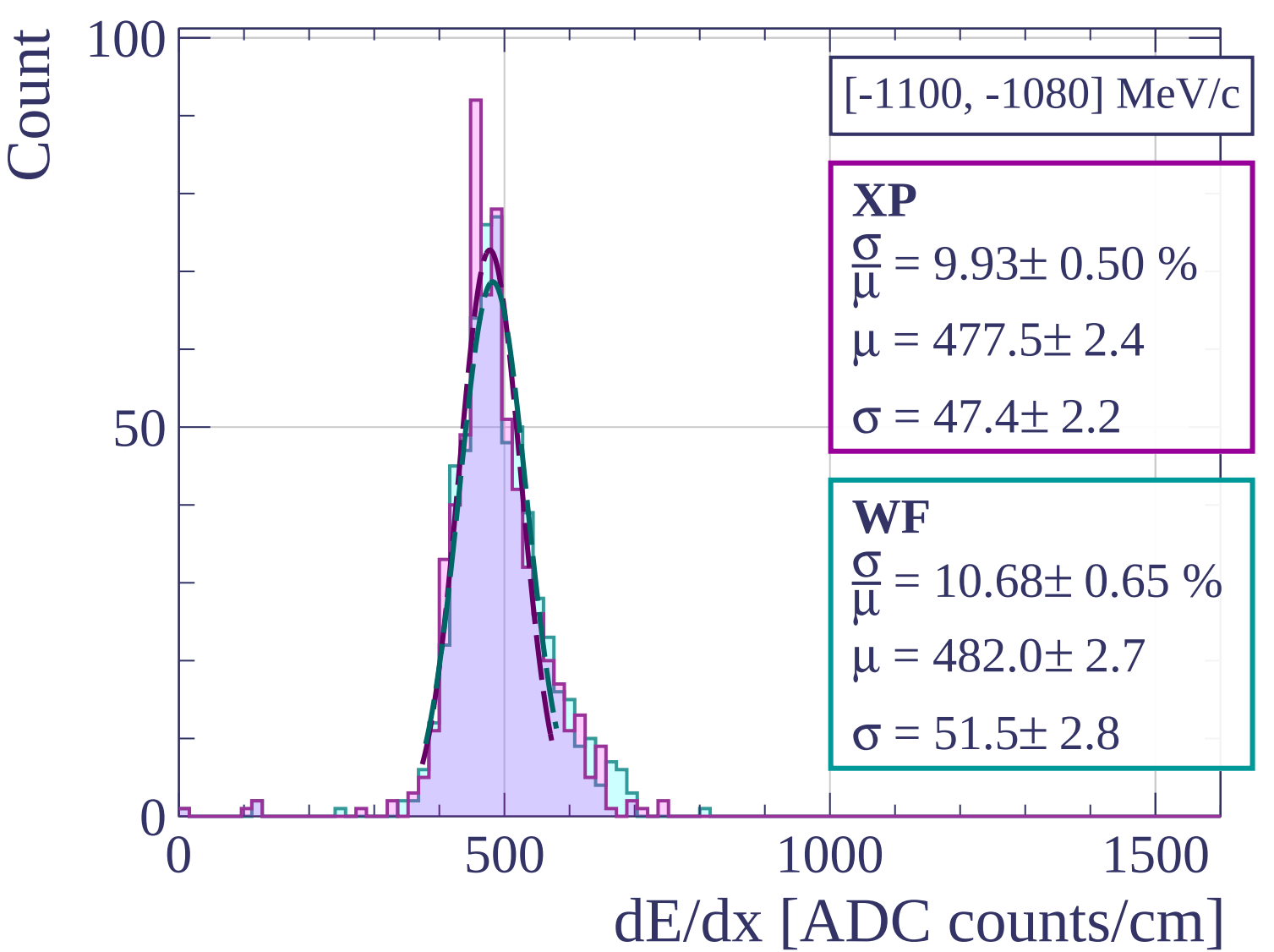
Count



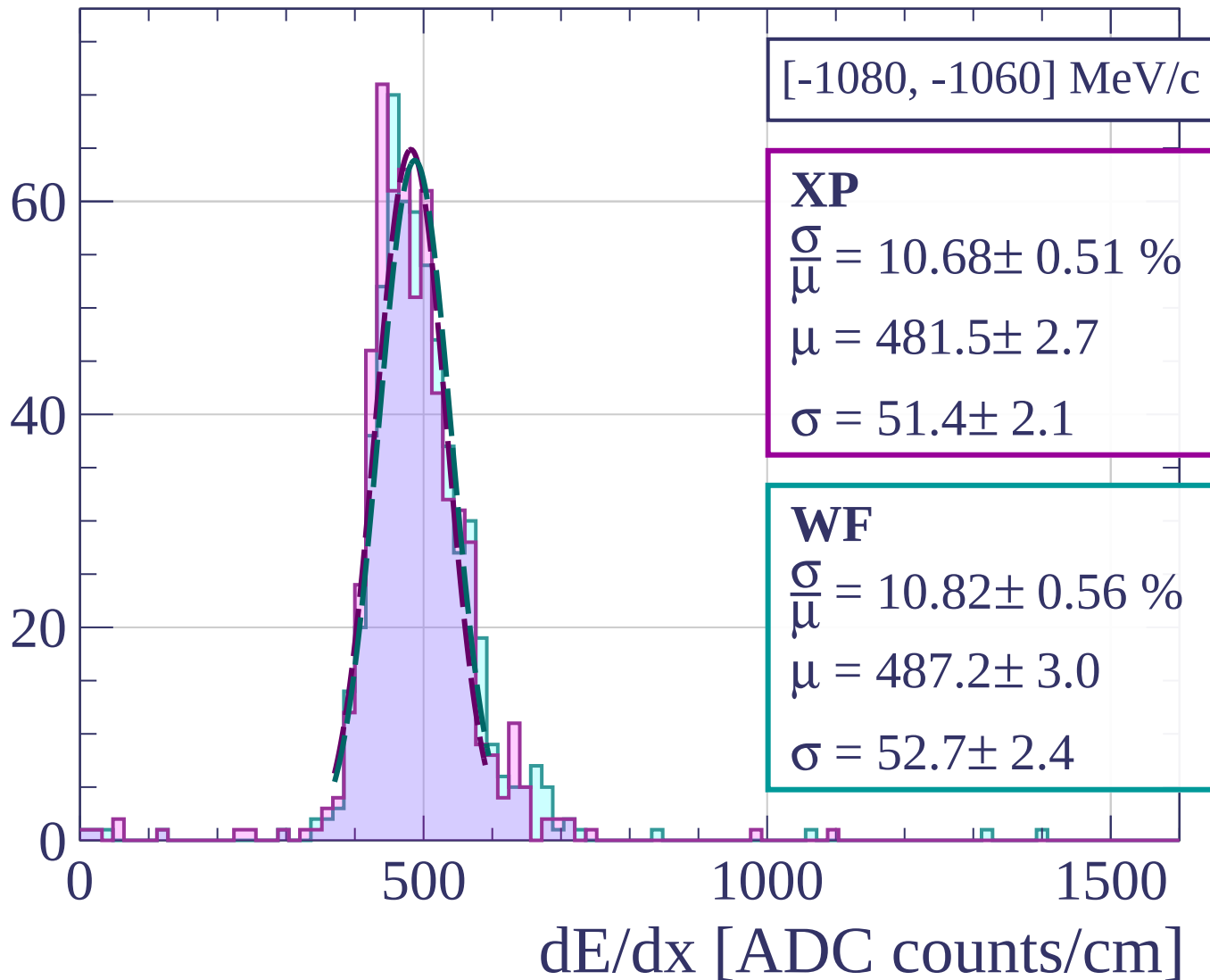


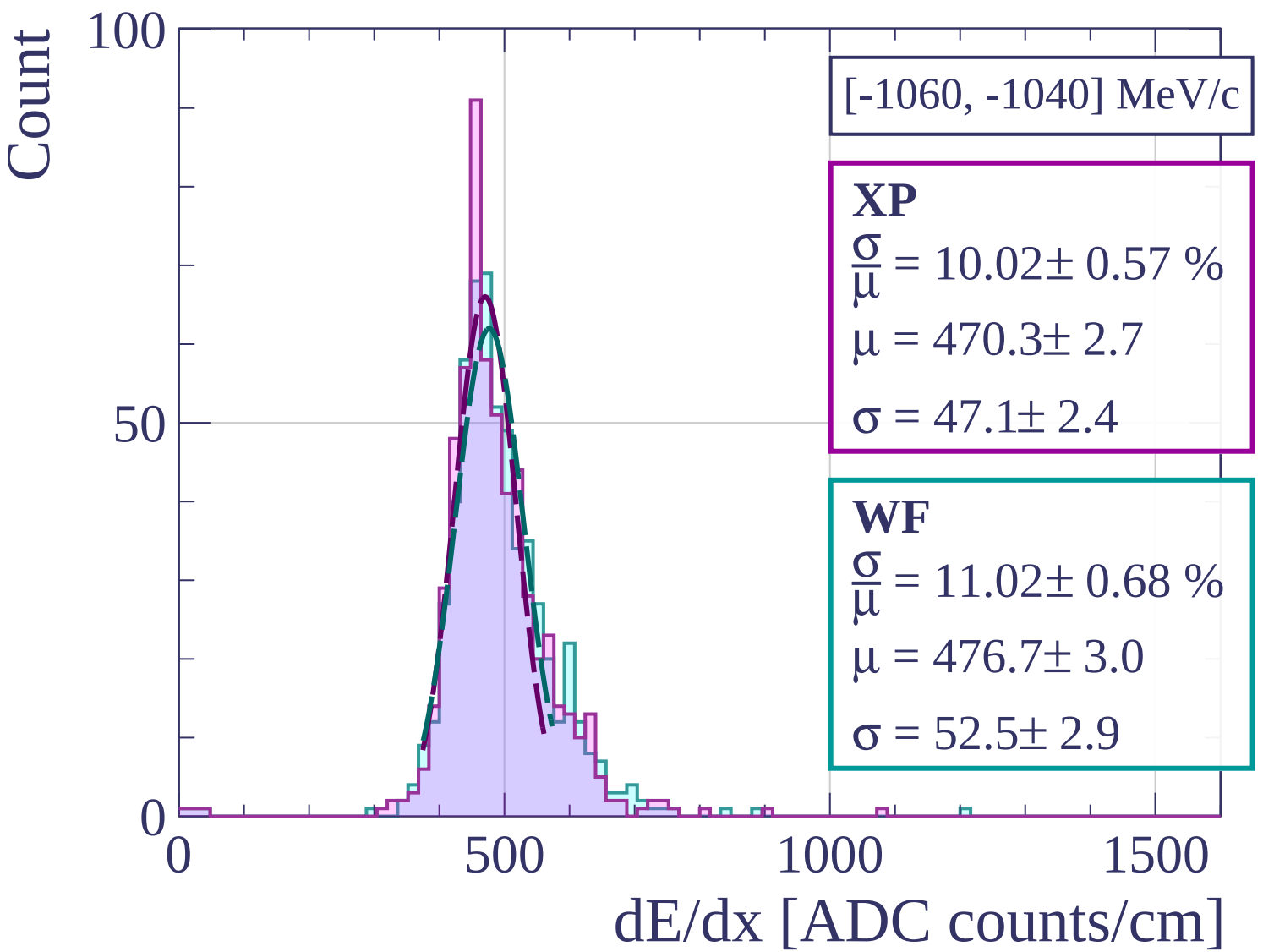
Count



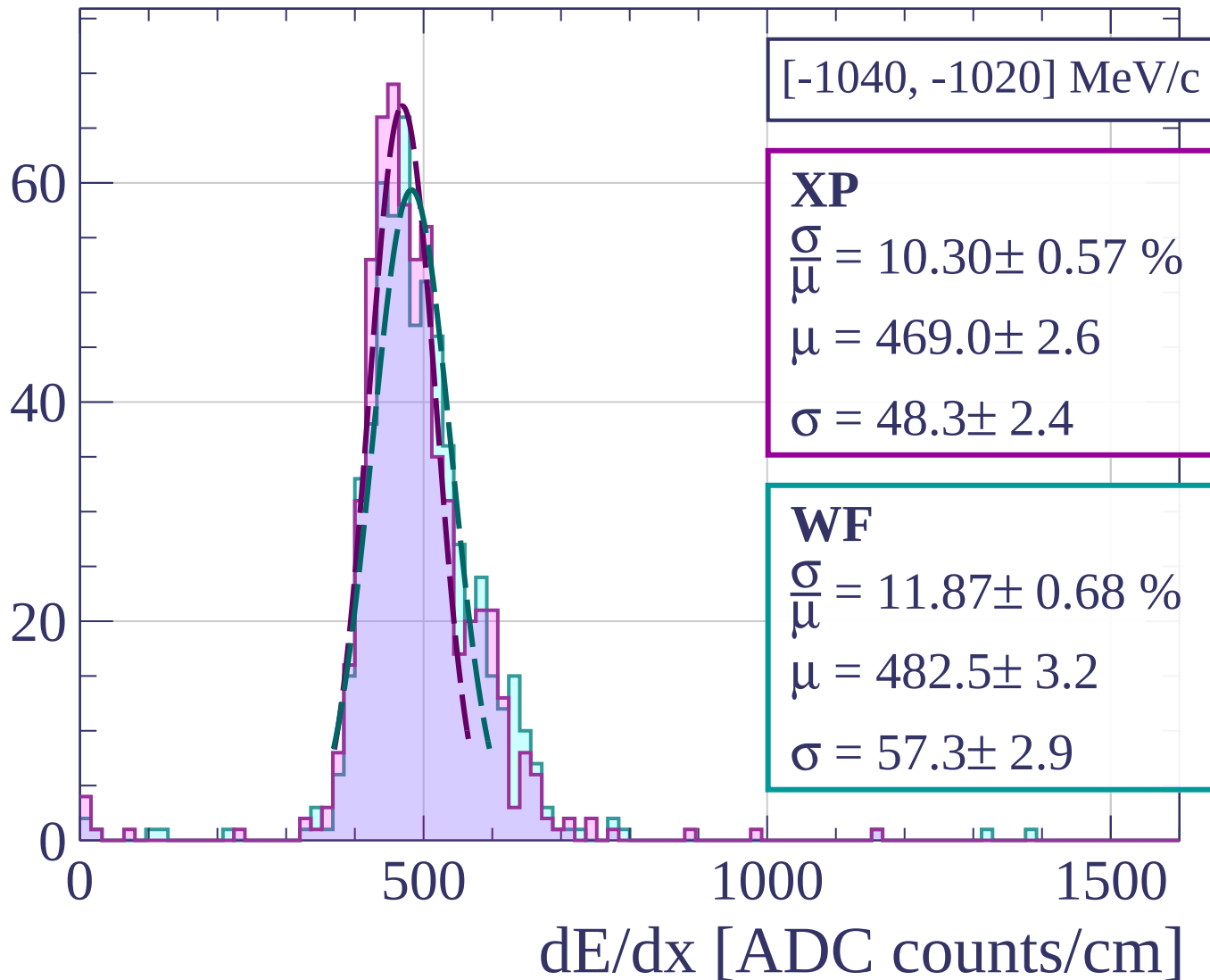


Count

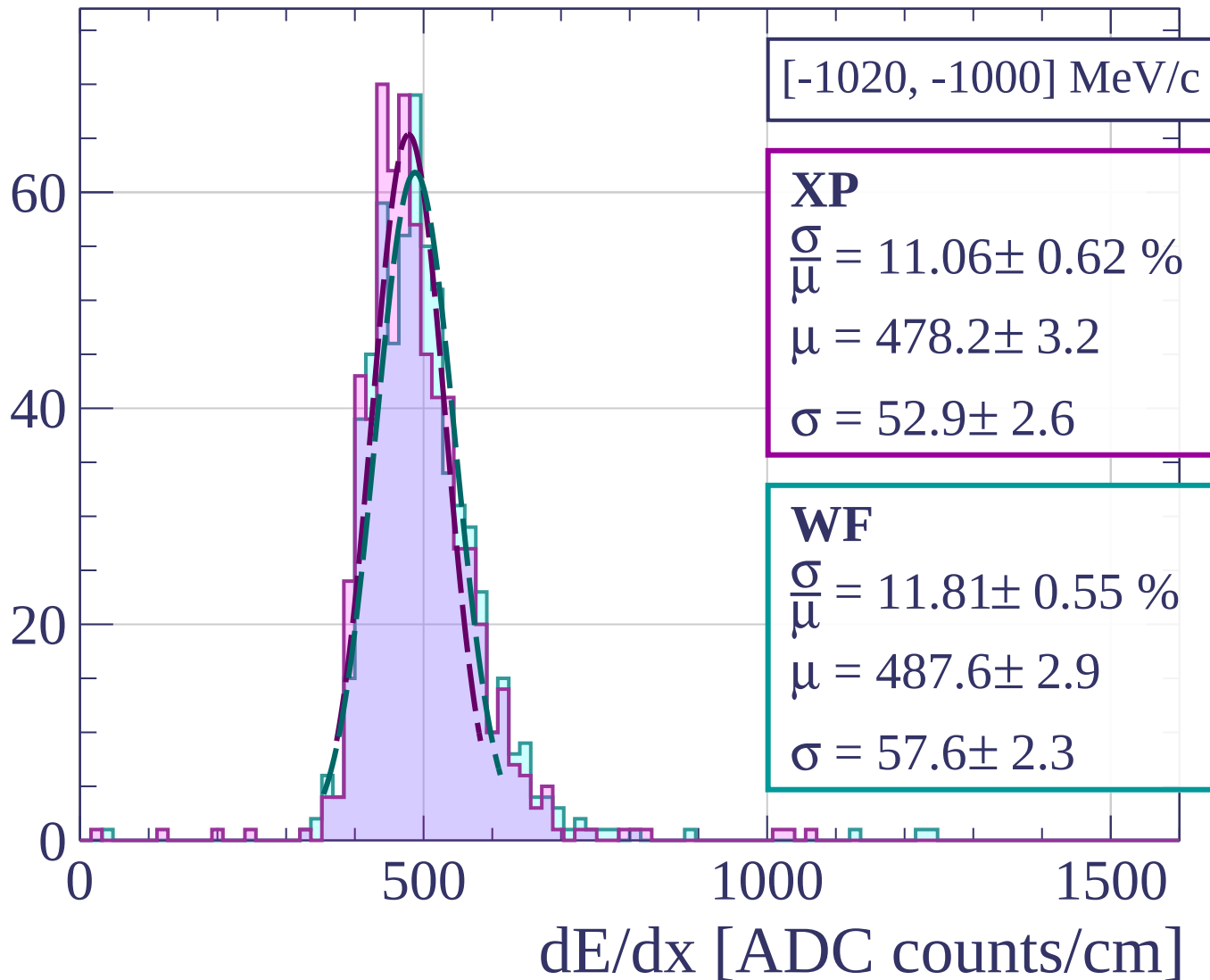




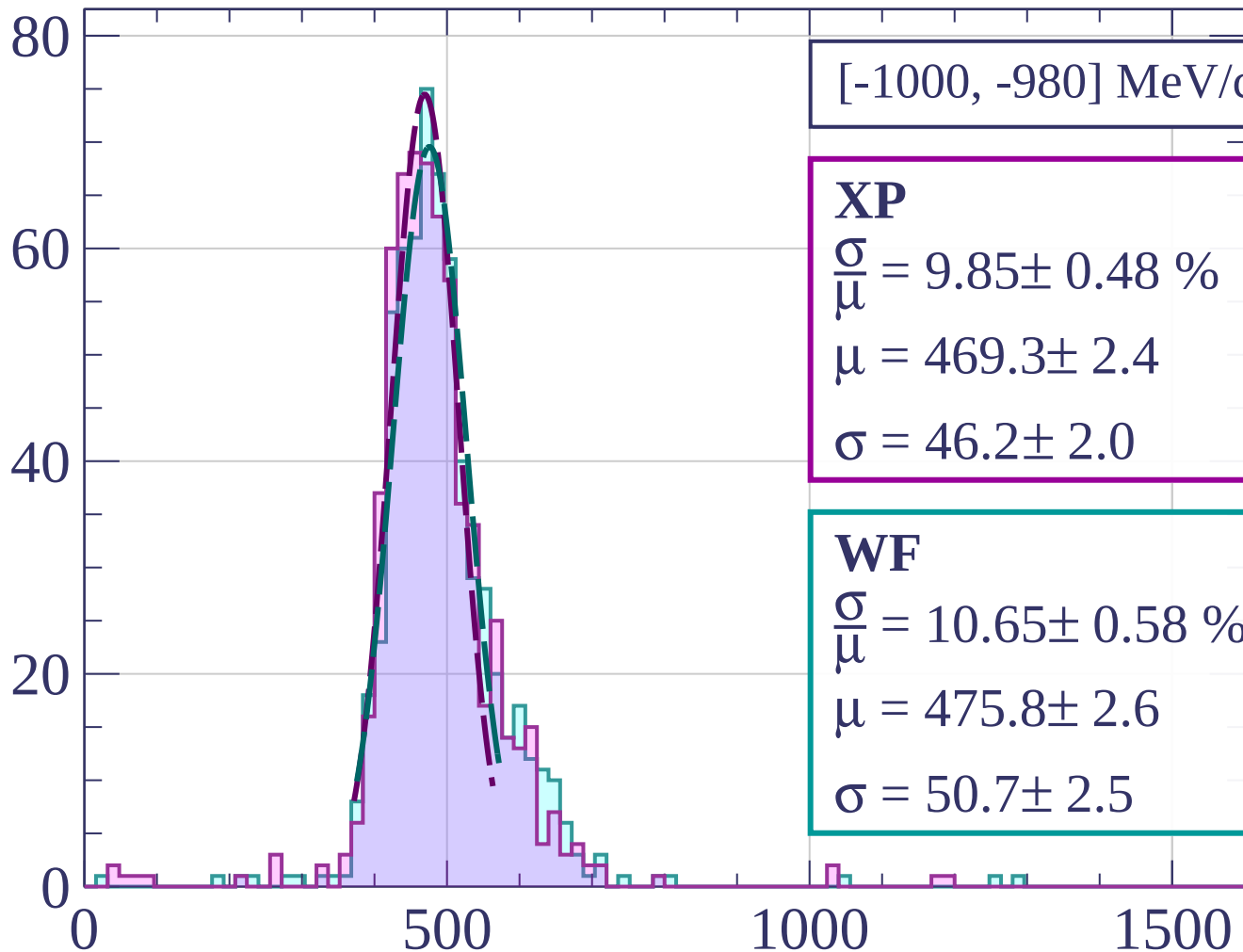
Count



Count



Count



[-1000, -980] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.85 \pm 0.48 \%$$

$$\mu = 469.3 \pm 2.4$$

$$\sigma = 46.2 \pm 2.0$$

WF

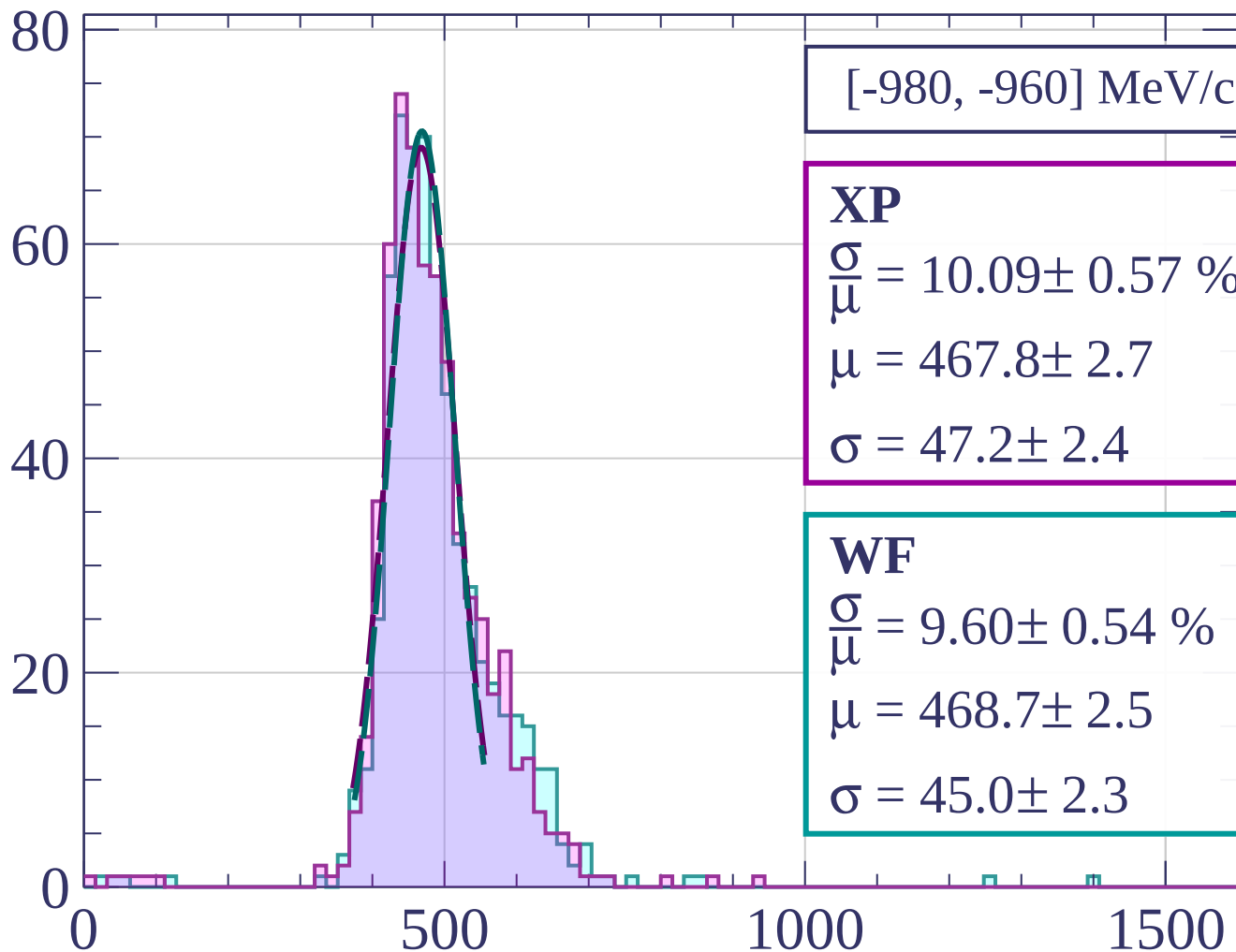
$$\frac{\sigma}{\mu} = 10.65 \pm 0.58 \%$$

$$\mu = 475.8 \pm 2.6$$

$$\sigma = 50.7 \pm 2.5$$

dE/dx [ADC counts/cm]

Count



[-980, -960] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.09 \pm 0.57 \%$$

$$\mu = 467.8 \pm 2.7$$

$$\sigma = 47.2 \pm 2.4$$

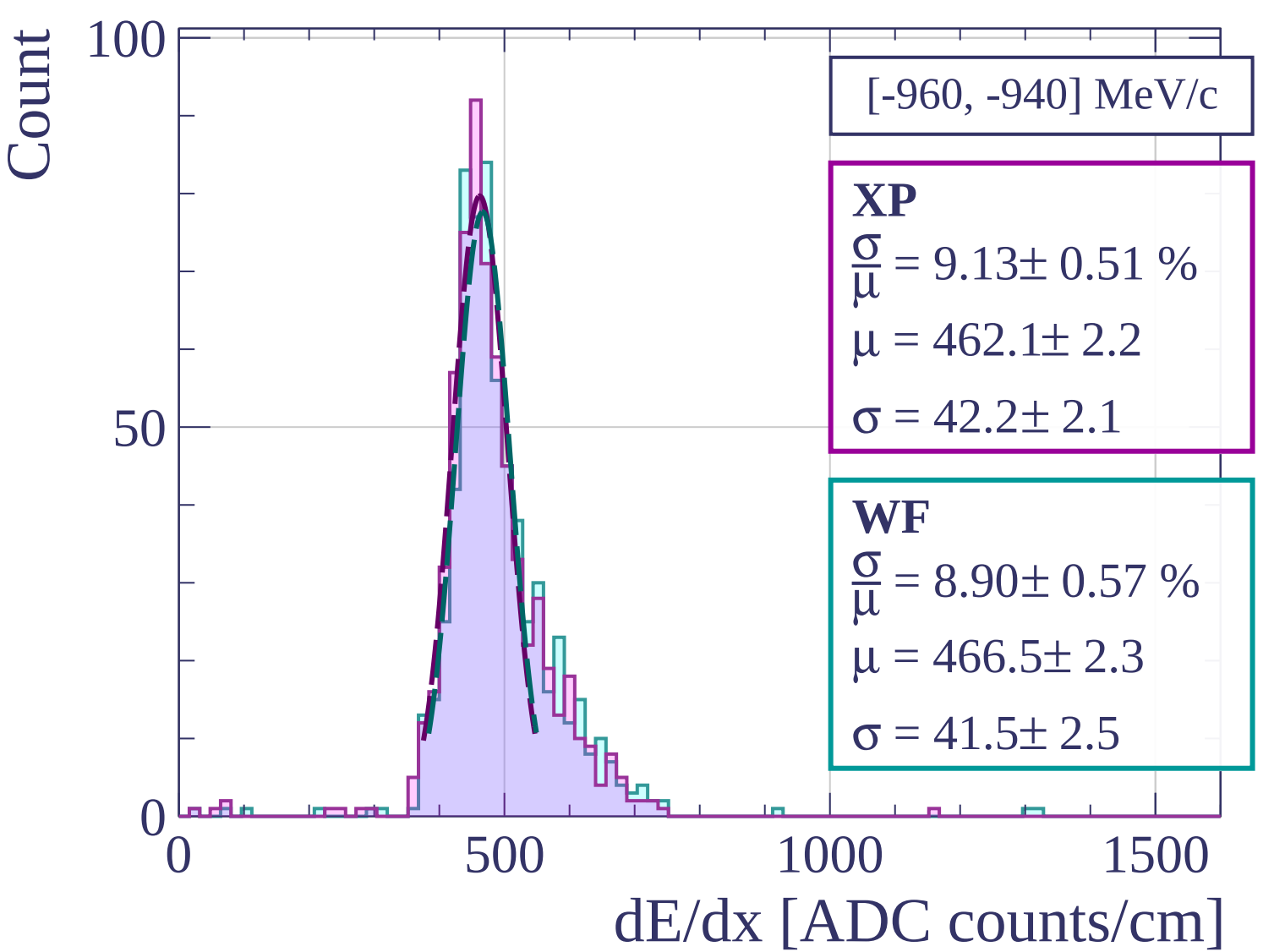
WF

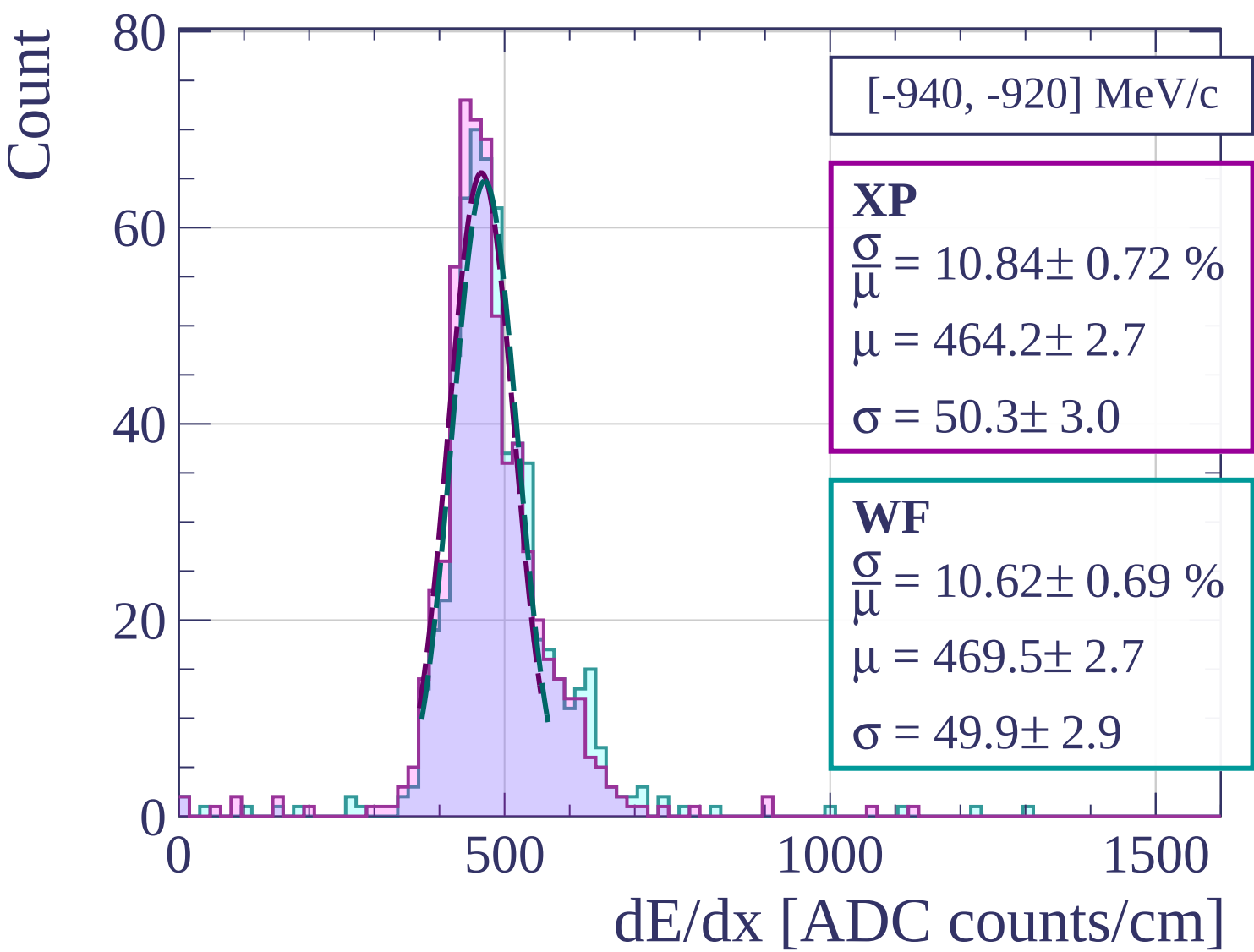
$$\frac{\sigma}{\mu} = 9.60 \pm 0.54 \%$$

$$\mu = 468.7 \pm 2.5$$

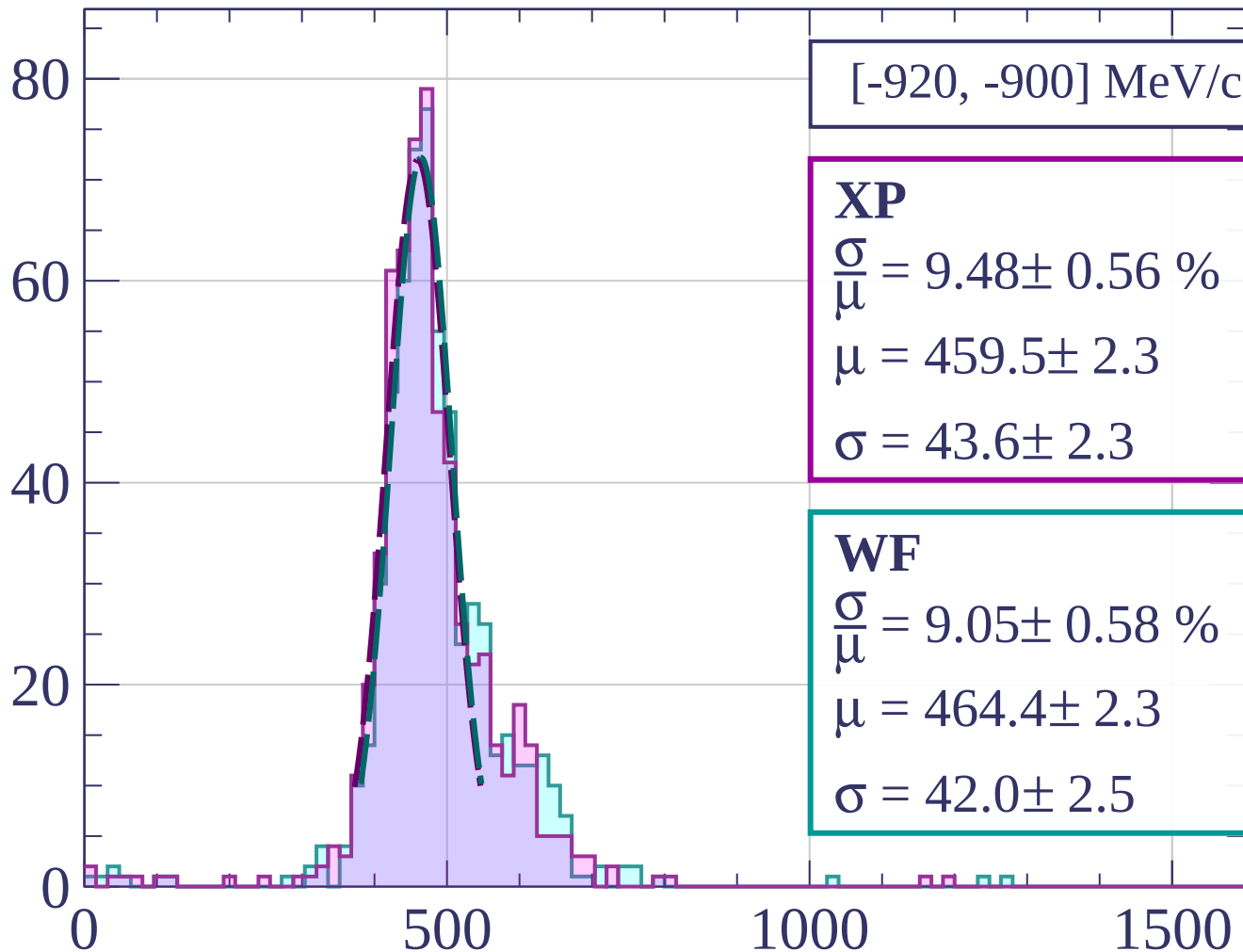
$$\sigma = 45.0 \pm 2.3$$

dE/dx [ADC counts/cm]





Count



[-920, -900] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.48 \pm 0.56 \%$$

$$\mu = 459.5 \pm 2.3$$

$$\sigma = 43.6 \pm 2.3$$

WF

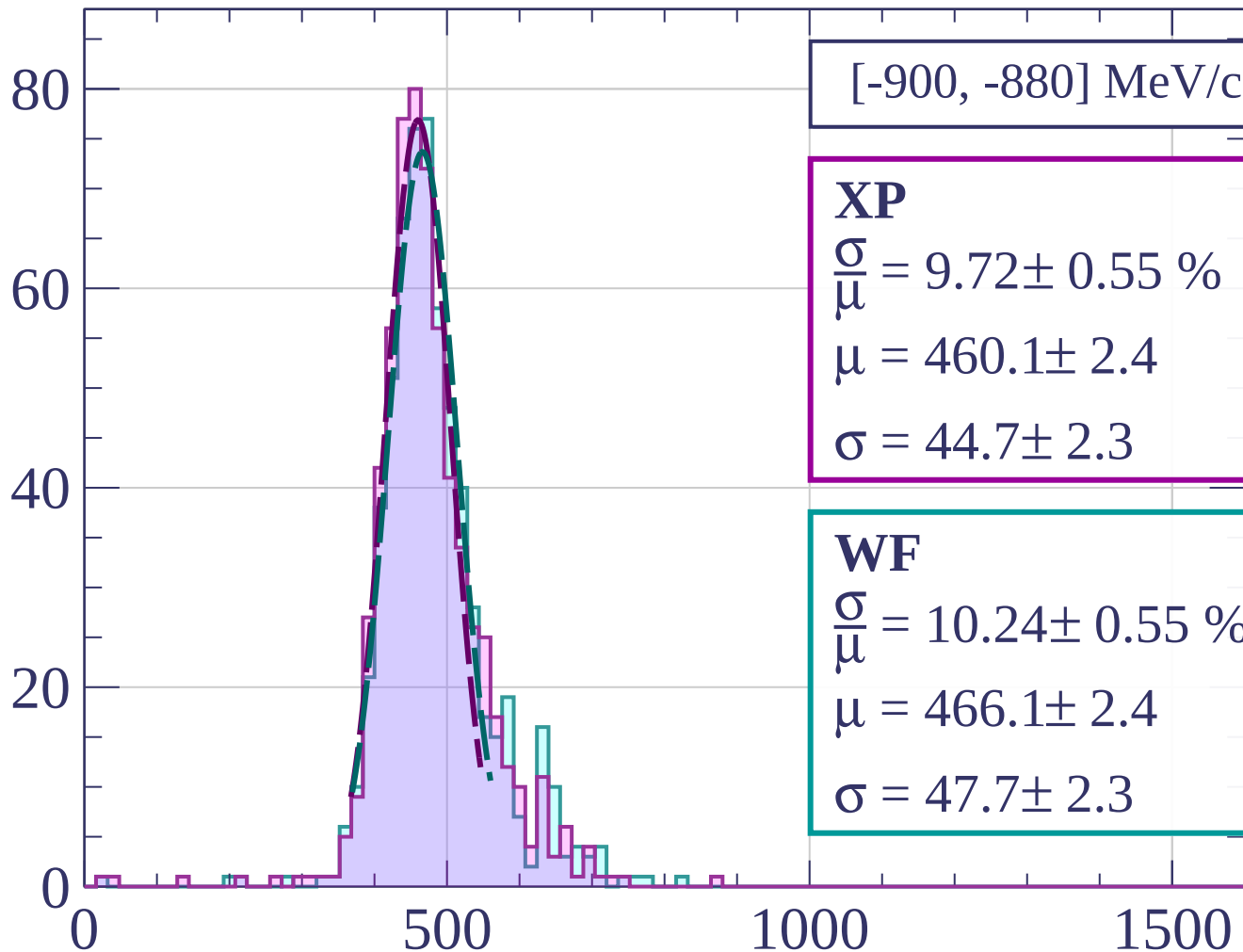
$$\frac{\sigma}{\mu} = 9.05 \pm 0.58 \%$$

$$\mu = 464.4 \pm 2.3$$

$$\sigma = 42.0 \pm 2.5$$

dE/dx [ADC counts/cm]

Count



[-900, -880] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.72 \pm 0.55 \%$$

$$\mu = 460.1 \pm 2.4$$

$$\sigma = 44.7 \pm 2.3$$

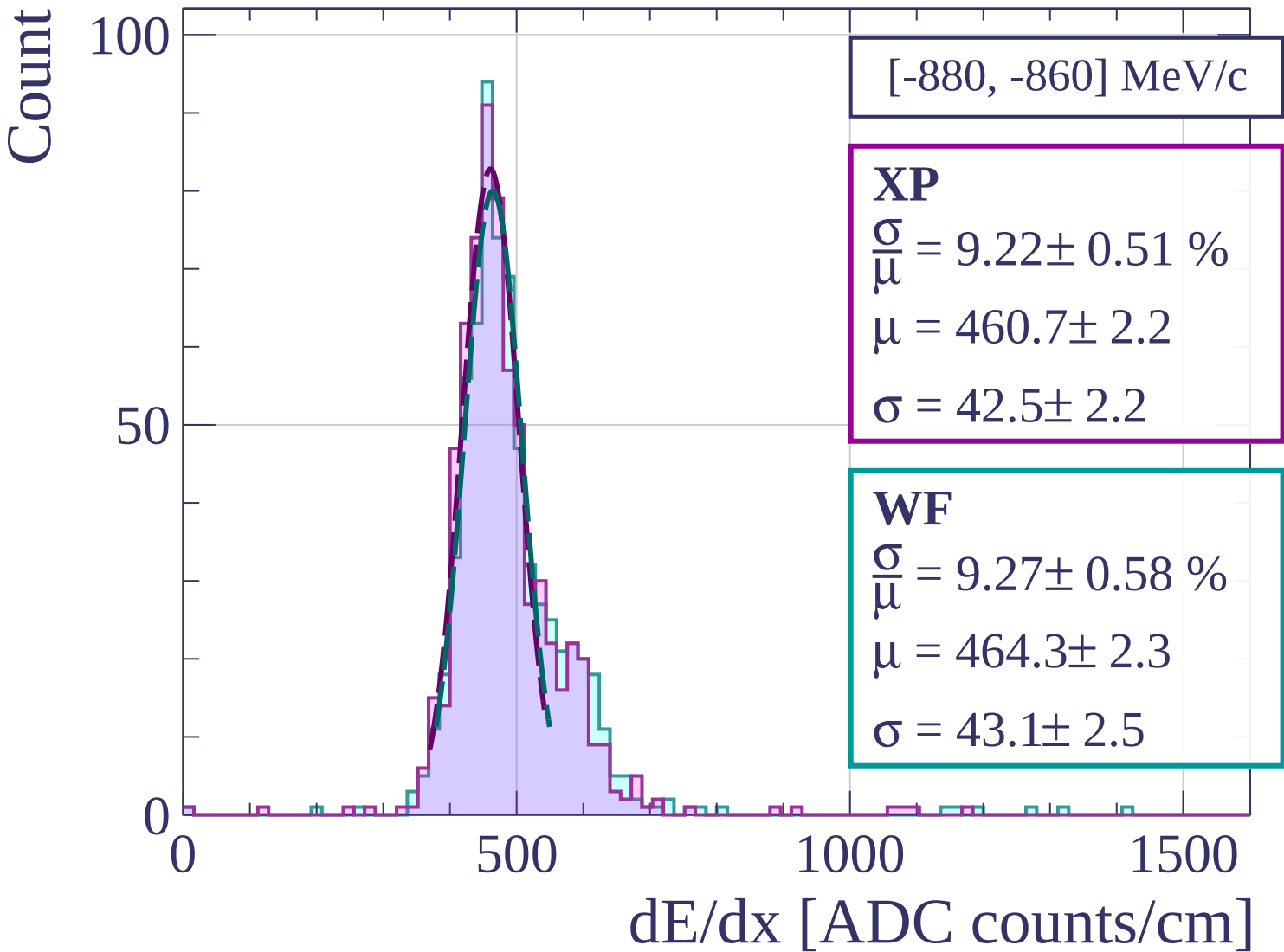
WF

$$\frac{\sigma}{\mu} = 10.24 \pm 0.55 \%$$

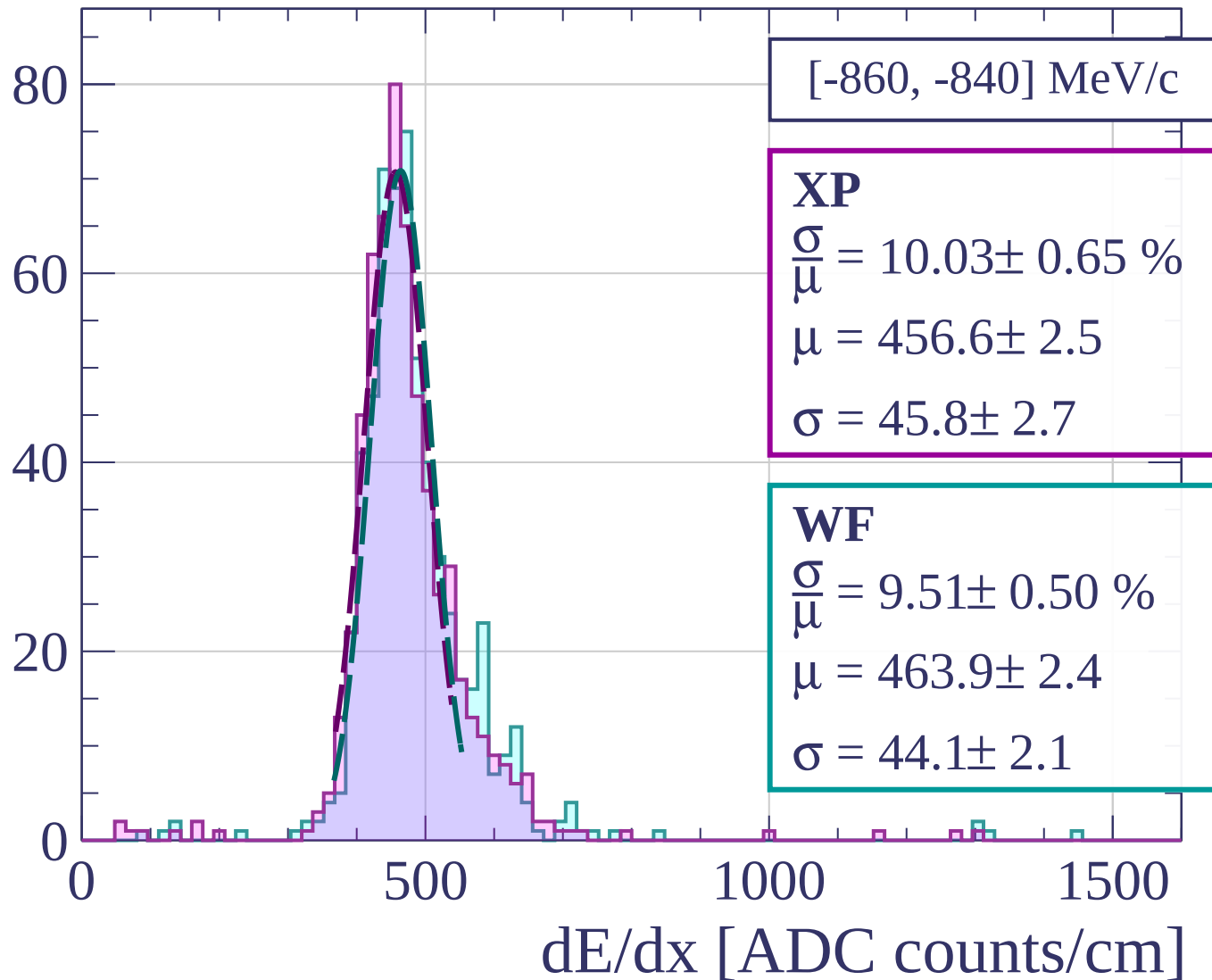
$$\mu = 466.1 \pm 2.4$$

$$\sigma = 47.7 \pm 2.3$$

dE/dx [ADC counts/cm]



Count



Count

[-840, -820] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.88 \pm 0.49 \%$$

$$\mu = 452.3 \pm 2.2$$

$$\sigma = 40.2 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 9.19 \pm 0.50 \%$$

$$\mu = 456.6 \pm 2.2$$

$$\sigma = 42.0 \pm 2.1$$

80

60

40

20

0

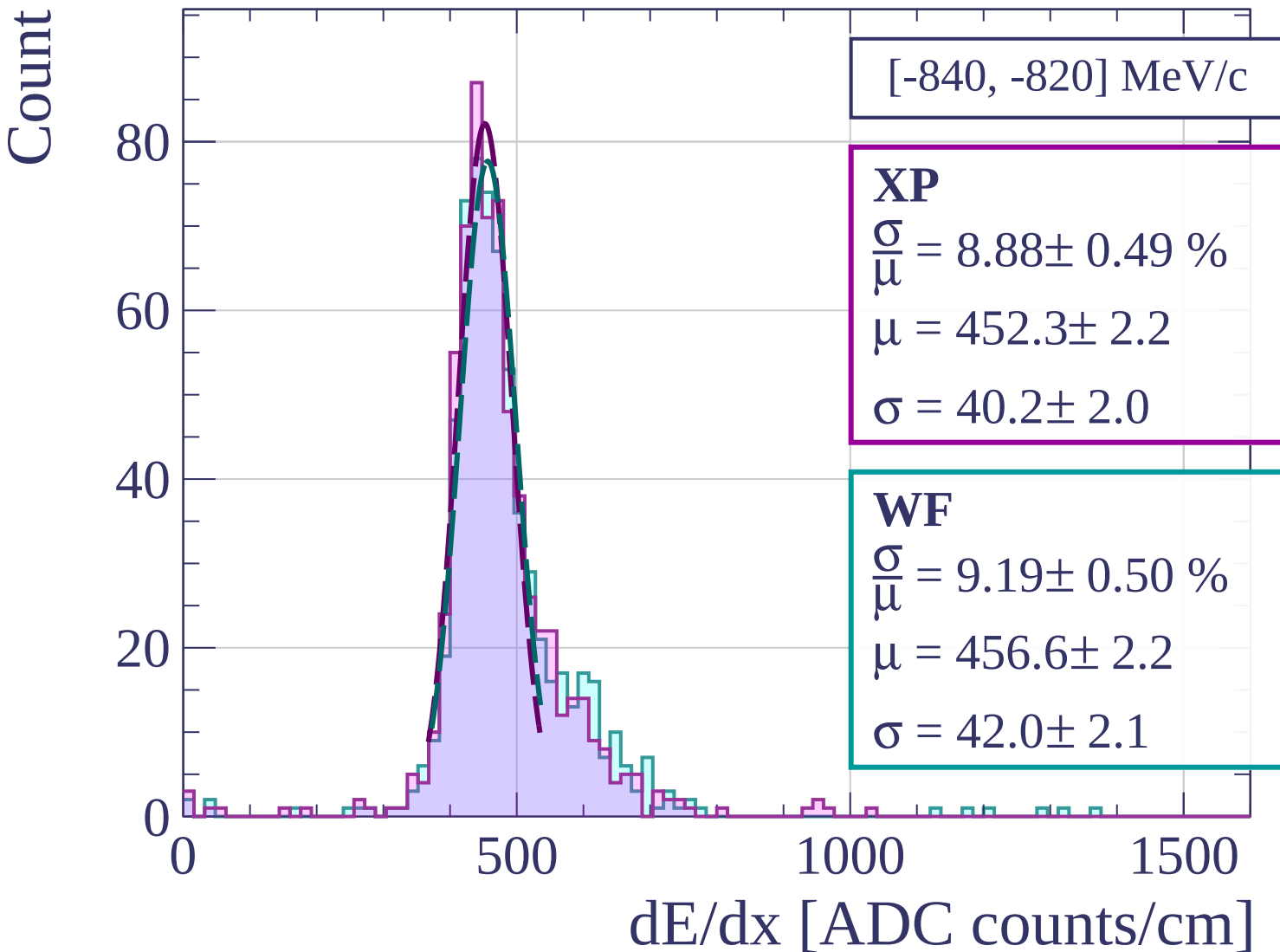
0

500

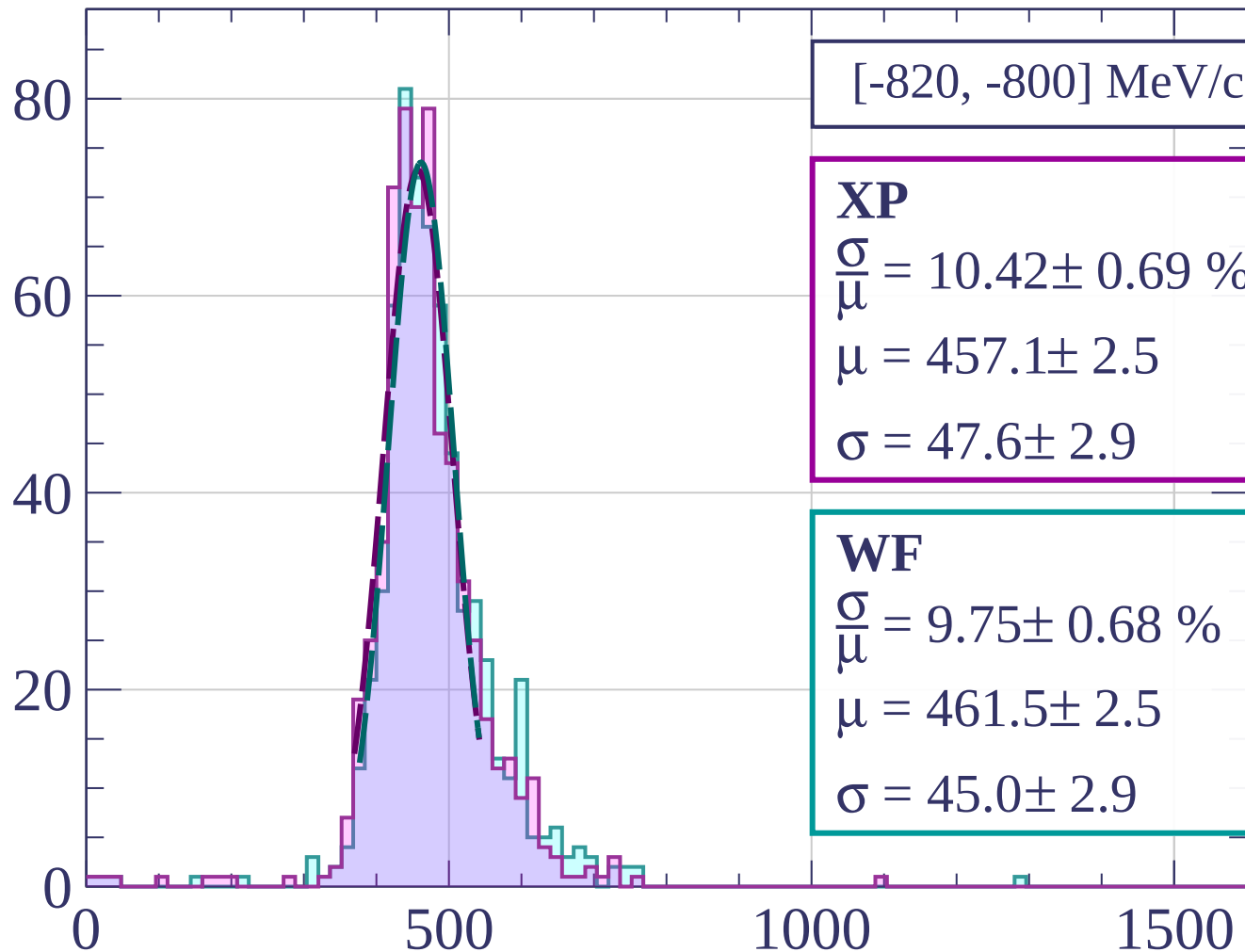
1000

1500

dE/dx [ADC counts/cm]



Count



[-820, -800] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.42 \pm 0.69 \%$$

$$\mu = 457.1 \pm 2.5$$

$$\sigma = 47.6 \pm 2.9$$

WF

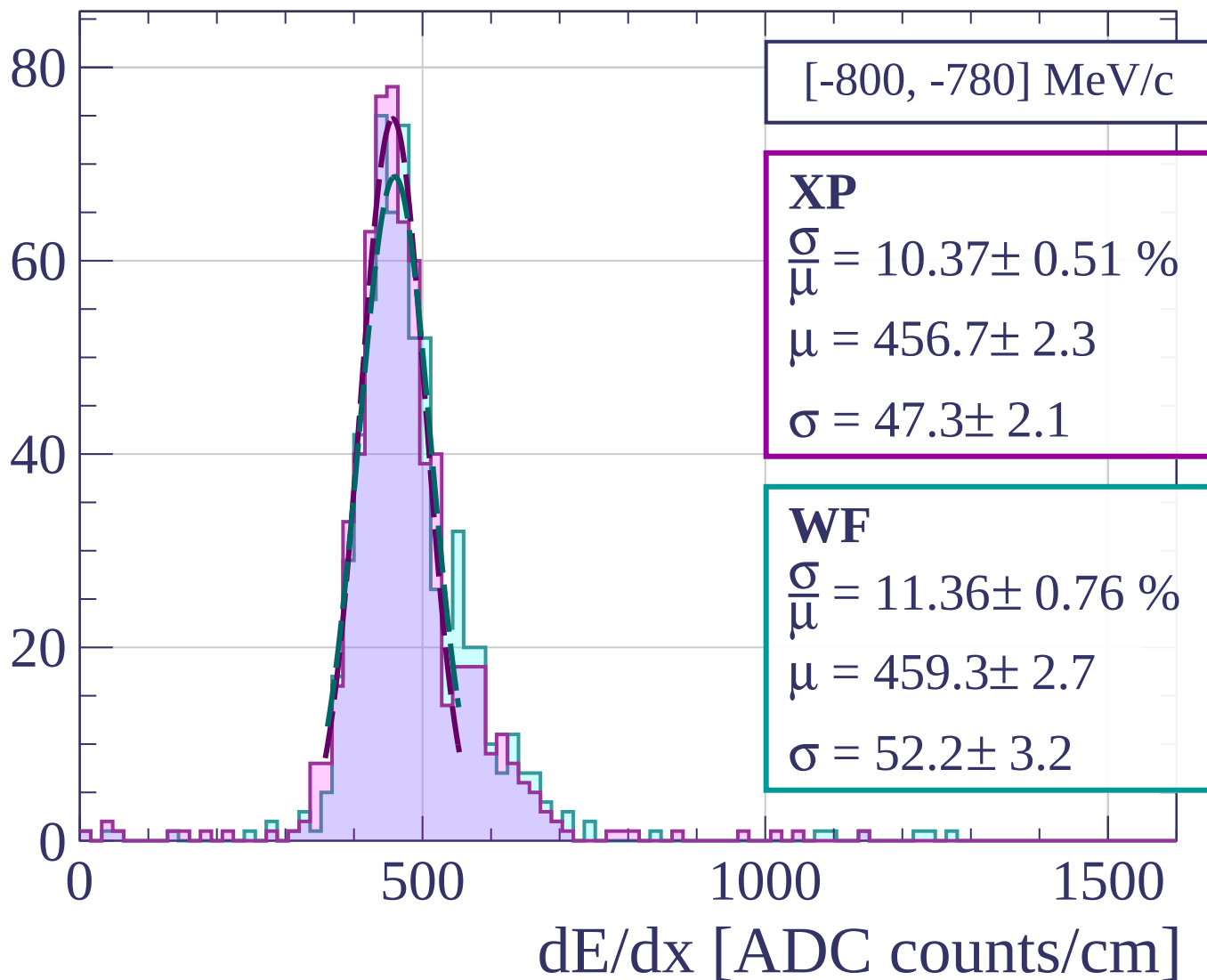
$$\frac{\sigma}{\mu} = 9.75 \pm 0.68 \%$$

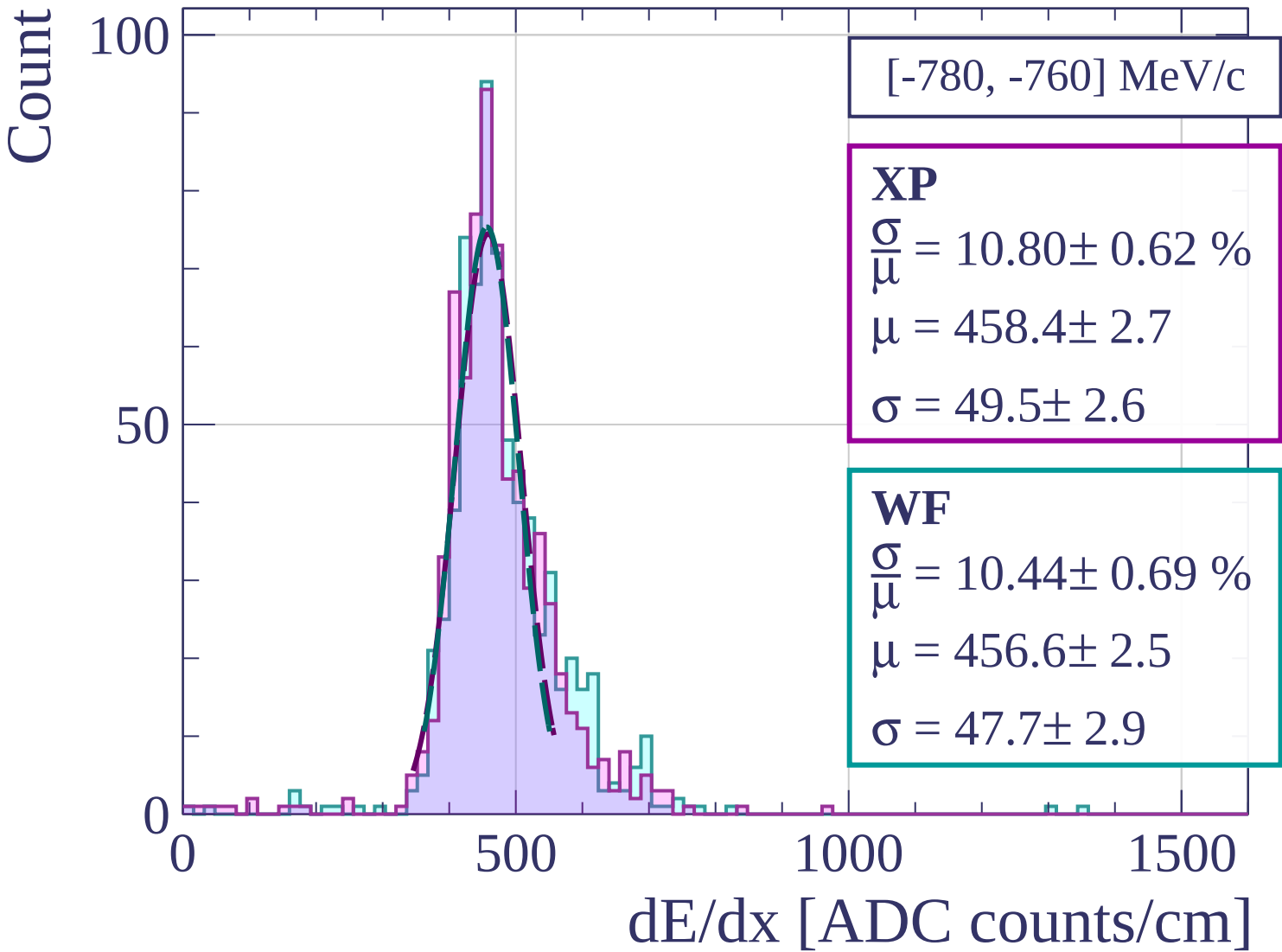
$$\mu = 461.5 \pm 2.5$$

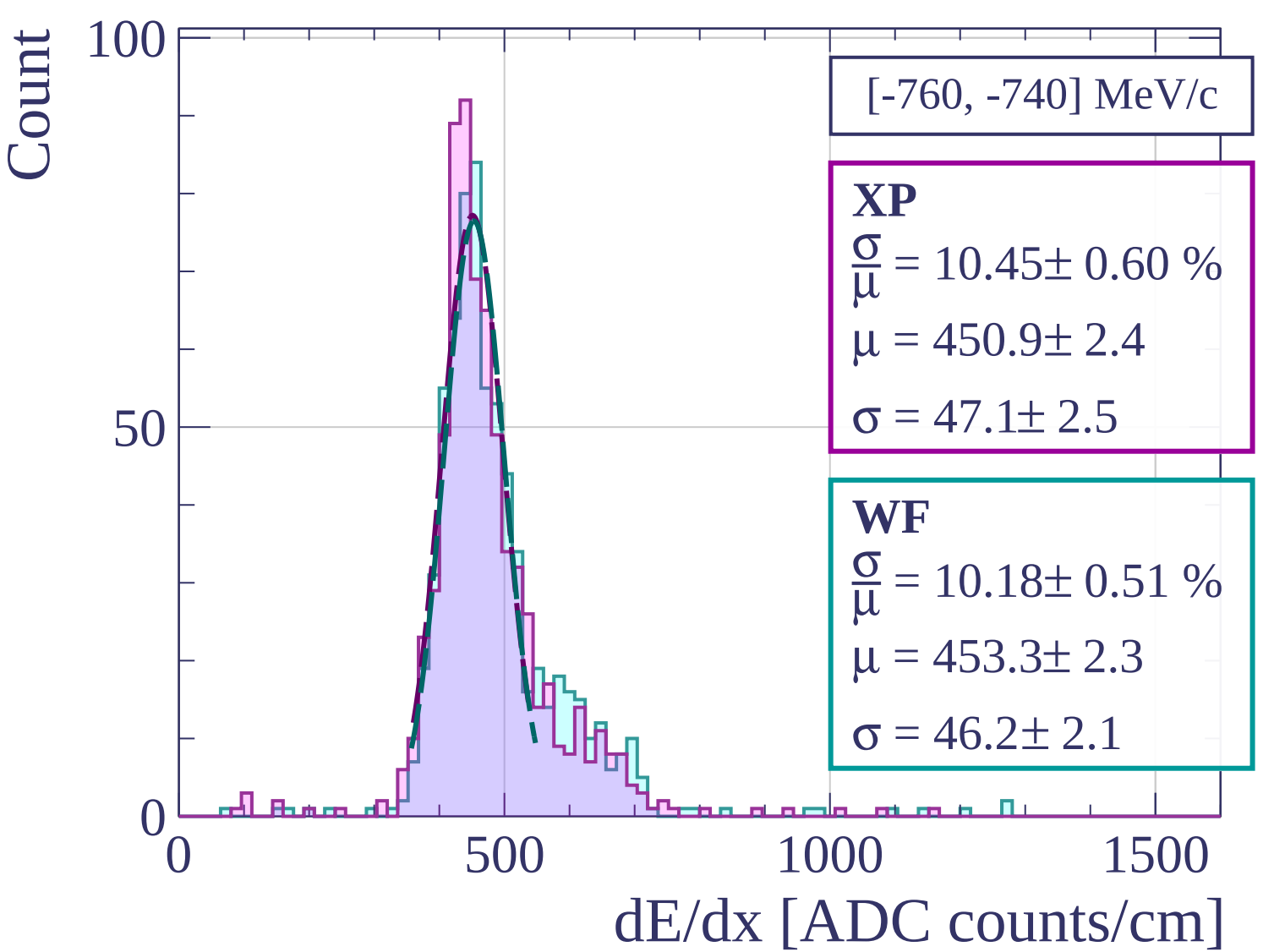
$$\sigma = 45.0 \pm 2.9$$

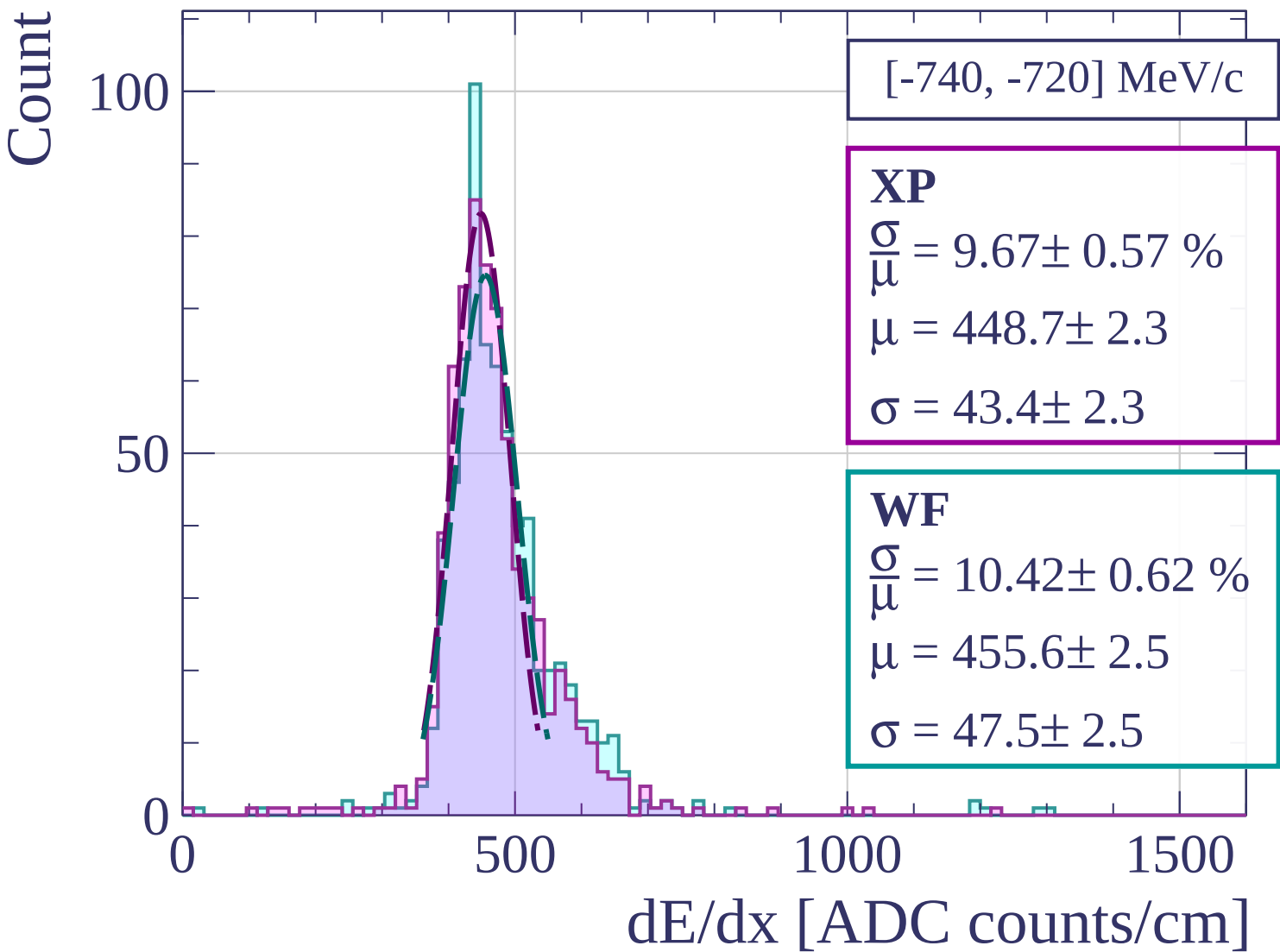
dE/dx [ADC counts/cm]

Count

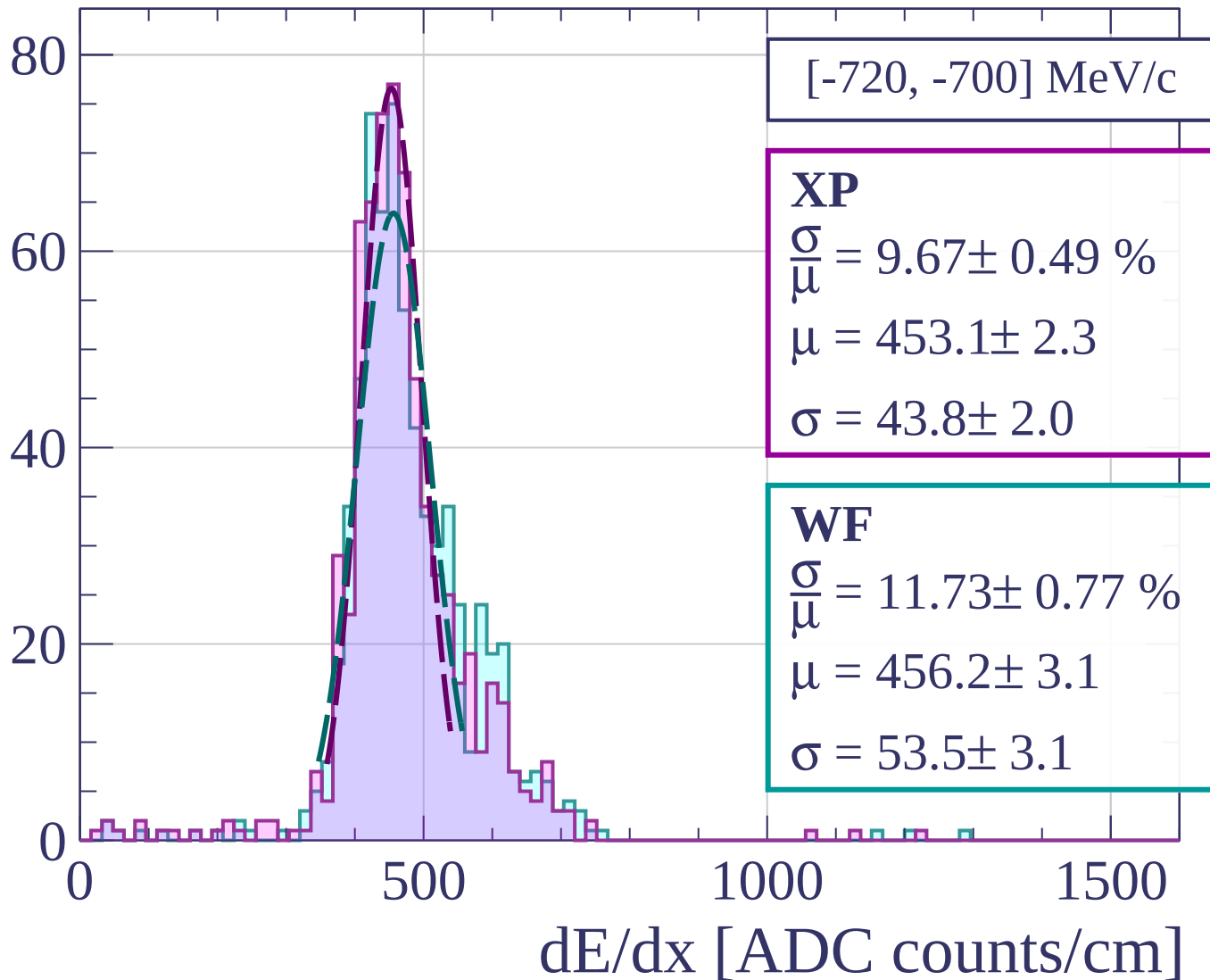




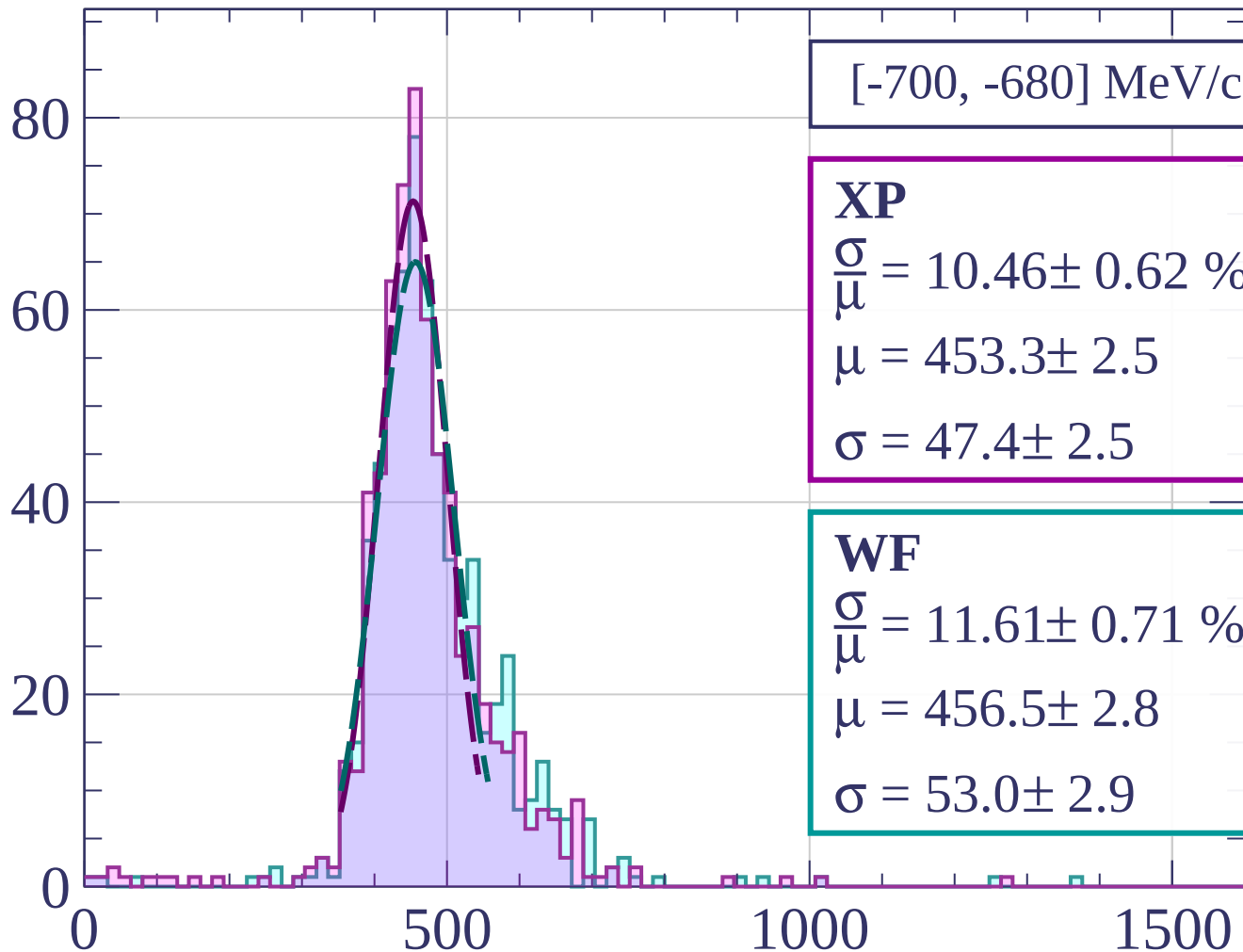




Count



Count



[-700, -680] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.46 \pm 0.62 \%$$

$$\mu = 453.3 \pm 2.5$$

$$\sigma = 47.4 \pm 2.5$$

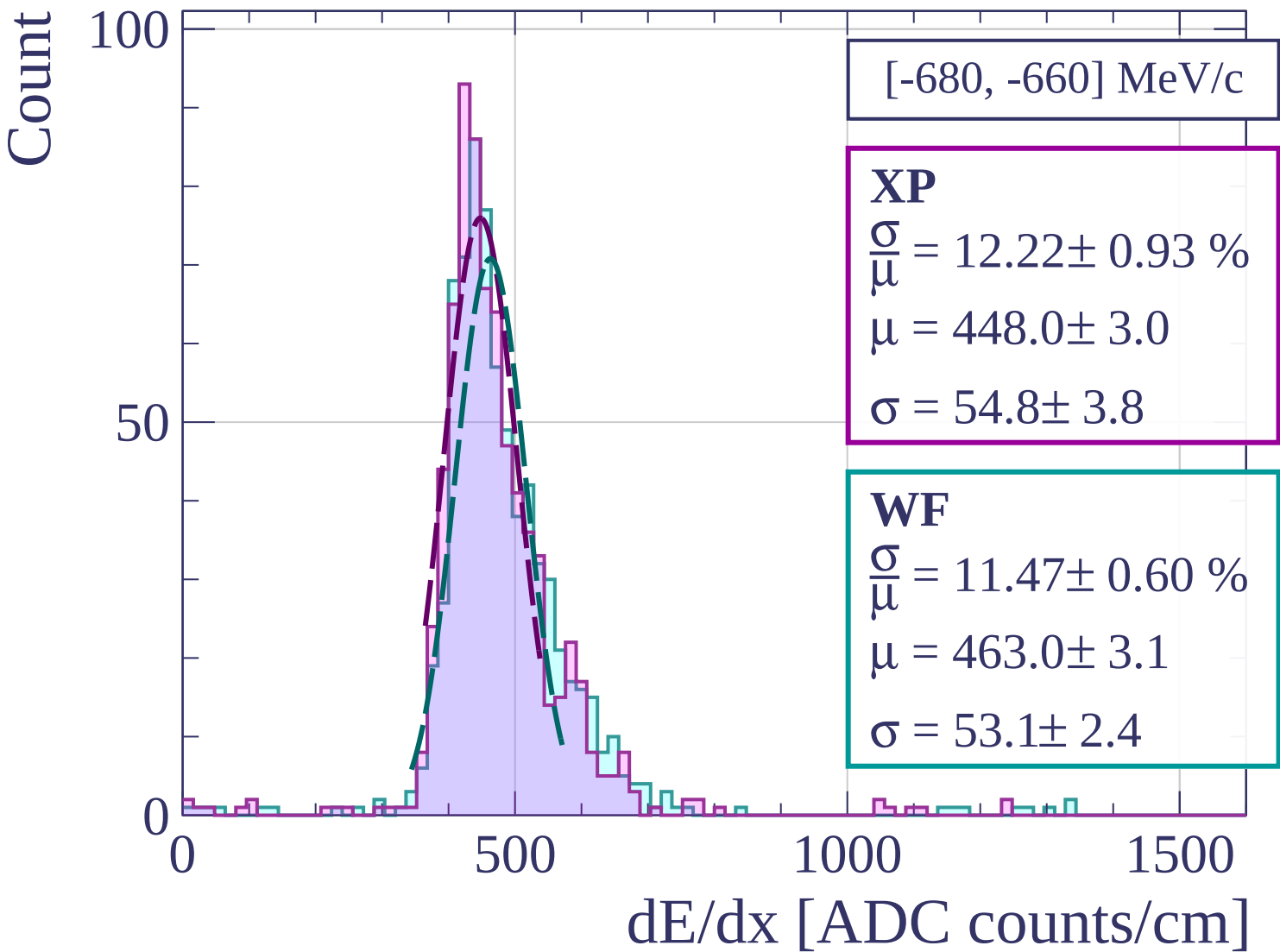
WF

$$\frac{\sigma}{\mu} = 11.61 \pm 0.71 \%$$

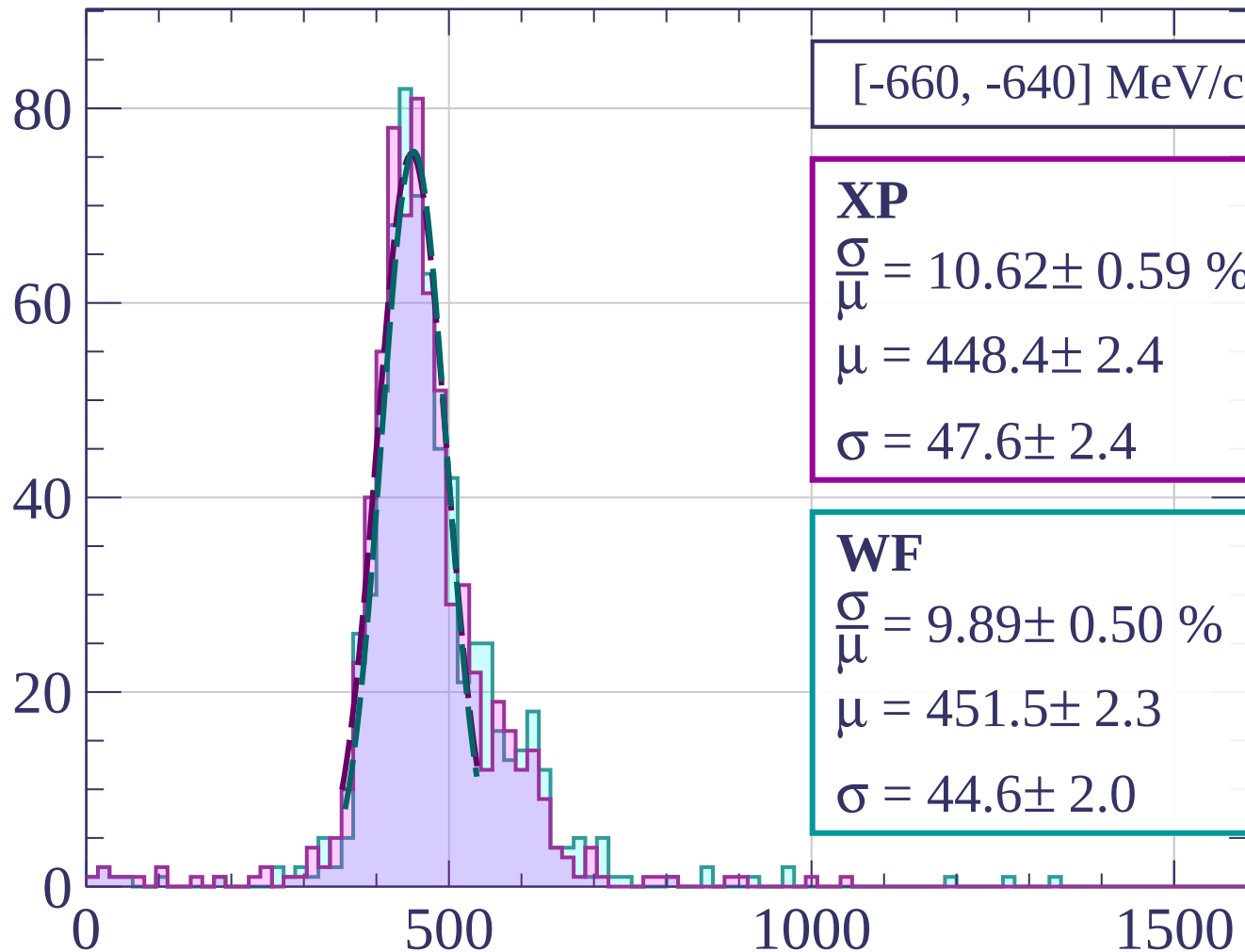
$$\mu = 456.5 \pm 2.8$$

$$\sigma = 53.0 \pm 2.9$$

dE/dx [ADC counts/cm]



Count



[-660, -640] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.62 \pm 0.59 \%$$

$$\mu = 448.4 \pm 2.4$$

$$\sigma = 47.6 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 9.89 \pm 0.50 \%$$

$$\mu = 451.5 \pm 2.3$$

$$\sigma = 44.6 \pm 2.0$$

dE/dx [ADC counts/cm]

Count

[-640, -620] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.53 \pm 0.57 \%$$

$$\mu = 441.8 \pm 2.1$$

$$\sigma = 42.1 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 9.79 \pm 0.60 \%$$

$$\mu = 445.0 \pm 2.3$$

$$\sigma = 43.6 \pm 2.5$$

0

500

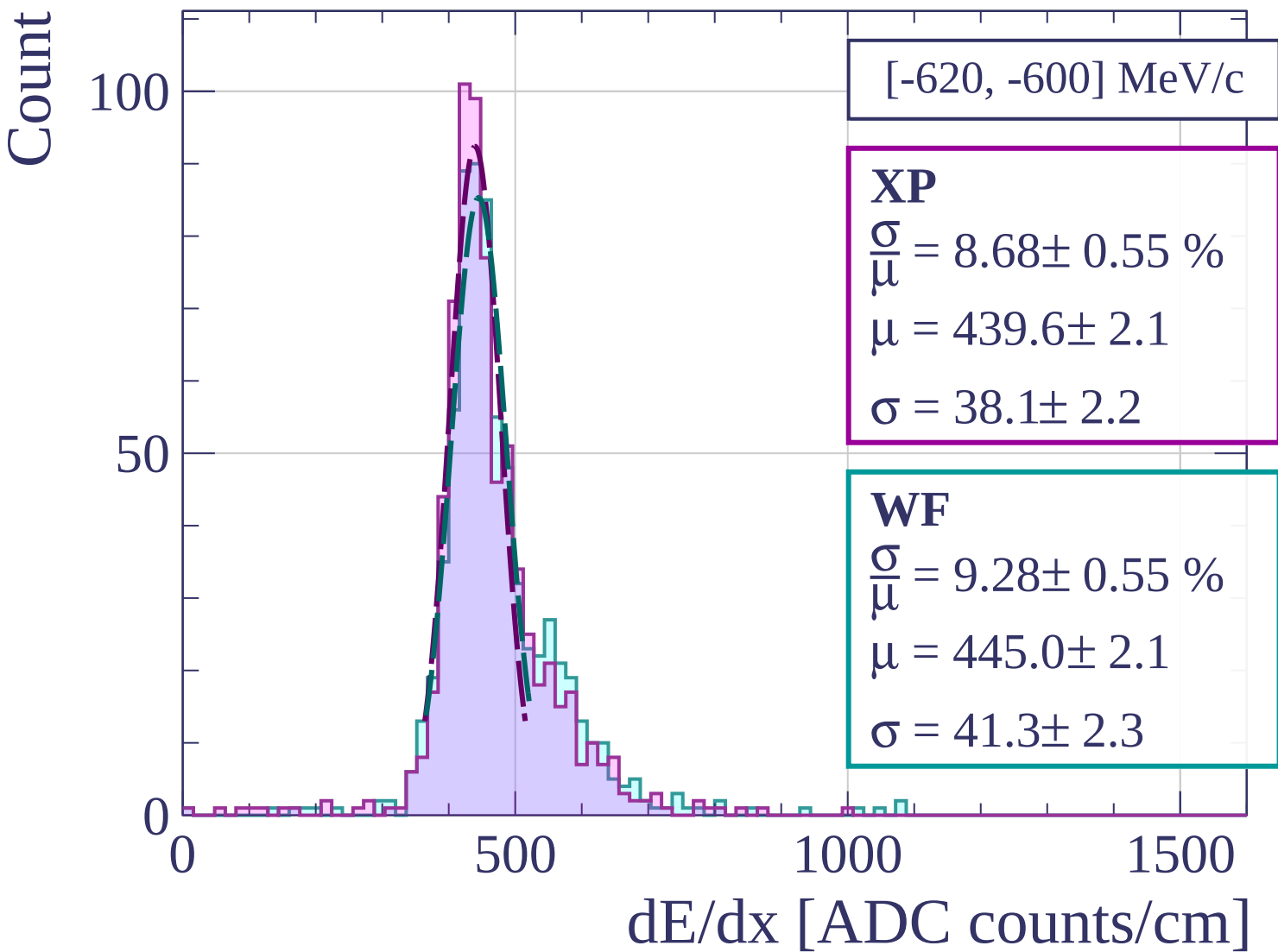
1000

1500

dE/dx [ADC counts/cm]

100

50



Count

[-600, -580] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.83 \pm 0.60 \%$$

$$\mu = 445.8 \pm 2.4$$

$$\sigma = 48.3 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 10.63 \pm 0.57 \%$$

$$\mu = 451.6 \pm 2.4$$

$$\sigma = 48.0 \pm 2.3$$

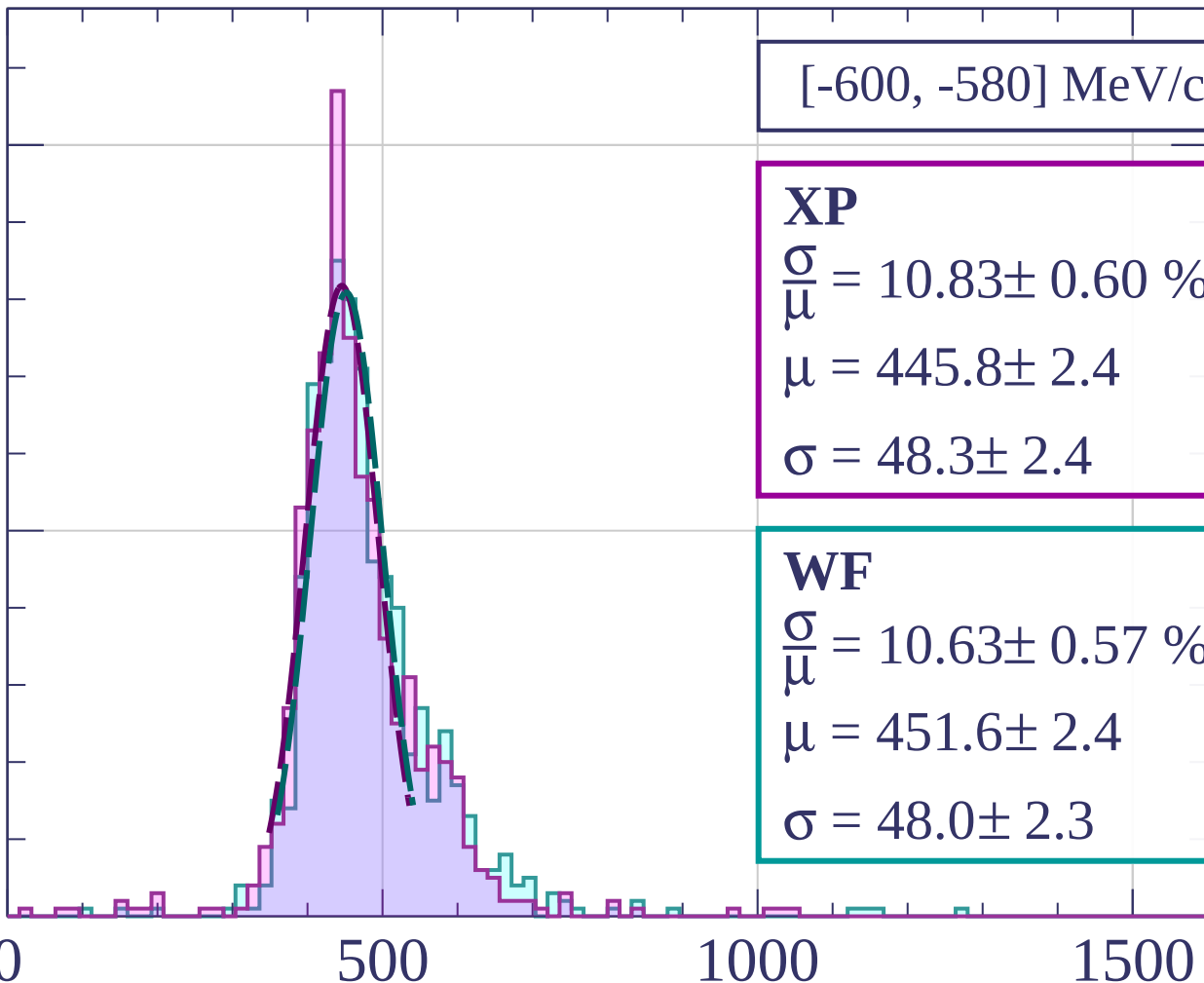
0

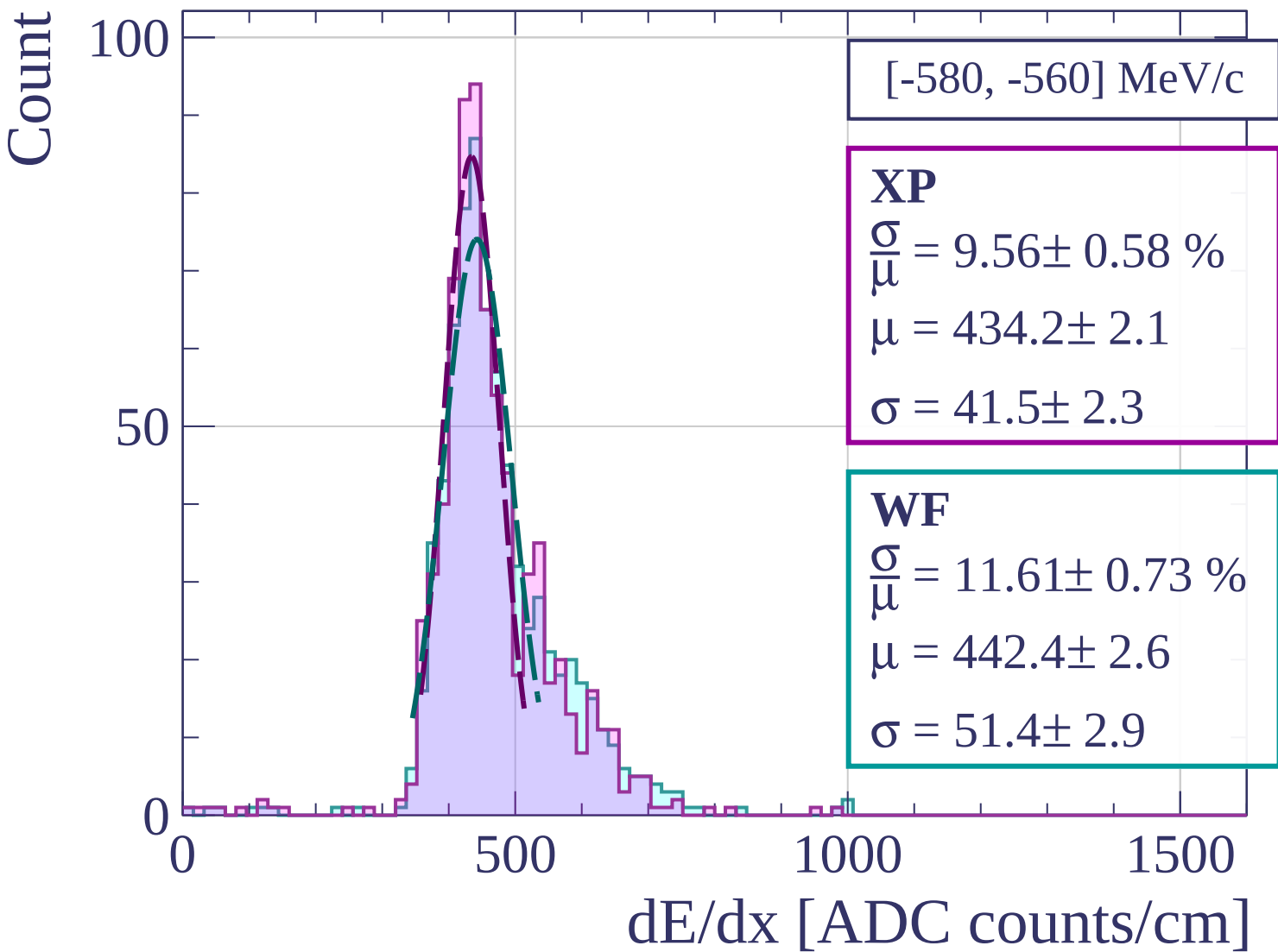
500

1000

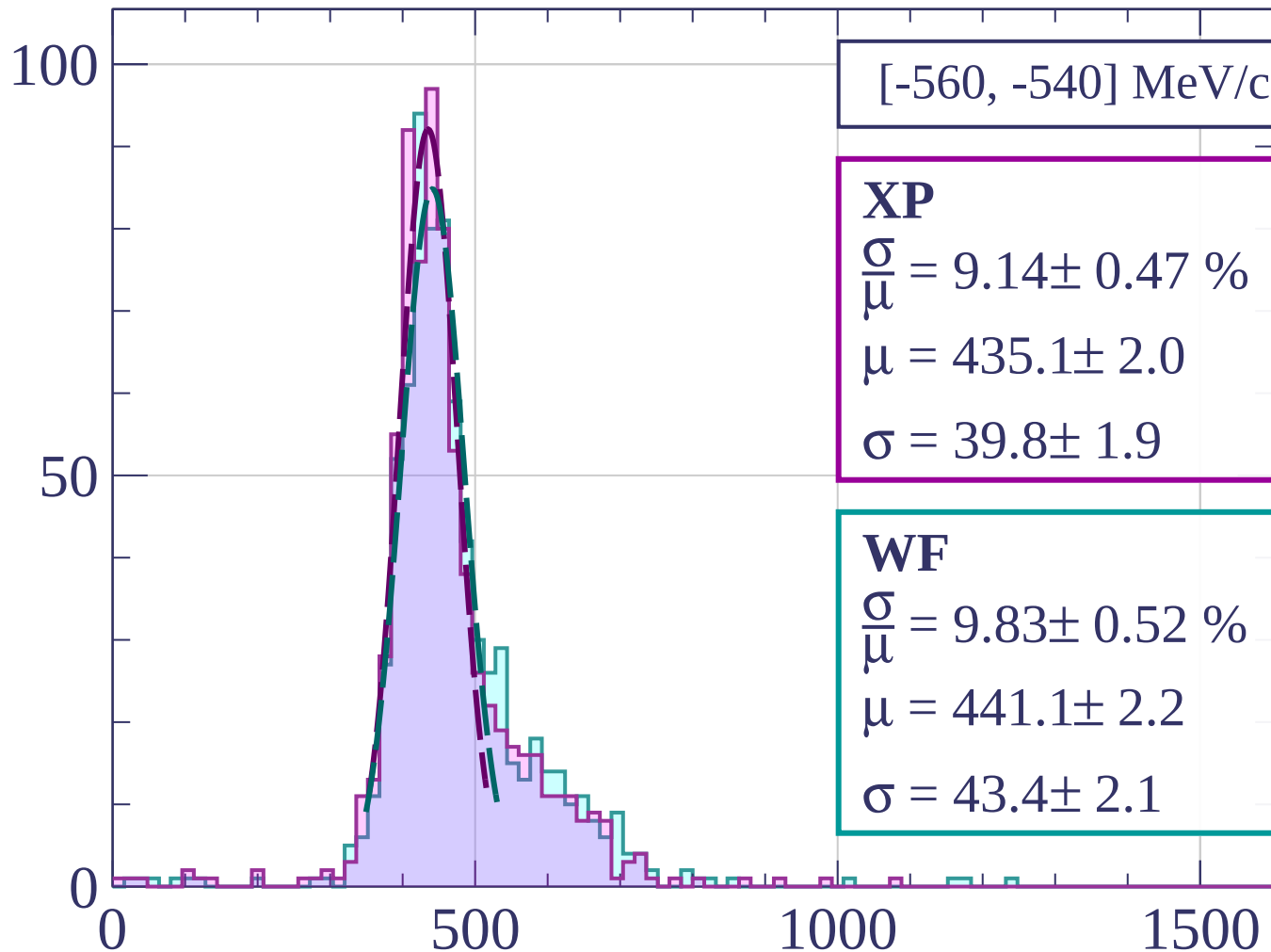
1500

dE/dx [ADC counts/cm]





Count



dE/dx [ADC counts/cm]

Count

[-540, -520] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.62 \pm 0.53 \%$$

$$\mu = 434.3 \pm 2.1$$

$$\sigma = 41.8 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 10.25 \pm 0.53 \%$$

$$\mu = 438.2 \pm 2.2$$

$$\sigma = 44.9 \pm 2.1$$

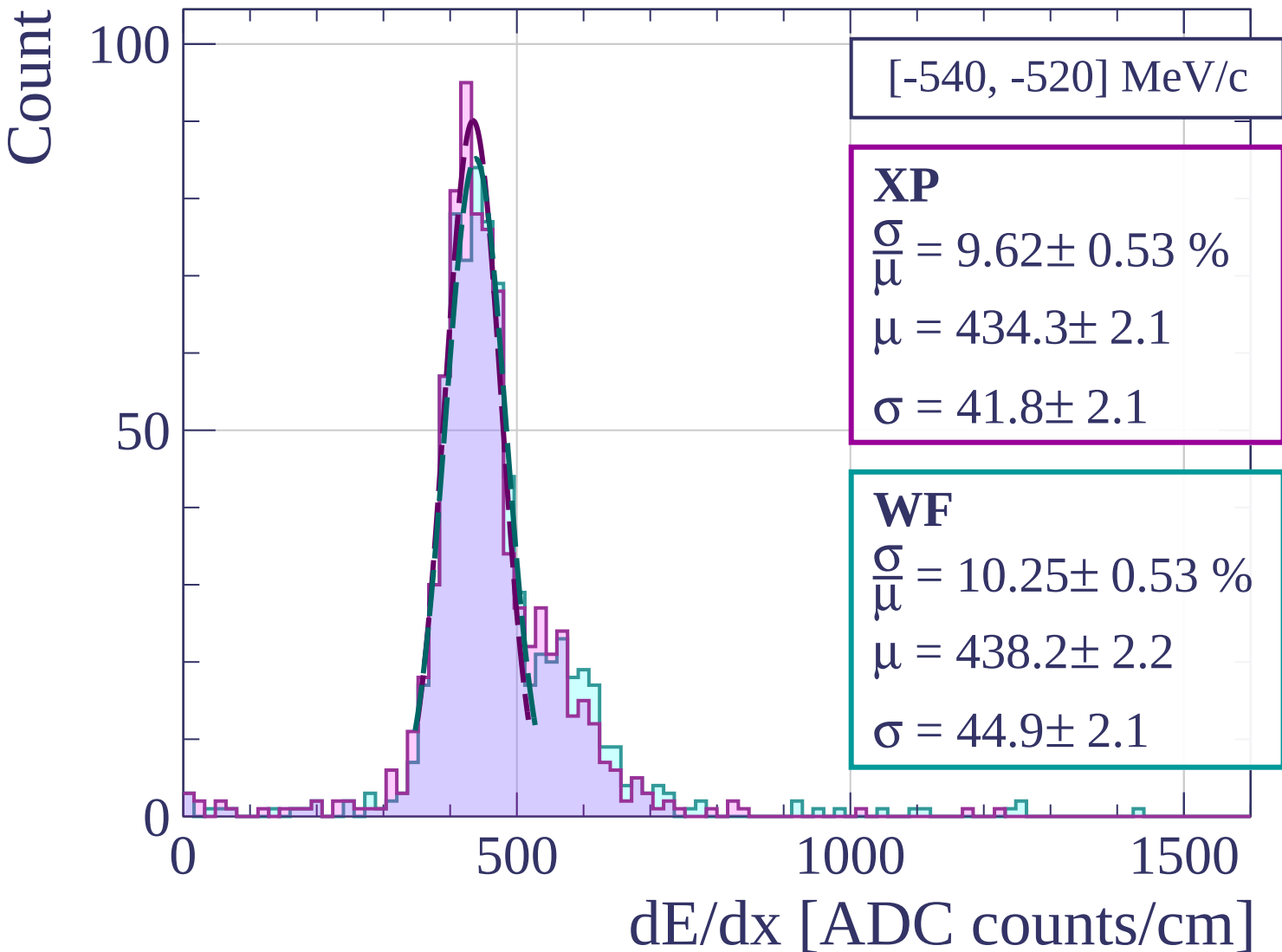
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-520, -500] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.68 \pm 0.55 \%$$

$$\mu = 429.4 \pm 2.0$$

$$\sigma = 41.6 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 9.40 \pm 0.50 \%$$

$$\mu = 435.2 \pm 2.0$$

$$\sigma = 40.9 \pm 2.0$$

0

50

100

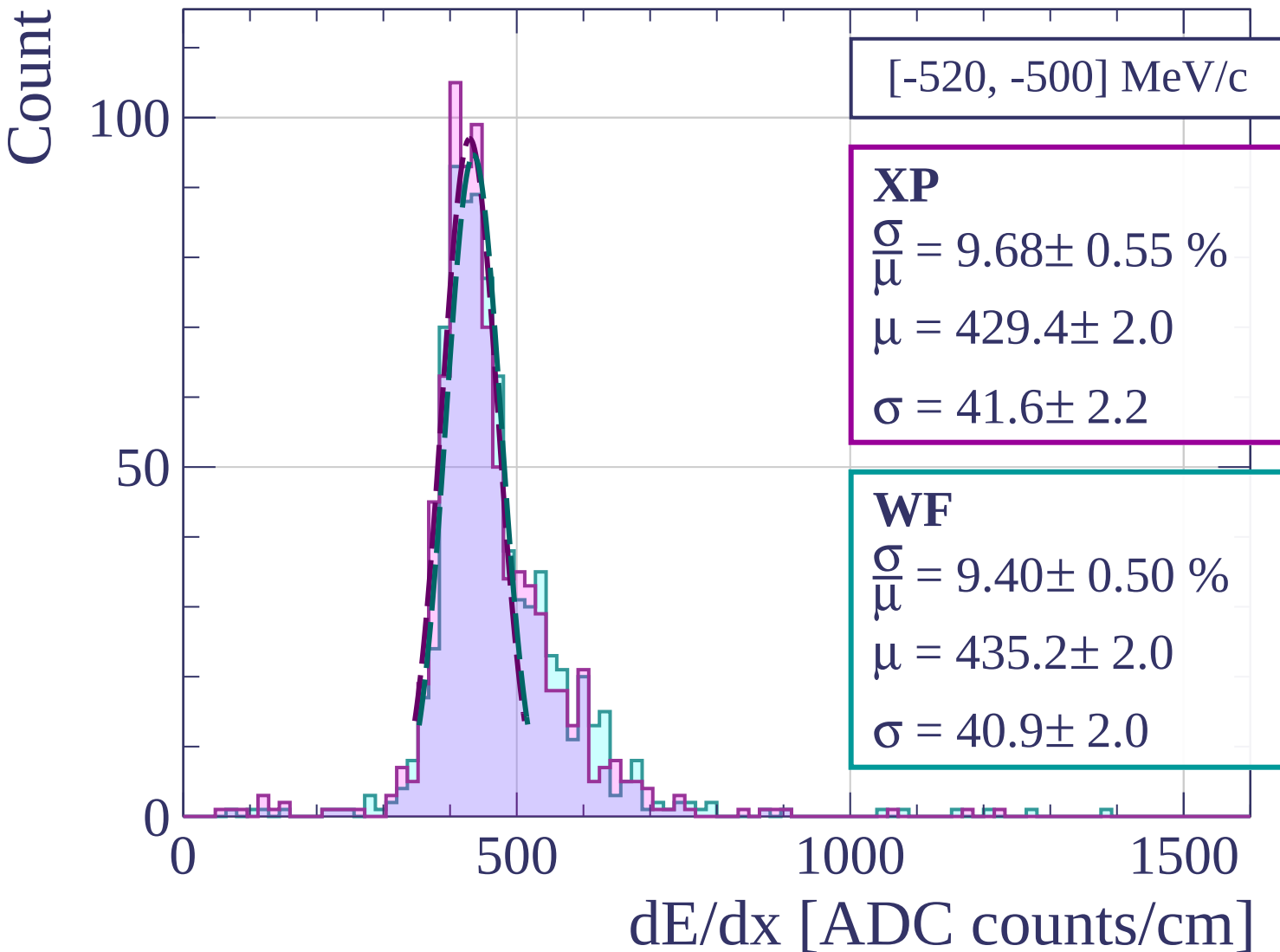
0

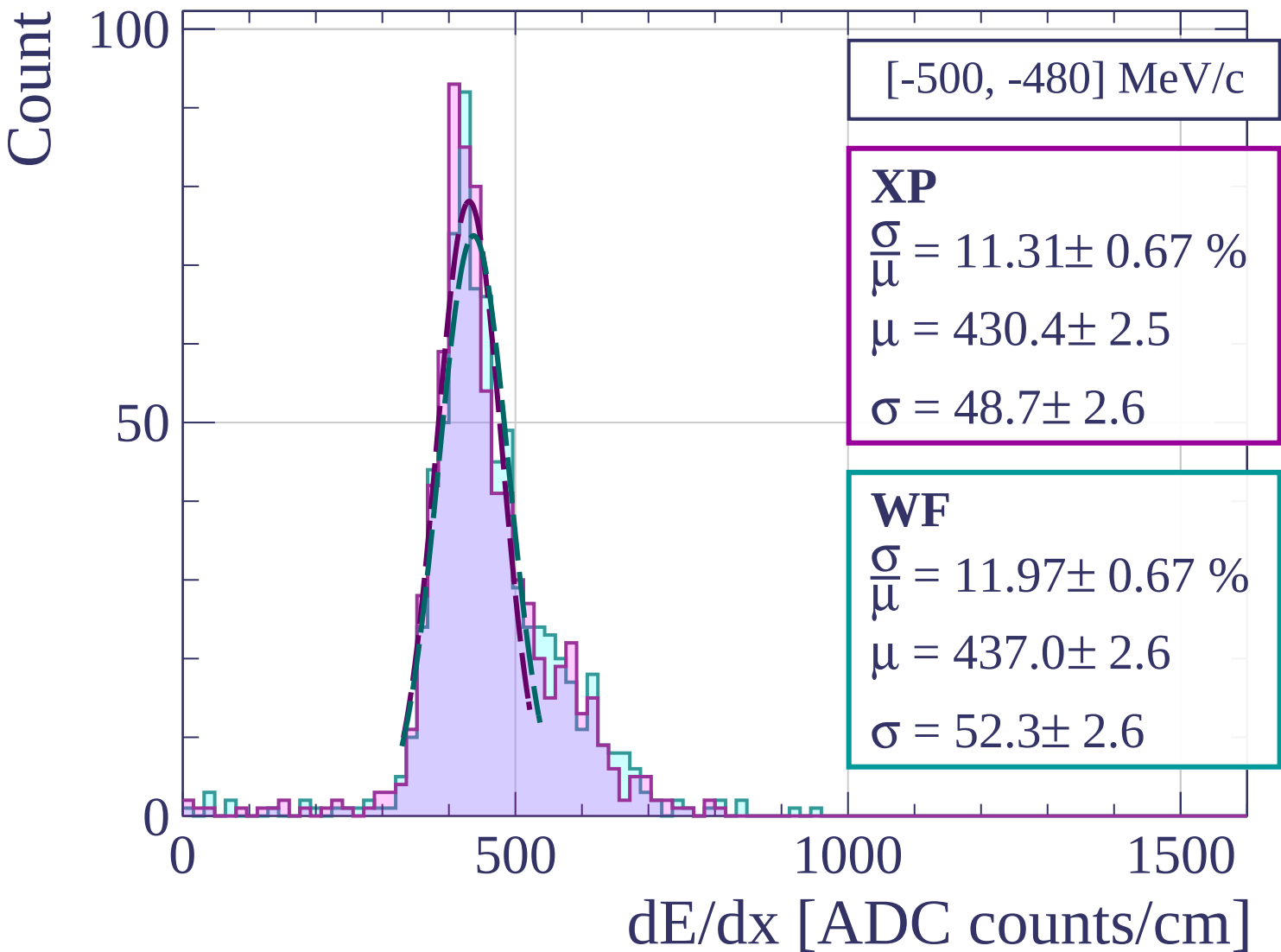
500

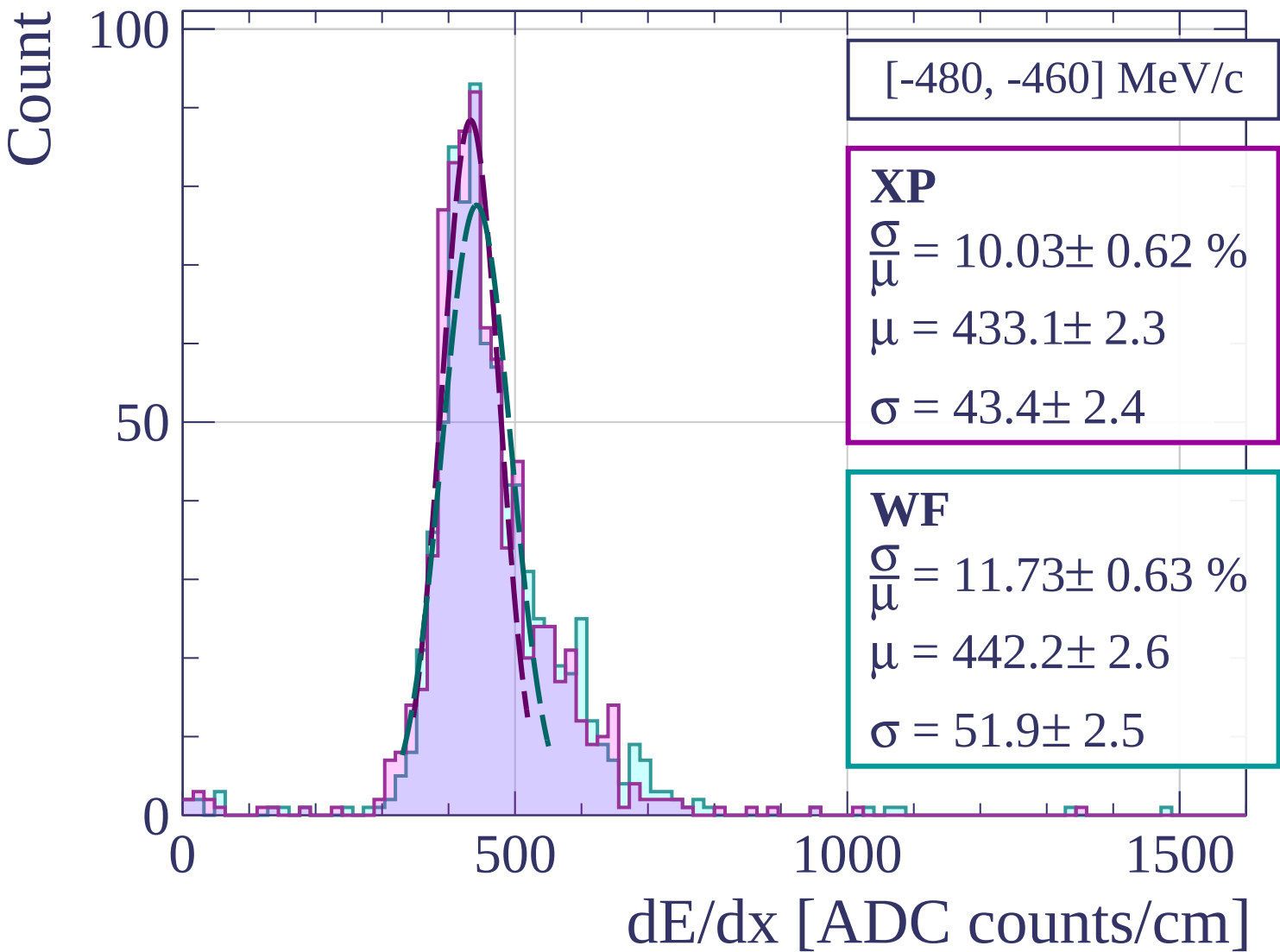
1000

1500

dE/dx [ADC counts/cm]







Count

[-460, -440] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.60 \pm 0.51 \%$$

$$\mu = 429.6 \pm 1.9$$

$$\sigma = 41.3 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 10.48 \pm 0.52 \%$$

$$\mu = 436.4 \pm 2.2$$

$$\sigma = 45.7 \pm 2.1$$

0

500

1000

1500

dE/dx [ADC counts/cm]

100

50

Count

[-440, -420] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.86 \pm 0.56 \%$$

$$\mu = 432.0 \pm 2.2$$

$$\sigma = 42.6 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 11.20 \pm 0.65 \%$$

$$\mu = 442.6 \pm 2.7$$

$$\sigma = 49.6 \pm 2.6$$

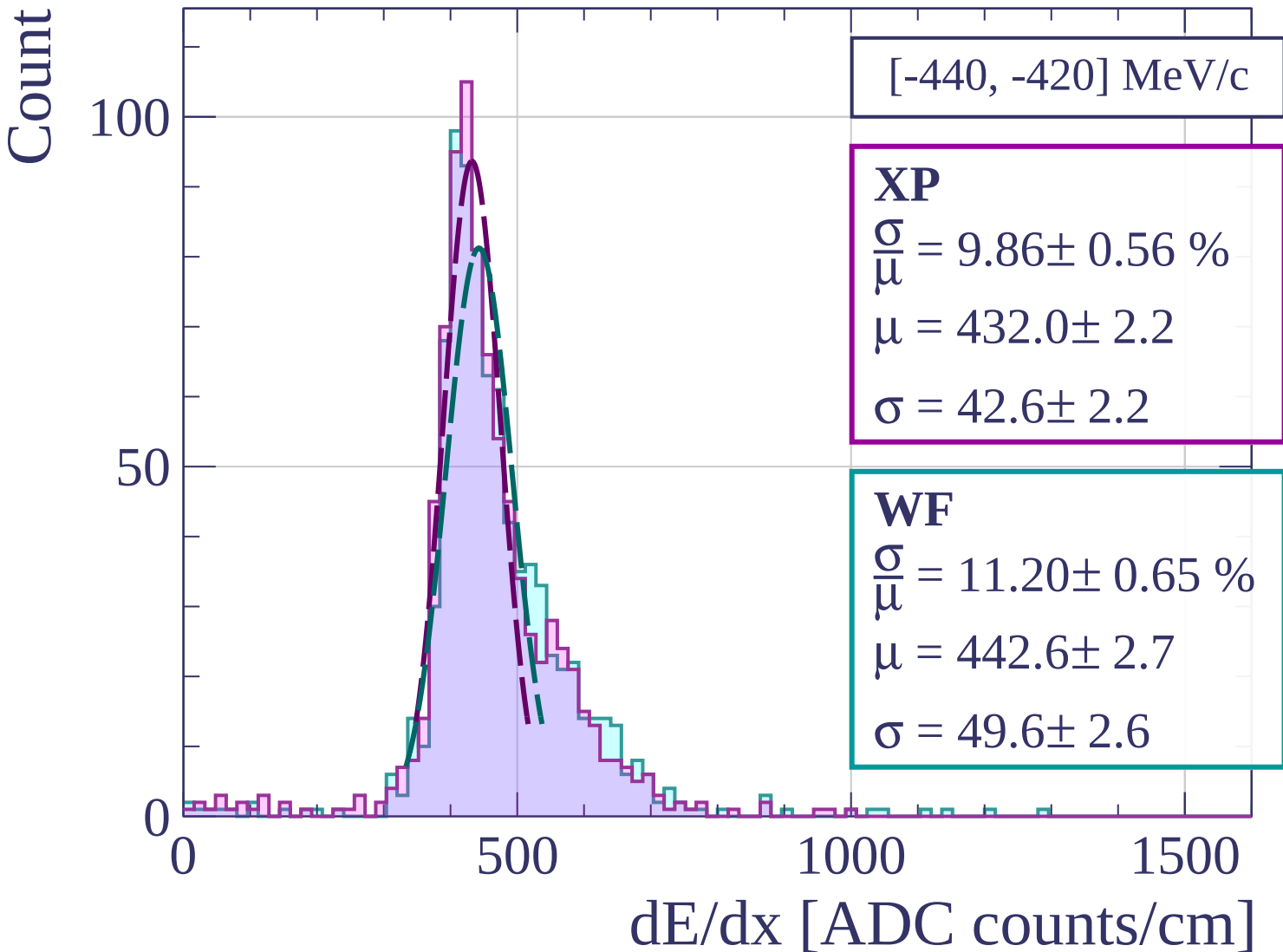
0

500

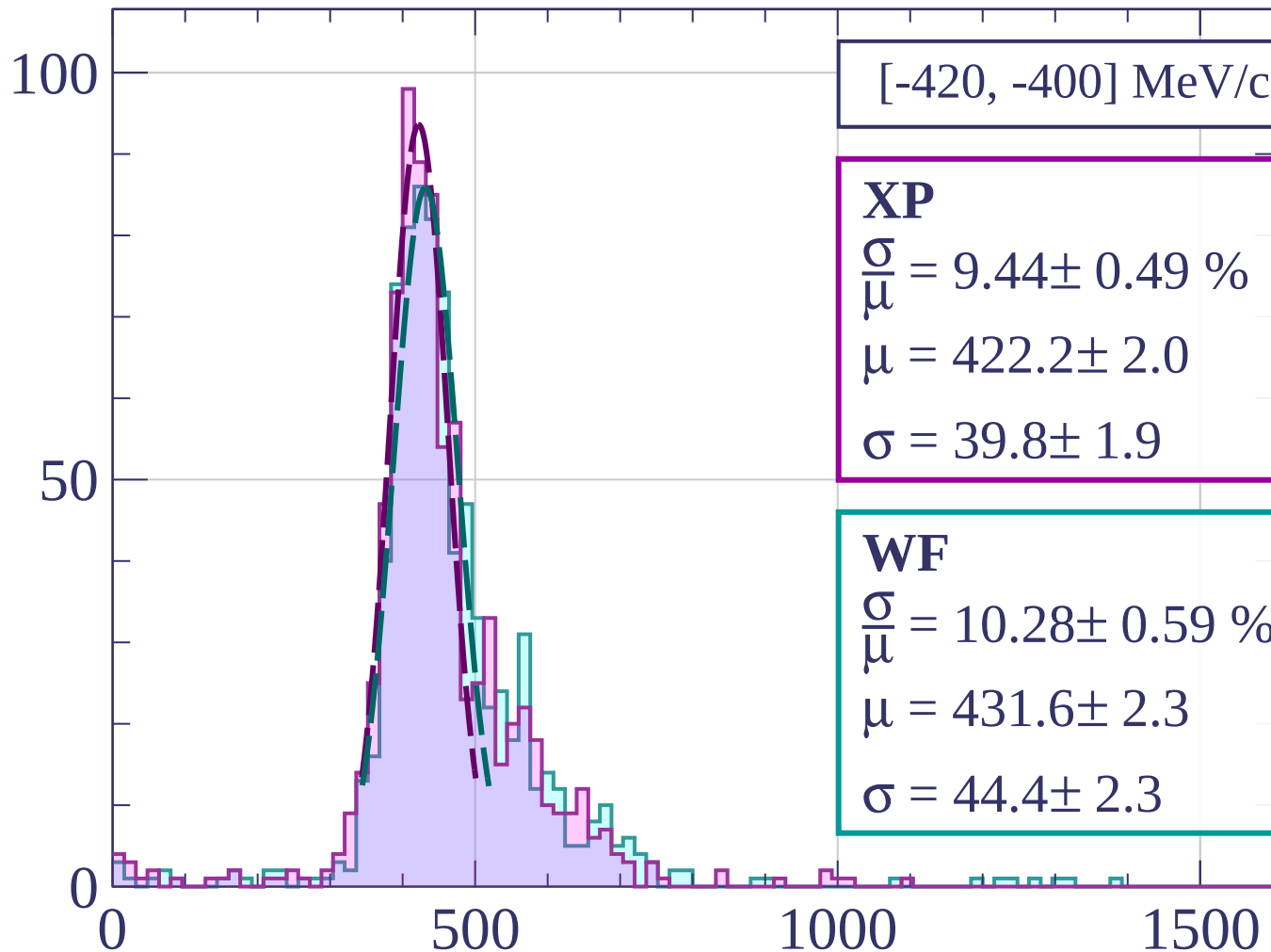
1000

1500

dE/dx [ADC counts/cm]



Count



[-420, -400] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.44 \pm 0.49 \%$$

$$\mu = 422.2 \pm 2.0$$

$$\sigma = 39.8 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 10.28 \pm 0.59 \%$$

$$\mu = 431.6 \pm 2.3$$

$$\sigma = 44.4 \pm 2.3$$

dE/dx [ADC counts/cm]

Count

[-400, -380] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.01 \pm 0.56 \%$$

$$\mu = 427.2 \pm 2.1$$

$$\sigma = 42.8 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 10.64 \pm 0.54 \%$$

$$\mu = 437.1 \pm 2.3$$

$$\sigma = 46.5 \pm 2.1$$

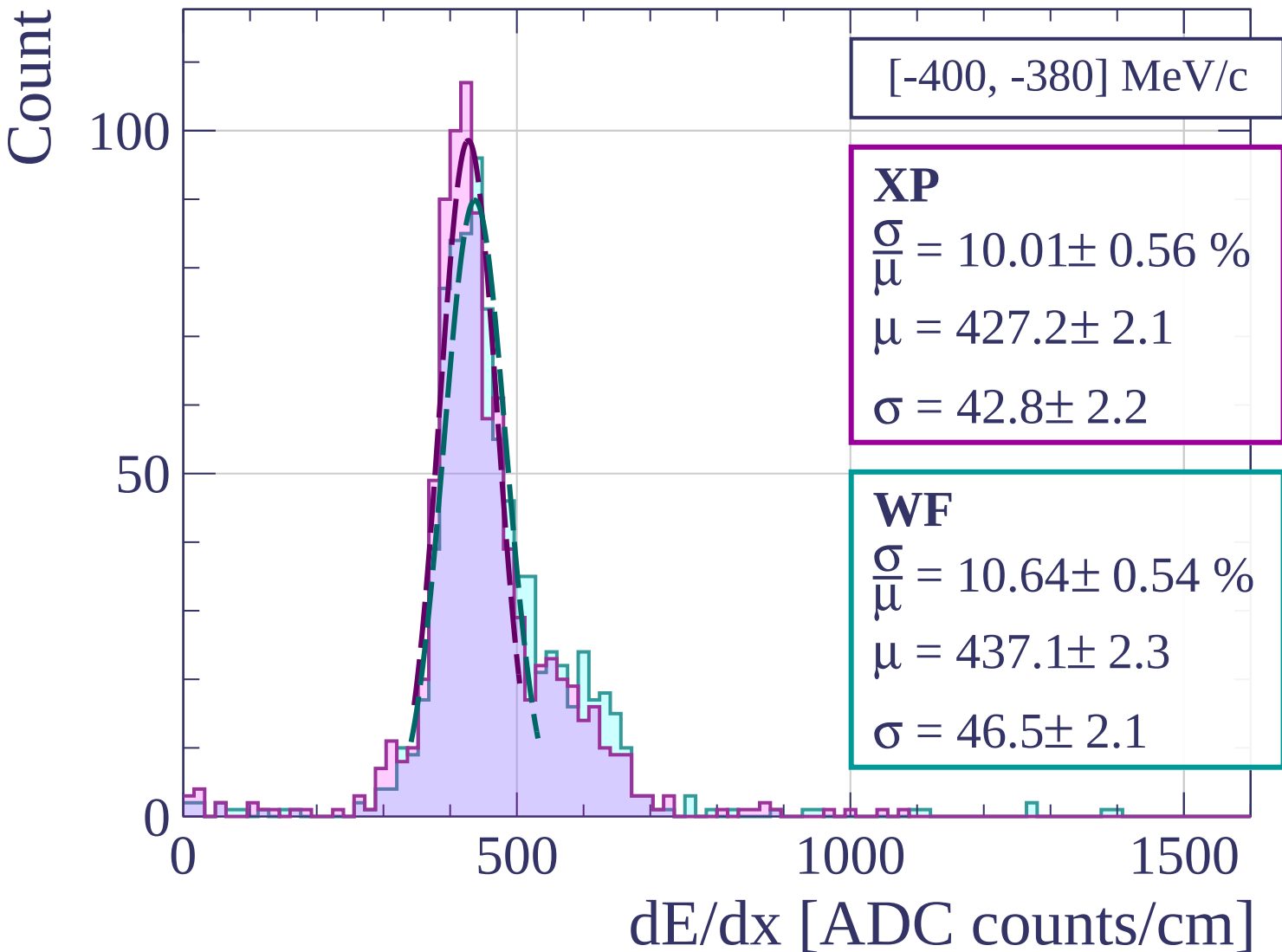
0

500

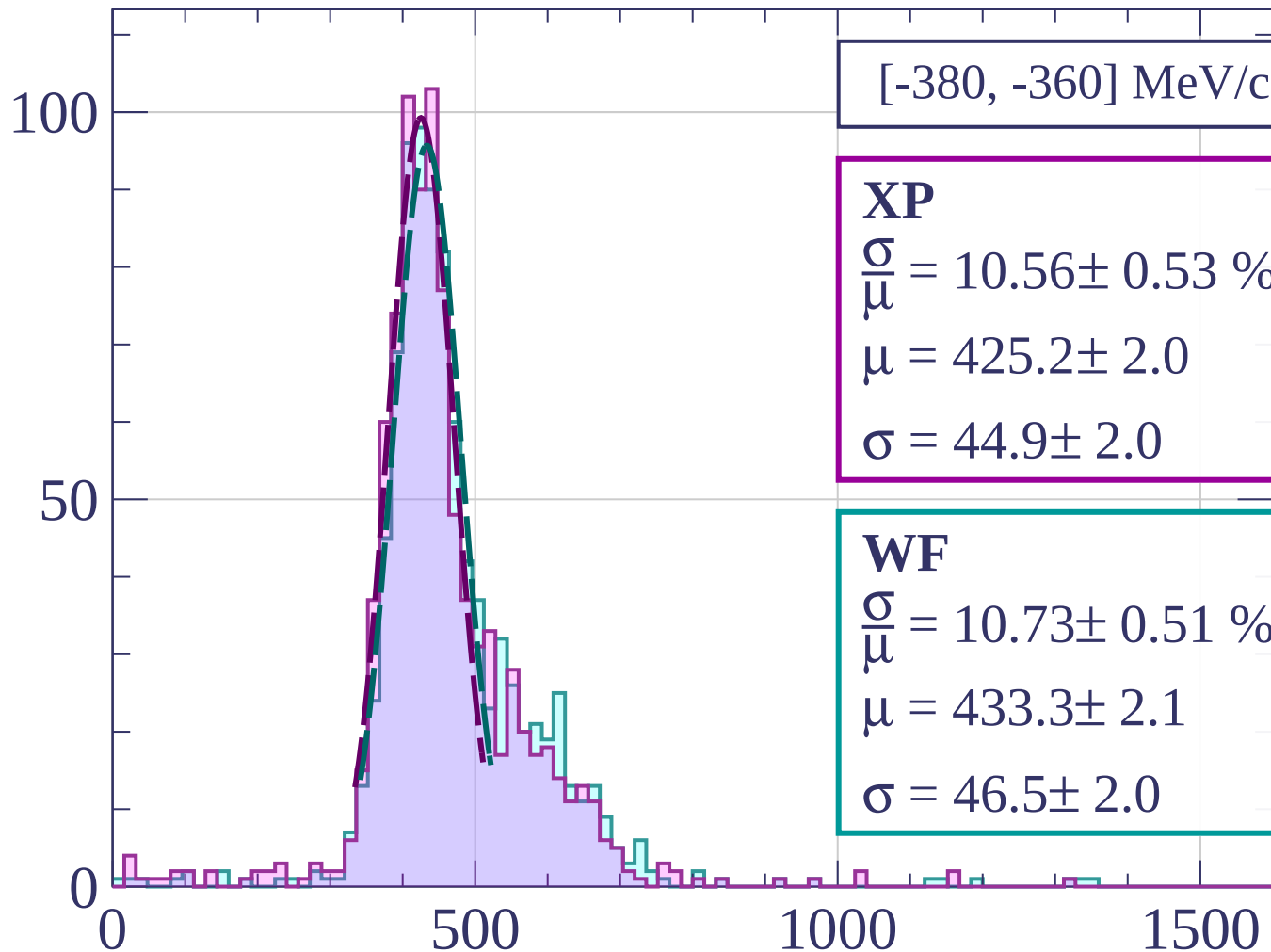
1000

1500

dE/dx [ADC counts/cm]



Count



[-380, -360] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.56 \pm 0.53 \%$$

$$\mu = 425.2 \pm 2.0$$

$$\sigma = 44.9 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 10.73 \pm 0.51 \%$$

$$\mu = 433.3 \pm 2.1$$

$$\sigma = 46.5 \pm 2.0$$

dE/dx [ADC counts/cm]

Count

[-360, -340] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.12 \pm 0.54 \%$$

$$\mu = 426.0 \pm 2.0$$

$$\sigma = 43.1 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 9.69 \pm 0.56 \%$$

$$\mu = 430.6 \pm 2.0$$

$$\sigma = 41.7 \pm 2.2$$

100

50

0

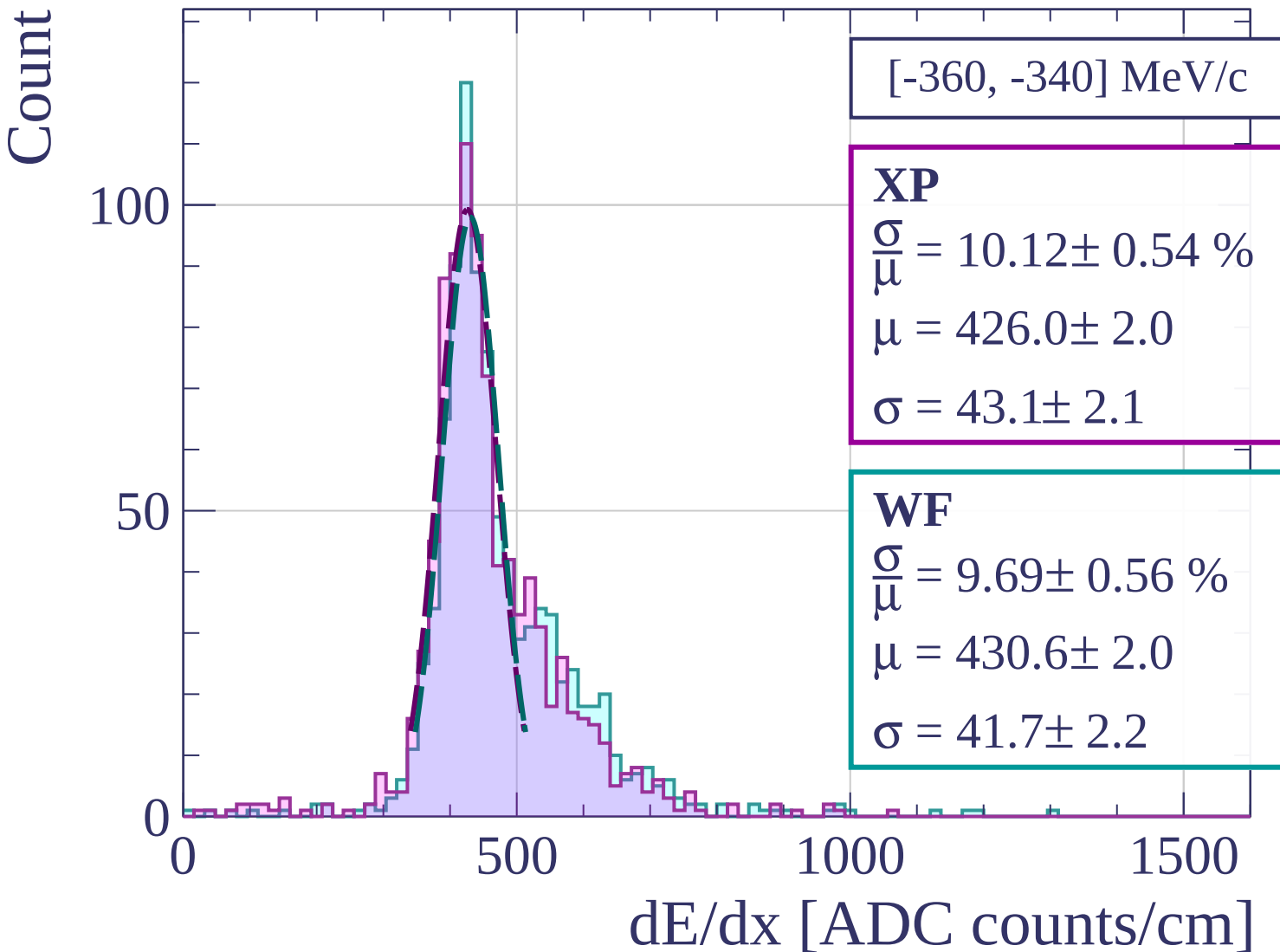
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-340, -320] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.09 \pm 0.60 \%$$

$$\mu = 424.3 \pm 2.1$$

$$\sigma = 47.1 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 11.59 \pm 0.66 \%$$

$$\mu = 430.7 \pm 2.3$$

$$\sigma = 49.9 \pm 2.6$$

0

50

100

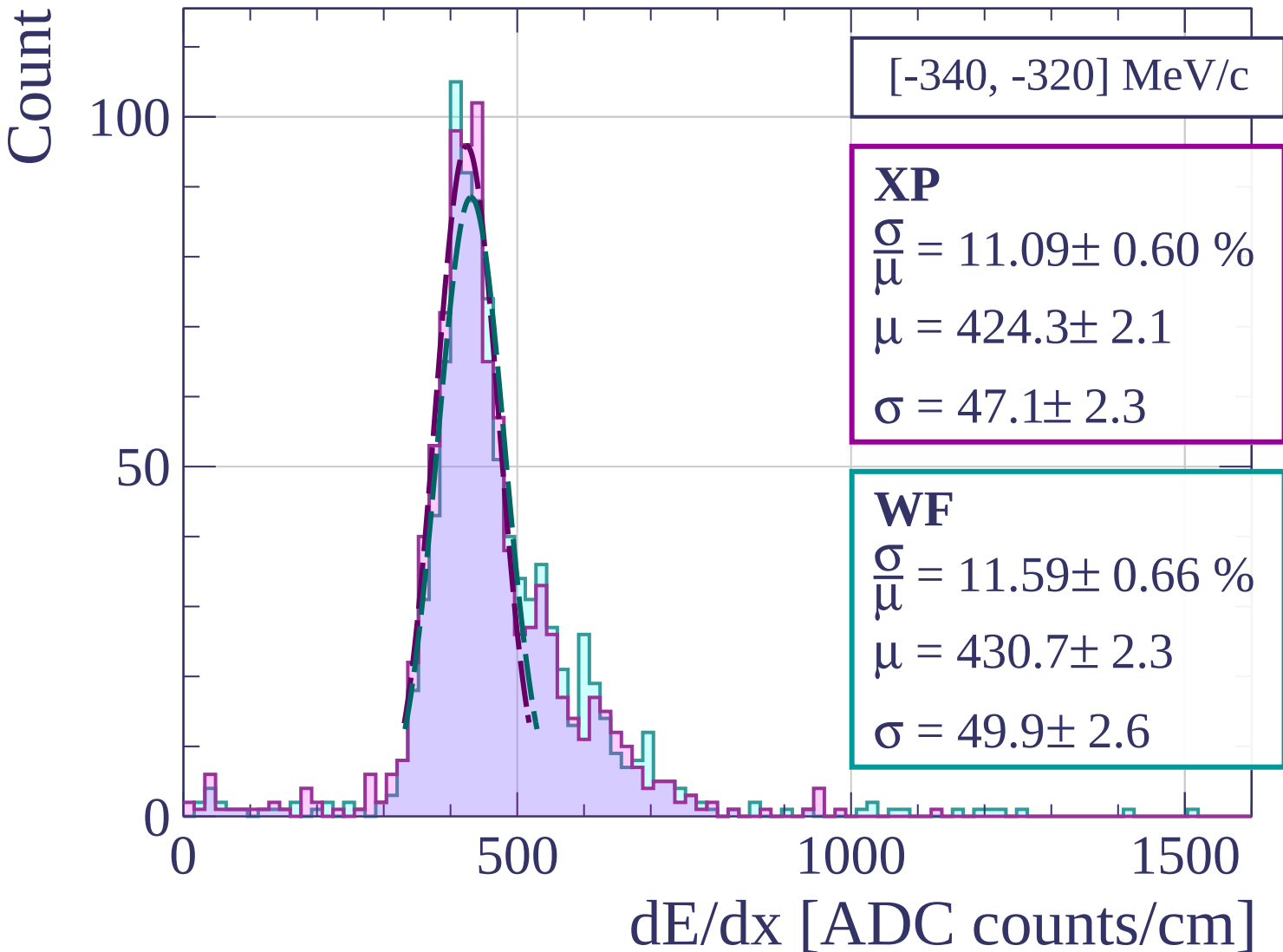
0

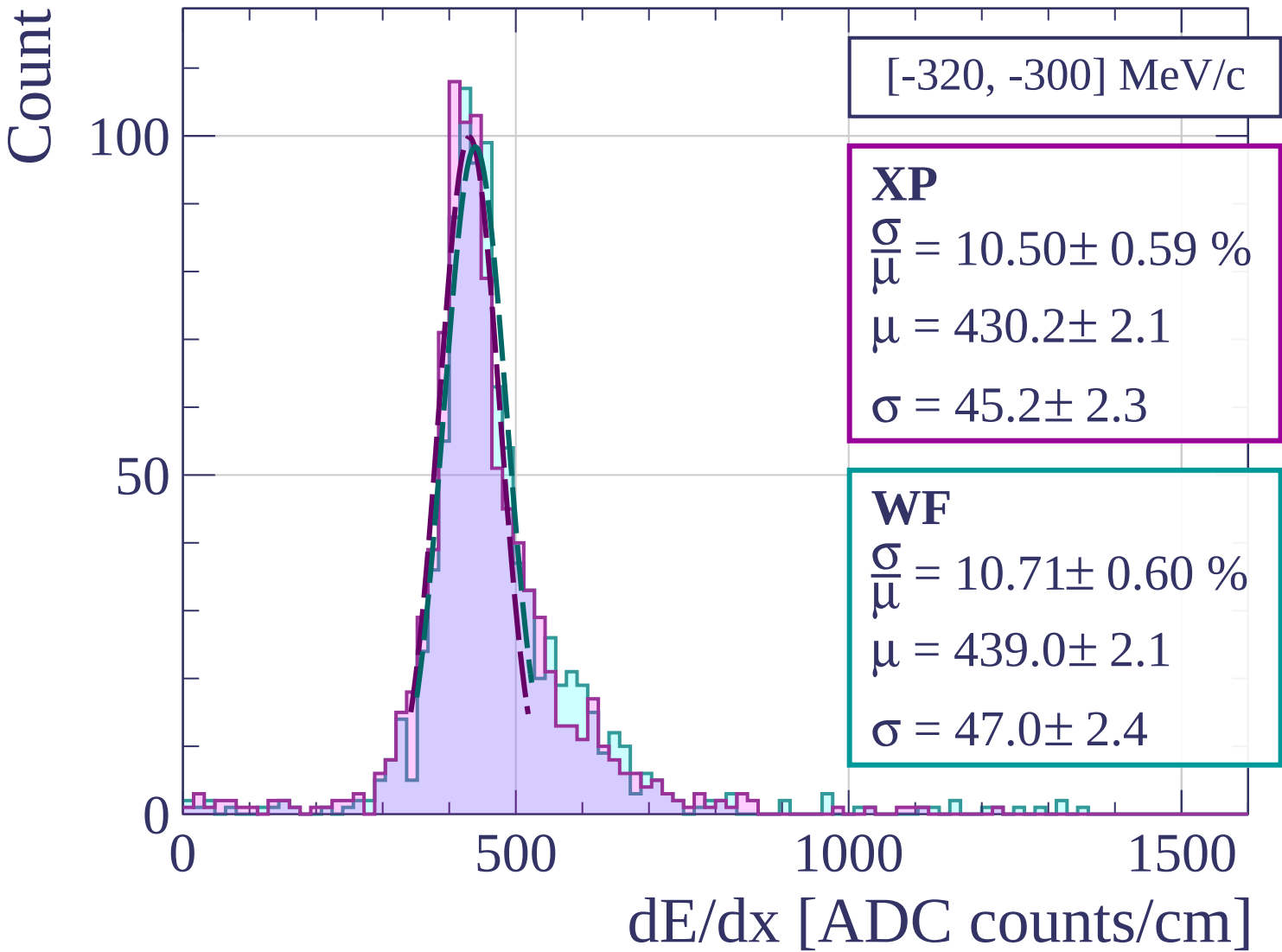
500

1000

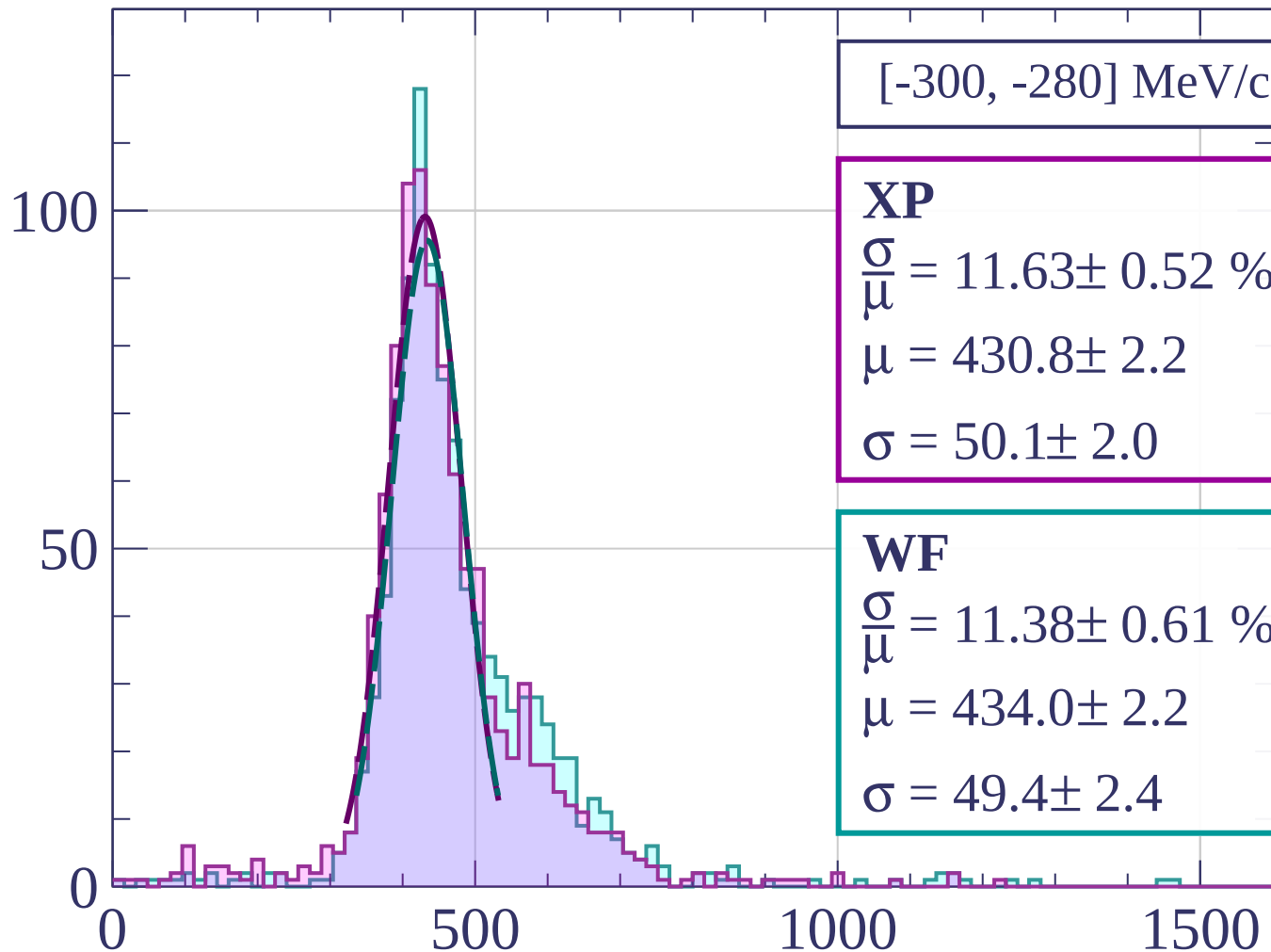
1500

dE/dx [ADC counts/cm]





Count



[-300, -280] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.63 \pm 0.52 \%$$

$$\mu = 430.8 \pm 2.2$$

$$\sigma = 50.1 \pm 2.0$$

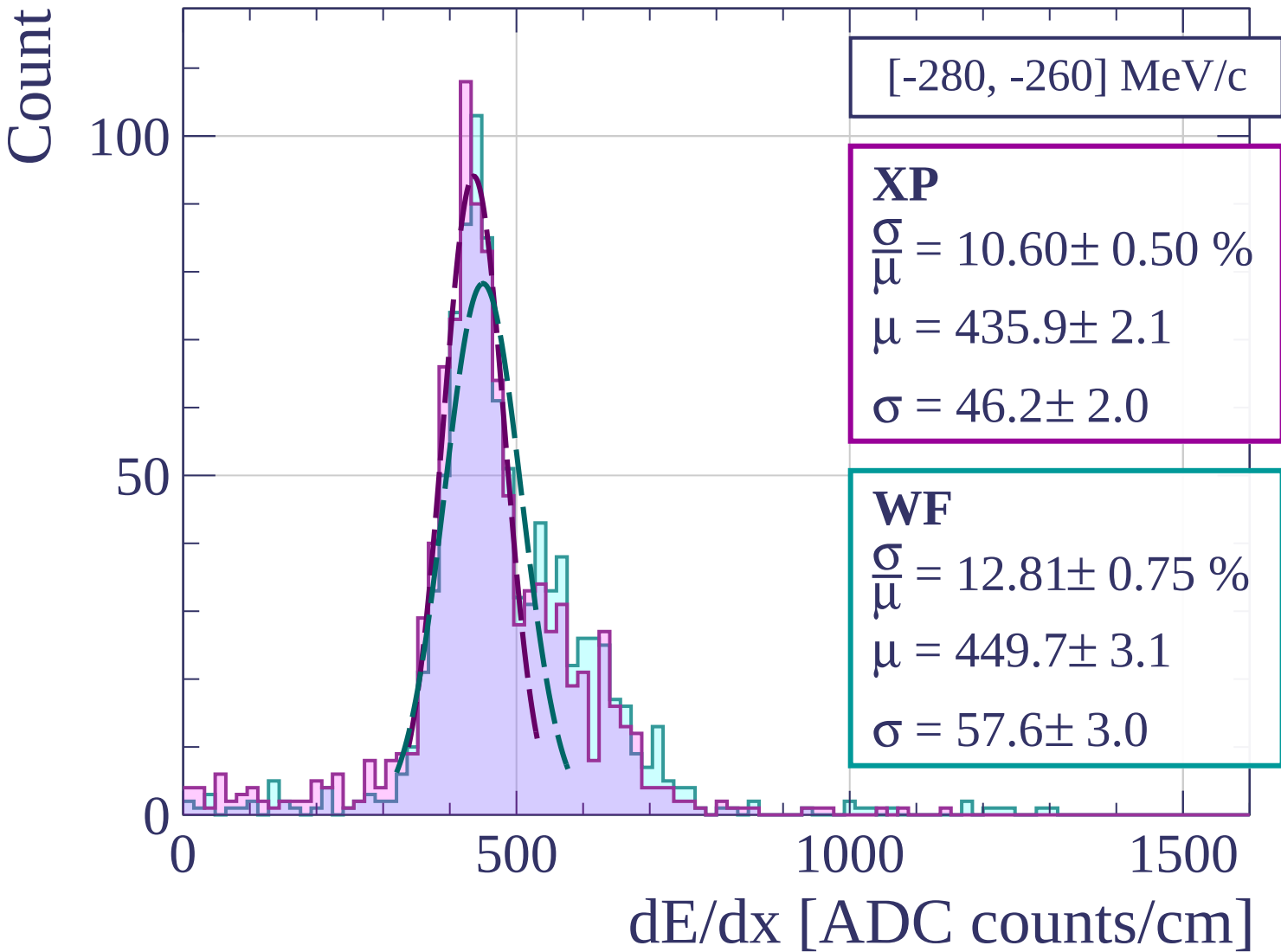
WF

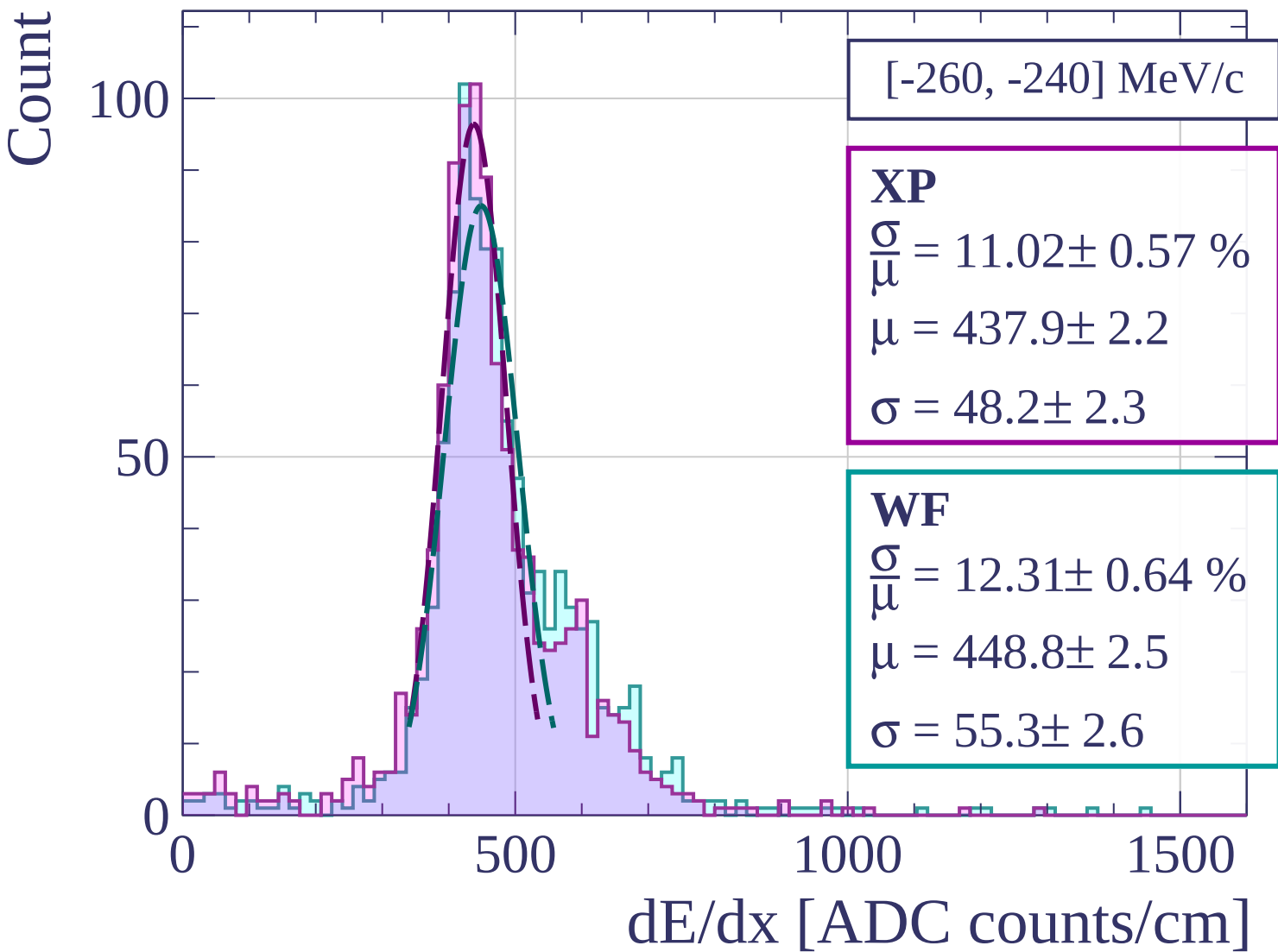
$$\frac{\sigma}{\mu} = 11.38 \pm 0.61 \%$$

$$\mu = 434.0 \pm 2.2$$

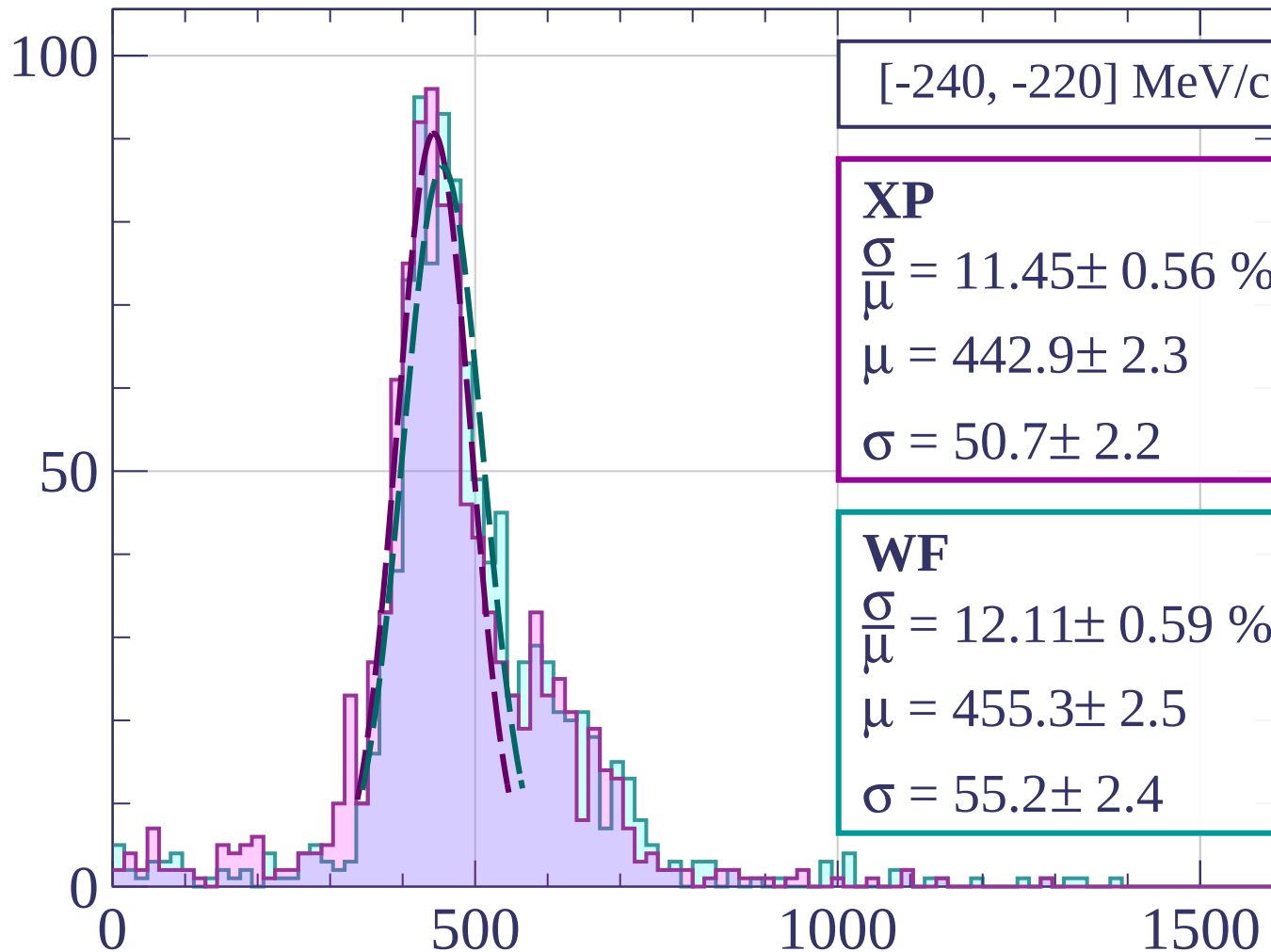
$$\sigma = 49.4 \pm 2.4$$

dE/dx [ADC counts/cm]





Count



[-240, -220] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.45 \pm 0.56 \%$$

$$\mu = 442.9 \pm 2.3$$

$$\sigma = 50.7 \pm 2.2$$

WF

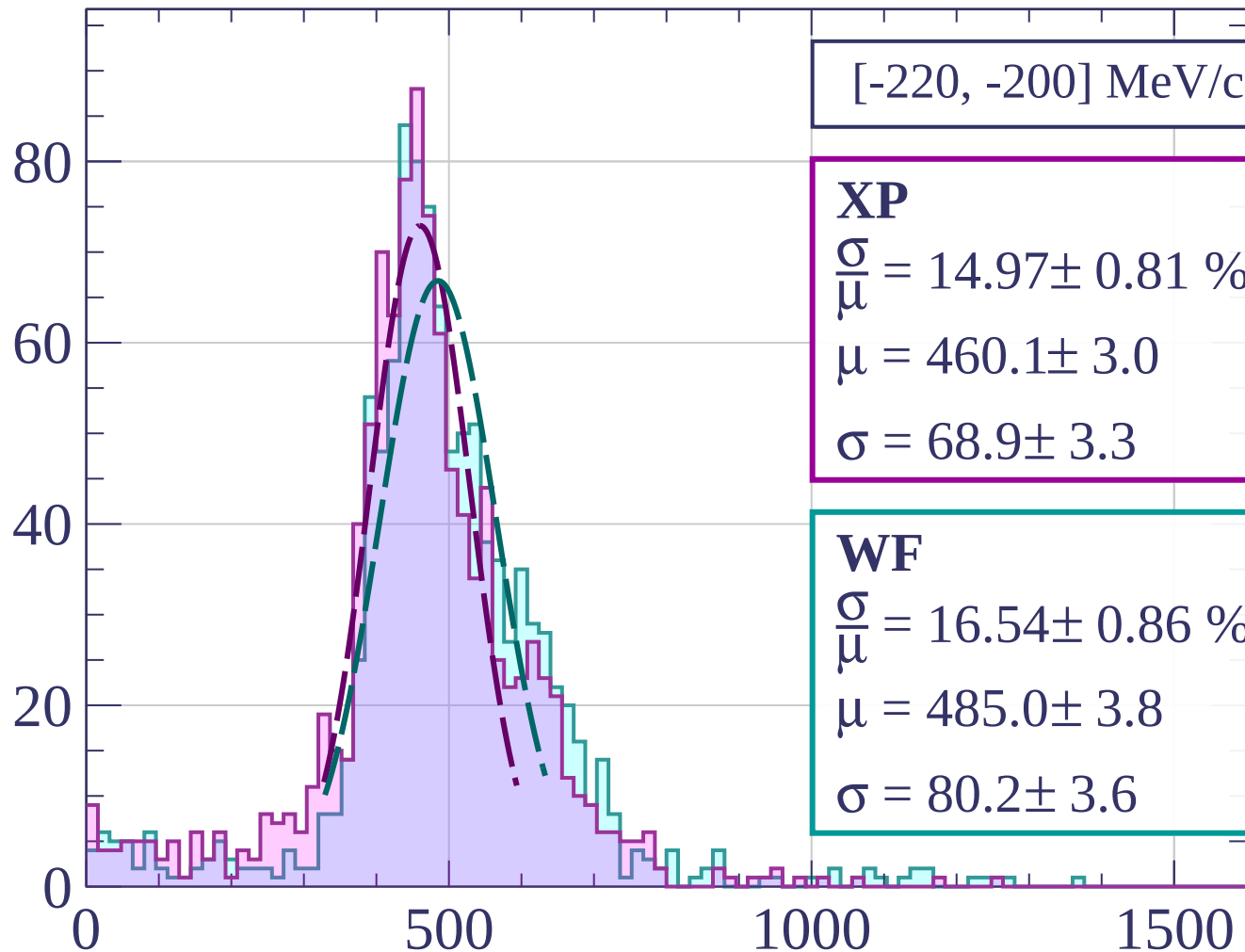
$$\frac{\sigma}{\mu} = 12.11 \pm 0.59 \%$$

$$\mu = 455.3 \pm 2.5$$

$$\sigma = 55.2 \pm 2.4$$

dE/dx [ADC counts/cm]

Count



[-220, -200] MeV/c

XP

$$\frac{\sigma}{\mu} = 14.97 \pm 0.81 \%$$

$$\mu = 460.1 \pm 3.0$$

$$\sigma = 68.9 \pm 3.3$$

WF

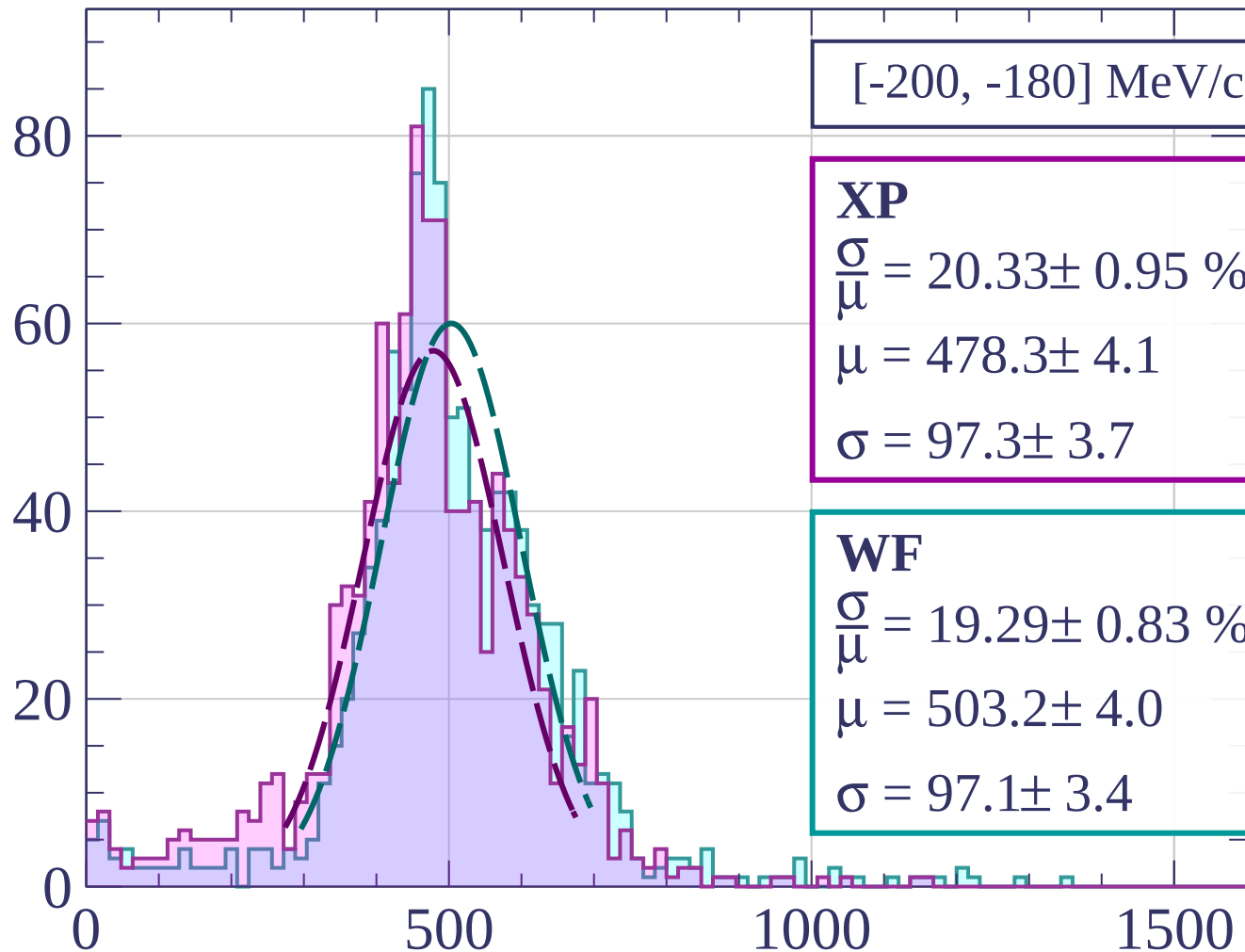
$$\frac{\sigma}{\mu} = 16.54 \pm 0.86 \%$$

$$\mu = 485.0 \pm 3.8$$

$$\sigma = 80.2 \pm 3.6$$

dE/dx [ADC counts/cm]

Count



[-200, -180] MeV/c

XP

$$\frac{\sigma}{\mu} = 20.33 \pm 0.95 \%$$

$$\mu = 478.3 \pm 4.1$$

$$\sigma = 97.3 \pm 3.7$$

WF

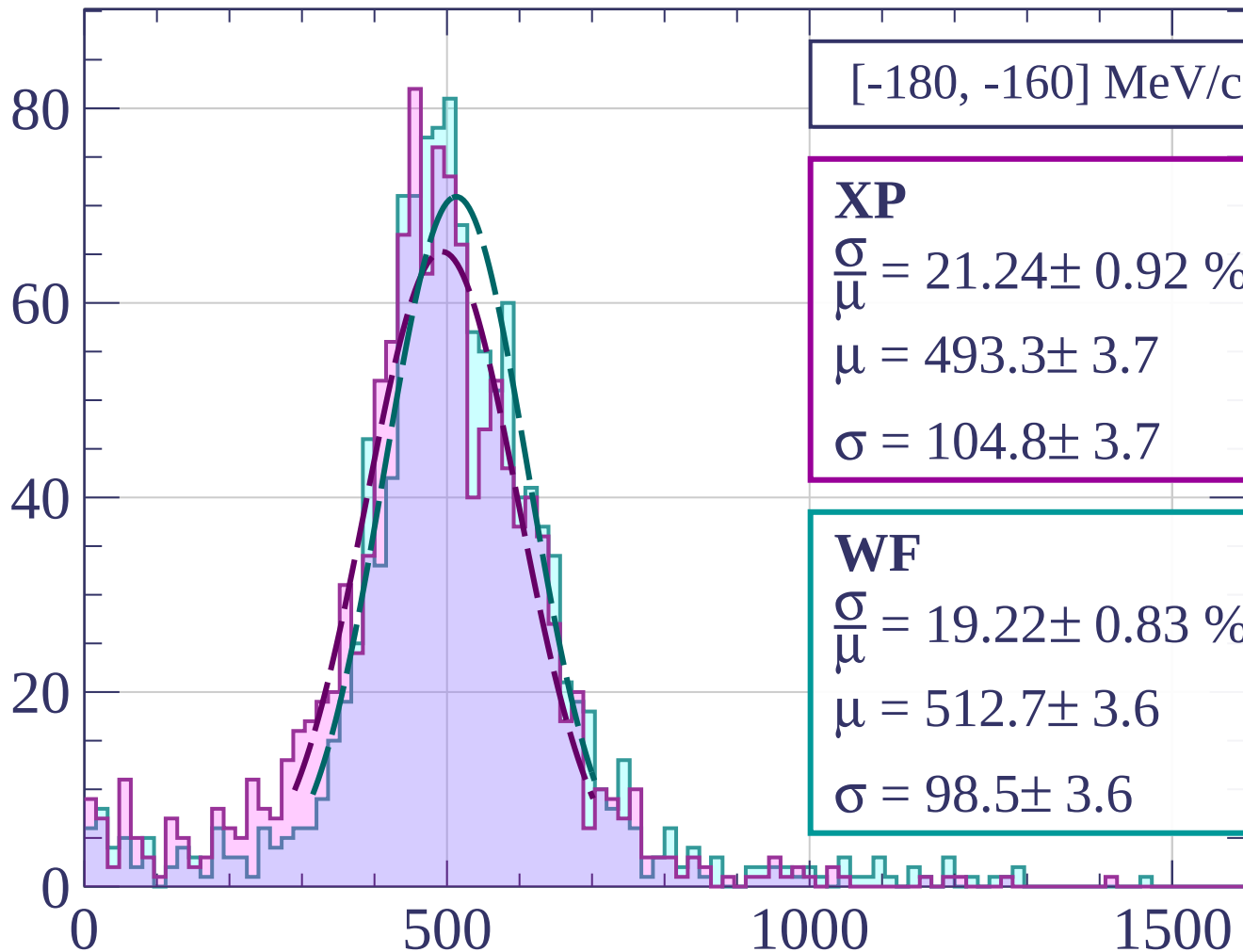
$$\frac{\sigma}{\mu} = 19.29 \pm 0.83 \%$$

$$\mu = 503.2 \pm 4.0$$

$$\sigma = 97.1 \pm 3.4$$

dE/dx [ADC counts/cm]

Count



$[-180, -160] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 21.24 \pm 0.92 \%$$

$$\mu = 493.3 \pm 3.7$$

$$\sigma = 104.8 \pm 3.7$$

WF

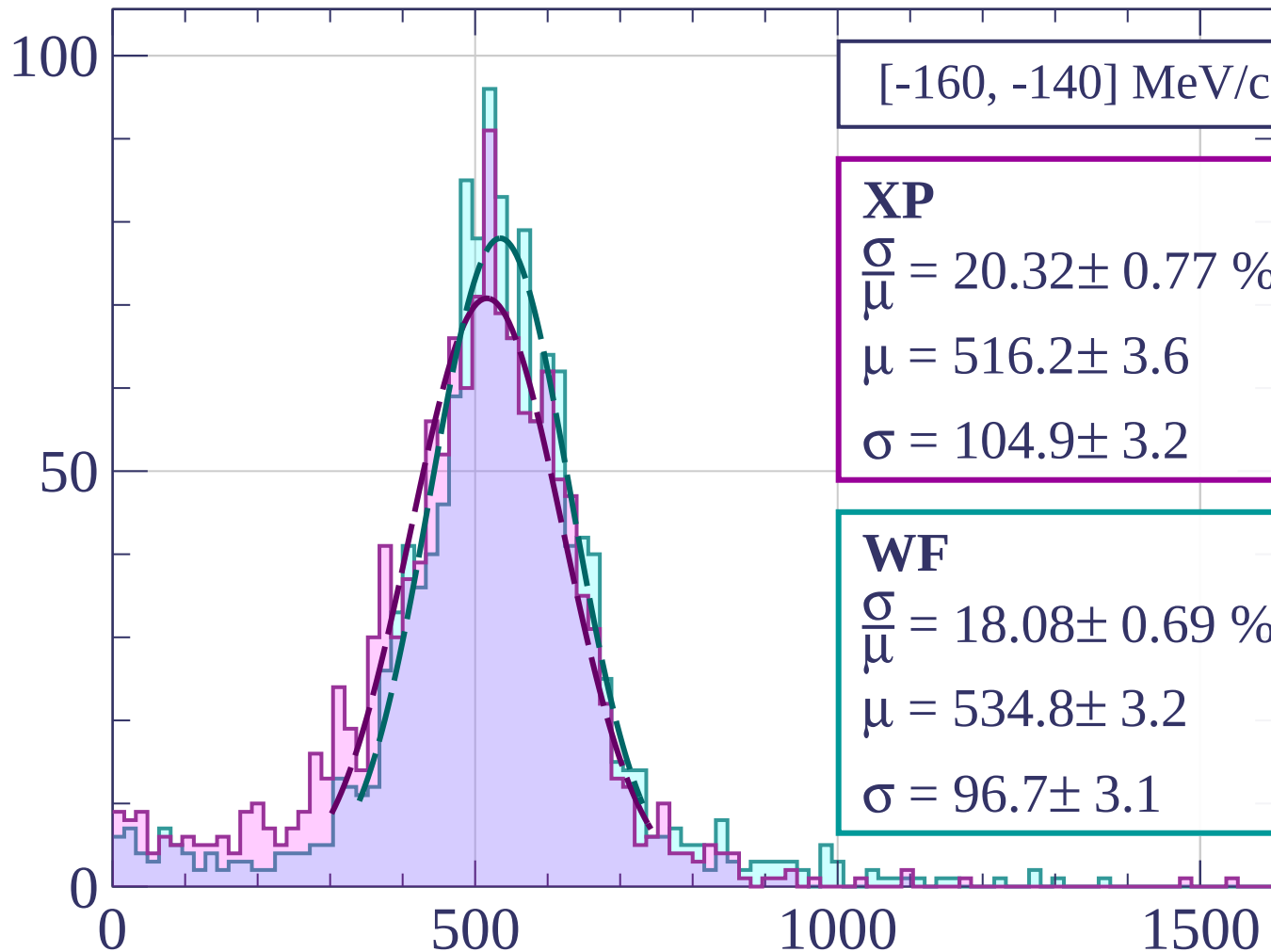
$$\frac{\sigma}{\mu} = 19.22 \pm 0.83 \%$$

$$\mu = 512.7 \pm 3.6$$

$$\sigma = 98.5 \pm 3.6$$

dE/dx [ADC counts/cm]

Count



[-160, -140] MeV/c

XP

$$\frac{\sigma}{\mu} = 20.32 \pm 0.77 \%$$

$$\mu = 516.2 \pm 3.6$$

$$\sigma = 104.9 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 18.08 \pm 0.69 \%$$

$$\mu = 534.8 \pm 3.2$$

$$\sigma = 96.7 \pm 3.1$$

dE/dx [ADC counts/cm]

Count

[-140, -120] MeV/c

XP

$$\frac{\sigma}{\mu} = 21.23 \pm 0.86 \%$$

$$\mu = 519.8 \pm 4.2$$

$$\sigma = 110.4 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 18.57 \pm 0.74 \%$$

$$\mu = 544.1 \pm 3.5$$

$$\sigma = 101.1 \pm 3.4$$

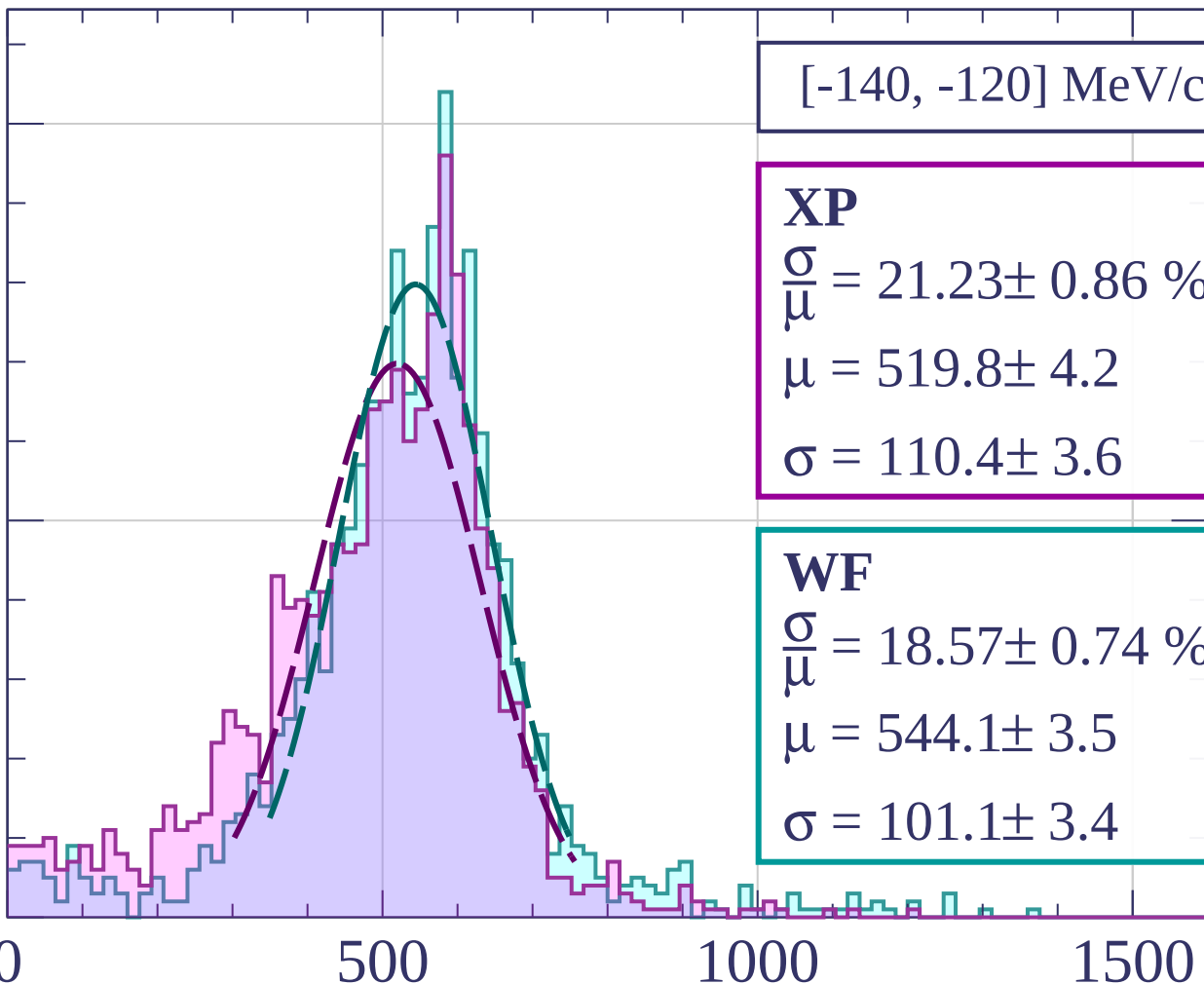
0

500

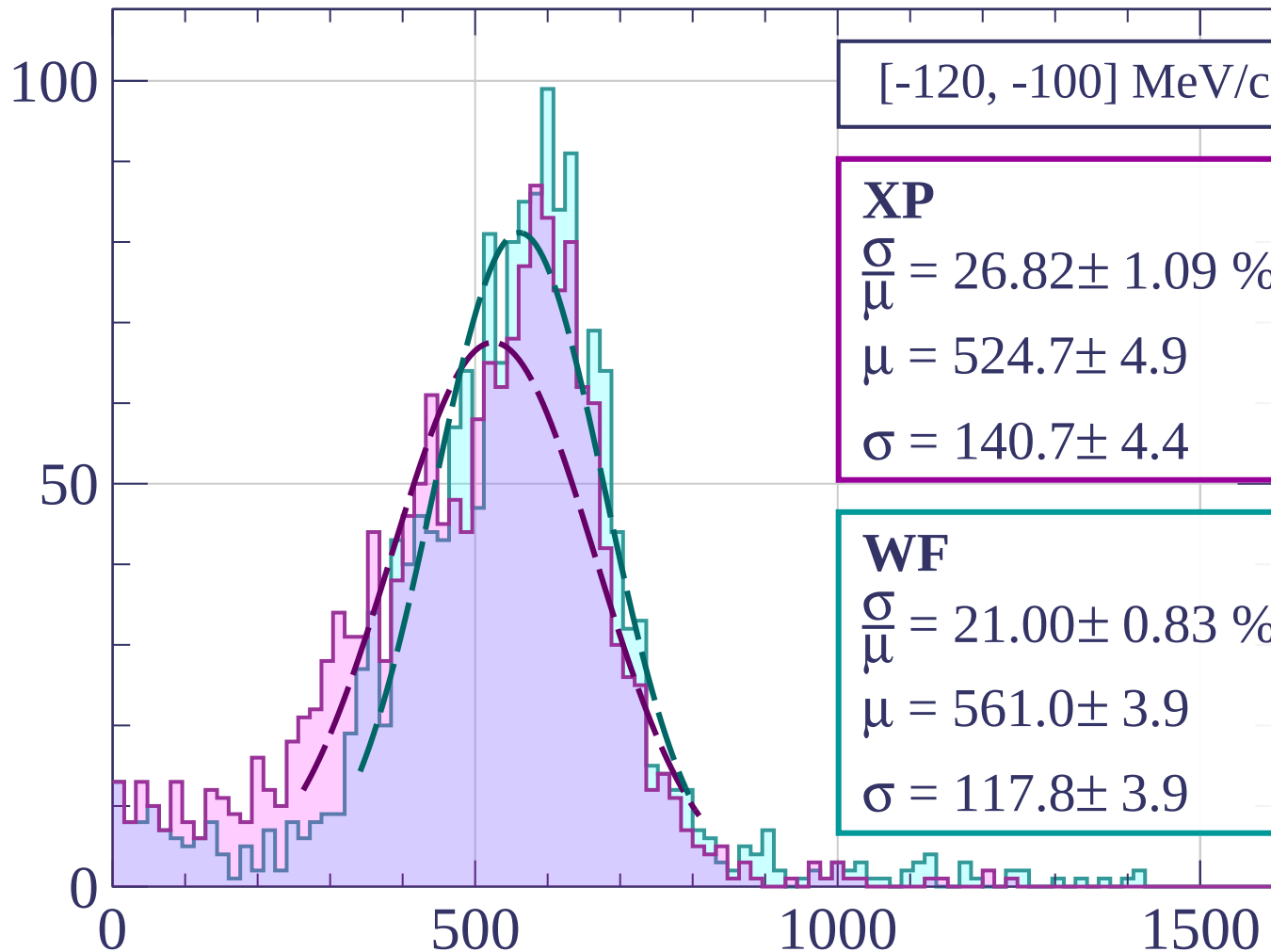
1000

1500

dE/dx [ADC counts/cm]



Count



[-120, -100] MeV/c

XP

$$\frac{\sigma}{\mu} = 26.82 \pm 1.09 \%$$

$$\mu = 524.7 \pm 4.9$$

$$\sigma = 140.7 \pm 4.4$$

WF

$$\frac{\sigma}{\mu} = 21.00 \pm 0.83 \%$$

$$\mu = 561.0 \pm 3.9$$

$$\sigma = 117.8 \pm 3.9$$

dE/dx [ADC counts/cm]

Count

[-100, -80] MeV/c

XP

$$\frac{\sigma}{\mu} = 25.59 \pm 0.95 \%$$

$$\mu = 529.5 \pm 4.2$$

$$\sigma = 135.5 \pm 4.0$$

WF

$$\frac{\sigma}{\mu} = 20.40 \pm 0.69 \%$$

$$\mu = 561.7 \pm 3.3$$

$$\sigma = 114.6 \pm 3.2$$

100

50

0

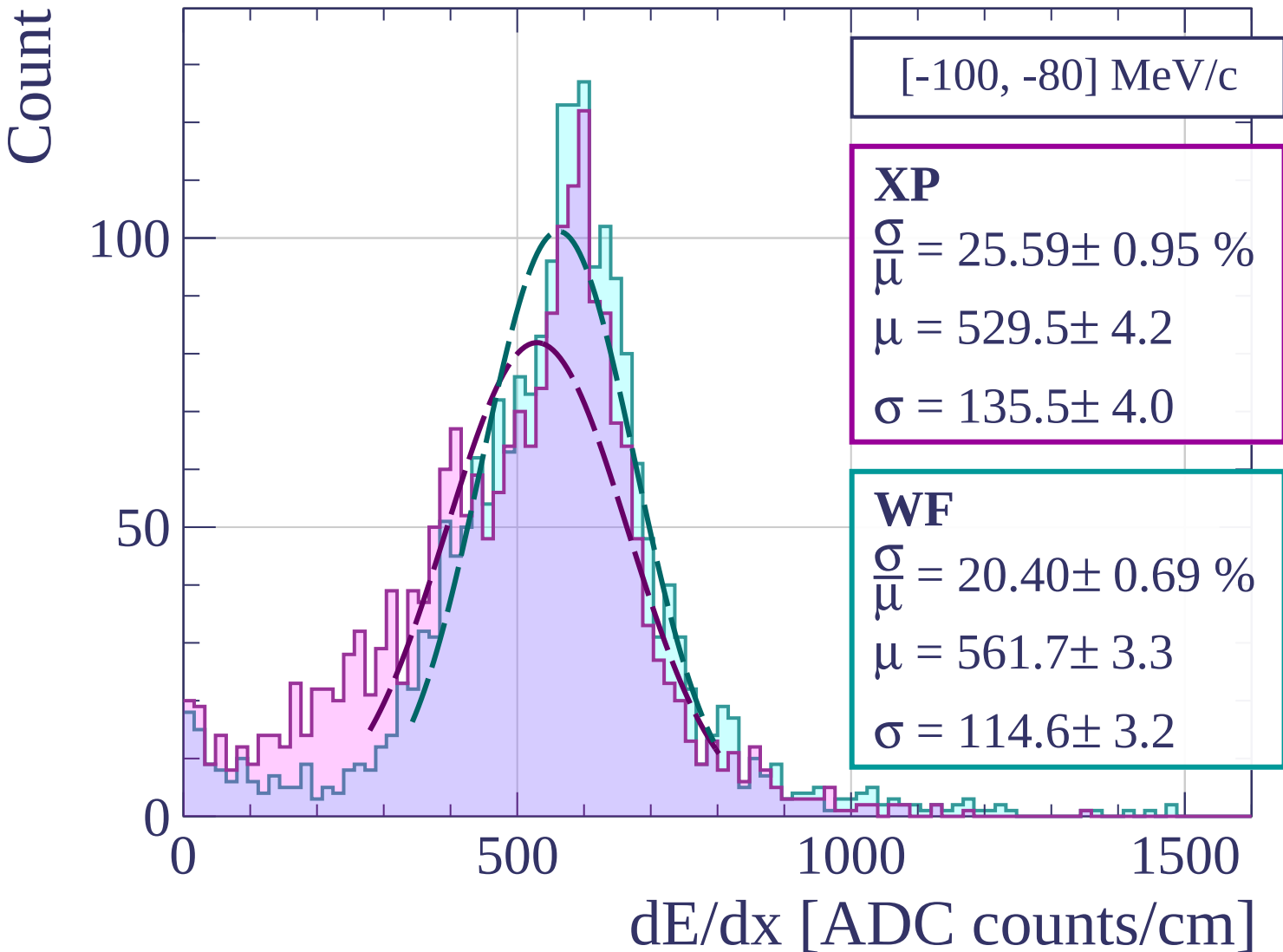
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-80, -60] MeV/c

XP

$$\frac{\sigma}{\mu} = 24.83 \pm 0.81 \%$$

$$\mu = 515.7 \pm 3.9$$

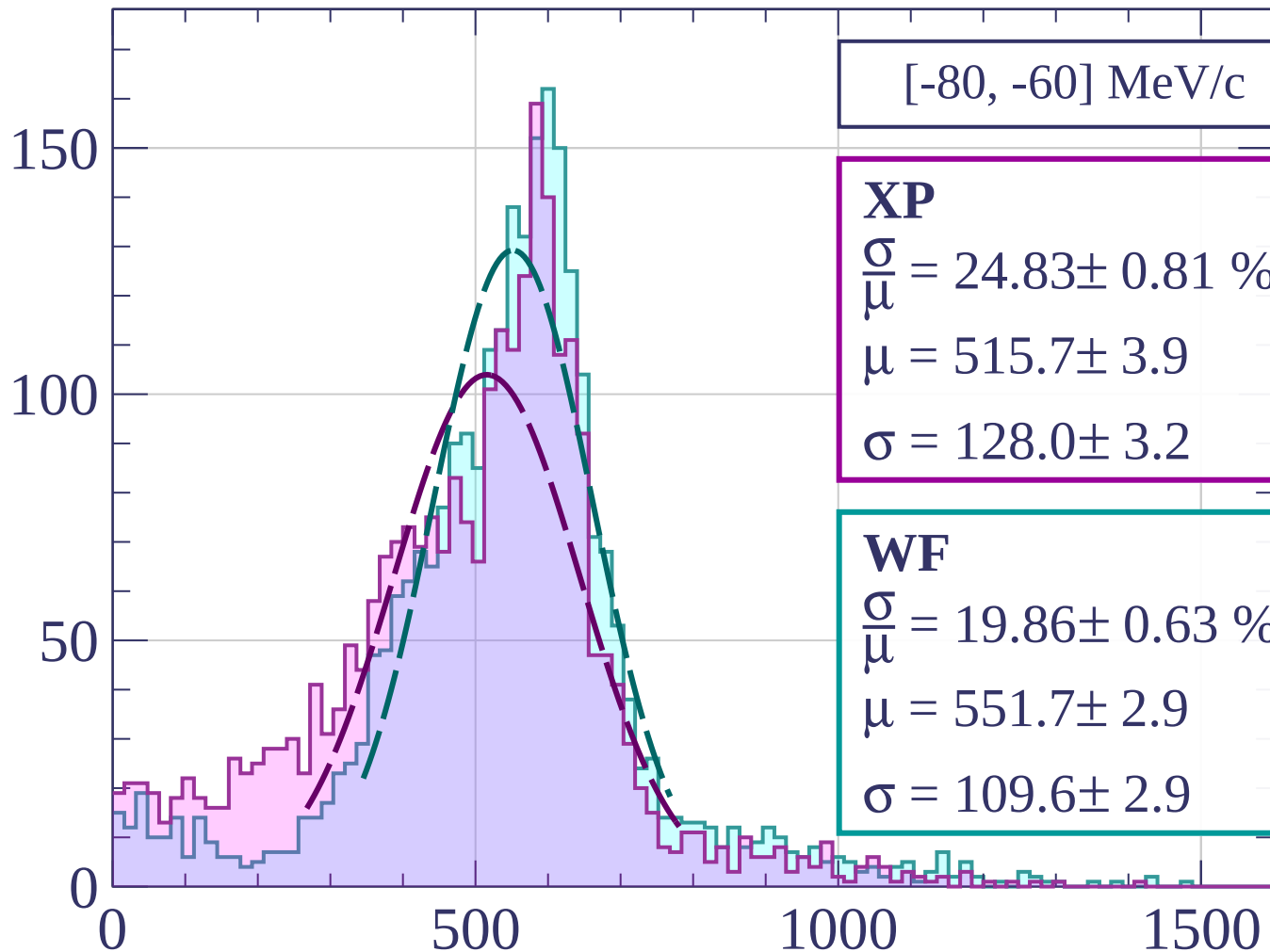
$$\sigma = 128.0 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 19.86 \pm 0.63 \%$$

$$\mu = 551.7 \pm 2.9$$

$$\sigma = 109.6 \pm 2.9$$



dE/dx [ADC counts/cm]

Count

[-60, -40] MeV/c

XP

$$\frac{\sigma}{\mu} = 20.14 \pm 0.67 \%$$

$$\mu = 532.4 \pm 3.0$$

$$\sigma = 107.2 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 16.53 \pm 0.45 \%$$

$$\mu = 558.7 \pm 2.1$$

$$\sigma = 92.4 \pm 2.2$$

200

100

0

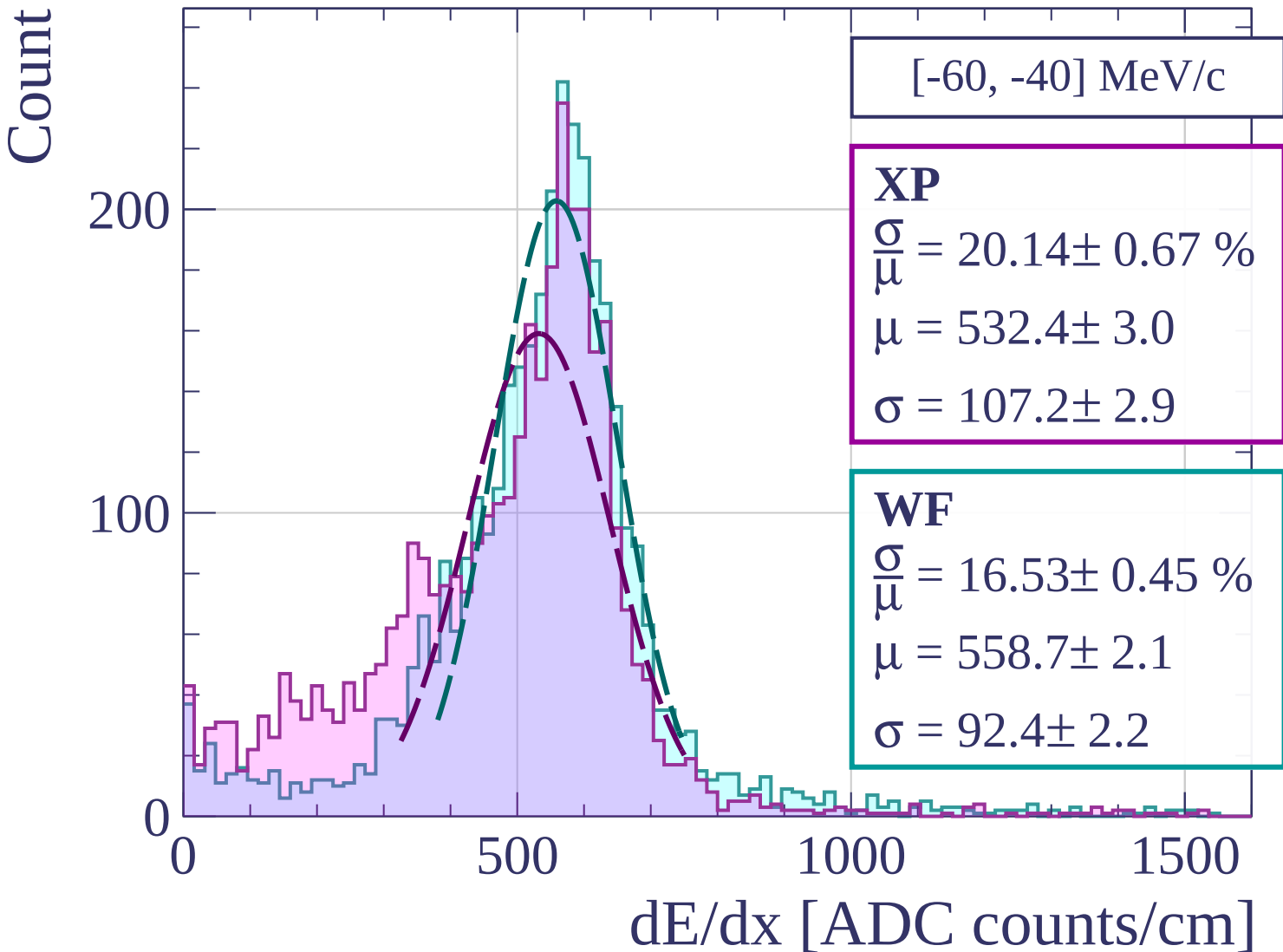
0

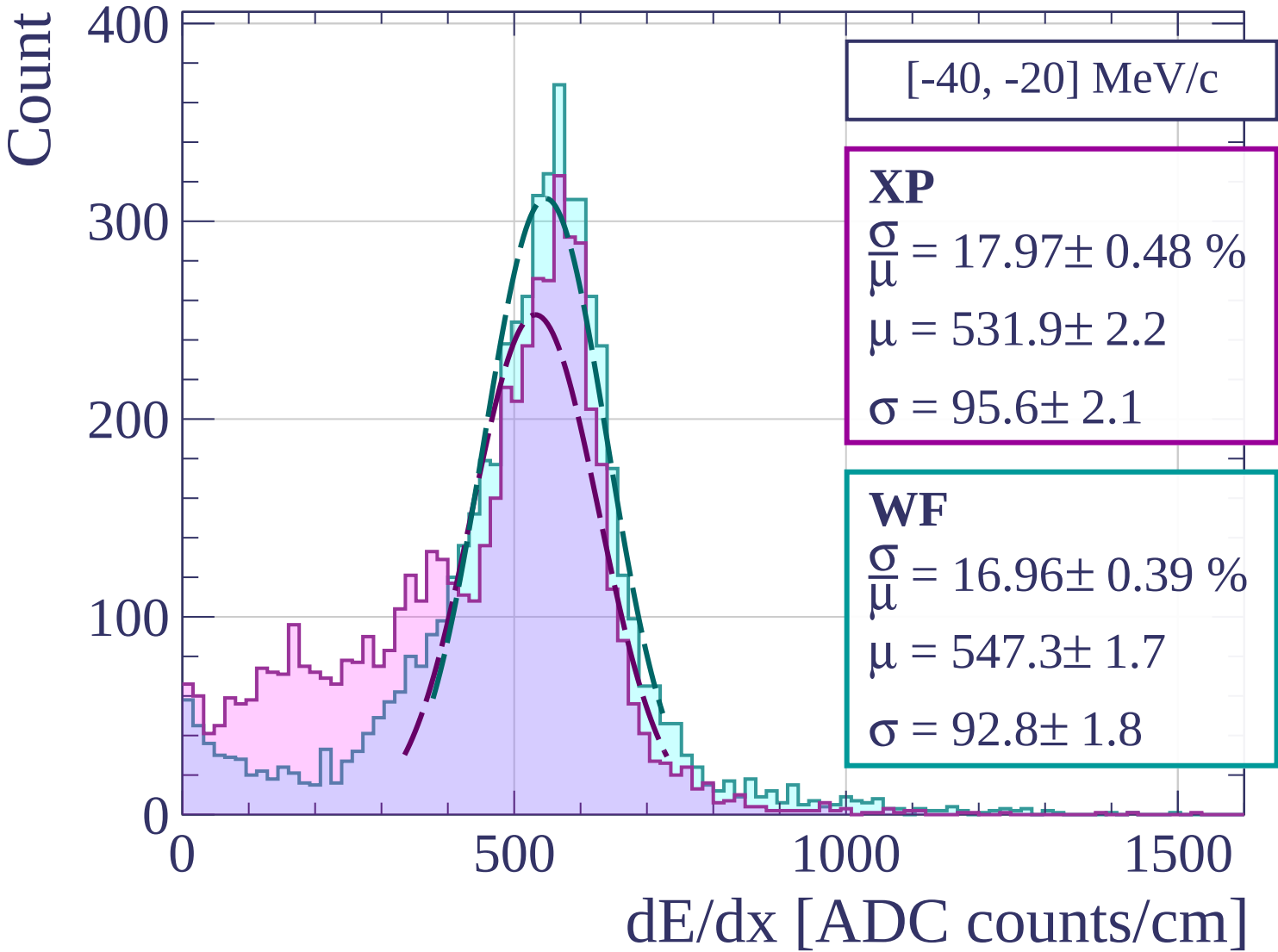
500

1000

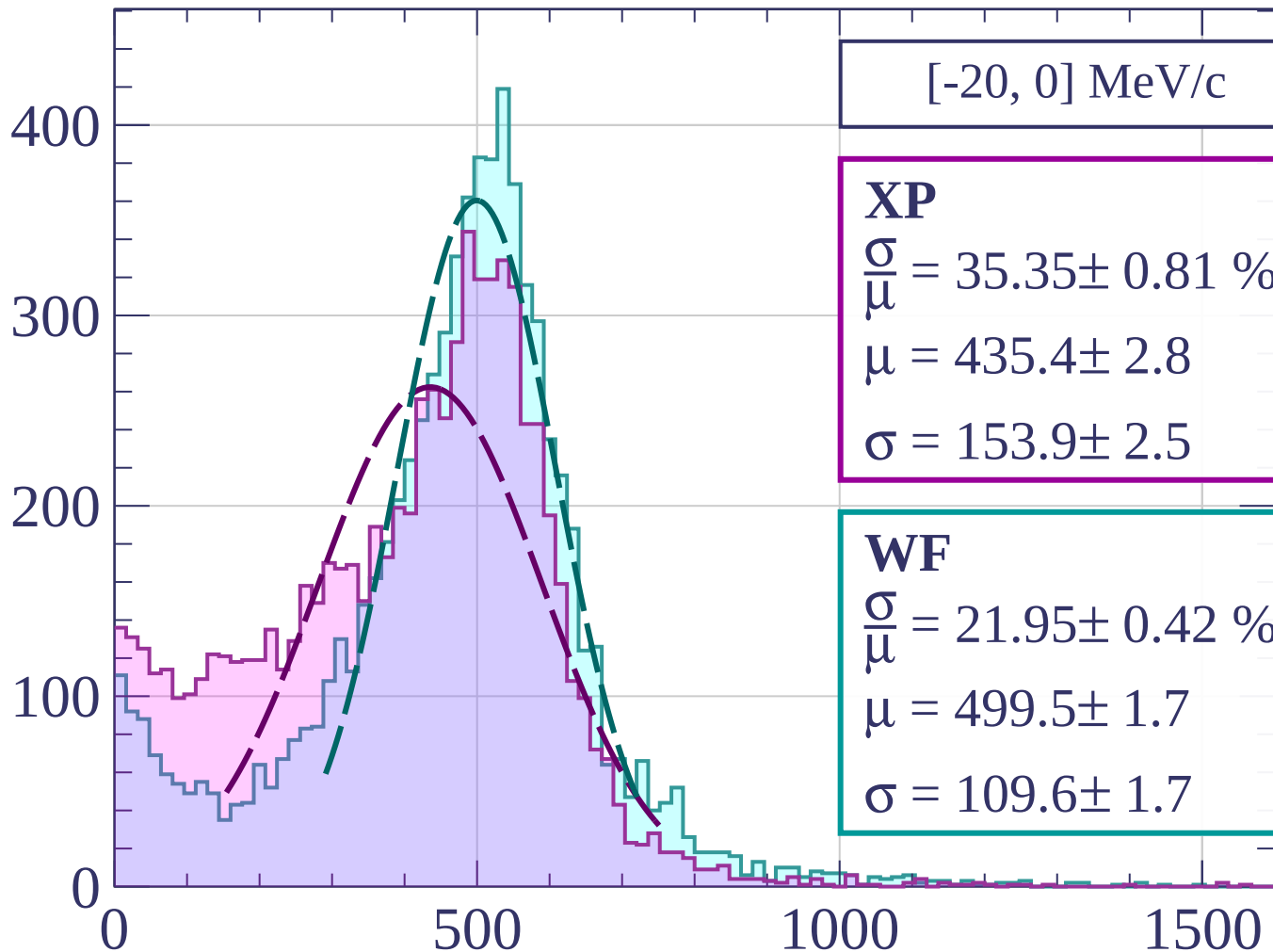
1500

dE/dx [ADC counts/cm]





Count



[-20, 0] MeV/c

XP

$$\frac{\sigma}{\mu} = 35.35 \pm 0.81 \%$$

$$\mu = 435.4 \pm 2.8$$

$$\sigma = 153.9 \pm 2.5$$

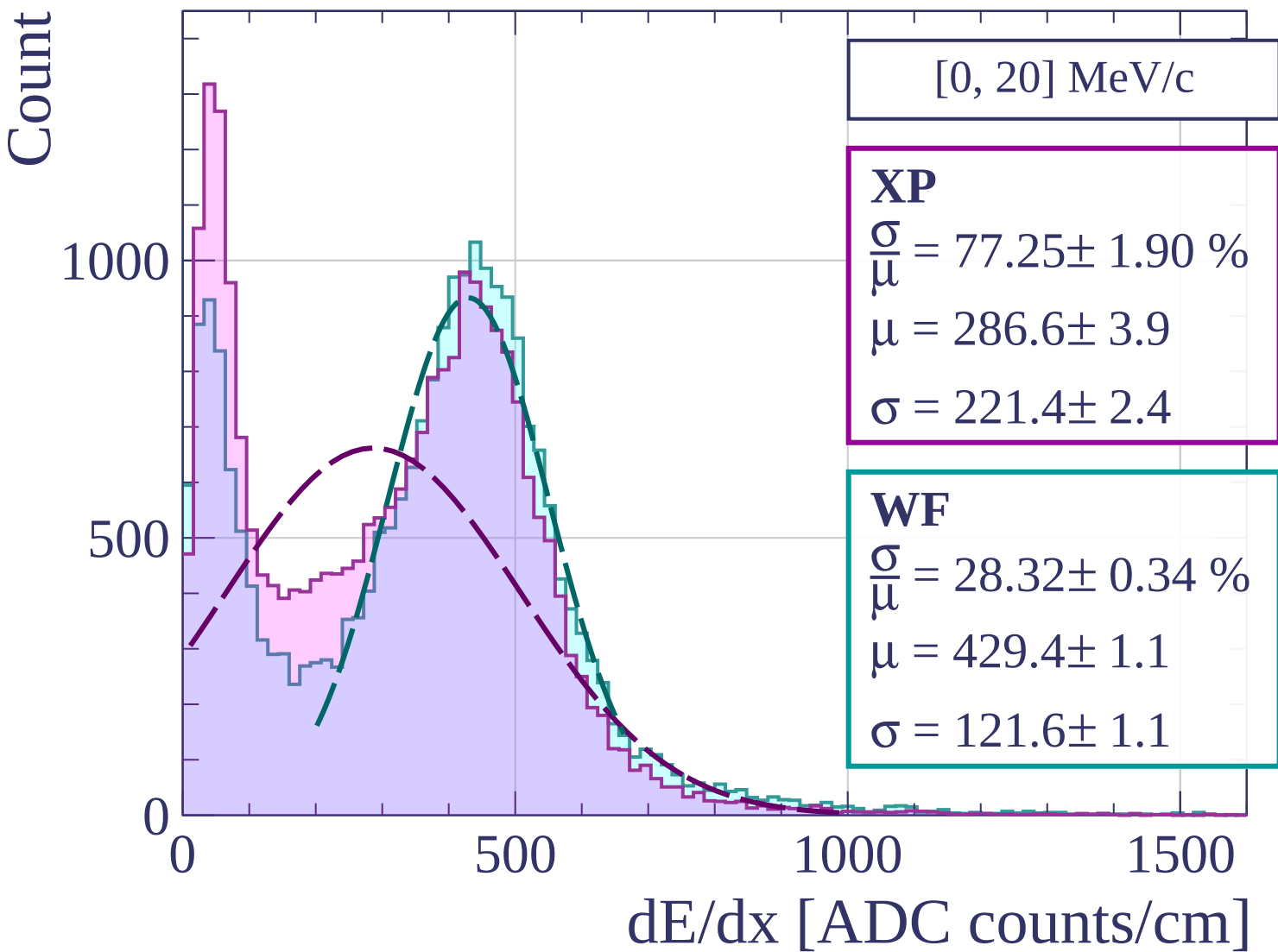
WF

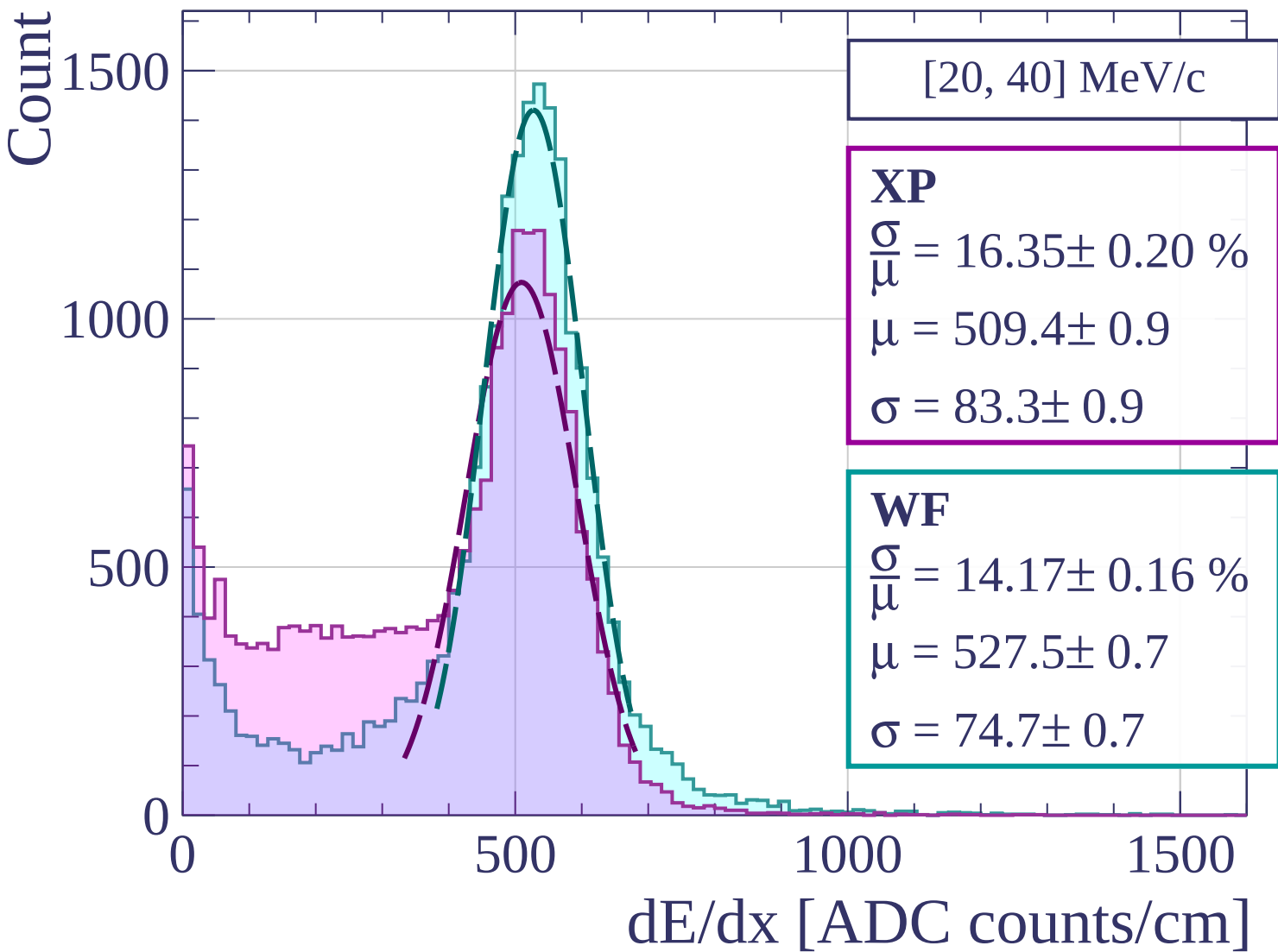
$$\frac{\sigma}{\mu} = 21.95 \pm 0.42 \%$$

$$\mu = 499.5 \pm 1.7$$

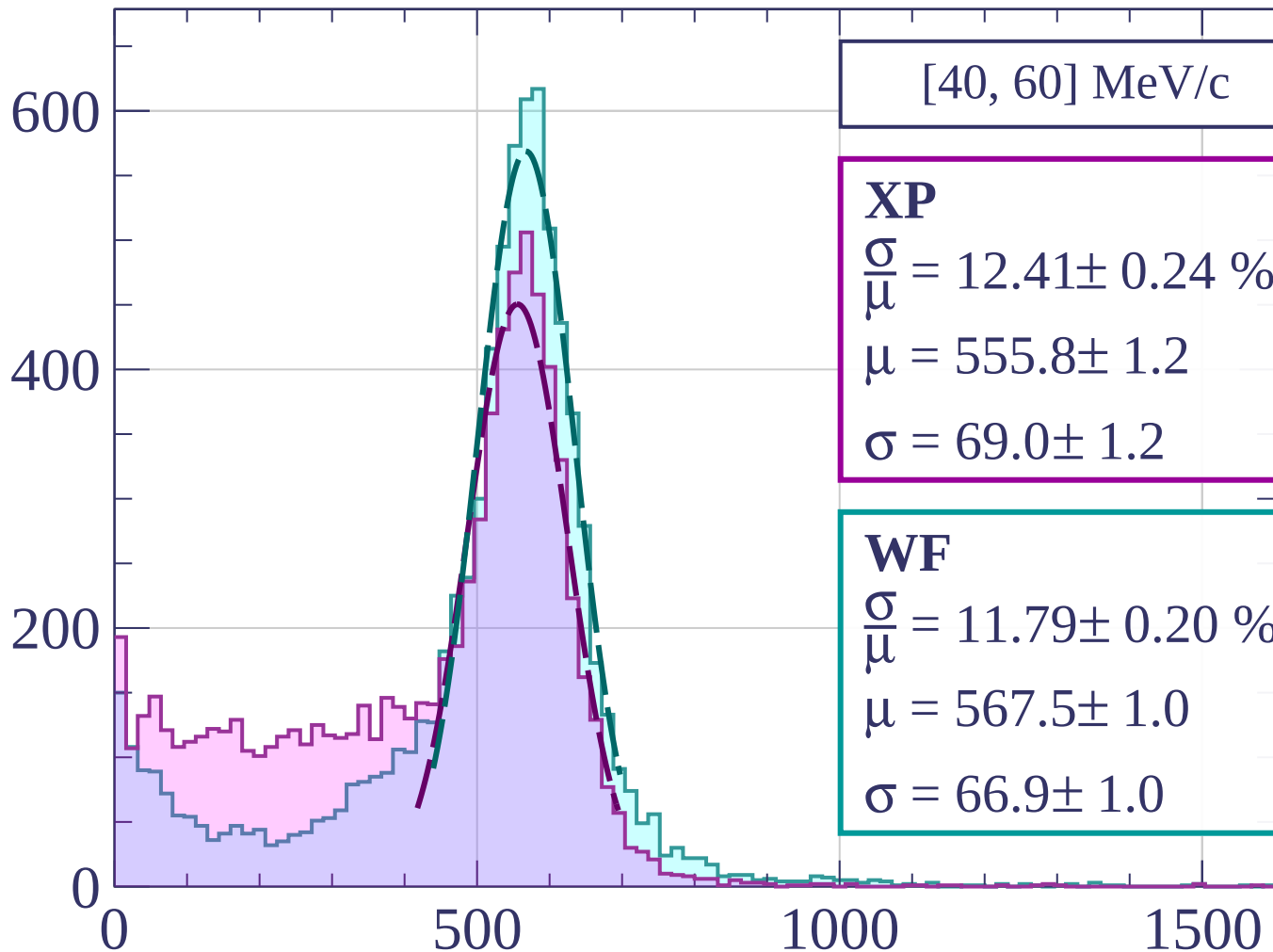
$$\sigma = 109.6 \pm 1.7$$

dE/dx [ADC counts/cm]





Count



[40, 60] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.41 \pm 0.24 \%$$

$$\mu = 555.8 \pm 1.2$$

$$\sigma = 69.0 \pm 1.2$$

WF

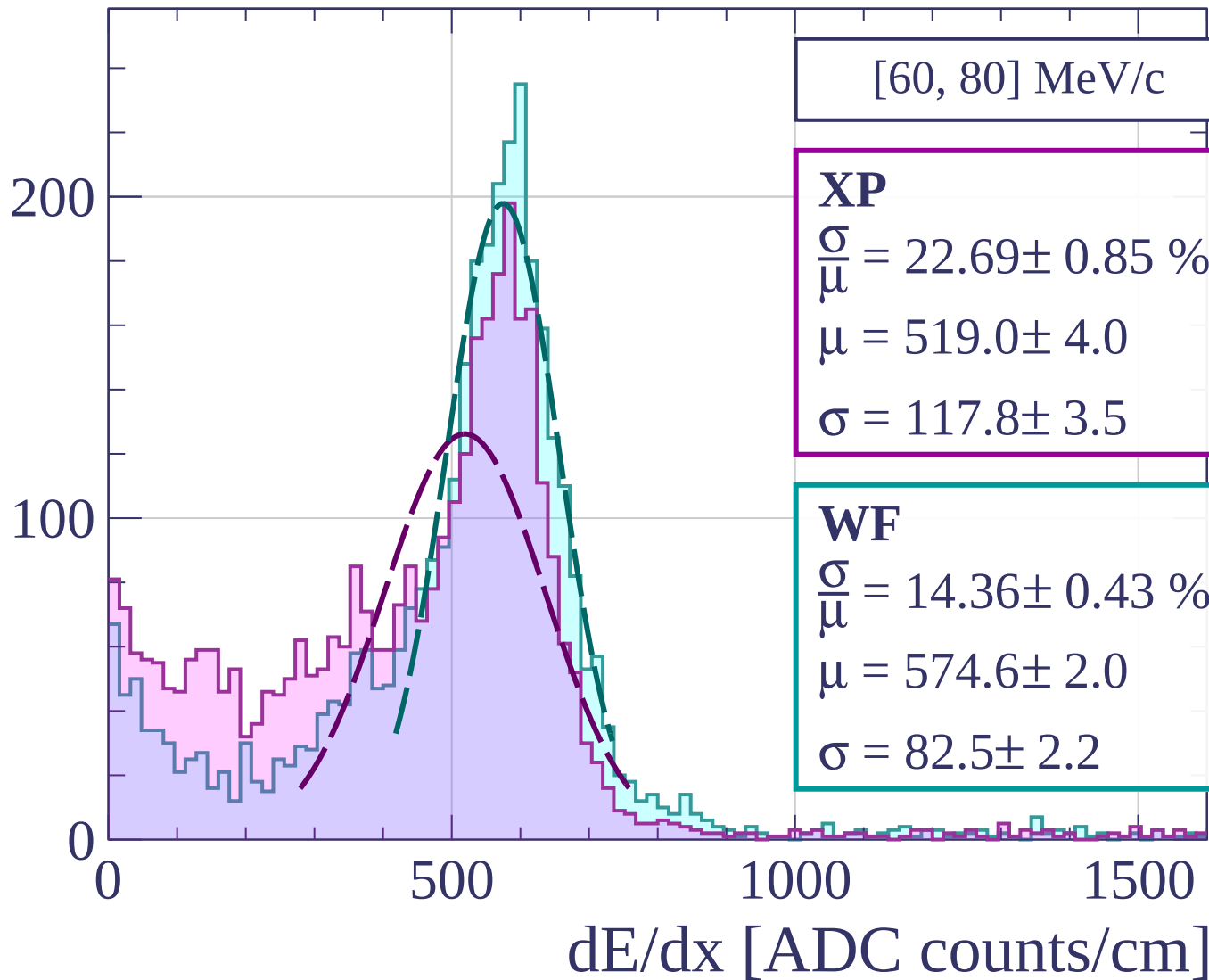
$$\frac{\sigma}{\mu} = 11.79 \pm 0.20 \%$$

$$\mu = 567.5 \pm 1.0$$

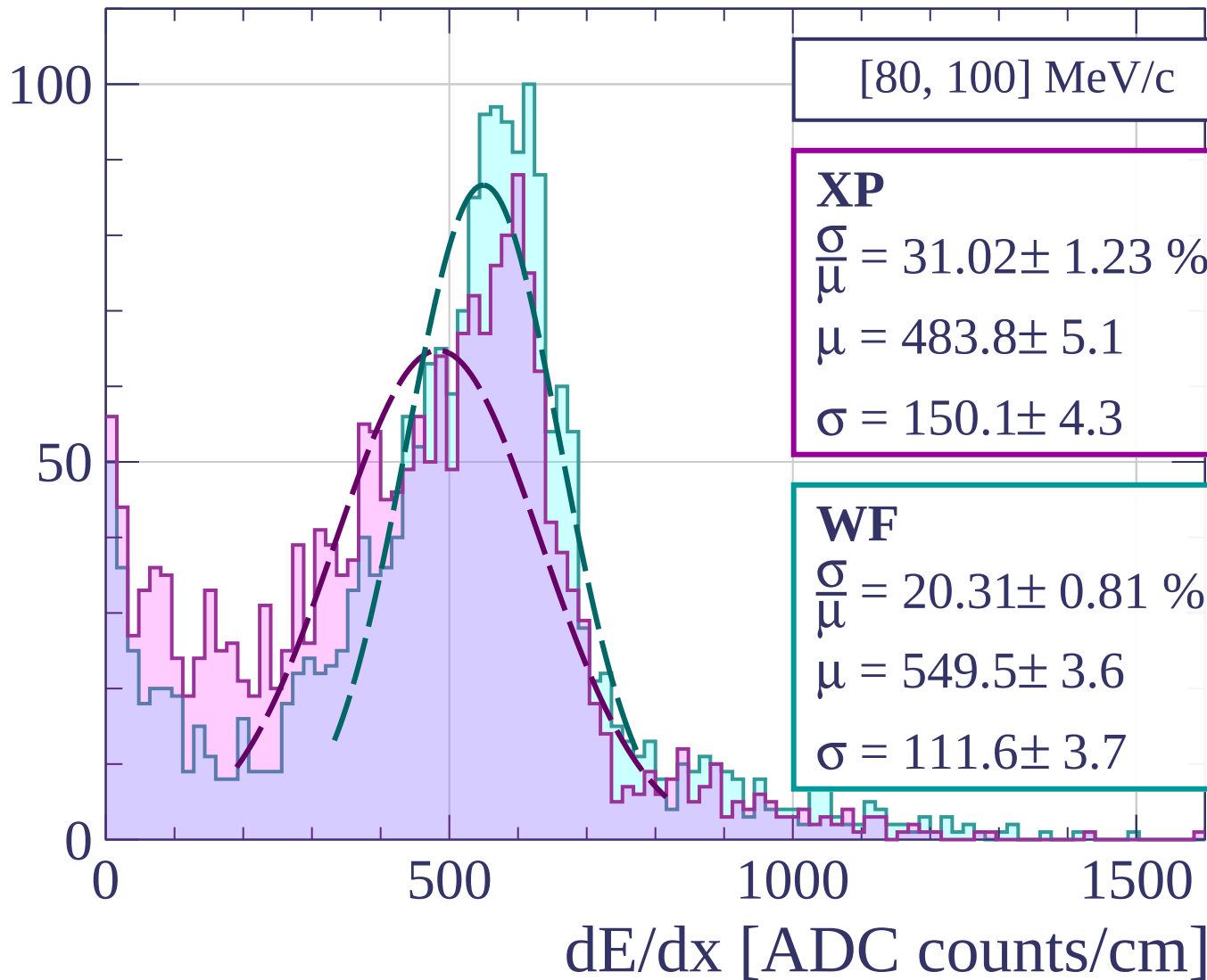
$$\sigma = 66.9 \pm 1.0$$

dE/dx [ADC counts/cm]

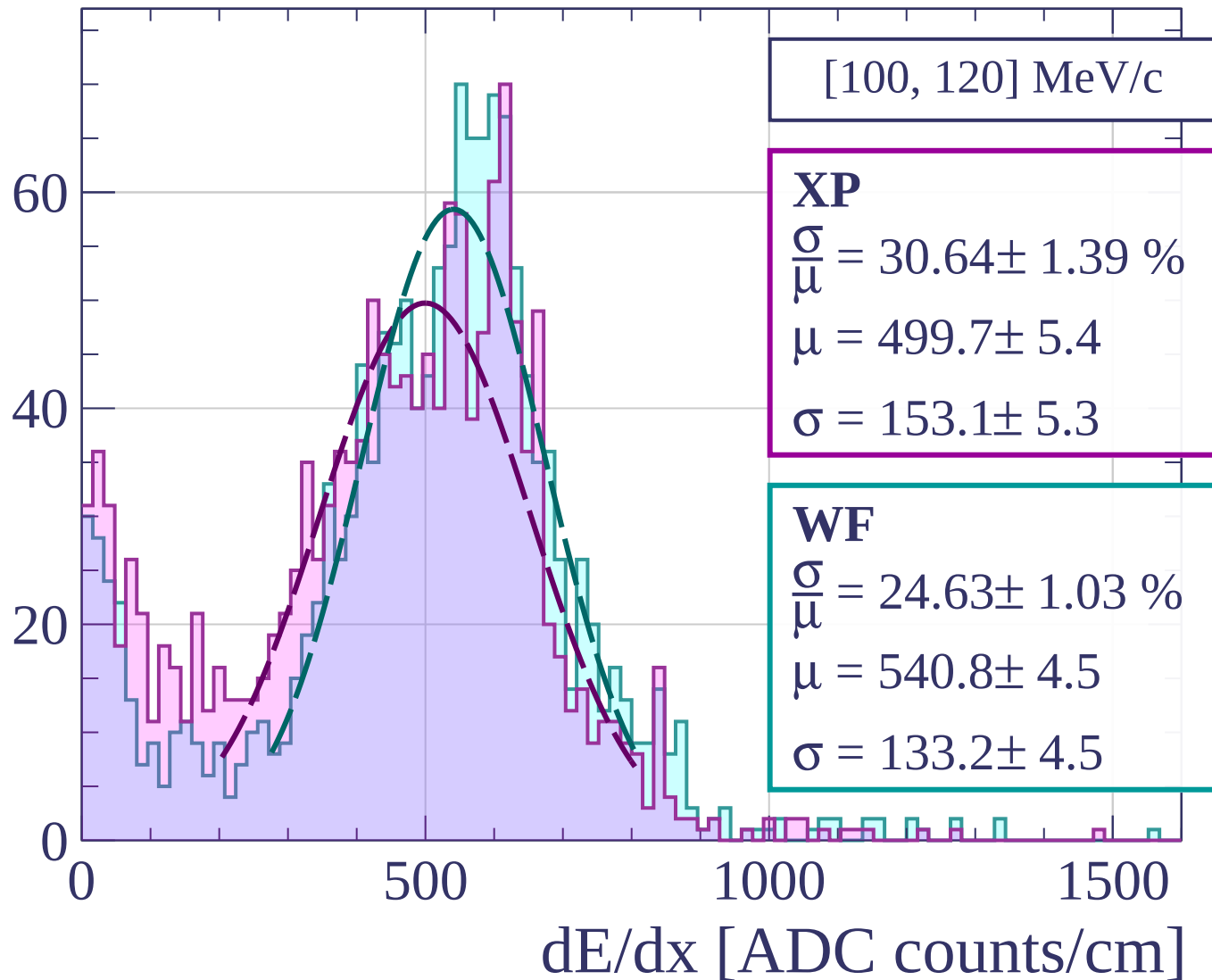
Count



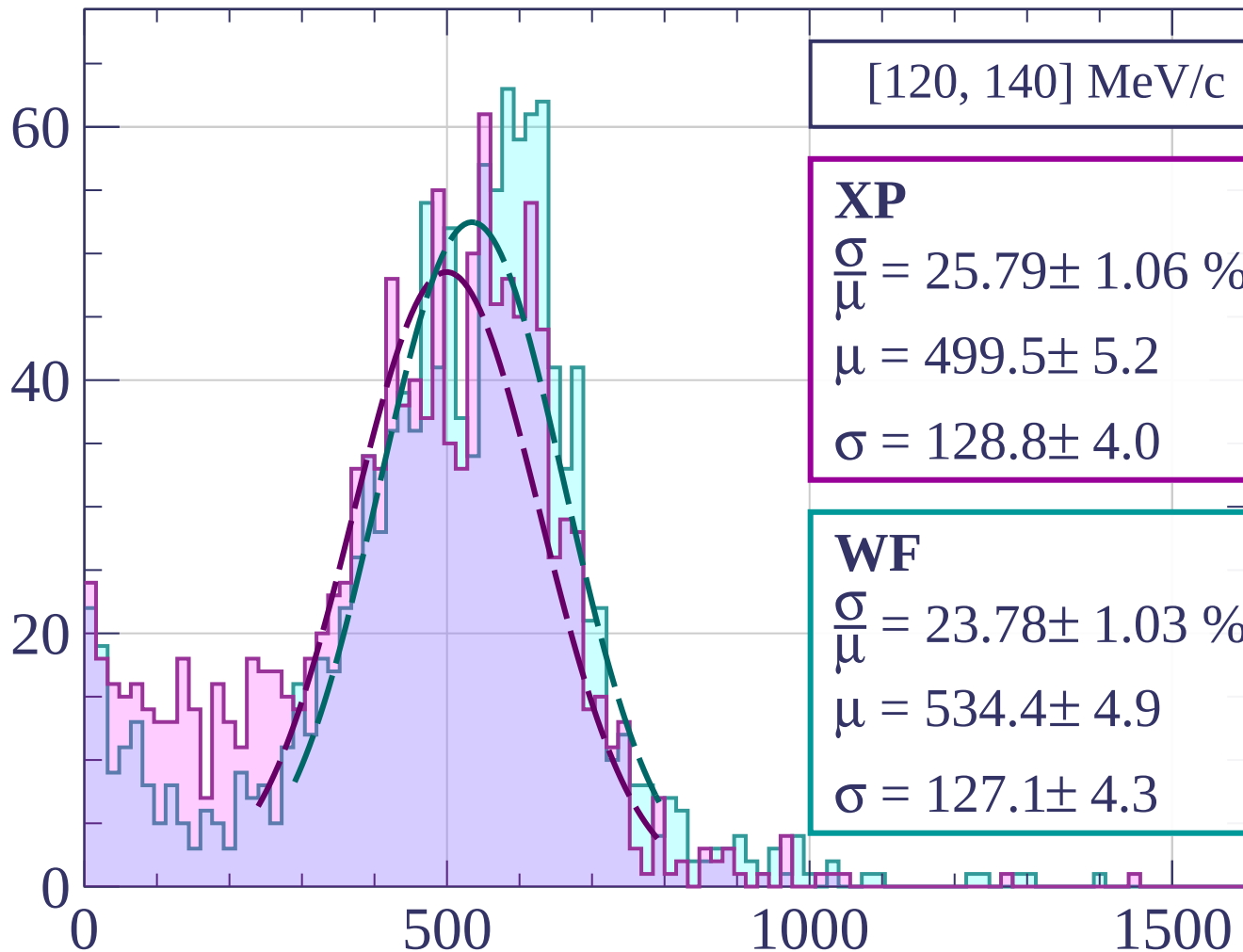
Count



Count



Count



[120, 140] MeV/c

XP

$$\frac{\sigma}{\mu} = 25.79 \pm 1.06 \%$$

$$\mu = 499.5 \pm 5.2$$

$$\sigma = 128.8 \pm 4.0$$

WF

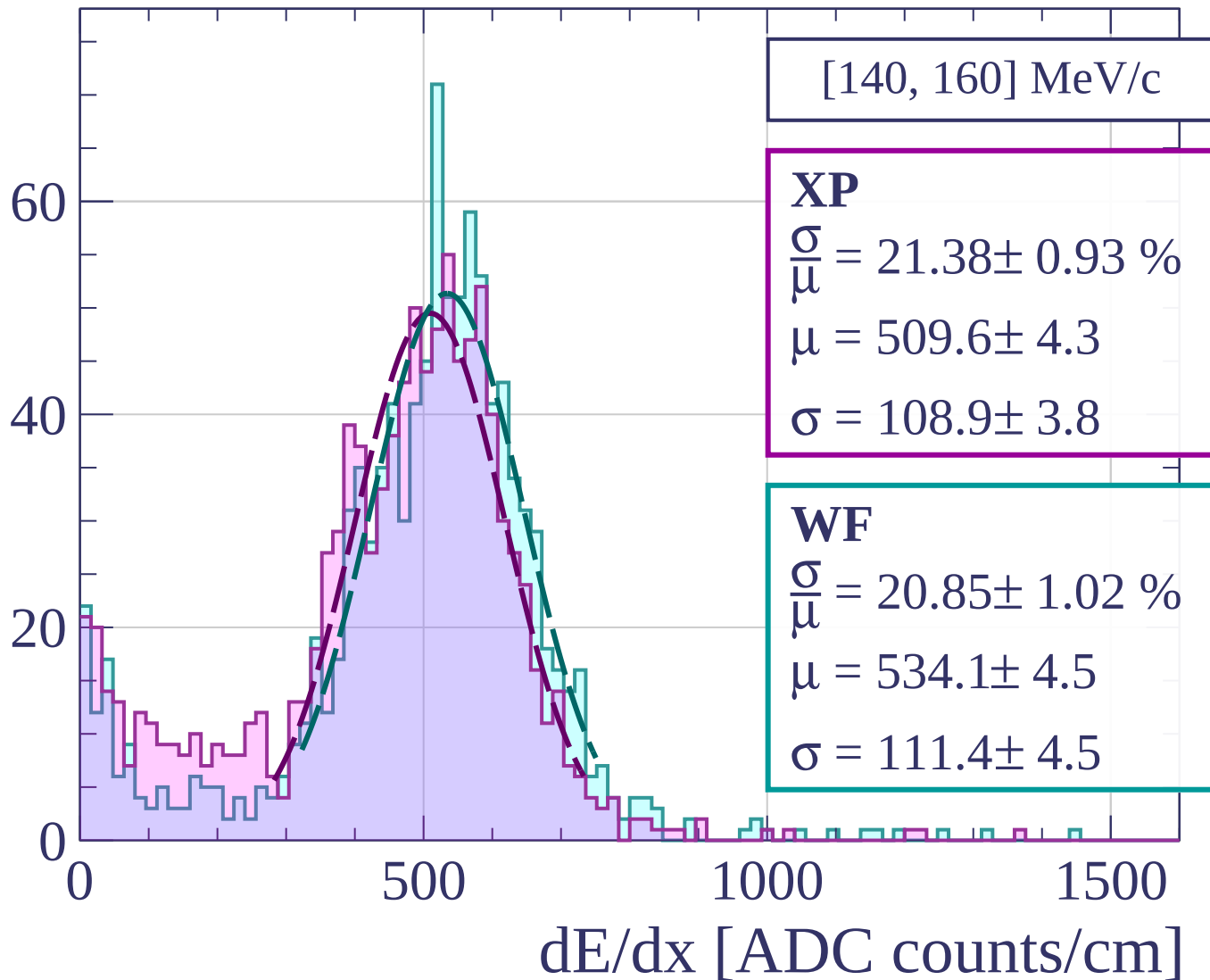
$$\frac{\sigma}{\mu} = 23.78 \pm 1.03 \%$$

$$\mu = 534.4 \pm 4.9$$

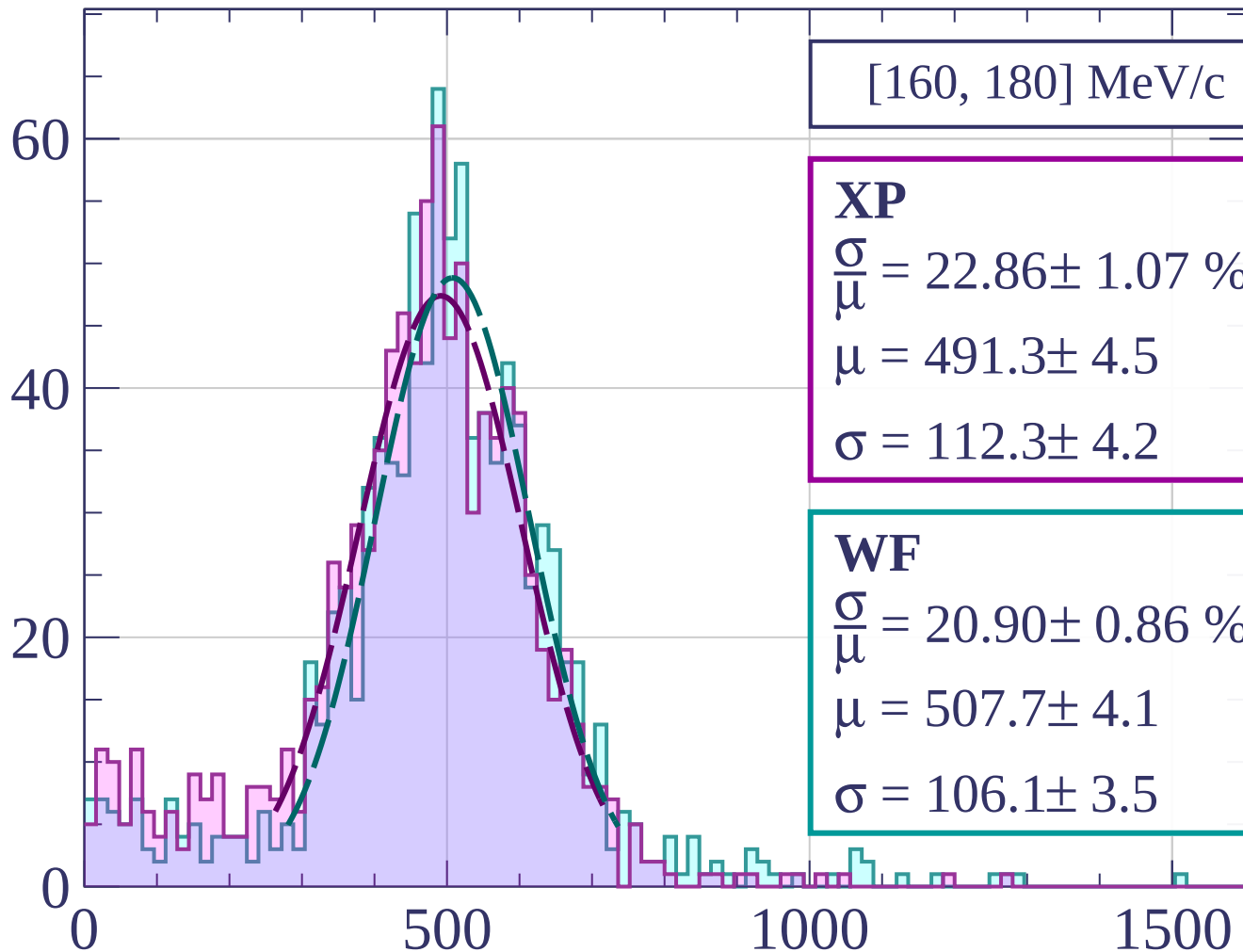
$$\sigma = 127.1 \pm 4.3$$

dE/dx [ADC counts/cm]

Count

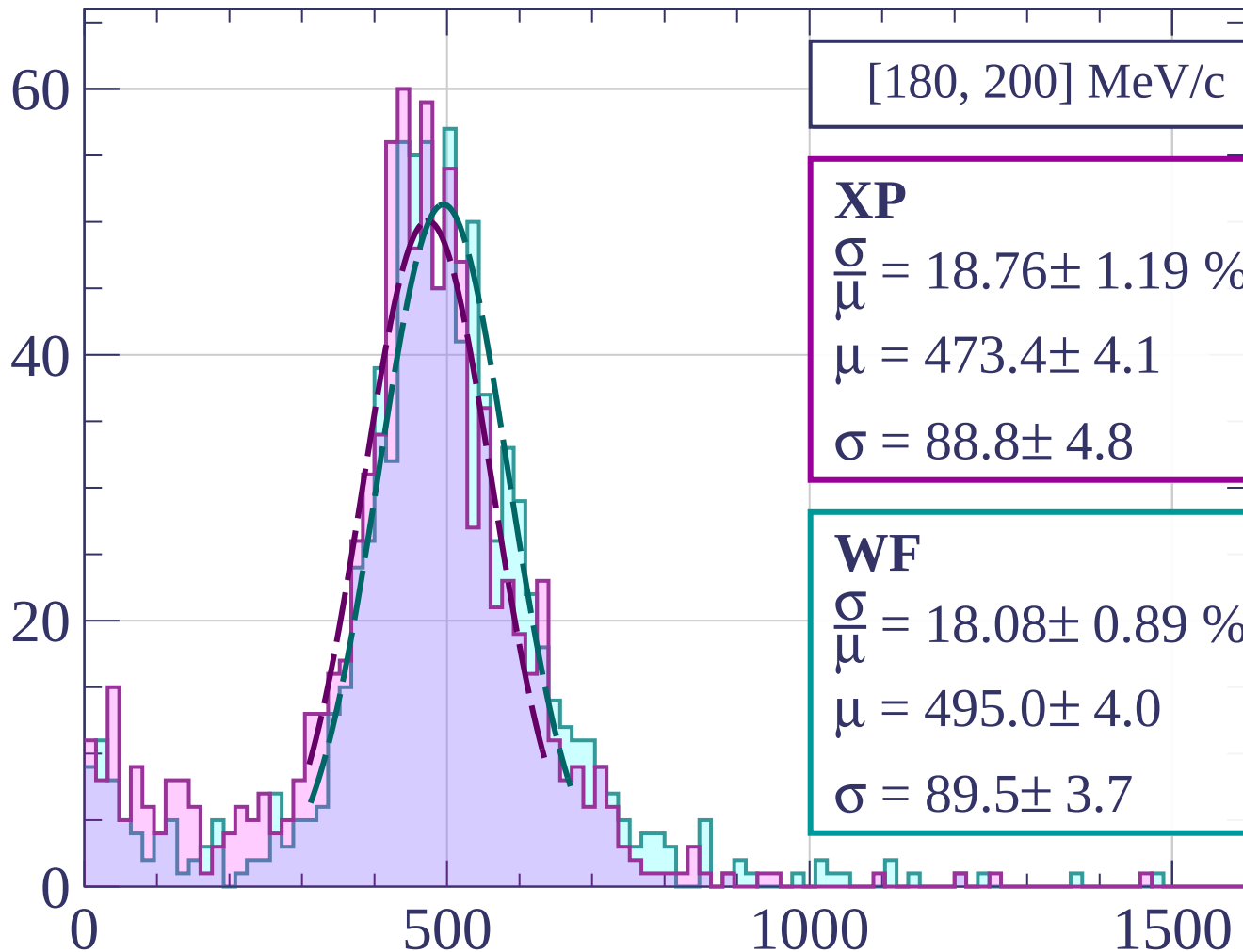


Count



dE/dx [ADC counts/cm]

Count



[180, 200] MeV/c

XP

$$\frac{\sigma}{\mu} = 18.76 \pm 1.19 \%$$

$$\mu = 473.4 \pm 4.1$$

$$\sigma = 88.8 \pm 4.8$$

WF

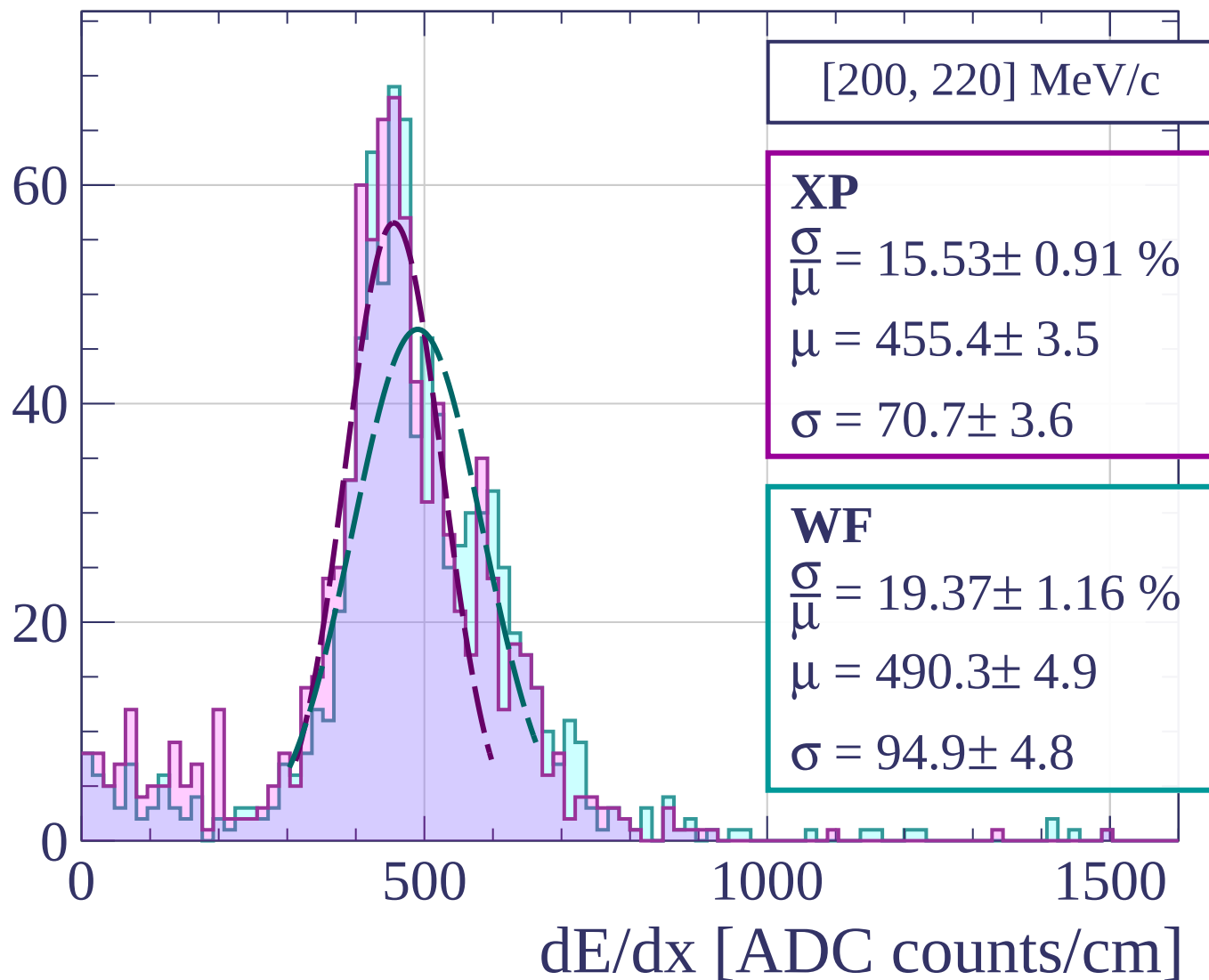
$$\frac{\sigma}{\mu} = 18.08 \pm 0.89 \%$$

$$\mu = 495.0 \pm 4.0$$

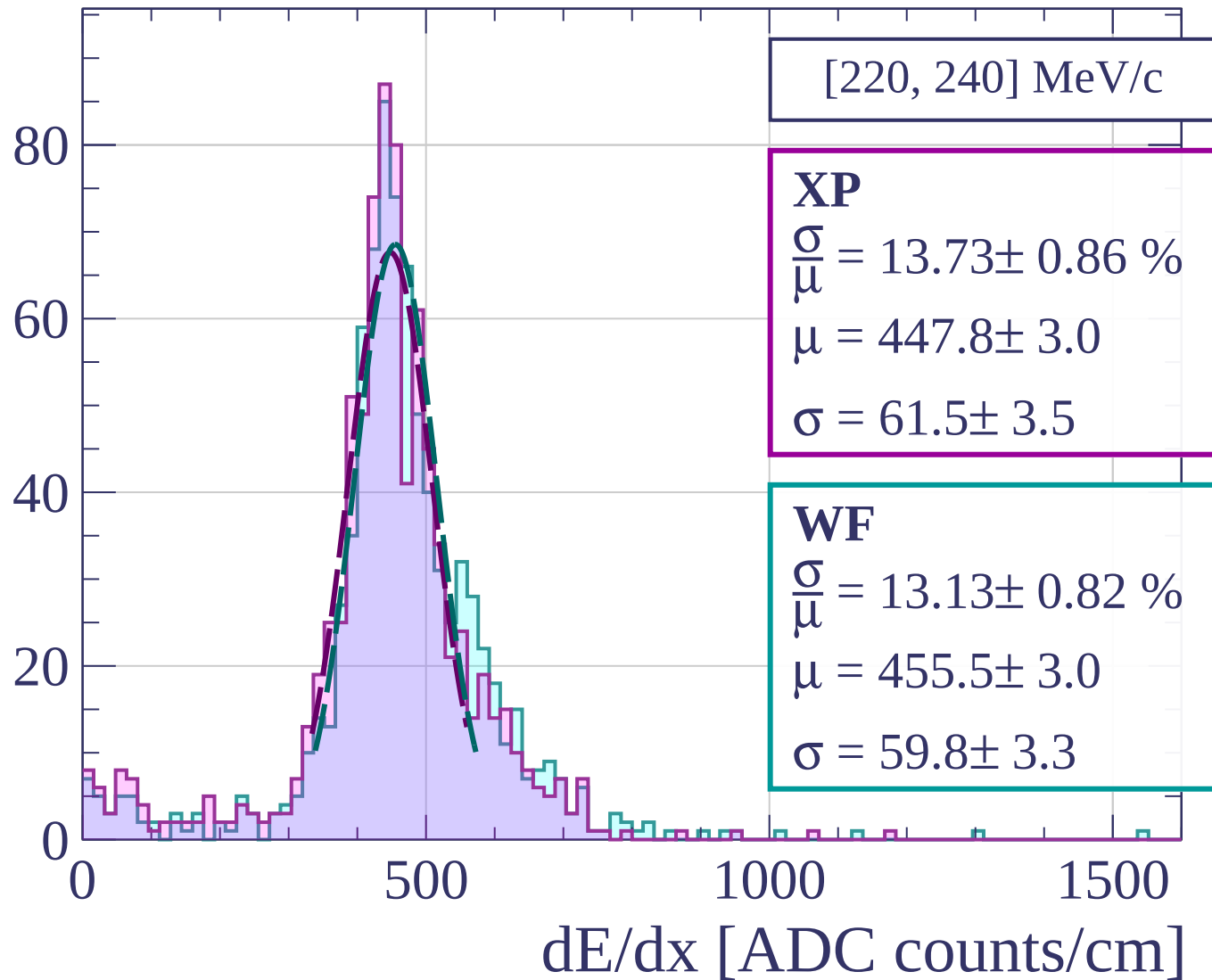
$$\sigma = 89.5 \pm 3.7$$

dE/dx [ADC counts/cm]

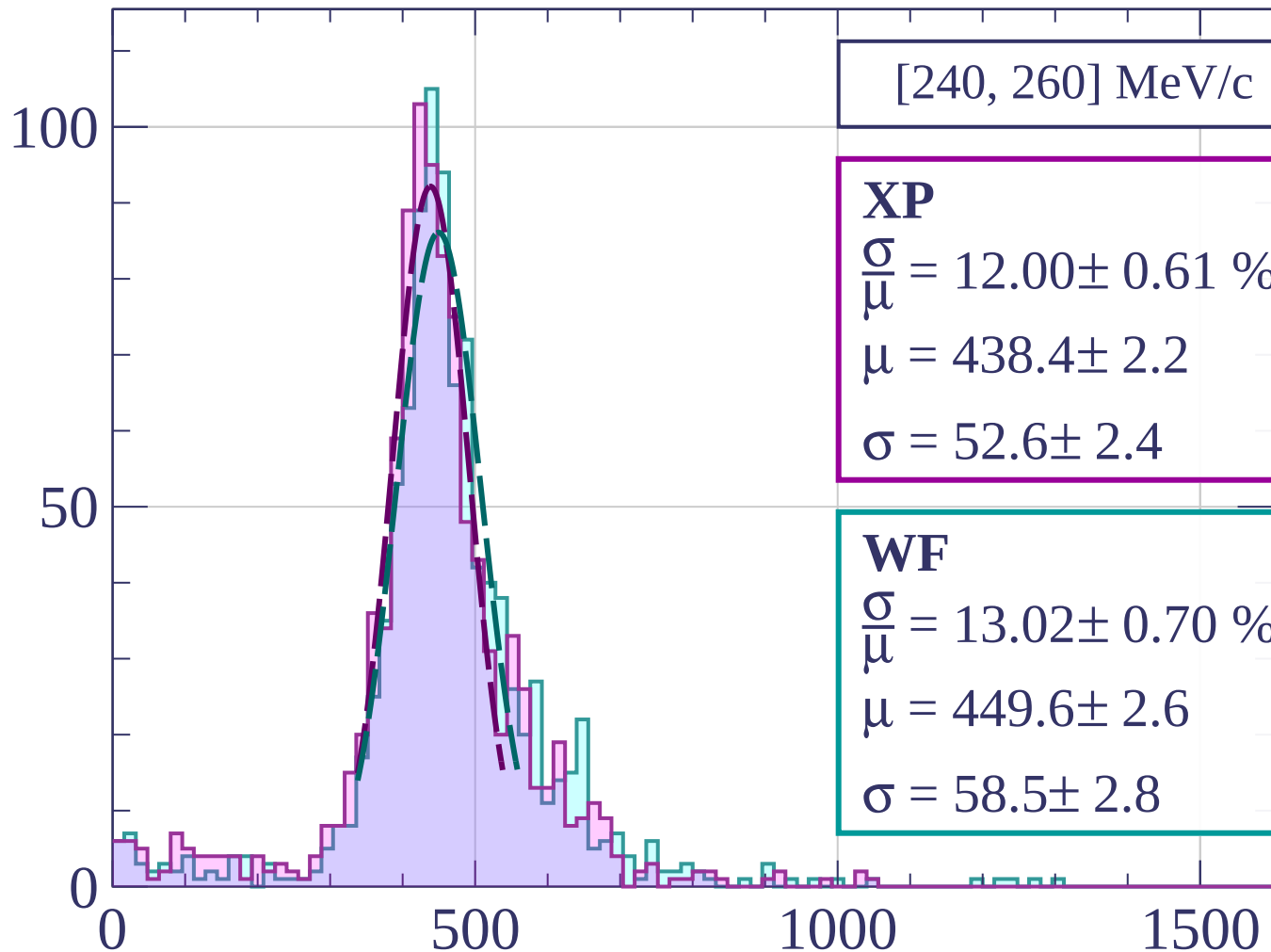
Count



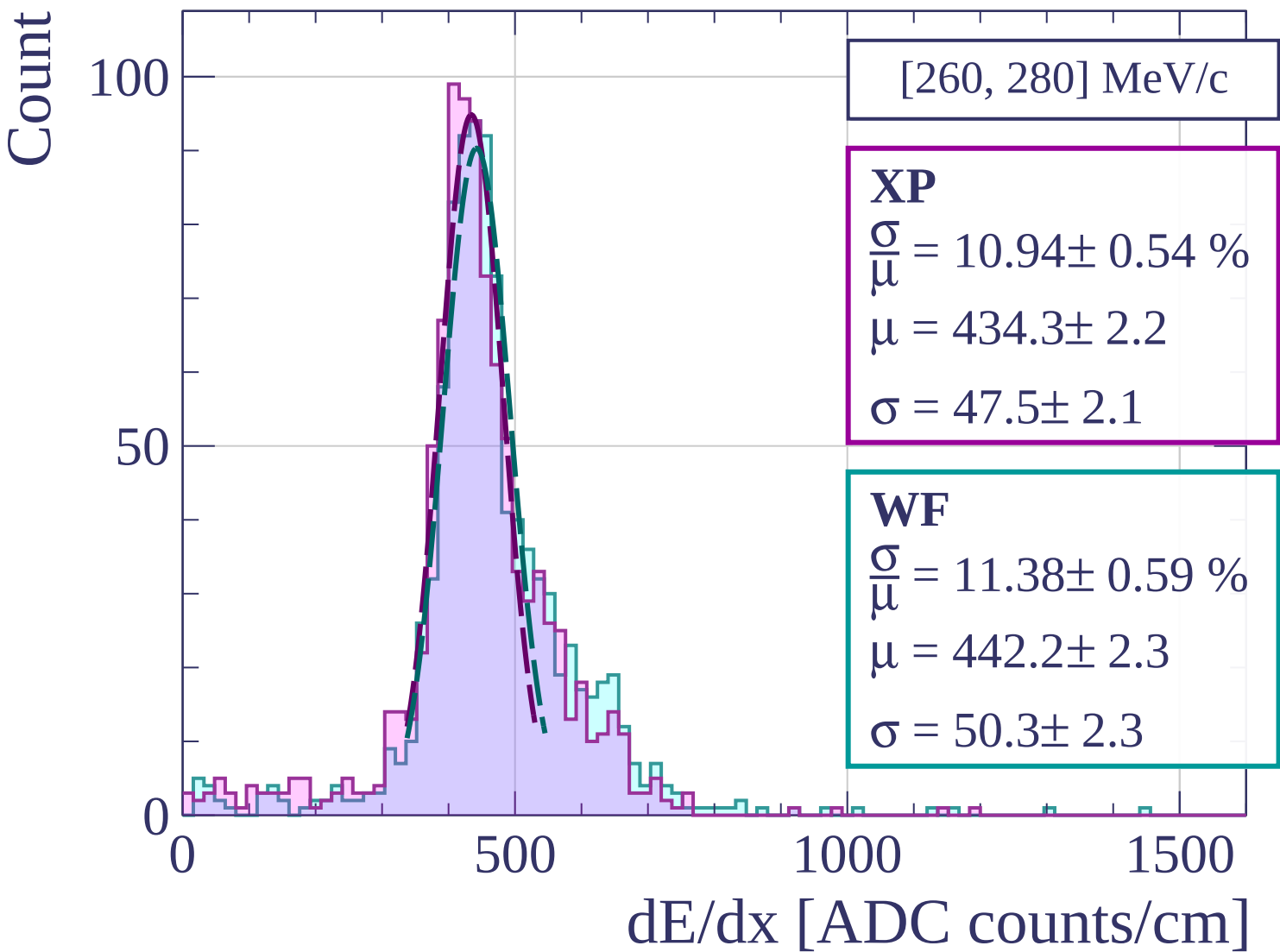
Count



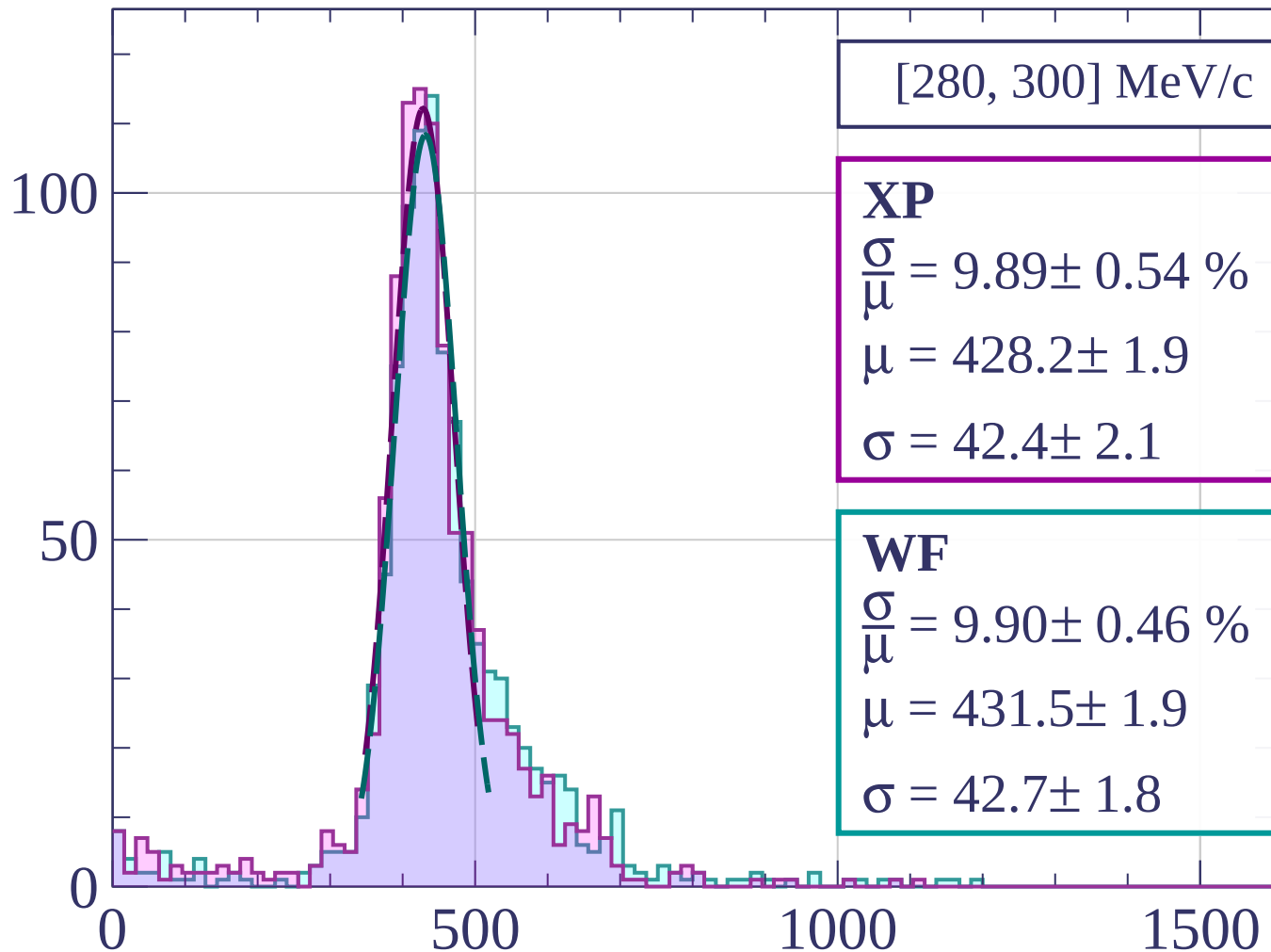
Count



dE/dx [ADC counts/cm]



Count



[280, 300] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.89 \pm 0.54 \%$$

$$\mu = 428.2 \pm 1.9$$

$$\sigma = 42.4 \pm 2.1$$

WF

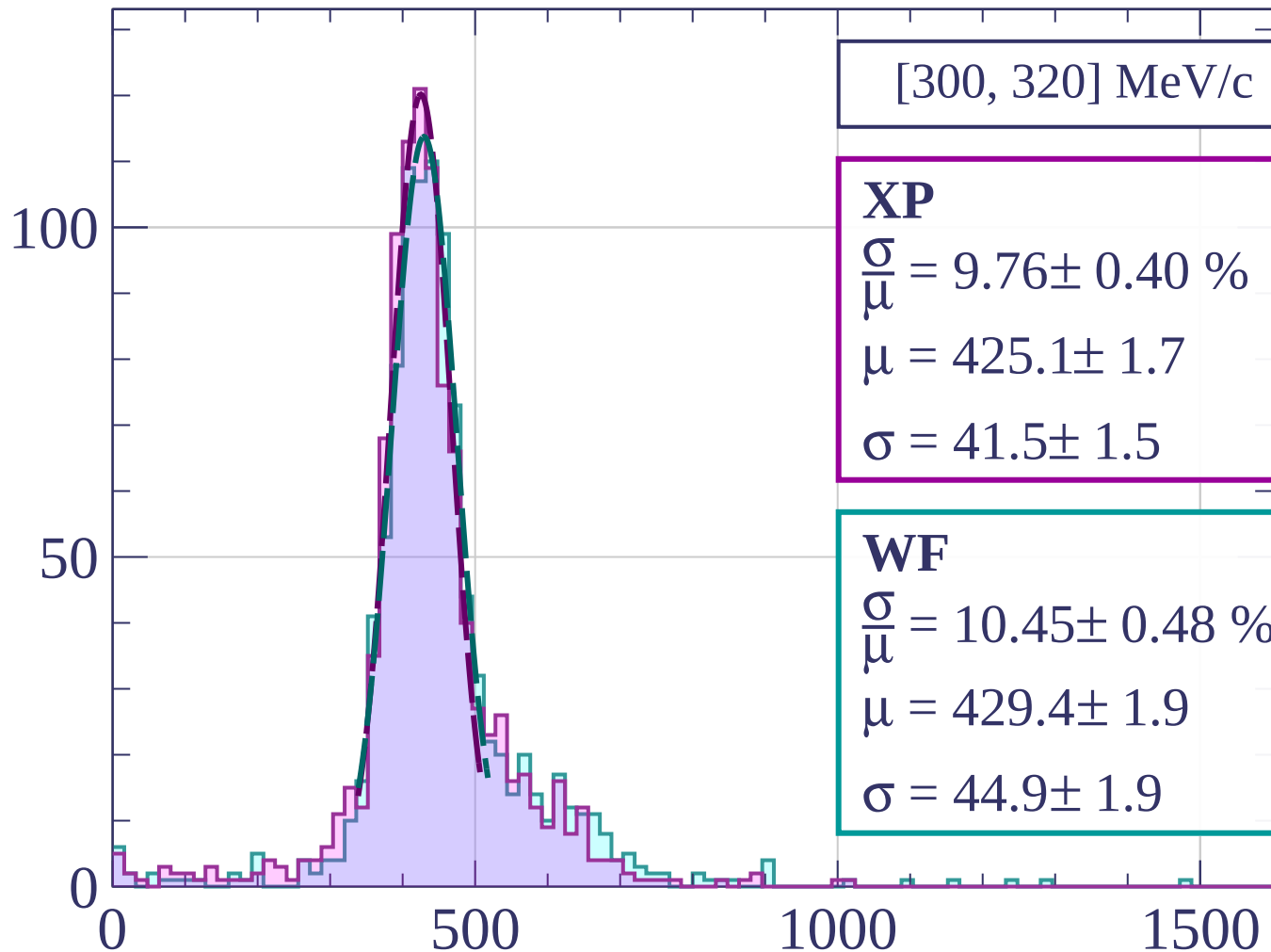
$$\frac{\sigma}{\mu} = 9.90 \pm 0.46 \%$$

$$\mu = 431.5 \pm 1.9$$

$$\sigma = 42.7 \pm 1.8$$

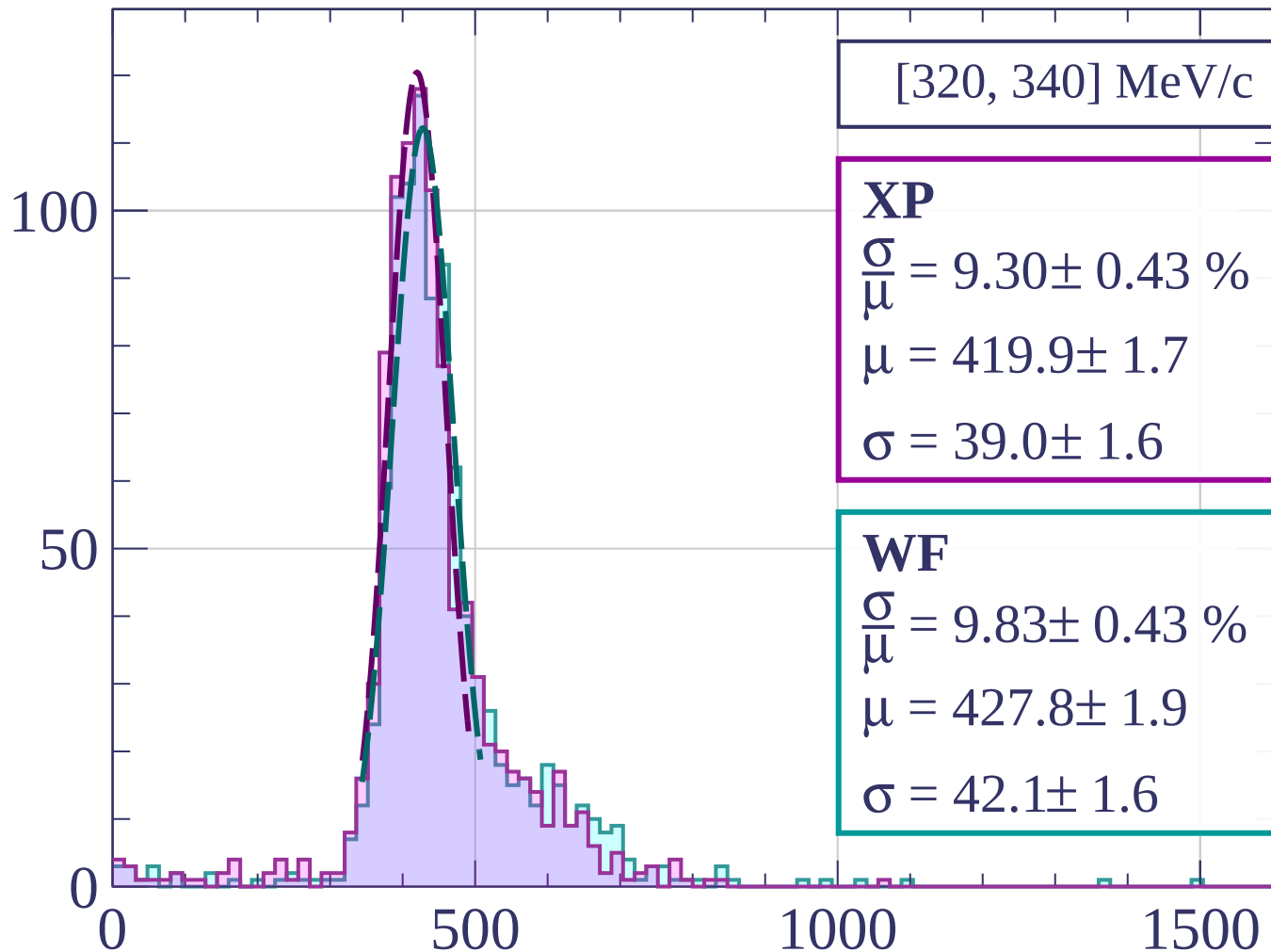
dE/dx [ADC counts/cm]

Count



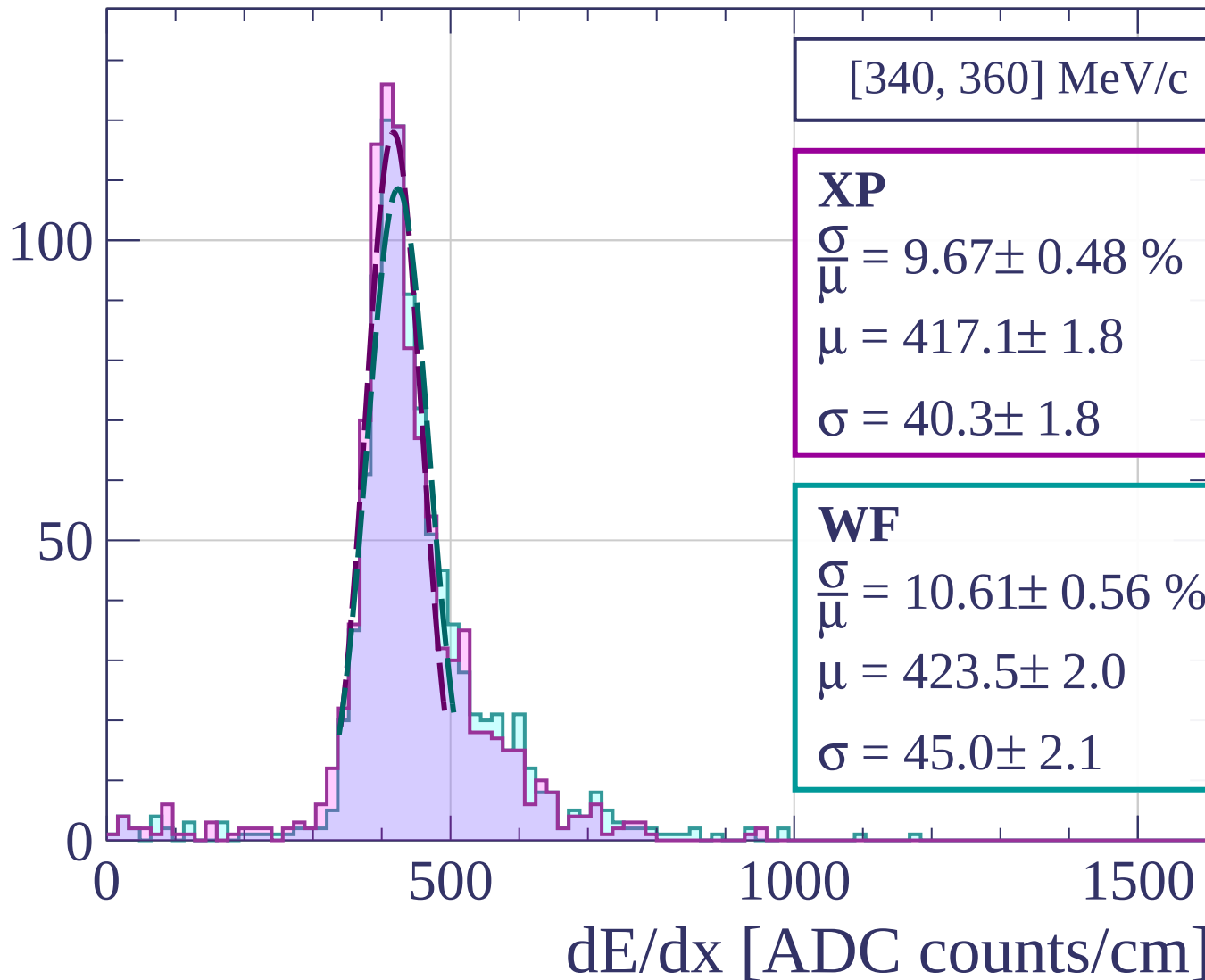
dE/dx [ADC counts/cm]

Count

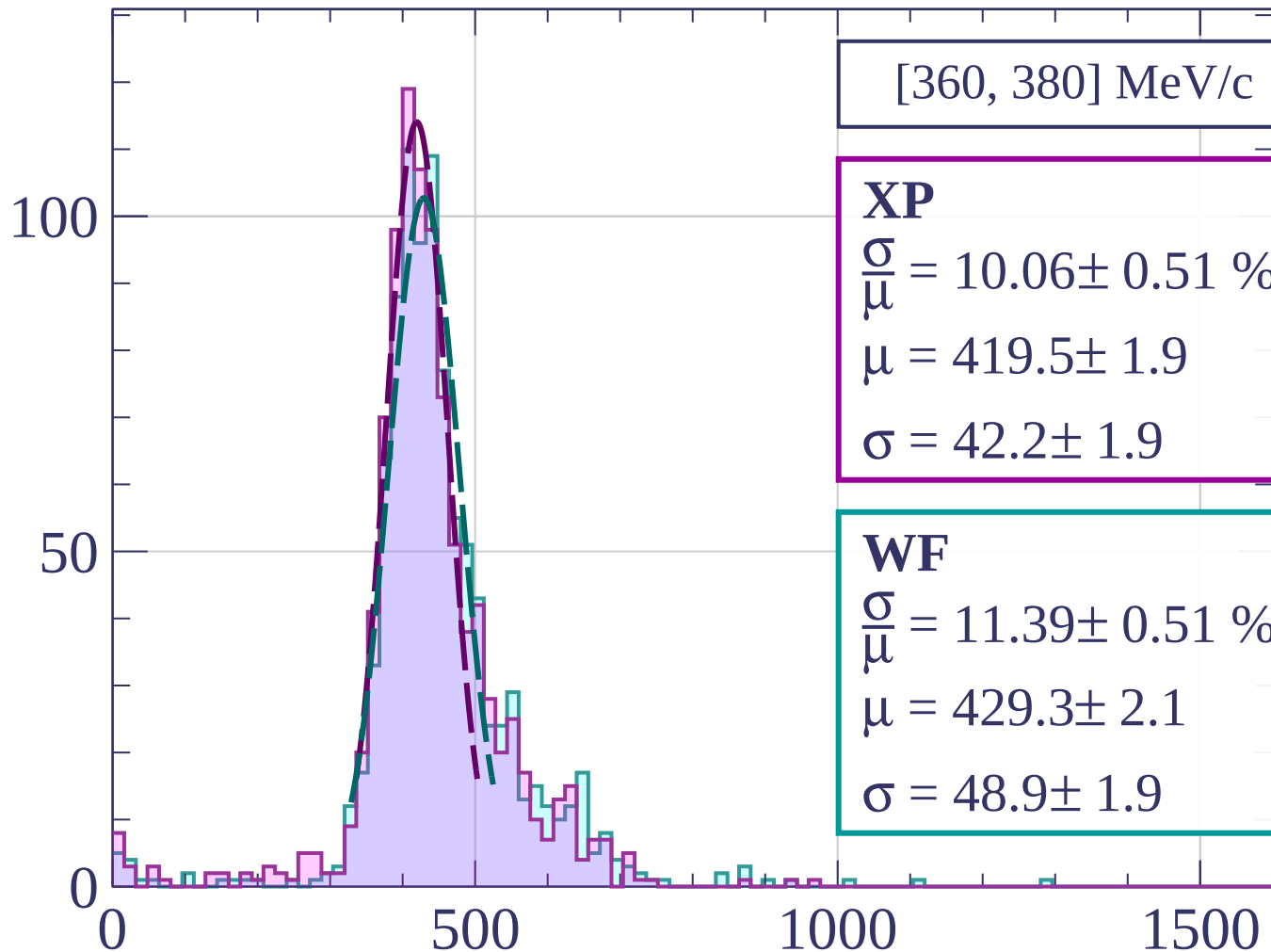


dE/dx [ADC counts/cm]

Count



Count



[360, 380] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.06 \pm 0.51 \%$$

$$\mu = 419.5 \pm 1.9$$

$$\sigma = 42.2 \pm 1.9$$

WF

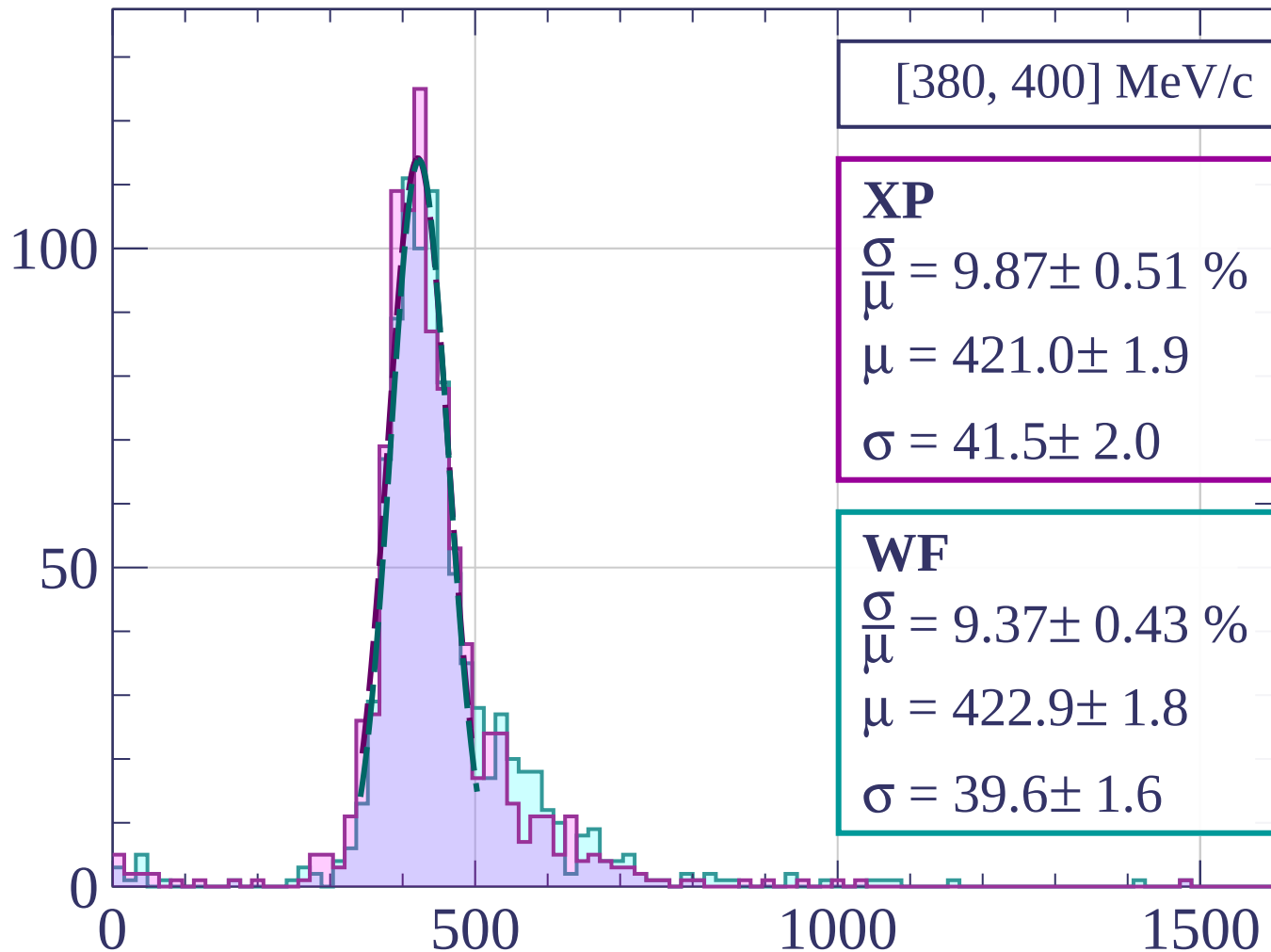
$$\frac{\sigma}{\mu} = 11.39 \pm 0.51 \%$$

$$\mu = 429.3 \pm 2.1$$

$$\sigma = 48.9 \pm 1.9$$

dE/dx [ADC counts/cm]

Count



[380, 400] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.87 \pm 0.51 \%$$

$$\mu = 421.0 \pm 1.9$$

$$\sigma = 41.5 \pm 2.0$$

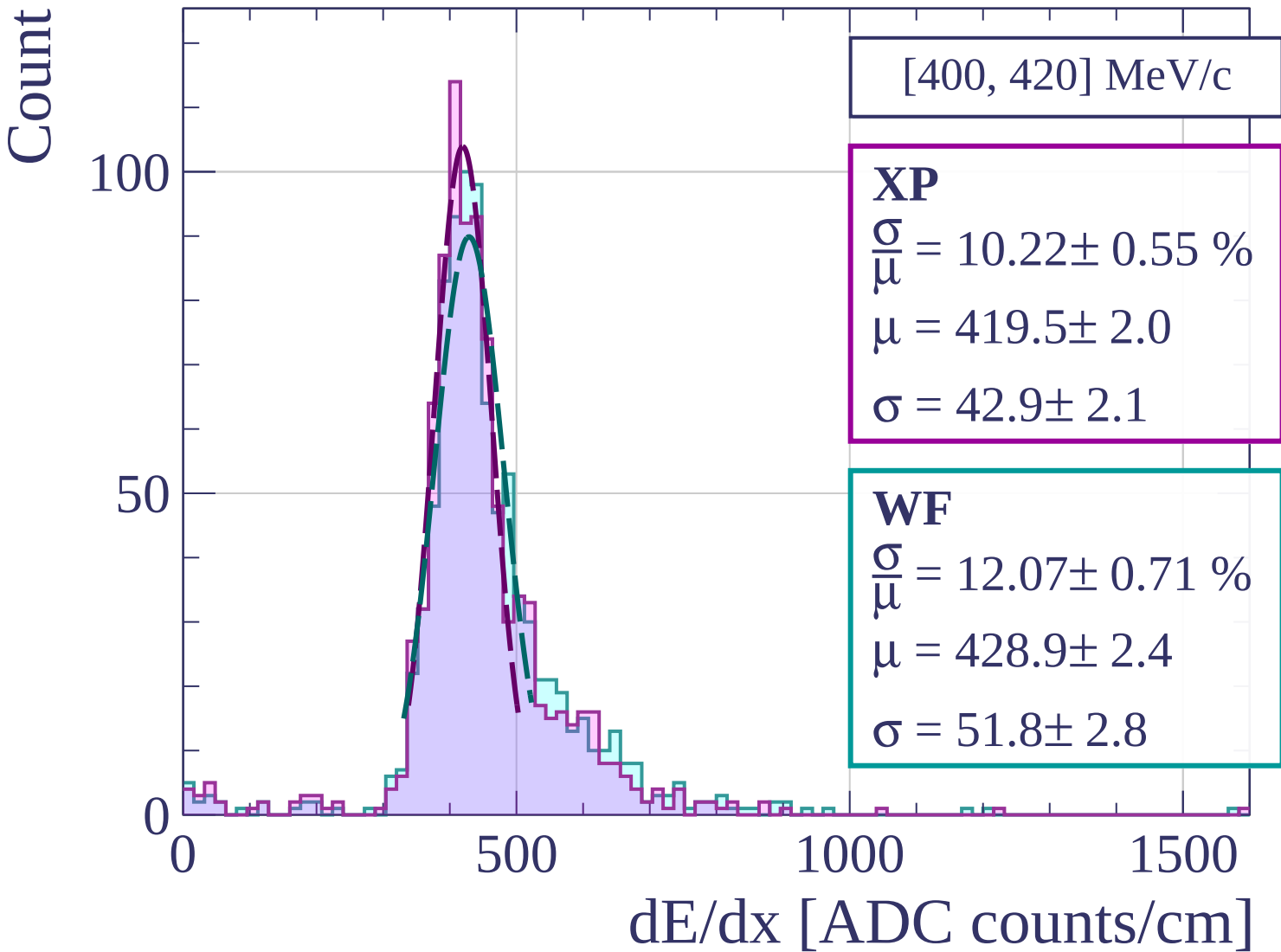
WF

$$\frac{\sigma}{\mu} = 9.37 \pm 0.43 \%$$

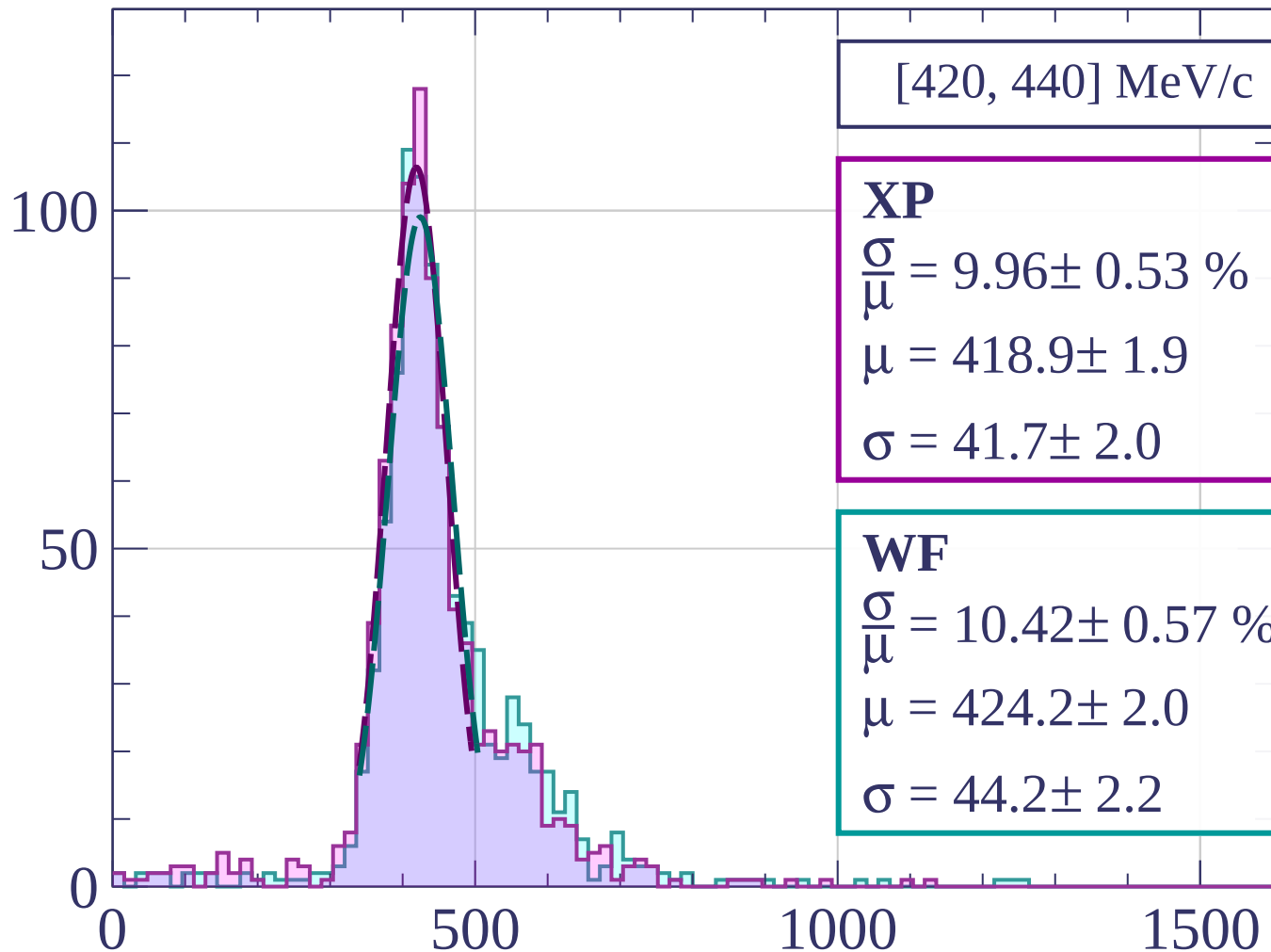
$$\mu = 422.9 \pm 1.8$$

$$\sigma = 39.6 \pm 1.6$$

dE/dx [ADC counts/cm]



Count



[420, 440] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.96 \pm 0.53 \%$$

$$\mu = 418.9 \pm 1.9$$

$$\sigma = 41.7 \pm 2.0$$

WF

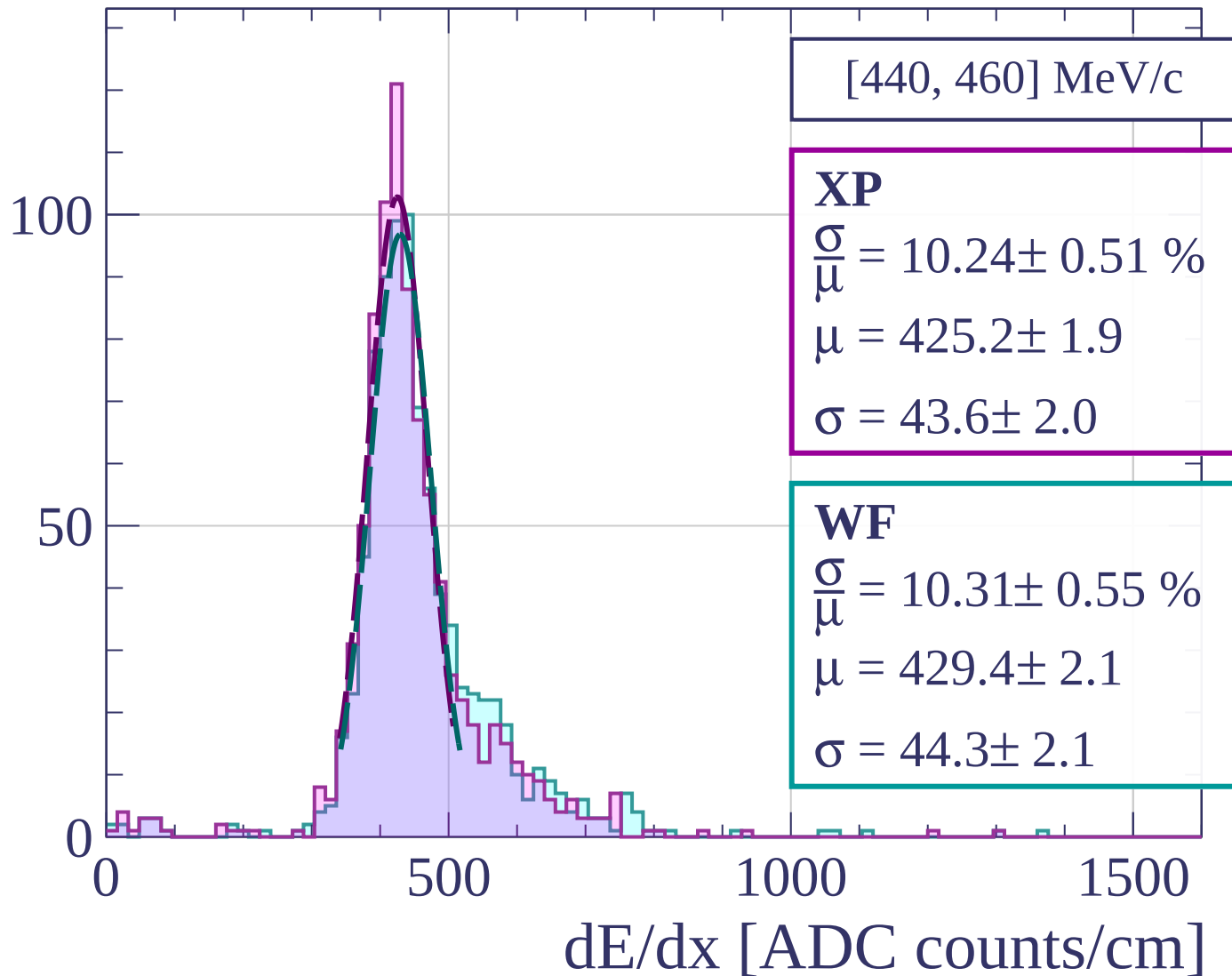
$$\frac{\sigma}{\mu} = 10.42 \pm 0.57 \%$$

$$\mu = 424.2 \pm 2.0$$

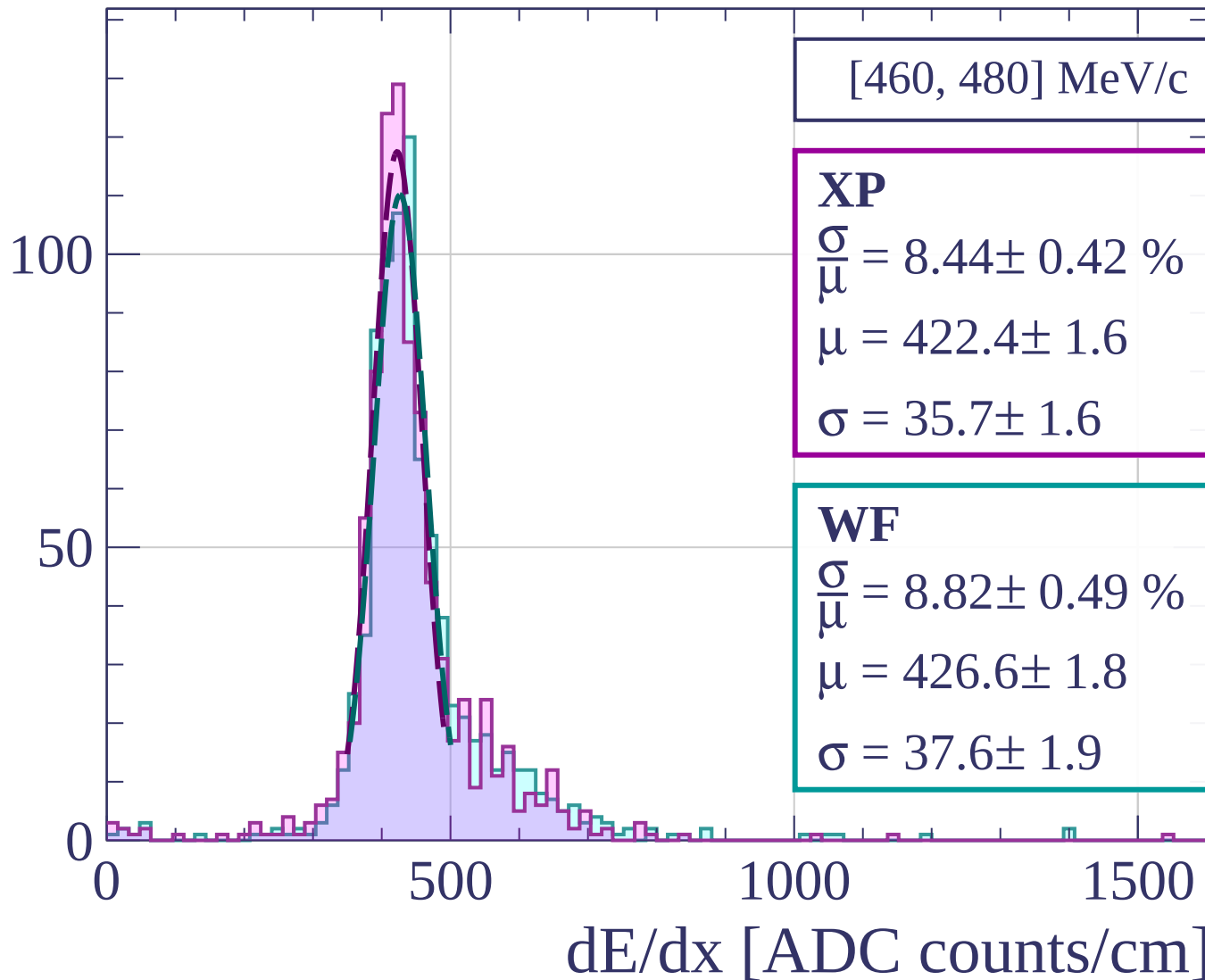
$$\sigma = 44.2 \pm 2.2$$

dE/dx [ADC counts/cm]

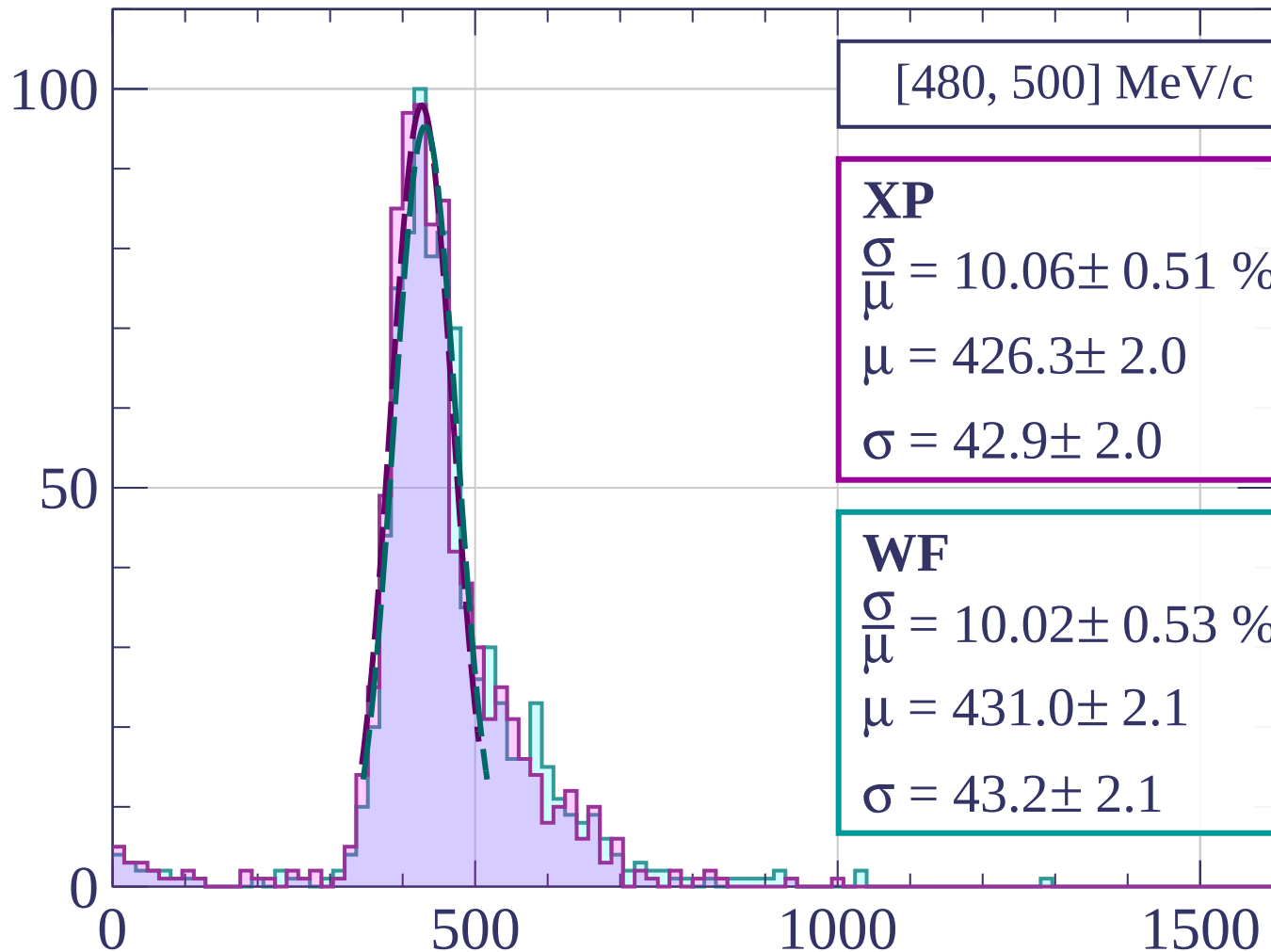
Count



Count



Count



[480, 500] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.06 \pm 0.51 \%$$

$$\mu = 426.3 \pm 2.0$$

$$\sigma = 42.9 \pm 2.0$$

WF

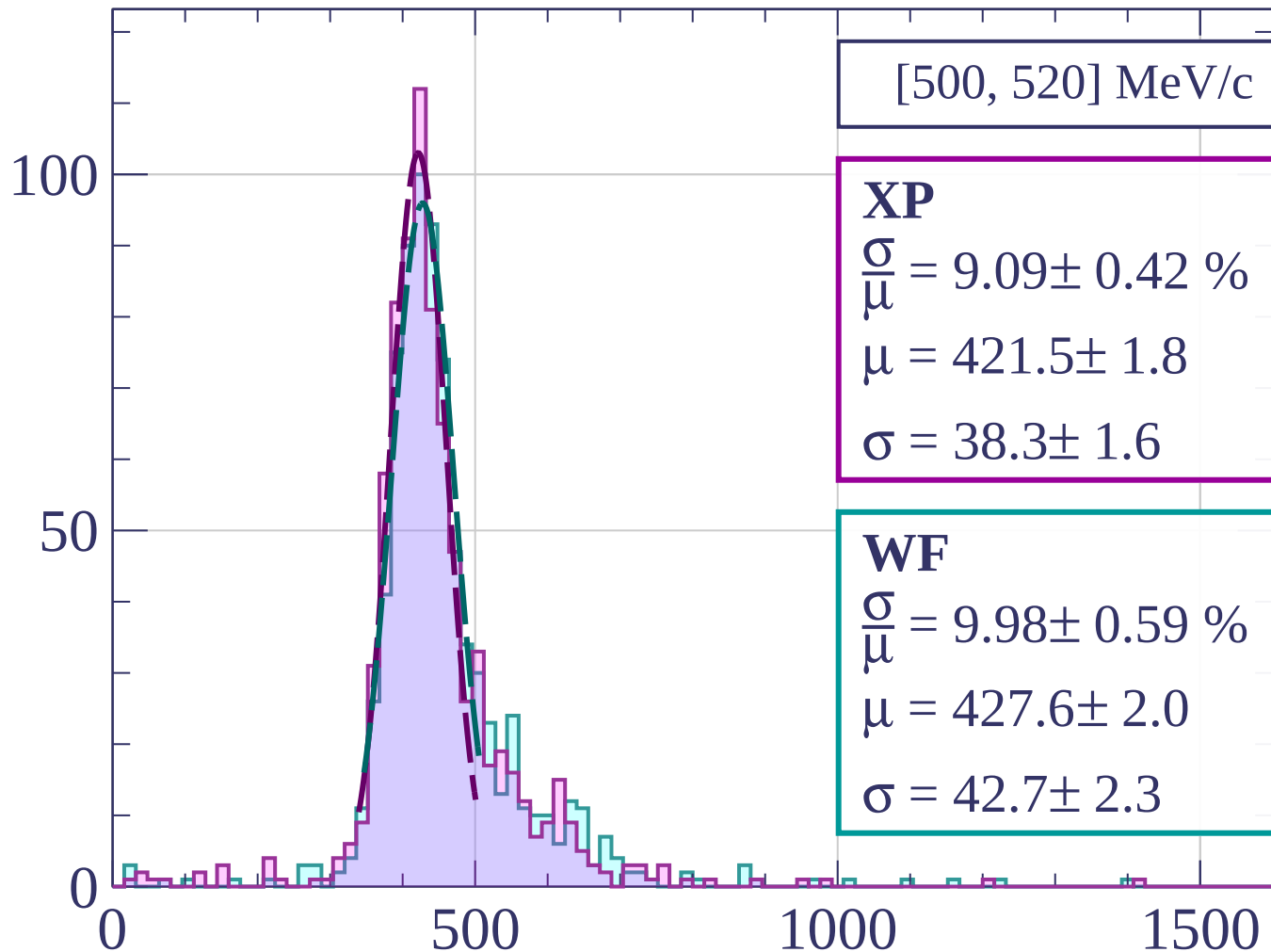
$$\frac{\sigma}{\mu} = 10.02 \pm 0.53 \%$$

$$\mu = 431.0 \pm 2.1$$

$$\sigma = 43.2 \pm 2.1$$

dE/dx [ADC counts/cm]

Count



[500, 520] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.09 \pm 0.42 \%$$

$$\mu = 421.5 \pm 1.8$$

$$\sigma = 38.3 \pm 1.6$$

WF

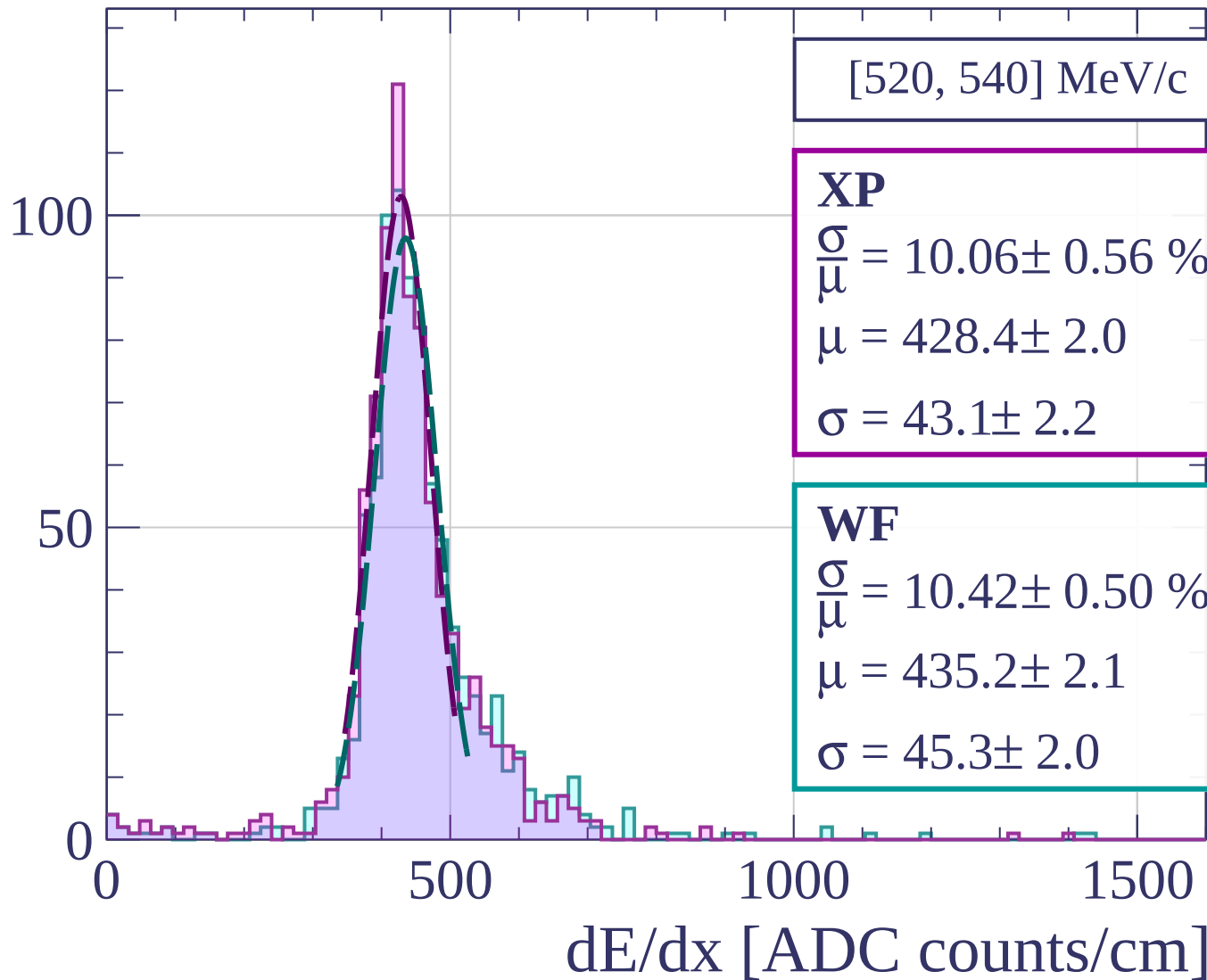
$$\frac{\sigma}{\mu} = 9.98 \pm 0.59 \%$$

$$\mu = 427.6 \pm 2.0$$

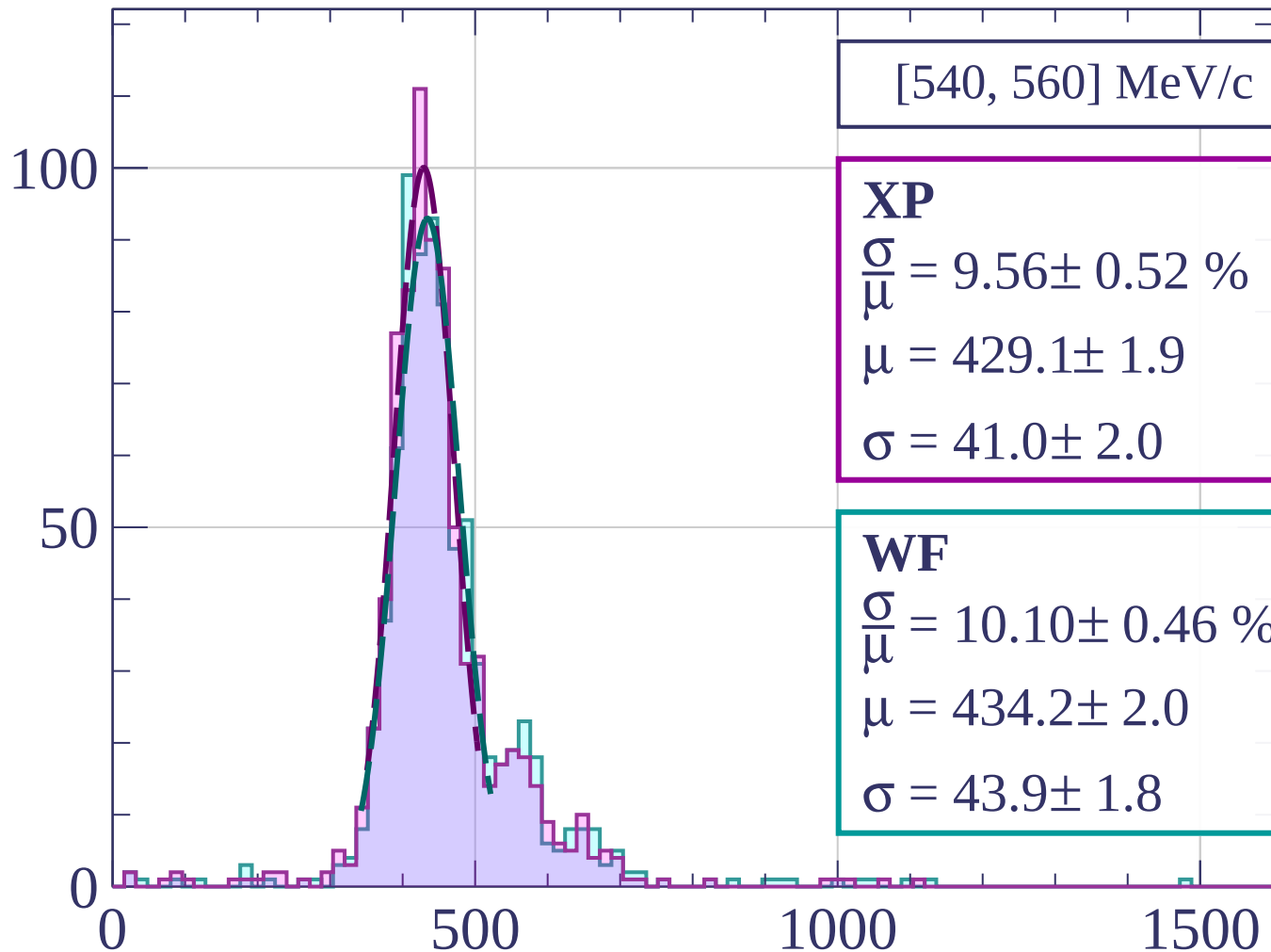
$$\sigma = 42.7 \pm 2.3$$

dE/dx [ADC counts/cm]

Count



Count



[540, 560] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.56 \pm 0.52 \%$$

$$\mu = 429.1 \pm 1.9$$

$$\sigma = 41.0 \pm 2.0$$

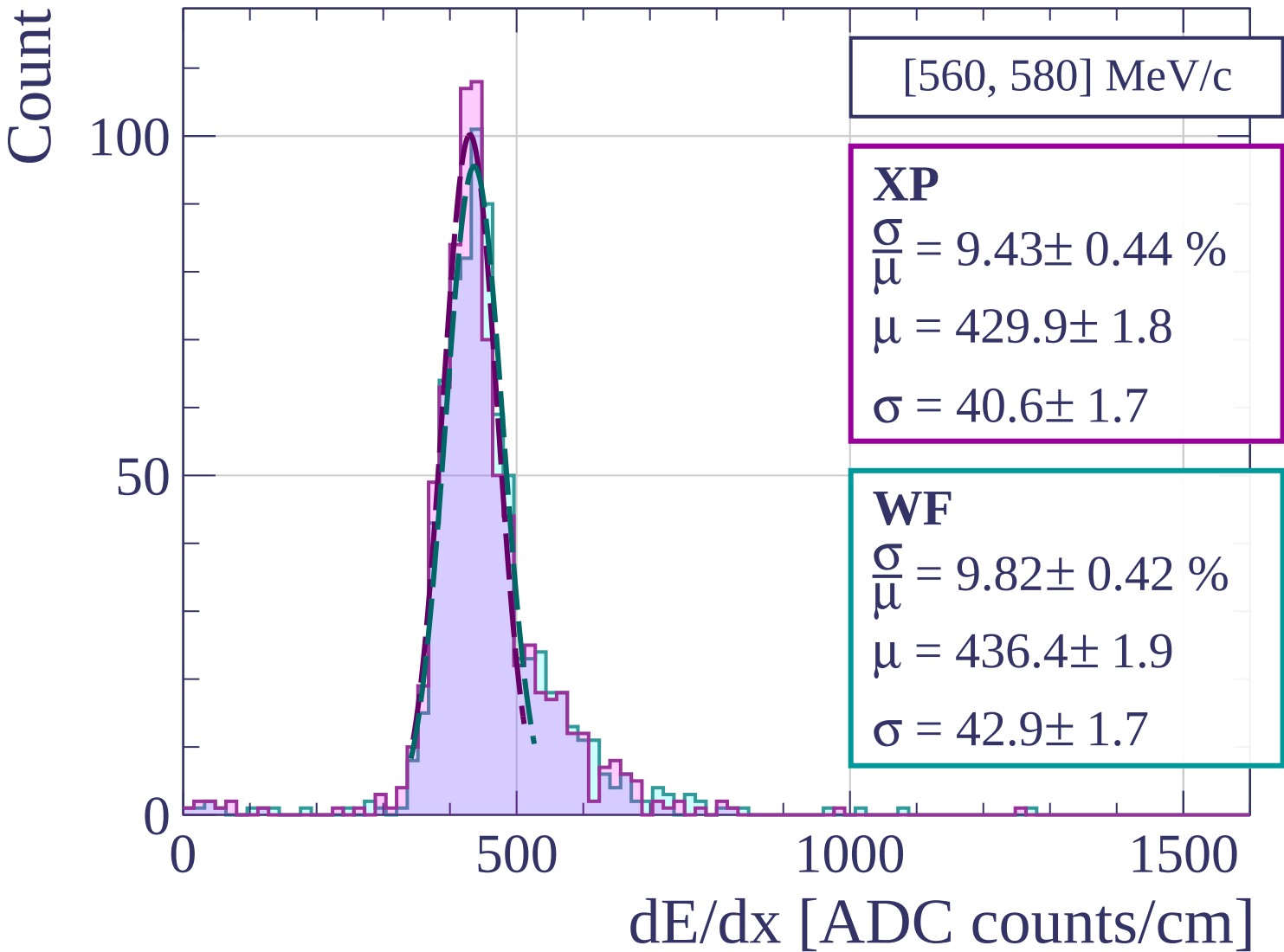
WF

$$\frac{\sigma}{\mu} = 10.10 \pm 0.46 \%$$

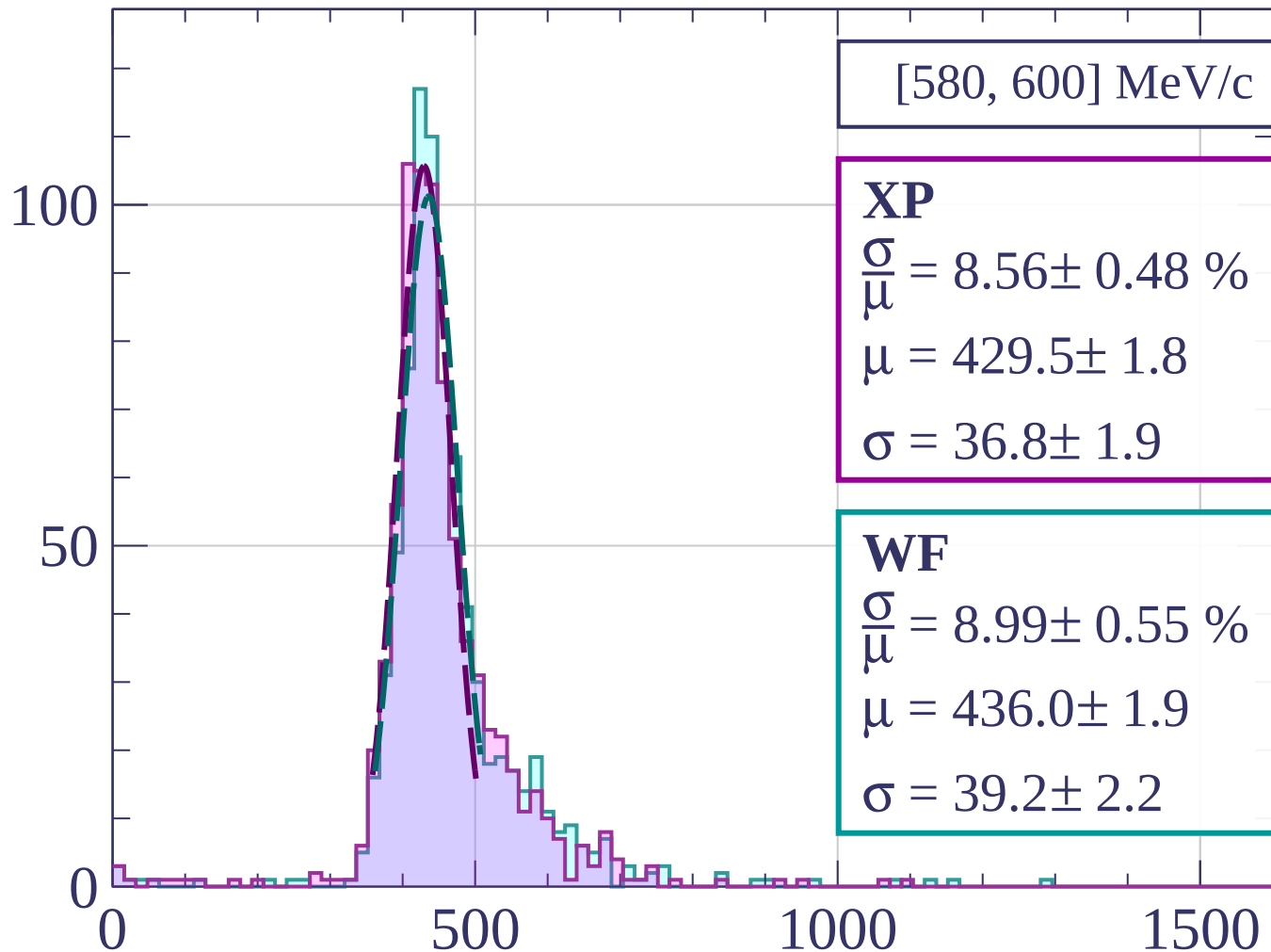
$$\mu = 434.2 \pm 2.0$$

$$\sigma = 43.9 \pm 1.8$$

dE/dx [ADC counts/cm]

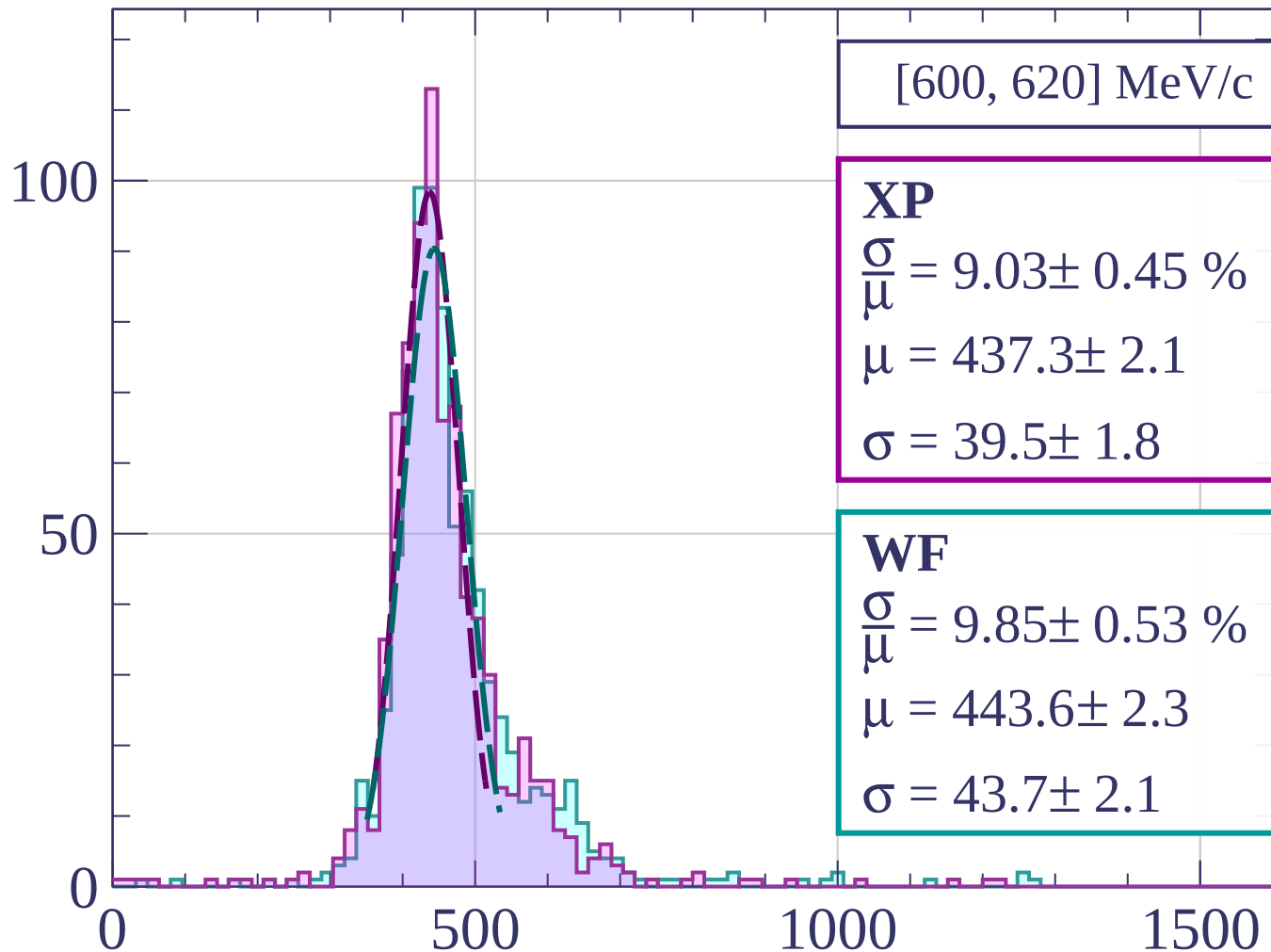


Count



dE/dx [ADC counts/cm]

Count



[600, 620] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.03 \pm 0.45 \%$$

$$\mu = 437.3 \pm 2.1$$

$$\sigma = 39.5 \pm 1.8$$

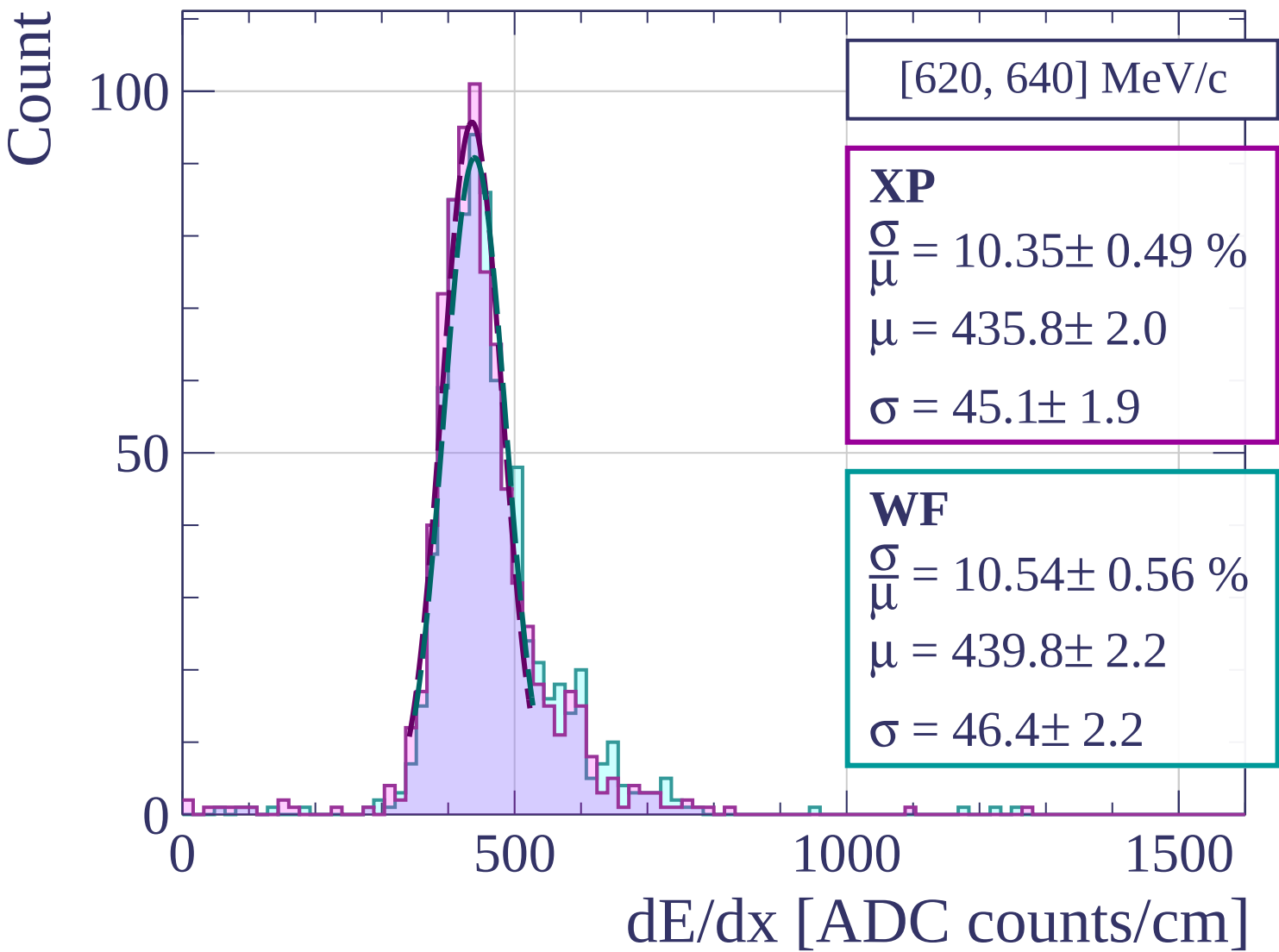
WF

$$\frac{\sigma}{\mu} = 9.85 \pm 0.53 \%$$

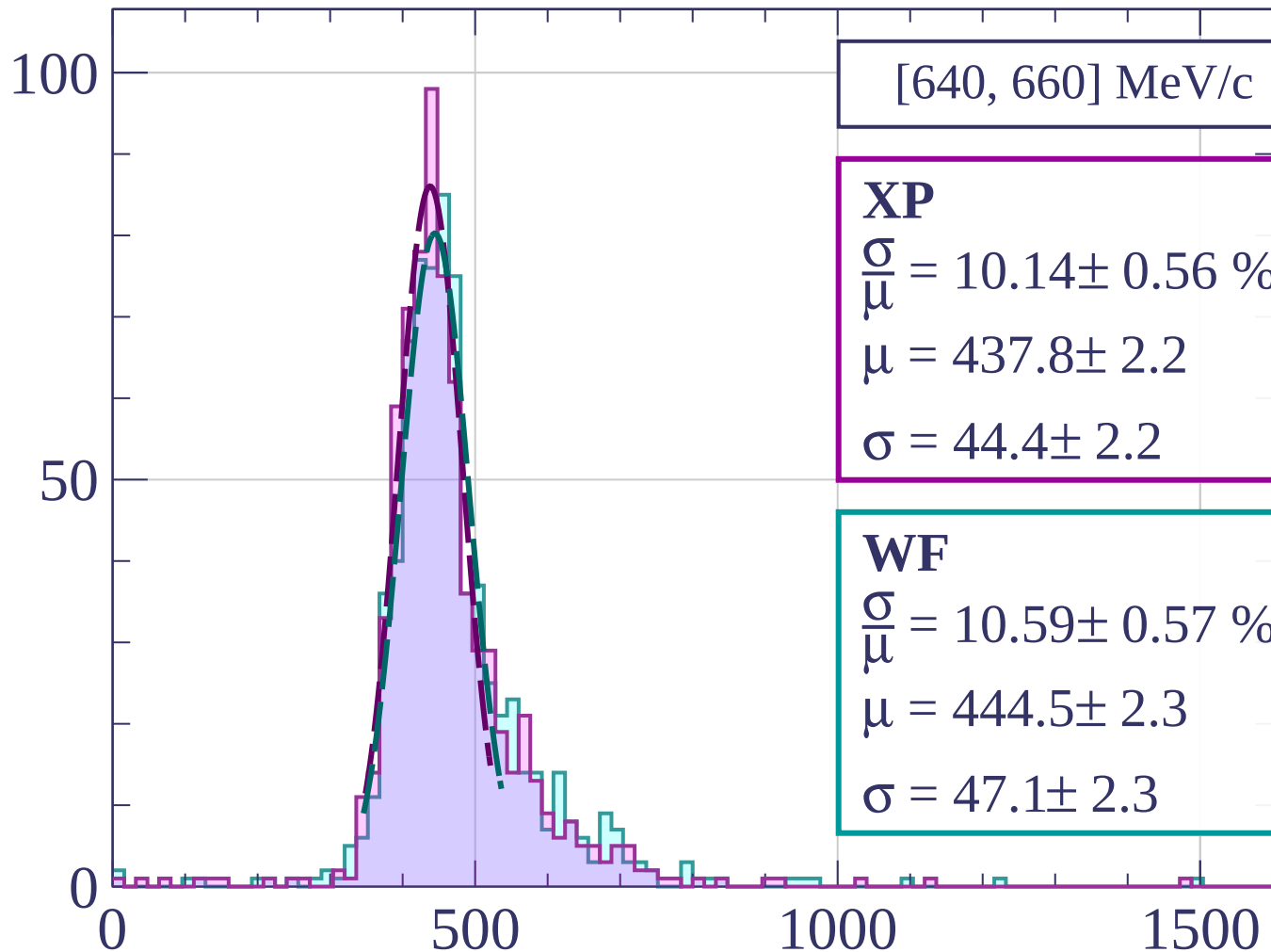
$$\mu = 443.6 \pm 2.3$$

$$\sigma = 43.7 \pm 2.1$$

dE/dx [ADC counts/cm]



Count



[640, 660] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.14 \pm 0.56 \%$$

$$\mu = 437.8 \pm 2.2$$

$$\sigma = 44.4 \pm 2.2$$

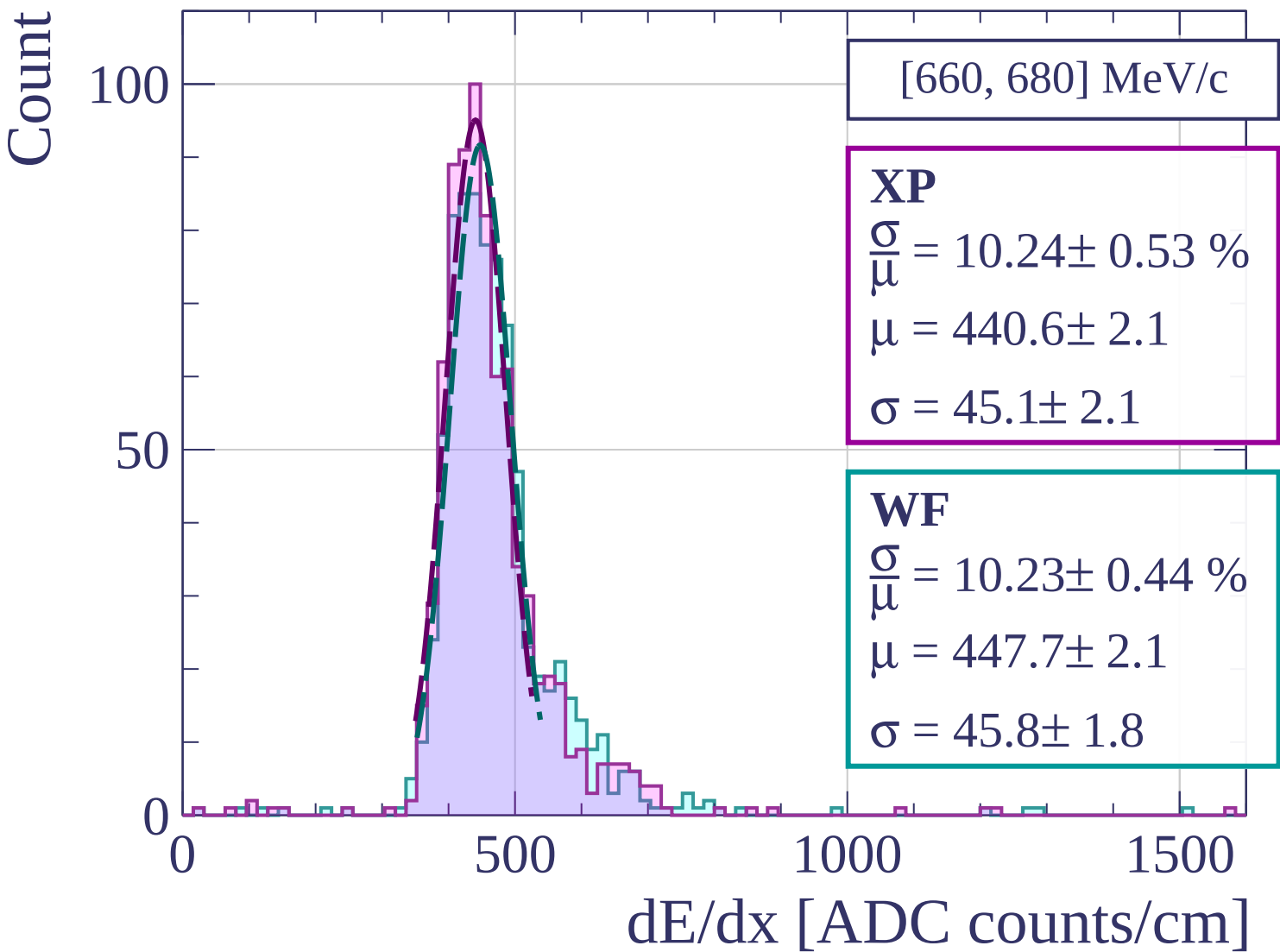
WF

$$\frac{\sigma}{\mu} = 10.59 \pm 0.57 \%$$

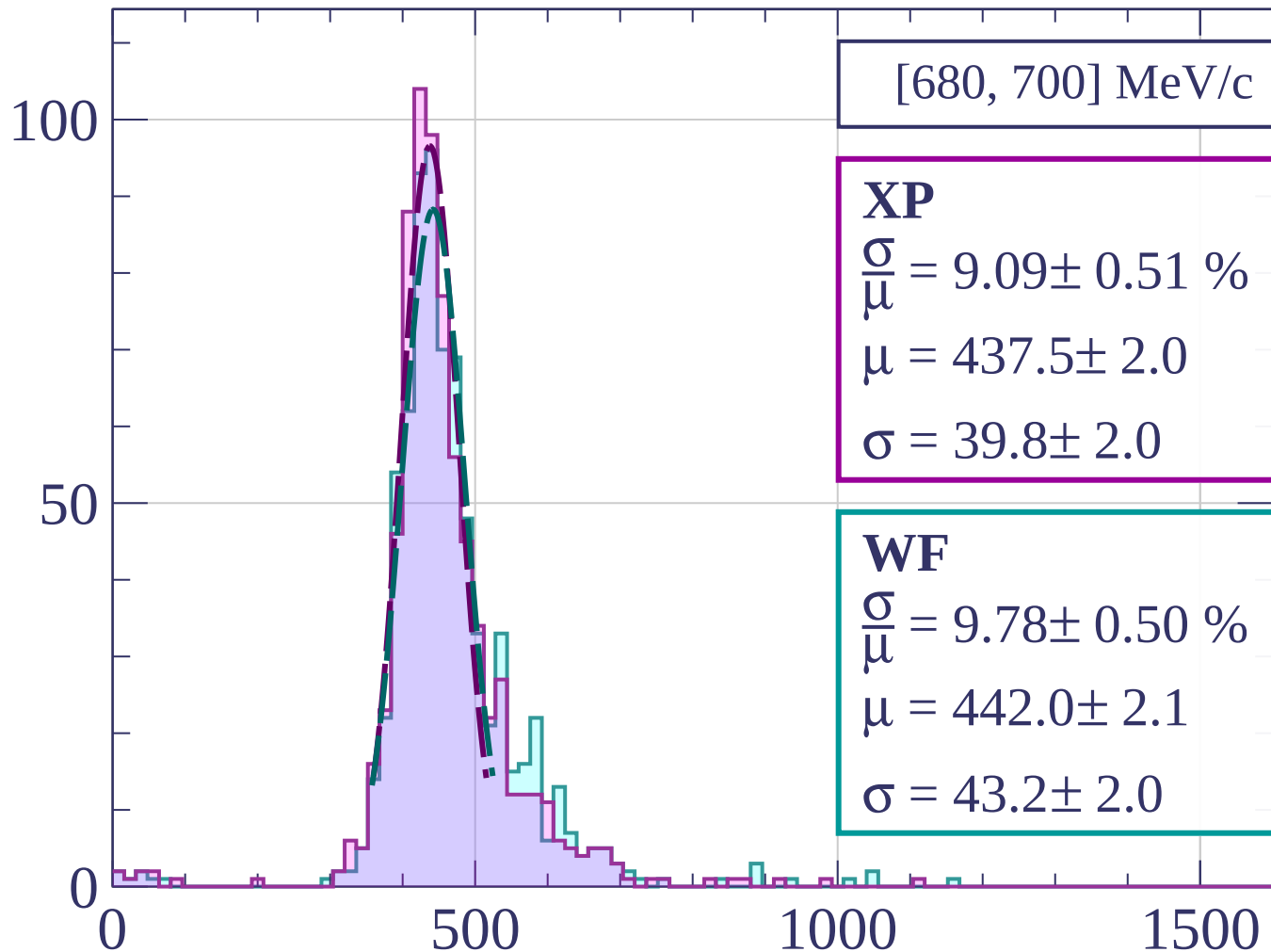
$$\mu = 444.5 \pm 2.3$$

$$\sigma = 47.1 \pm 2.3$$

dE/dx [ADC counts/cm]

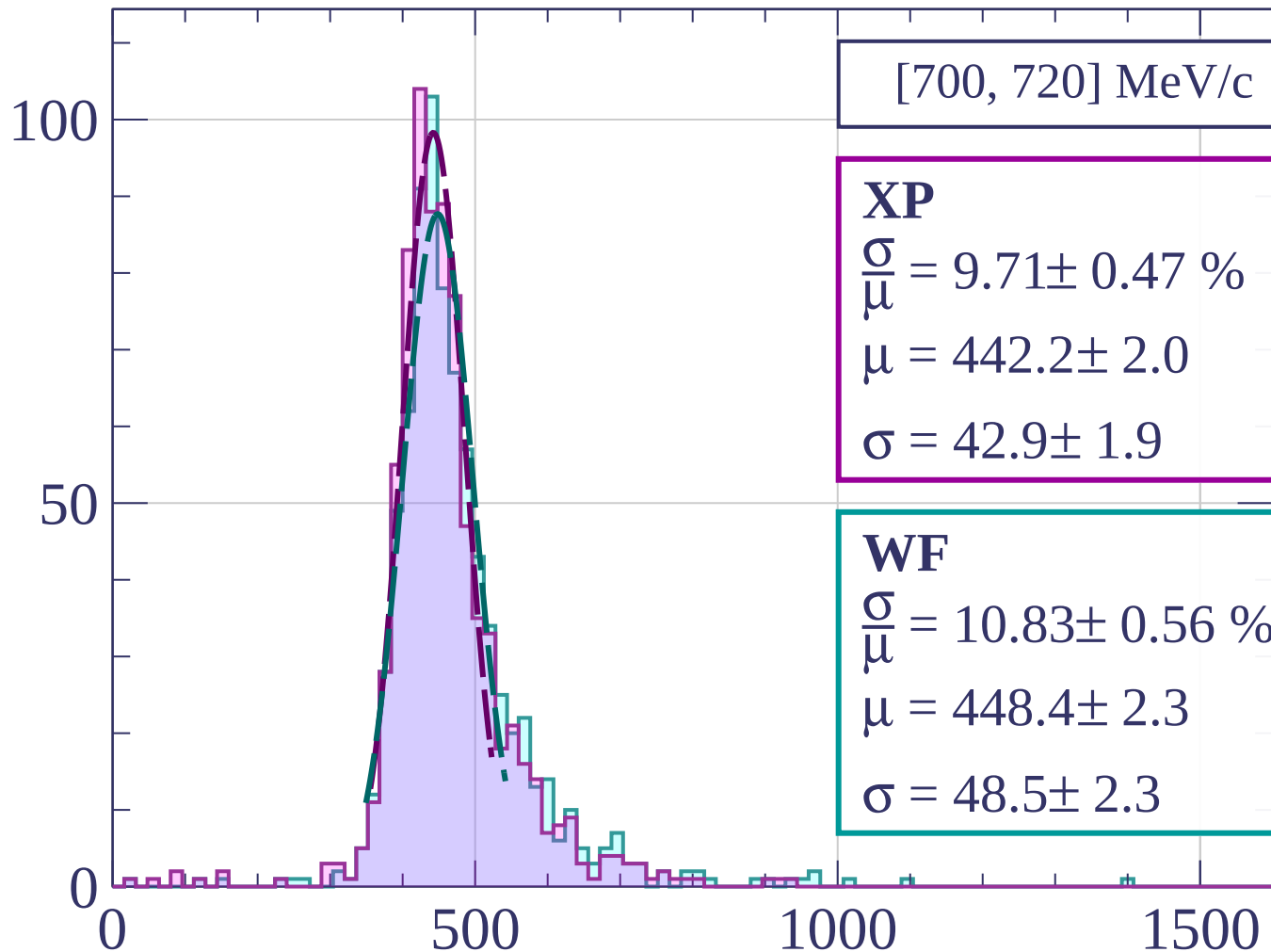


Count



dE/dx [ADC counts/cm]

Count



[700, 720] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.71 \pm 0.47 \%$$

$$\mu = 442.2 \pm 2.0$$

$$\sigma = 42.9 \pm 1.9$$

WF

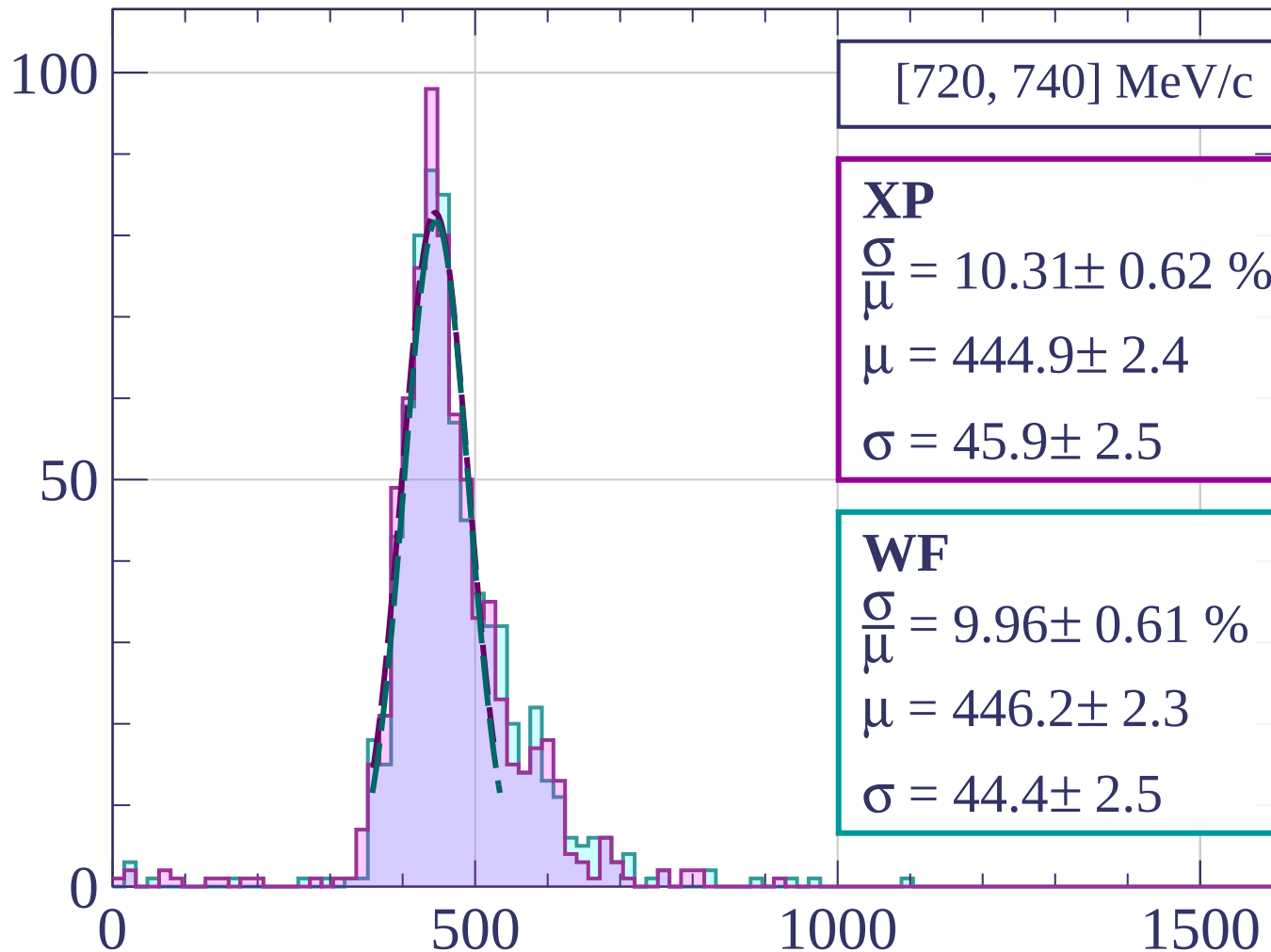
$$\frac{\sigma}{\mu} = 10.83 \pm 0.56 \%$$

$$\mu = 448.4 \pm 2.3$$

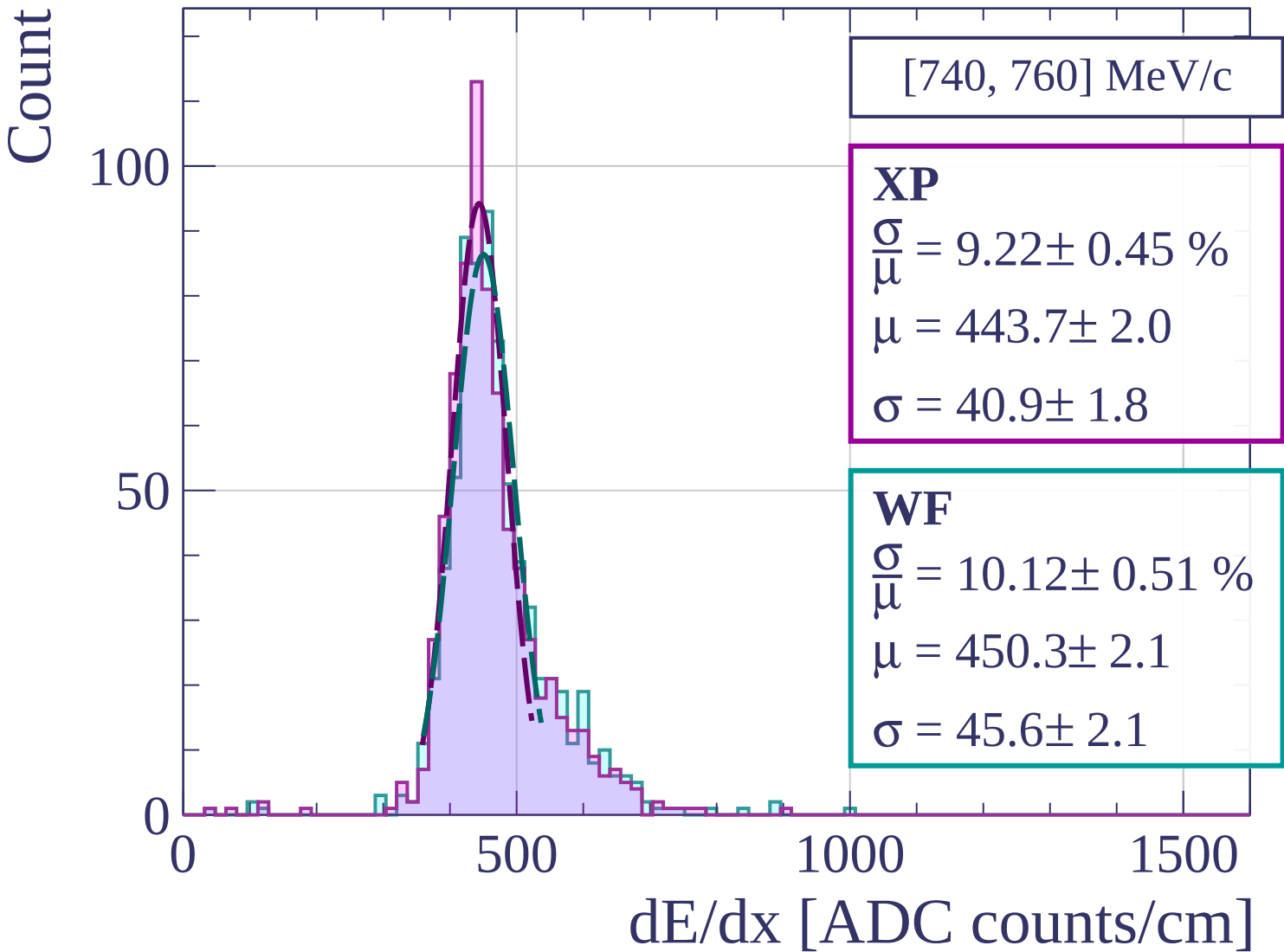
$$\sigma = 48.5 \pm 2.3$$

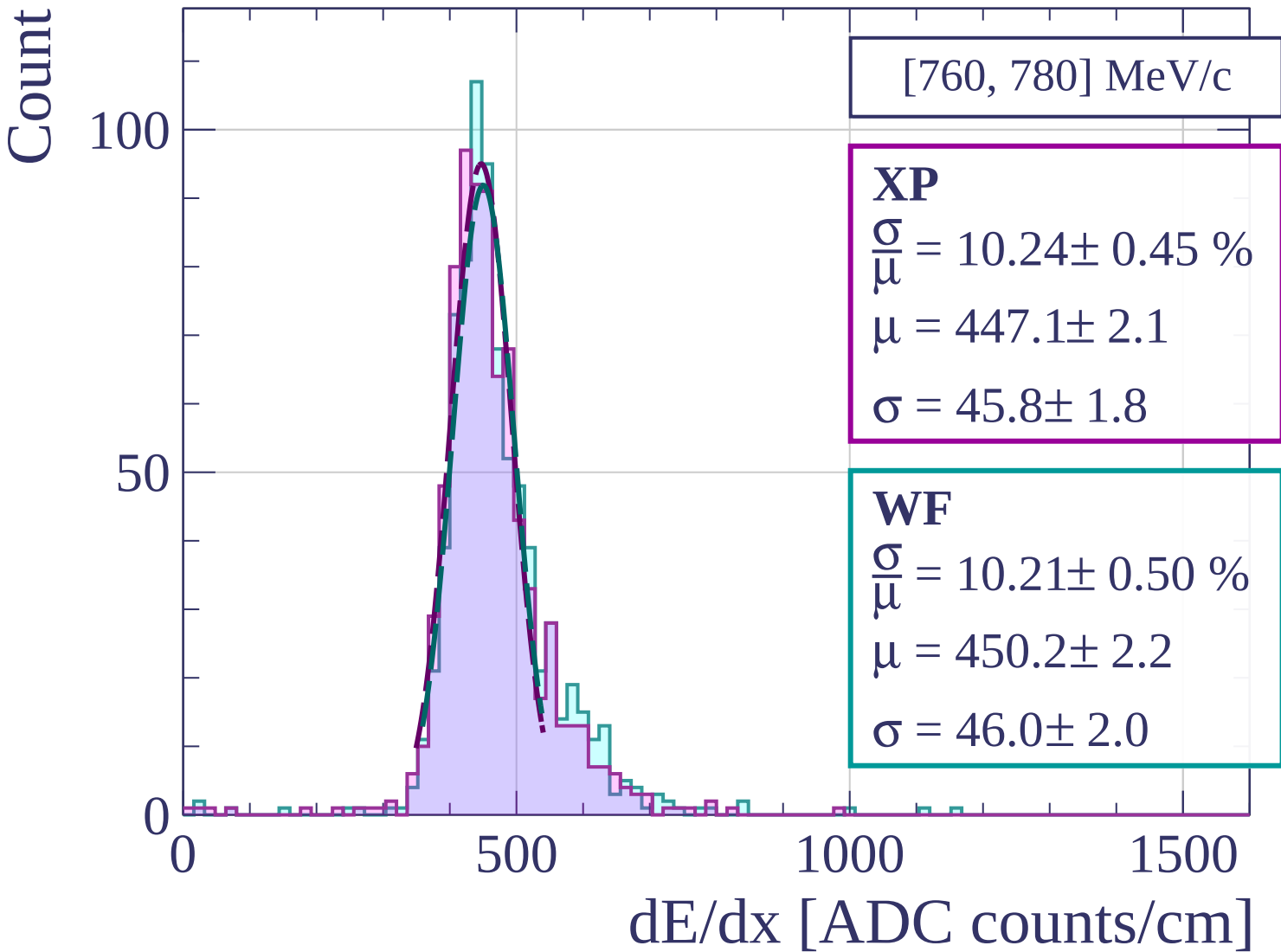
dE/dx [ADC counts/cm]

Count

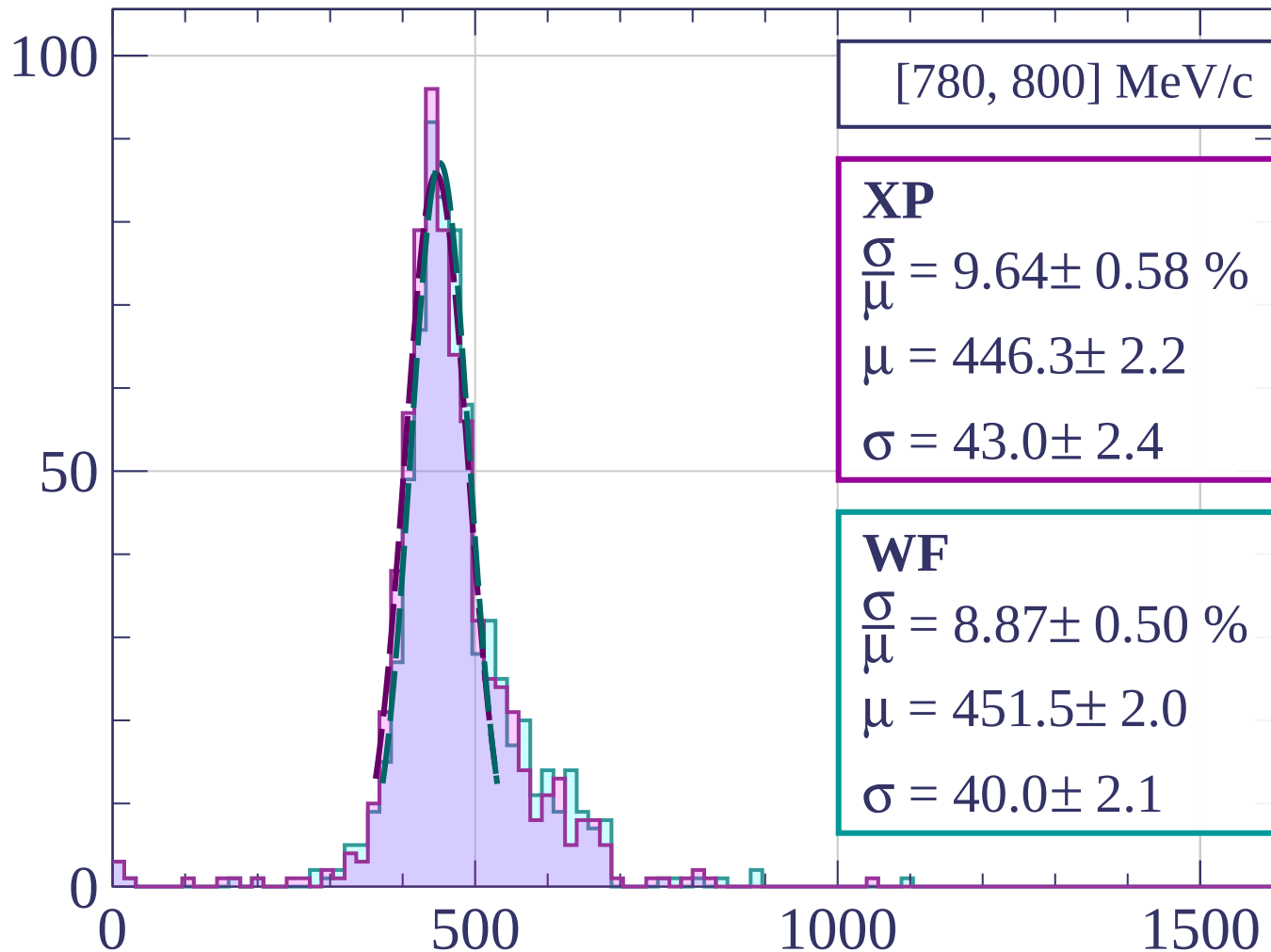


dE/dx [ADC counts/cm]





Count



[780, 800] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.64 \pm 0.58 \%$$

$$\mu = 446.3 \pm 2.2$$

$$\sigma = 43.0 \pm 2.4$$

WF

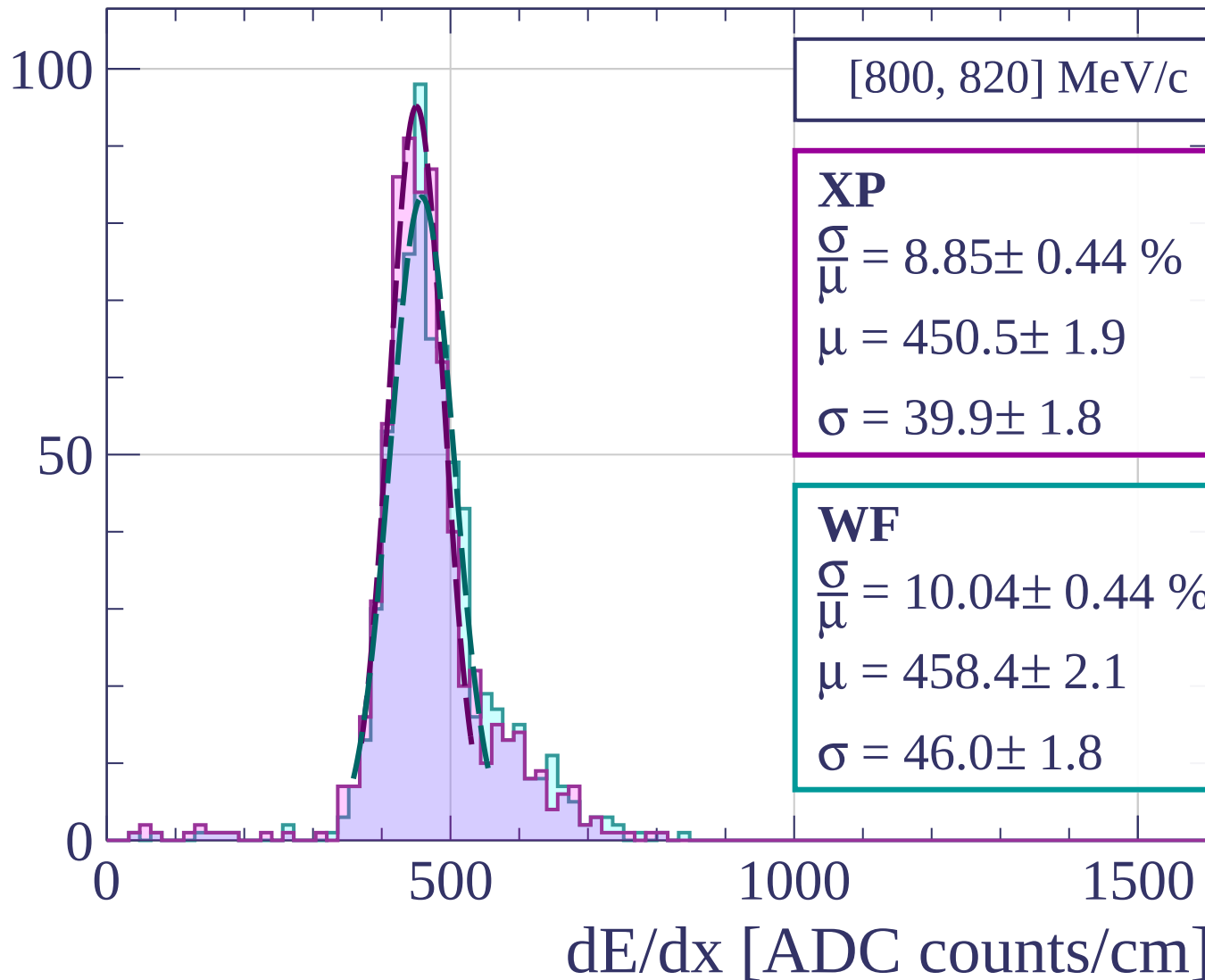
$$\frac{\sigma}{\mu} = 8.87 \pm 0.50 \%$$

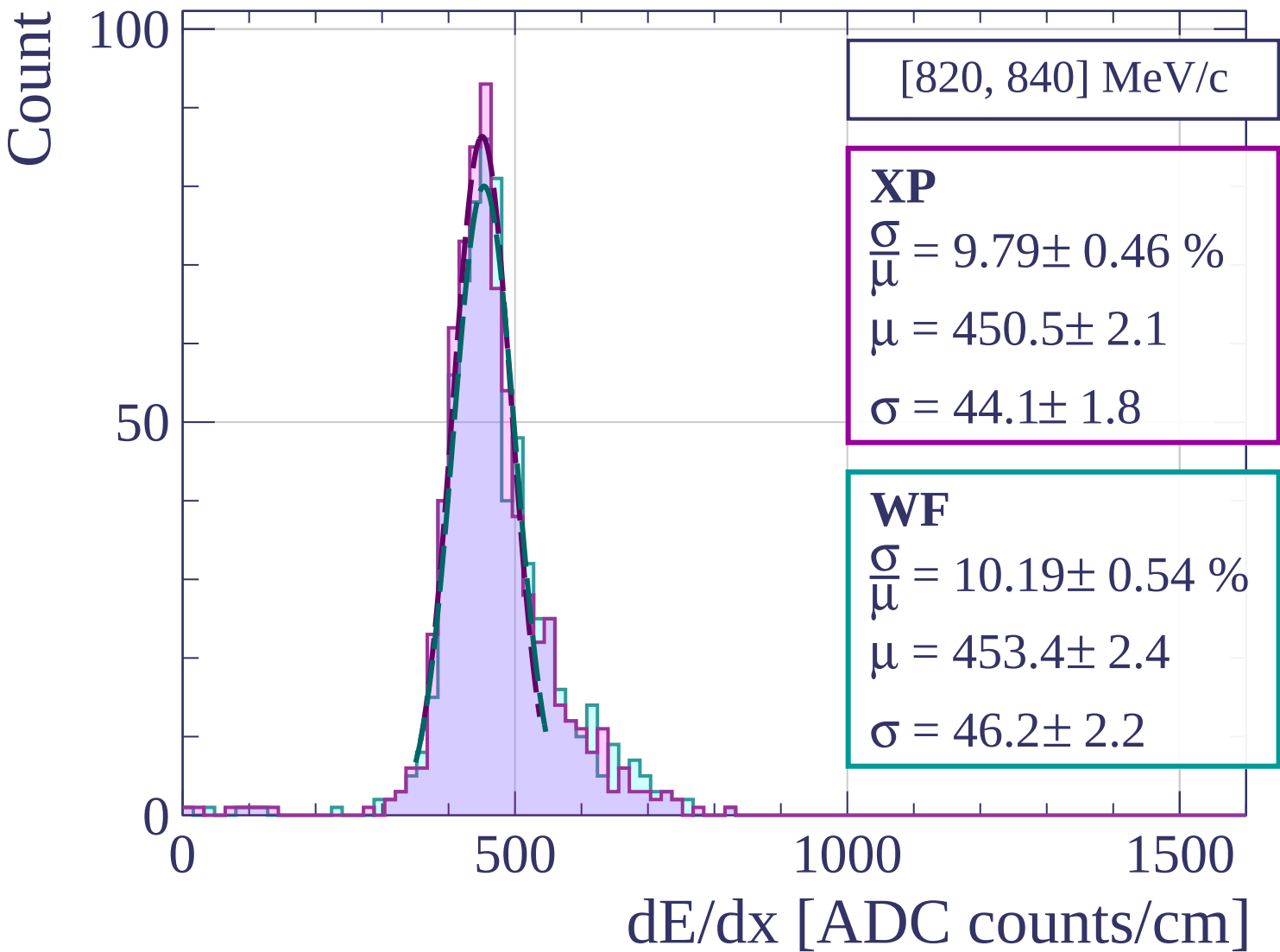
$$\mu = 451.5 \pm 2.0$$

$$\sigma = 40.0 \pm 2.1$$

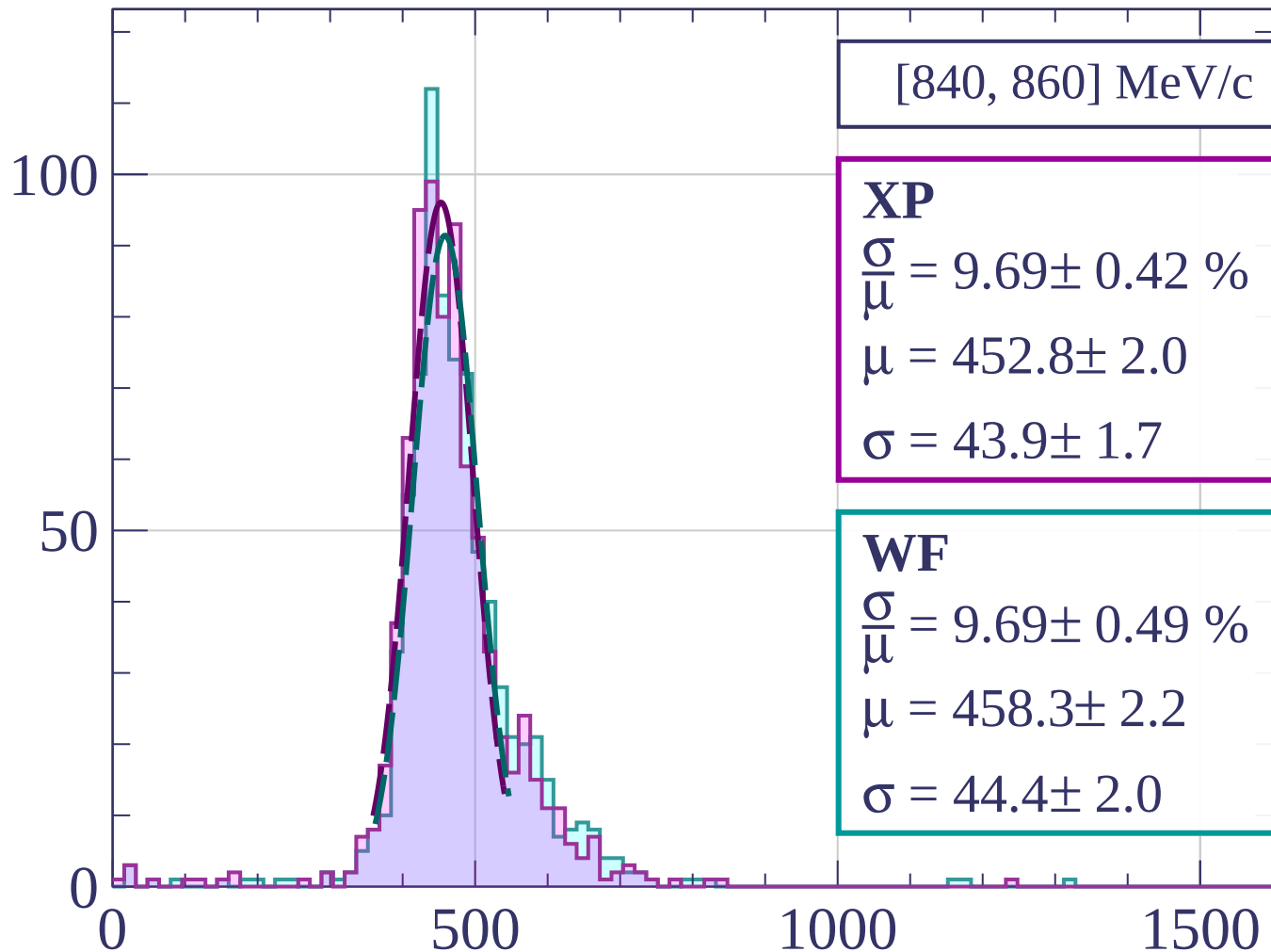
dE/dx [ADC counts/cm]

Count





Count



[840, 860] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.69 \pm 0.42 \%$$

$$\mu = 452.8 \pm 2.0$$

$$\sigma = 43.9 \pm 1.7$$

WF

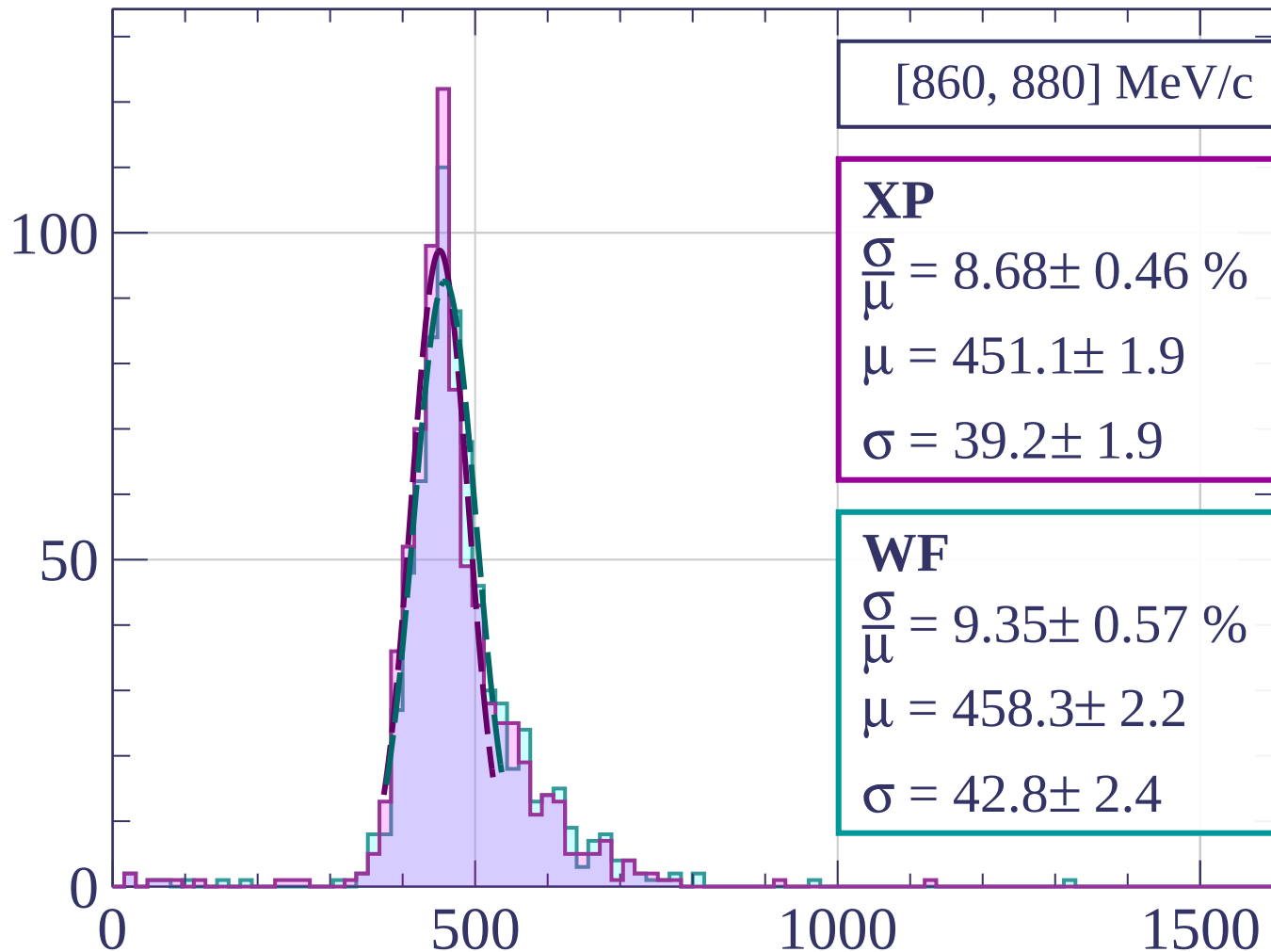
$$\frac{\sigma}{\mu} = 9.69 \pm 0.49 \%$$

$$\mu = 458.3 \pm 2.2$$

$$\sigma = 44.4 \pm 2.0$$

dE/dx [ADC counts/cm]

Count



[860, 880] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.68 \pm 0.46 \%$$

$$\mu = 451.1 \pm 1.9$$

$$\sigma = 39.2 \pm 1.9$$

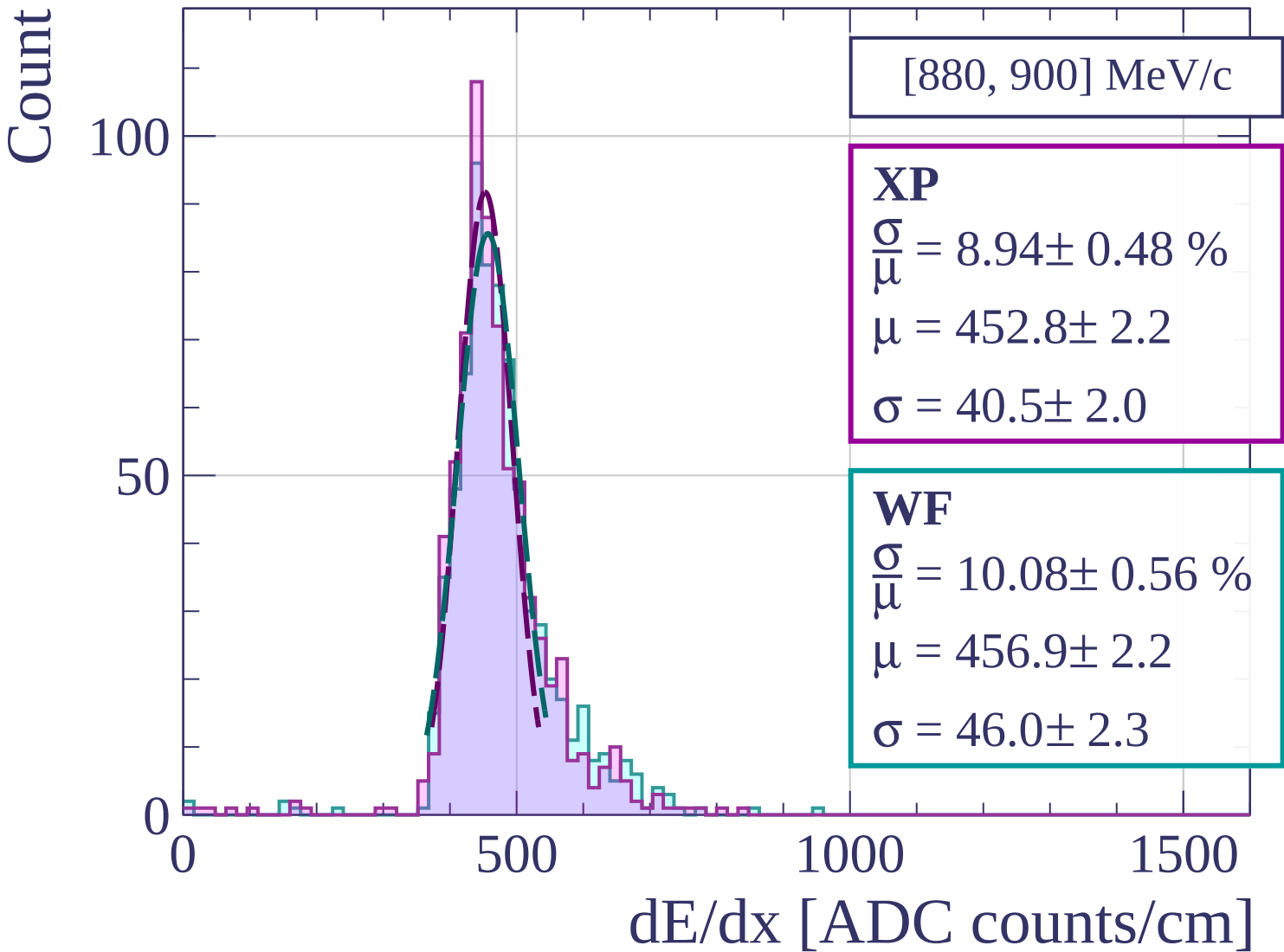
WF

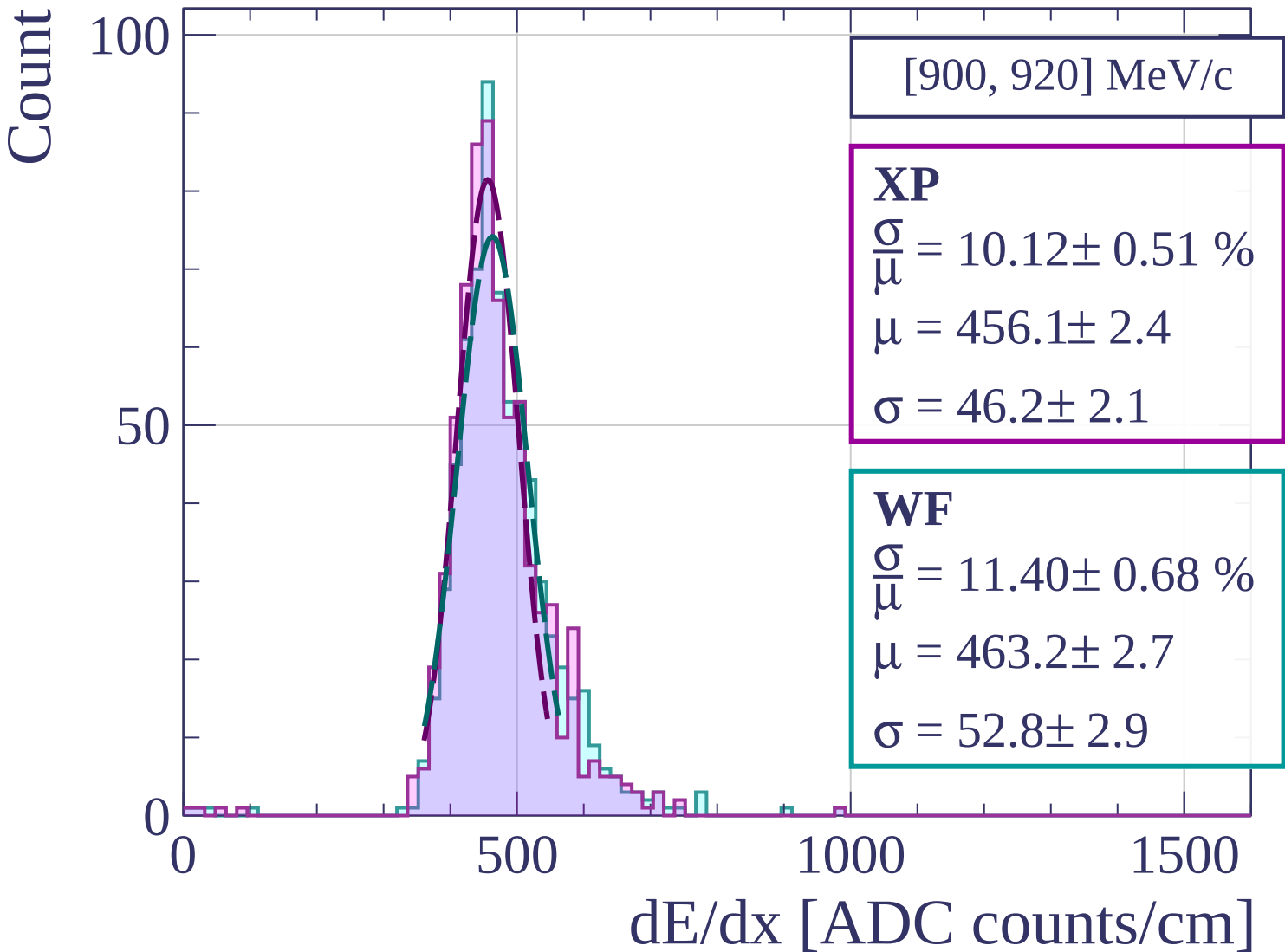
$$\frac{\sigma}{\mu} = 9.35 \pm 0.57 \%$$

$$\mu = 458.3 \pm 2.2$$

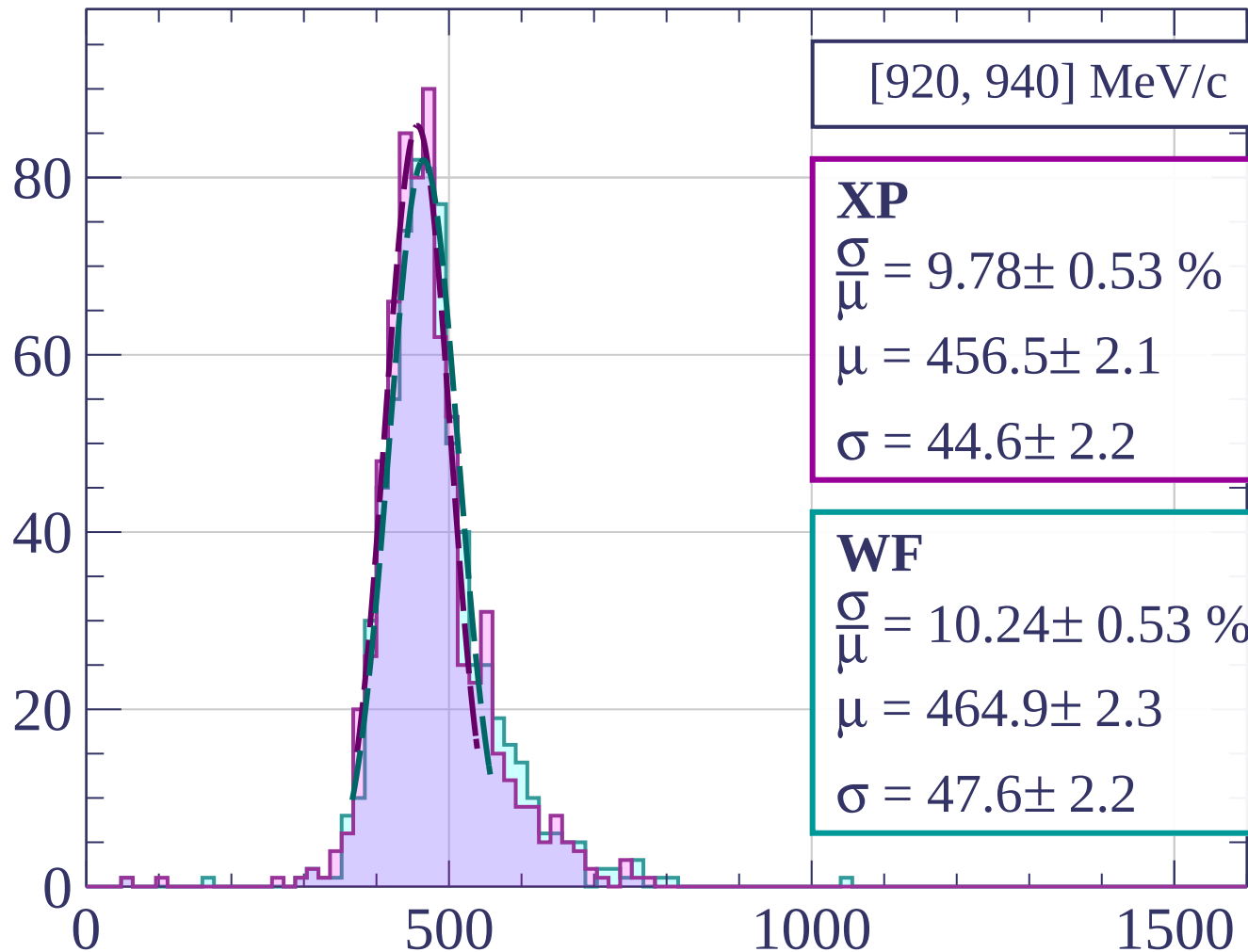
$$\sigma = 42.8 \pm 2.4$$

dE/dx [ADC counts/cm]





Count



[920, 940] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.78 \pm 0.53 \%$$

$$\mu = 456.5 \pm 2.1$$

$$\sigma = 44.6 \pm 2.2$$

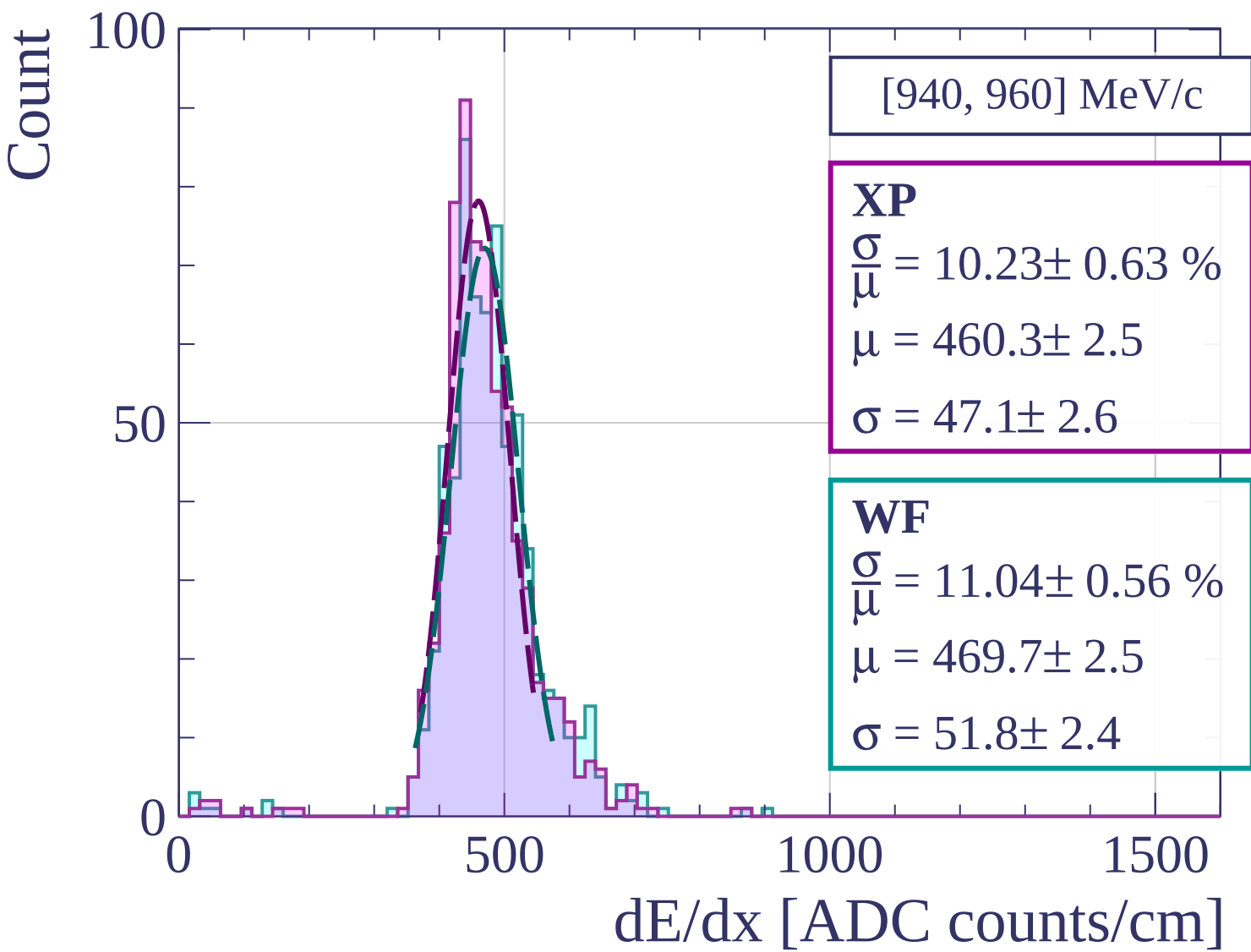
WF

$$\frac{\sigma}{\mu} = 10.24 \pm 0.53 \%$$

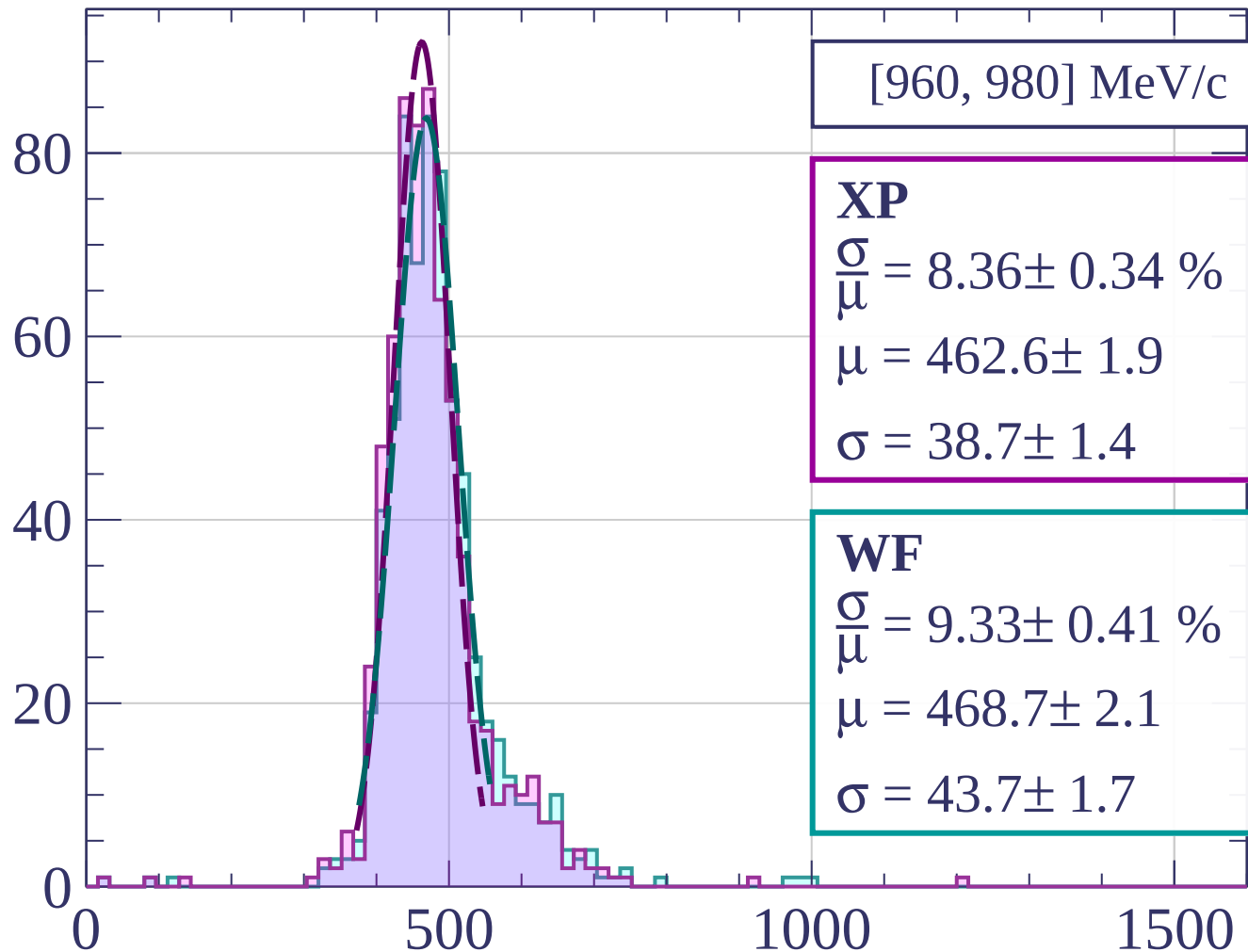
$$\mu = 464.9 \pm 2.3$$

$$\sigma = 47.6 \pm 2.2$$

dE/dx [ADC counts/cm]

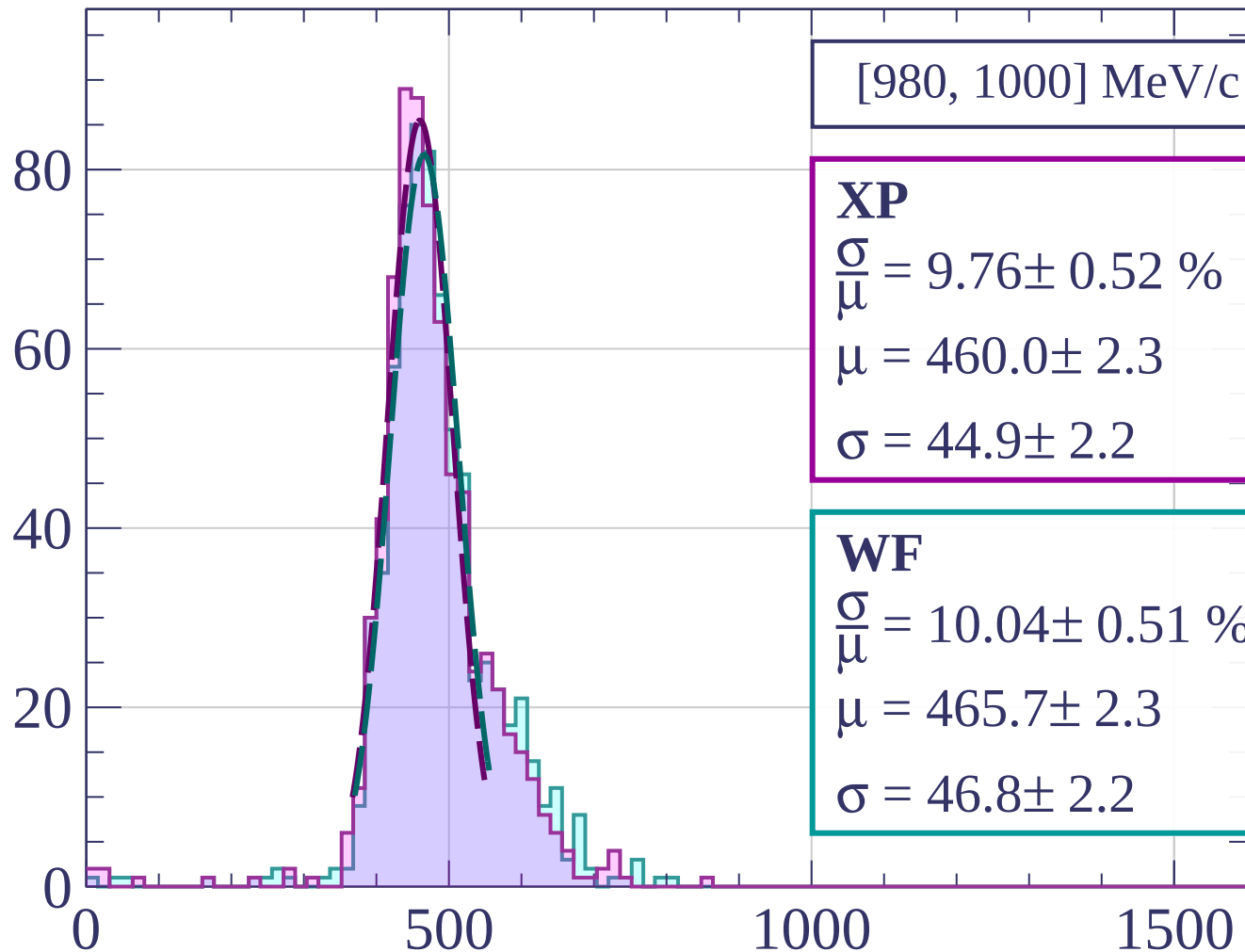


Count



dE/dx [ADC counts/cm]

Count



[980, 1000] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.76 \pm 0.52 \%$$

$$\mu = 460.0 \pm 2.3$$

$$\sigma = 44.9 \pm 2.2$$

WF

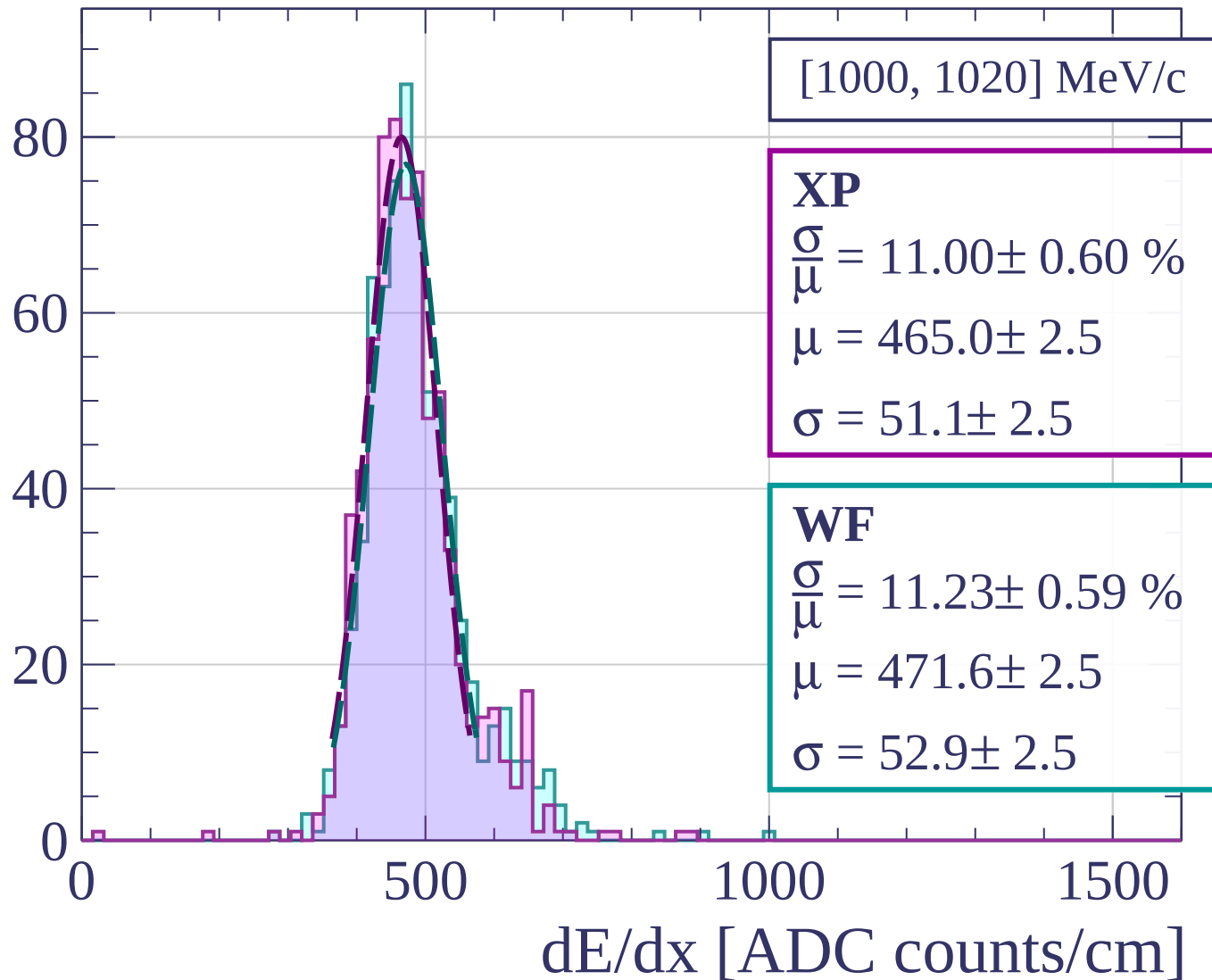
$$\frac{\sigma}{\mu} = 10.04 \pm 0.51 \%$$

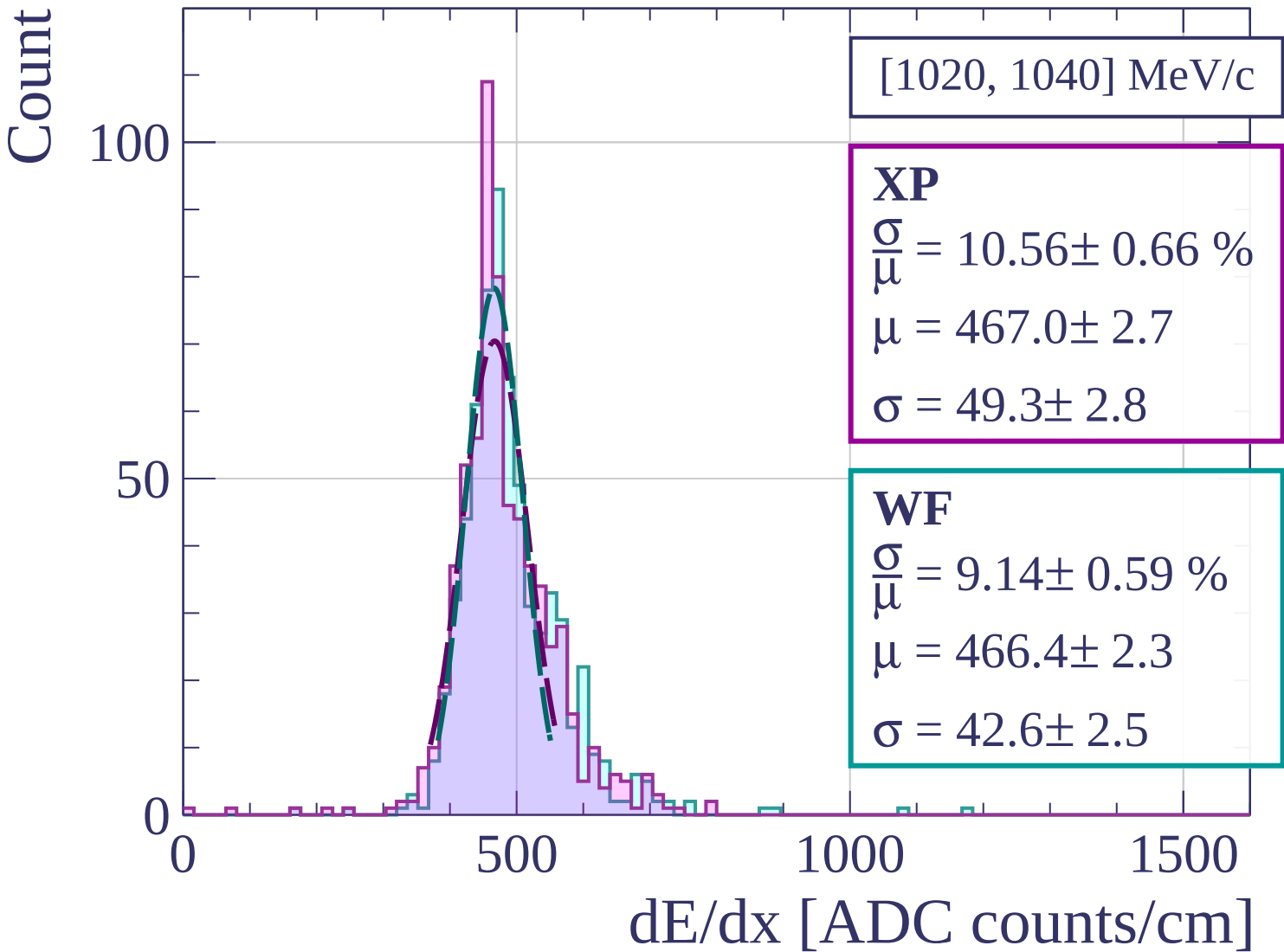
$$\mu = 465.7 \pm 2.3$$

$$\sigma = 46.8 \pm 2.2$$

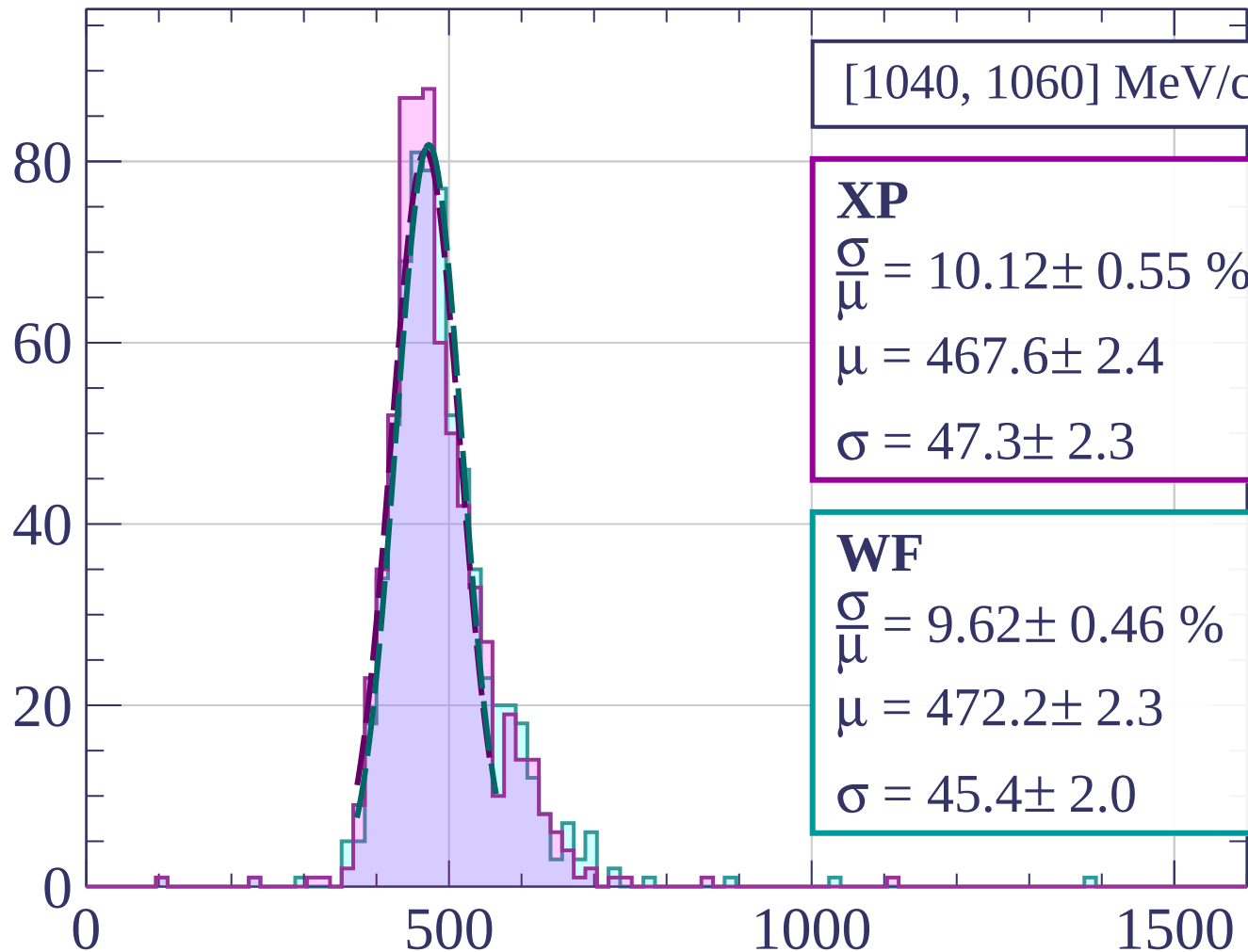
dE/dx [ADC counts/cm]

Count





Count



[1040, 1060] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.12 \pm 0.55 \%$$

$$\mu = 467.6 \pm 2.4$$

$$\sigma = 47.3 \pm 2.3$$

WF

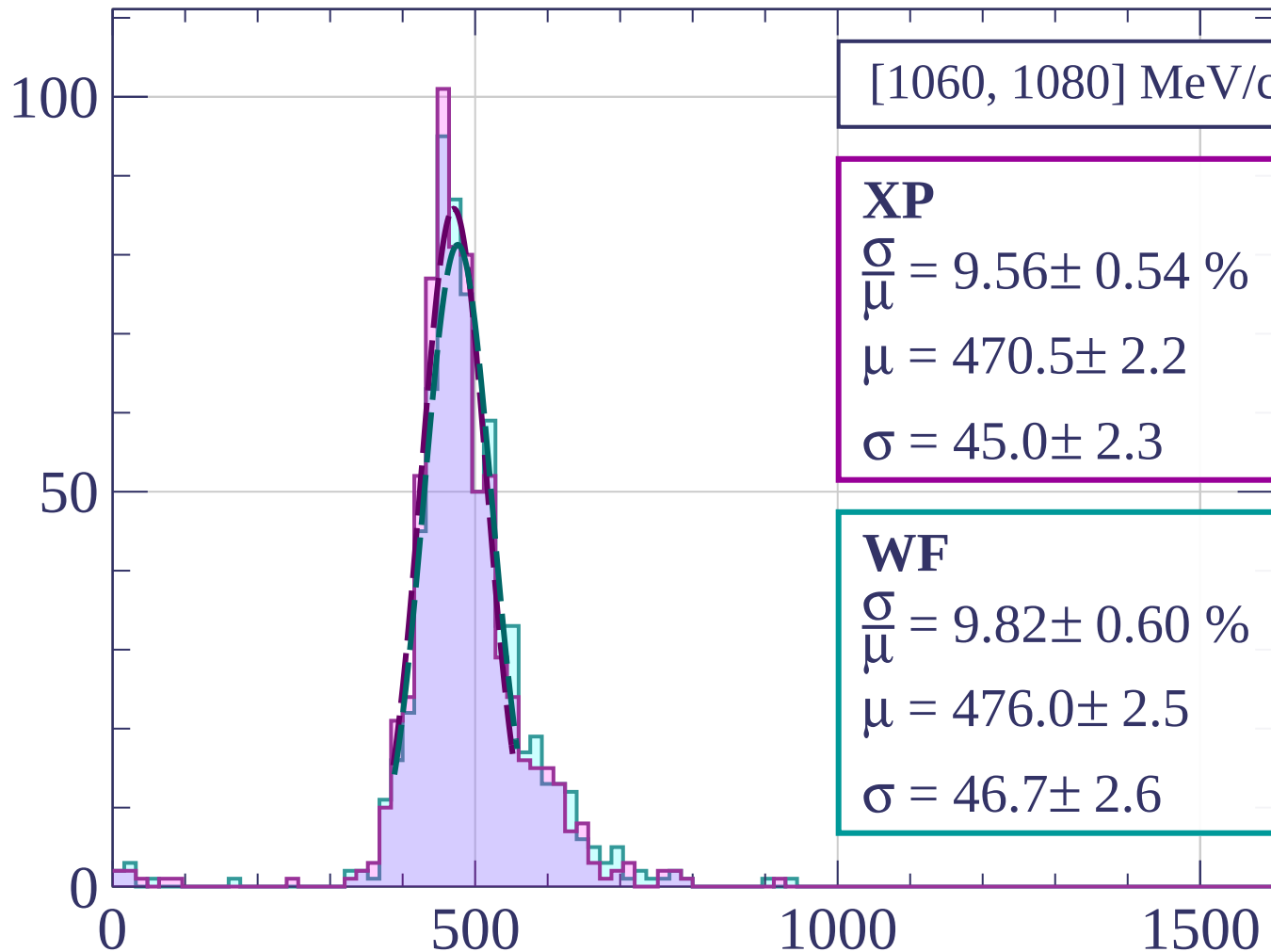
$$\frac{\sigma}{\mu} = 9.62 \pm 0.46 \%$$

$$\mu = 472.2 \pm 2.3$$

$$\sigma = 45.4 \pm 2.0$$

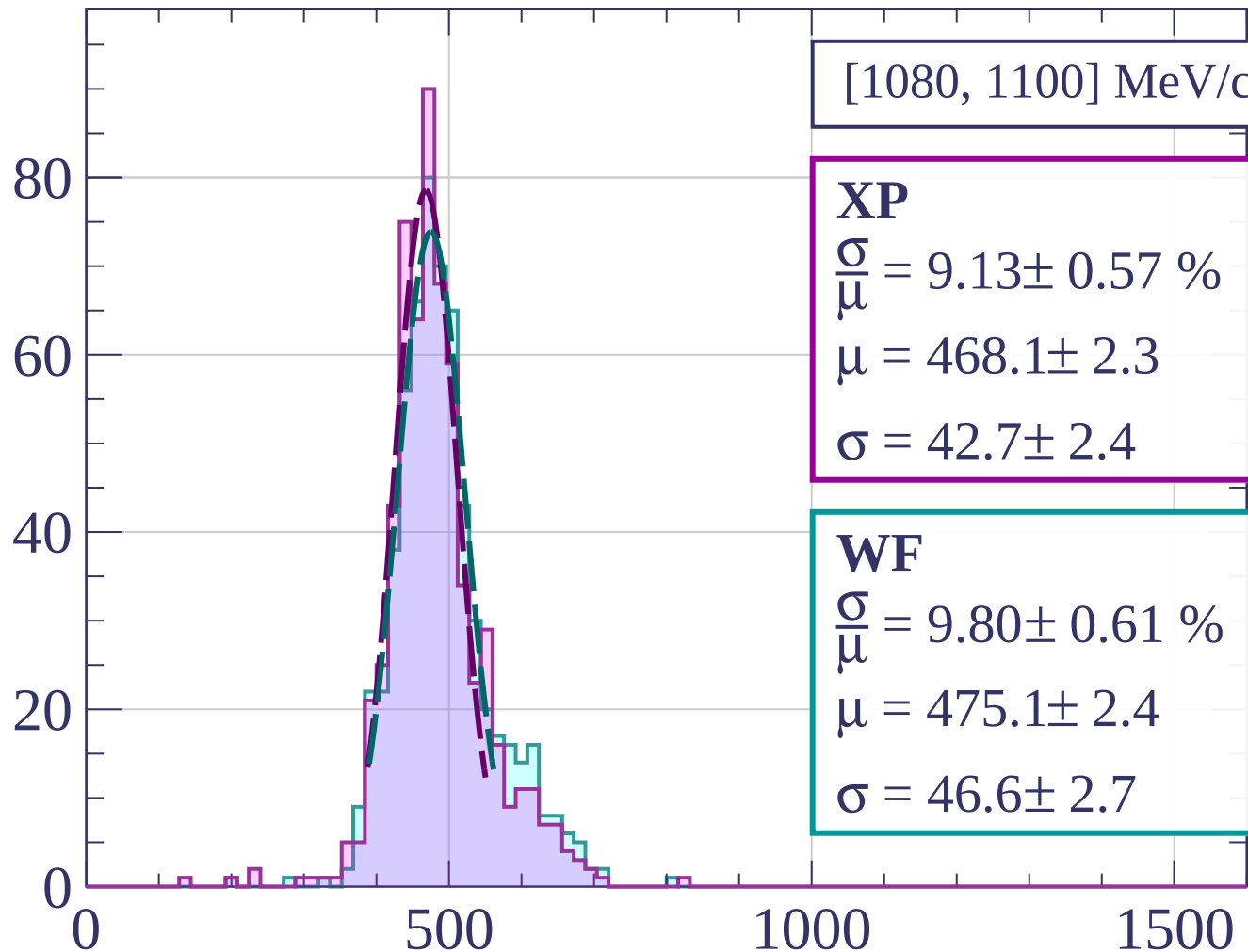
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[1080, 1100] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.57 \%$$

$$\mu = 468.1 \pm 2.3$$

$$\sigma = 42.7 \pm 2.4$$

WF

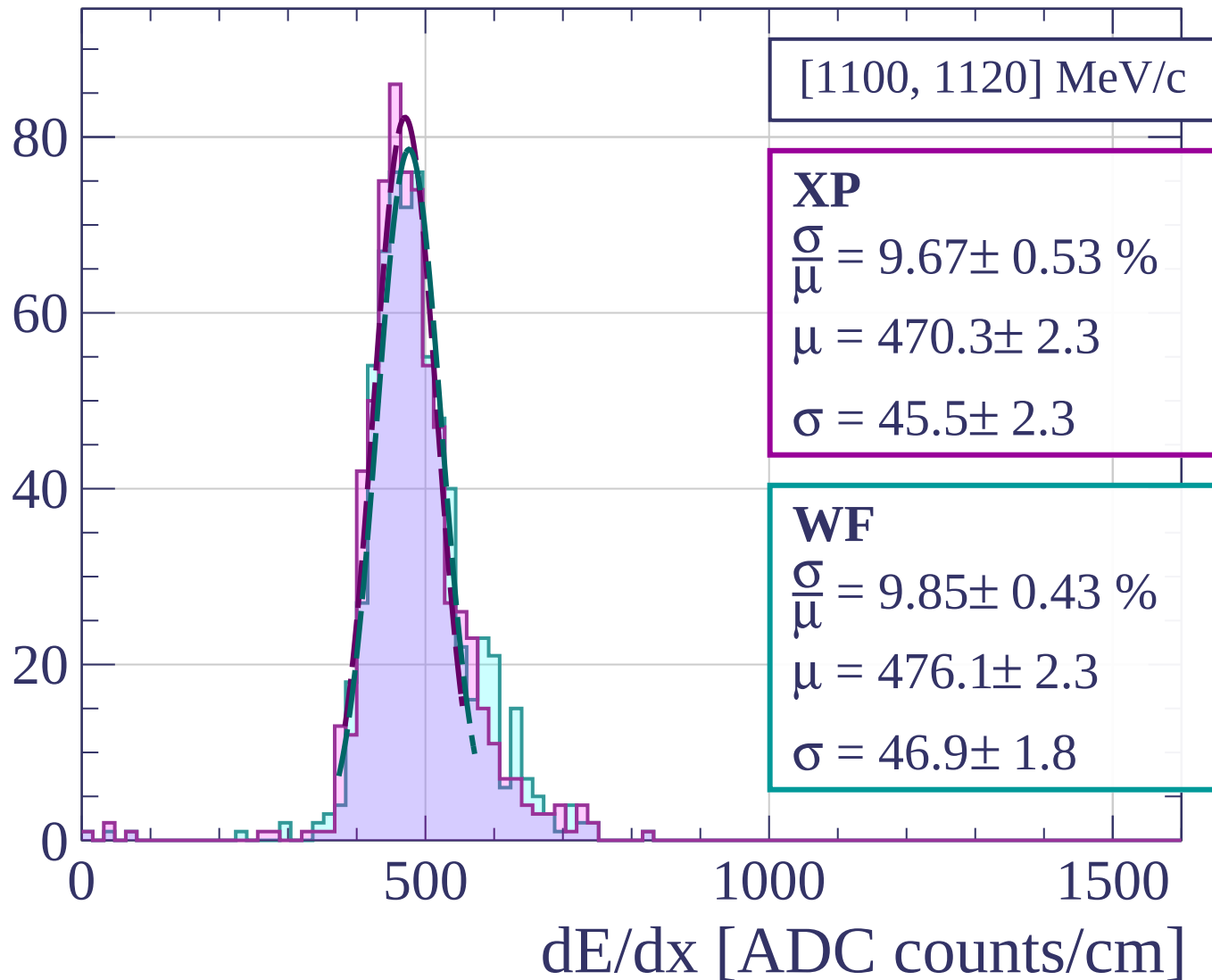
$$\frac{\sigma}{\mu} = 9.80 \pm 0.61 \%$$

$$\mu = 475.1 \pm 2.4$$

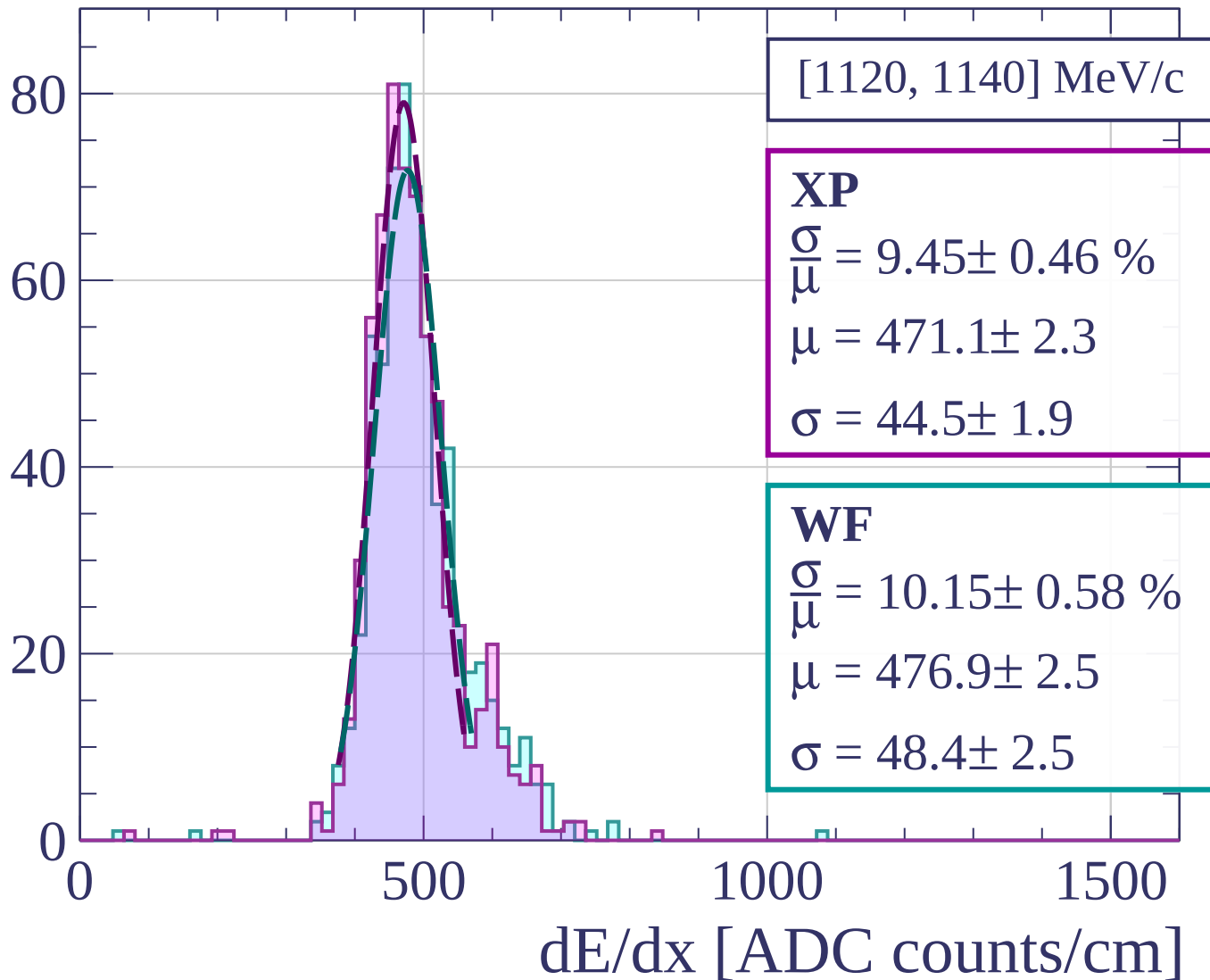
$$\sigma = 46.6 \pm 2.7$$

dE/dx [ADC counts/cm]

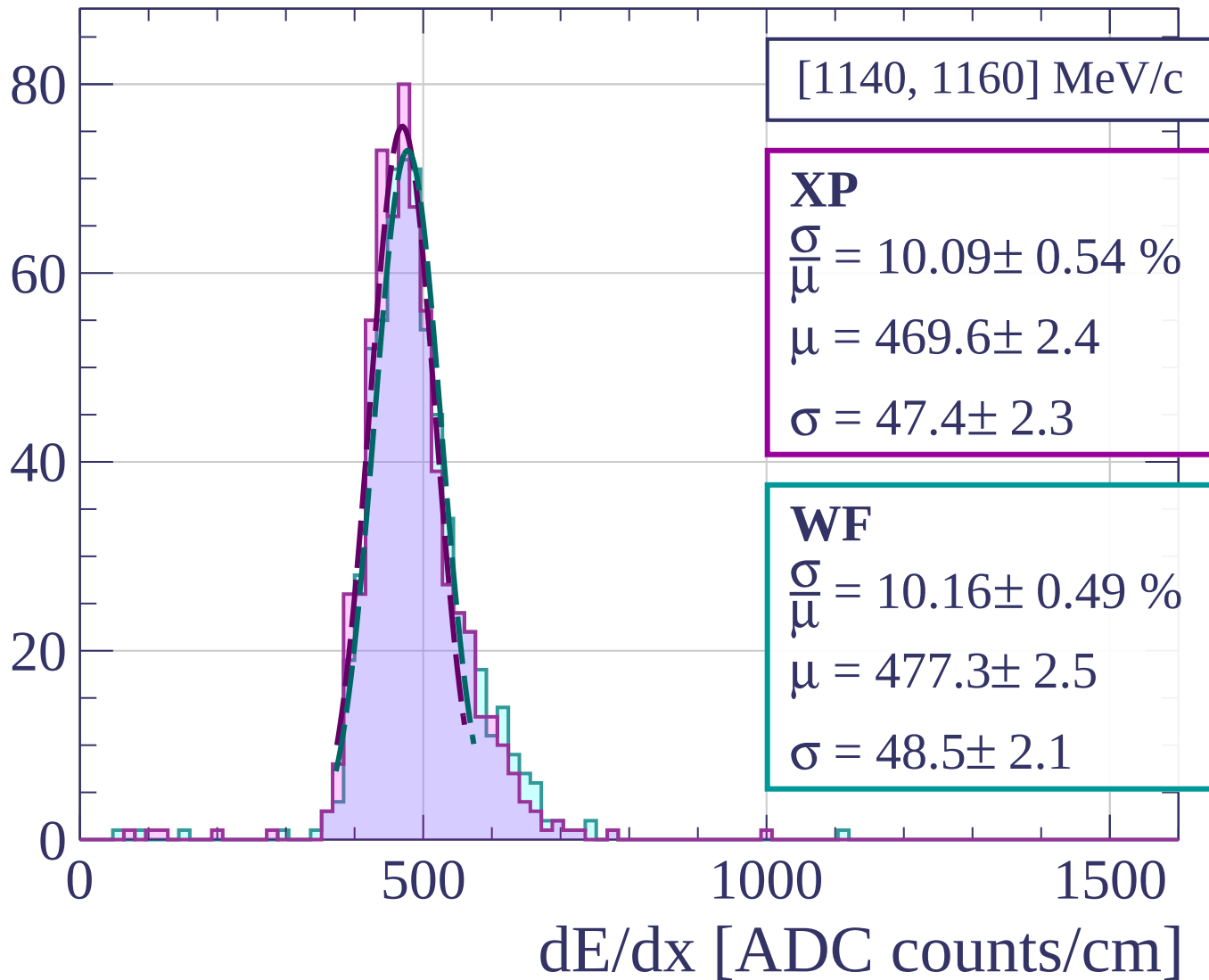
Count



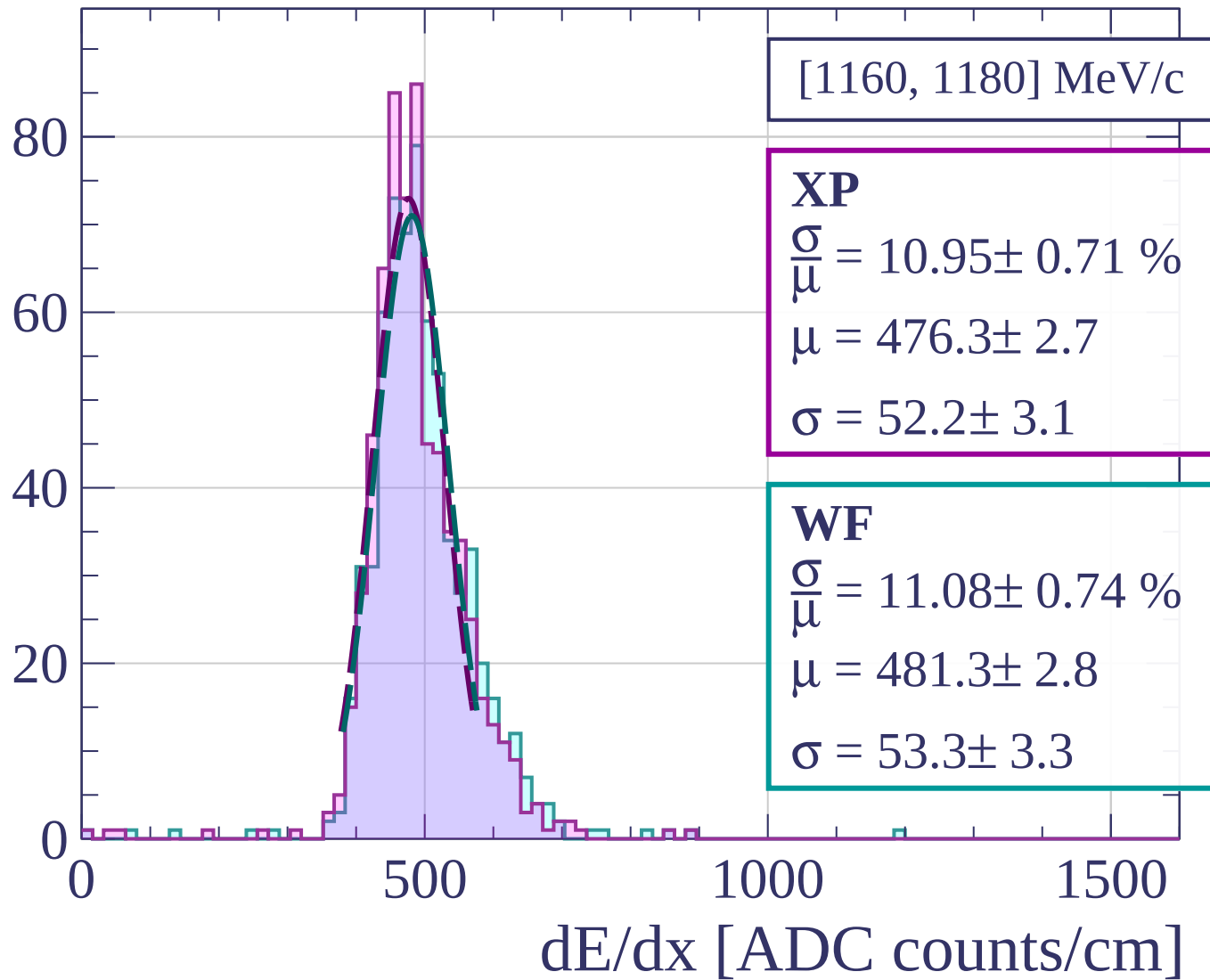
Count



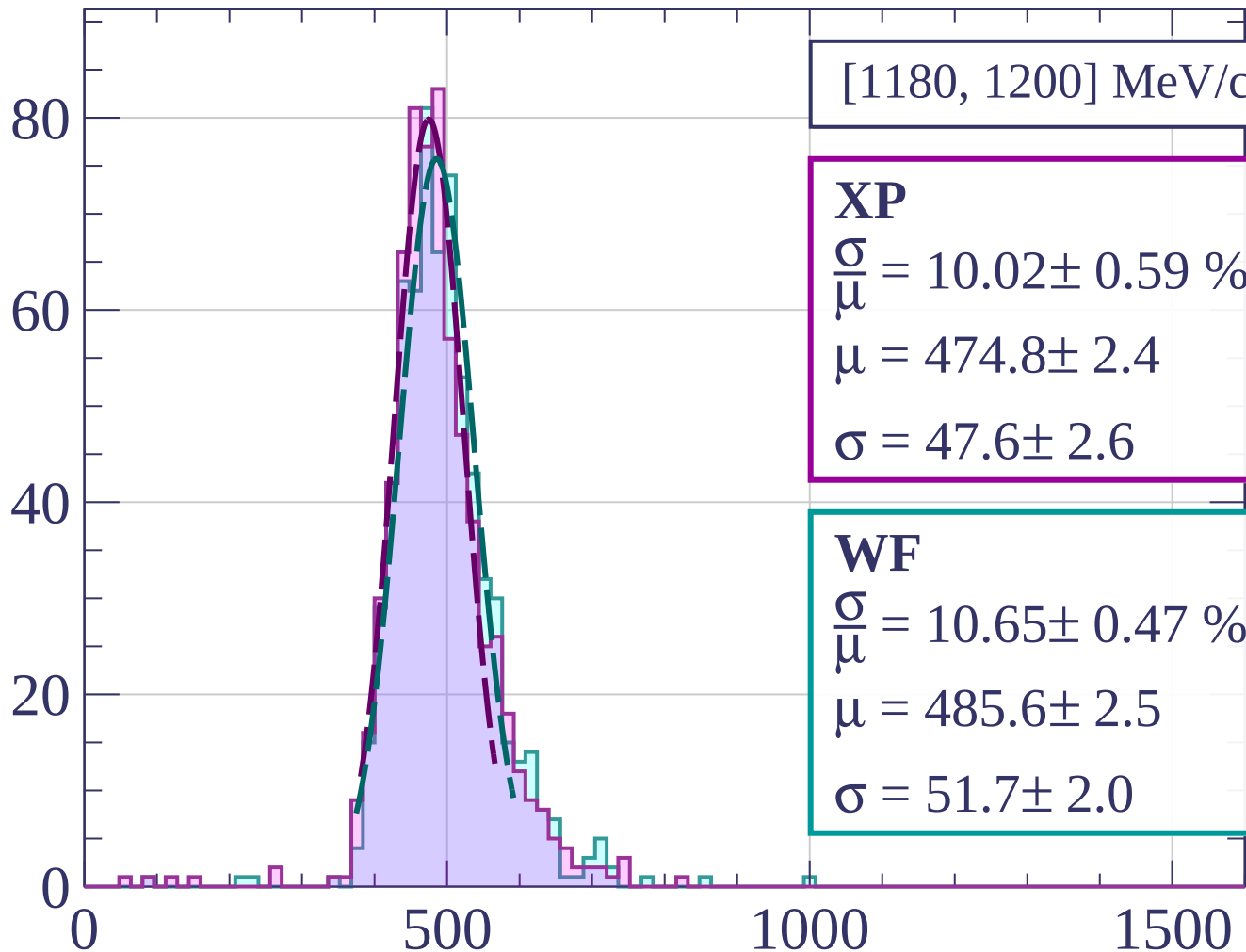
Count



Count



Count



[1180, 1200] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 10.02 \pm 0.59 \%$$

$$\mu = 474.8 \pm 2.4$$

$$\sigma = 47.6 \pm 2.6$$

WF

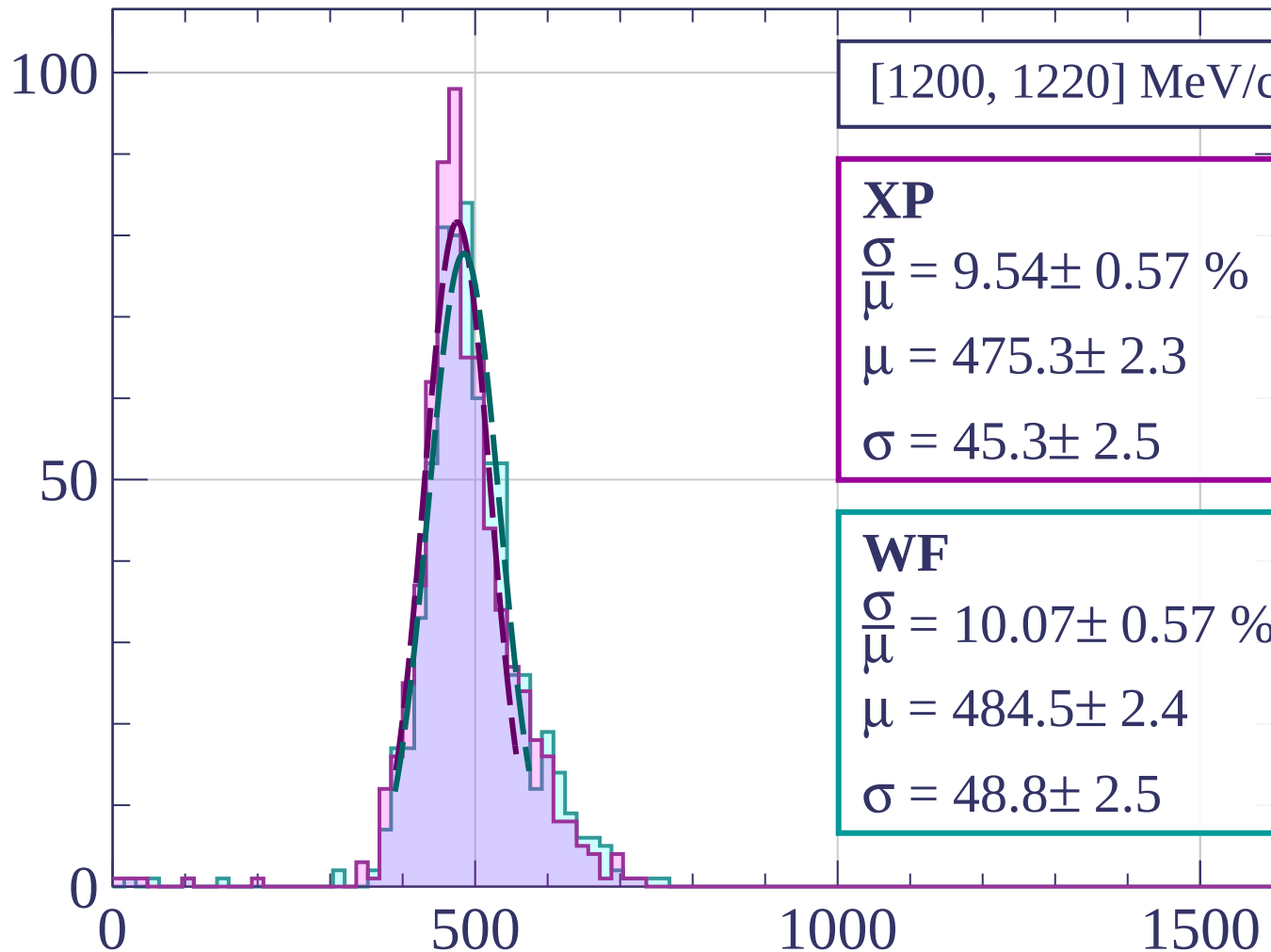
$$\frac{\sigma}{\bar{\mu}} = 10.65 \pm 0.47 \%$$

$$\mu = 485.6 \pm 2.5$$

$$\sigma = 51.7 \pm 2.0$$

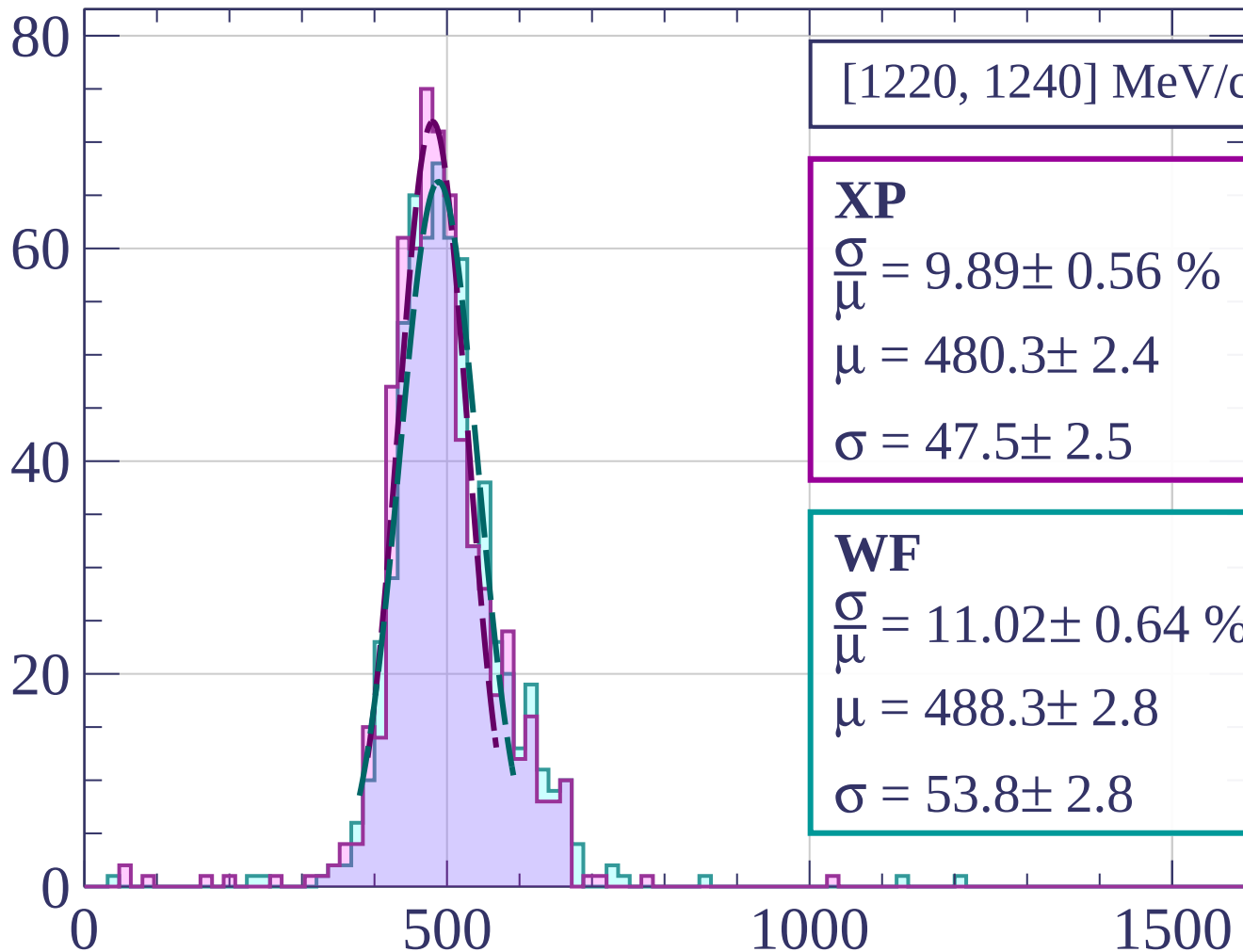
dE/dx [ADC counts/cm]

Count



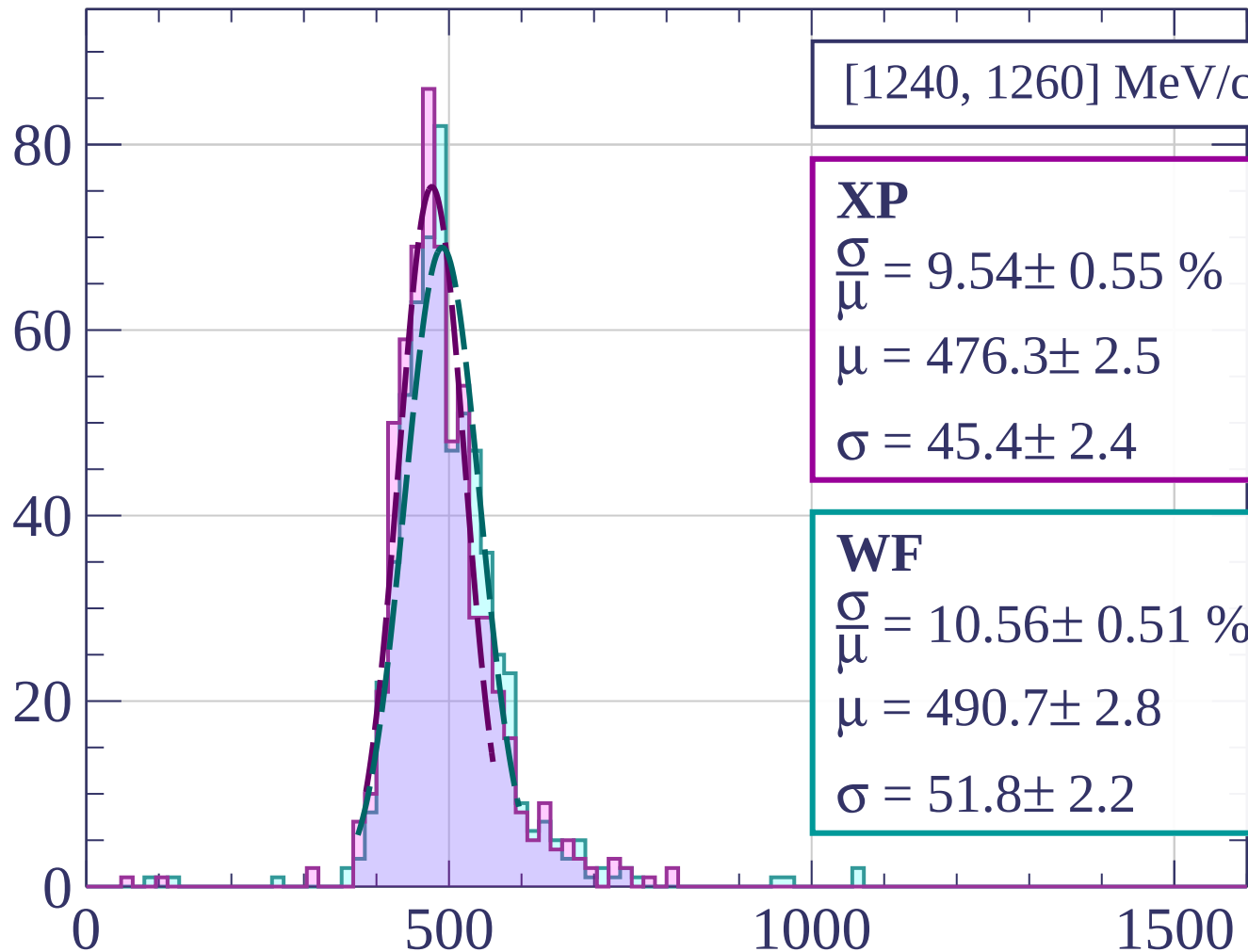
dE/dx [ADC counts/cm]

Count



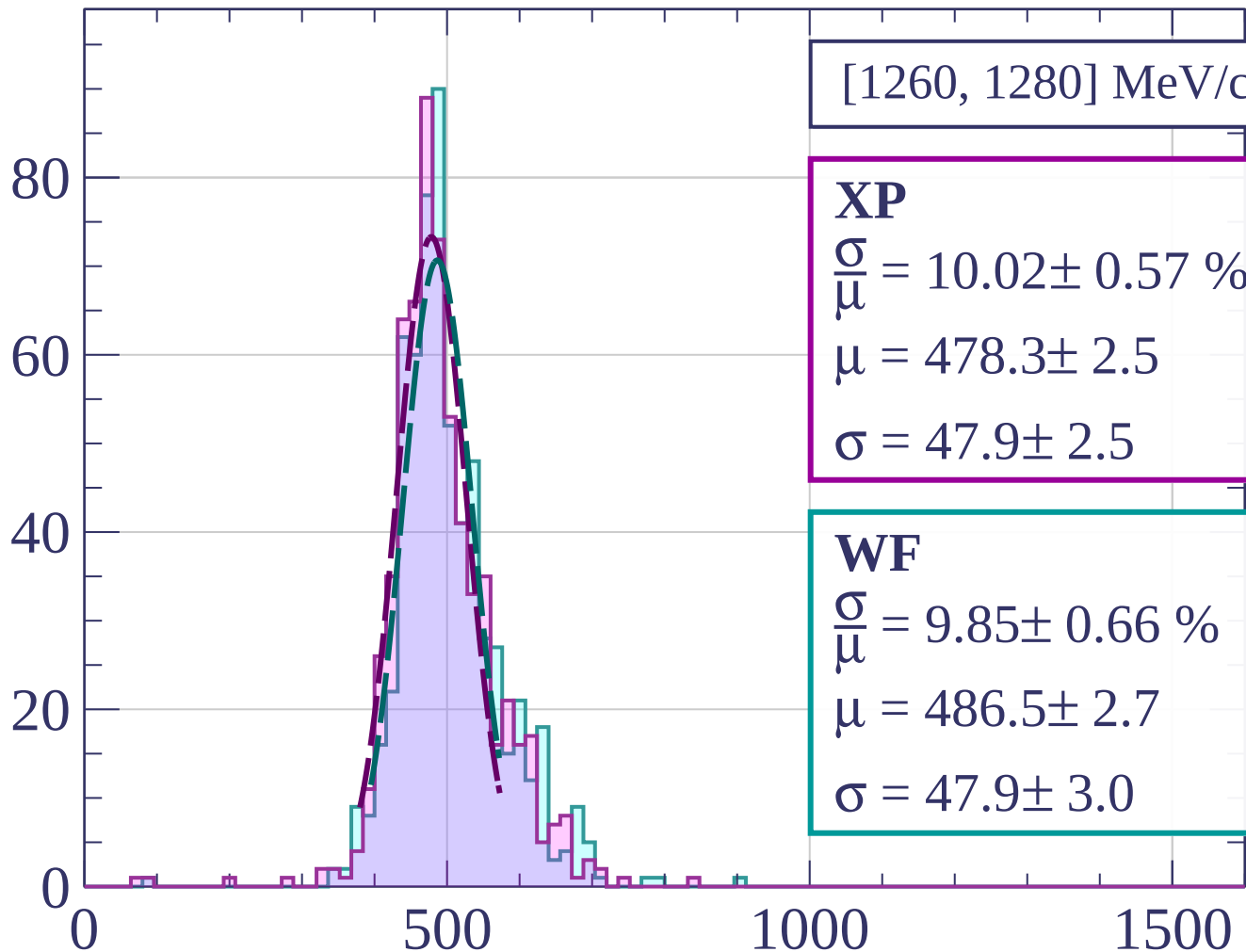
dE/dx [ADC counts/cm]

Count



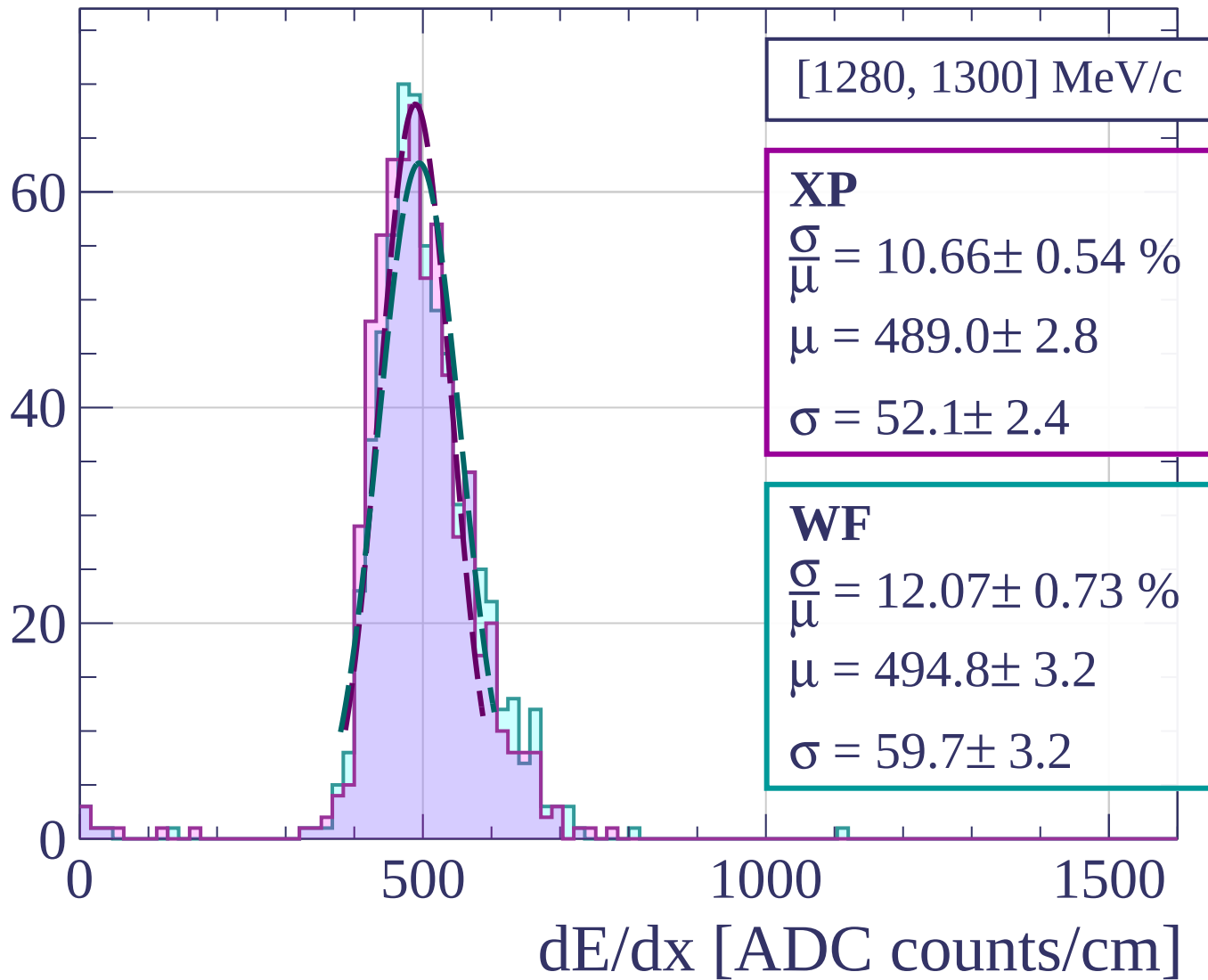
dE/dx [ADC counts/cm]

Count

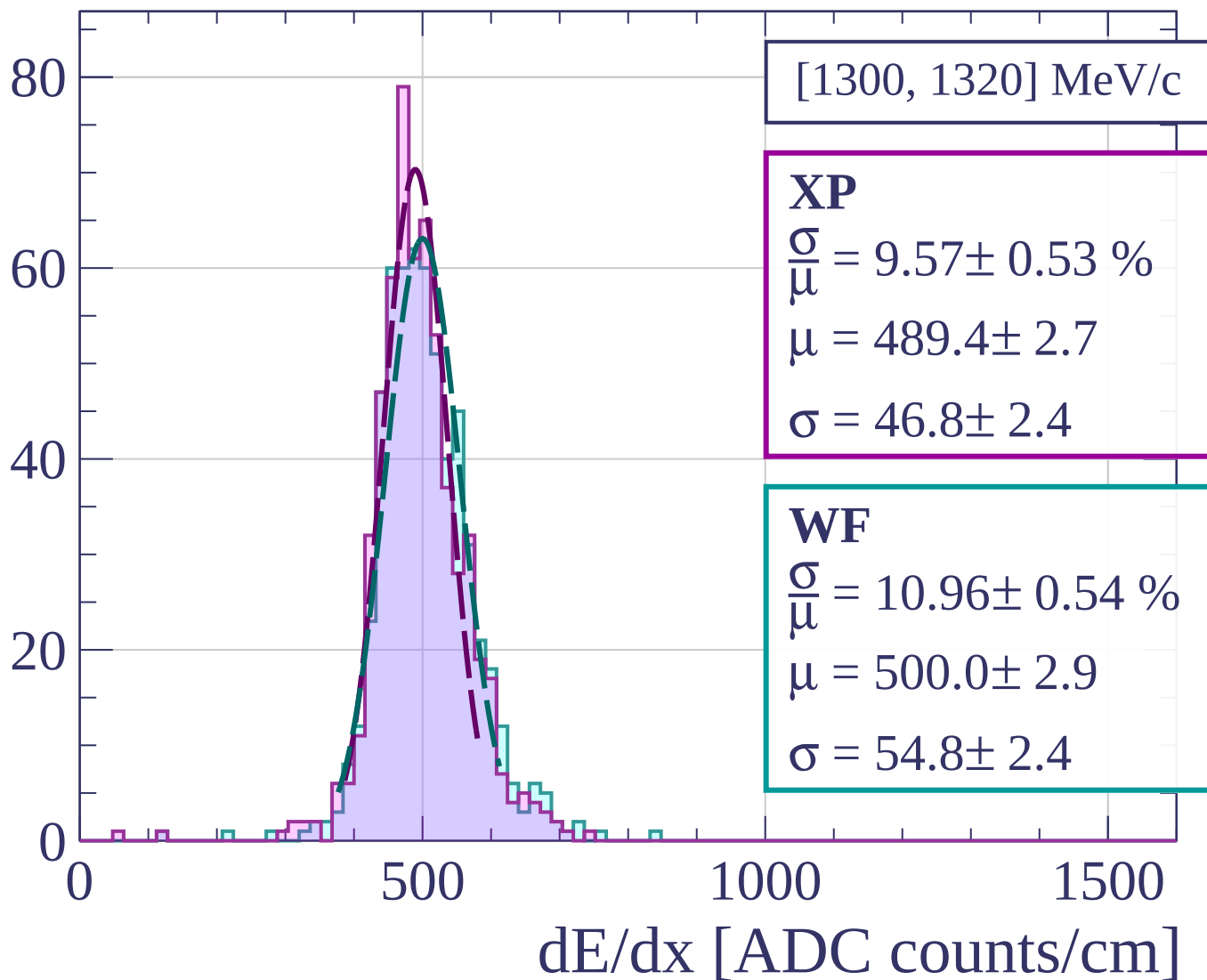


dE/dx [ADC counts/cm]

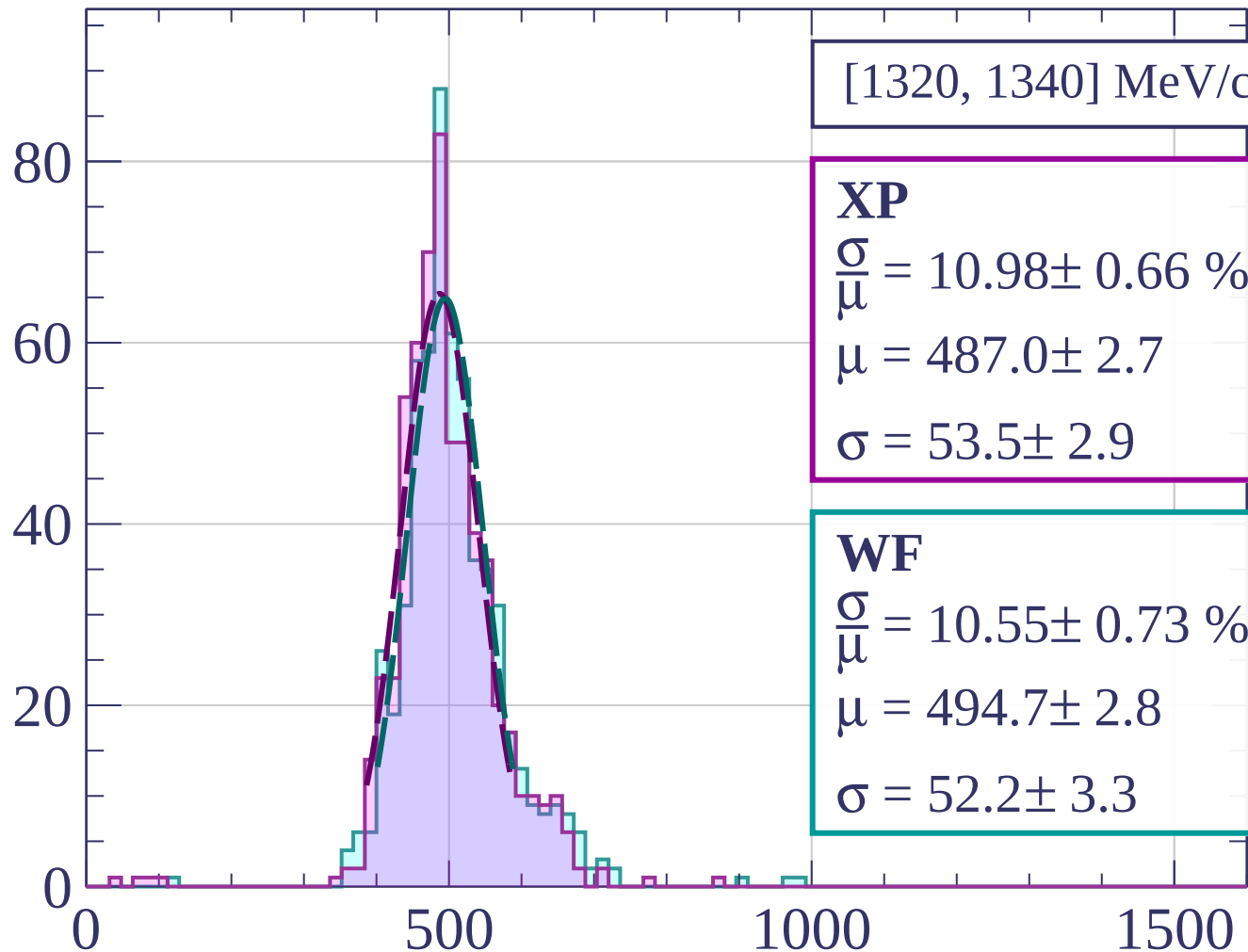
Count



Count

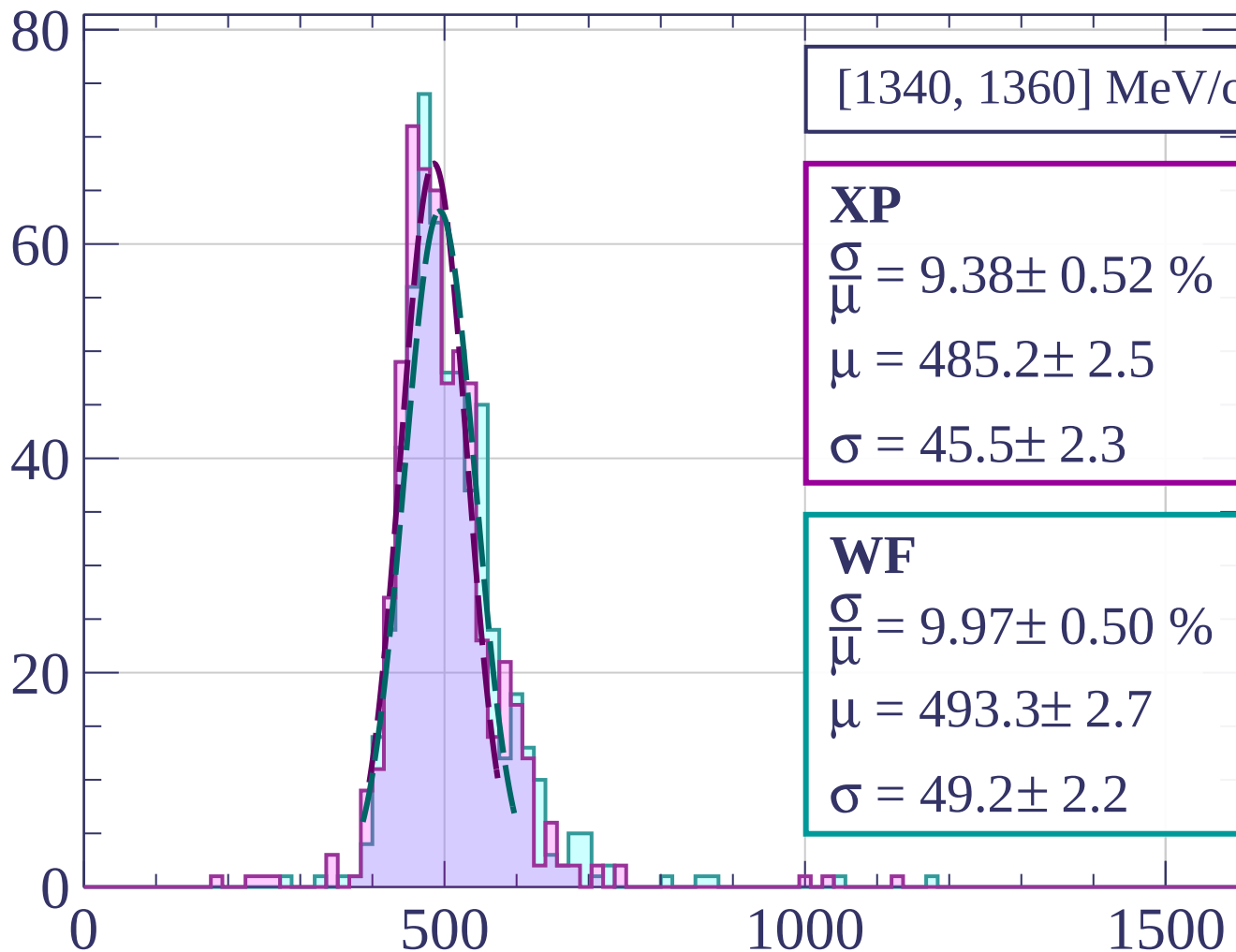


Count



dE/dx [ADC counts/cm]

Count



[1340, 1360] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.38 \pm 0.52 \%$$

$$\mu = 485.2 \pm 2.5$$

$$\sigma = 45.5 \pm 2.3$$

WF

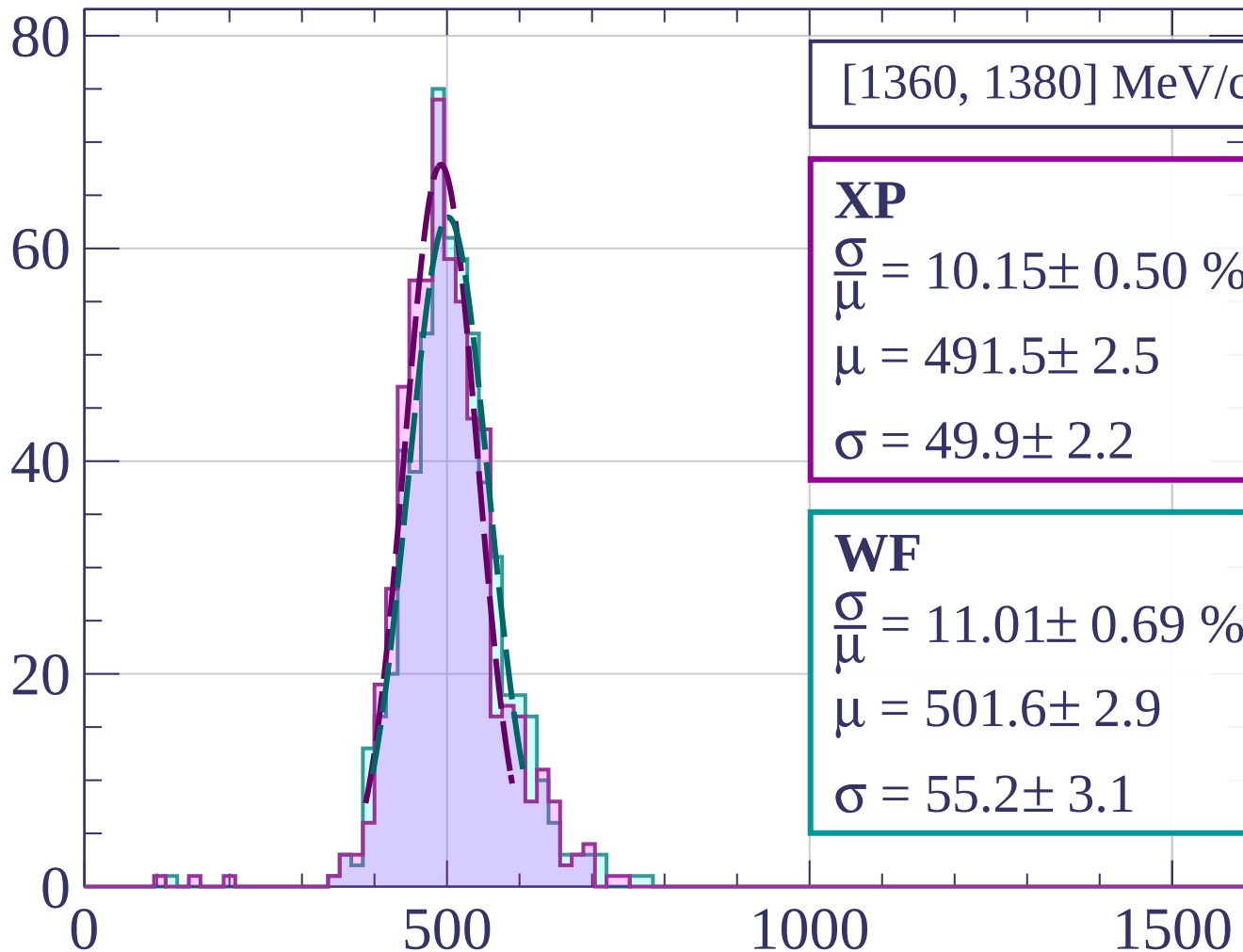
$$\frac{\sigma}{\mu} = 9.97 \pm 0.50 \%$$

$$\mu = 493.3 \pm 2.7$$

$$\sigma = 49.2 \pm 2.2$$

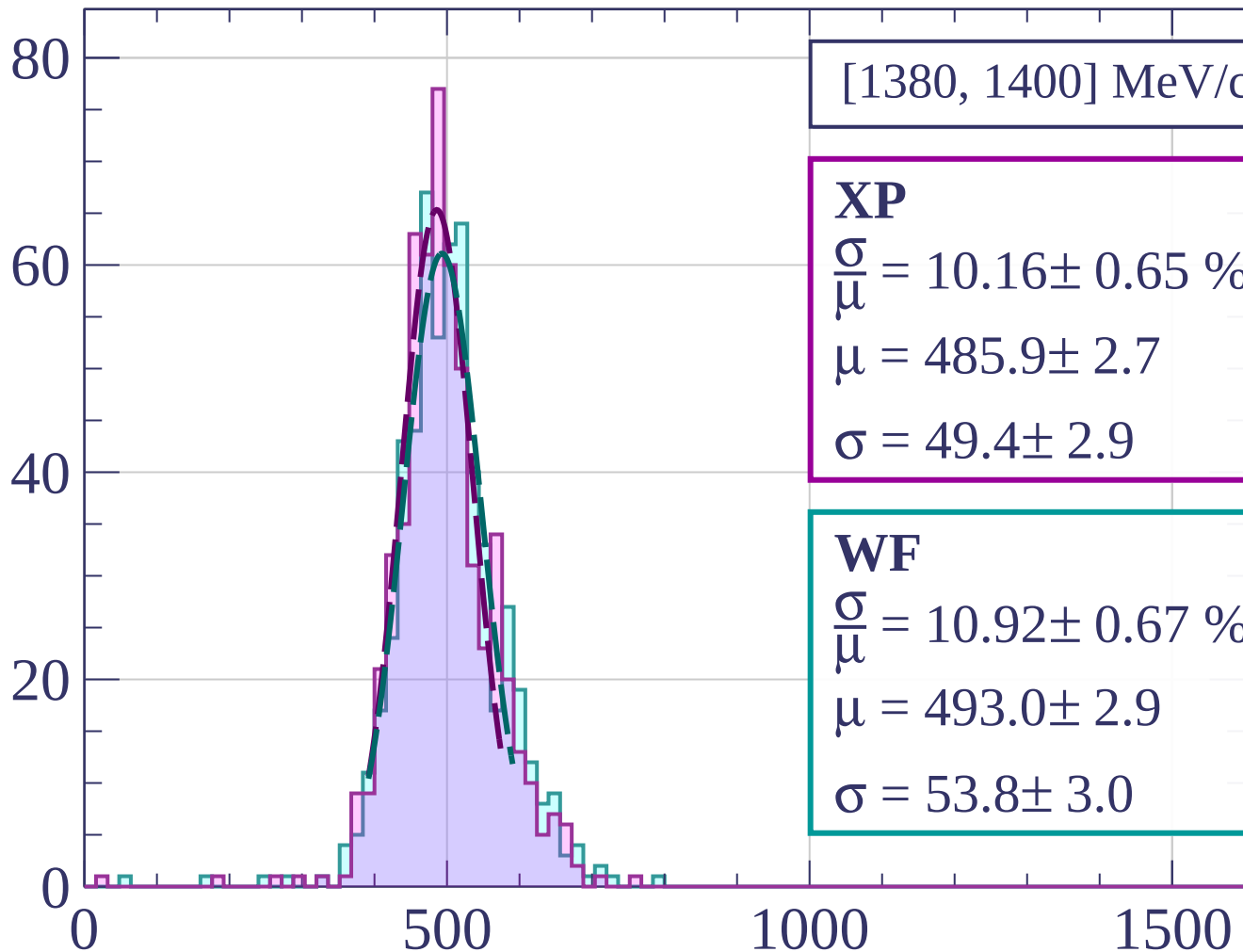
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[1380, 1400] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.16 \pm 0.65 \%$$

$$\mu = 485.9 \pm 2.7$$

$$\sigma = 49.4 \pm 2.9$$

WF

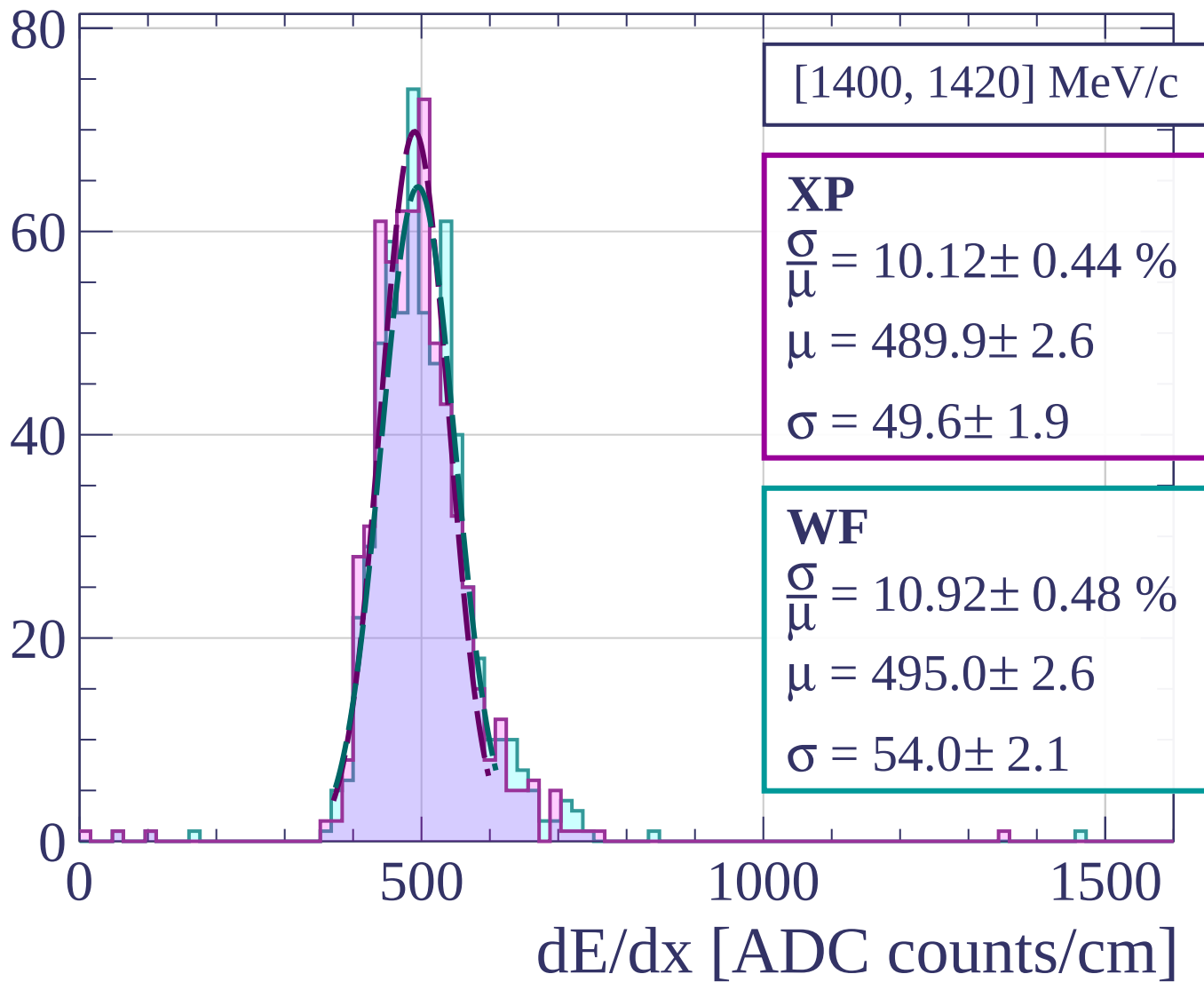
$$\frac{\sigma}{\mu} = 10.92 \pm 0.67 \%$$

$$\mu = 493.0 \pm 2.9$$

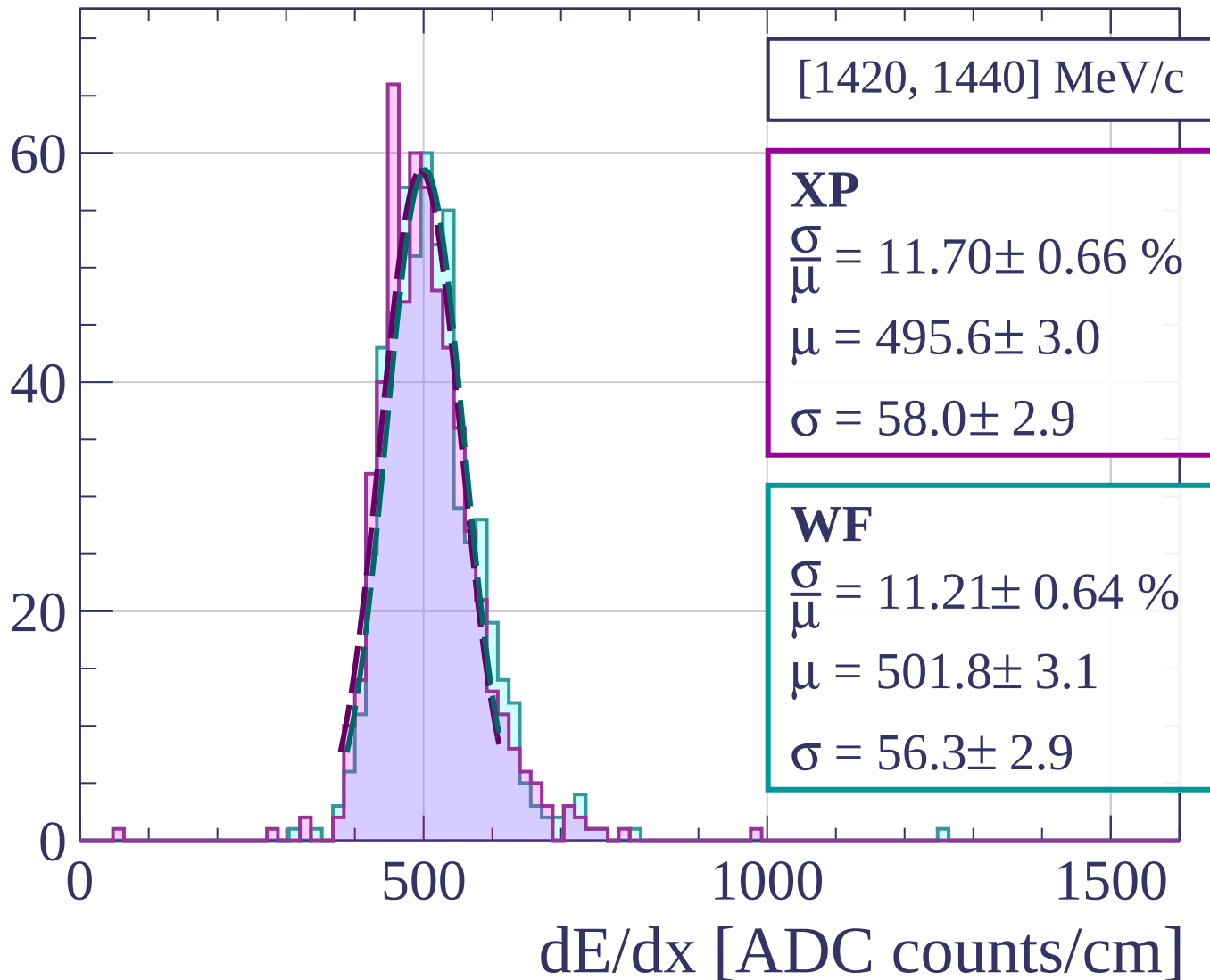
$$\sigma = 53.8 \pm 3.0$$

dE/dx [ADC counts/cm]

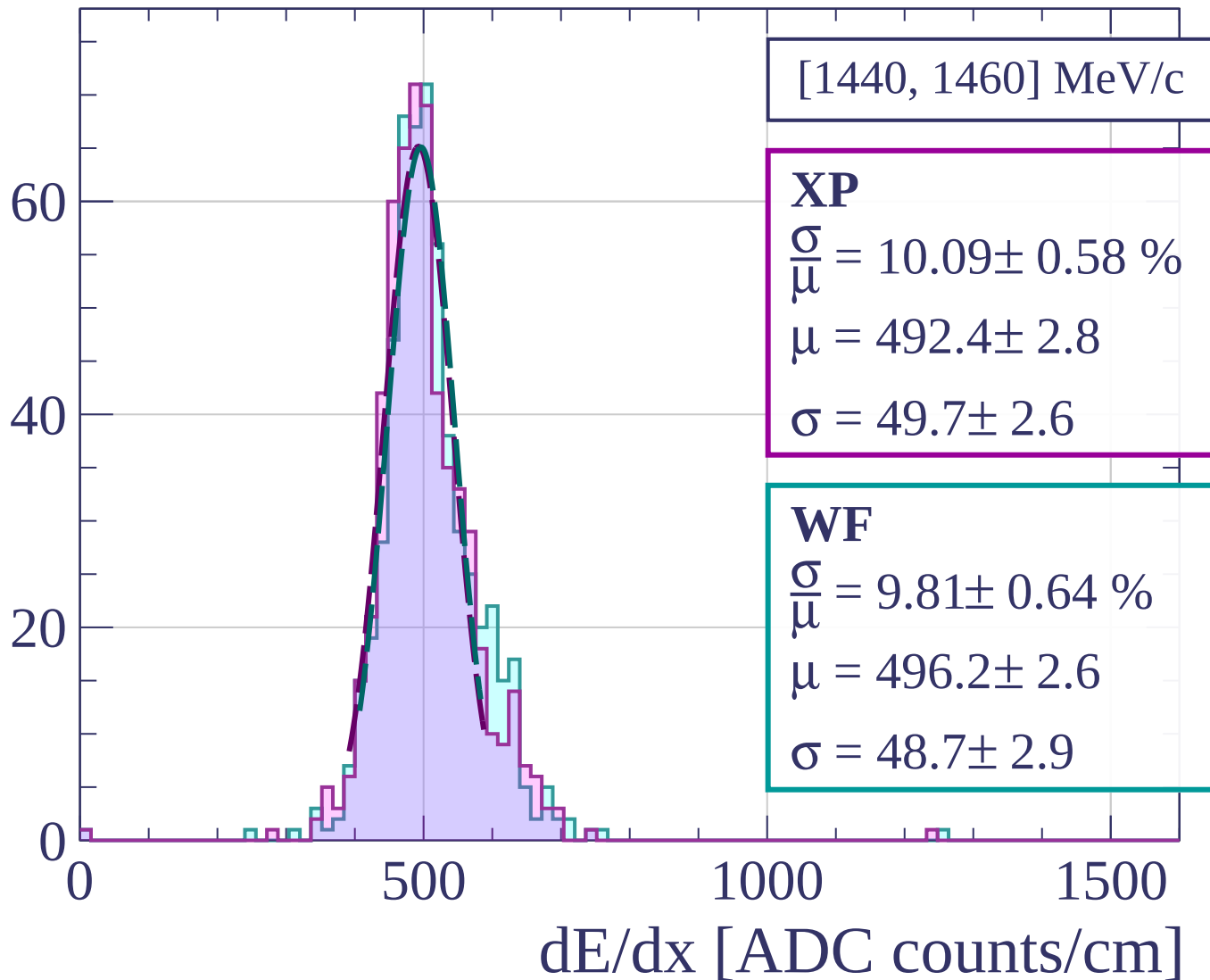
Count

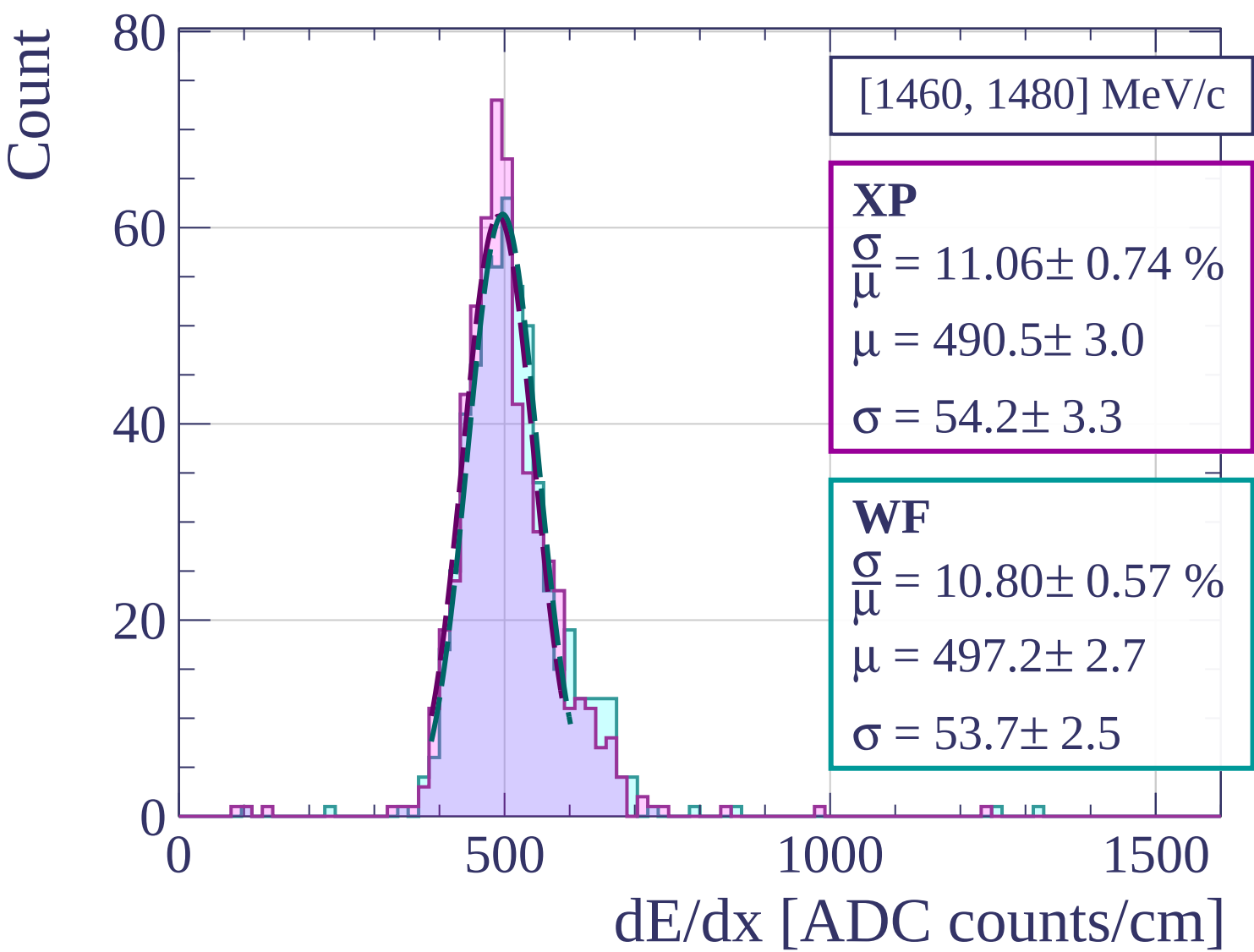


Count

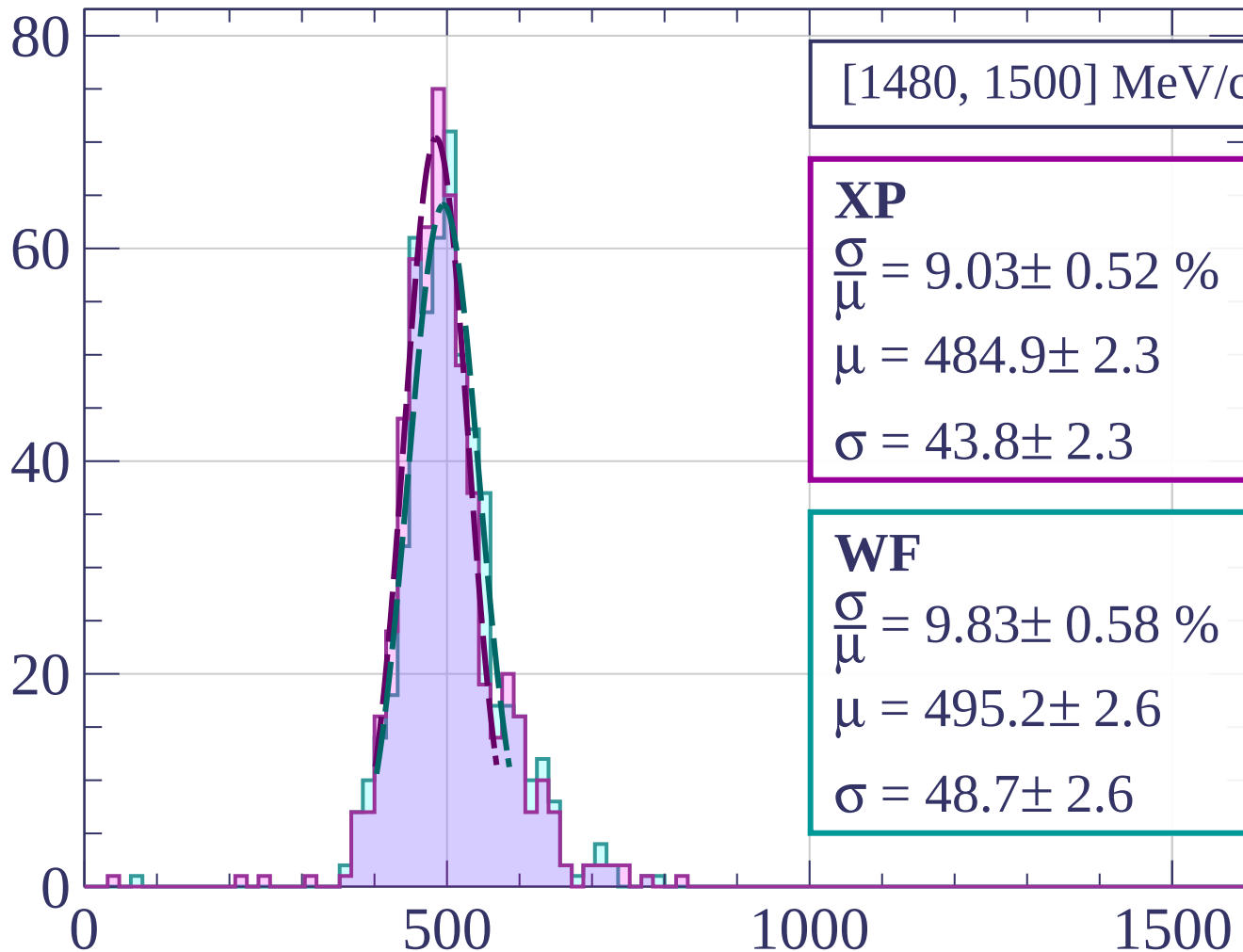


Count



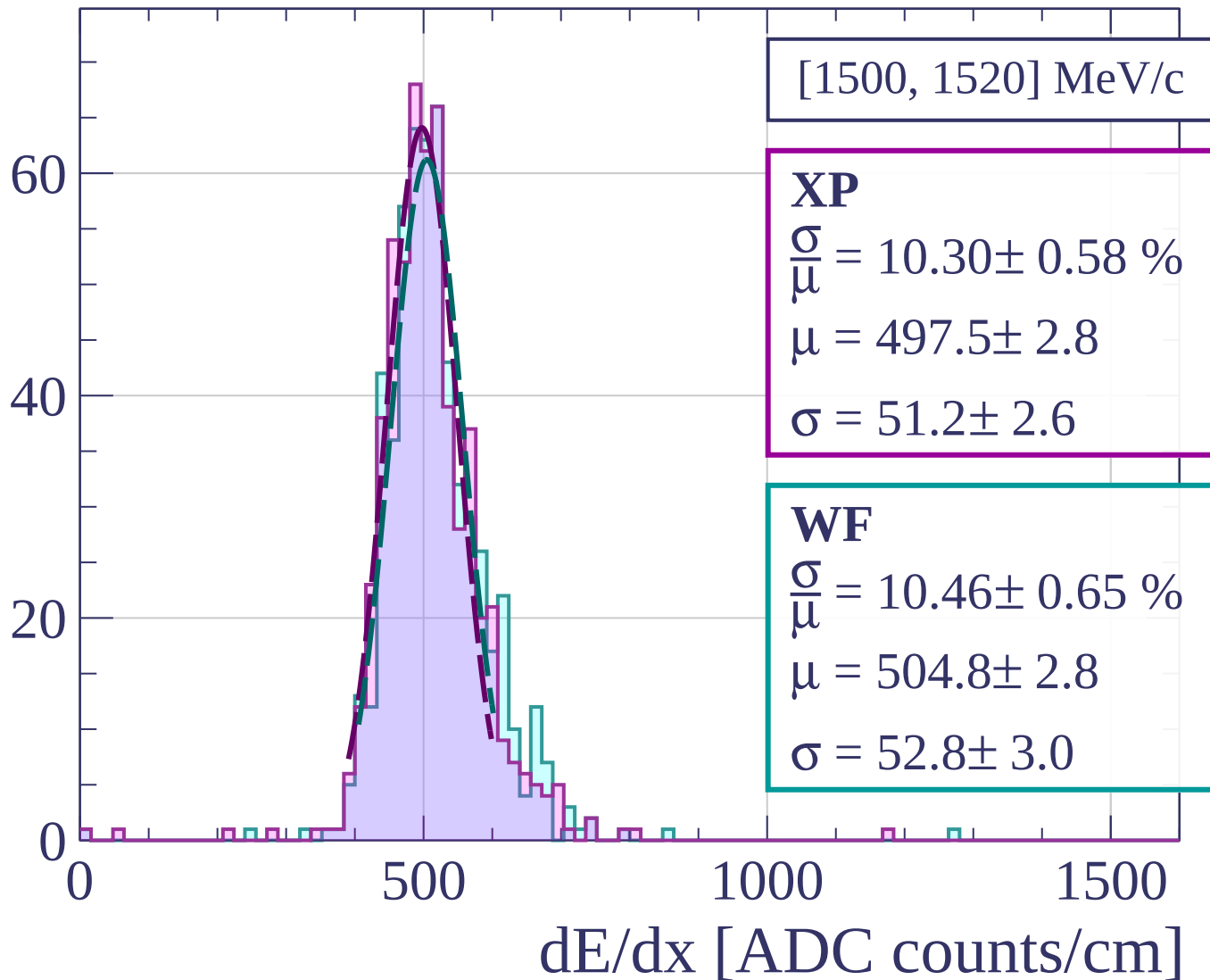


Count

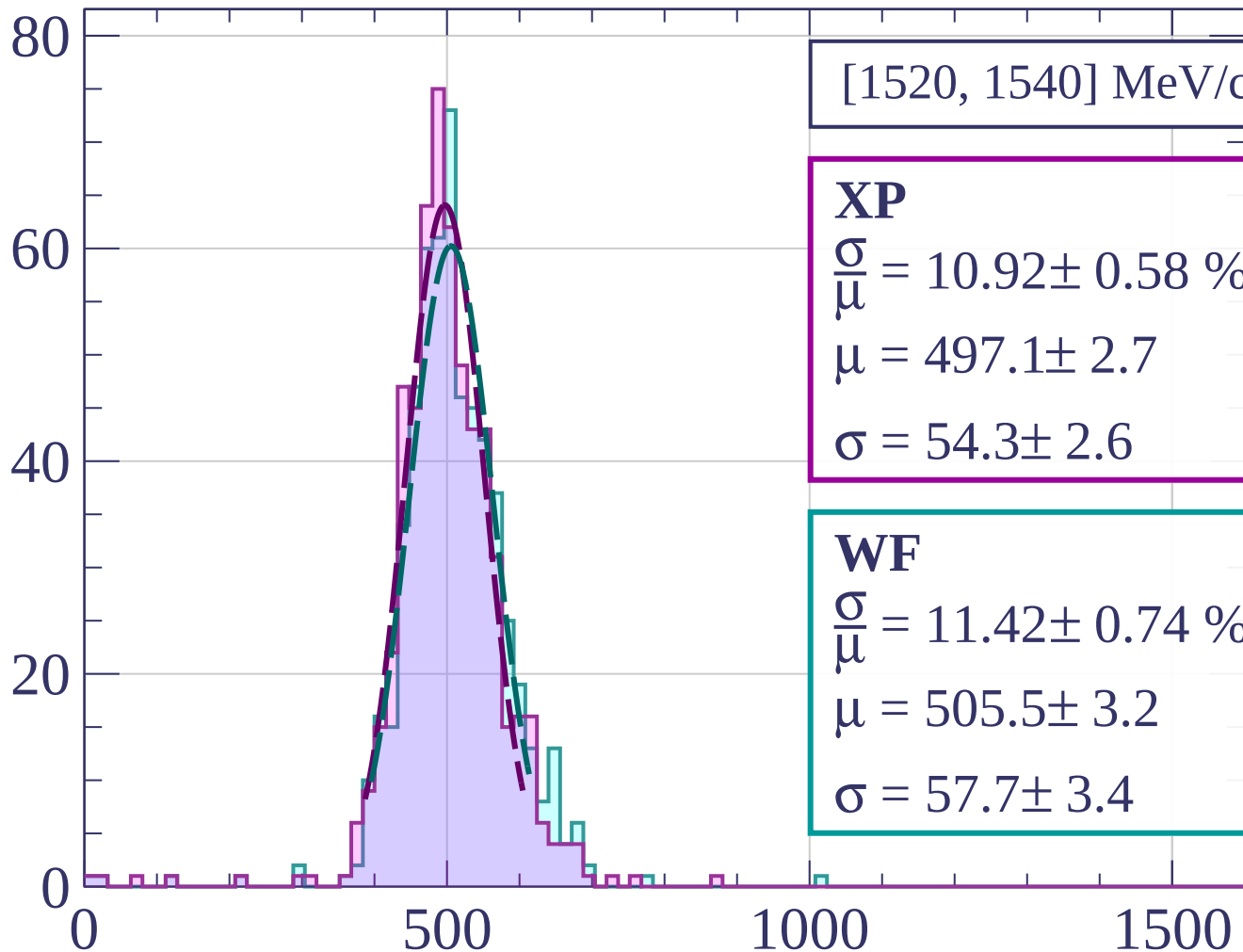


dE/dx [ADC counts/cm]

Count

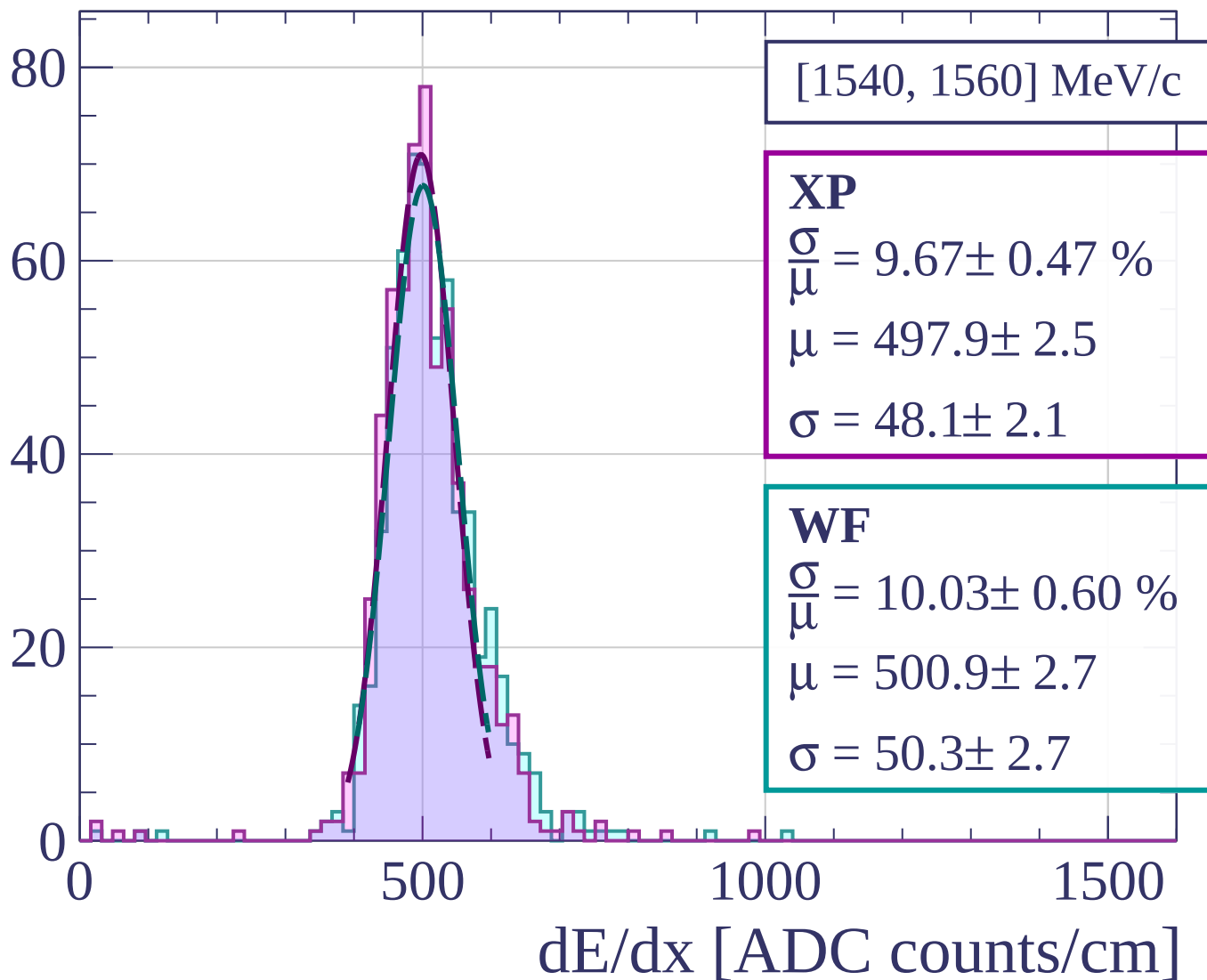


Count

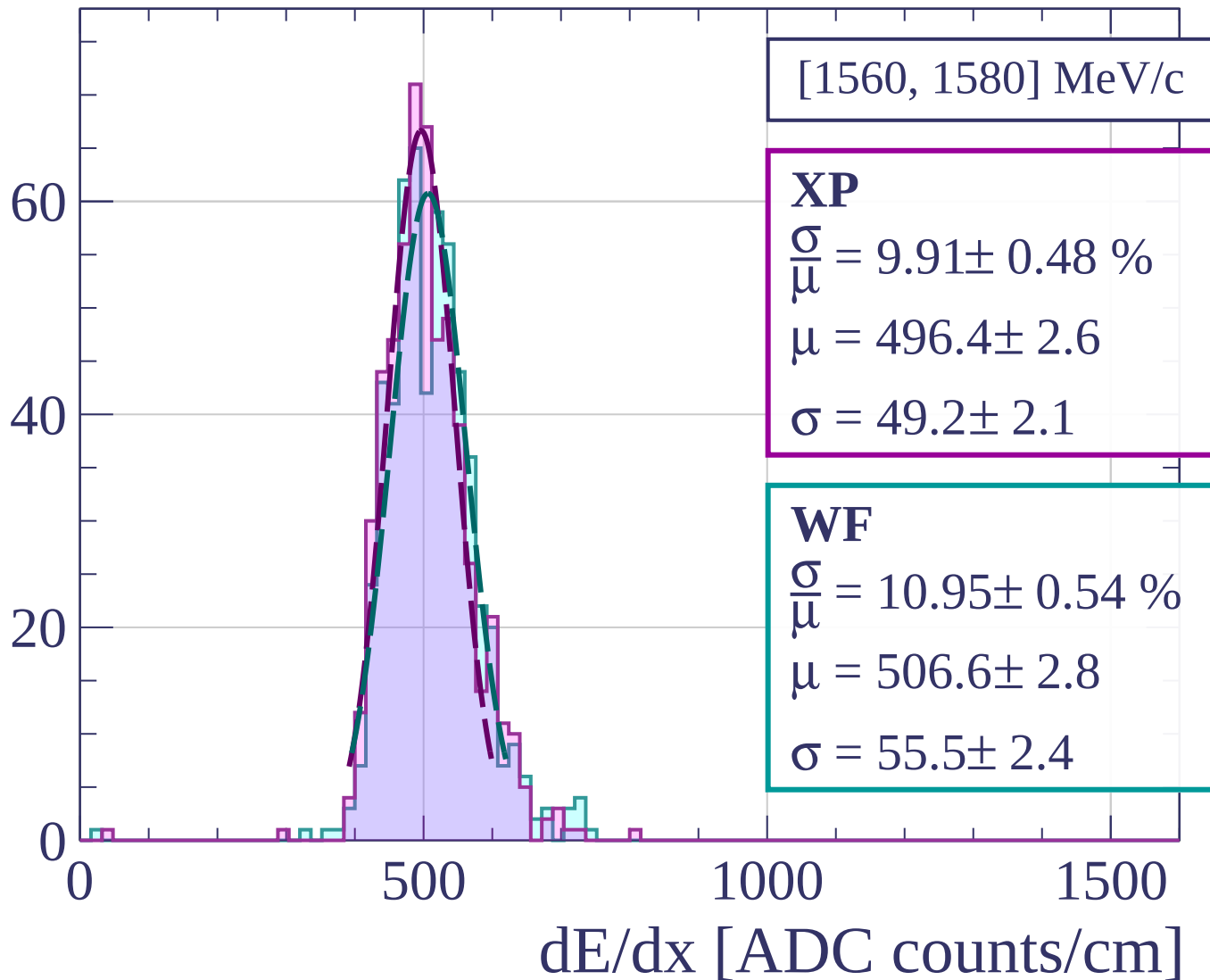


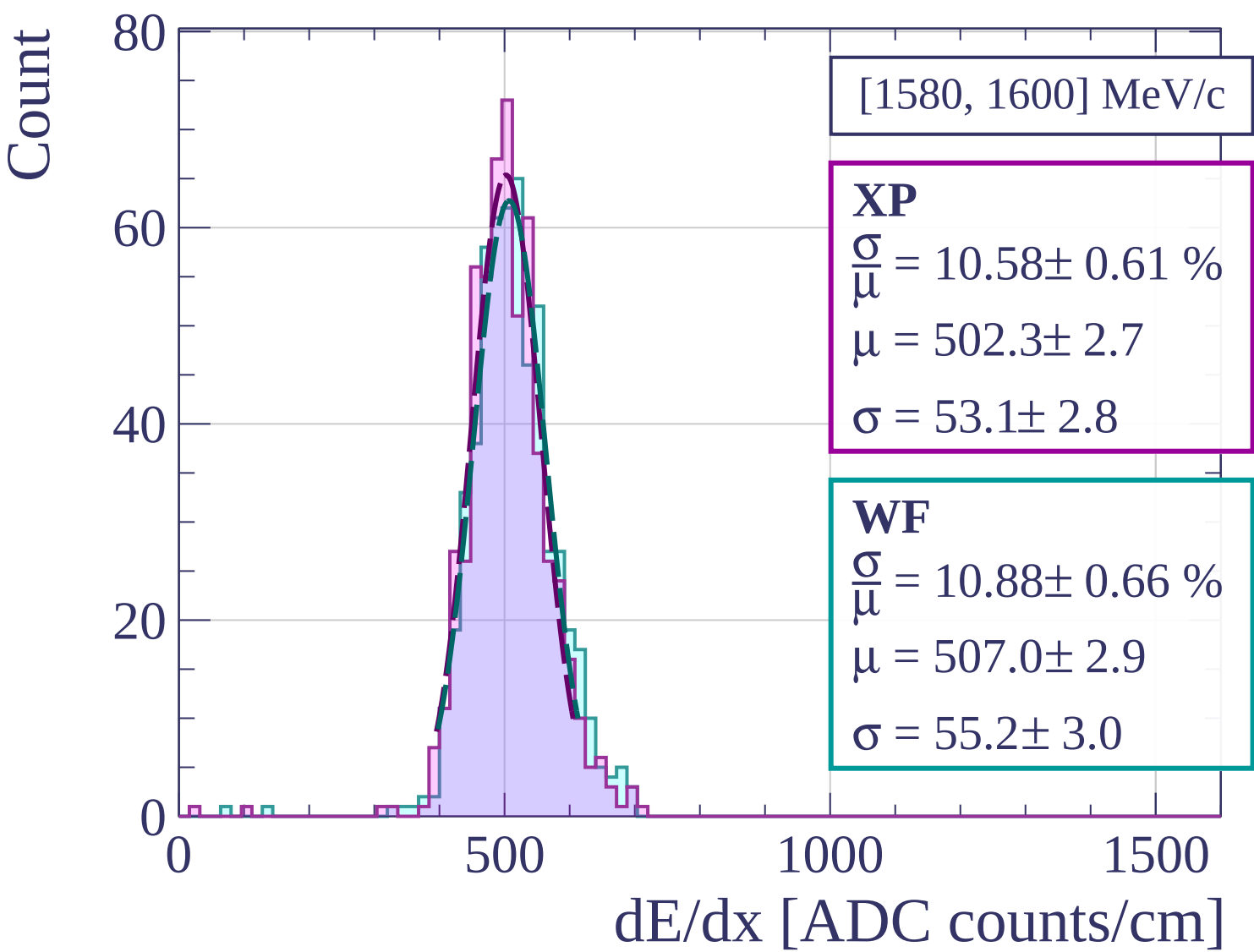
dE/dx [ADC counts/cm]

Count

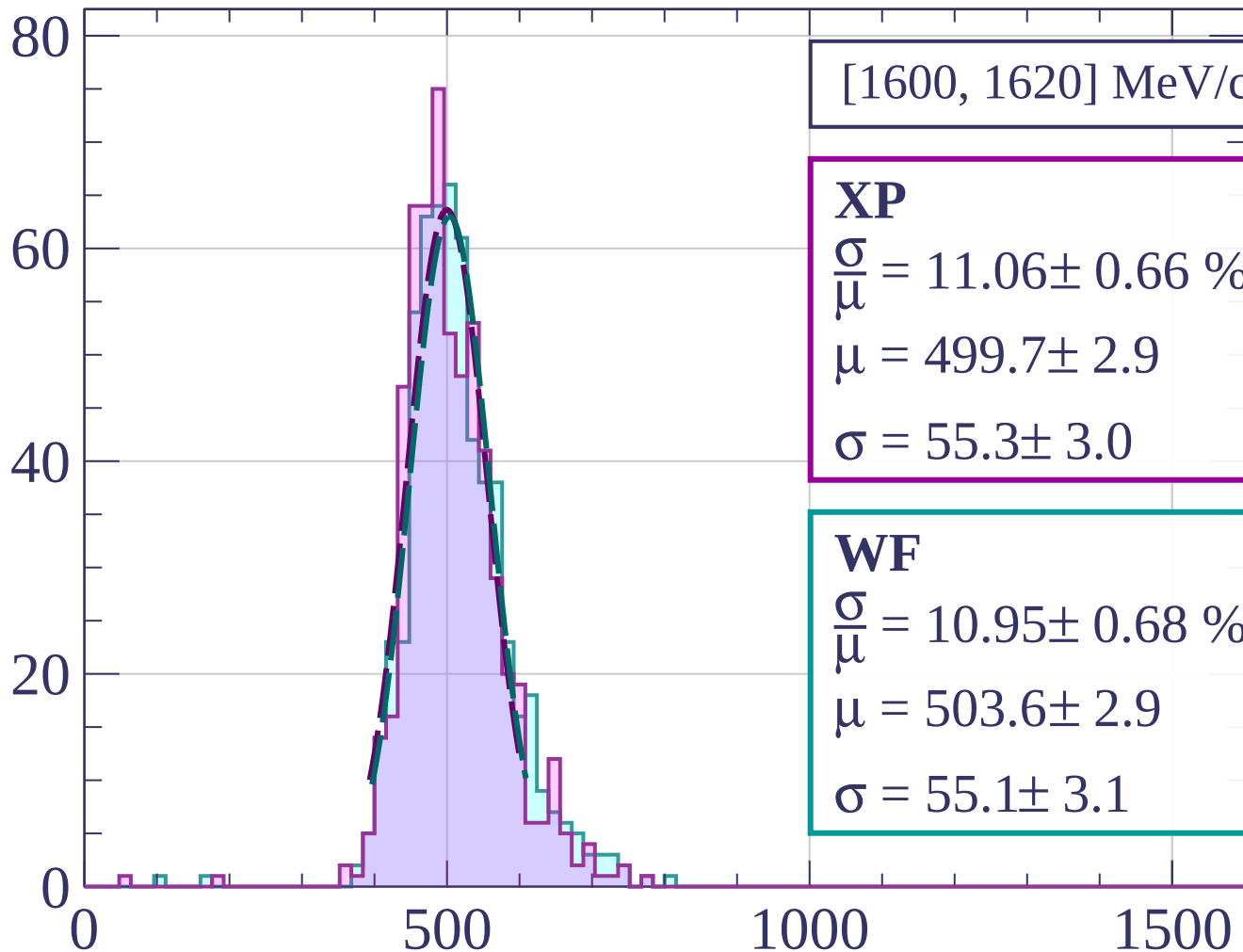


Count



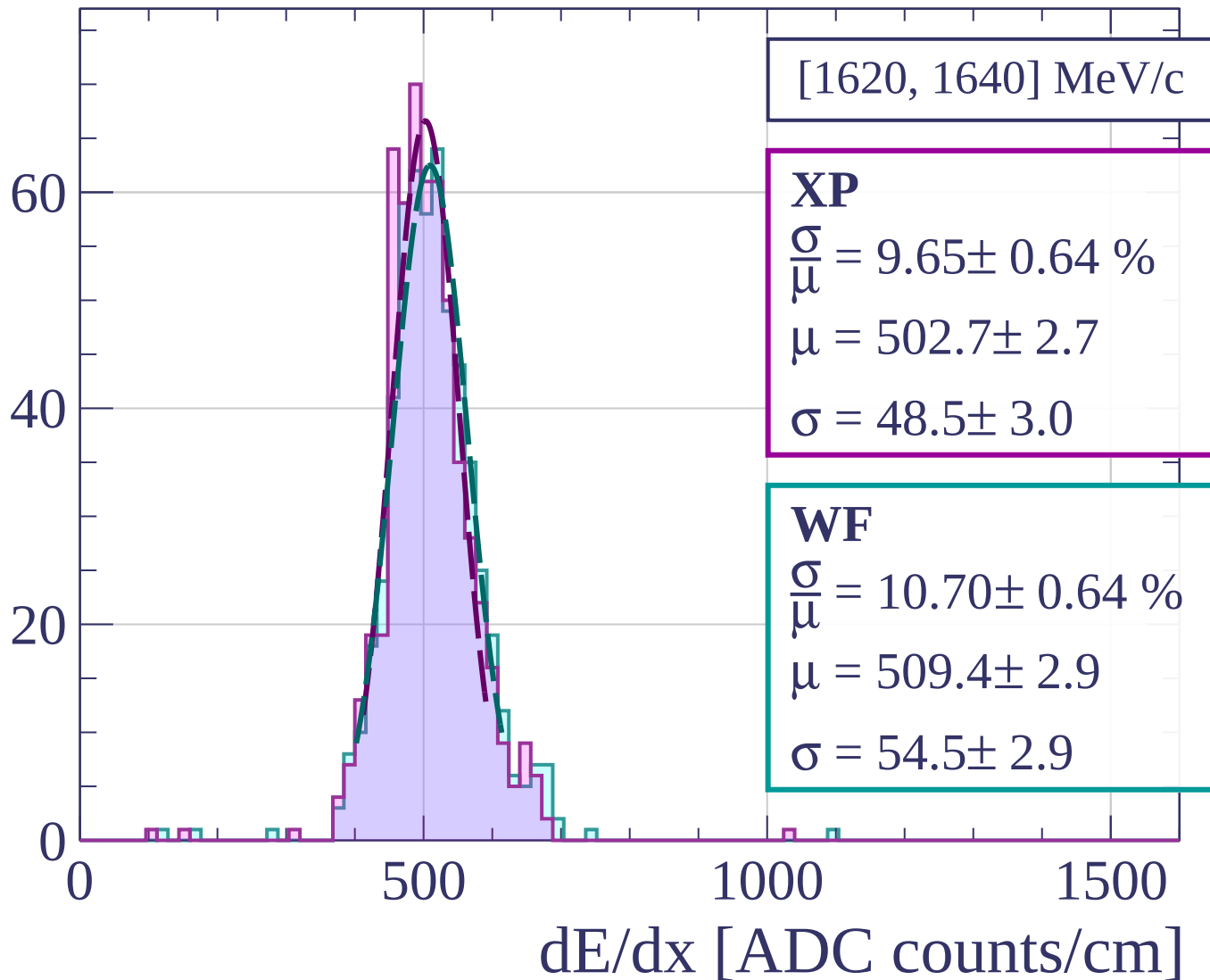


Count

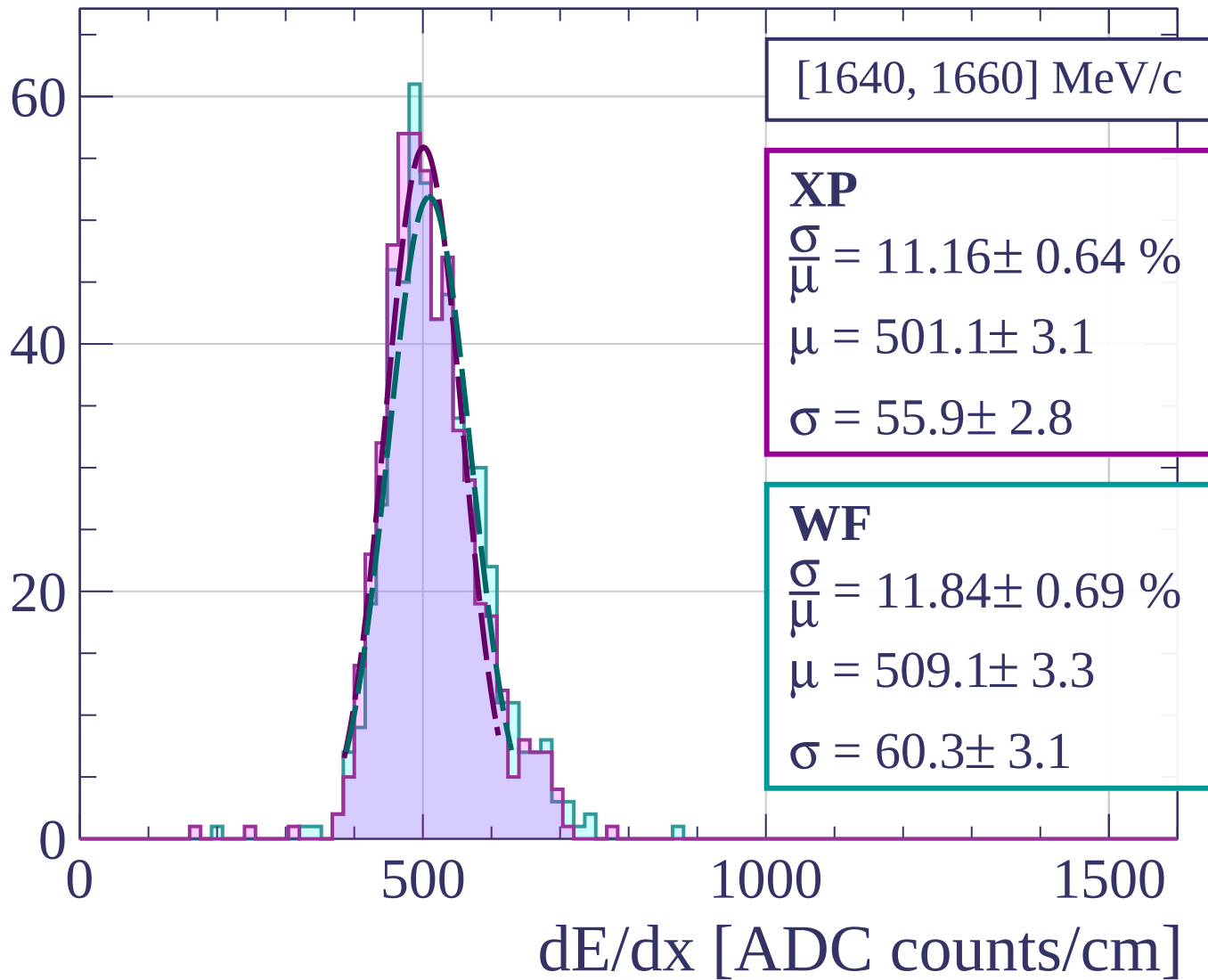


dE/dx [ADC counts/cm]

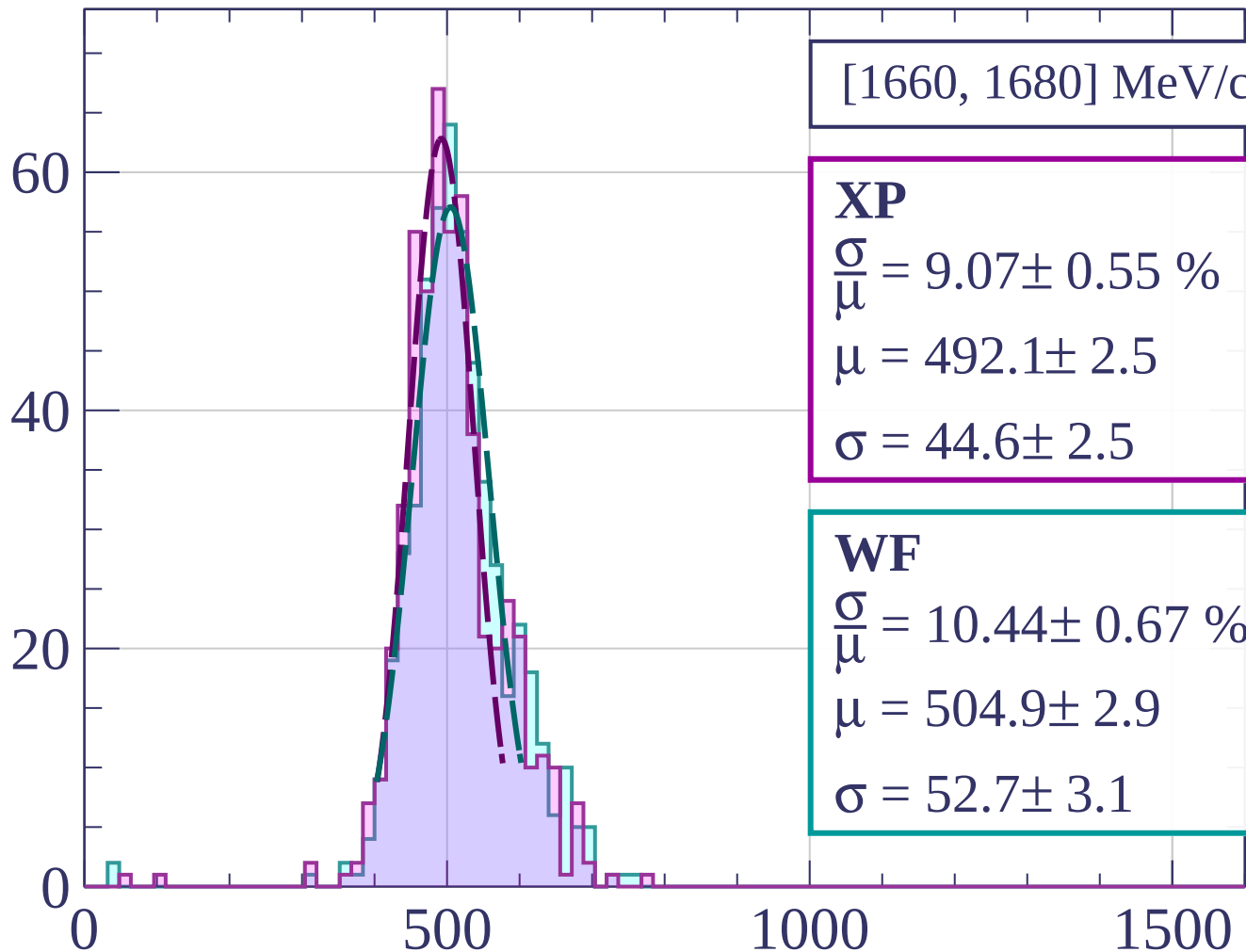
Count



Count



Count



[1660, 1680] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.07 \pm 0.55 \%$$

$$\mu = 492.1 \pm 2.5$$

$$\sigma = 44.6 \pm 2.5$$

WF

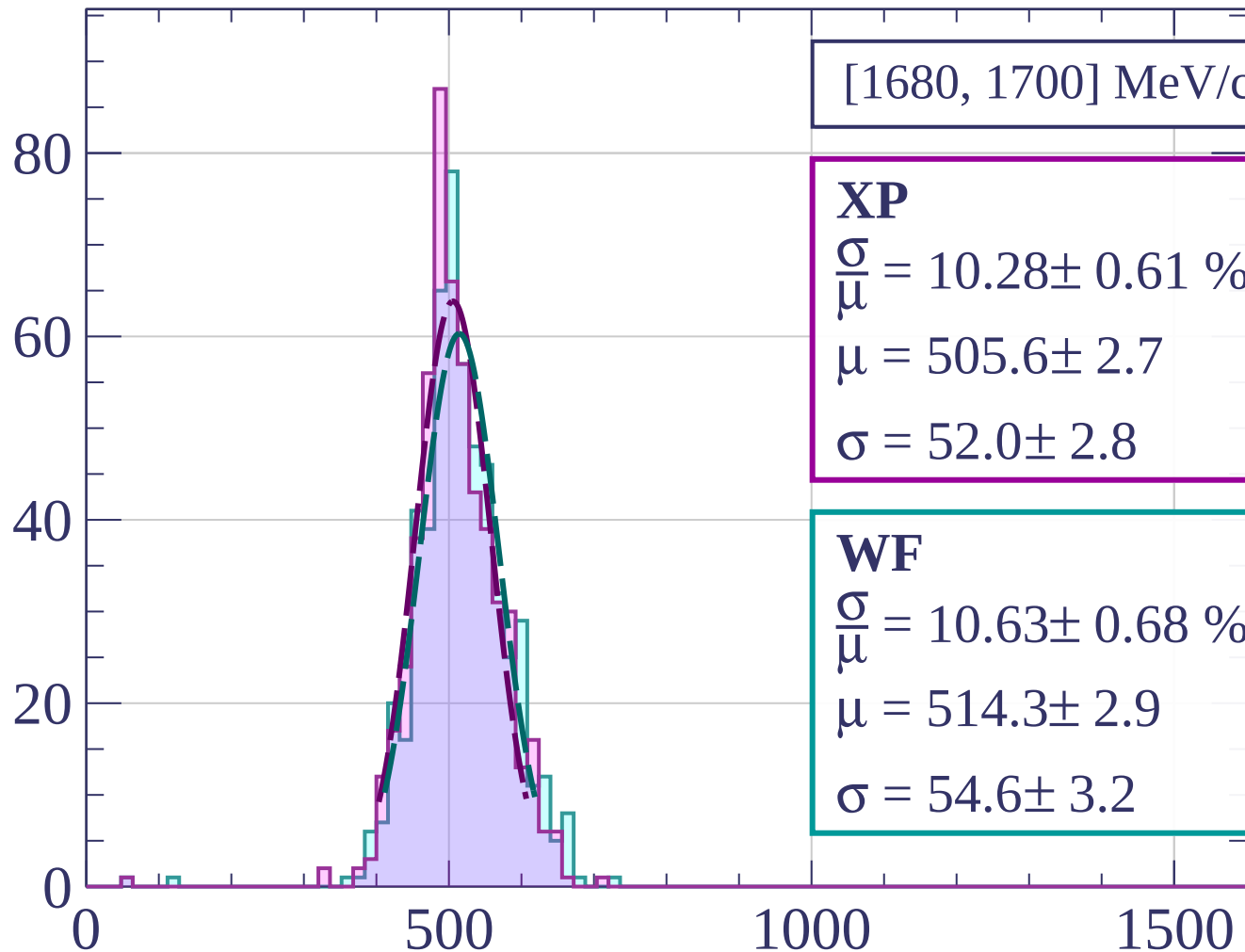
$$\frac{\sigma}{\mu} = 10.44 \pm 0.67 \%$$

$$\mu = 504.9 \pm 2.9$$

$$\sigma = 52.7 \pm 3.1$$

dE/dx [ADC counts/cm]

Count



[1680, 1700] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.28 \pm 0.61 \%$$

$$\mu = 505.6 \pm 2.7$$

$$\sigma = 52.0 \pm 2.8$$

WF

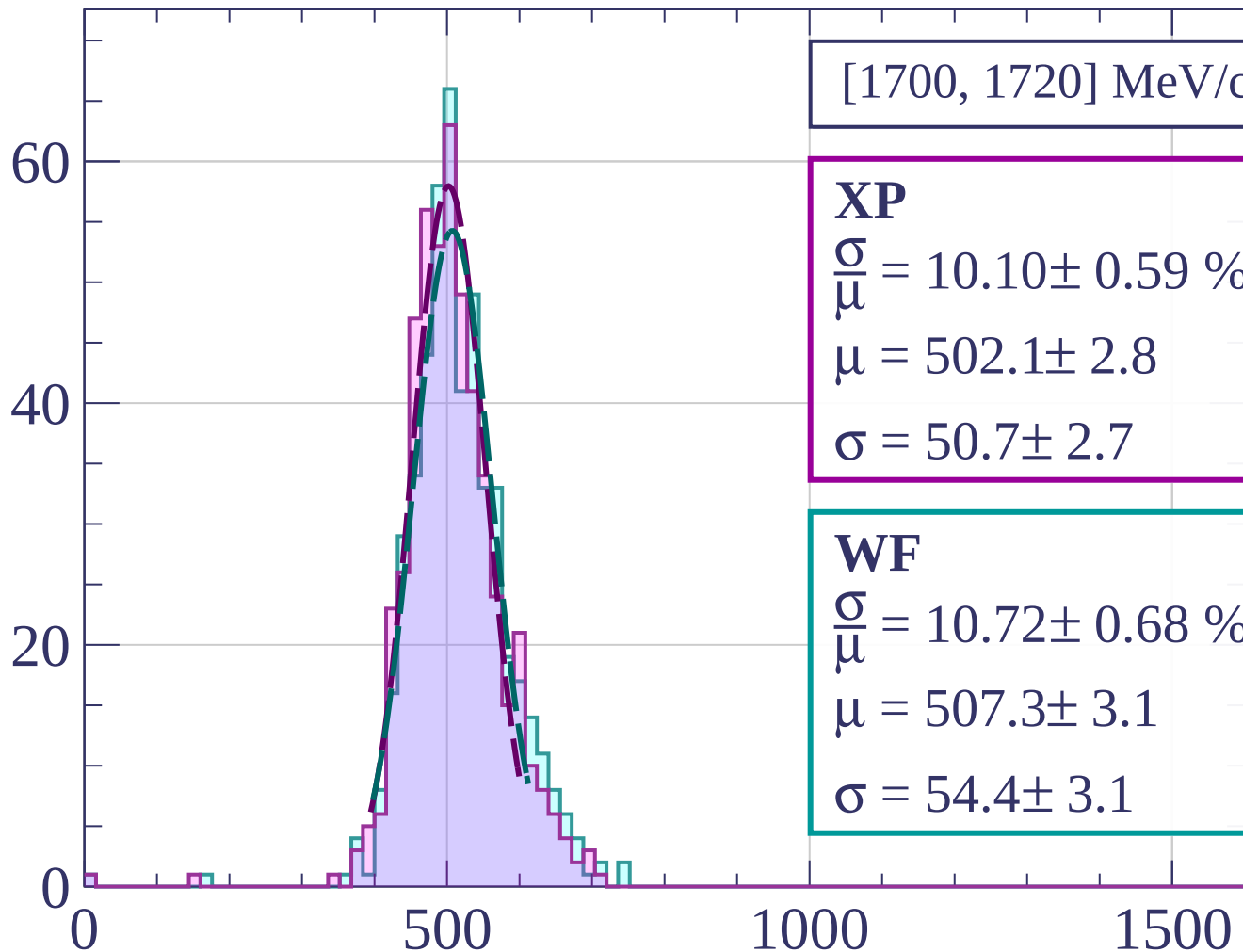
$$\frac{\sigma}{\mu} = 10.63 \pm 0.68 \%$$

$$\mu = 514.3 \pm 2.9$$

$$\sigma = 54.6 \pm 3.2$$

dE/dx [ADC counts/cm]

Count



[1700, 1720] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.10 \pm 0.59 \%$$

$$\mu = 502.1 \pm 2.8$$

$$\sigma = 50.7 \pm 2.7$$

WF

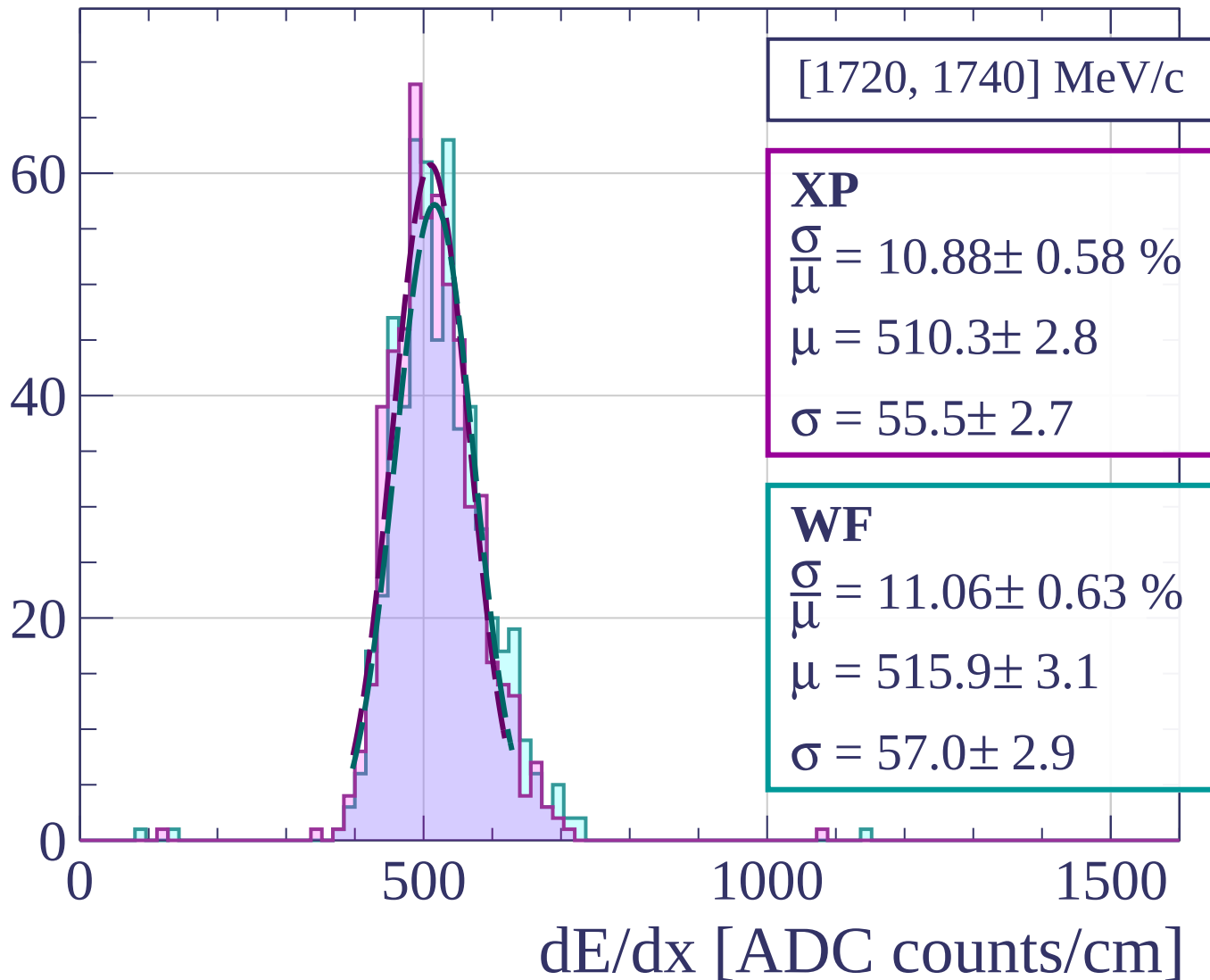
$$\frac{\sigma}{\mu} = 10.72 \pm 0.68 \%$$

$$\mu = 507.3 \pm 3.1$$

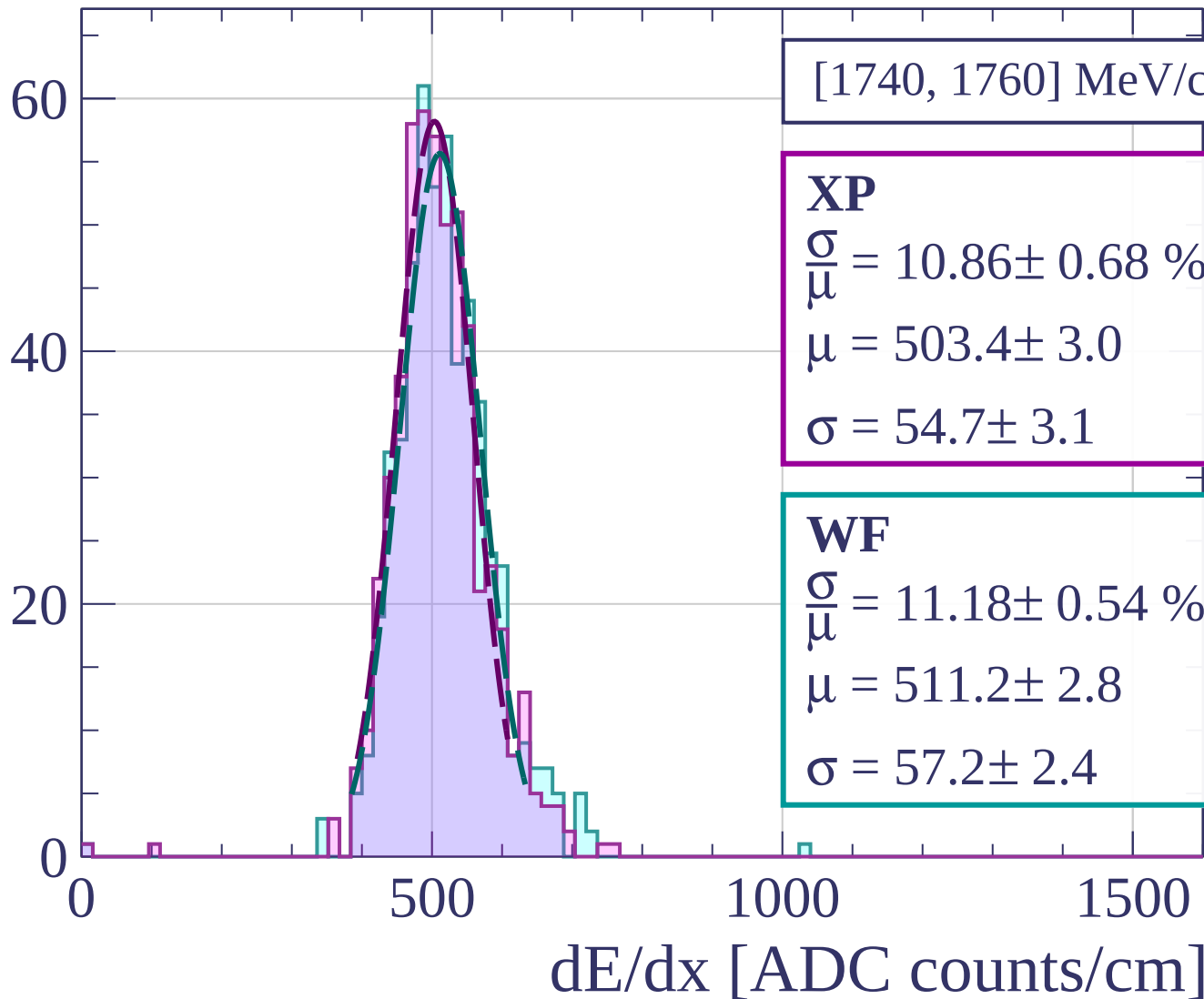
$$\sigma = 54.4 \pm 3.1$$

dE/dx [ADC counts/cm]

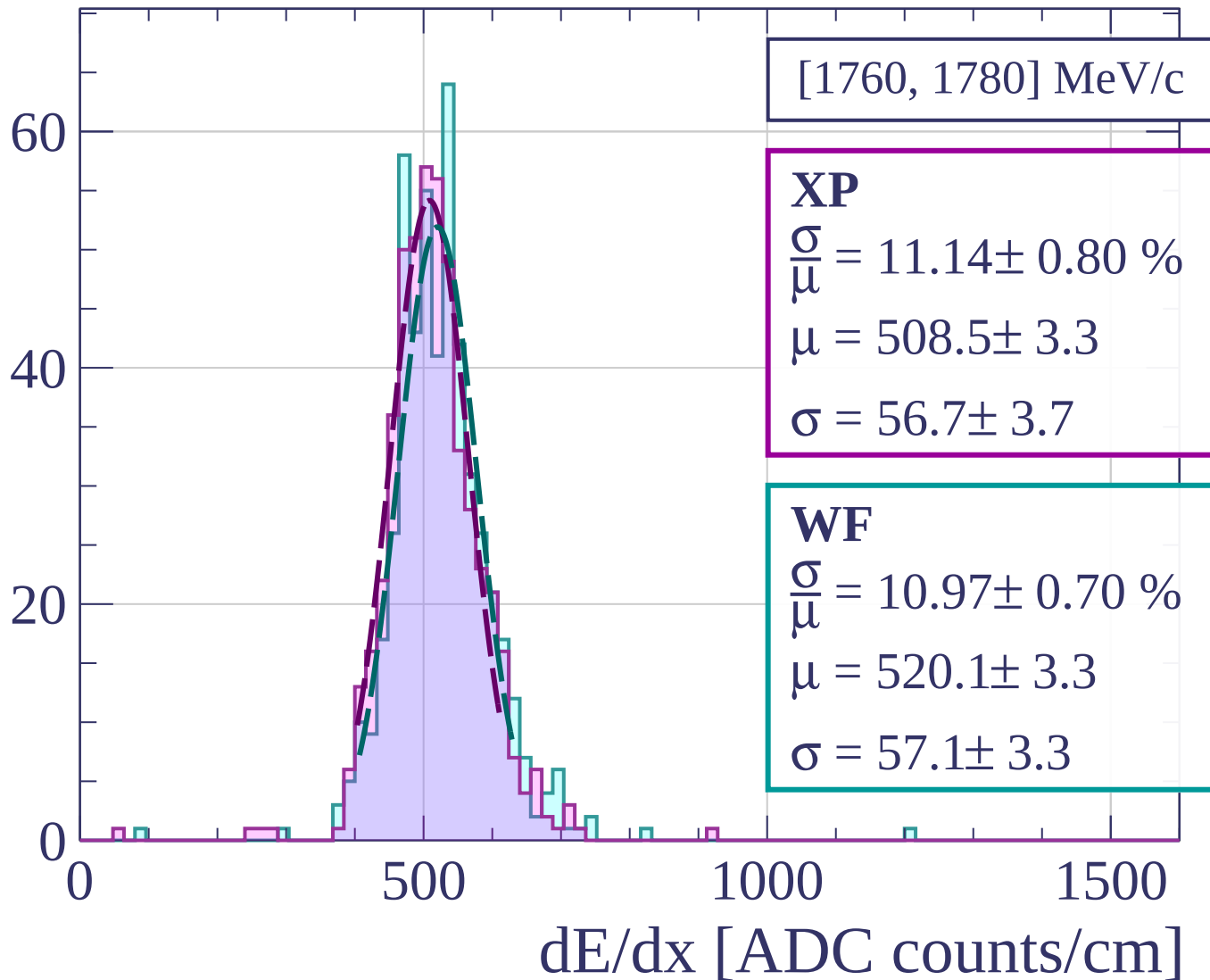
Count



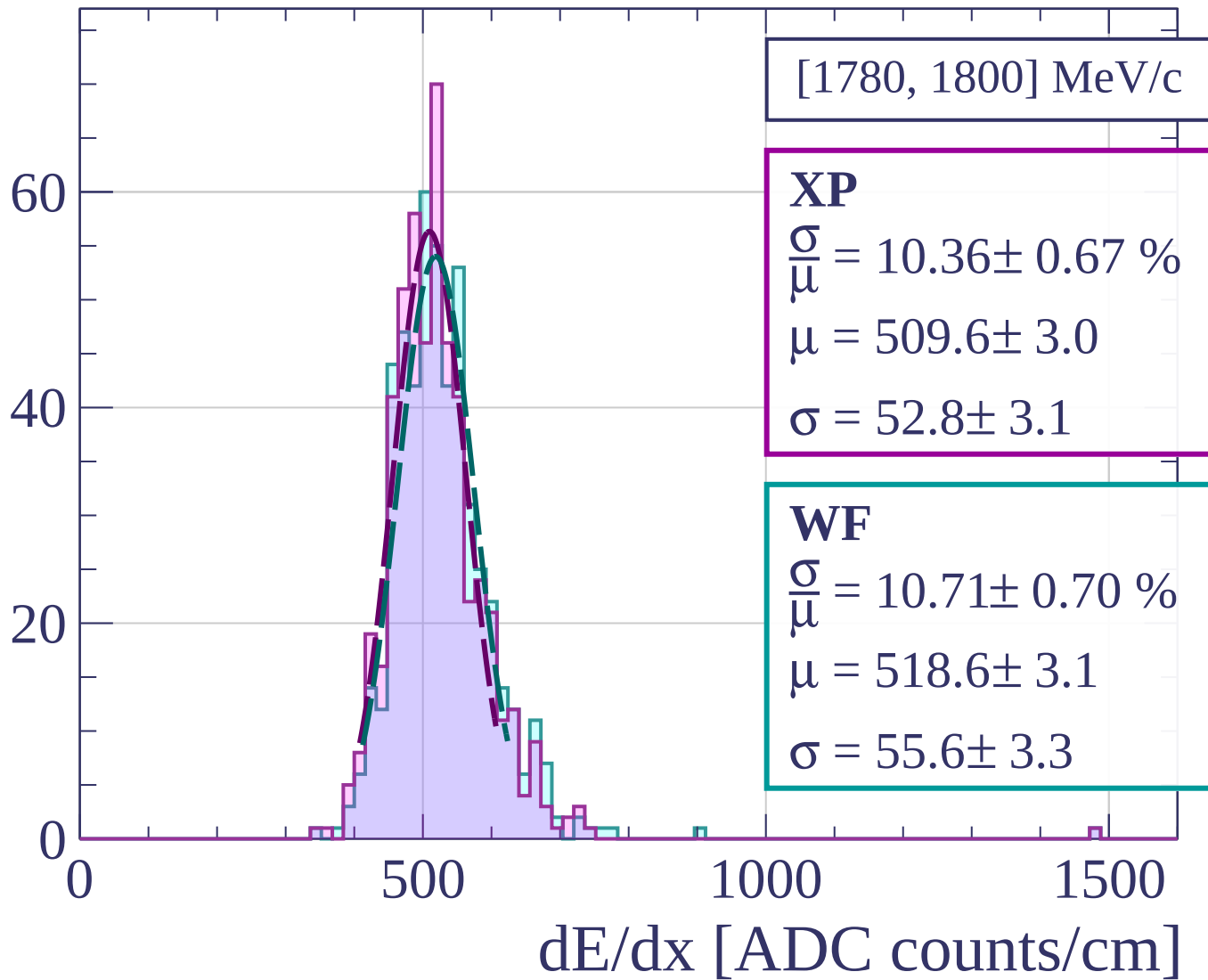
Count



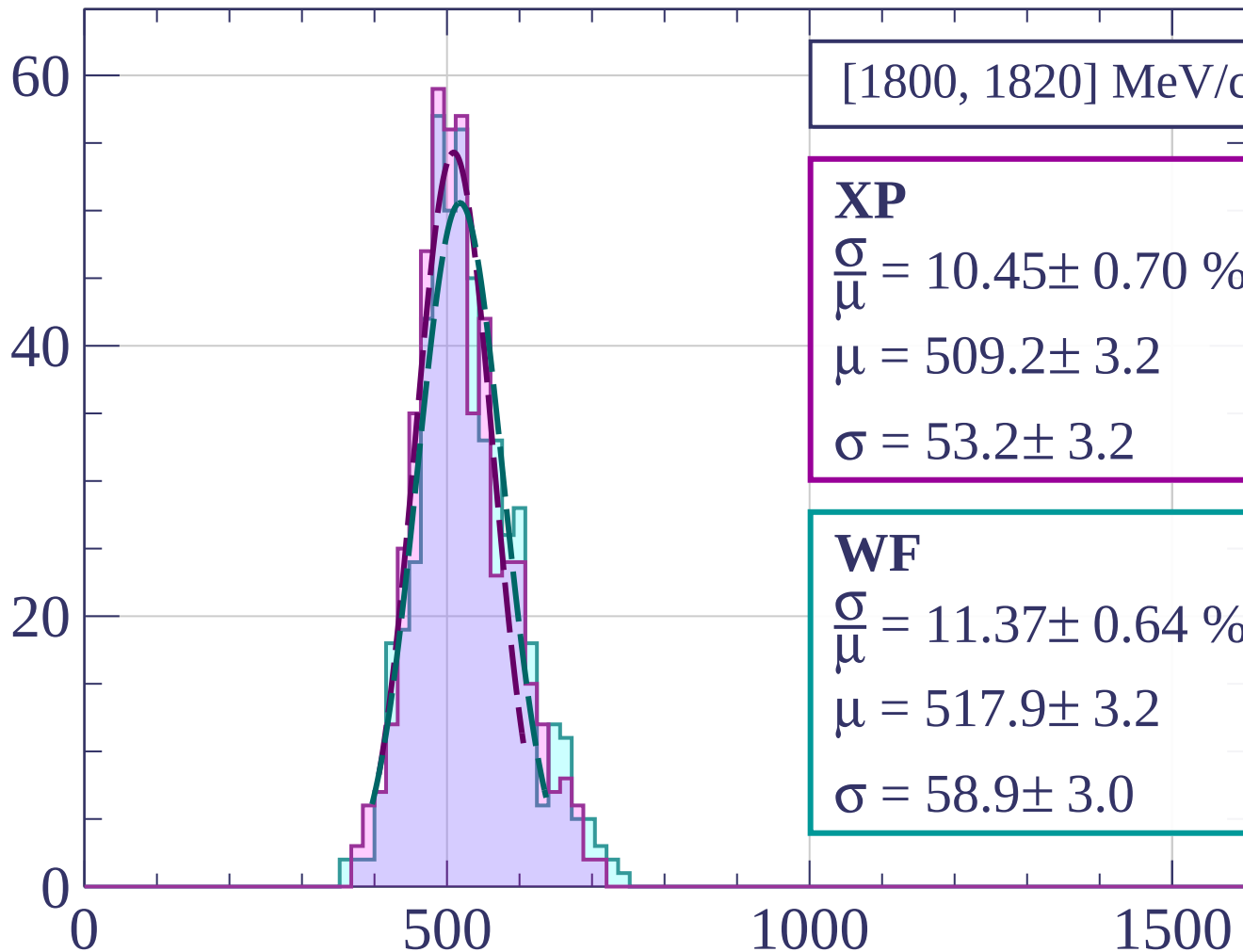
Count



Count

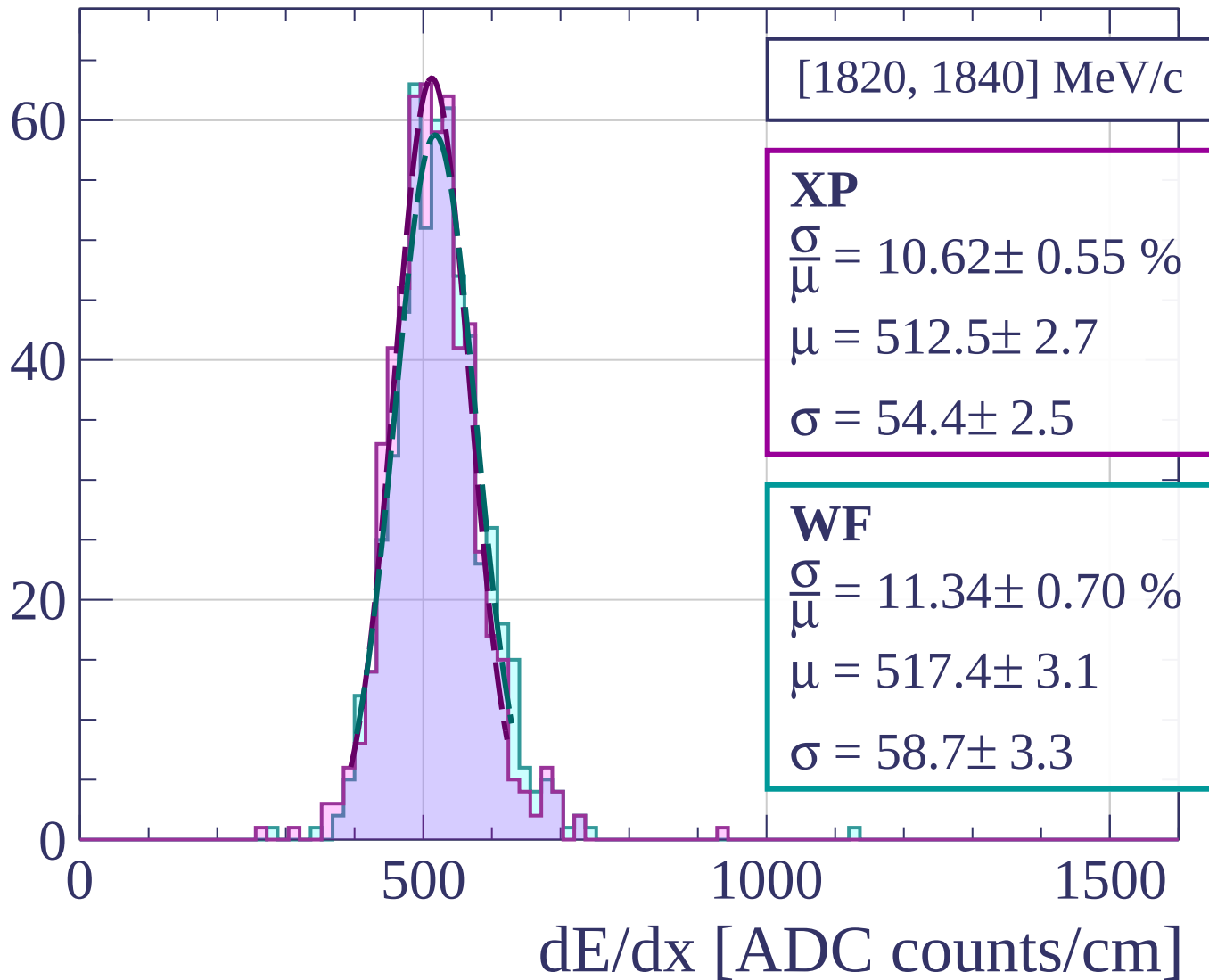


Count

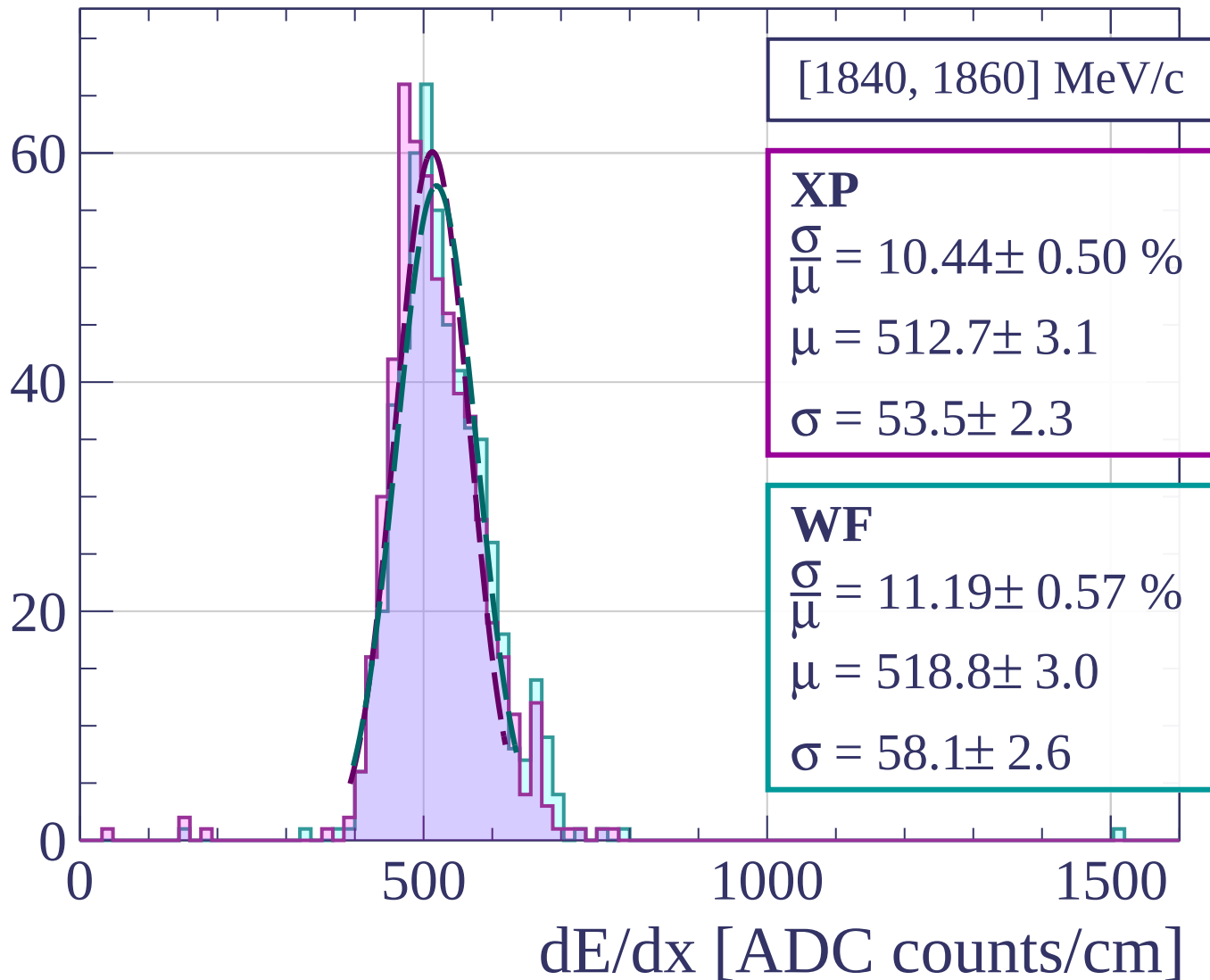


dE/dx [ADC counts/cm]

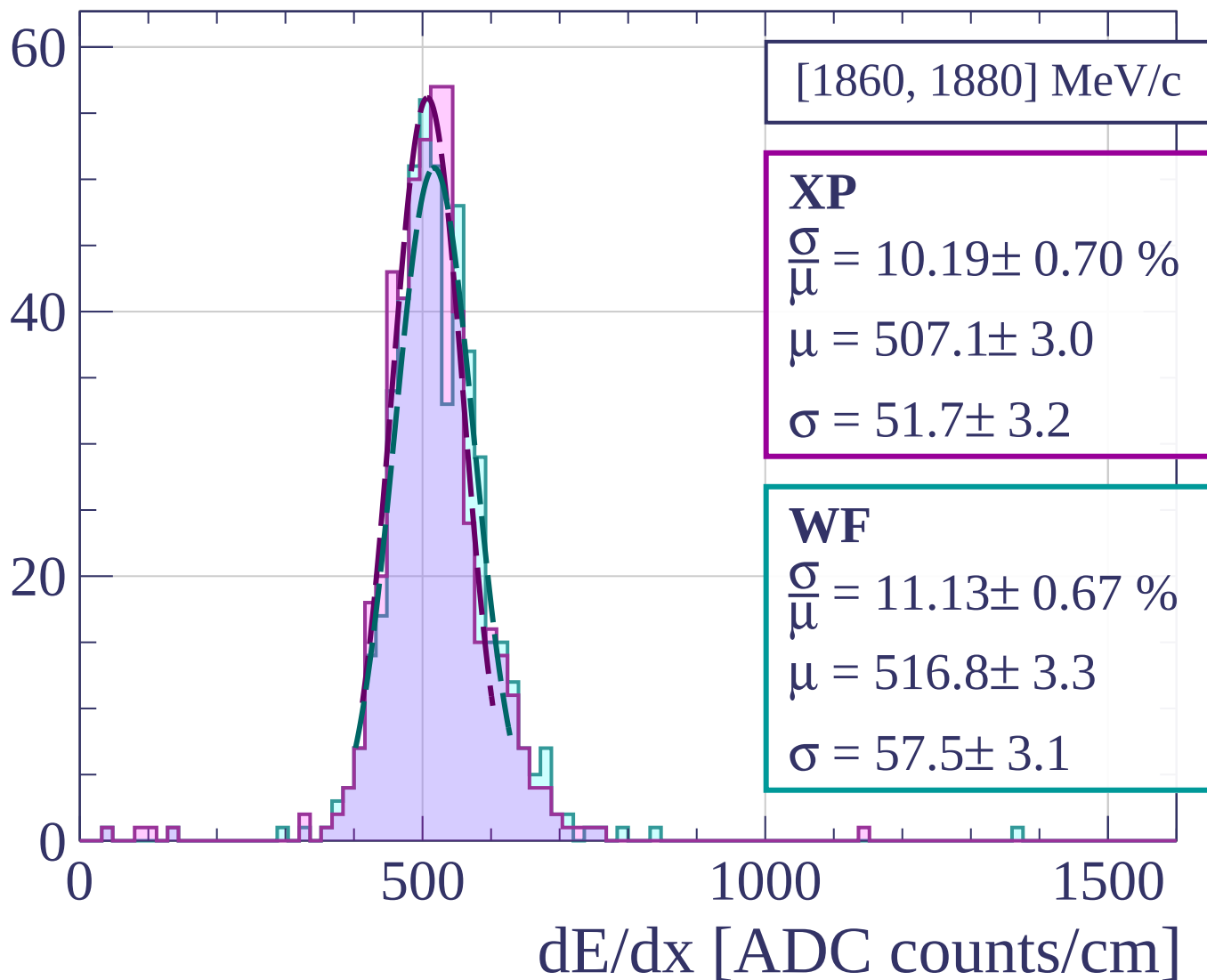
Count



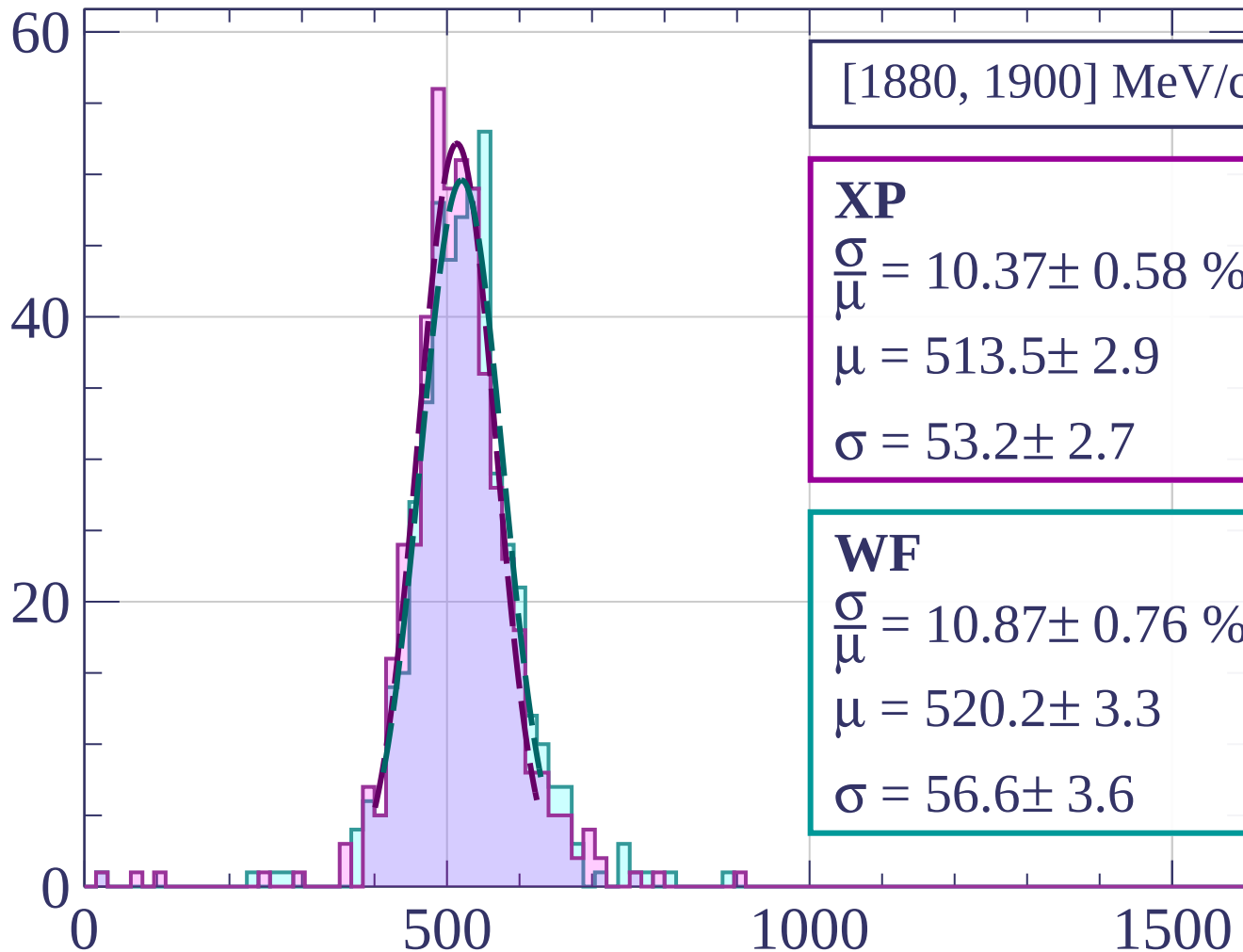
Count



Count



Count



[1880, 1900] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.37 \pm 0.58 \%$$

$$\mu = 513.5 \pm 2.9$$

$$\sigma = 53.2 \pm 2.7$$

WF

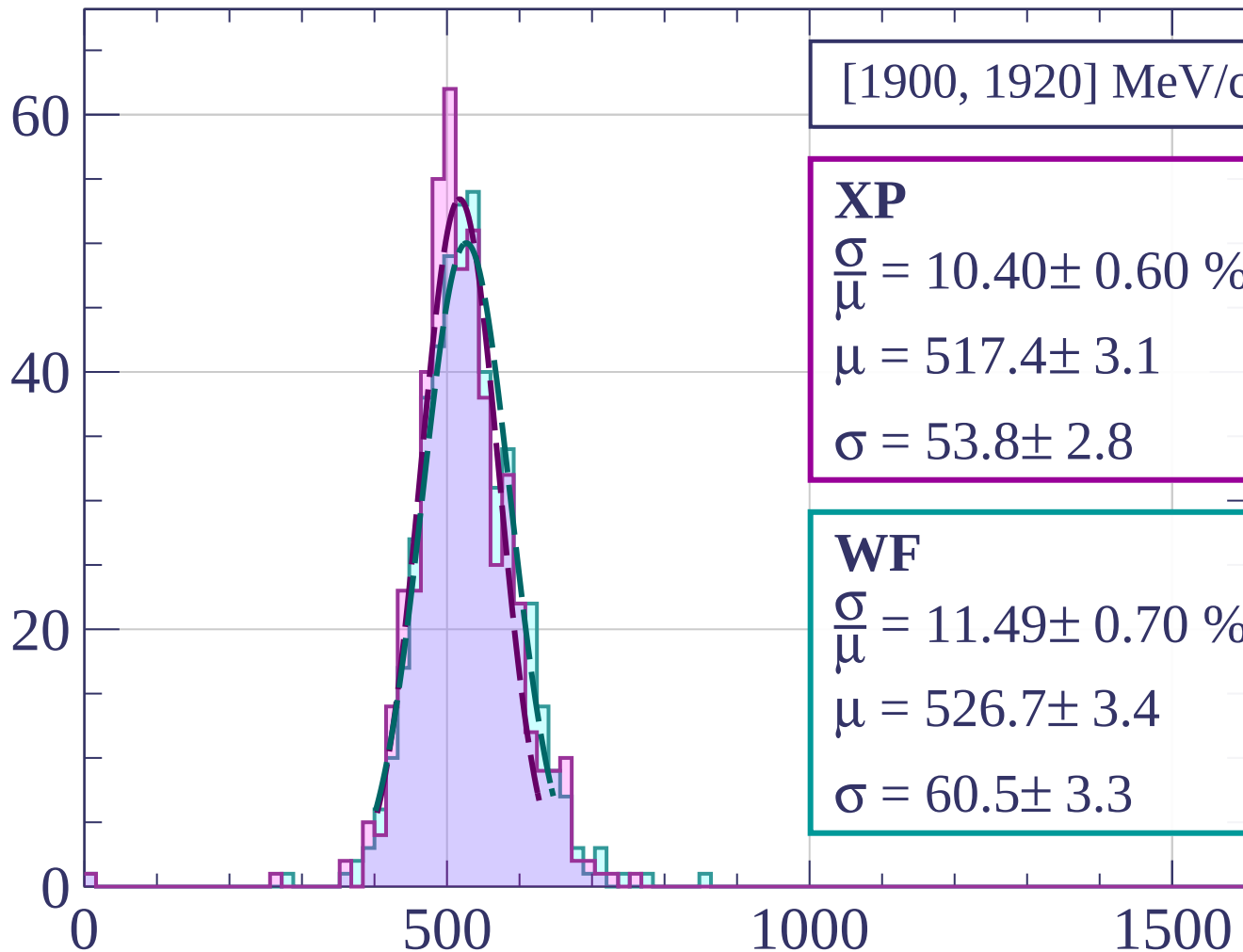
$$\frac{\sigma}{\mu} = 10.87 \pm 0.76 \%$$

$$\mu = 520.2 \pm 3.3$$

$$\sigma = 56.6 \pm 3.6$$

dE/dx [ADC counts/cm]

Count



[1900, 1920] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.40 \pm 0.60 \%$$

$$\mu = 517.4 \pm 3.1$$

$$\sigma = 53.8 \pm 2.8$$

WF

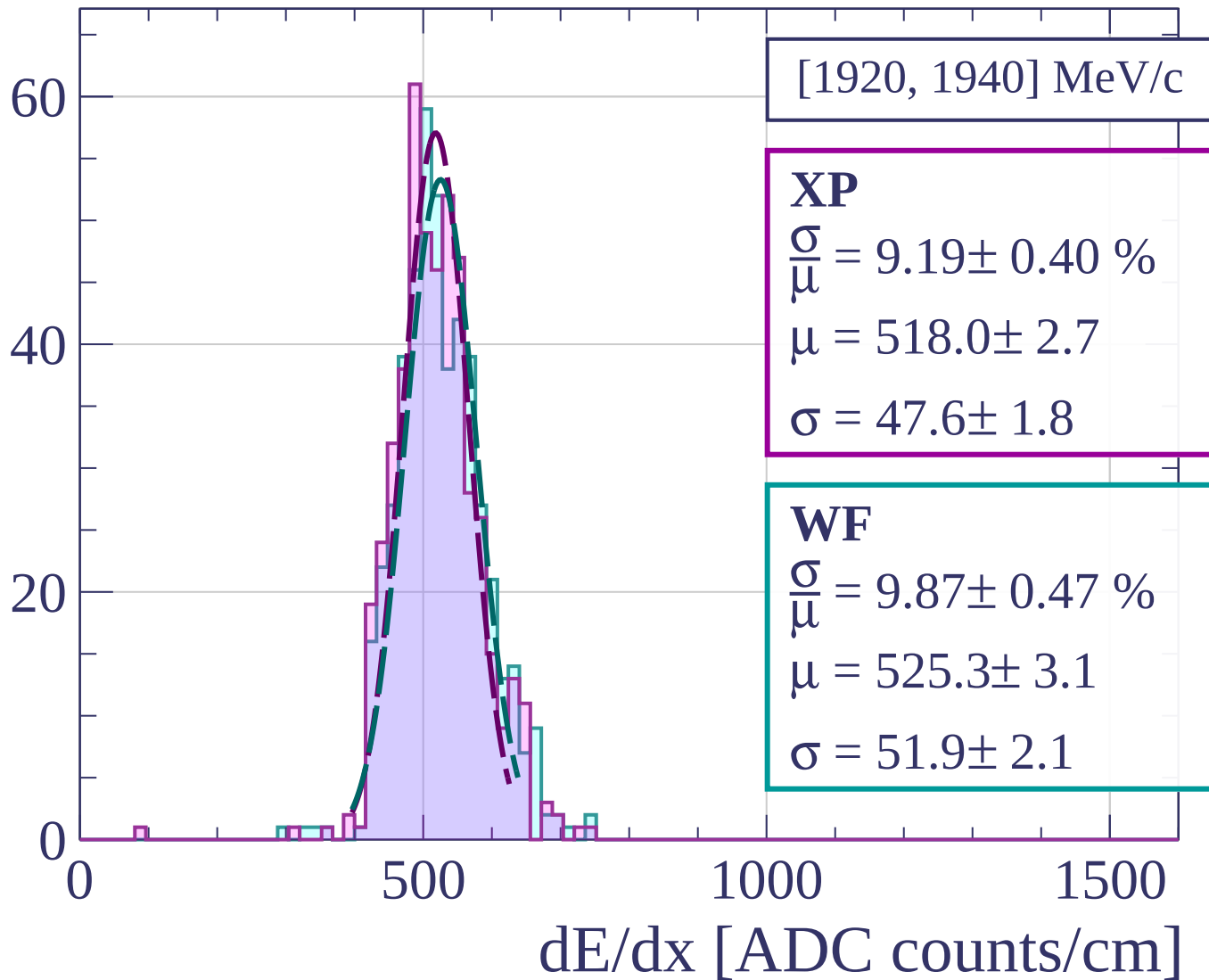
$$\frac{\sigma}{\mu} = 11.49 \pm 0.70 \%$$

$$\mu = 526.7 \pm 3.4$$

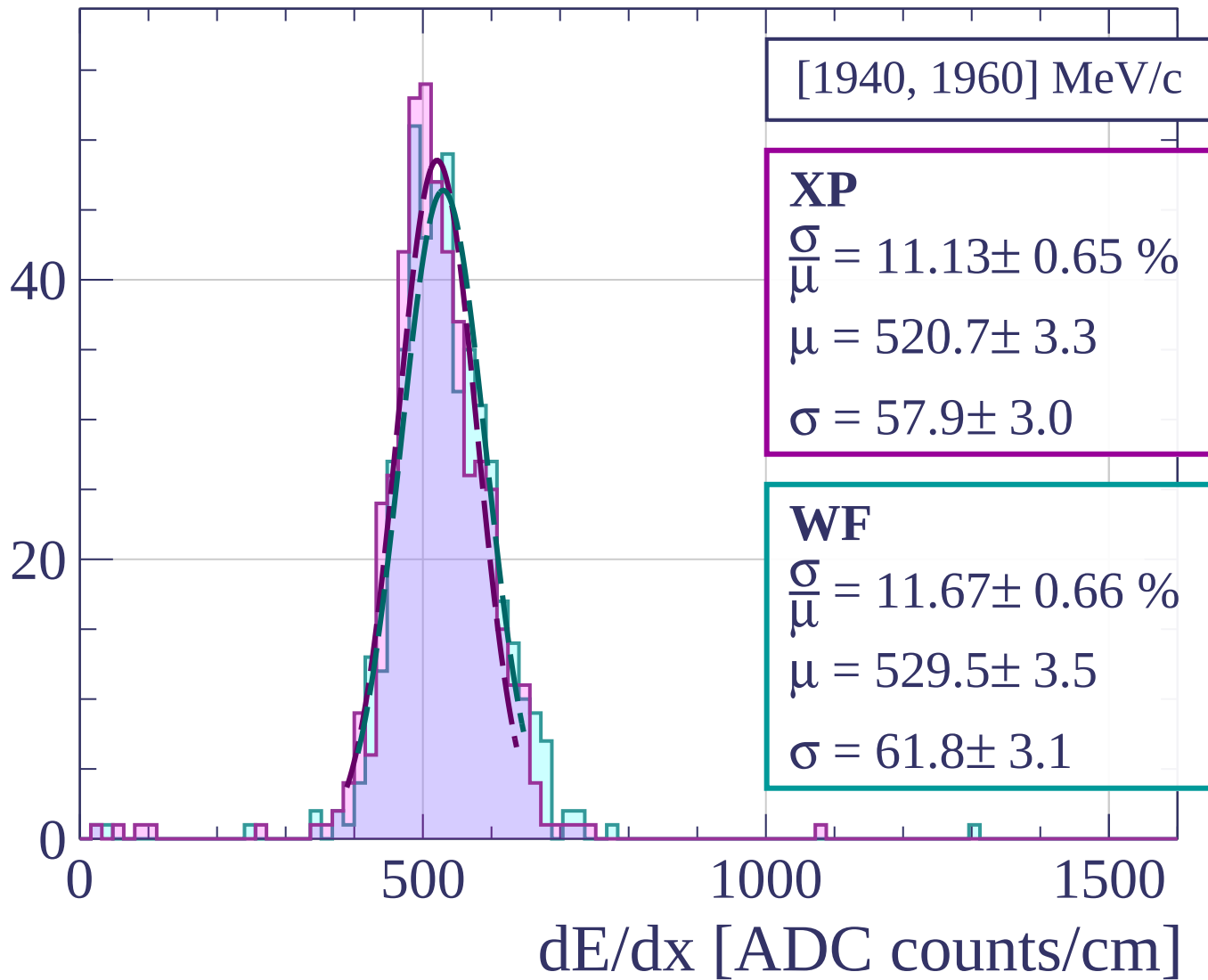
$$\sigma = 60.5 \pm 3.3$$

dE/dx [ADC counts/cm]

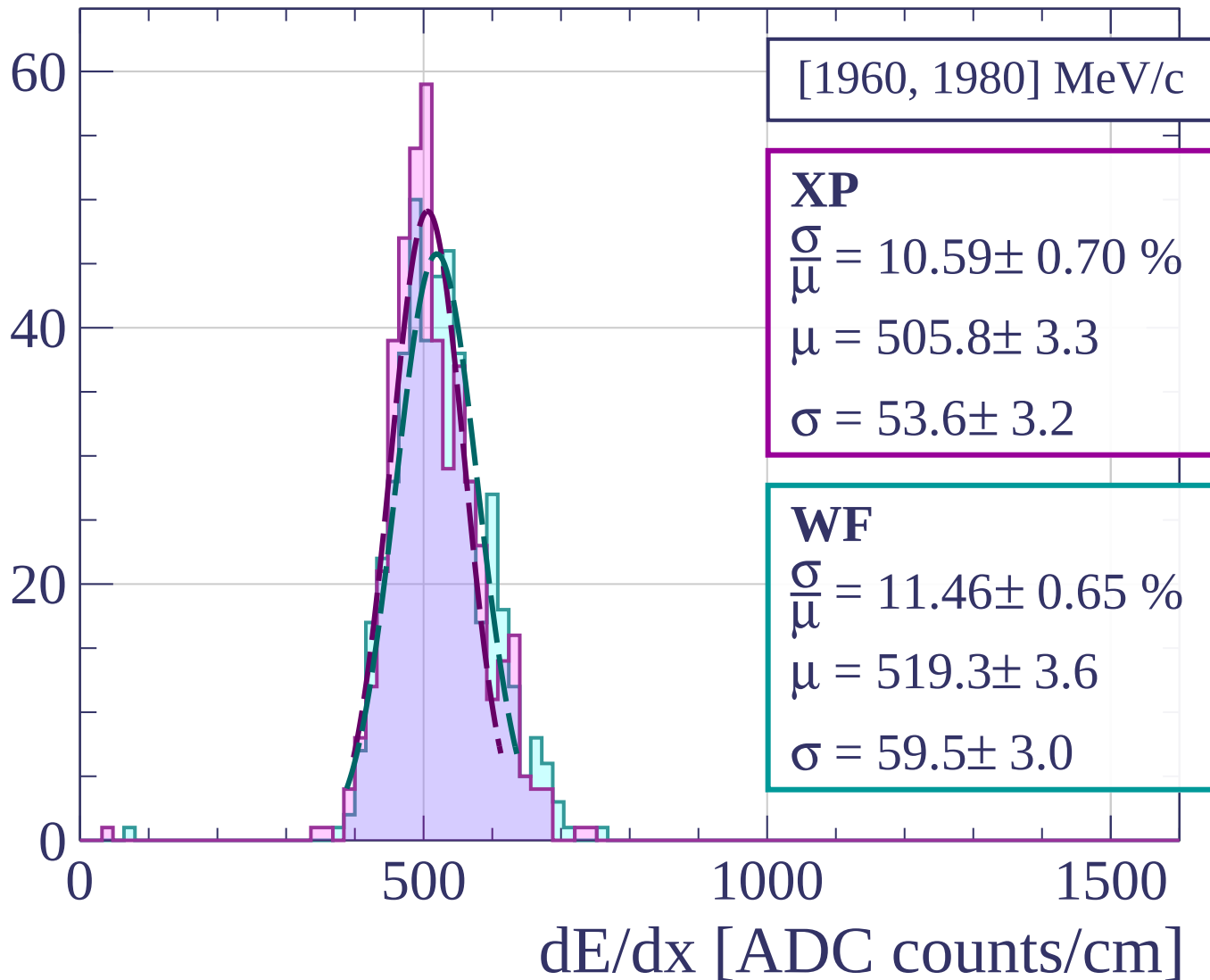
Count



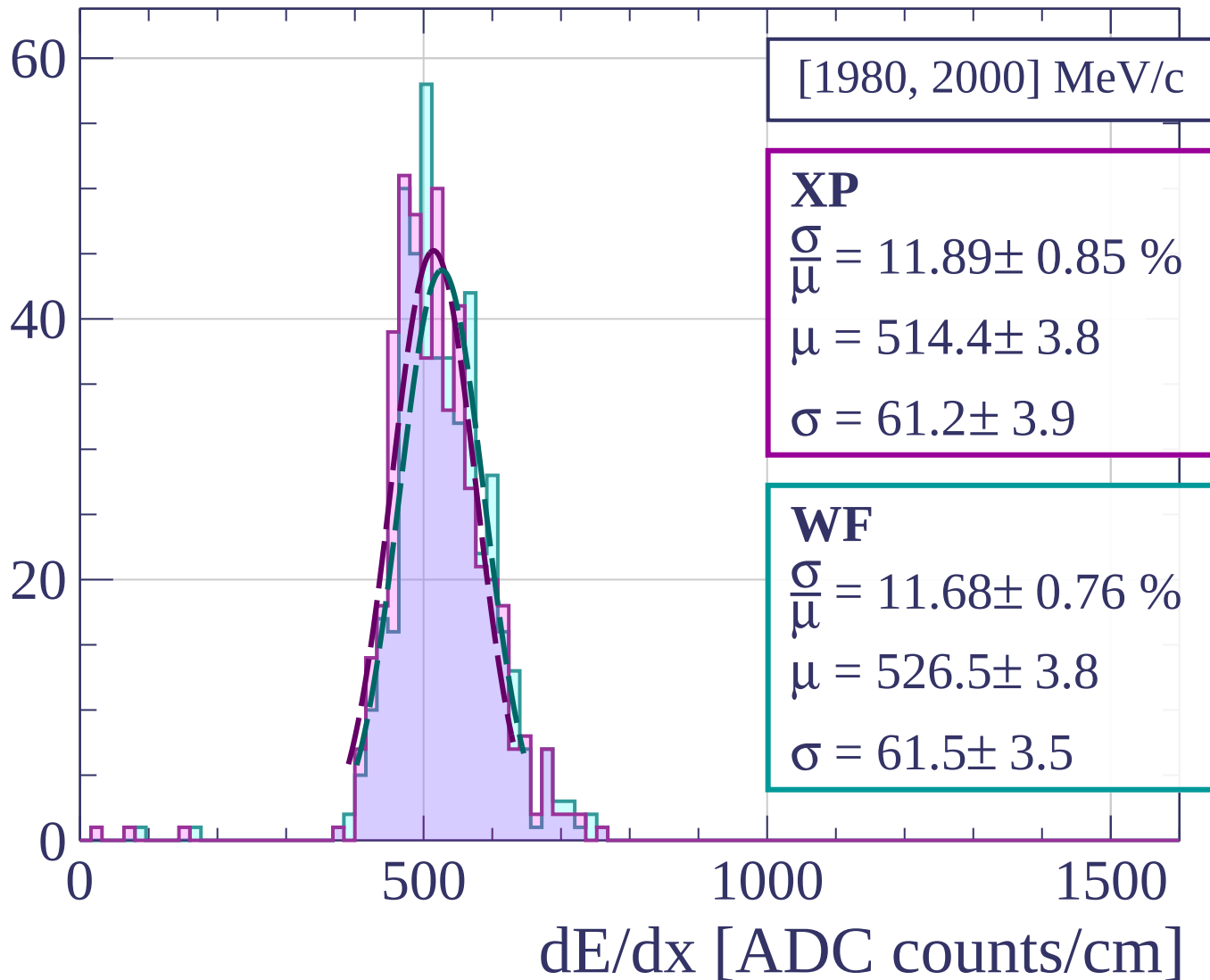
Count



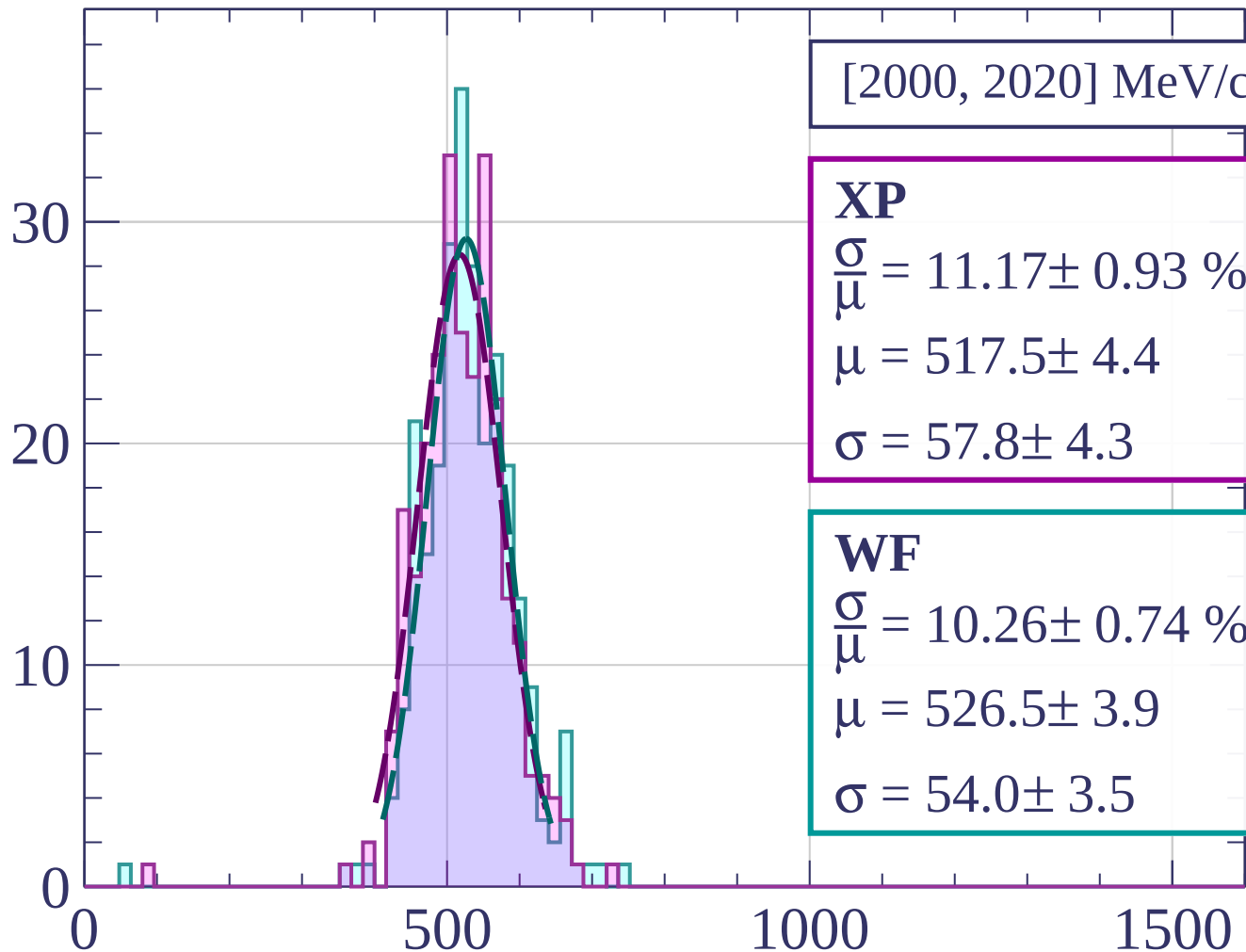
Count



Count



Count



dE/dx [ADC counts/cm]

Count

$\varphi \in [0^\circ, 3^\circ]$

XP

$$\frac{\sigma}{\mu} = -29.24 \pm 0.73 \%$$

$$\mu = -4611.2 \pm 306.6$$

$$\sigma = 1348.4 \pm 56.0$$

WF

$$\frac{\sigma}{\mu} = -50.47 \pm 2.63 \%$$

$$\mu = -3252.8 \pm 336.9$$

$$\sigma = 1641.6 \pm 255.6$$

100

50

0

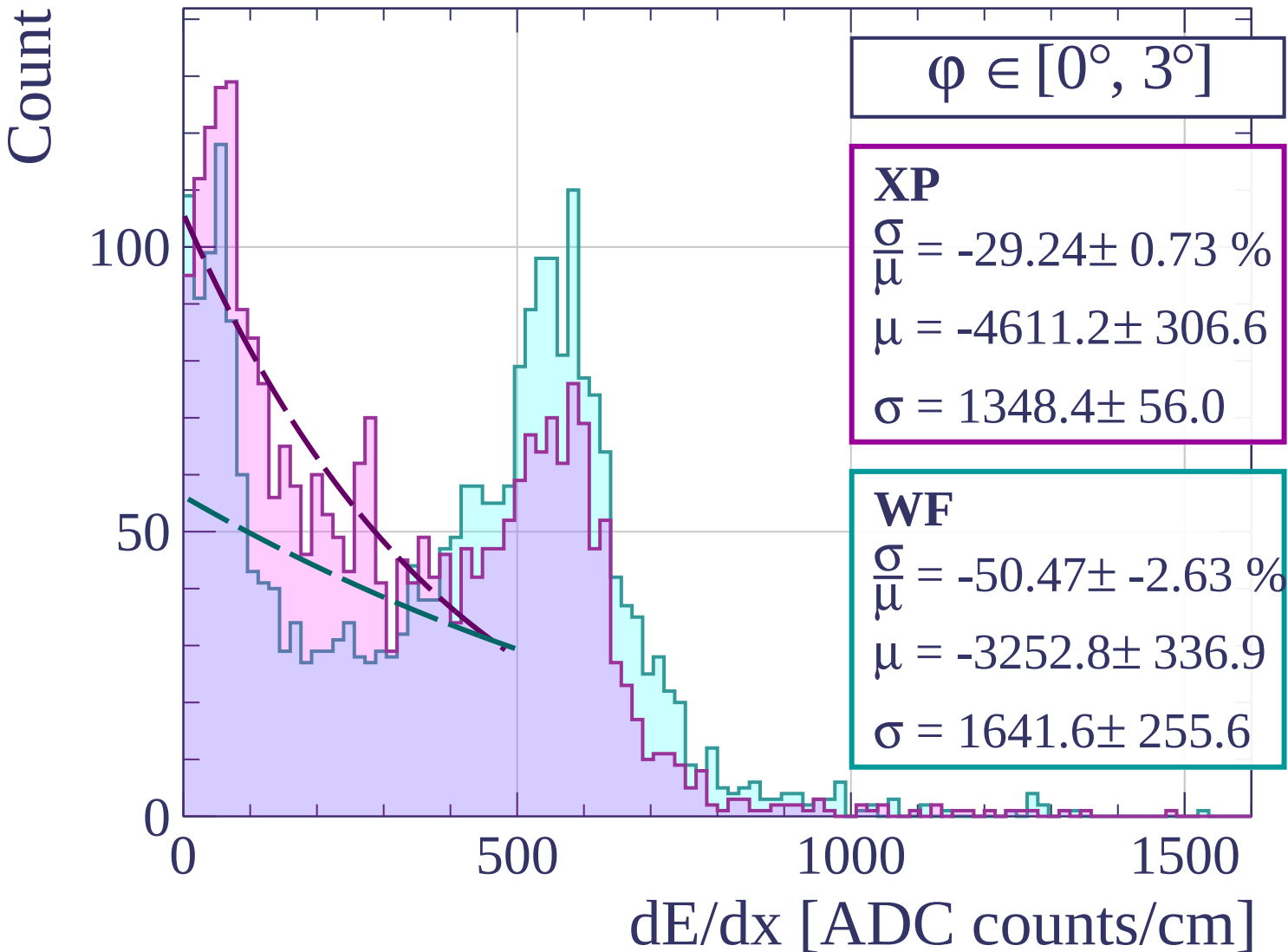
0

500

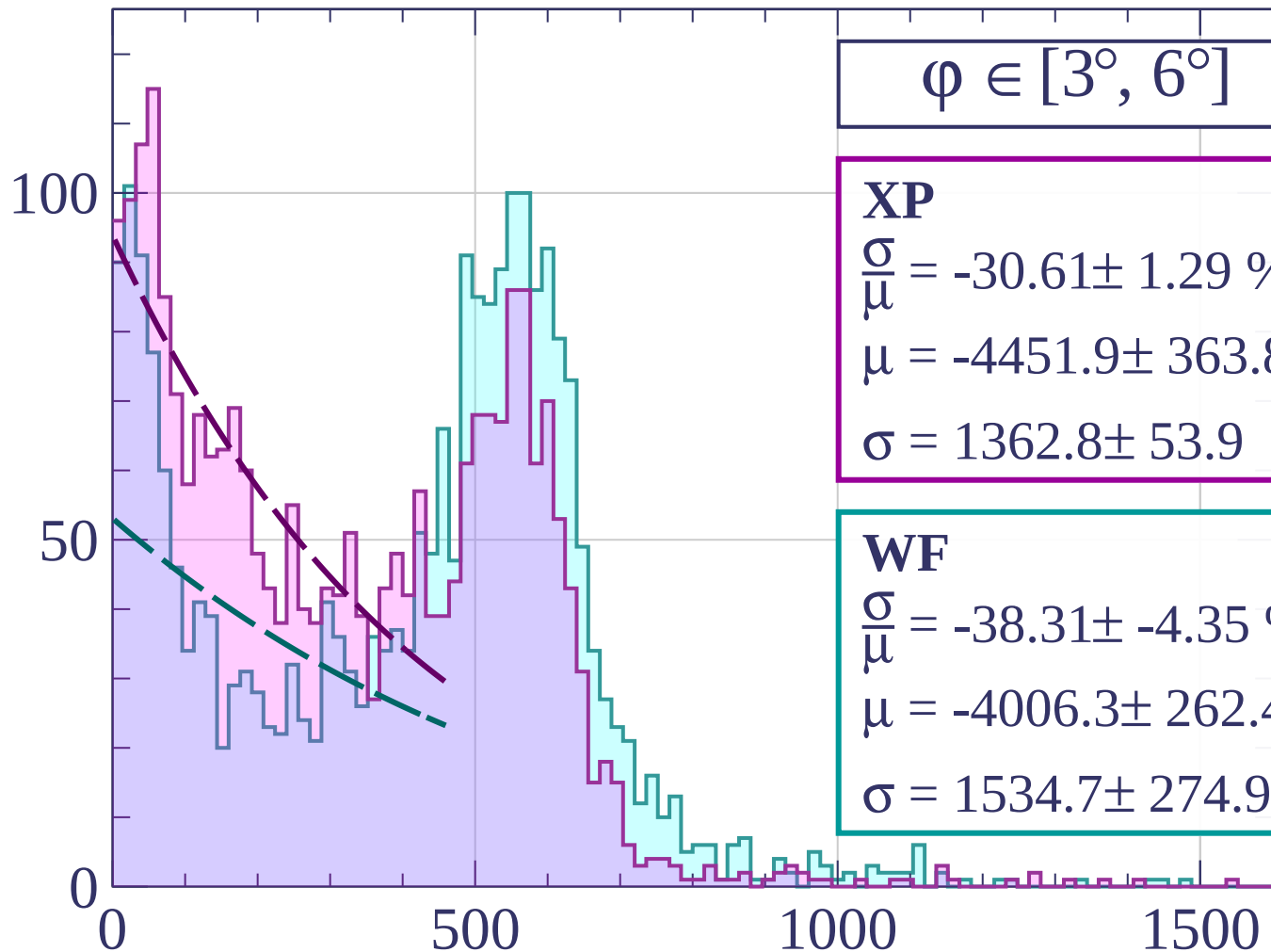
1000

1500

dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

$\varphi \in [6^\circ, 9^\circ]$

XP

$$\frac{\sigma}{\mu} = -34.69 \pm 3.45 \%$$

$$\mu = -3670.8 \pm 448.9$$

$$\sigma = 1273.6 \pm 282.5$$

WF

$$\frac{\sigma}{\mu} = 21.94 \pm 0.88 \%$$

$$\mu = 526.0 \pm 3.6$$

$$\sigma = 115.4 \pm 3.8$$

0

500

1000

1500

dE/dx [ADC counts/cm]

100

50

Count

$$\varphi \in [9^\circ, 12^\circ]$$

XP

$$\frac{\sigma}{\mu} = -36.20 \pm 17.86 \%$$

$$\mu = -3842.0 \pm 510.9$$

$$\sigma = 1390.6 \pm 871.2$$

WF

$$\frac{\sigma}{\mu} = 21.14 \pm 0.82 \%$$

$$\mu = 530.7 \pm 3.4$$

$$\sigma = 112.2 \pm 3.6$$

100

50

0

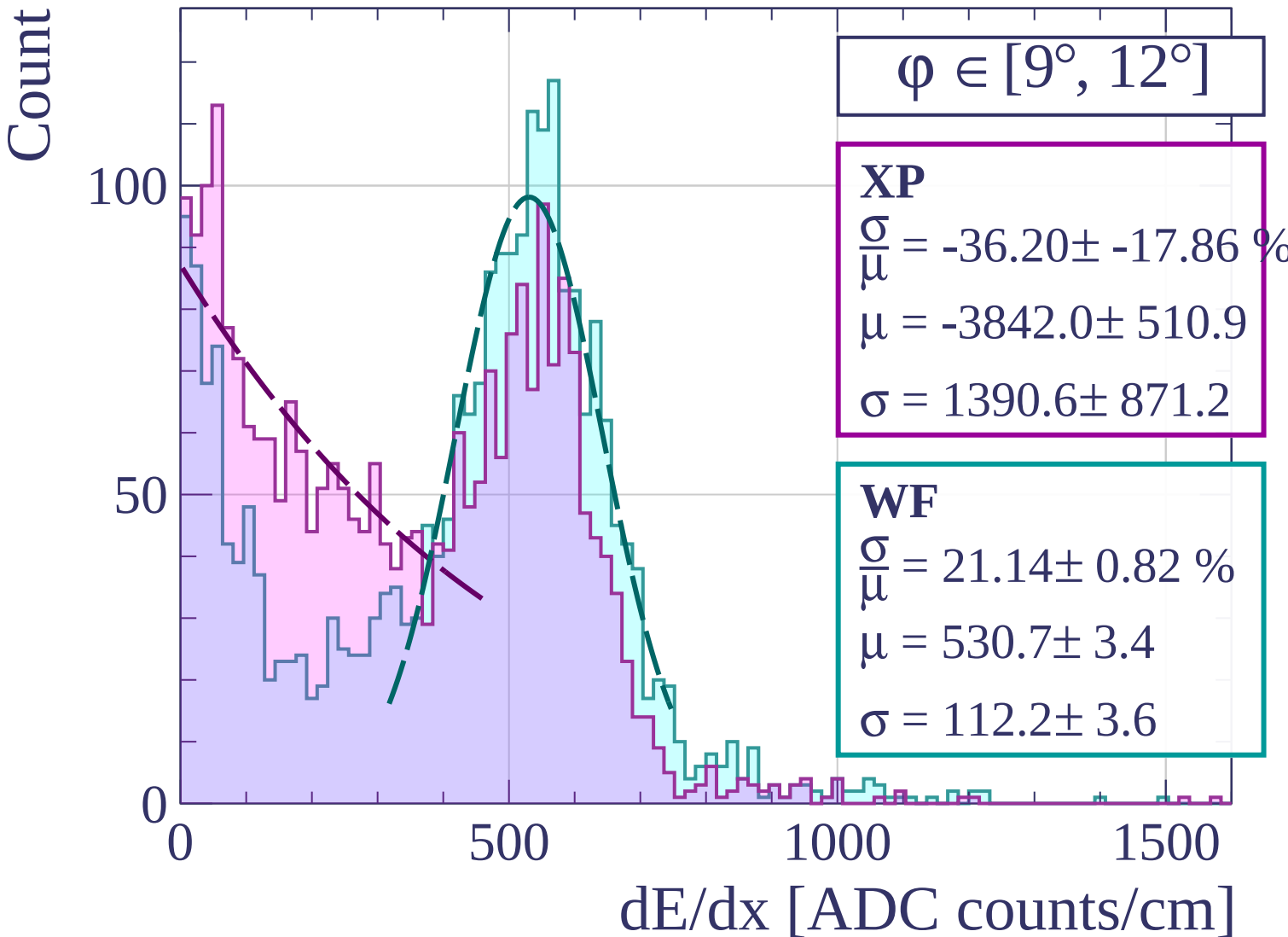
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [12^\circ, 15^\circ]$

XP

$$\frac{\sigma}{\mu} = 177.61 \pm 38.55 \%$$

$$\mu = 177.5 \pm 29.8$$

$$\sigma = 315.2 \pm 15.6$$

WF

$$\frac{\sigma}{\mu} = 20.41 \pm 0.74 \%$$

$$\mu = 533.1 \pm 3.2$$

$$\sigma = 108.8 \pm 3.3$$

100

50

0

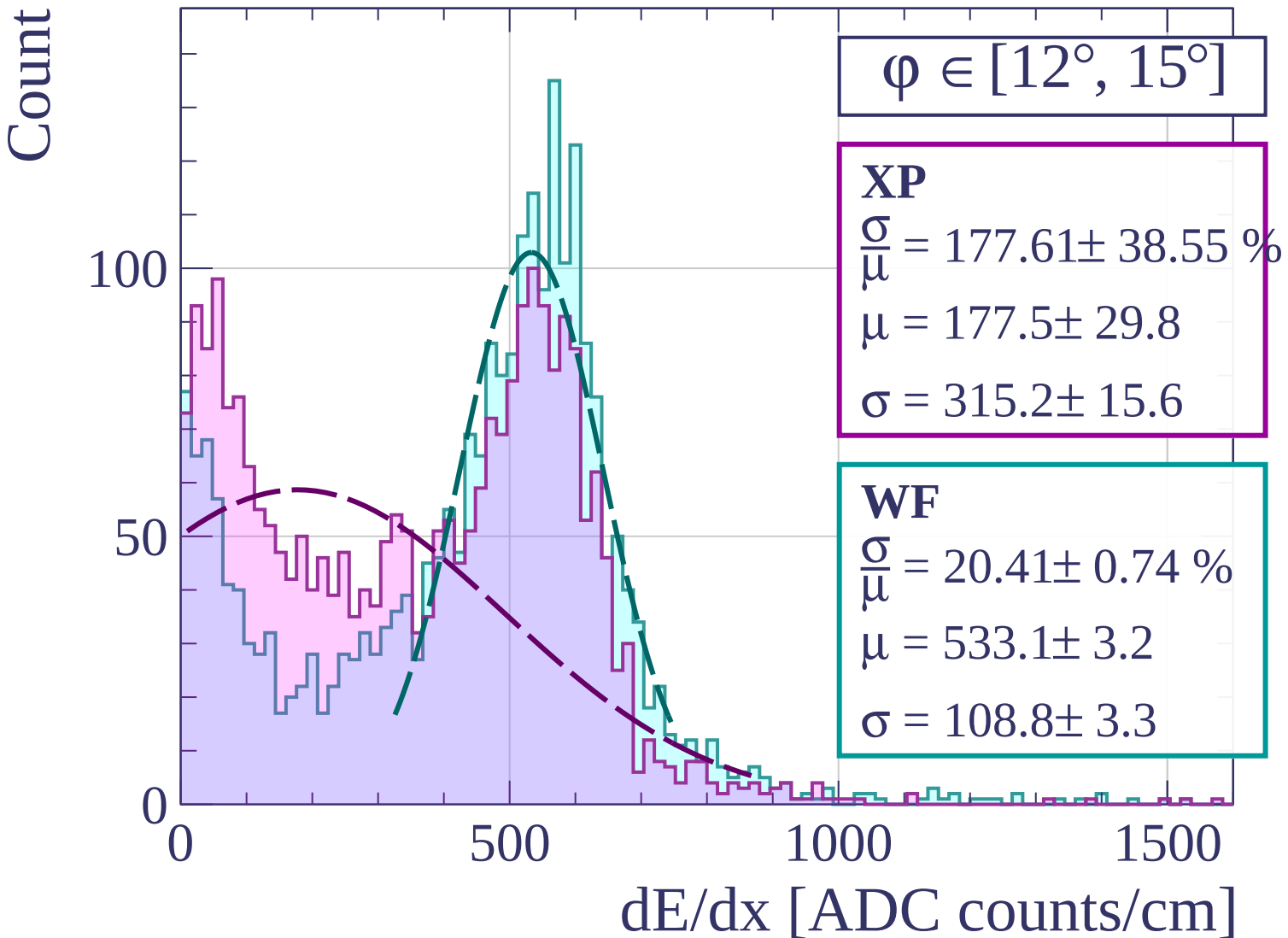
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\varphi \in [15^\circ, 18^\circ]$

XP

$$\frac{\sigma}{\mu} = 24.07 \pm 0.95 \%$$

$$\mu = 501.3 \pm 4.1$$

$$\sigma = 120.7 \pm 3.8$$

WF

$$\frac{\sigma}{\mu} = 19.99 \pm 0.62 \%$$

$$\mu = 530.6 \pm 2.9$$

$$\sigma = 106.0 \pm 2.7$$

100

50

0

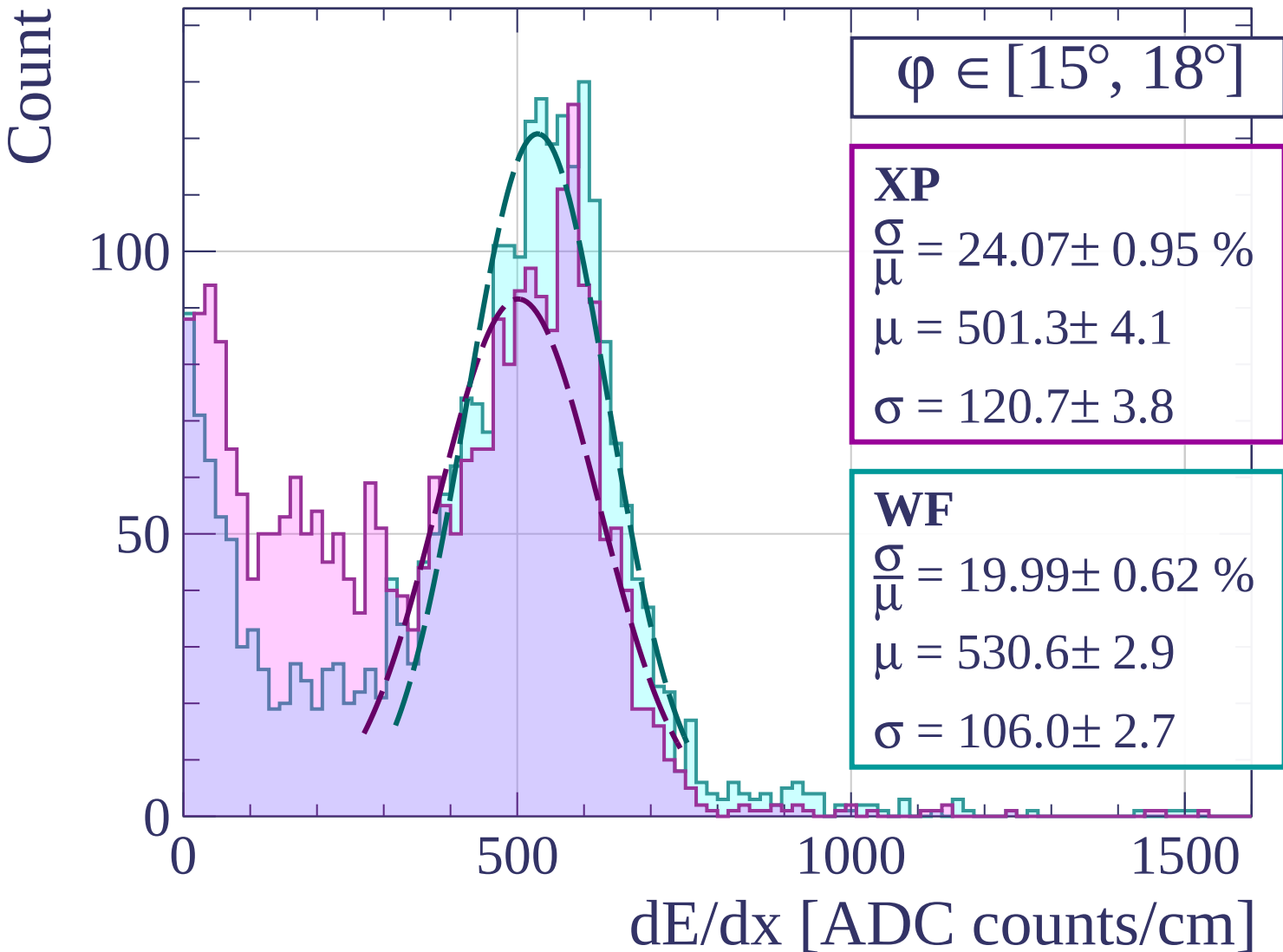
0

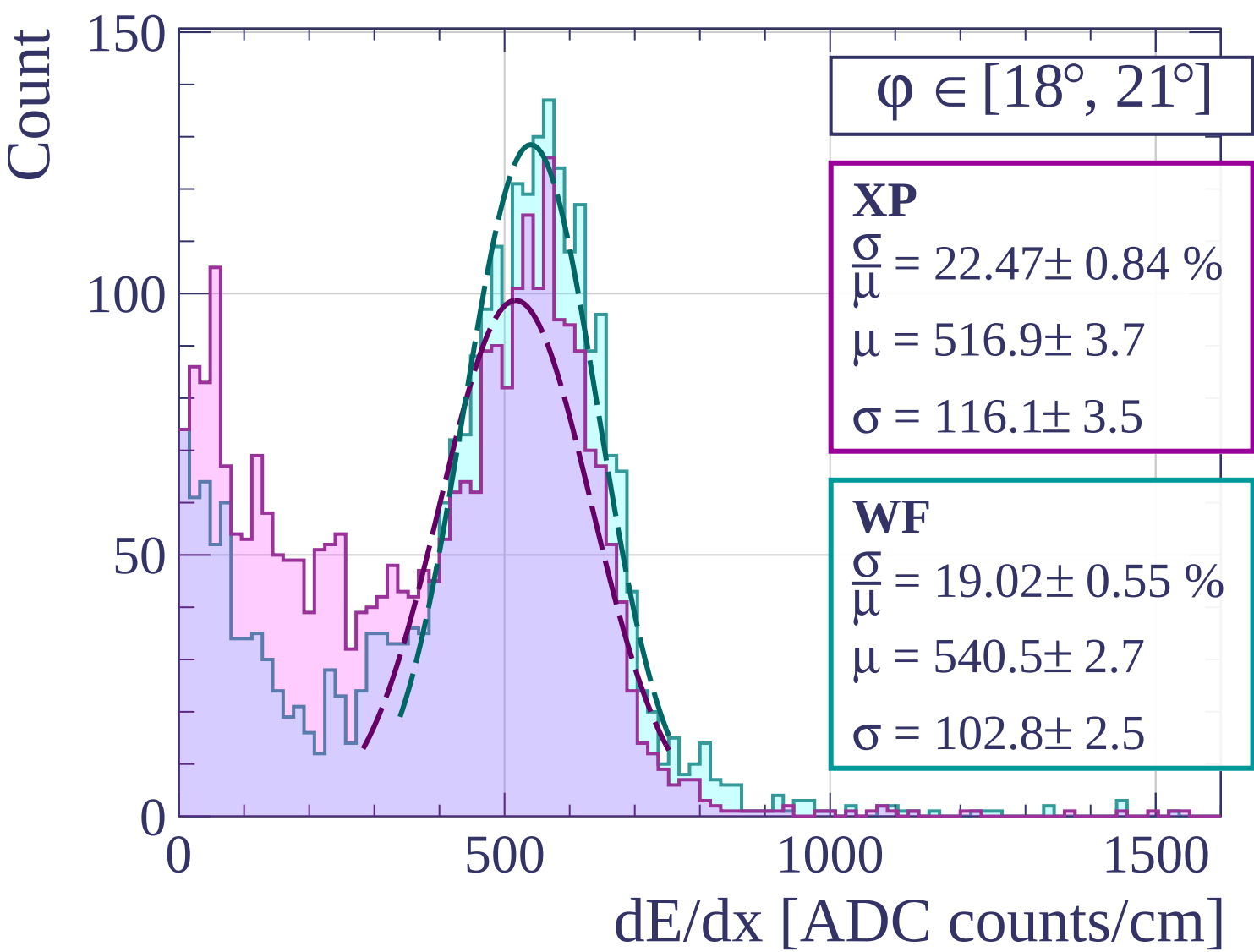
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\varphi \in [21^\circ, 24^\circ]$

XP

$$\frac{\sigma}{\mu} = 27.18 \pm 0.94 \%$$

$$\mu = 500.6 \pm 3.8$$

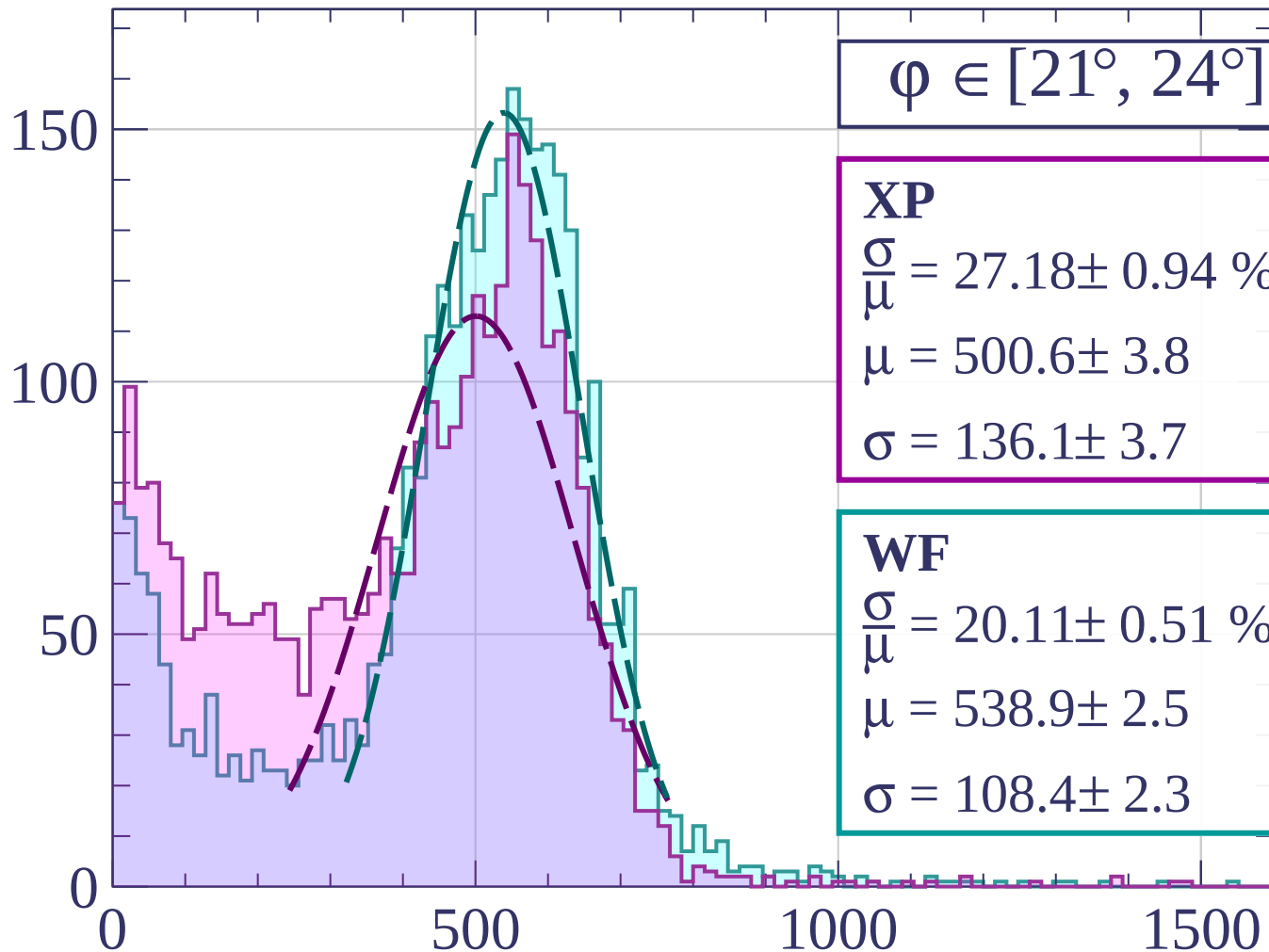
$$\sigma = 136.1 \pm 3.7$$

WF

$$\frac{\sigma}{\mu} = 20.11 \pm 0.51 \%$$

$$\mu = 538.9 \pm 2.5$$

$$\sigma = 108.4 \pm 2.3$$



dE/dx [ADC counts/cm]

Count

$\phi \in [24^\circ, 27^\circ]$

XP

$$\frac{\sigma}{\mu} = 19.78 \pm 0.57 \%$$

$$\mu = 525.7 \pm 2.6$$

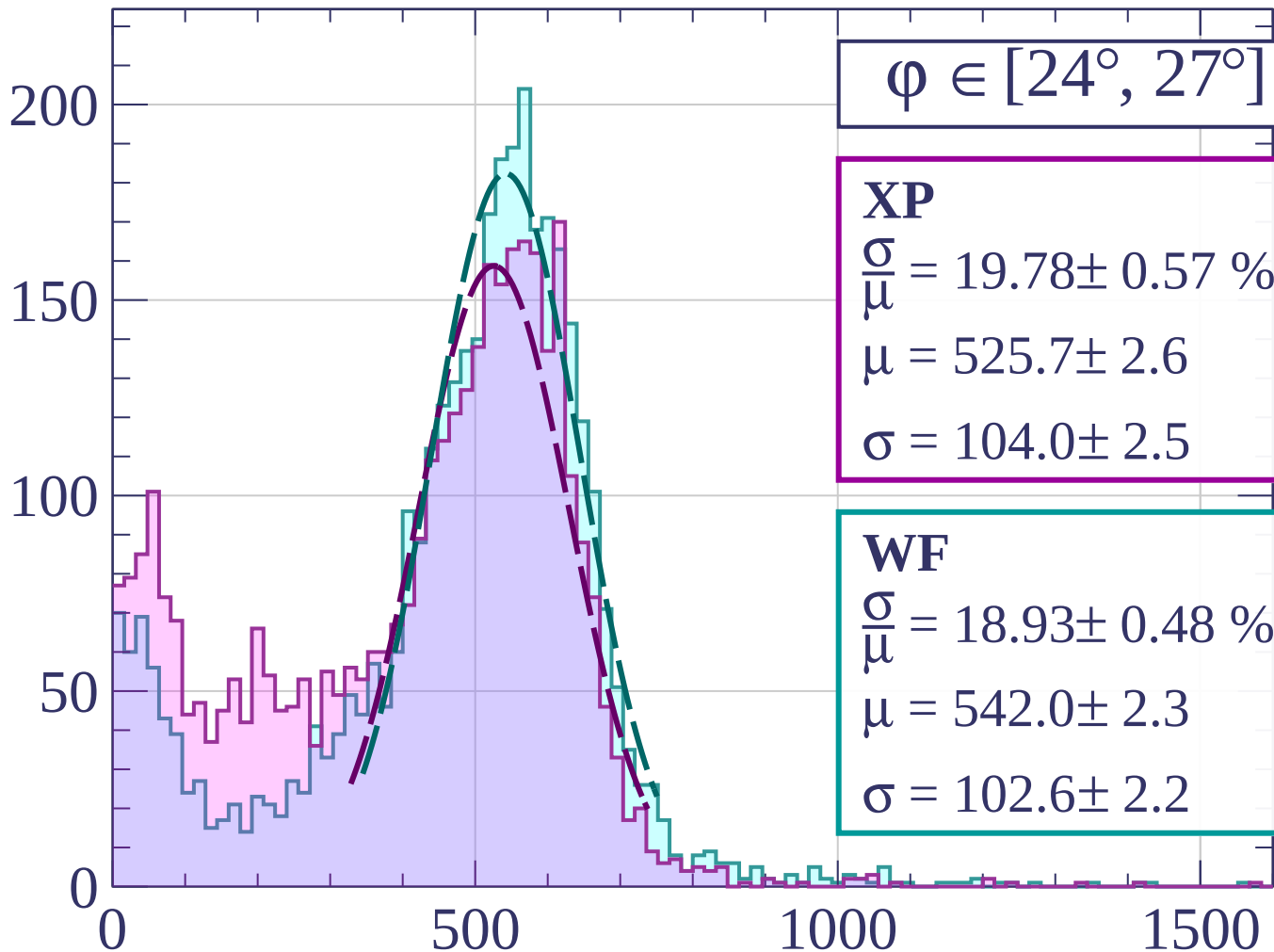
$$\sigma = 104.0 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 18.93 \pm 0.48 \%$$

$$\mu = 542.0 \pm 2.3$$

$$\sigma = 102.6 \pm 2.2$$



dE/dx [ADC counts/cm]

Count

$\phi \in [27^\circ, 30^\circ]$

XP

$$\frac{\sigma}{\mu} = 19.41 \pm 0.46 \%$$

$$\mu = 522.9 \pm 2.2$$

$$\sigma = 101.5 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 18.48 \pm 0.39 \%$$

$$\mu = 534.1 \pm 1.9$$

$$\sigma = 98.7 \pm 1.7$$

200

100

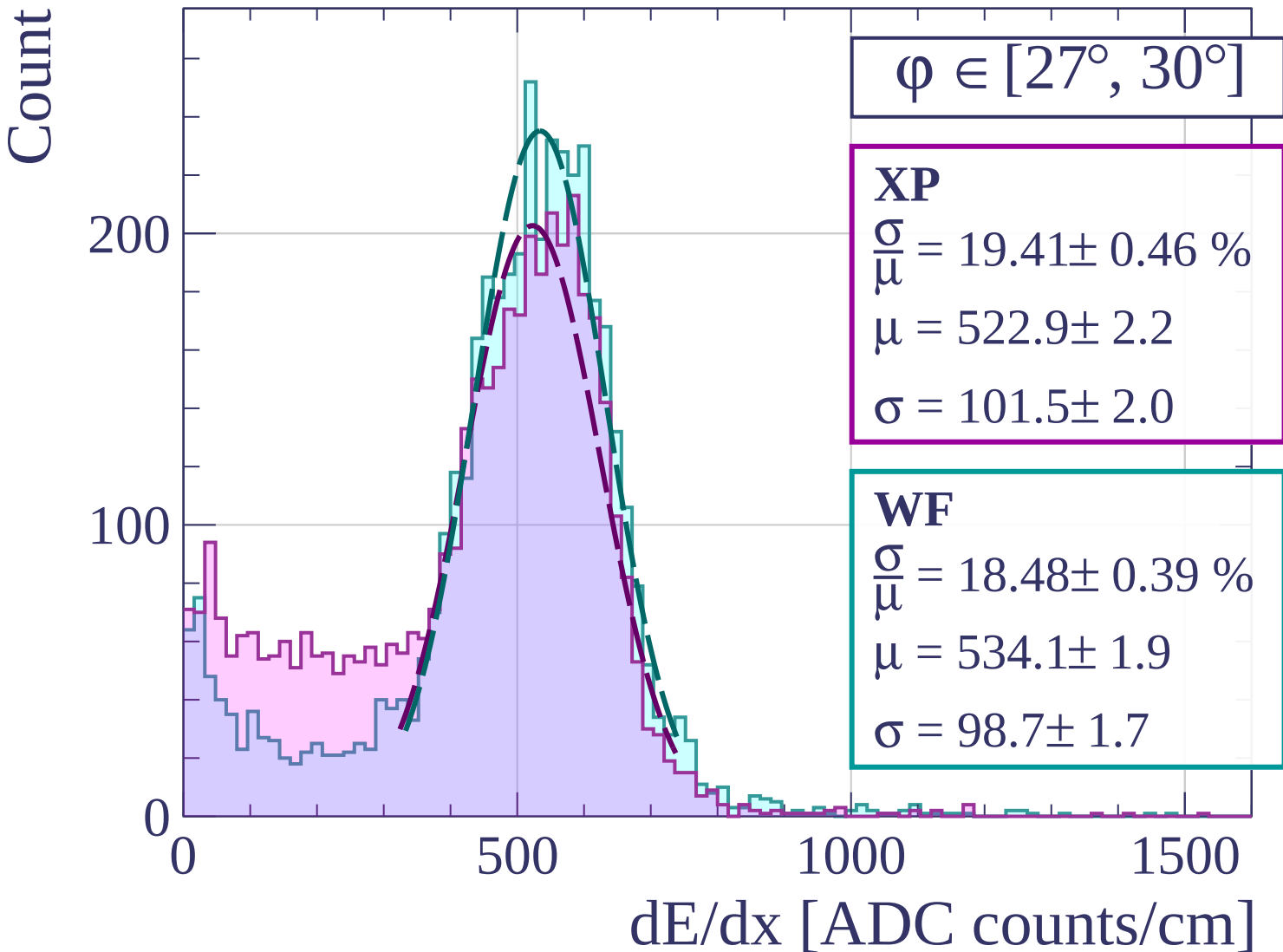
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [30^\circ, 33^\circ]$

XP

$$\frac{\sigma}{\mu} = 20.06 \pm 0.43 \%$$

$$\mu = 523.4 \pm 2.0$$

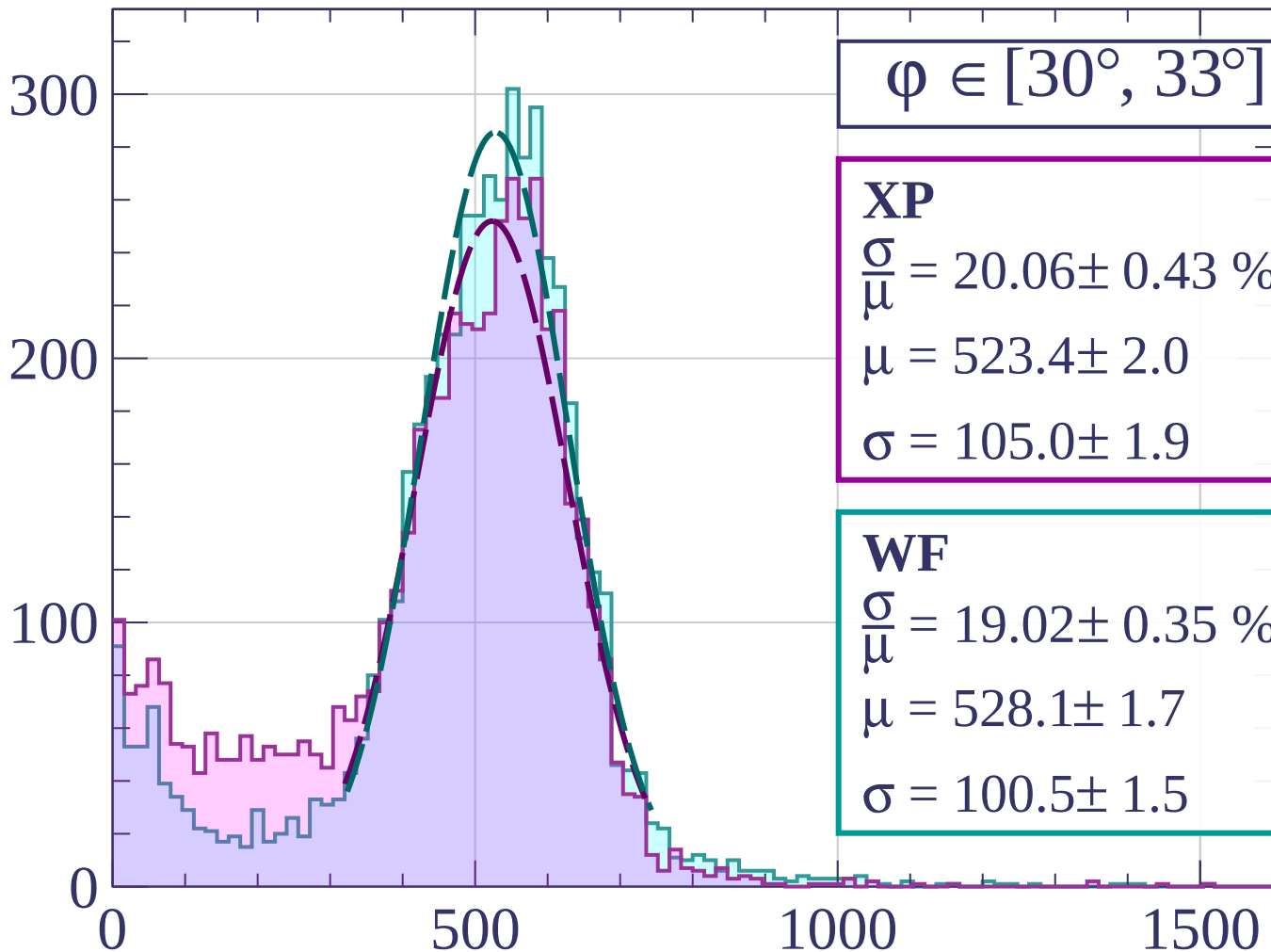
$$\sigma = 105.0 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 19.02 \pm 0.35 \%$$

$$\mu = 528.1 \pm 1.7$$

$$\sigma = 100.5 \pm 1.5$$



dE/dx [ADC counts/cm]

Count

$\varphi \in [33^\circ, 36^\circ]$

XP

$$\frac{\sigma}{\mu} = 18.64 \pm 0.37 \%$$

$$\mu = 529.2 \pm 1.8$$

$$\sigma = 98.6 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 18.09 \pm 0.33 \%$$

$$\mu = 535.3 \pm 1.6$$

$$\sigma = 96.8 \pm 1.5$$

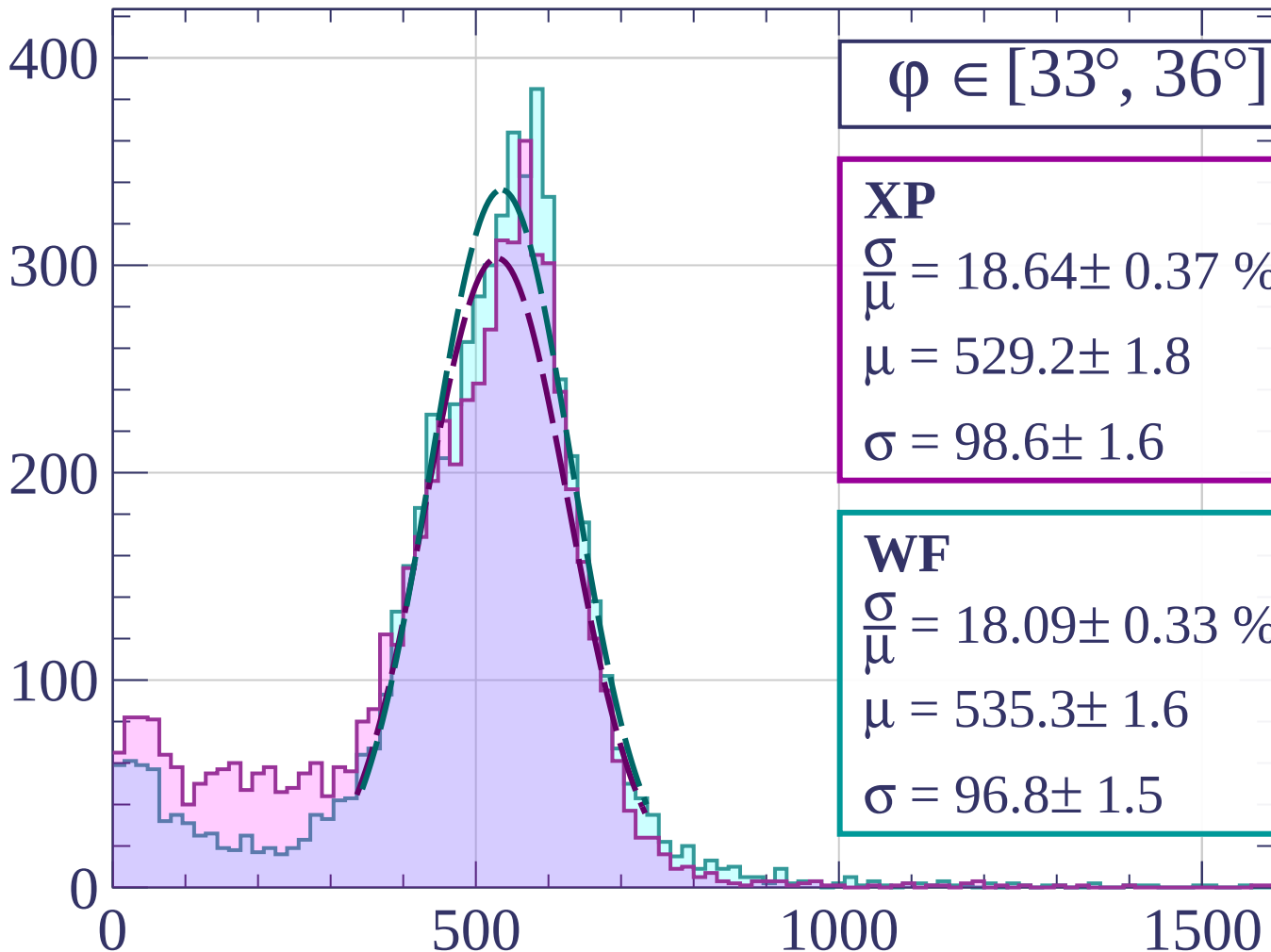
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [36^\circ, 39^\circ]$

XP

$$\frac{\sigma}{\mu} = 18.22 \pm 0.30 \%$$

$$\mu = 531.7 \pm 1.5$$

$$\sigma = 96.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 17.55 \pm 0.29 \%$$

$$\mu = 538.5 \pm 1.4$$

$$\sigma = 94.5 \pm 1.3$$

400

300

200

100

0

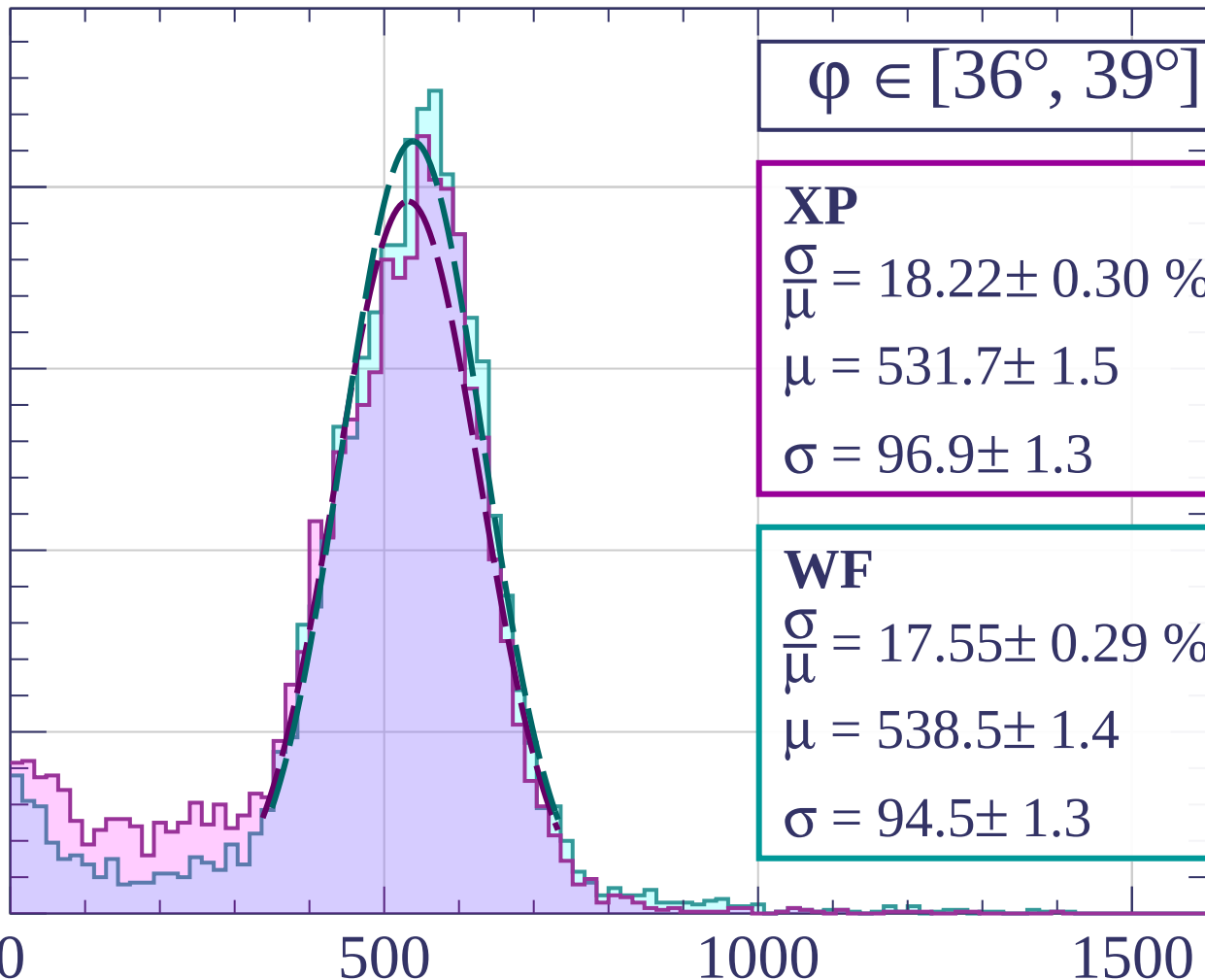
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [39^\circ, 42^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.53 \pm 0.25 \%$$

$$\mu = 531.6 \pm 1.3$$

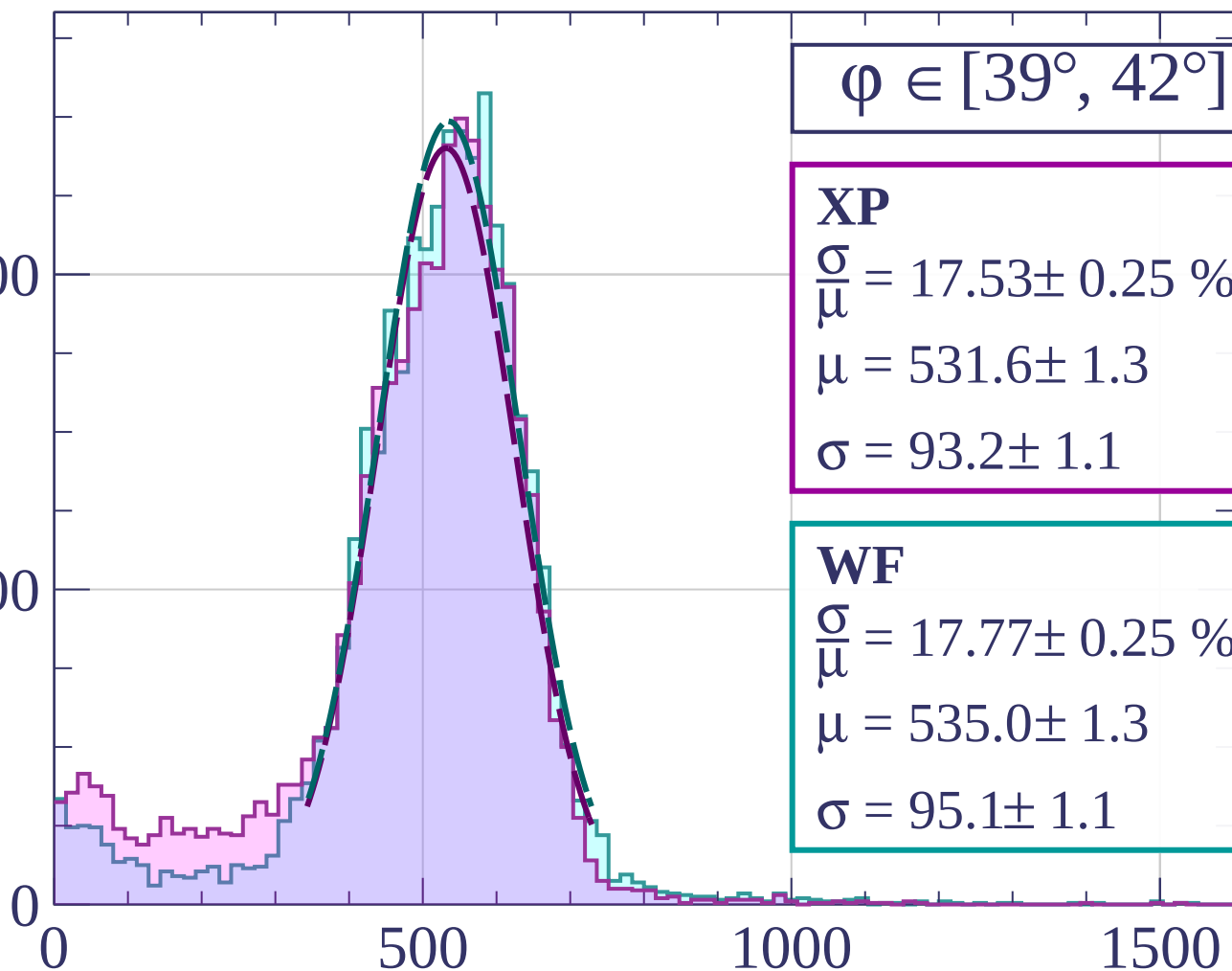
$$\sigma = 93.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 17.77 \pm 0.25 \%$$

$$\mu = 535.0 \pm 1.3$$

$$\sigma = 95.1 \pm 1.1$$



dE/dx [ADC counts/cm]

Count

$\phi \in [42^\circ, 45^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.97 \pm 0.24 \%$$

$$\mu = 531.1 \pm 1.2$$

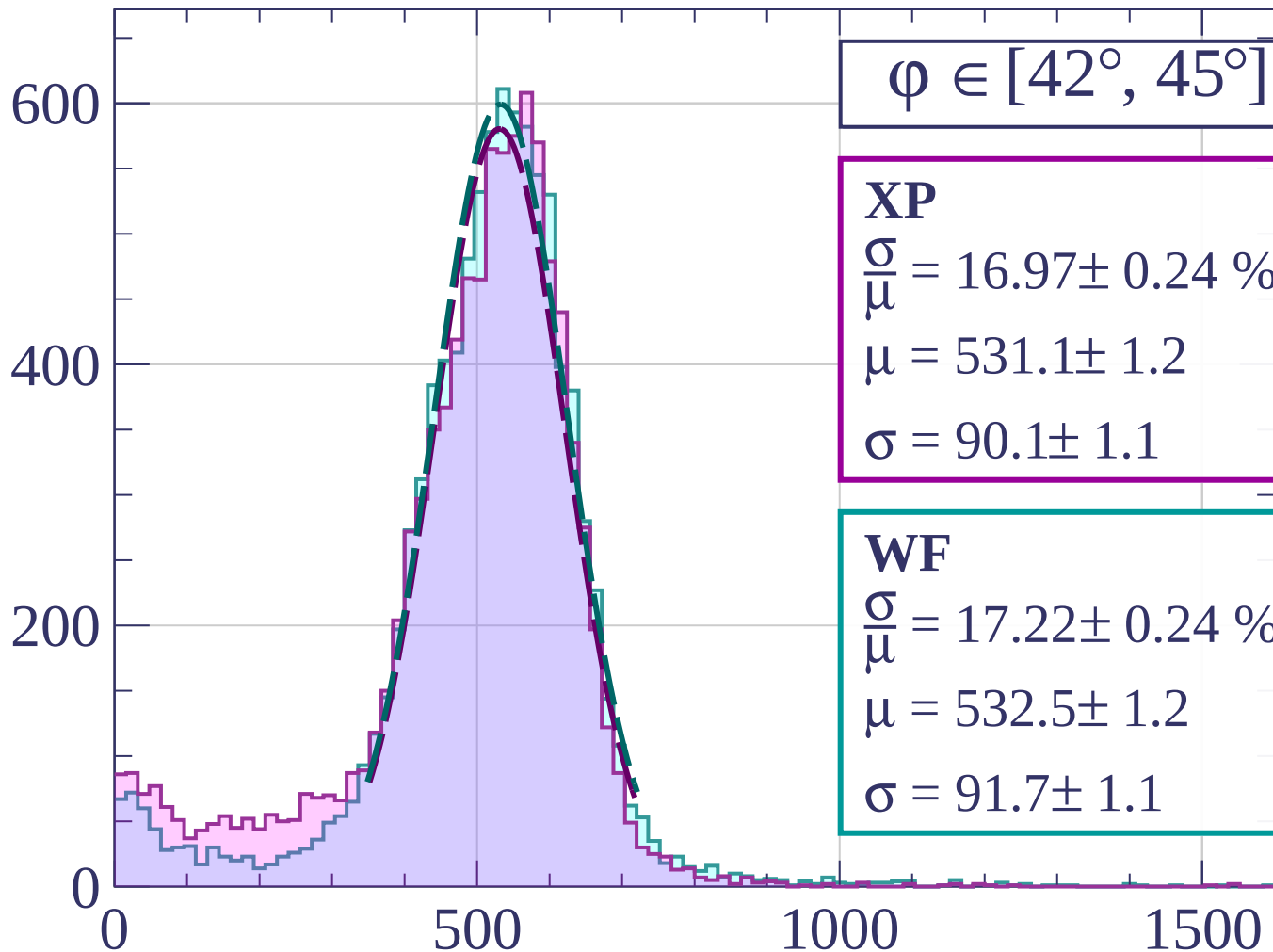
$$\sigma = 90.1 \pm 1.1$$

WF

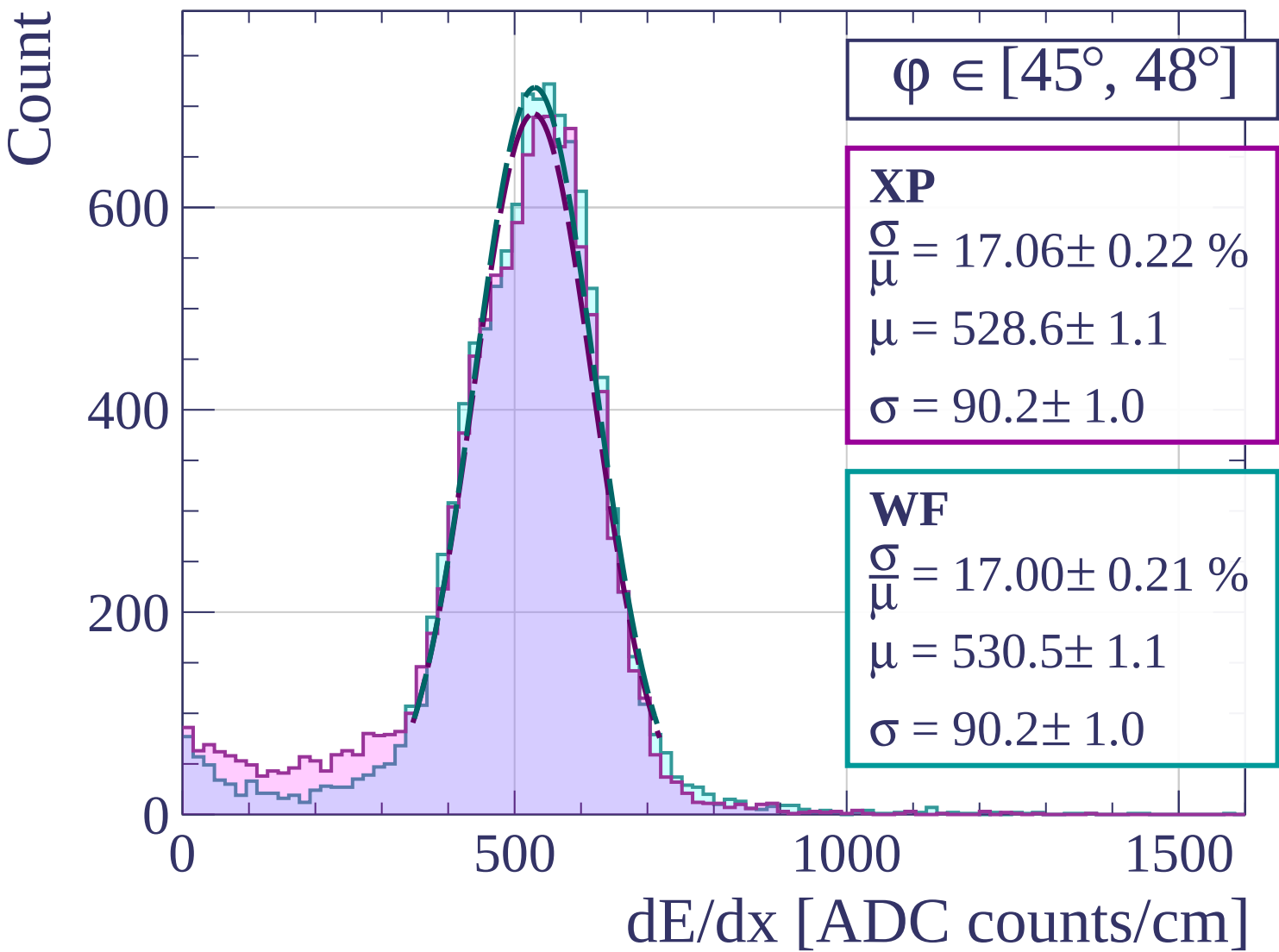
$$\frac{\sigma}{\mu} = 17.22 \pm 0.24 \%$$

$$\mu = 532.5 \pm 1.2$$

$$\sigma = 91.7 \pm 1.1$$



dE/dx [ADC counts/cm]



Count

$\varphi \in [48^\circ, 51^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.68 \pm 0.19 \%$$

$$\mu = 528.0 \pm 1.0$$

$$\sigma = 88.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 16.67 \pm 0.19 \%$$

$$\mu = 527.1 \pm 1.0$$

$$\sigma = 87.9 \pm 0.9$$

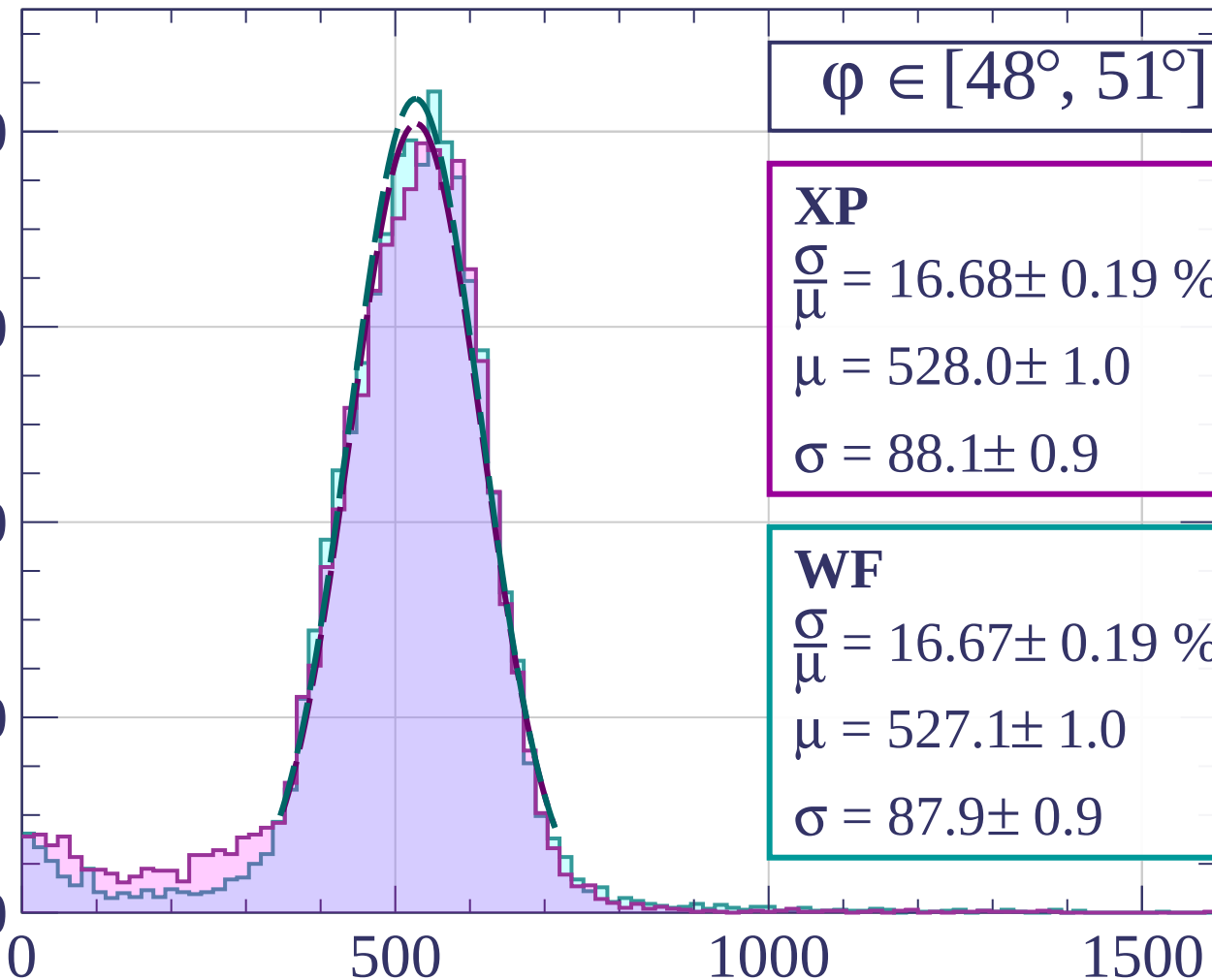
800

600

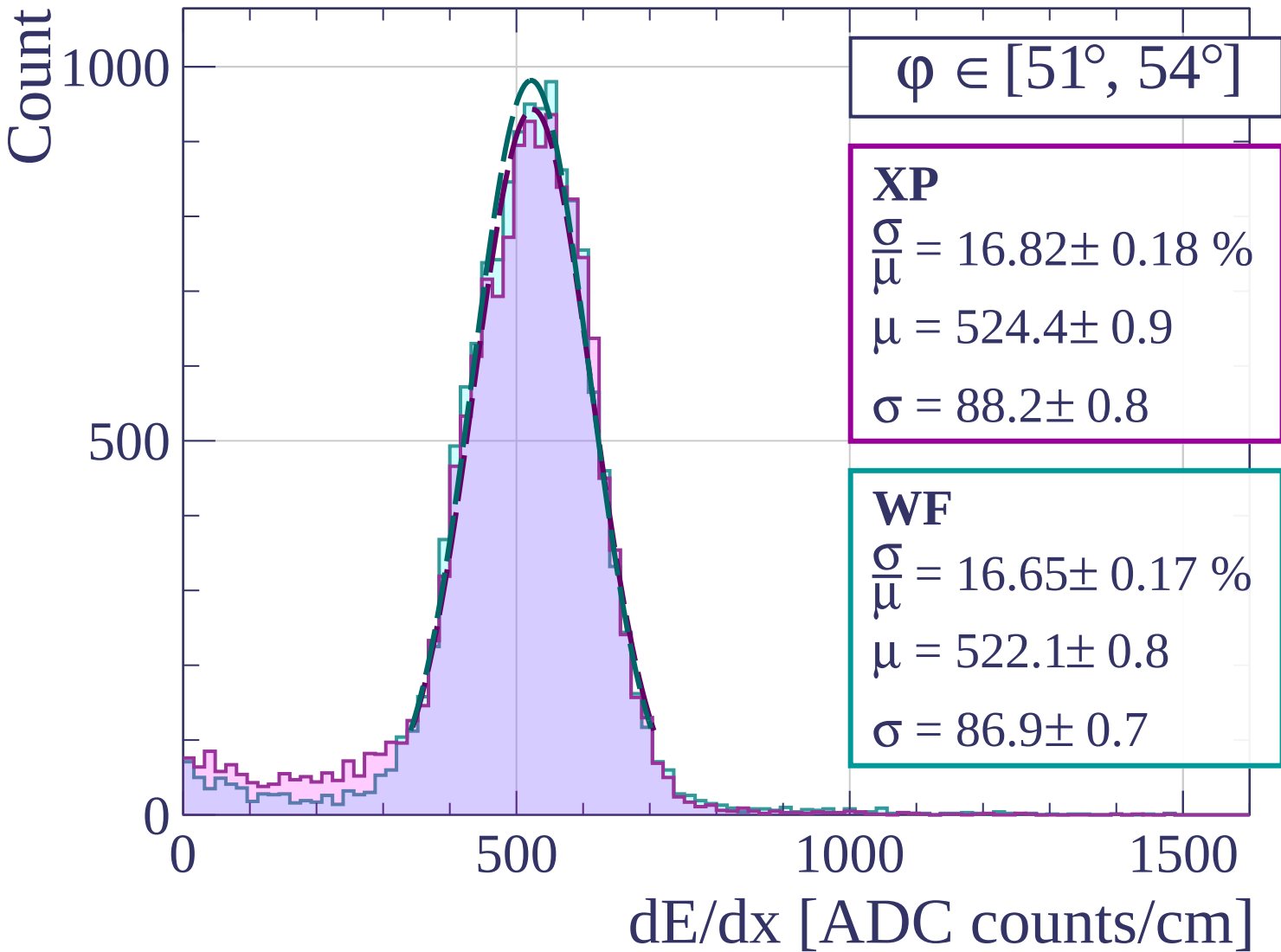
400

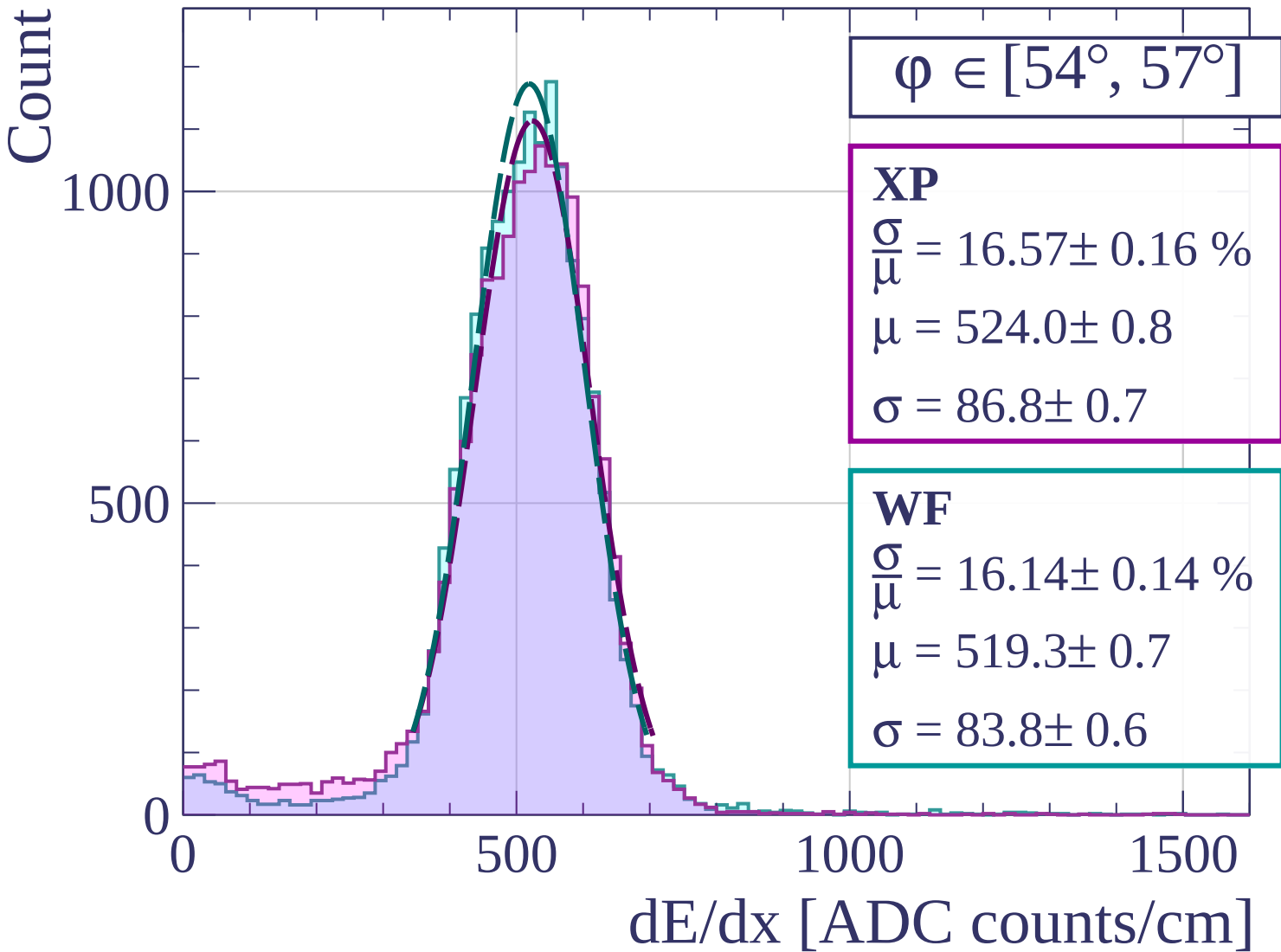
200

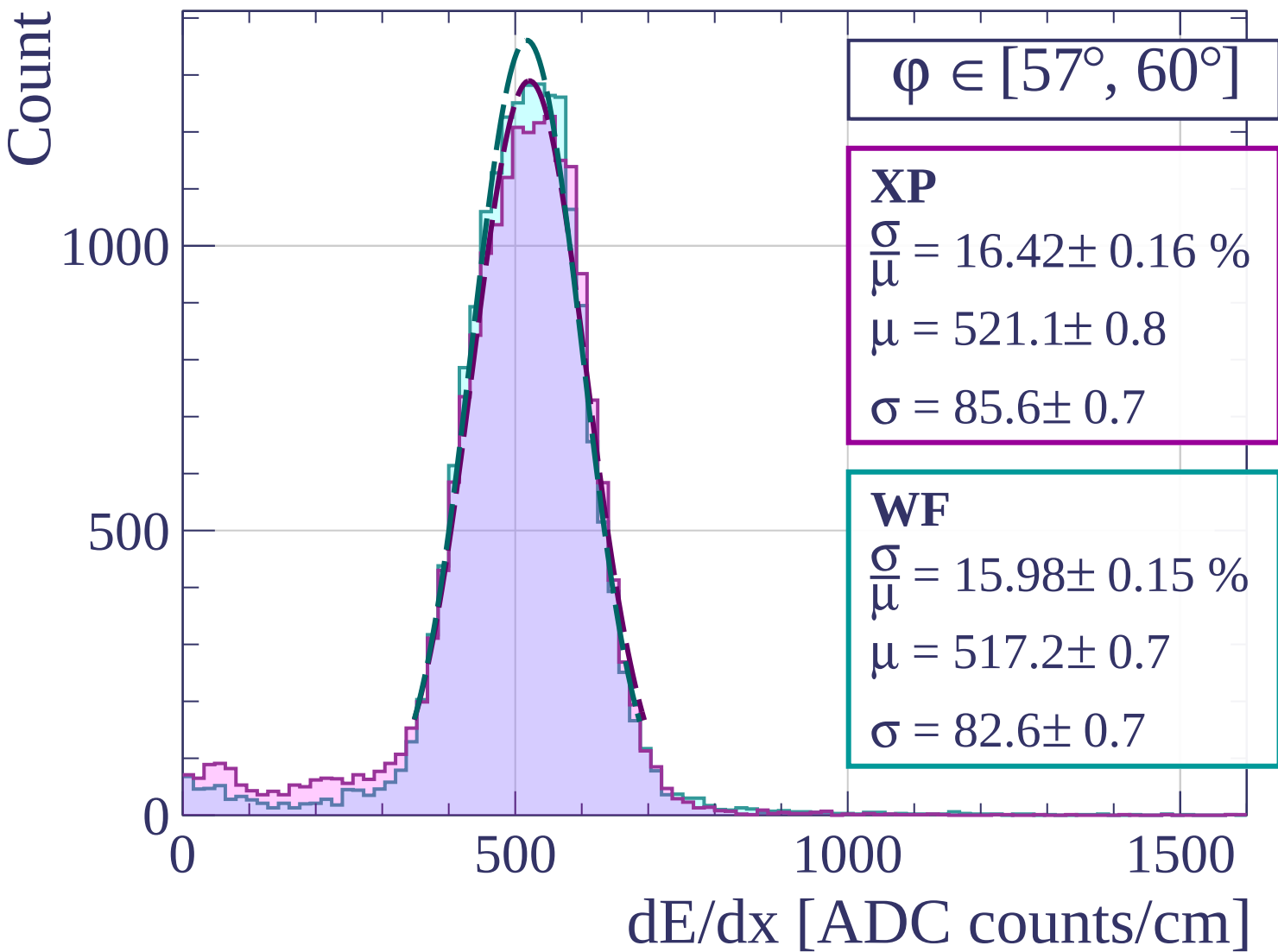
0

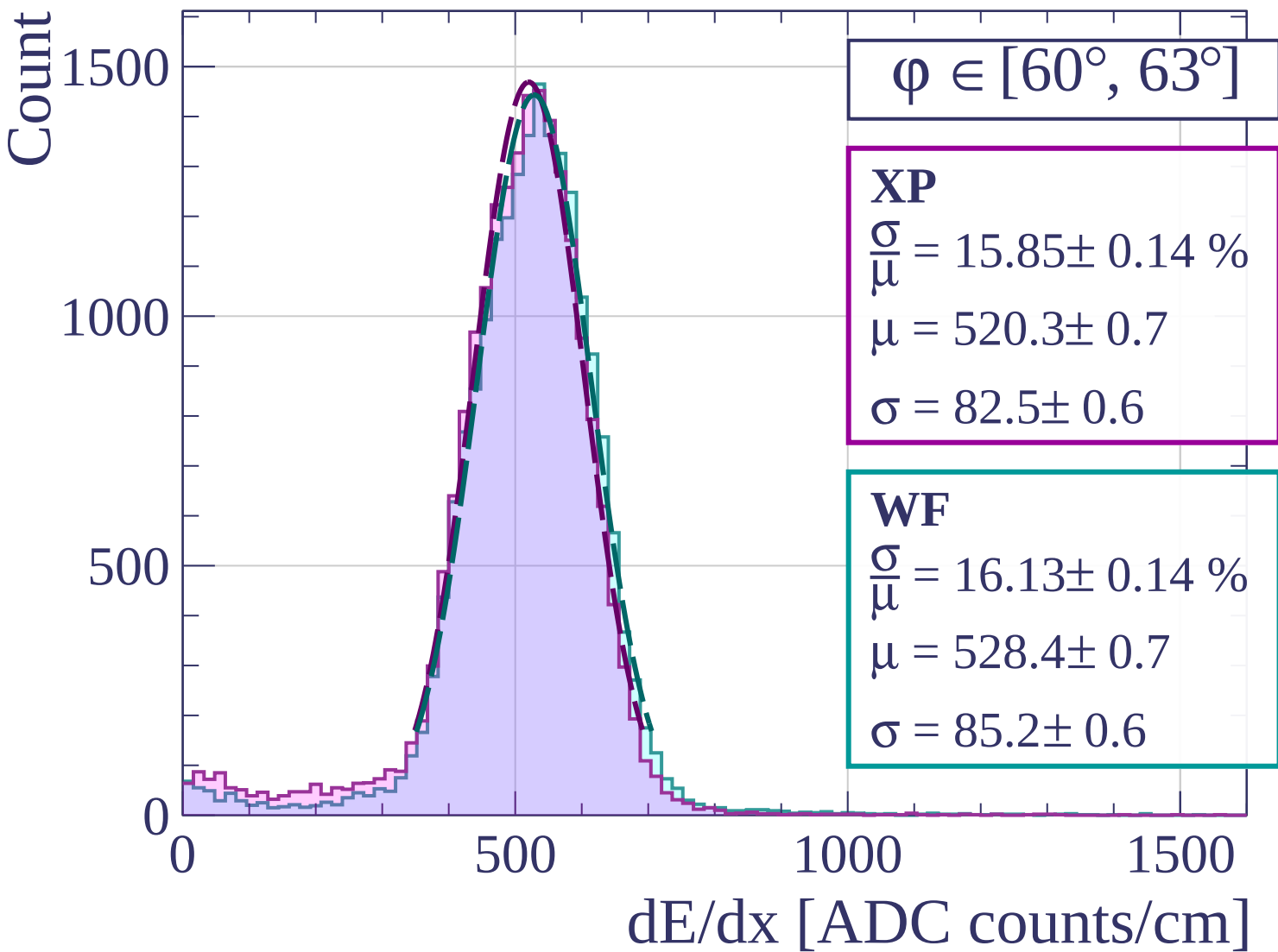


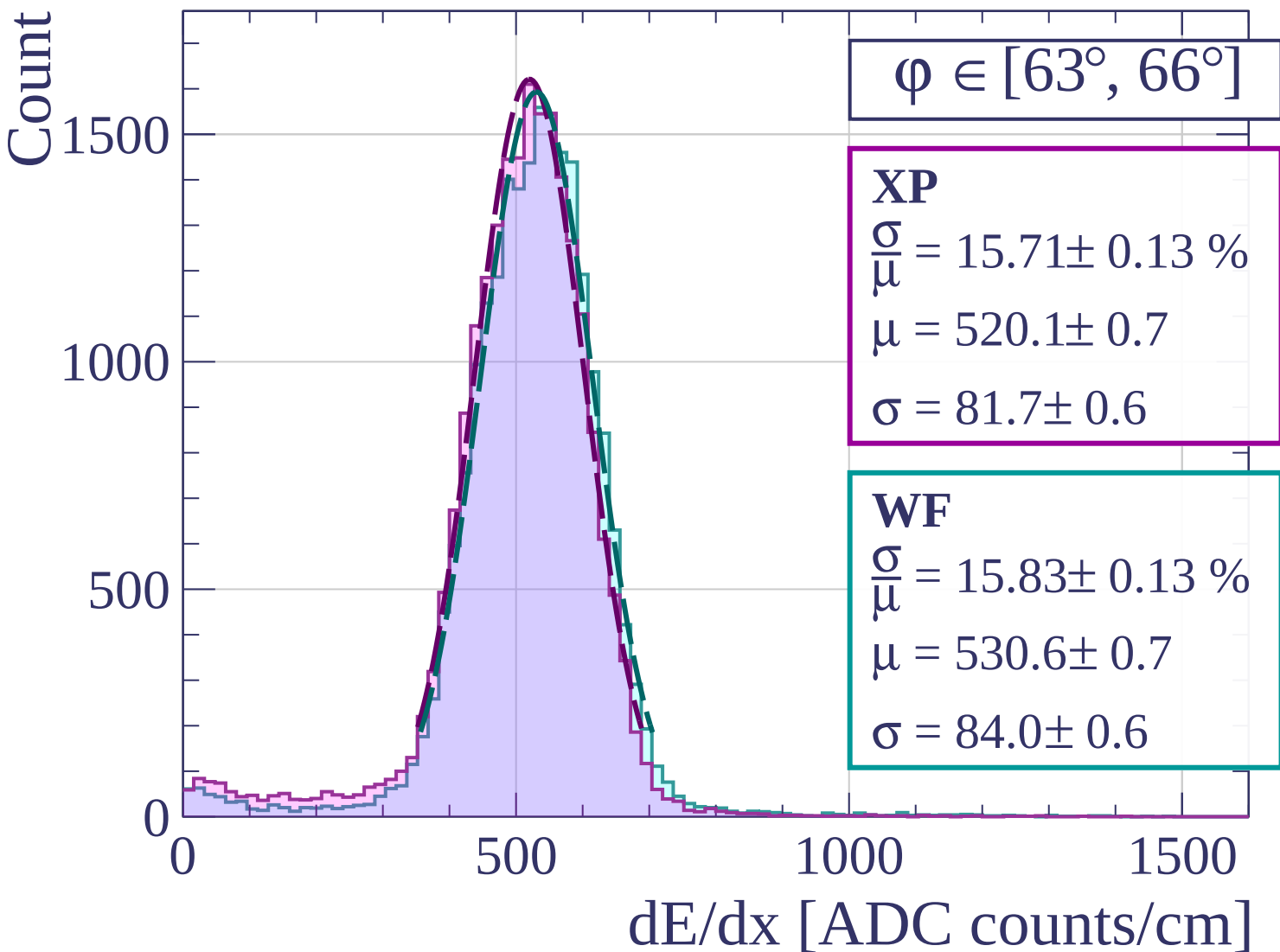
dE/dx [ADC counts/cm]

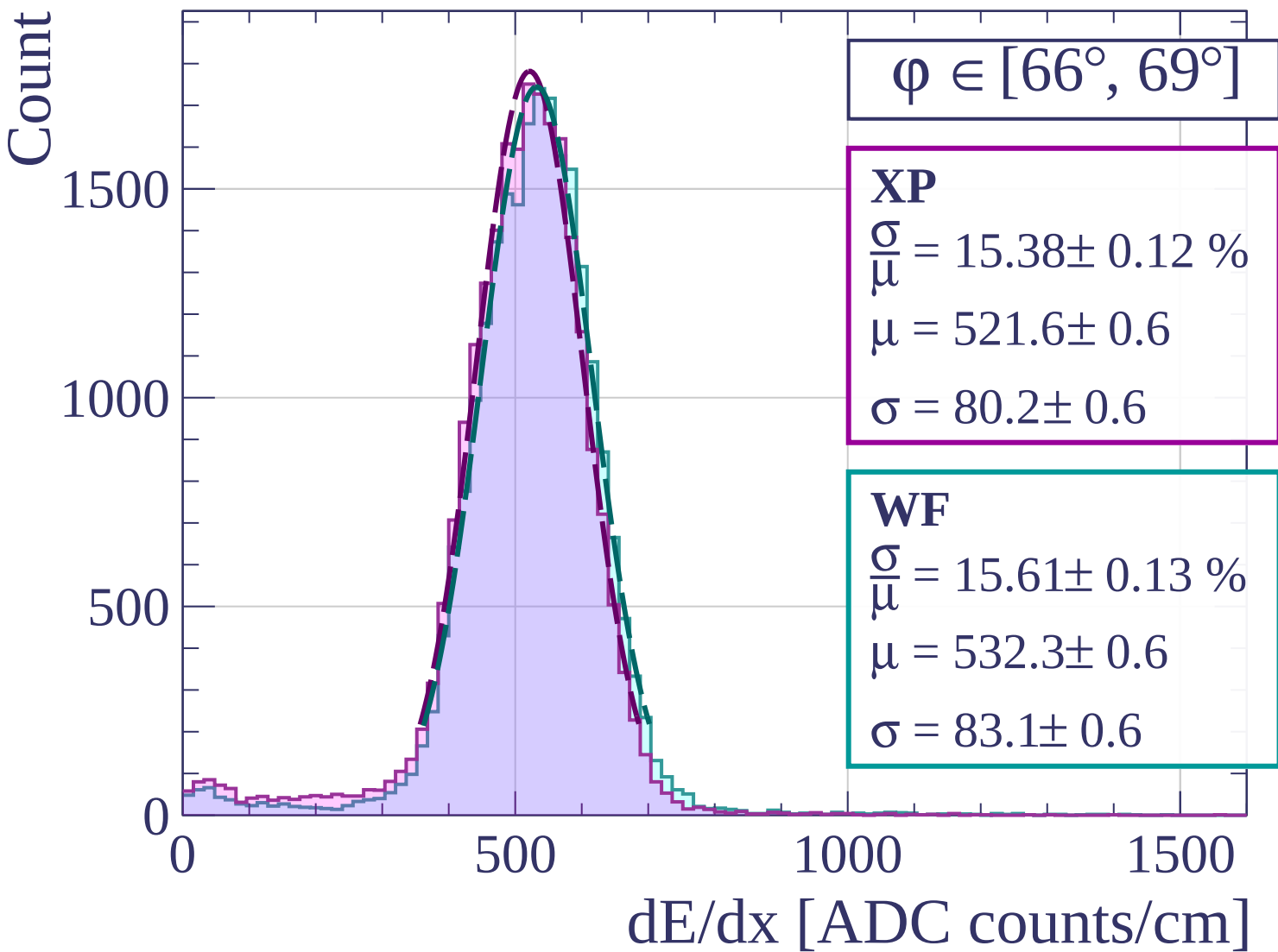


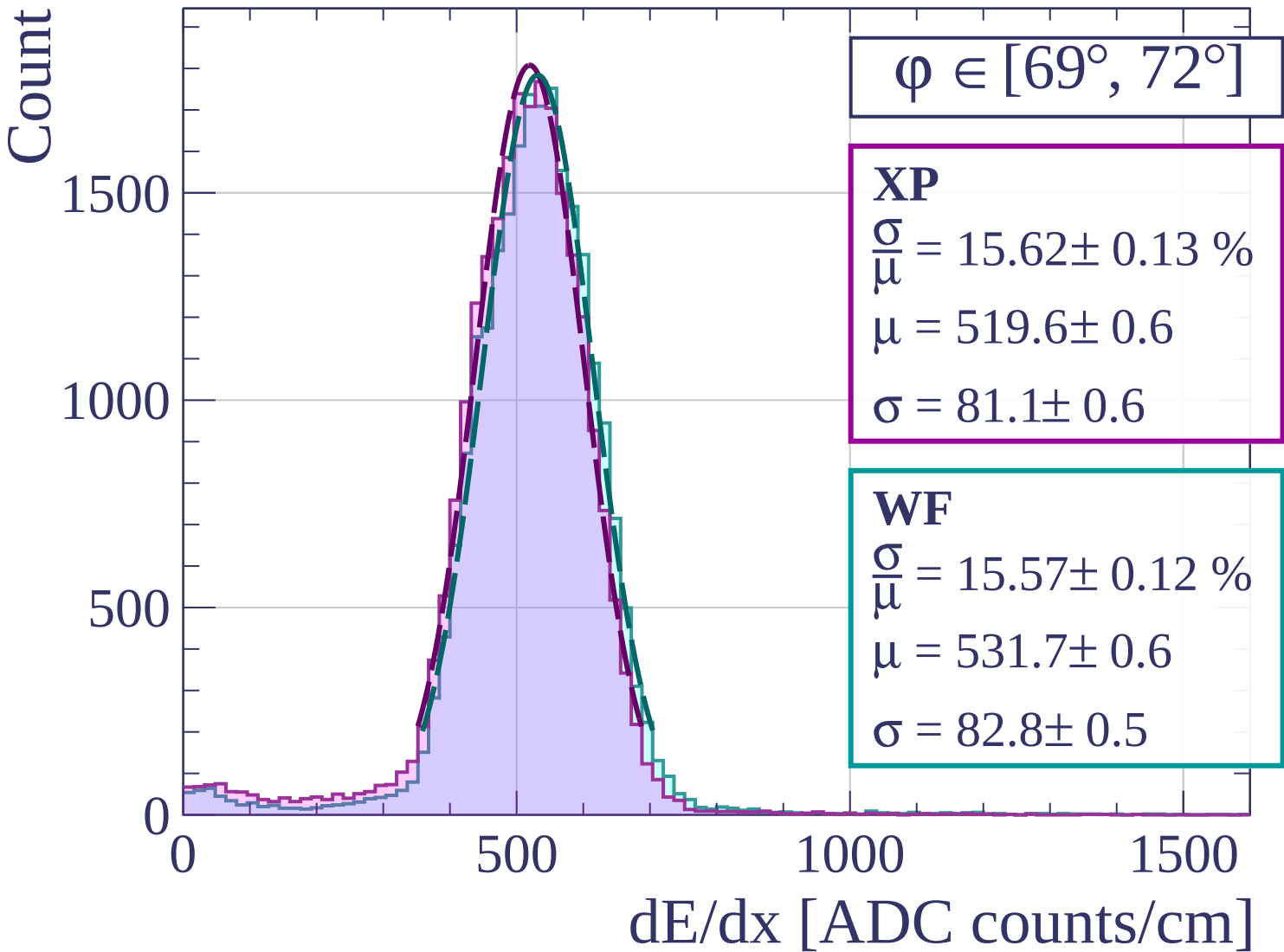


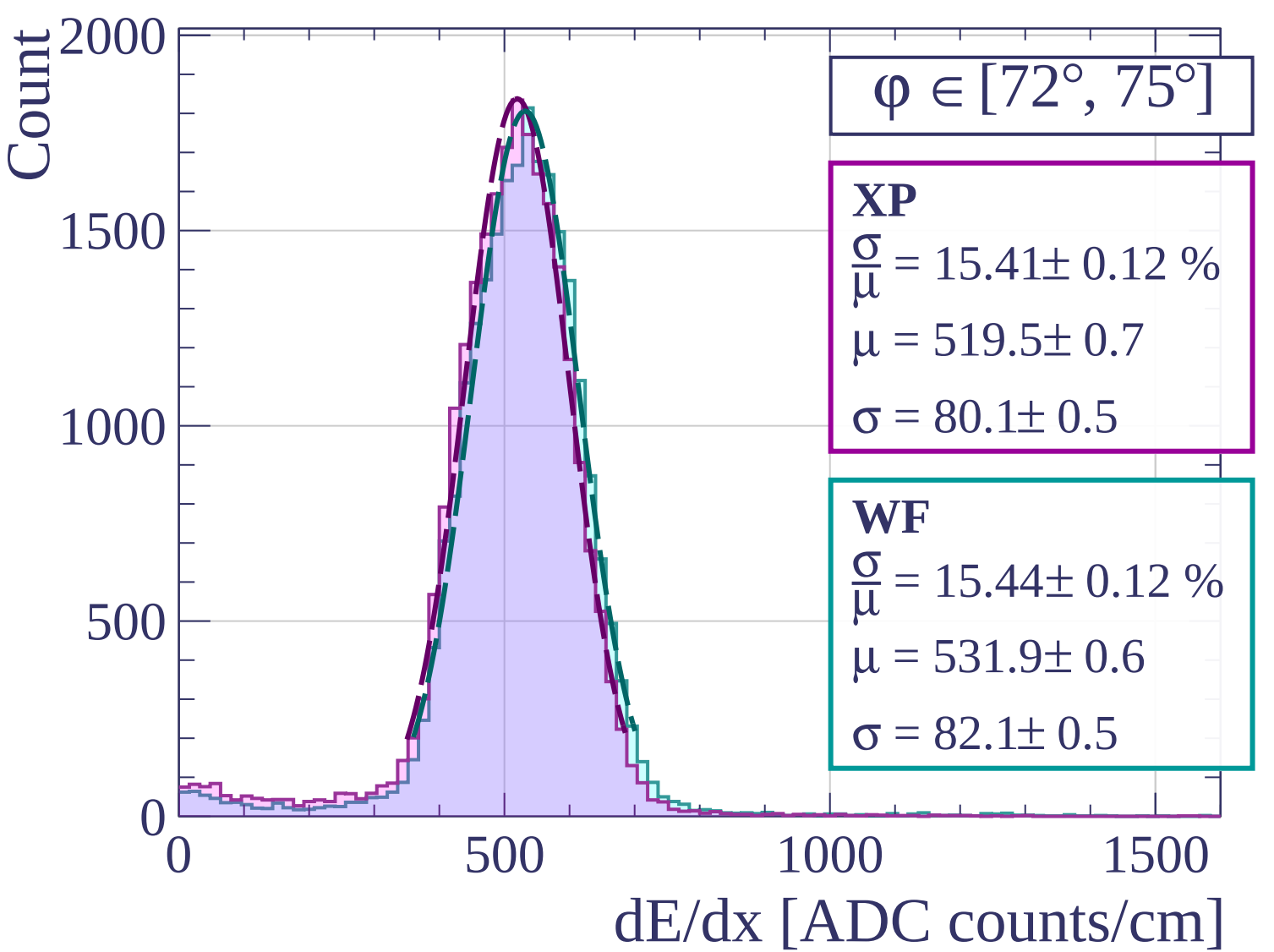


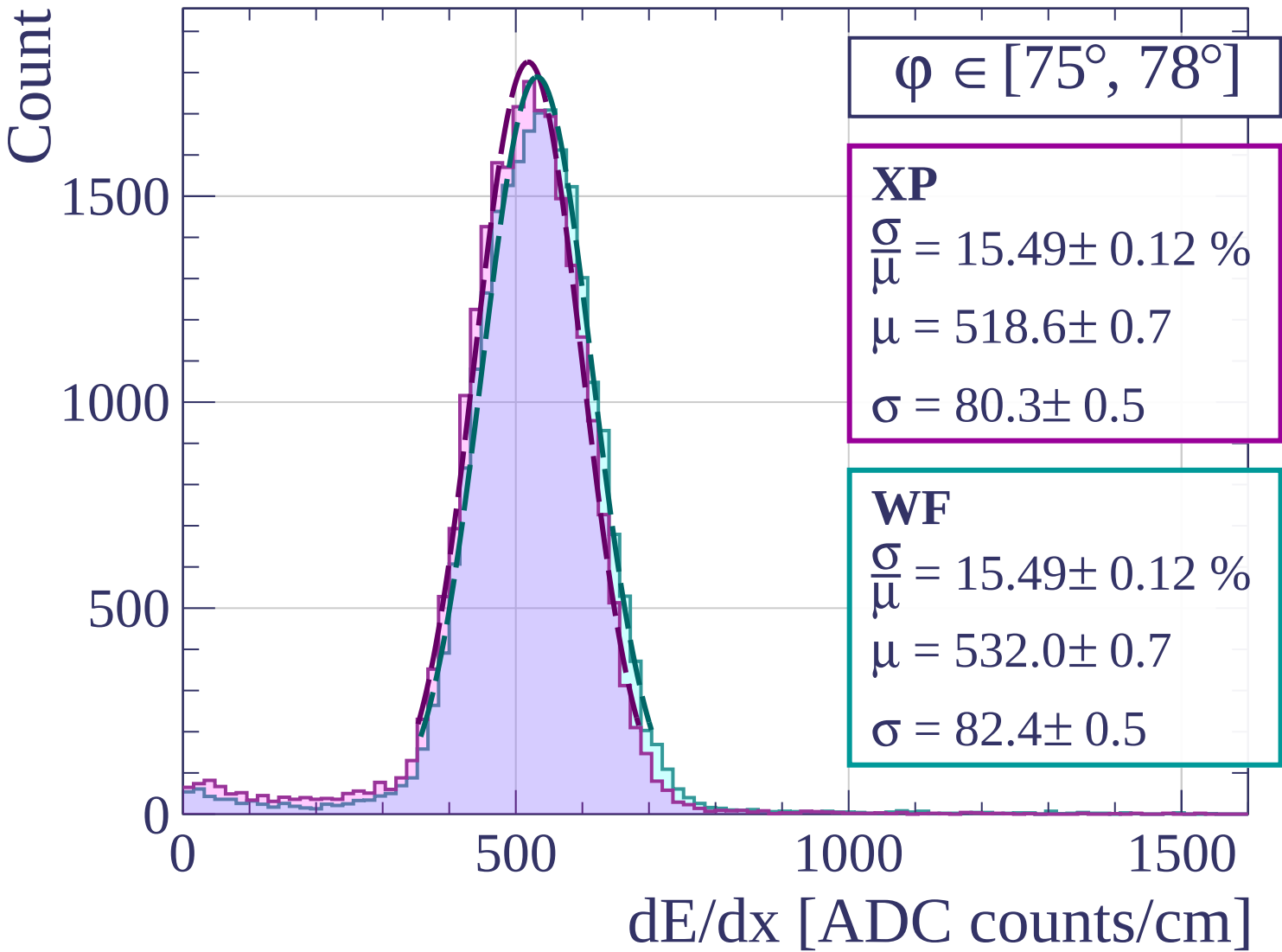


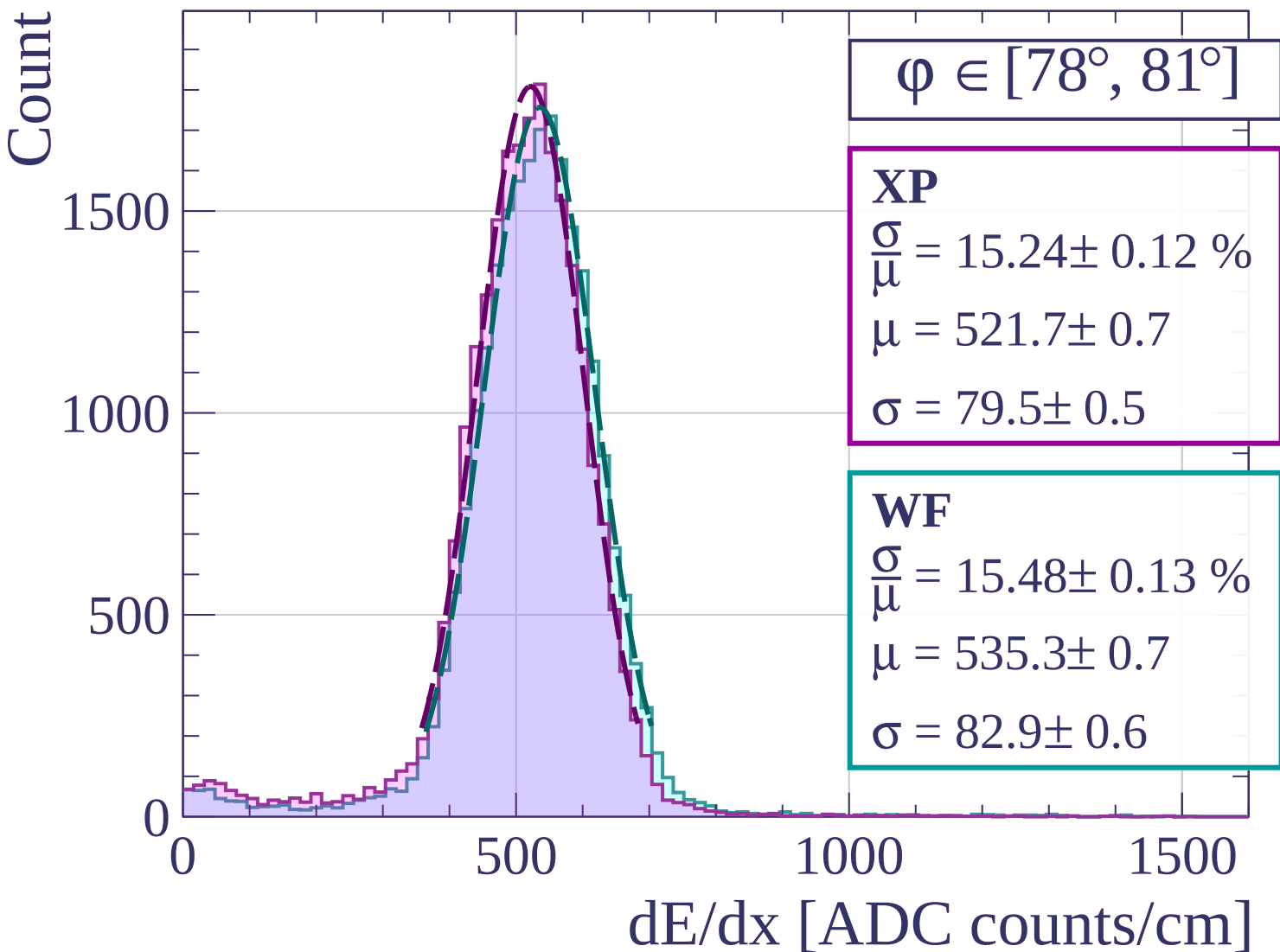


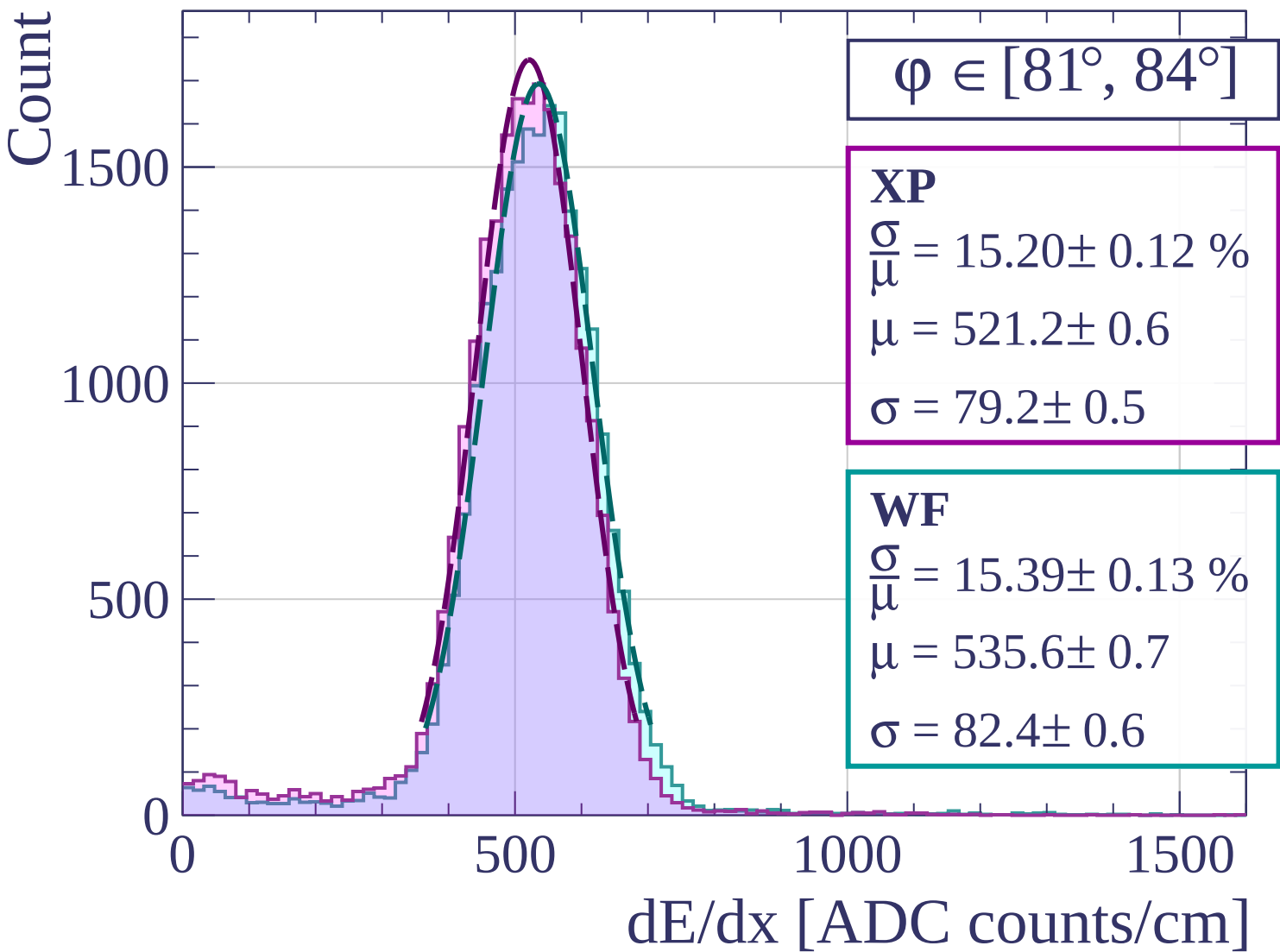


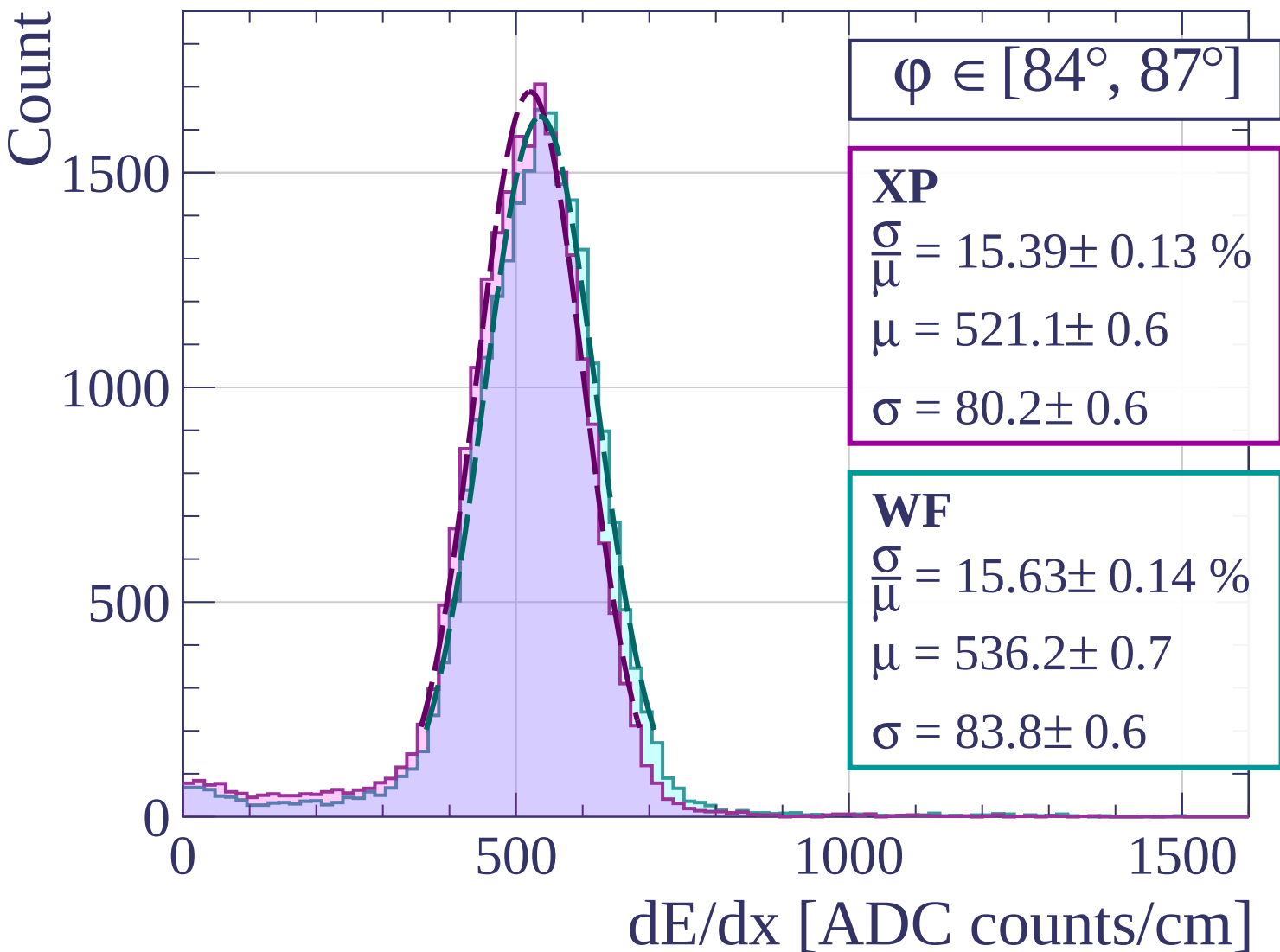


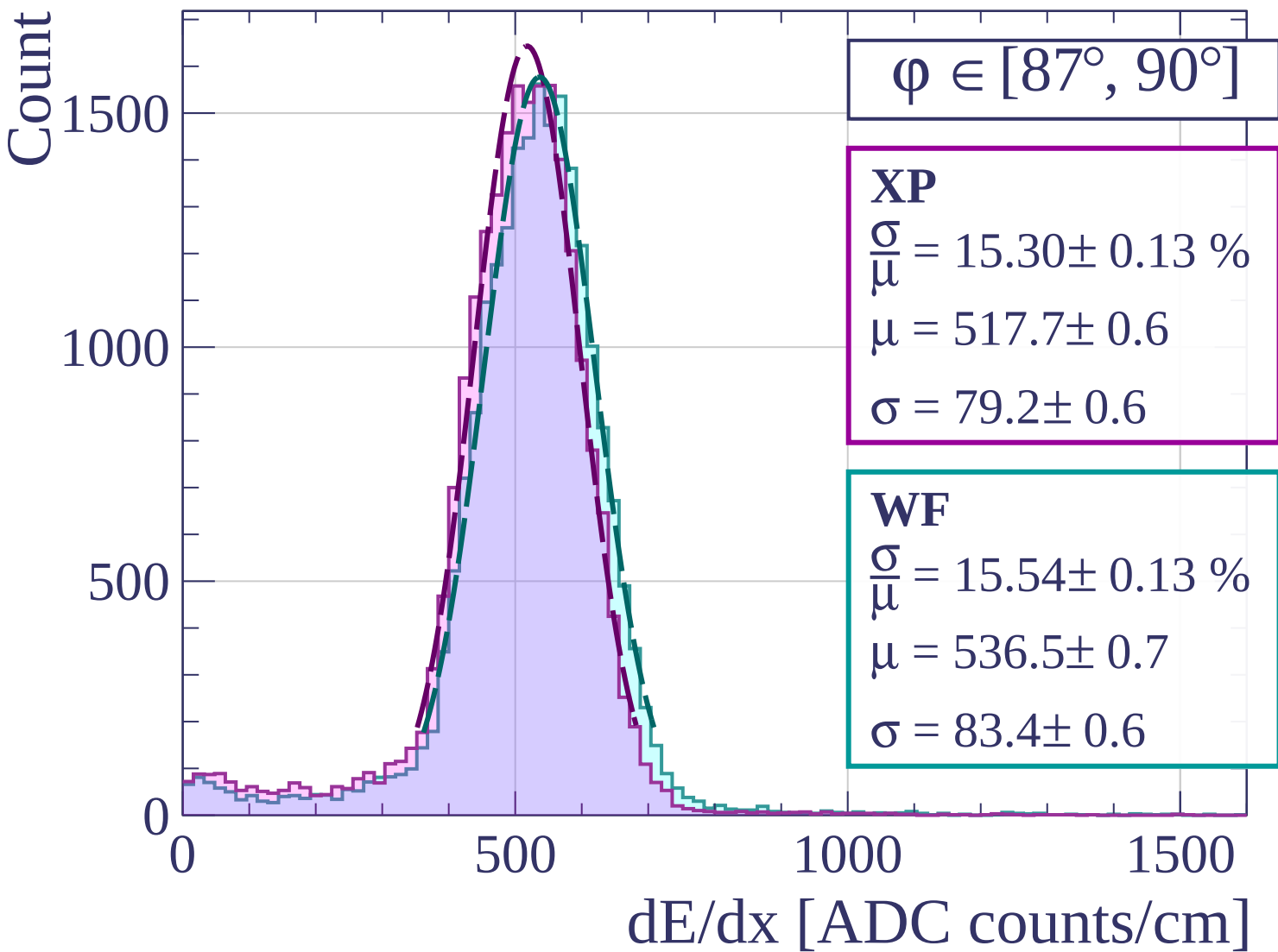


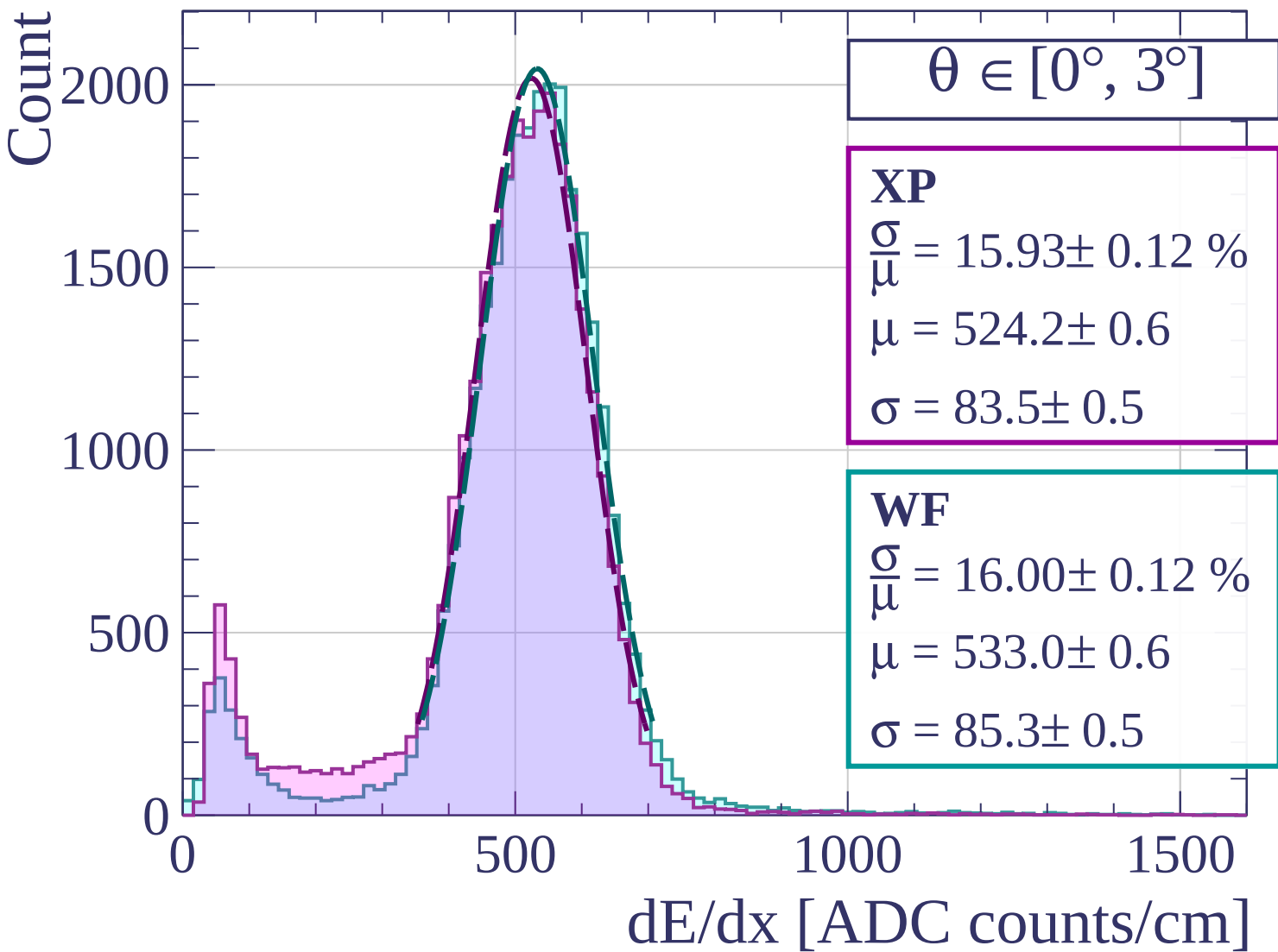


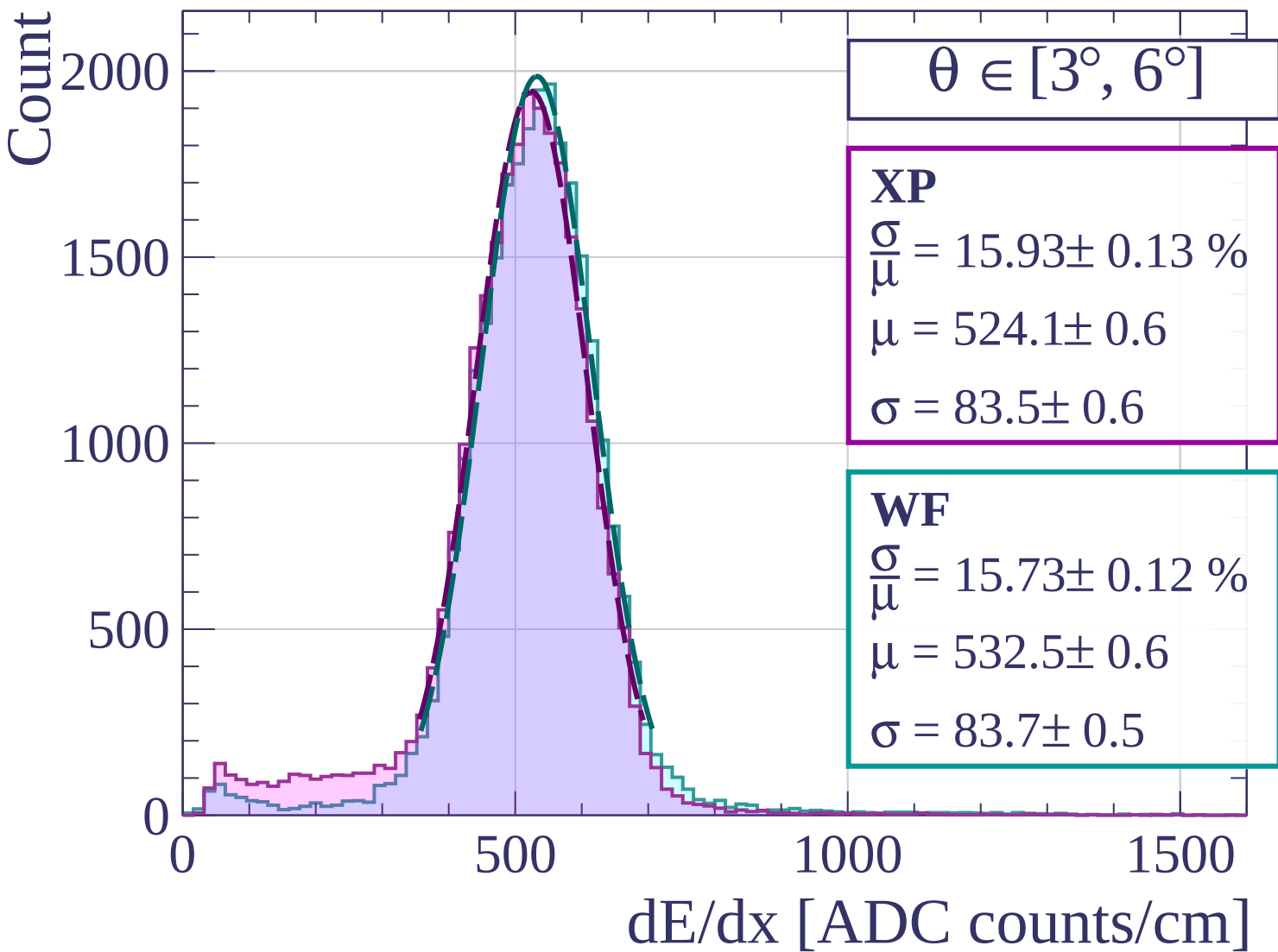


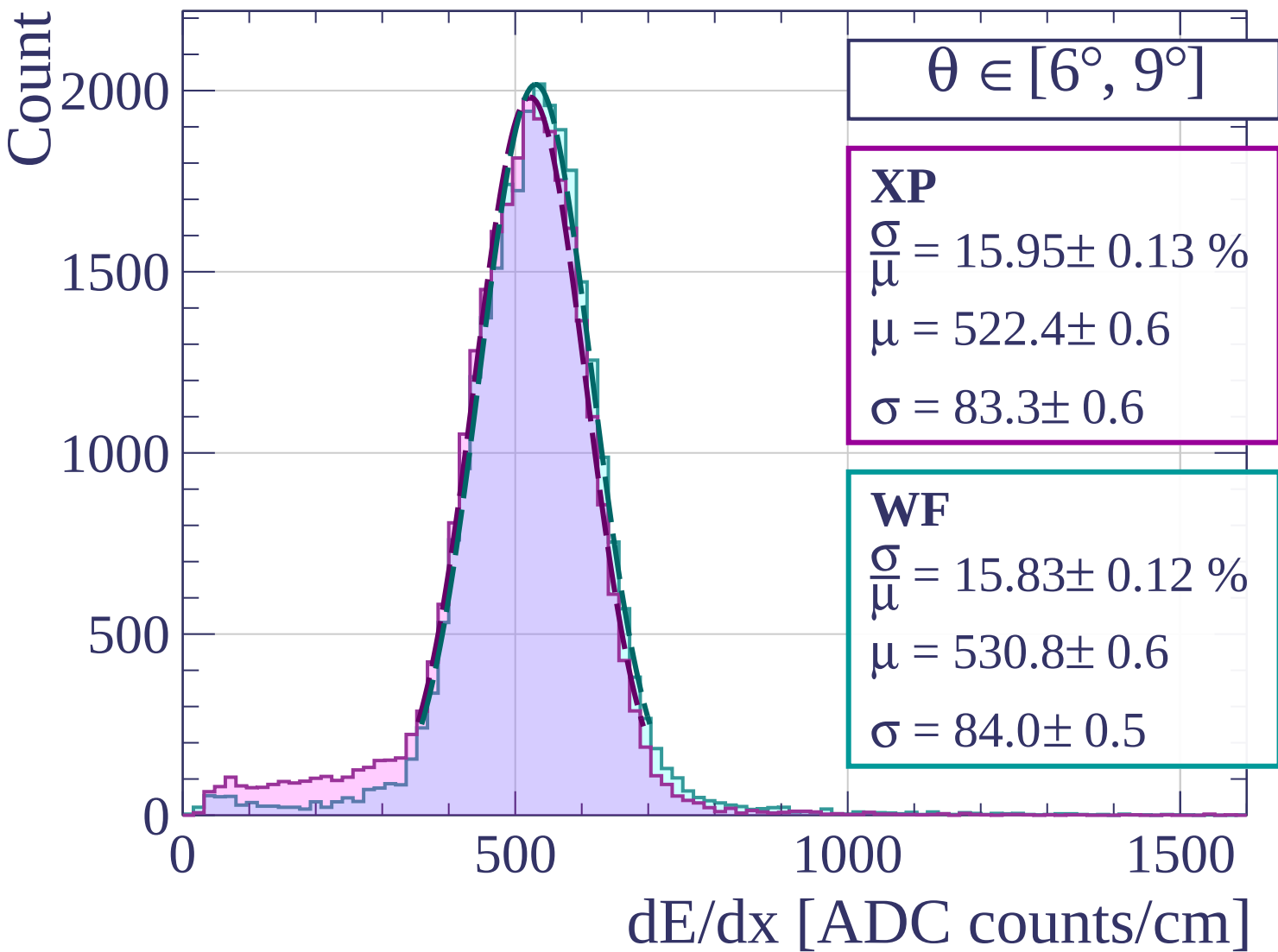


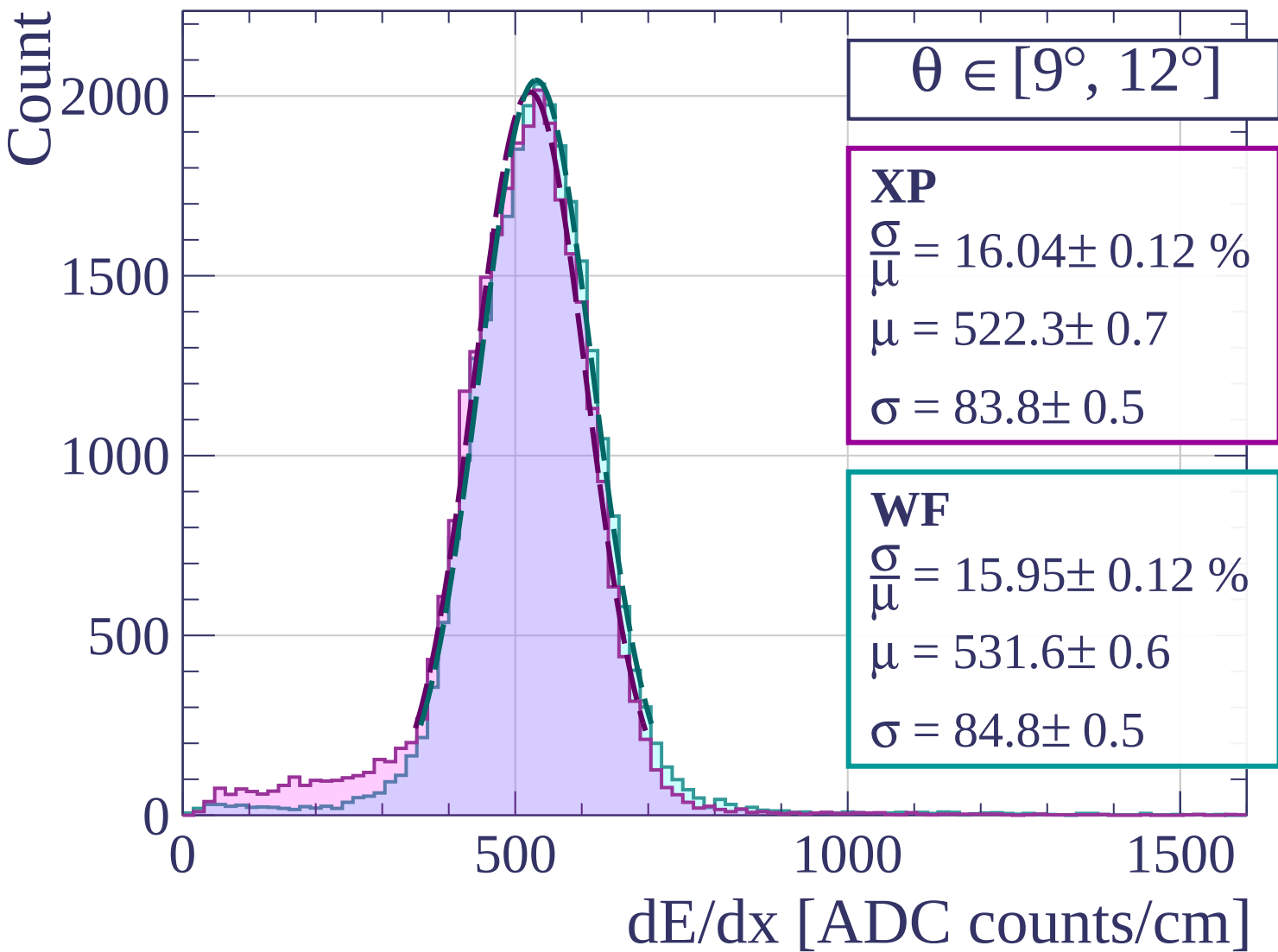


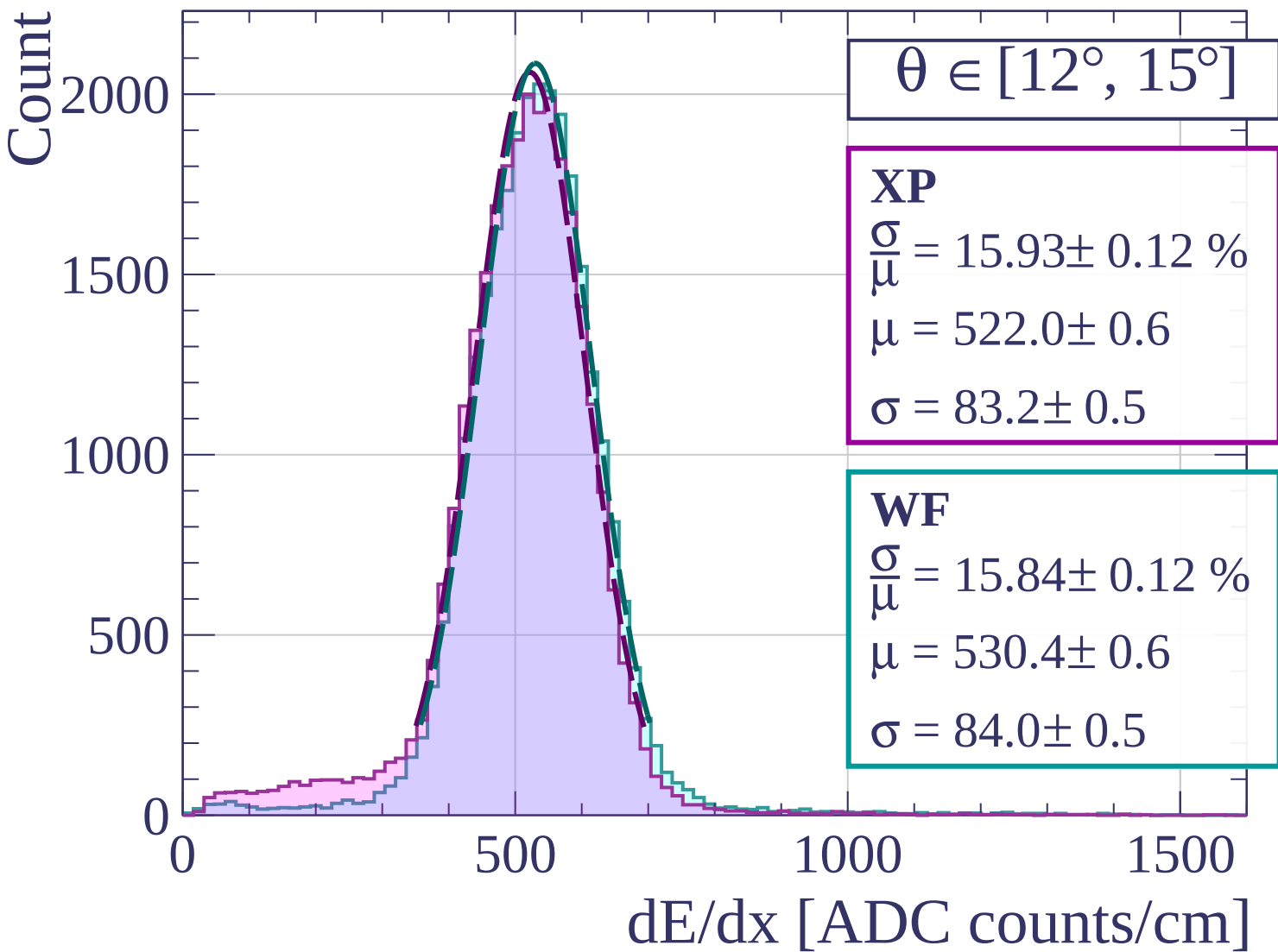


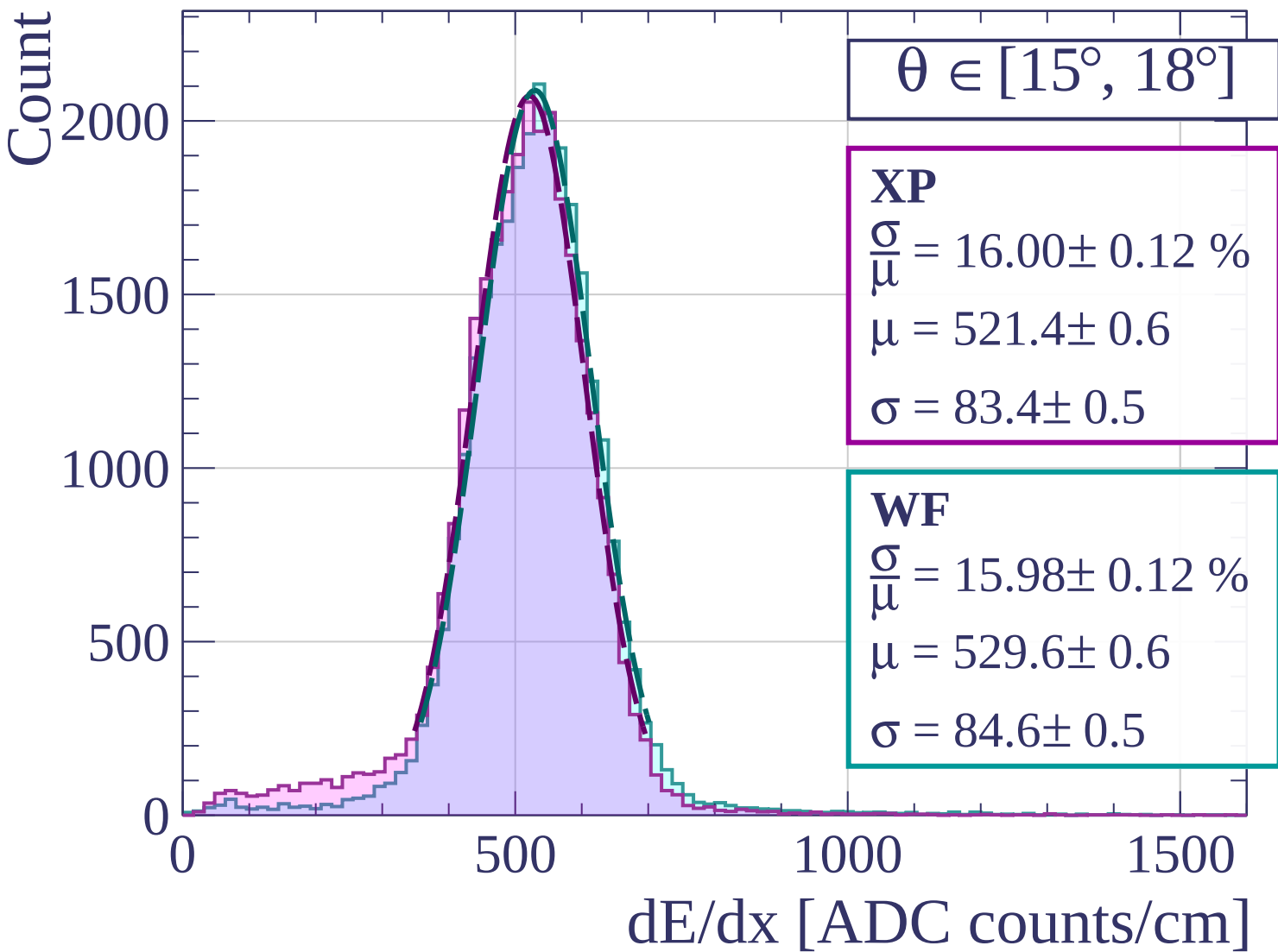


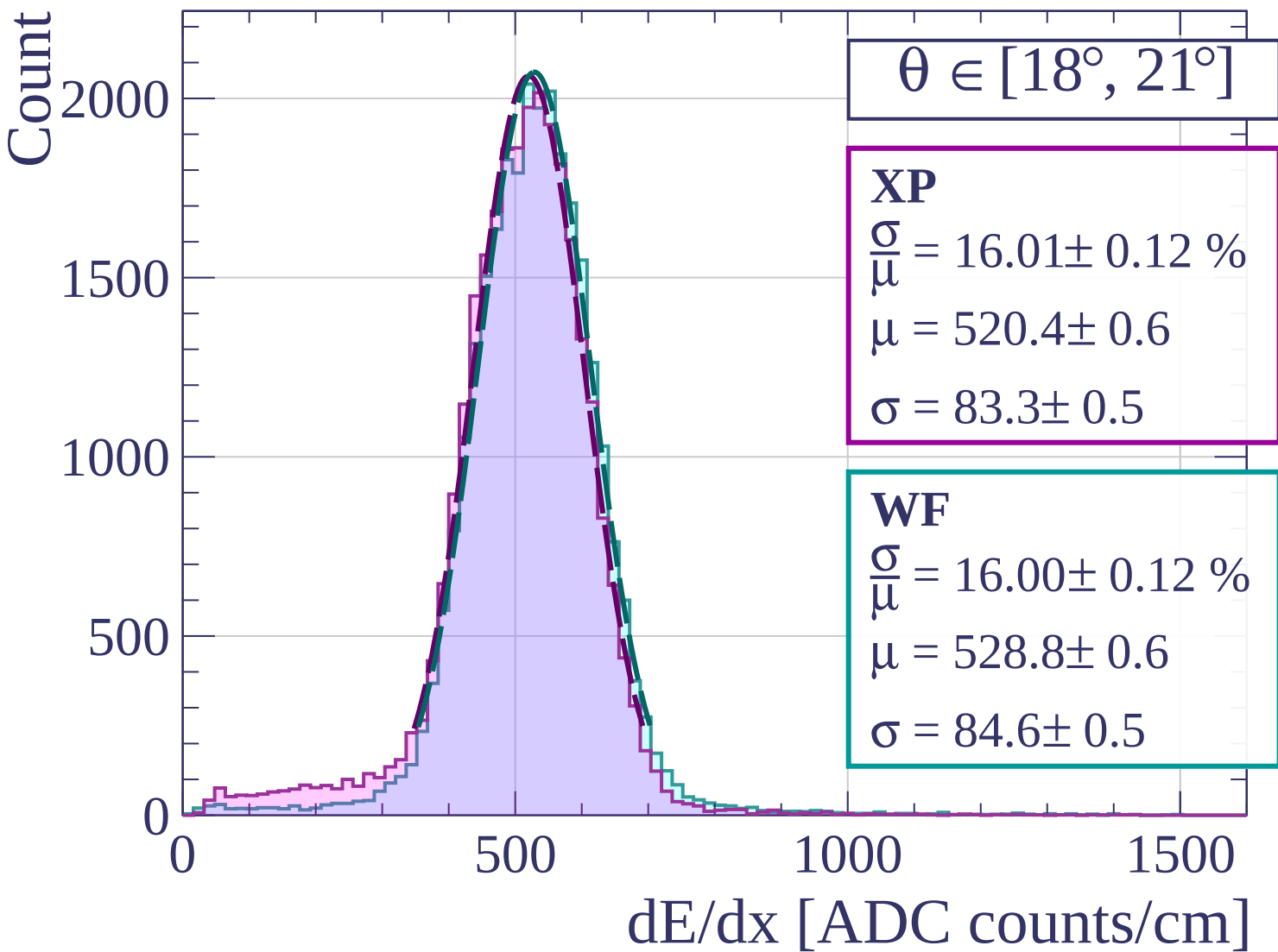


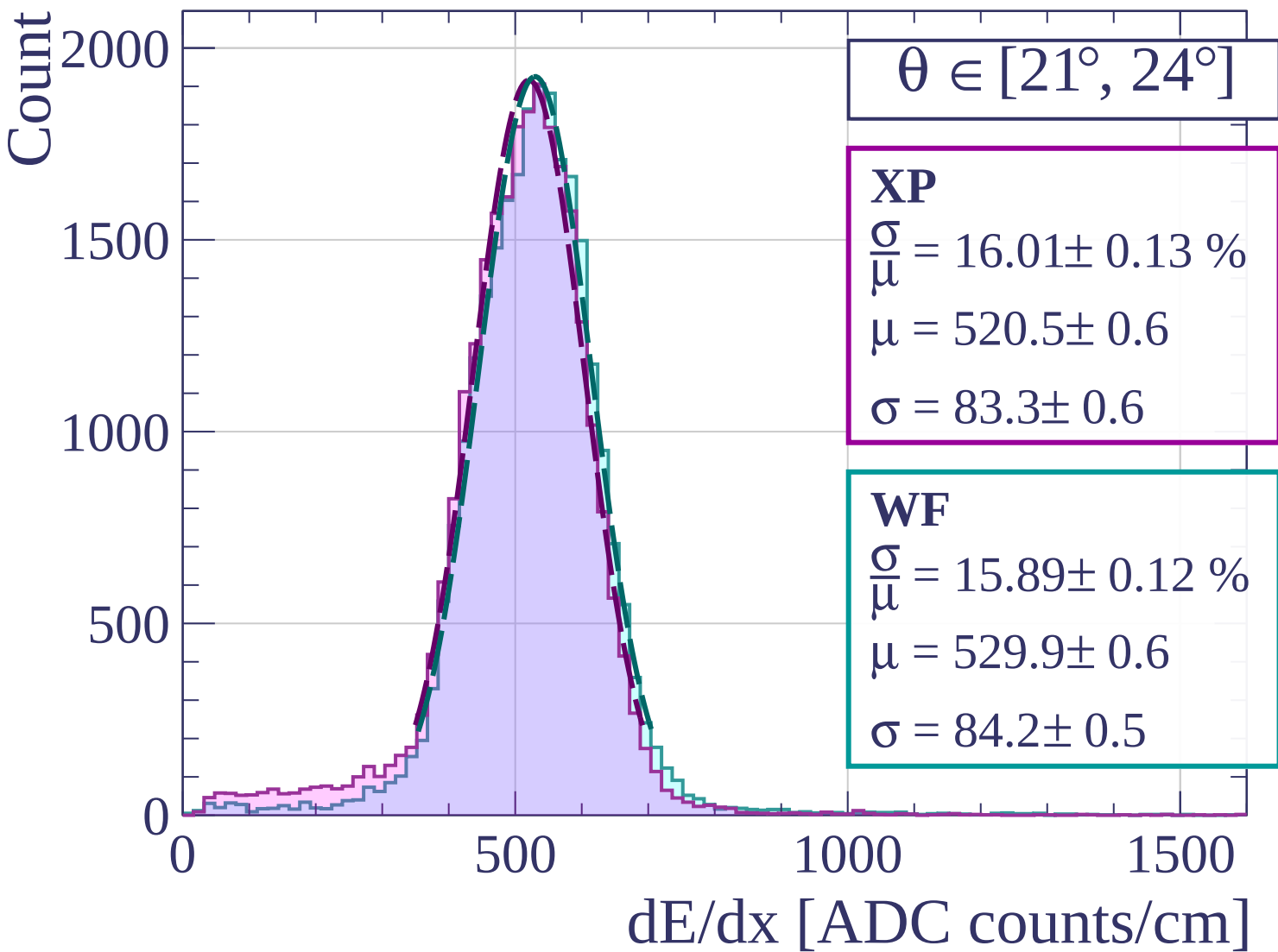


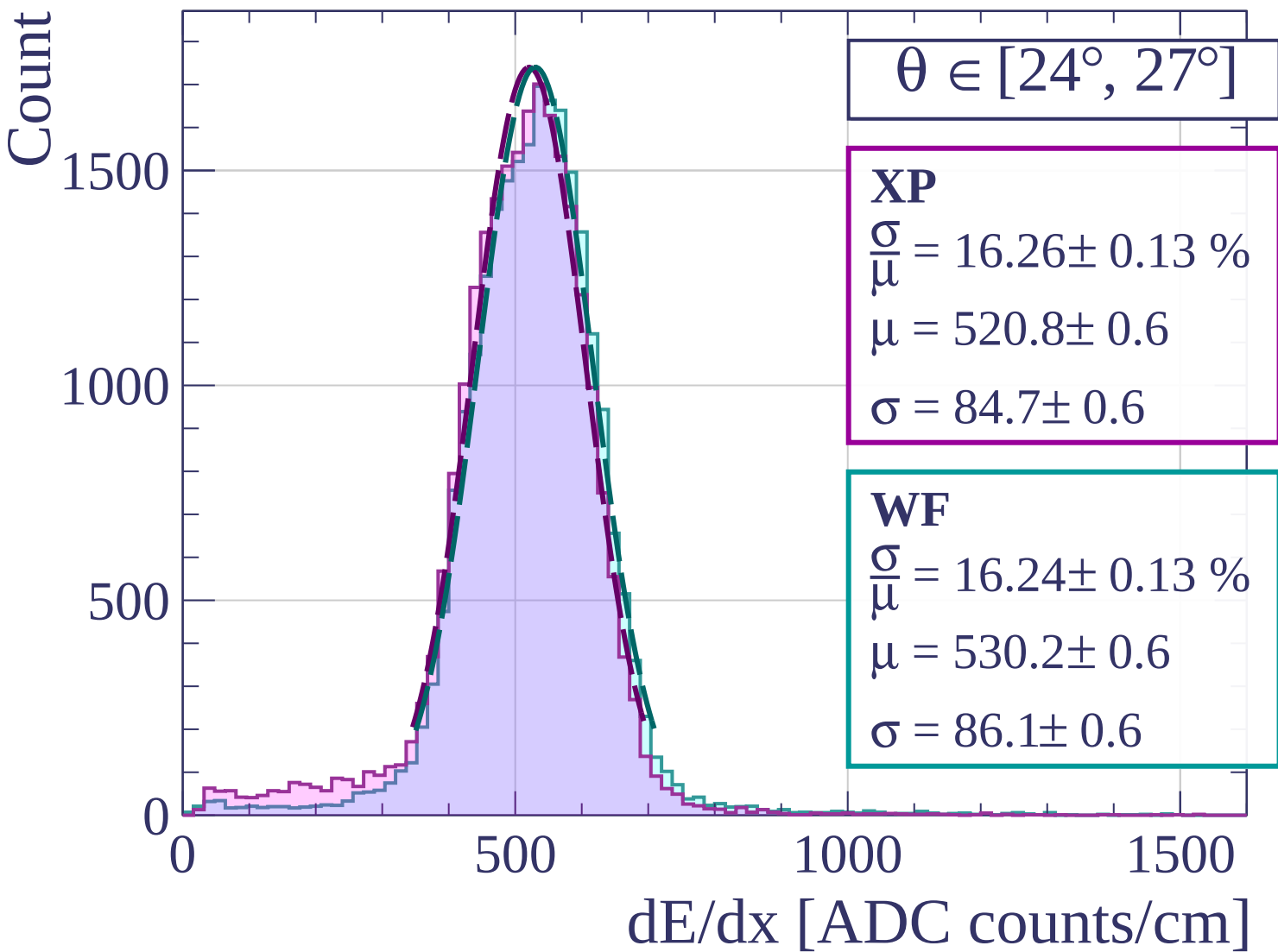


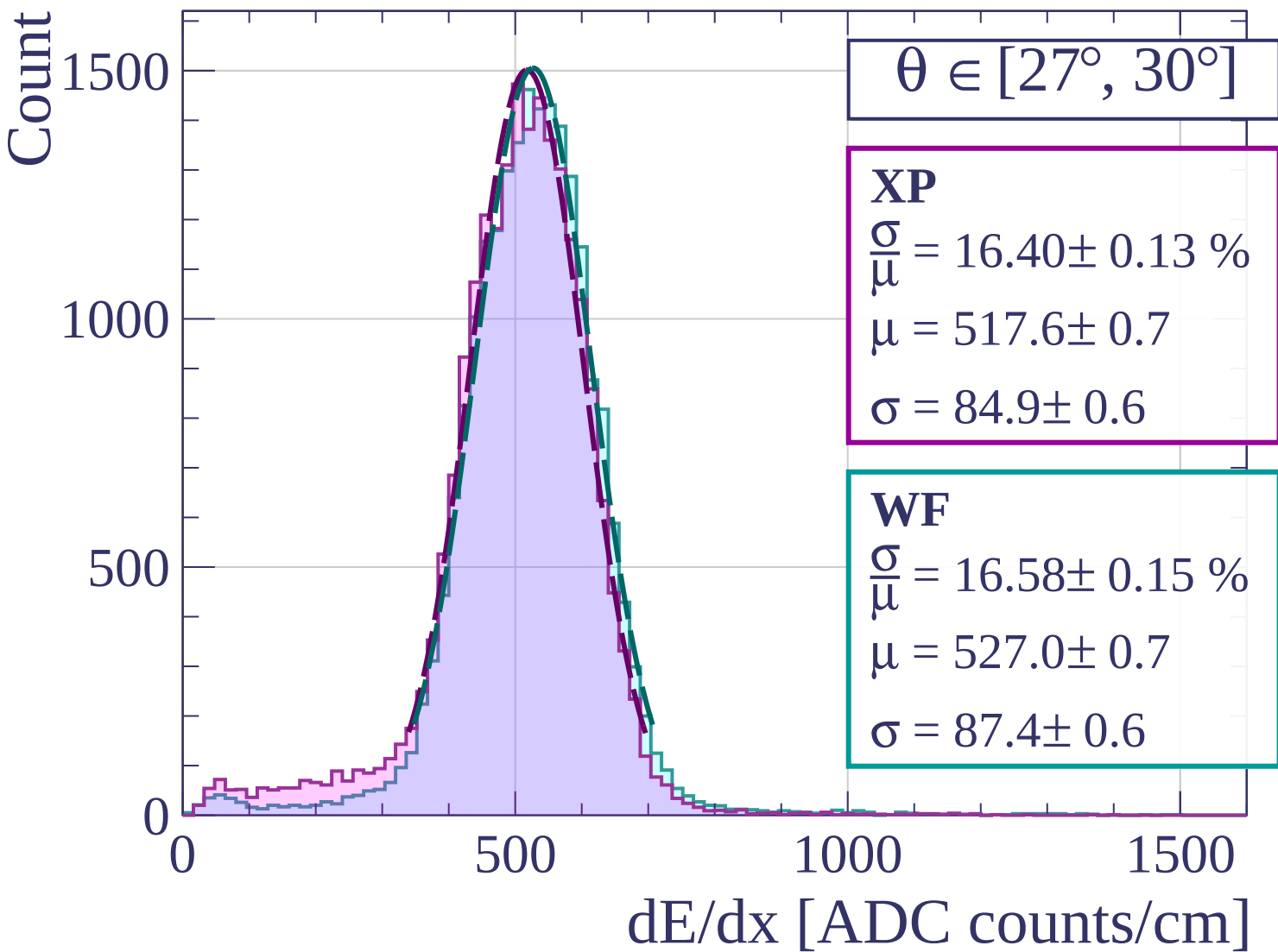


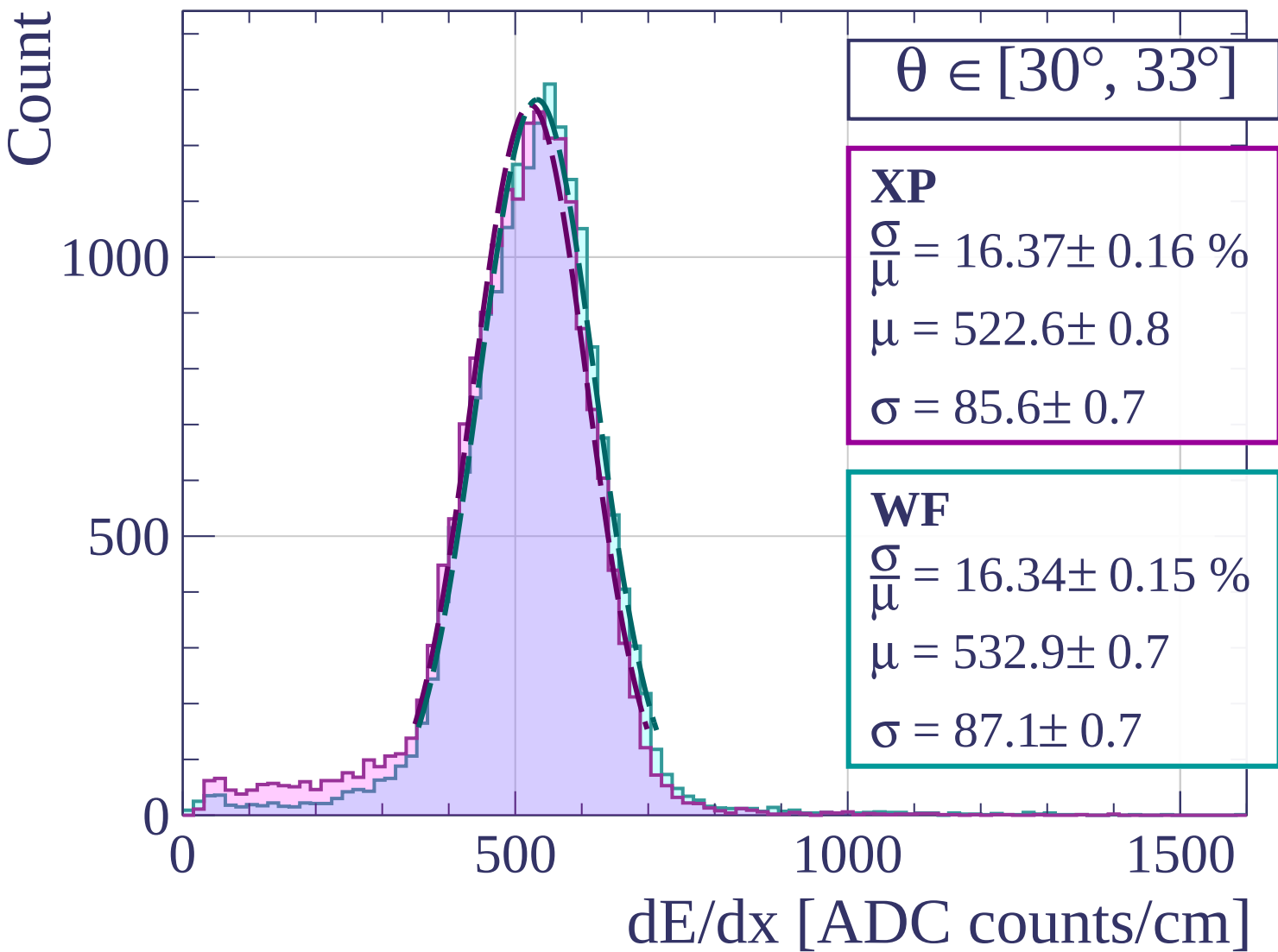


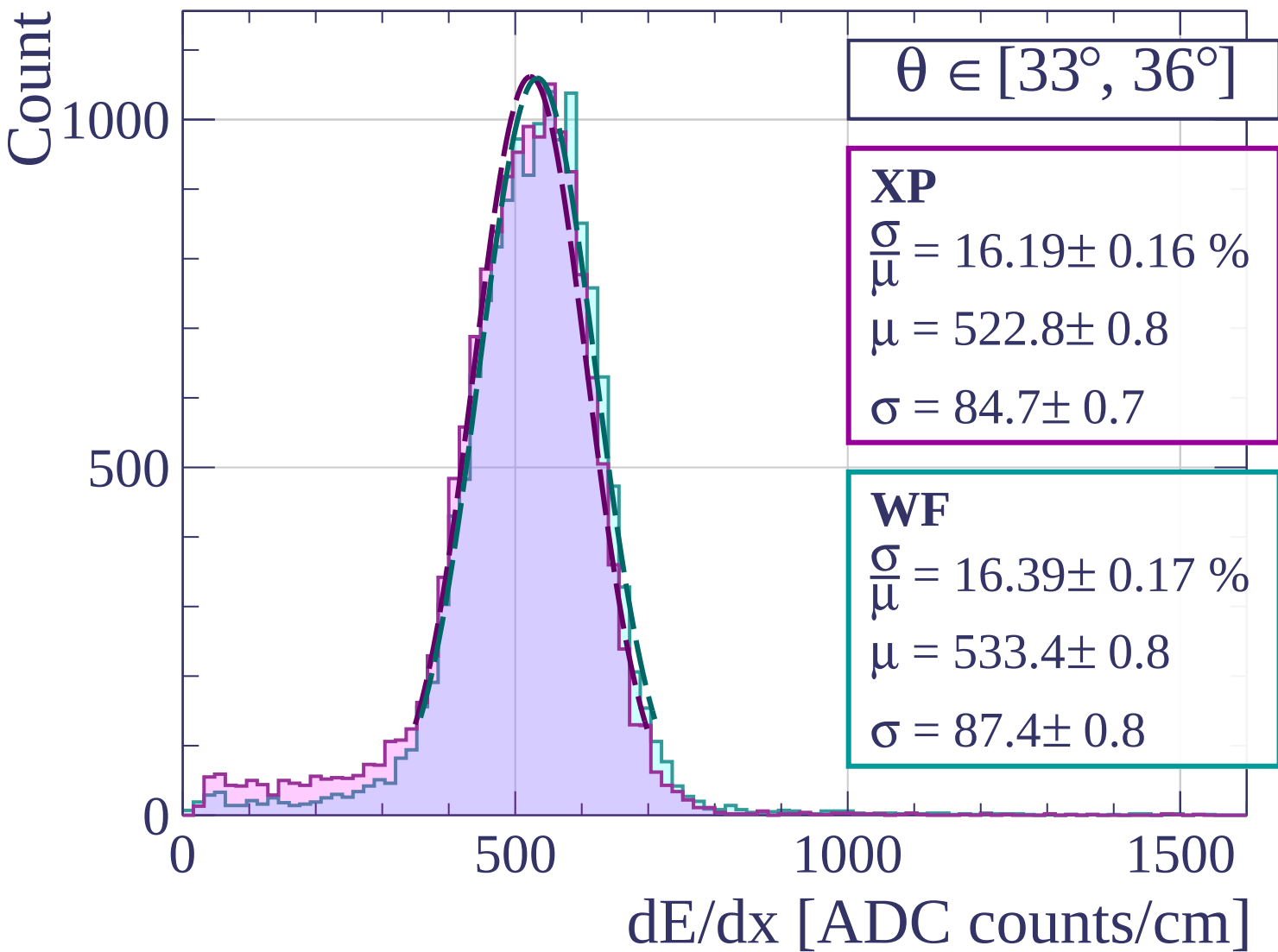


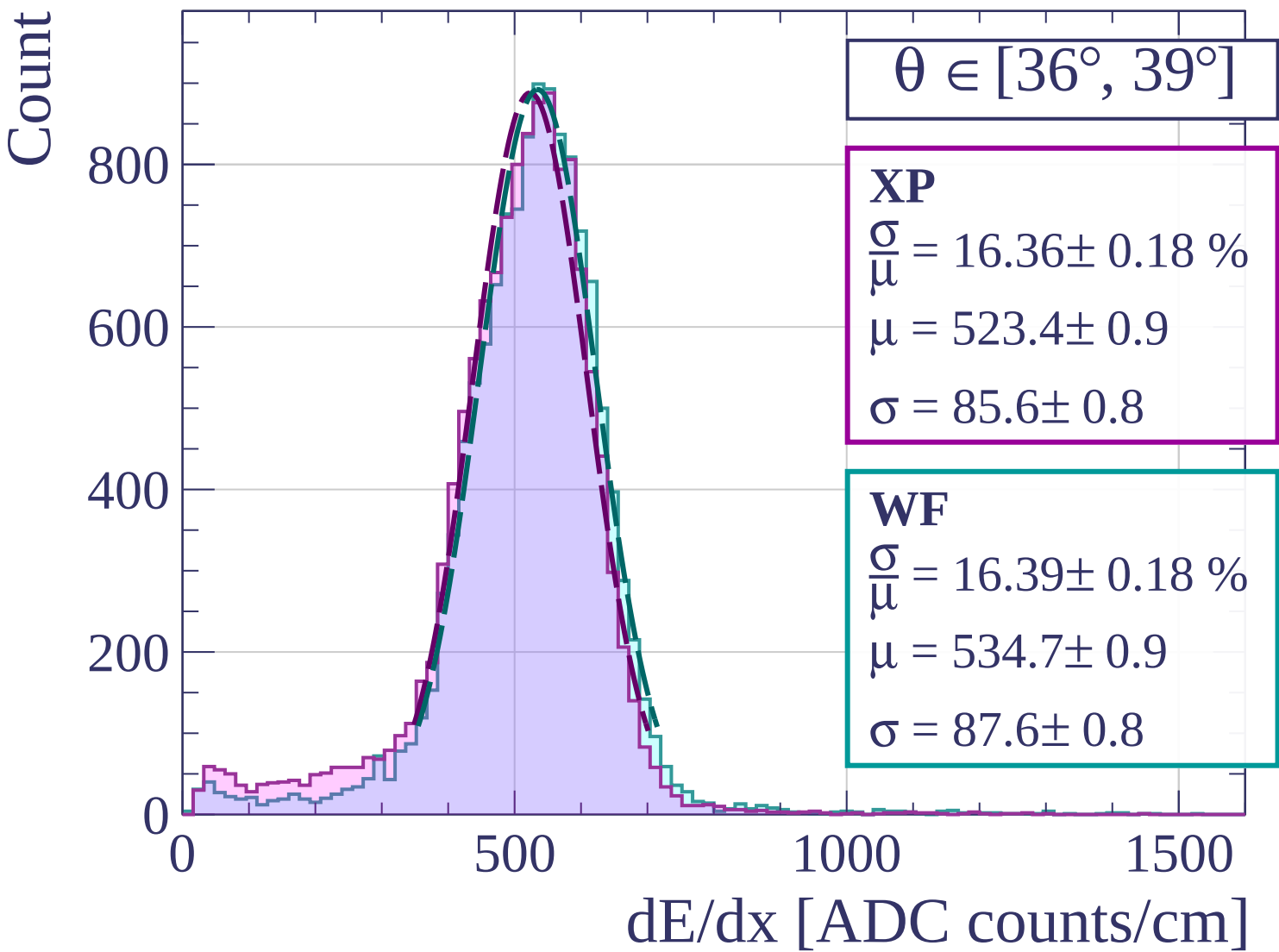












Count

$\theta \in [39^\circ, 42^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.68 \pm 0.23 \%$$

$$\mu = 525.7 \pm 1.1$$

$$\sigma = 87.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 16.78 \pm 0.22 \%$$

$$\mu = 536.2 \pm 1.1$$

$$\sigma = 90.0 \pm 1.0$$

600

400

200

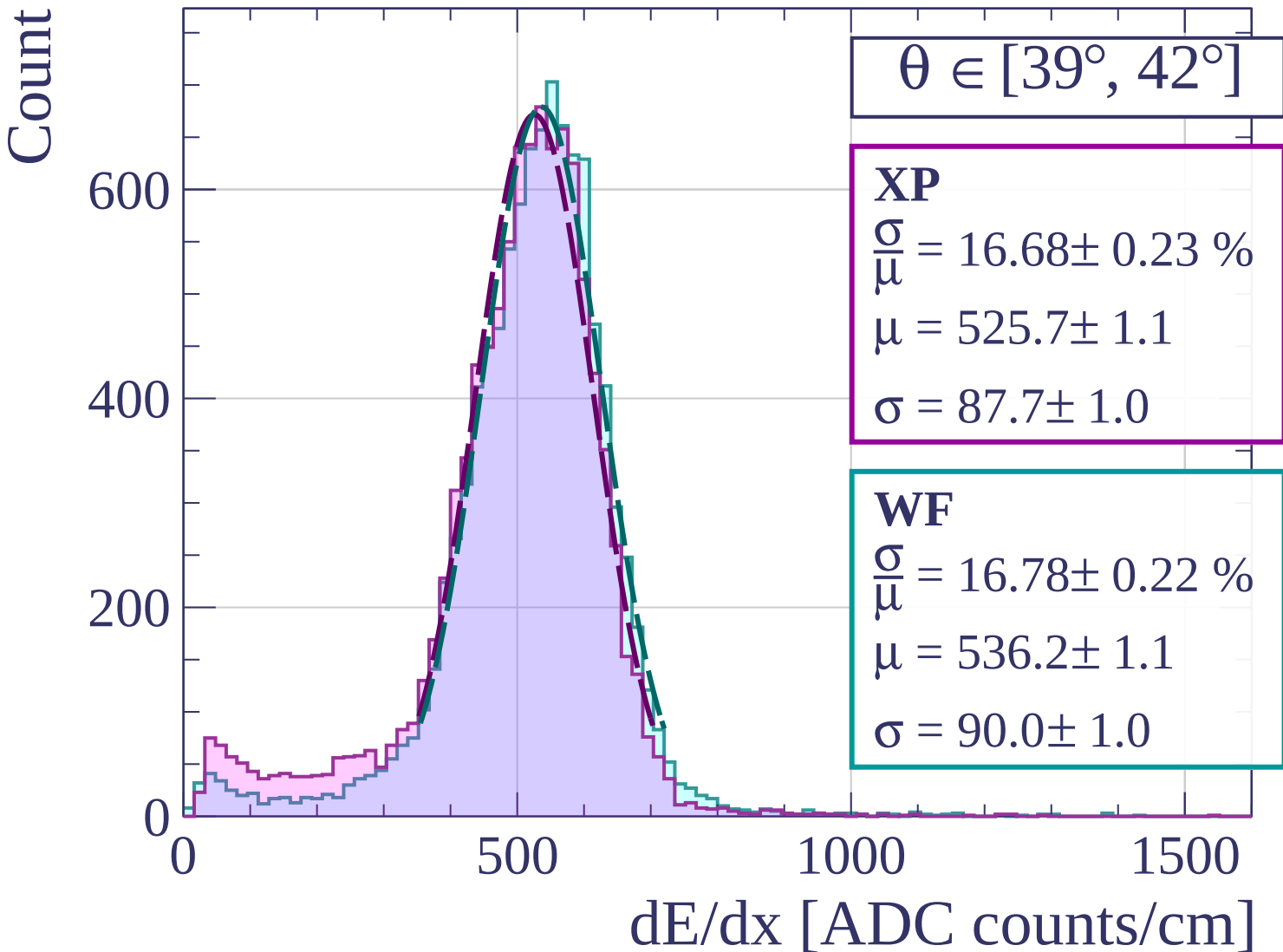
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [42^\circ, 45^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.36 \pm 0.24 \%$$

$$\mu = 526.3 \pm 1.2$$

$$\sigma = 86.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 16.89 \pm 0.25 \%$$

$$\mu = 535.6 \pm 1.2$$

$$\sigma = 90.5 \pm 1.1$$

400

200

0

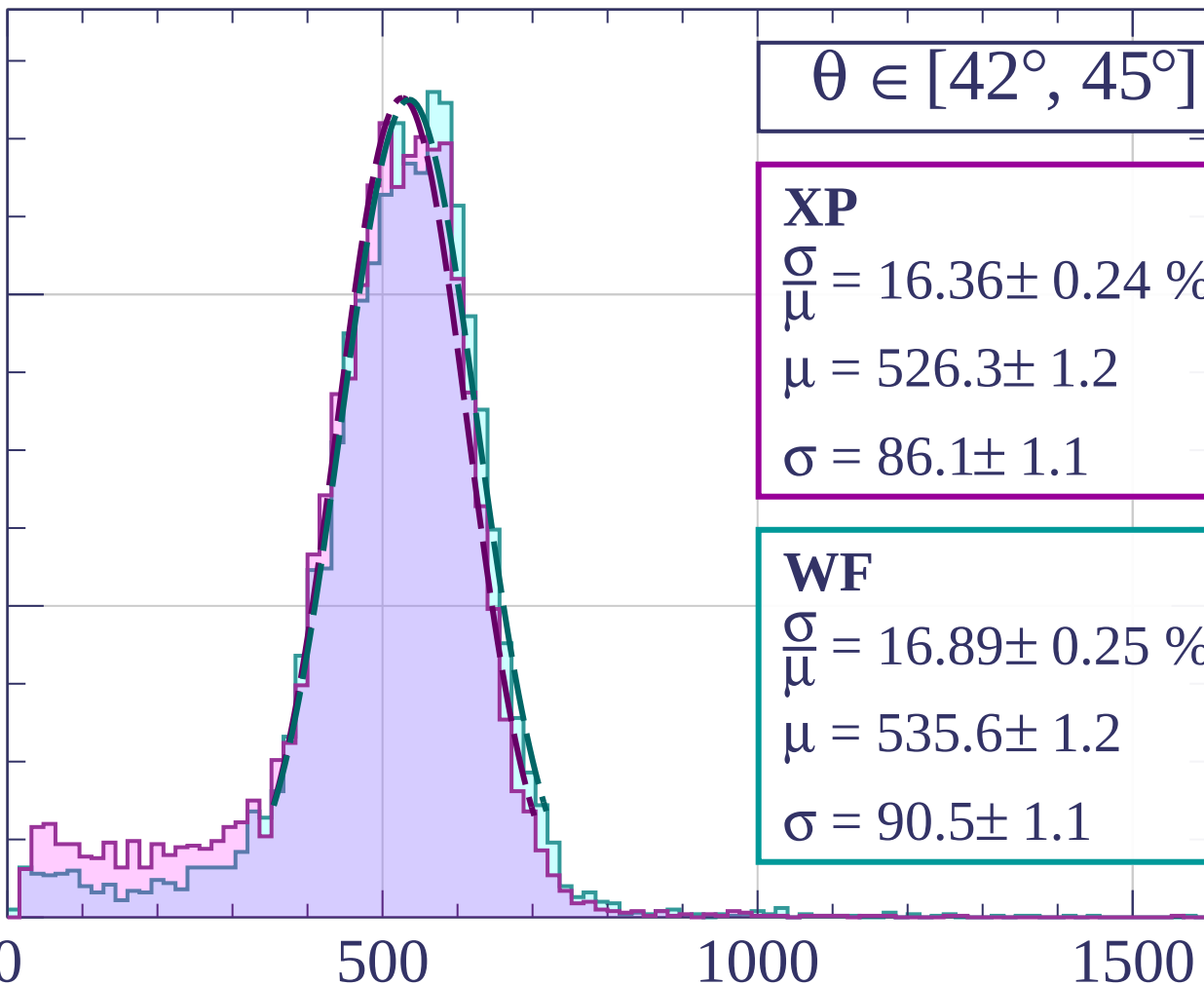
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [45^\circ, 48^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.47 \pm 0.26 \%$$

$$\mu = 527.8 \pm 1.3$$

$$\sigma = 86.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 16.94 \pm 0.27 \%$$

$$\mu = 537.7 \pm 1.4$$

$$\sigma = 91.1 \pm 1.2$$

400

300

200

100

0

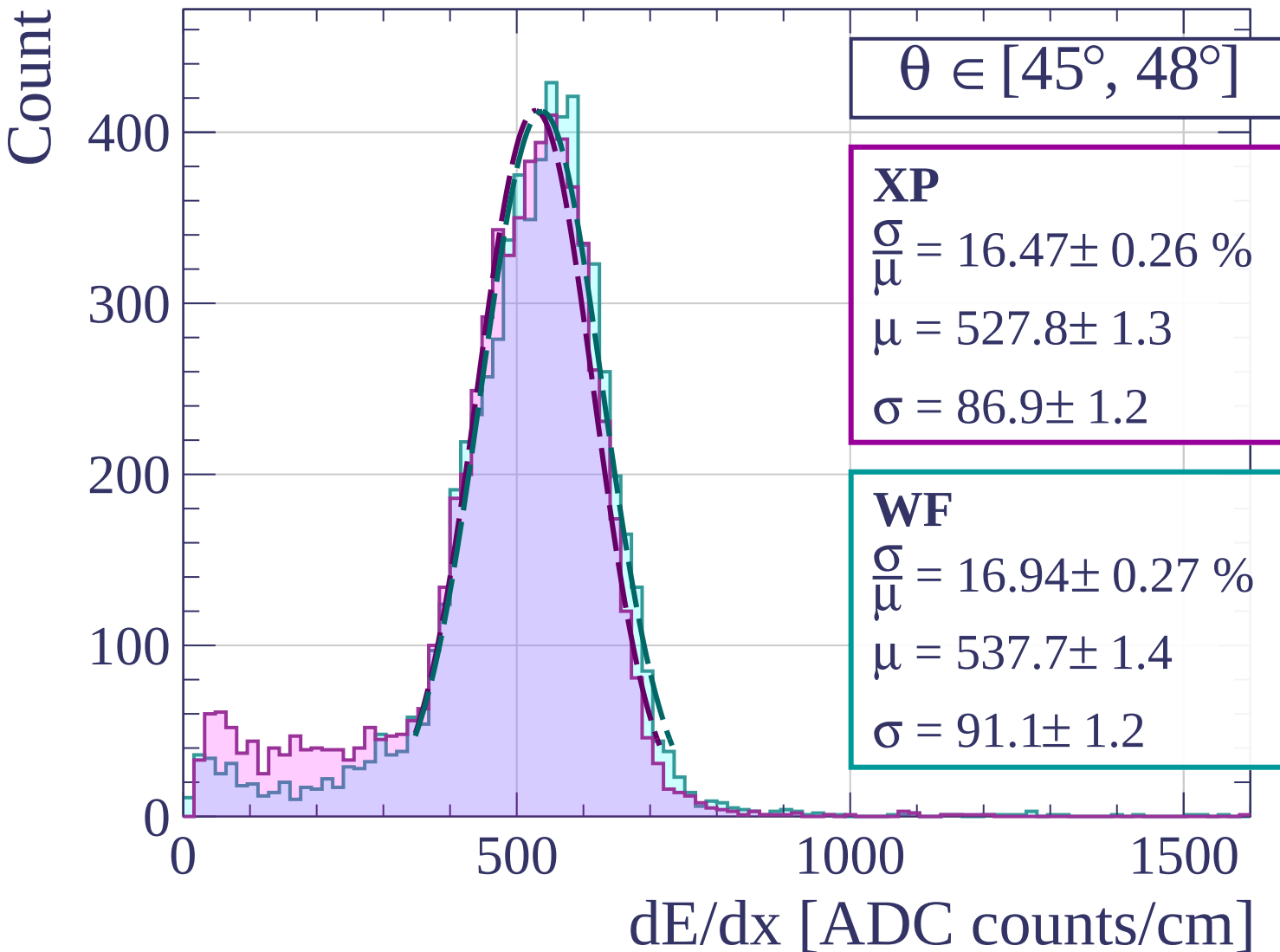
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [48^\circ, 51^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.42 \pm 0.31 \%$$

$$\mu = 528.5 \pm 1.6$$

$$\sigma = 86.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 17.08 \pm 0.32 \%$$

$$\mu = 538.5 \pm 1.6$$

$$\sigma = 92.0 \pm 1.4$$

300

200

100

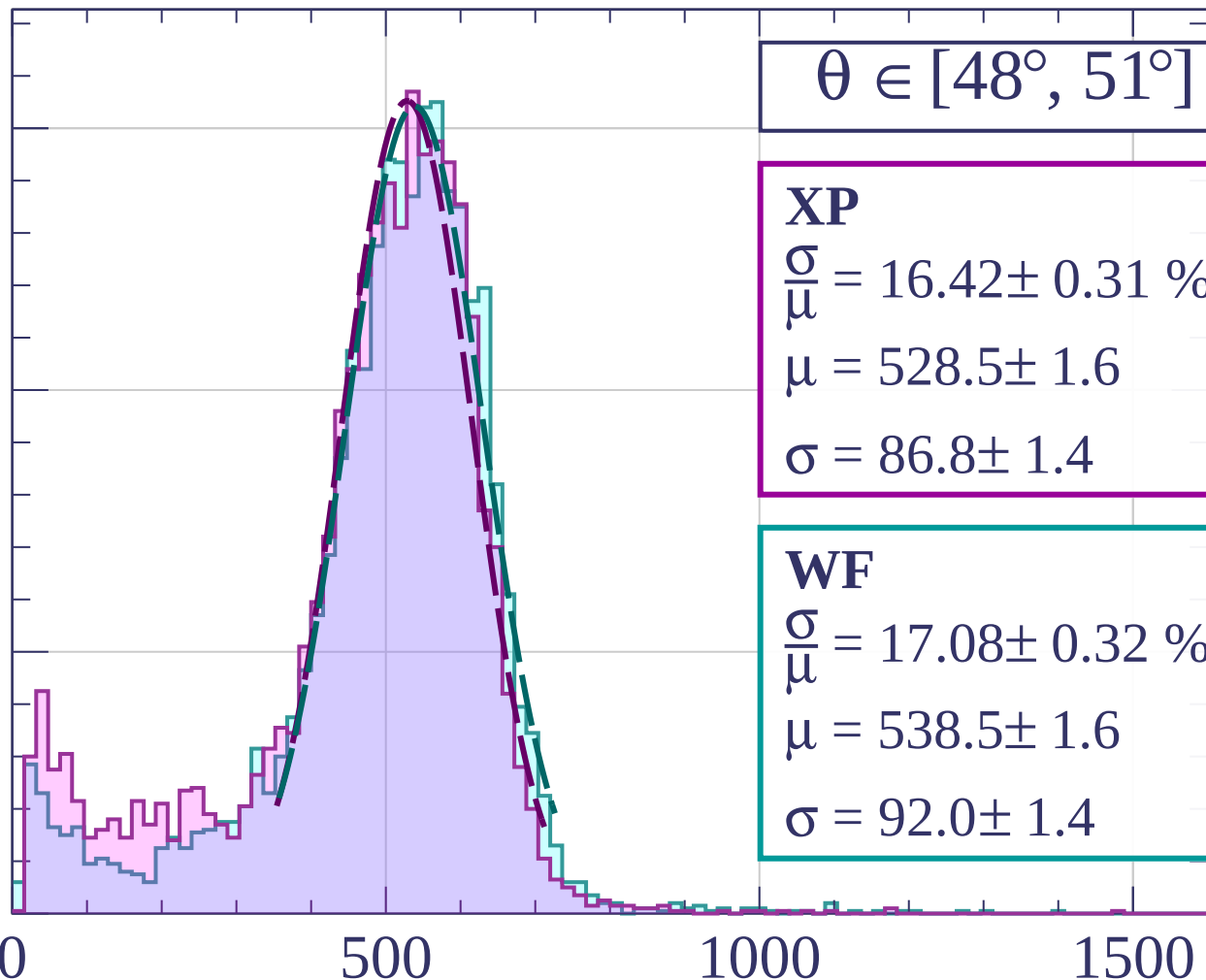
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [51^\circ, 54^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.96 \pm 0.42 \%$$

$$\mu = 528.2 \pm 2.0$$

$$\sigma = 89.6 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 17.46 \pm 0.42 \%$$

$$\mu = 537.1 \pm 2.0$$

$$\sigma = 93.8 \pm 1.9$$

200

100

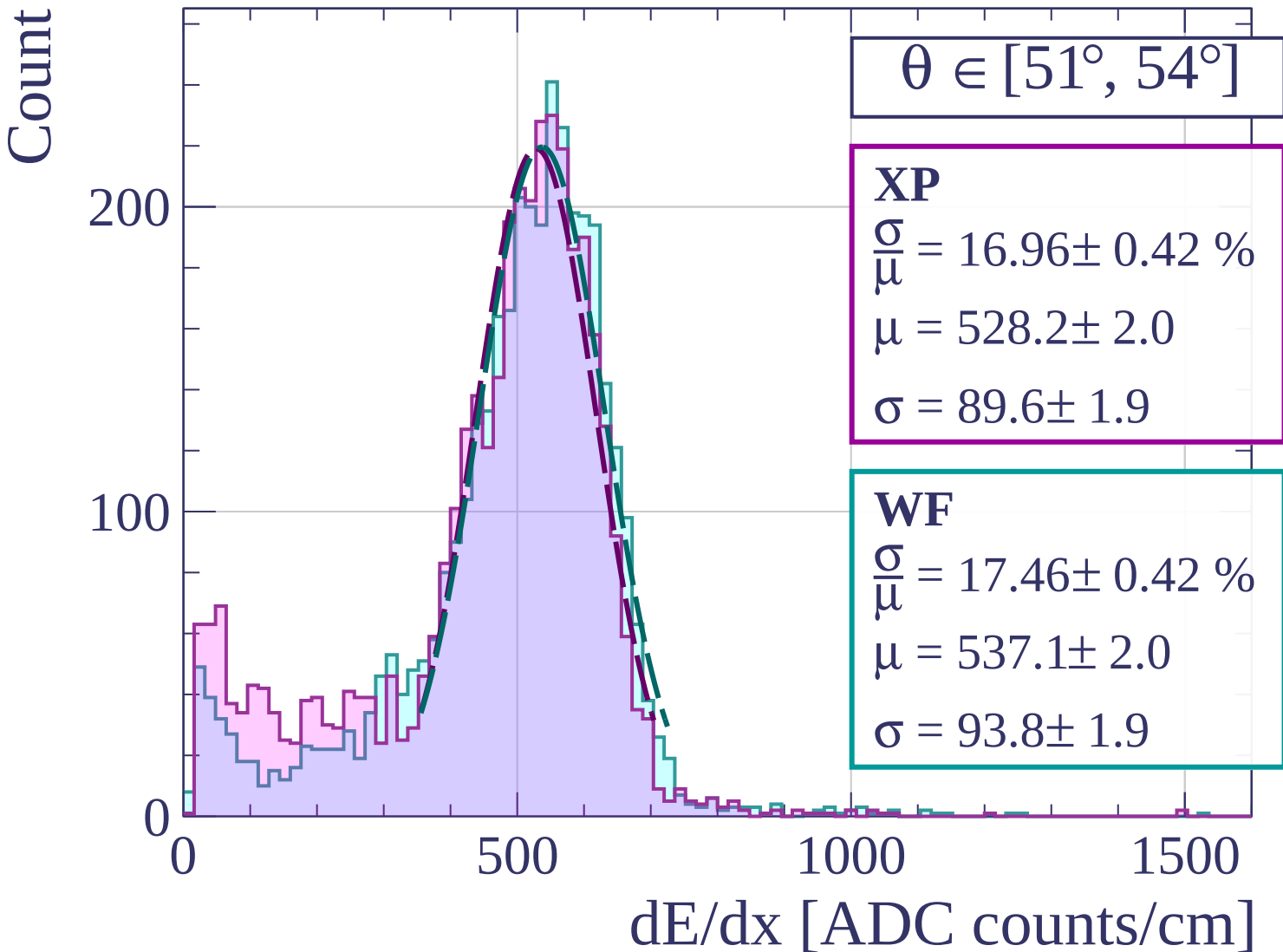
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [54^\circ, 57^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.71 \pm 0.52 \%$$

$$\mu = 531.3 \pm 2.3$$

$$\sigma = 88.8 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 17.45 \pm 0.52 \%$$

$$\mu = 538.3 \pm 2.5$$

$$\sigma = 93.9 \pm 2.4$$

150

100

50

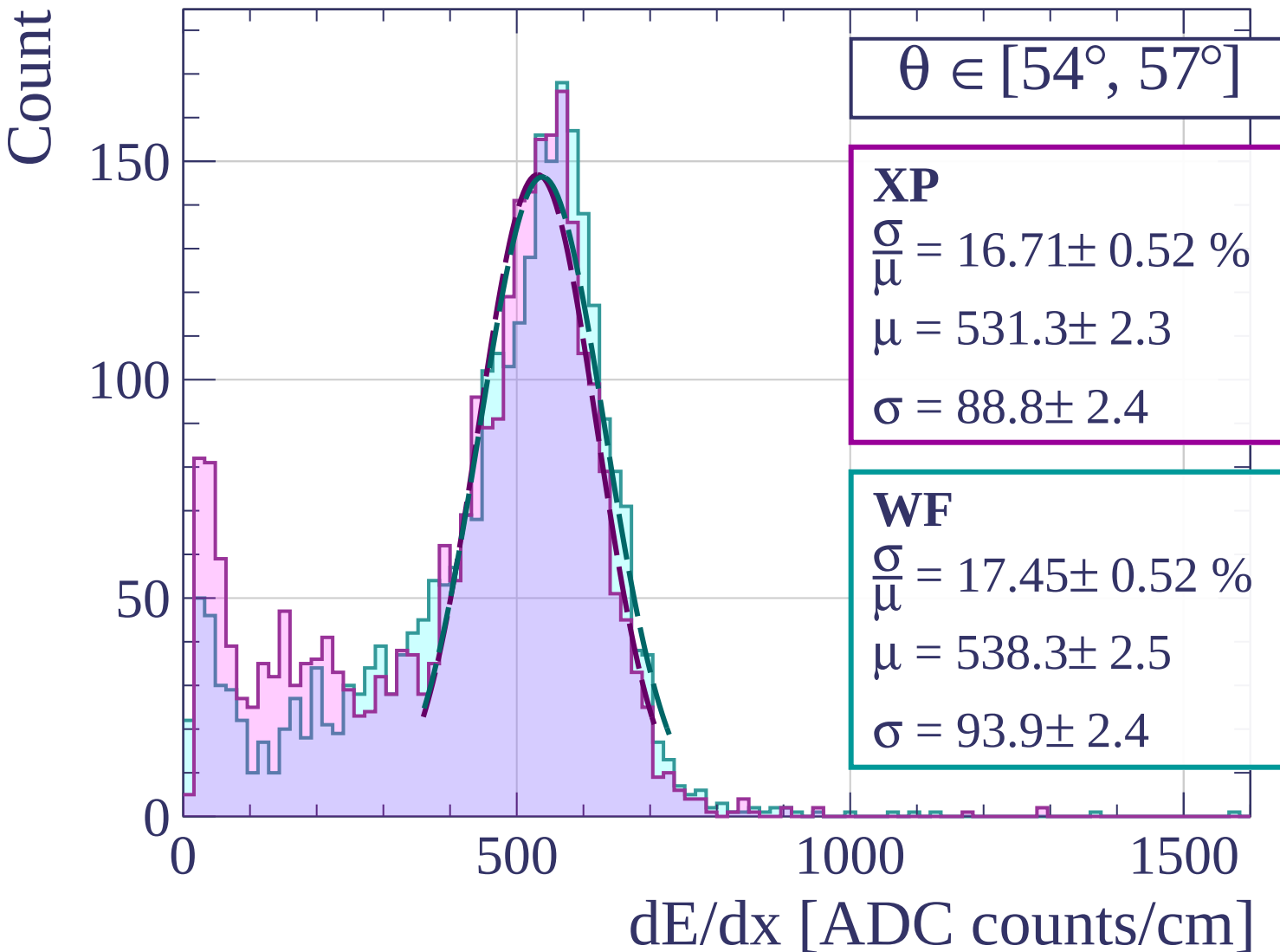
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [57^\circ, 60^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.13 \pm 0.65 \%$$

$$\mu = 525.8 \pm 2.9$$

$$\sigma = 90.1 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 19.14 \pm 0.63 \%$$

$$\mu = 527.4 \pm 3.0$$

$$\sigma = 100.9 \pm 2.8$$

100

50

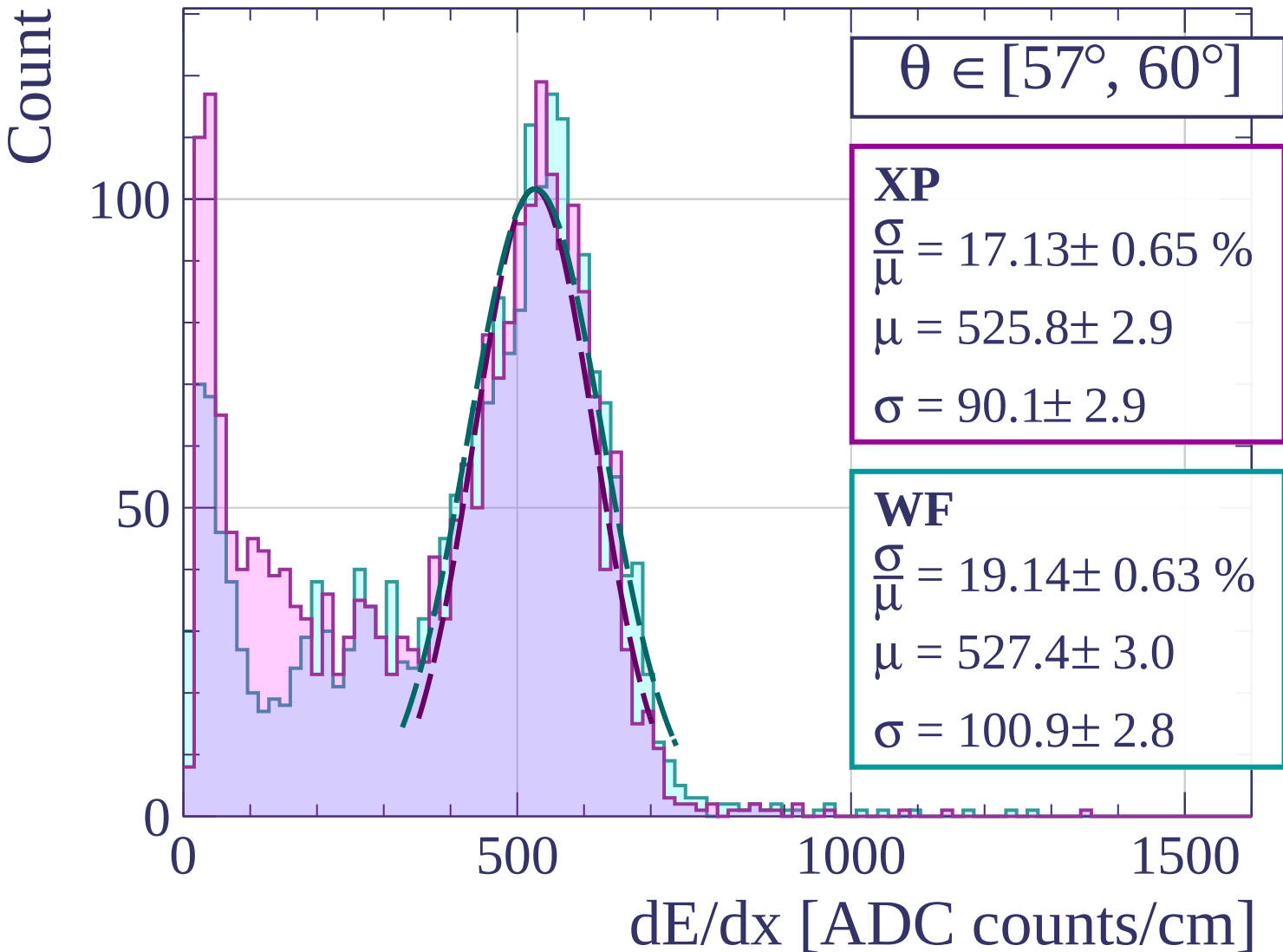
0

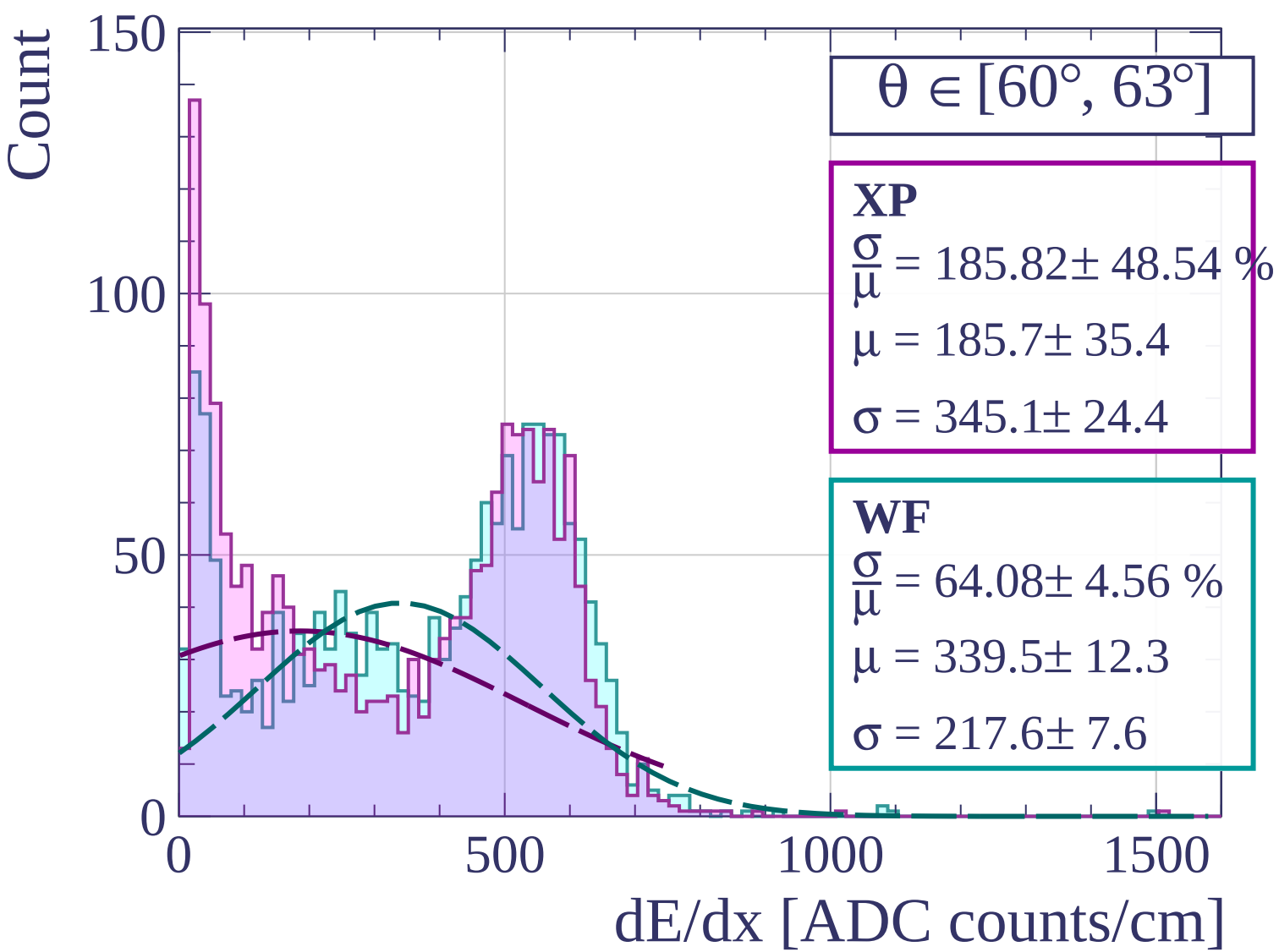
500

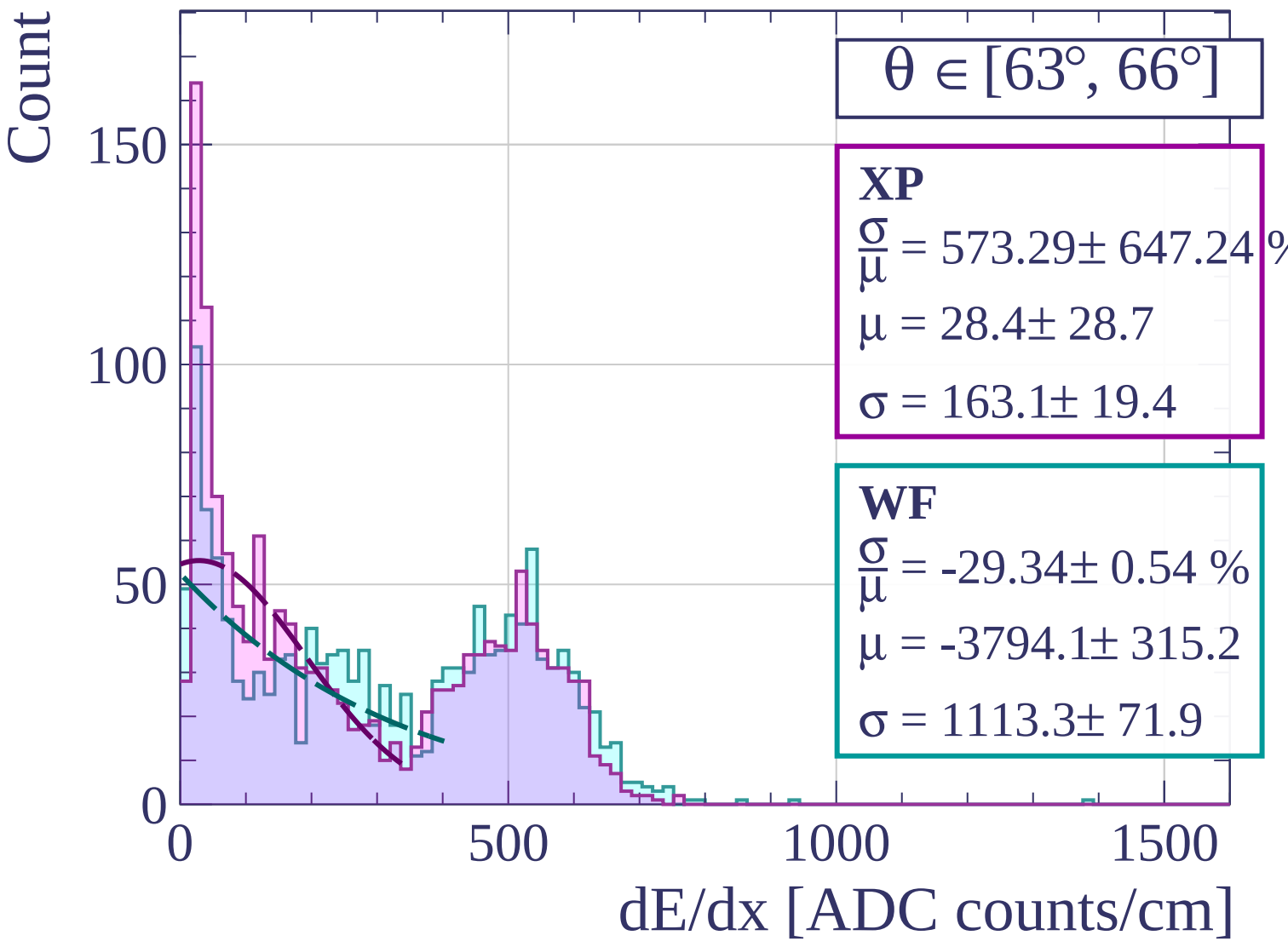
1000

1500

dE/dx [ADC counts/cm]







Count

$\theta \in [66^\circ, 69^\circ]$

XP

$$\frac{\sigma}{\mu} = 645.29 \pm 861.62 \%$$

$$\mu = 23.0 \pm 27.9$$

$$\sigma = 148.7 \pm 18.2$$

WF

$$\frac{\sigma}{\mu} = -31.22 \pm 1.64 \%$$

$$\mu = -2706.5 \pm 439.3$$

$$\sigma = 845.0 \pm 92.7$$

200

150

100

50

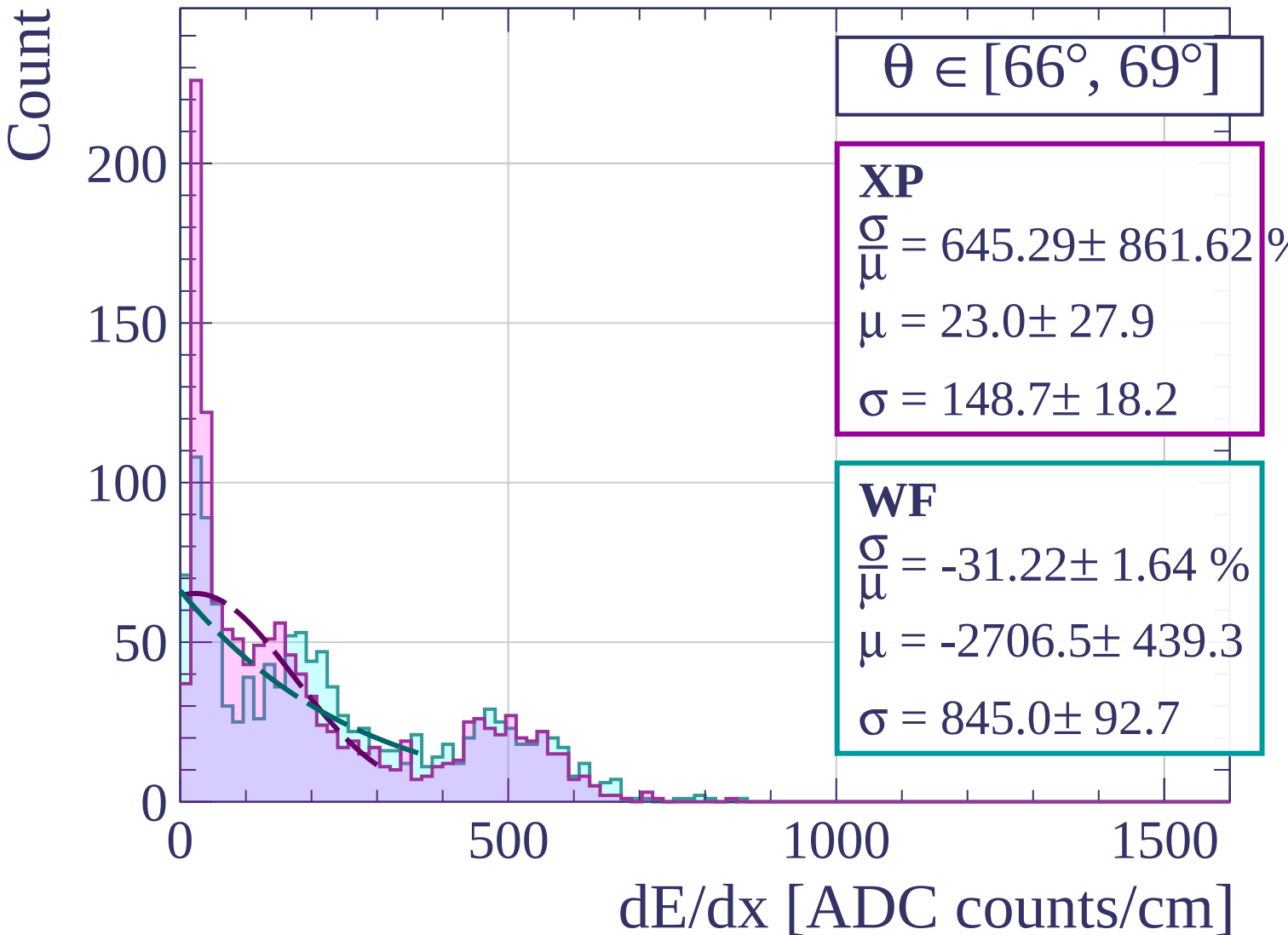
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$$\theta \in [69^\circ, 72^\circ]$$

XP

$$\frac{\sigma}{\mu} = -64.72 \pm 6.09 \%$$

$$\mu = -331.2 \pm 51.8$$

$$\sigma = 214.4 \pm 13.4$$

WF

$$\frac{\sigma}{\mu} = -27.51 \pm 1.33 \%$$

$$\mu = -2120.0 \pm 266.0$$

$$\sigma = 583.3 \pm 45.1$$

200

100

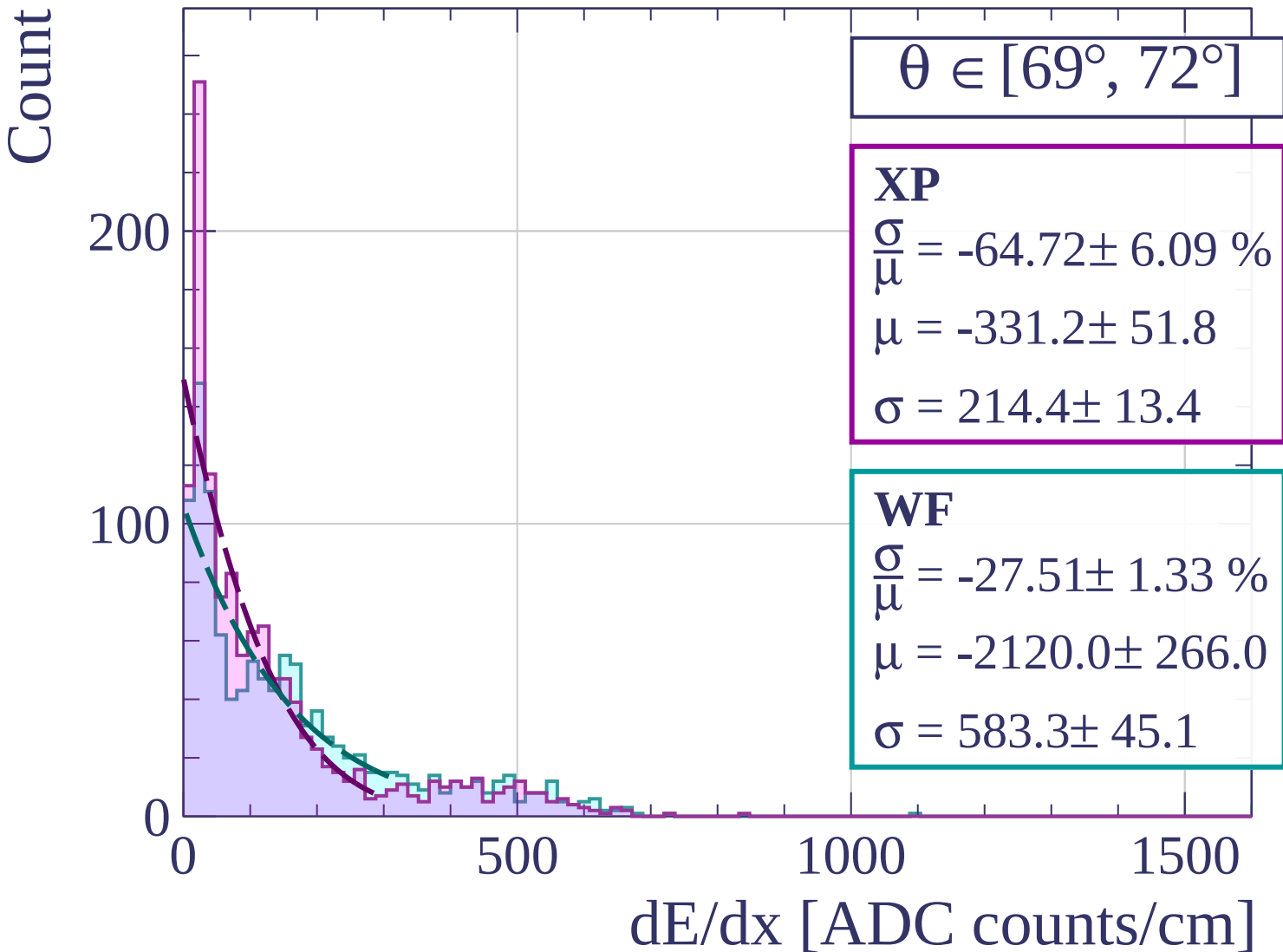
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [72^\circ, 75^\circ]$

XP

$$\frac{\sigma}{\mu} = -31.13 \pm 4.33 \%$$

$$\mu = -889.1 \pm 240.9$$

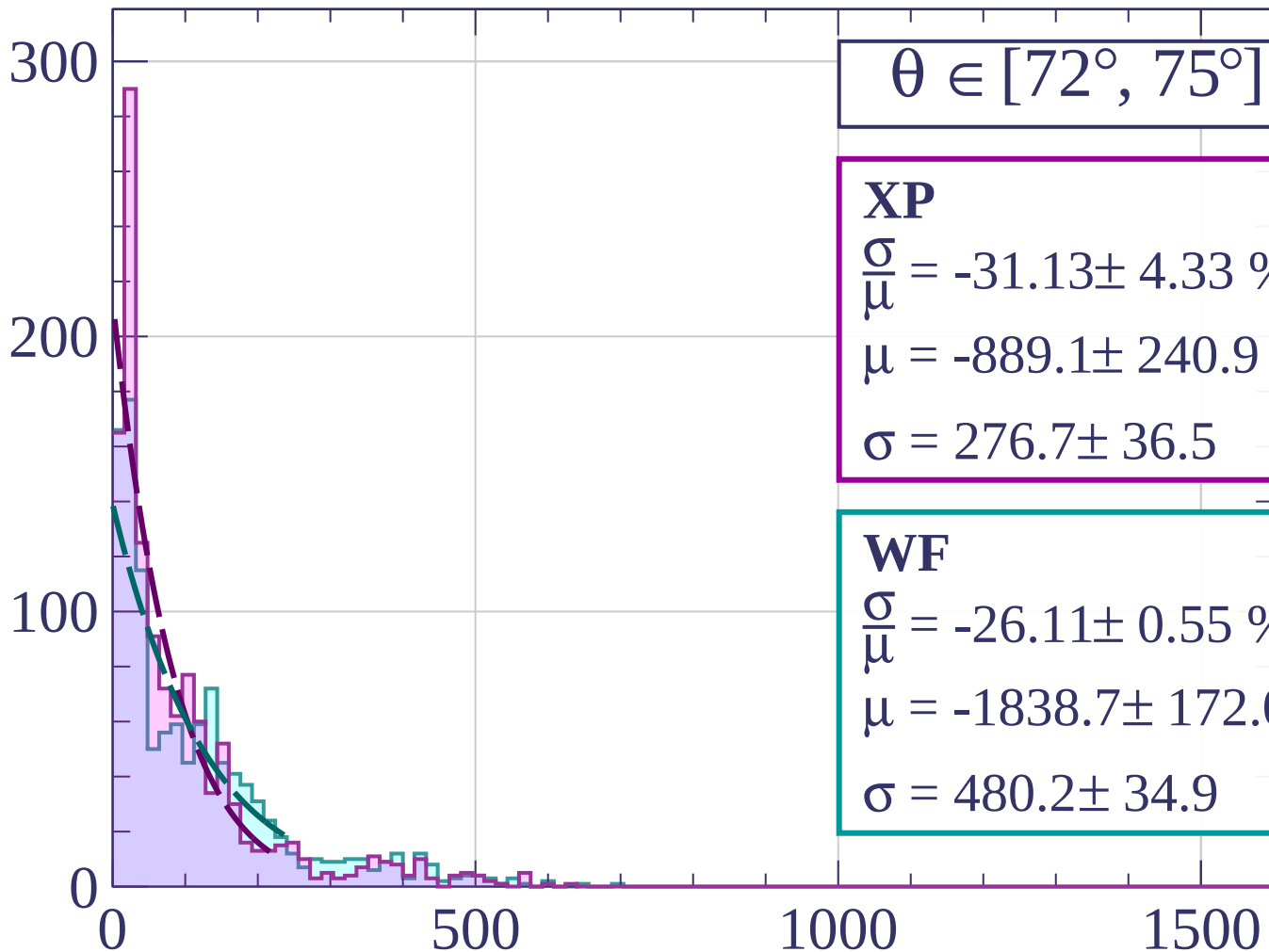
$$\sigma = 276.7 \pm 36.5$$

WF

$$\frac{\sigma}{\mu} = -26.11 \pm 0.55 \%$$

$$\mu = -1838.7 \pm 172.0$$

$$\sigma = 480.2 \pm 34.9$$



dE/dx [ADC counts/cm]

Count

$\theta \in [75^\circ, 78^\circ]$

XP

$$\frac{\sigma}{\mu} = -24.24 \pm 0.21 \%$$

$$\mu = -936.0 \pm 63.6$$

$$\sigma = 226.8 \pm 13.5$$

WF

$$\frac{\sigma}{\mu} = -23.11 \pm 0.33 \%$$

$$\mu = -1344.0 \pm 111.6$$

$$\sigma = 310.6 \pm 21.3$$

300

200

100

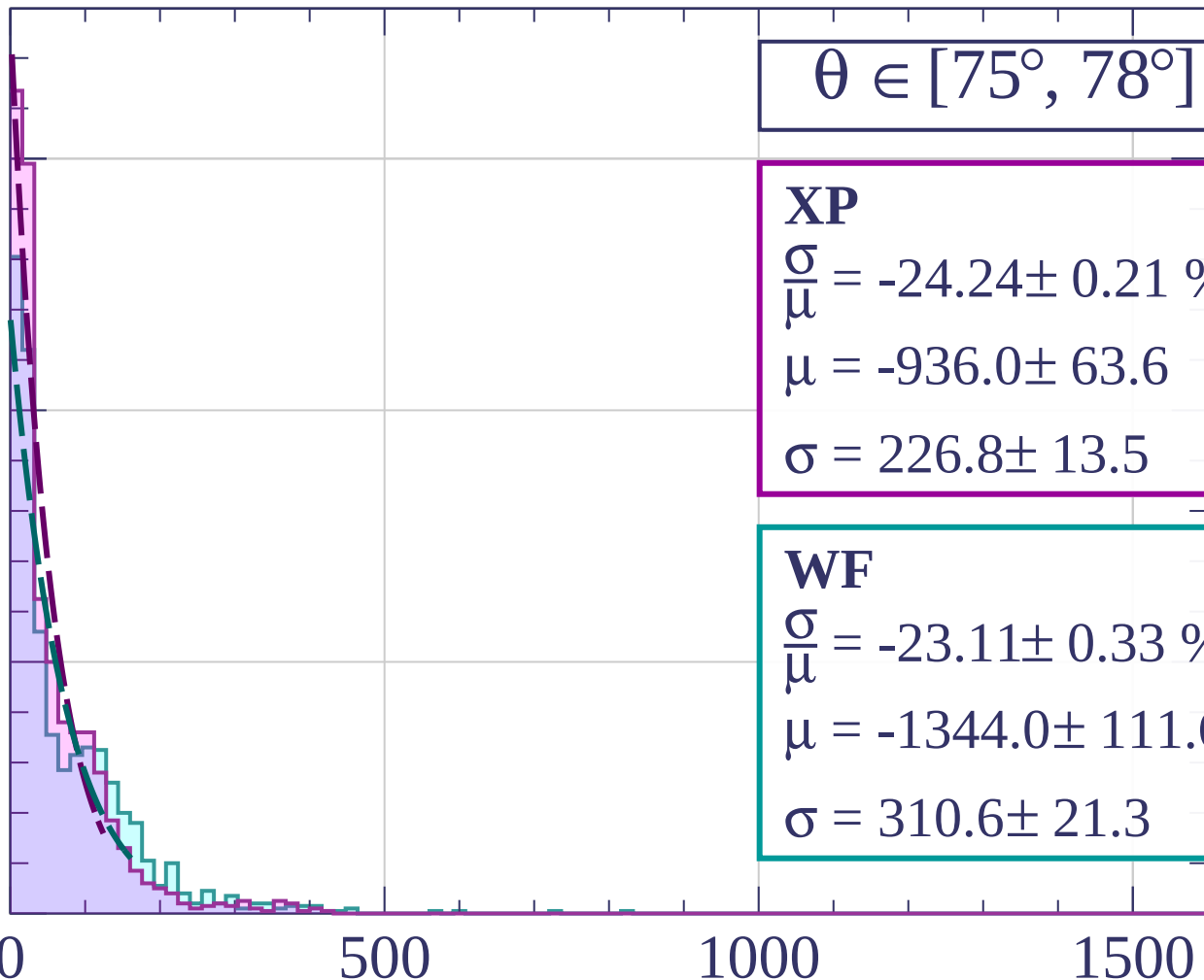
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$$\theta \in [78^\circ, 81^\circ]$$

XP

$$\frac{\sigma}{\mu} = -24.28 \pm 0.01 \%$$

$$\mu = -609.6 \pm 24.9$$

$$\sigma = 148.0 \pm 5.9$$

WF

$$\frac{\sigma}{\mu} = -742.58 \pm 573360$$

$$\mu = -27.2 \pm 210296.4$$

$$\sigma = 202.2 \pm 229.3$$

400

200

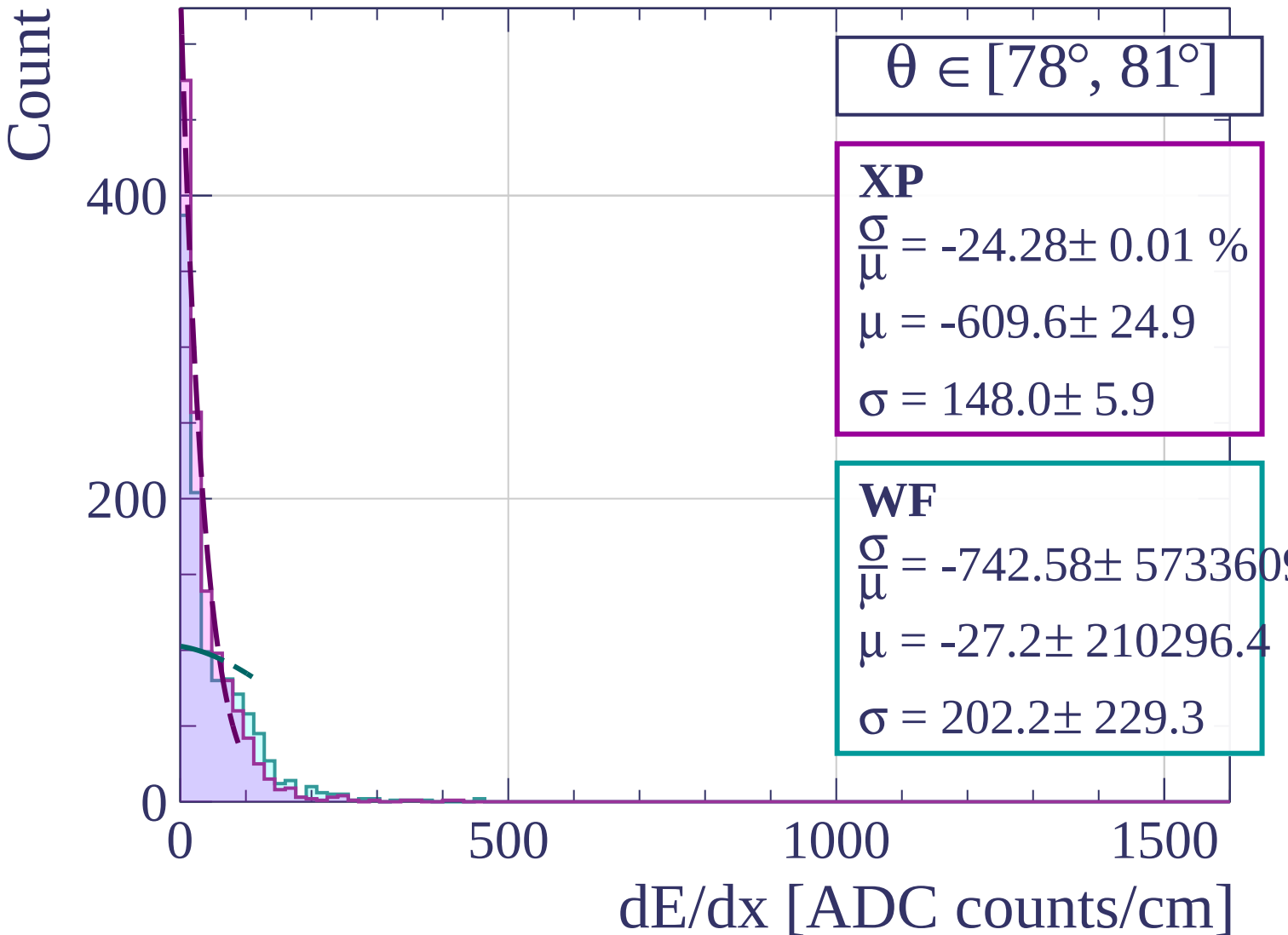
0

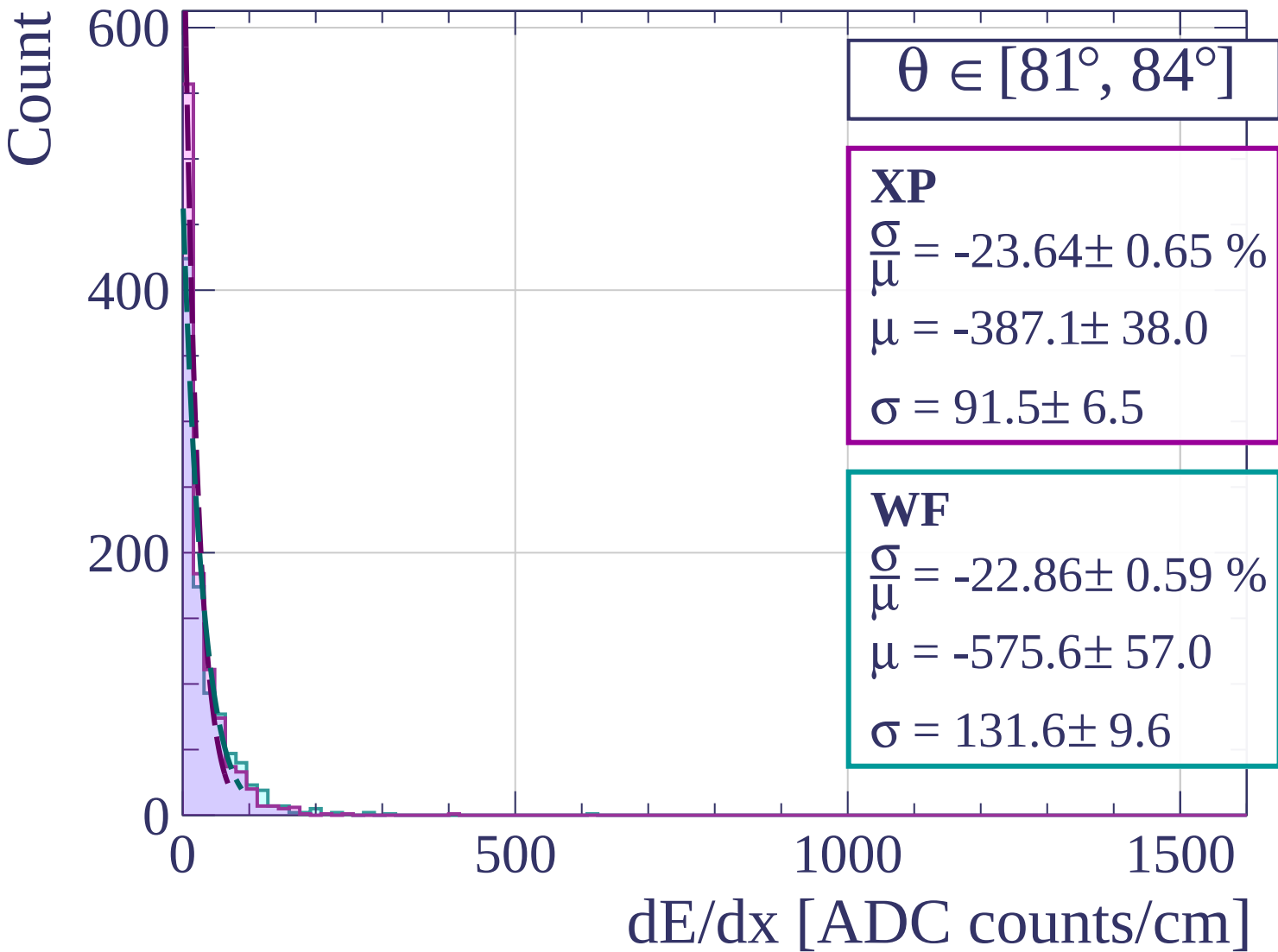
500

1000

1500

dE/dx [ADC counts/cm]





Count

$$\theta \in [84^\circ, 87^\circ]$$

XP

$$\frac{\sigma}{\mu} = 38.20 \pm 52.02 \%$$

$$\mu = 6.9 \pm 4.0$$

$$\sigma = 2.6 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = -23.68 \pm 0.66 \%$$

$$\mu = -253.8 \pm 28.6$$

$$\sigma = 60.1 \pm 5.1$$

400

200

0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

100

50

0

0

500

1000

1500

dE/dx [ADC counts/cm]

$\theta \in [87^\circ, 90^\circ]$

XP

$$\frac{\sigma}{\mu} = 3.42 \pm 6.09 \%$$

$$\mu = 7.7 \pm 0.3$$

$$\sigma = 0.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 3.42 \pm 6.75 \%$$

$$\mu = 7.7 \pm 0.3$$

$$\sigma = 0.3 \pm 0.5$$

